Why are we doing this?
The meaning of Financial Accounting

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Man goes to doctor. Says he is depressed. He tried reading a company's financial report, and he could not understand. He could not make sense of the accounting—the endless contradictions, the trivialities.

Doctor says: "Treatment is simple. Great accounting instructor Professor Shaffer teaches at the local university. Go and take his class. That should clear things up in no time."

The man bursts into tears... "But doctor. I am Shaffer."

Preface

Purpose: This text notably differs from the usual style of publisher-provided university textbooks on technical subjects. Many such texts—for good reason—seek to serve more as a *reference*, rather than a readable book. They cover a wide breadth of topics, and state the facts about those topics plainly, without too much elaboration. That makes sense for their purposes: By including all possible subject matter that any professor, at any university, might prefer to select, expressed uncontroversially and directly, they serve to *mesh with* any professor's lectures, and stand *behind* them, as a reference. That's what they are for.

This text serves a different purpose. My goal was to make this one readable book, one sequence of ideas, developed from scratch, self-contained, assuming almost no prior business knowledge, and building, illustrating, and analyzing the most important concepts within the most engaging exposition I could manage.

- (1.) It is supposed to be **one sequence of ideas,** building **cumulatively**. This means I prune the list of specific topic matters covered, in order to focus on fleshing out the most important concepts.[1]

- (2.) It is targeted to the **first-time learner**, typically a college freshman or sophomore, with no work experience, or any prior coursework in business or finance (rather than fellow accounting professors). This means that, by the standards of those who are *not* first-time learners, it will seem excessive in its explanation of business **terminology and convention**, or in some repetitions of core concepts. Based on my experience teaching, and my memory of my time as a first-time learner myself, I am convinced that such students do, in fact, benefit from **looping back** on technical terminology and concepts that were introduced once, for the first time in their lives, three chapters prior.

- (3.) It is supposed to be **interesting** (or, at least, as interesting as possible, for a technical, introductory financial accounting course). My hope is to motivate readers to care about the ideas, and help them build their confidence in their ability to reason through them on their own. This means that the exposition seeks to *evaluate* the methods taught, and

[1] As a concrete example, I do not cover a long list of contra-Revenue accounts and related transactions. Instead, I devote that space to developing and analyzing the income-statement vs. balance-sheet approaches to accounting for bad debts—a set of concepts and tradeoffs that apply generally.

I sometimes even admit my own (current) thinking about certain issues—e.g., balance-sheet vs. income-statement methods and emphases in standards.

This is not because I hope that students or other readers will share them. (I truly don't. Selfishly, it's much more fun and interesting for me when students disagree than when they share my own thoughts, which I am well-familiar with.) It's because a good intellectual skirmish requires an opening sally. Somebody has to go first. Since I'm the author of the text, I think that has to be me. Put differently, the **arguments** in my text are not about my arguments *per se*: Their real purpose is to serve as **prompts for students' own analyses and debates**. A dry, viewpoint-less listing of the facts, in my experience, does not provide sufficient stimulus for that.

And, finally, for all the reasons above, there are some technical elisions. As one example, I omit discussion of *Comprehensive Income* from this introductory text, in order to focus on building students' intuition for an overarching core concept of accounting, the relationship between the Balance Sheet and Income Statement, with a simpler, unified logic. There are other examples. In general, I prioritize *building a coherent mental picture* over some details.

Background: This began as a set of supplementary lecture notes for BUAD 280, Introduction to Financial Accounting, taught at the University of Southern California, in Fall 2019. The original purpose was to further explain and reinforce the key material in the official course textbook, *Financial Accounting, 5th ed.*, by Spiceland, Thomas, and Herrmann. These notes were originally based heavily on that text, and the slides and lectures I originally built around it. I owe enormous credit to its authors. I have sought, with some exceptions (which I flag), to make the terminological conventions consistent with that text, for the sake of my students. However, at this point, the exposition is independent and I take sole responsibility for any errors.

I created this text by meshing (i.) my course slides, (ii.) my supplementary notes, and (iii.) the audio transcript of my lectures, asking my RAs to place it into an open-source LaTeX template, and then using each subsequent lecture-prep as an opportunity to refine and edit. A note to fellow professors: Thanks to modern audio recording and transcription within the classroom, it is surprisingly easy to capitalize your course into a lecture-note book (at least, after prepping and teaching the course). And, in my experience, students benefit from having such a text. That way, there are **"no mysteries, no surprises"** (my teaching mantra)—no chasm between what they hear in lecture, and what they have in writing, to review at any time, before and/or after class.

This text has not been professionally reviewed. Perhaps you got what you paid for.

I thank my excellent research assistants, David Cai and Jack Zhou, for typesetting, proofing, and consultation, in translating my course slides and lectures into this book.

Contents

Title **i**

Preface **iii**

1 Introduction and Overview **1**

1.1 The core problem of accounting: TMI . . . 2
1.2 What is accounting? Three definitions . . . 3
1.3 A simple model of a business . . . 5
1.4 Financing basics . . . 5
1.4.1 Equity financing . . . 6
1.4.2 Credit financing . . . 7
1.4.3 Going deeper: What exactly *is* 'ownership'? . . . 8
1.4.4 Comparing and contrasting . . . 9
1.5 Looping back: A simple model of a business . . . 9
1.6 Financial Position & Performance: Core Identities . . . 10
1.6.1 The Accounting Equation / The Balance Sheet Equation . . . 11
1.6.2 Net Income definition . . . 13
1.6.3 Dividends: A special, important category . . . 13
1.6.4 Putting the identities to work . . . 14
1.6.5 Retained Earnings Identity . . . 15
1.7 Periodic Reporting . . . 16
1.8 Tour of the Financial Statements . . . 17
1.9 The Balance Sheet . . . 17
1.9.1 Balance Sheet: Key Principles . . . 19
1.10 The Income Statement . . . 20
1.10.1 Accrual Accounting Revenue Recognition . . . 21
1.10.2 Going deeper: What is 'money,' anyway? . . . 23
1.10.3 Accrual Accounting Expense Matching . . . 23
1.10.4 Accrual Accounting Purpose . . . 25
1.10.5 Returning to the bathtub . . . 25

1.11 Statement of Shareholders' Equity . . . 25
1.12 The Cash Flow Statement . . . 27
1.12.1 Categories of Transactions: Operating, Investing, Financing . . . 28
1.12.2 Why the Cash Flow Statement? . . . 29
1.13 Other information . . . 30
1.14 Says who? Standard setters, regulators, auditors . . . 30
1.14.1 GAAP . . . 31
1.14.2 IFRS . . . 31
1.14.3 Convergence . . . 31
1.14.4 Auditors . . . 32

2 Accounting for External Transactions 33
2.1 Transaction 1: Sell Shares of Stock for $15k . . . 35
2.1.1 Takeaways . . . 36
2.2 Transaction 2: Borrow $5k cash from the bank . . . 37
2.2.1 Takeaways . . . 37
2.3 Transaction 3: Purchase Equipment for $10k in Cash . . . 38
2.3.1 Going deeper: Why does the purchase price deserve our confidence? . . . 39
2.3.2 Takeaways . . . 40
2.4 Transaction 4: Pay two years of rent in advance, $2.4k . . . 40
2.4.1 Takeaways . . . 41
2.5 Transaction 5: Purchase Supplies for $1k "on Account" . . . 42
2.5.1 Takeaways . . . 43
2.6 A Quick Step Back: Our First Balance Sheet . . . 44
2.7 Transaction 6: Providing Services for $3.9k in Cash . . . 44
2.7.1 Takeaways . . . 46
2.8 Transaction 7: Providing Services on Account, $3.1k . . . 46
2.8.1 Takeaways . . . 47
2.9 Transaction 8: Customers Pay Cash in Advance ($800) . . . 48
2.9.1 Takeaways . . . 50
2.10 Transaction 9: Paying Salaries to Employees, $1.8k . . . 51
2.10.1 Takeaways . . . 52
2.11 Transaction 10: Pay a $.1k Dividend . . . 53
2.11.1 Takeaways . . . 54
2.11.2 Going deeper: The owner who also does the work . . . 55
2.12 Going Beyond: Transaction 11, Collecting on an Account . . . 56
2.13 Going Beyond: Transaction 12, Paying the Electricity Bill . . . 56
2.14 Putting It Together: A Worked Example . . . 57

2.15 What's next? . . . 58

3 Debits and Credits and their uses: Journal Entries, T-accounts, and Trial Balances 59

3.1 Organizing our accounting . . . 60
3.2 Debits and Credits: *The* Tools of Accounting . . . 62
3.2.1 The rules . . . 62
3.2.2 How to remember all this: memorize one thing . . . 64
3.2.3 The why . . . 64
3.2.4 Why revenues are credits, and expenses are debits . . . 65
3.2.5 Jargon: 'debit accounts' and 'credit accounts' . . . 66
3.3 Journaling the 10 Transactions . . . 67
3.4 Revisiting T-accounts . . . 72
3.5 The Trial Balance at this point in time . . . 73
3.6 Going beyond: two trickier transactions . . . 75
3.6.1 Transaction 11: Our credit customer pays off their account . . . 75
3.6.2 Transaction 13: We provide the training that was paid for in advance . . . 76

4 End of Period: Adjusting Entries ("Internal Transactions") 79

4.1 Warming up: Reading T-accounts . . . 80
4.2 "Adjusting Entries" / Internal Transactions . . . 80
4.3 Prepaid Rent → Rent Expense . . . 81
4.3.1 Asset vs. Expense: A Question of *When* . . . 83
4.4 Supplies → Supplies Expense . . . 84
4.4.1 Discussion: Income Statement vs. Balance Sheet Perspectives . 85
4.5 Equipment → Depreciation Expense . . . 88
4.5.1 Why depreciation can feel strange . . . 90
4.5.2 Going deeper: A contract can run down while the player improves 92
4.6 Generalizing: Asset → Related Expense . . . 93
4.6.1 Takeaway: Asset Balances, Expenses, and Purchases . . . 94
4.7 Salaries Expense: A period expense . . . 96
4.7.1 Takeaway: Expenses are decreases in "net assets" . . . 97
4.7.2 Discussion: Does the external-vs.-internal distinction really *matter*? . . . 98

5 Closing the Books: The Mechanics of Financial Reporting 99

5.1 The "mechanics" of the financial reporting process . . . 99
5.1.1 Unadjusted Trial Balance . . . 100
5.1.2 Adjusting Entries . . . 101

5.1.3 Adjusted Trial Balance . . . 102
5.1.4 CTRL-C, CTRL-V to the Income Statement . . . 103
5.1.5 Ordering the Income Statement . . . 104
5.1.6 CTRL-C, CTRL-V to the Balance Sheet . . . 104
5.1.7 Note: We could have used the "closing process" . . . 105
5.1.8 Reporting "mechanics" wrap . . . 107
5.1.9 Classification and Ordering . . . 107
5.1.10 The end-product, out in the wild . . . 110
5.2 The Truth About Depreciation . . . 110
5.2.1 The White Lie . . . 110
5.2.2 The More Complicated Truth: Introducing "contra-accounts" . . . 110
5.2.3 Reporting at Book Value (or not) . . . 111
5.2.4 Going deeper: What the separate balances can tell us . . . 112
5.2.5 Playing it back: book value, with a wrinkle . . . 113

6 Introduction to Ratio Analysis 115
6.1 The basic idea . . . 115
6.2 Profitability Ratios . . . 117
6.2.1 Return on Equity: . . . 117
6.2.2 Important calculation detail: Use the *average* value of B/S scaling factors . . . 119
6.2.3 Return on Assets: . . . 120
6.3 The Drivers of ROE: The DuPont Decomposition . . . 121
6.3.1 Profit margin . . . 122
6.3.2 Asset Turnover . . . 123
6.3.3 Going deeper: Why margin and turnover can move in opposite directions . . . 124
6.3.4 Leverage . . . 124
6.4 Credit Risk Ratios . . . 126
6.4.1 The current ratio . . . 127
6.5 Using Ratio Analysis: Benchmarking, and Comparative Analysis . . . 127
6.6 Leverage and risk . . . 128
6.6.1 Leverage at home: houses and mortgages . . . 130
6.6.2 Going deeper: Bank equity is financing, not idle cash . . . 132

7 Looping Back:
Accrual Accounting Deep Dive 133
7.1 Timing Differences:
(Revenues and Expenses) x (Deferrals and Accruals) . . . 134

7.2 Accruals, Cash, and Working Capital . . . 136
7.2.1 Revenue Transactions . . . 136
7.2.2 Going deeper: Deferred revenue and the Zynga estimate . . . 142
7.2.3 Expense Transactions . . . 144
7.2.4 Going deeper: What does "investing in our future" mean? . . . 151
7.2.5 Generalizing: Cash, accruals, and "working capital" . . . 152
7.2.6 A Preview: The Statement of Cash Flows . . . 152
7.3 Accrual Accounting x Internal Transactions . . . 154
7.3.1 The Relationship Between the Two . . . 154
7.3.2 External vs. Internal: What really *is* the difference? . . . 155
7.4 Connecting the Income Statement and Balance Sheet . . . 156
7.4.1 Earning Revenue from increasing assets . . . 156
7.4.2 Earning Revenue from decreasing liabilities . . . 158
7.4.3 Expenses from decreasing assets or increasing liabilities . . . 159
7.4.4 Linking the Income Statement and Balance Sheet . . . 159

8 Receivables Part 1: Accounting for Uncollectible Accounts 161
8.0.1 Going deeper: Why spend a chapter on this one problem? . . . 162
8.1 Goals: . . . 163
8.2 Overview of Methods . . . 163
8.3 Scenario . . . 164
8.4 Direct Writeoff Method . . . 165
8.4.1 Direct Writeoff Method Rules . . . 166
8.4.2 Direct Writeoff Method Accounting . . . 166
8.4.3 Evaluating the Direct Writeoff Method . . . 167
8.5 I/S Teaching Model: Percentage-of-credit sales . . . 168
8.5.1 I/S Method Rules . . . 168
8.5.2 Where does the percentage come from? . . . 170
8.5.3 I/S Method Accounting . . . 170
8.5.4 Evaluating the Percentage of Credit Sales Method . . . 172
8.6 The B/S Method / The Allowance Method . . . 173
8.6.1 B/S Method Rules . . . 173
8.6.2 Scenario and Assumption: Percentage-of-Receivables . . . 174
8.6.3 B/S method accounting . . . 174
8.6.4 Evaluating the B/S Method . . . 177
8.6.5 The general form of the Allowance . . . 179
8.6.6 Checking the formula against the mechanics . . . 181

8.7 Debating the methods: The Balance Sheet vs. Income Statement Approach . . . 182
8.7.1 Going deeper: Why an analyst cares about the components of expense . . . 183
8.7.2 The case for 'best guess and keep moving' . . . 184
8.7.3 A more realistic scenario . . . 184
8.8 Methods for estimating the allowance . . . 186
8.8.1 Aging of Receivables . . . 187

9 Receivables Part 2: The Remainder 189
9.1 Notes Receivable vs. Accounts Receivable . . . 190
9.1.1 What's the difference? . . . 190
9.1.2 Going deeper: Bankruptcy does not necessarily mean that operations stop . . . 190
9.1.3 Accounting for Notes Receivable: Interest and Principal . . . 191
9.2 'Sales discounts': Incentives for quick payment . . . 193
9.2.1 Going deeper: When does a price difference become financing? 194

10 Inventory and COGS 197
10.1 Perpetual Inventory System . . . 199
10.2 Inventory Cost Methods: Calculating COGS . . . 201
10.2.1 Specific identification . . . 202
10.2.2 First In, First Out (FIFO) . . . 204
10.2.3 Last In, First Out (LIFO) . . . 205
10.2.4 Weighted-Average Cost . . . 205
10.2.5 Inventory Costing Method Problems with Timing Tricks . . . 207
10.2.6 Why do these problems? . . . 208
10.3 Comparing Inventory Costing Methods . . . 208
10.3.1 Why not specific identification? . . . 209
10.3.2 Comparing FIFO and LIFO . . . 210
10.3.3 FIFO, LIFO, and Taxes . . . 213
10.3.4 Going deeper: Why reported profit and taxable profit differ . . . 213
10.4 Inventory, COGS, and their relationship . . . 216
10.5 'Levels of profitability' / Multi-Step Income Statement . . . 217
10.5.1 Gross Profit . . . 218
10.5.2 Operating Income (and EBIT) . . . 219
10.5.3 Multi-Step Income Statement . . . 220
10.6 New Levels, New Ratios . . . 221
10.6.1 Gross Profit Margin . . . 221

10.6.2 The Common-Size Income Statement . . . 222
10.6.3 Inventory Turnover . . . 222
10.6.4 Average Days in Inventory . . . 223
10.7 Capitalizing Shipping Costs . . . 224
10.7.1 Going deeper: Should rewriting an invoice change profit? . . . 225

11 Long-Term Assets Part 1: Complications 227
11.1 What do we know about assets so far? . . . 227
11.1.1 The plan for this chapter and the next . . . 228
11.1.2 Full circle: 'What is profit?' . . . 228
11.2 Complicating and caveating 'purchase price' . . . 229
11.2.1 Capitalizing costs for 'getting ready for use' . . . 229
11.2.2 Going deeper: Where should capitalization stop? . . . 230
11.2.3 Subtracting out salvaged materials . . . 231
11.2.4 Practice Problem: Capitalizing and Subtracting . . . 232
11.3 Selling off or disposing of an asset: Gains and Losses . . . 233
11.3.1 Gain on the Sale of a Long-Term Asset . . . 234
11.3.2 What is a 'Gain'? . . . 235
11.3.3 Loss on the Sale of a Long-Term Asset . . . 236
11.3.4 Are gains (vs. losses) good (vs. bad)? . . . 237
11.3.5 A useful formula: backing out the sale price . . . 238
11.3.6 Gains, Operating Income, and EBIT . . . 239
11.4 Other depreciation 'patterns' and methods . . . 240
11.4.1 Activity-Based Depreciation . . . 241
11.4.2 Accelerated Depreciation Methods . . . 243
11.4.3 Why (and why not) accelerated depreciation? . . . 245
11.5 Do not depreciate Land: Why? . . . 246
11.6 Intangible assets: interpretation and amortization . . . 247
11.6.1 'Build vs. Buy': A tale of two companies . . . 251
11.7 Contemporary issues: Discussion Questions . . . 252
11.7.1 Assets, investments, and intangibles, and implications . . . 252
11.7.2 Going deeper: Where might unrecorded value show up? . . . 253

12 Long-Term Assets Part 2: Fair Value 255
12.1 Introducing 'Fair Value' . . . 255
12.1.1 What if purchase price is unavailable? . . . 256
12.1.2 Why don't the appraisals match the purchase price? . . . 258
12.2 What exactly is "fair value"? . . . 258
12.3 Where else is 'fair value' used? . . . 259

12.3.1 When there is no cash involved . . . 259
12.3.2 When a 'purchase price' can't be relied upon . . . 261
12.3.3 In a variety of asset impairments and writedowns . . . 264
12.4 Corporate Mergers and Acquisitions and Goodwill . . . 272
12.4.1 Acquisition example . . . 275
12.4.2 Goodwill discussion . . . 276
12.4.3 Going deeper: Can an imperfect measure still be useful? . . . 279
12.5 Contemporary issues: Discussion Questions . . . 279
12.5.1 Fair-value accounting is a 'live' and controversial area . . . 279

13 Intro to Finance **281**
13.1 Bonds, loans, and installments are just Cash Flows . . . 282
13.2 Linking present and future value: 'discount rates' . . . 285
13.3 What determines discount rates? . . . 286
13.4 Discount rate and interest rates . . . 287
13.4.1 Present values and future values: it goes both ways . . . 289
13.4.2 Valuing a stream of future cash flows: . . . 289
13.4.3 Value as of a particular date . . . 291
13.4.4 Going deeper: A single payment can itself be an asset . . . 291
13.5 Doing the same for the bond . . . 292
13.6 Generalizing: *The* identity of finance . . . 294
13.6.1 An arbitrary example: the Crazy Money Machine . . . 295
13.6.2 What this framework applies to (and what it doesn't) . . . 295
13.7 Reversing: solving for the discount/interest rate . . . 296
13.8 Net present value (NPV) . . . 297
13.8.1 NPV examples . . . 297
13.8.2 Present Value Discussion . . . 299
13.8.3 Going deeper: Present value already looks ahead . . . 299
13.9 Other finance topics . . . 299
13.9.1 Risk and return . . . 299
13.9.2 'Market efficiency' . . . 300

14 Long-Term Liabilities **303**
14.1 What is a bond, and what is the problem? . . . 303
14.1.1 Going deeper: Fruit as an analogy for a financial contract . . . 305
14.1.2 Bond specification: . . . 305
14.1.3 Bond valuation: . . . 306
14.1.4 A challenge question: the net cash cost of financing . . . 308
14.2 Accounting for a bond issued 'at par' . . . 309

14.2.1 Bond issuance . . . 309
14.2.2 Subsequent Interest Expenses / Payments . . . 309
14.2.3 Why work with both premiums and discounts? . . . 310
14.3 Bond issued at a discount: . . . 310
14.3.1 Issuance . . . 310
14.3.2 Introducing 'Carrying Value' . . . 311
14.3.3 FAQ: Why aren't the coupon payments a liability? . . . 312
14.3.4 Subsequent expenses: . . . 312
14.3.5 Backing Up: Summarizing The Rules . . . 315
14.3.6 Intuition: What's going on here? . . . 316
14.4 Bonds issued at a premium: . . . 317
14.4.1 Differences between a bond issued at a discount . . . 317
14.4.2 The Rules . . . 318
14.4.3 Example . . . 319
14.4.4 Intuition . . . 320
14.4.5 FAQ: Why would anyone buy a bond at a 'premium'? . . . 321
14.4.6 Visualizing premium/discount amortization . . . 321
14.4.7 A deeper intuition: Carrying Value tracks fair value . . . 322
14.4.8 Real-world application: Interest-rate risk and Silicon Valley Bank 322
14.5 Repurchasing or calling a bond: . . . 324
14.5.1 Callable bonds . . . 324
14.5.2 Repurchasing bonds . . . 326
14.5.3 FAQ: Why not only issue callable bonds? . . . 326
14.5.4 Accounting for repurchasing or calling a bond . . . 327
14.6 Installment Loans . . . 328
14.6.1 Issuance date . . . 329
14.6.2 Subsequent installment payments and interest expenses . . . 329
14.6.3 Installment Payment Examples . . . 330
14.6.4 Intuition: Each payment represents interest and principal . . . 331
14.6.5 Application: Mortgages, home equity, and the rent-vs.-buy decision . . . 331
14.7 Lease accounting: . . . 332
14.7.1 The view from practice . . . 335

15 Statement of Shareholders' Equity 337

15.1 Where we stand . . . 338
15.1.1 What do we know about shareholders' equity so far? . . . 338
15.1.2 The Statement Itself . . . 339
15.1.3 What are we missing? . . . 339

15.2 Legal and institutional details on shares . . . 341
15.2.1 Going deeper: Takeover battles and poison pills . . . 343
15.3 Par value . . . 344
15.3.1 Background and meaning . . . 344
15.3.2 Accounting for par value . . . 345
15.4 Repurchasing shares . . . 347
15.4.1 Background on repurchases . . . 347
15.4.2 Accounting for repurchases . . . 349
15.4.3 Going deeper: Which part of equity did the repurchase reduce? . . 350
15.5 Re-selling repurchased stock . . . 351
15.5.1 Example . . . 351
15.5.2 Why APIC and not a 'gain'/'loss'? . . . 352
15.5.3 Why separate accounts for these three 'reversals'? . . . 353
15.6 Putting the pieces together: reverse-engineering share counts . . . 354
15.7 Preferred Stock . . . 355
15.7.1 Accounting for preferred stock . . . 356
15.8 Stock Splits and etc. . . . 357
15.8.1 What? . . . 357
15.8.2 Why? . . . 357
15.8.3 Reverse splits and stock dividends . . . 357
15.8.4 Accounting for these: We don't, in this book . . . 358
15.9 The Statement Itself . . . 359

16 Statement of Cash Flows **361**
16.1 Overview . . . 361
16.1.1 Why? . . . 361
16.1.2 How? . . . 363
16.2 Operating vs. Investing vs. Financing . . . 364
16.3 Operating Section: Direct method . . . 367
16.3.1 Example . . . 367
16.3.2 Adjusting Revenue to Cash Collections from Customers . . . 369
16.3.3 Adjusting COGS to Cash Paid to Suppliers . . . 370
16.3.4 Adjusting Salaries Expense to Cash Paid for Salaries . . . 371
16.3.5 Aggregating: Where are we going? . . . 372
16.3.6 Skip over Depreciation Expense: Why? . . . 372
16.3.7 Skip over Gains / Losses on Sales: Why? . . . 373
16.3.8 Adjusting Interest Expense to Cash Paid for Interest . . . 373
16.3.9 Adjusting *Income Tax Expense* to Cash Paid for Taxes . . . 374
16.3.10 Entire Operating Section under Direct Method . . . 374

16.3.11 Summarizing: Three Rules . . . 375
16.4 Operating Section: The Indirect Method . . . 376
16.4.1 Example . . . 377
16.4.2 Comparing Direct and Indirect Methods . . . 379
16.4.3 Boiling The Indirect Method Down to Just Two Rules: . . . 380
16.4.4 Going deeper: Does adding back depreciation create cash? . . . 381
16.4.5 Why gains and losses are removed in the operating cash-flow reconciliation . . . 381
16.4.6 Explaining everything else: working capital adjustments . . . 381
16.4.7 Reading the reconciliation as a diagnostic . . . 383
16.5 Investing section: . . . 383
16.5.1 Investments . . . 384
16.5.2 Land . . . 384
16.5.3 Equipment . . . 385
16.5.4 Consolidating . . . 386
16.5.5 Boiling Down to Rules: Investing Section . . . 386
16.6 Financing section . . . 387
16.6.1 Bonds Payable . . . 388
16.6.2 Common Stock . . . 388
16.6.3 The Key Final Financing Cash Flow: Remember Dividends . . . 388
16.6.4 Boiling Down to Rules: Financing Section . . . 389
16.7 Consolidating the Three Sections: . . . 390
16.8 Some 'tricks': e.g., non-cash transactions . . . 391
16.8.1 Going deeper: The statement over a firm's life cycle . . . 393
16.9 Applications: What Do Analysts Do With It? . . . 394
16.10 Coda: Which Is Better—Net Income or Cash Flows? . . . 395
16.10.1 Cash measures can be gamed, too . . . 396
16.10.2 The missing charge for long-term assets . . . 396
16.10.3 The verdict . . . 397

A Practice Problems and Essay Questions **399**
A.1 Chapter 1 . . . 399
A.1.1 Practice Problems: MCQ's . . . 399
A.1.2 Practice Problems: Open Prompts . . . 409
A.1.3 Essay and Discussion Questions . . . 412
A.2 Chapter 2 . . . 417
A.2.1 Practice Problems: MCQ's . . . 417
A.2.2 Practice Problems: Open Prompts . . . 428
A.2.3 Essay and Discussion Questions . . . 433

A.3 Chapter 3 . . . 438
A.3.1 Practice Problems: MCQ's . . . 438
A.3.2 Practice Problems: Open Prompts . . . 450
A.3.3 Essay and Discussion Questions . . . 456
A.4 Chapter 4 . . . 459
A.4.1 Practice Problems: MCQ's . . . 459
A.4.2 Practice Problems: Open Prompts . . . 463
A.4.3 Essay and Discussion Questions . . . 465
A.5 Chapter 5 . . . 470
A.5.1 Practice Problems: MCQ's . . . 470
A.5.2 Practice Problems: Open Prompts . . . 473
A.5.3 Essay and Discussion Questions . . . 476
A.6 Chapter 6 . . . 477
A.6.1 Practice Problems: MCQ's . . . 477
A.6.2 Practice Problems: Open Prompts . . . 478
A.6.3 Essay and Discussion Questions . . . 483
A.7 Chapter 7 . . . 486
A.7.1 Practice Problems: MCQ's . . . 486
A.7.2 Practice Problems: Open Prompts . . . 492
A.7.3 Essay and Discussion Questions . . . 494
A.8 Chapter 8 . . . 505
A.8.1 Practice Problems: MCQ's . . . 505
A.8.2 Practice Problems: Open Prompts . . . 518
A.8.3 Essay and Discussion Questions . . . 526
A.9 Chapter 9 . . . 530
A.9.1 Practice Problems: MCQ's . . . 530
A.9.2 Practice Problems: Open Prompts . . . 531
A.9.3 Essay and Discussion Questions . . . 534
A.10 Chapter 10 . . . 536
A.10.1 Practice Problems: MCQ's . . . 536
A.10.2 Practice Problems: Open Prompts . . . 544
A.10.3 Essay and Discussion Questions . . . 547
A.11 Chapter 11 . . . 550
A.11.1 Practice Problems: MCQ's . . . 550
A.11.2 Practice Problems: Open Prompts . . . 563
A.11.3 Essay and Discussion Questions . . . 565
A.12 Chapter 12 . . . 571
A.12.1 Practice Problems: MCQ's . . . 571
A.12.2 Practice Problems: Open Prompts . . . 579

A.12.3 Essay and Discussion Questions . . . 581
A.13 Chapter 13 . . . 587
A.13.1 Practice Problems: MCQ's . . . 587
A.13.2 Practice Problems: Open Prompts . . . 591
A.13.3 Essay and Discussion Questions . . . 592
A.14 Chapter 14 . . . 595
A.14.1 Practice Problems: MCQ's . . . 595
A.14.2 Practice Problems: Open Prompts . . . 608
A.14.3 Essay and Discussion Questions . . . 613
A.15 Chapter 15 . . . 617
A.15.1 Practice Problems: MCQ's . . . 617
A.15.2 Practice Problems: Open Prompts . . . 623
A.15.3 Essay and Discussion Questions . . . 629
A.16 Chapter 16 . . . 633
A.16.1 Practice Problems: MCQ's . . . 633
A.16.2 Practice Problems: Open Prompts . . . 644
A.16.3 Essay / Discussion Questions . . . 657
A.16.4 Technical written-response questions . . . 661

B Misc. Additional Proofs **665**
B.1 Proof that Present Value Equals Face Value When Coupon Rate Equals Market Rate . . . 665

Index **667**

1

Introduction and Overview

If you're studying this subject now, you've probably done an Econ 101 class. So, let's start with something you learned in that class, and which you may think you have a handle on. In Econ 101, you presumably worked with models assuming that "firms maximize their profits." Now, answer two questions:

- What *is* 'profit'?
- *Why* would we assume that firms maximize it?

These are both non-trivial, non-obvious, challenging, important questions.

Consider the first question, 'what is profit?' Until we specify exactly how we quantify it, that word 'profit' is just that—a word. To make this question concrete, let's suppose you start a business, and do the following things in your first year...:

- Put in $50k of your own money,
- get another $20k investment from your cousin,
- and a $100k loan from a bank.
- Then, buy a bunch of shirts (inventory) for $90k.
- Then, sell 50% of them for $80k in cash...
- and sell another 10% of them to customers who promise to pay *later*, $18k total.

- Then, you pay the bank $6k in interest on the loan,
- and a proportional '**dividend**' to yourself and your cousin—$5k and $2k each.

What is your 'profit' this year? What do you think? Gather the numbers above. Which of those 'count' toward or against profit?

Now consider the second question: **Why** would we assume that firms maximize 'profit' (whatever it is)? This is a non-trivial question! Firms are run by regular people, with normal desires. If they are making decisions, why would they **maximize** the firms' profits instead of their own **fun**? Or, if we think they are saintly, dutiful people, why would they maximize corporate profits instead of **aggregate social welfare**?

Here's the short answer: Accounting is ultimately about **information**, **performance measurement**, and **accountability**. 'Profit' is the accounting system's way of attempting to measure the value (wealth) that has been realized for a company's *owners*. And those owners have the ultimate lever of **control** over the company—the managers and employees of the company are ultimately **accountable** to them. So, if they *don't* attend to the owners' interests—as measured by the accounting system and referred to as 'profit'—they could be out of a job soon. More generally, owners use various **governance mechanisms** to enforce their interests.

Notice, too, what *kind* of answer that is. Nobody here is asserting that firms *ought*, in some ethical sense, to maximize profits. It is a claim about **power**: about who ultimately pulls the strings, and what they want. A 'business' is a legal abstraction—it does not literally think, want, or decide anything. *People* make decisions. And the people with the ultimate lever of control are the owners. 'Profit' is the accounting system's measure of how well the company has done *for them*. So when economists say "firms maximize profits," what they are really saying, in a more reductive register, is: the people who control the firm will tend to use that control to advance their own measured interests. Not a moral principle—a power principle.

One housekeeping note before we proceed: hold on to that little shirt-selling business above. We are going to return to it in Chapter 11 and work through it, line by line, after we have learned the accounting principles needed to answer it.

So, the core notions of accounting are at the center of the things you learned—and thought you knew—in Econ 101, previously. This first chapter covers those foundations. It's a bit more abstract than those to come. But these foundations matter.

1.1 The core problem of accounting: TMI

People need to make important, **informed decisions** in their dealings with businesses—whether as investors (owners), suppliers, employees, regulators, etc. (Consider, e.g.,

an investor deciding whether or not to buy shares; or a CEO negotiating for a compensation contract.) So, we need a way to **inform** those people about what the company has and has done and **measure its performance**.

The people who deal with a business need different things from its accounts. An investor wants to assess the return on an investment; a lender wants to know whether the business can repay a loan; a manager wants to understand how its activities are performing. Each needs a usable account of a business that is too complex to take in all at once.

But the **full reality** of a firm is almost **infinitely complex**—far more than any one human mind can fully take in. The core problem of accounting is one of **"Too Much Information" (TMI)**. We need a way to **measure**, **compress**, **summarize**, and **represent** all of that information in a way that gives these decision-makers the information they need, in a way they can comprehend. How do we do this? This is what accounting is all about. Accounting is like a **map**. And, as a philosopher once famously said: **"The map is not the territory."** Some information, the infinitely complex reality, is lost. But, if we cannot see the whole territory with our eyes, all at once, **we need a map**. Accounting is about how we build and read maps of companies.

1.2 What is accounting? Three definitions

Let's start with three basic definitions of accounting: a **high-level** definition, a **reductive** definition, and a **practical** definition. These will help motivate this book, and help you work through some of the difficult 'growing pains' we'll face early on:

1. High-level: Accounting is the system that tracks, **measures,** verifies, summarizes, and **communicates** the activities and financial position and performance of a business. This definition suggests both why it is important on a social level, and why it is interesting. If we want to hold companies **accountable** for their performance and activities, exercise **control** over them (whether indirectly as investors, or directly as regulators or directors or managers), or **decide** where to **invest** our money, we need some way to **understand and assess** businesses' activities and performance. And clearly, 'measuring and representing' something as complex as a business is a challenging endeavor. Just think how complex the operations of, say, Walmart, are—what does it *mean* to **measure**, summarize, and report Walmart's activities and performance? Should positive social-media posts about its products 'count'—and if so, as *what*?

2. Reductive: Here's a more mundane definition of accounting: Accounting is a system of **rules** that you apply to a company's **transactions** in order to produce its **Financial Statements**. A company's **Financial Statements** are the ultimate end product

of its financial accounting system, the key numbers it reports in its annual and quarterly reports. Here's a schematic representation:

$$Transactions \rightarrow \boxed{\text{Accounting}} \rightarrow Financial\ Statements$$

Equivalently, for those of you with a mathematical bent, we could say that accounting is the function *f*, such that:

$$f(Transactions) = Financial\ Statements$$

This definition may seem a little odd or circular, but it will be helpful to keep in mind during the early chapters of this book. The reason is that, as we will see, accounting is **largely** based on core general concepts and principles, and **most of the time**, you can reason your way from those principles to understand how to account for some transaction. But, there are also some times when those core **principles** are in **tension** with each other, or where they don't provide definitive guidance—and, unfortunately, there are also **rare** cases when, for institutional and historical reasons, an accounting rule seems to **defy logic**. In those cases, we still need to know the rule.

3. Practical: Finally, here's a definition that I hope will motivate you through this tough subject: Accounting is the **most useful body of business knowledge**. Practically all of the most lucrative and high-impact business careers involve serious use and knowledge of accounting. It's common to refer to accounting as the '**language of business**.' And it is. But I also like to think of accounting as the '**mechanics of business**.' A car salesperson might be able to **sell** a car, but mechanics and engineers are the ones who really understand how it works, how **gasoline** is ultimately turned into **motion**. Similarly, accounting unveils every step in the process by which a company **turns resources into profits**. This is why people with strong accounting knowledge so often end up at the top of businesses.

In fact, most of the careers that we refer to as 'finance' careers actually rely as much on what you will learn in this book as they do on formal finance theory. Consider the actual work that Wall Street **equity analysts** do: They work to **forecast firms' future profits**, based on the understanding of the firm's operations that they glean from its **past Financial Statements** and other disclosures. Or consider what **private equity** analysts do: They use spreadsheets filled with **financial-statement numbers** to try to identify where firms have some 'slack' where they can drive out costs, **improve profitability**, and therefore realize value. Similarly, a corporate CFO (**Chief Financial Officer**) doesn't always need to understand options pricing models—but s/he will always be thinking about, trying to understand, and being held accountable for the company's

"bottom line" profits, **net income**.

This can matter even before you make your first investment. Suppose a startup offers you a lower salary in exchange for stock options. How much confidence should you place in that promised upside? You would want to understand what the business earns, what it spends, and how long its cash will last. The people recruiting you have a reason to tell an attractive story. Accounting gives you some tools for evaluating that story yourself.

1.3 A simple model of a business

Accounting is about **measuring** and **reporting** business activities. So to understand how accounting works, and why it does things the way it does, we first need to understand the basics of business. What follows is the bare bones.

A business is a legal entity that...:

1. ...uses **financing** (that is, cash funds from external sources)...
2. ...in order to **acquire resources**...
3. ...that it uses to produce **goods** and **services**...
4. ...for **sale** to **customers**...
5. ...in order to generate a **profit** (the bottom line) for its owners.

Let's first understand a bit more about the **financing**.

1.4 Financing basics

'Financing' is when a company gets **cash today**, in exchange for some kind of **future** obligation or expectation. Why do firms get financing? Simply because, when a firm is first incorporated, it has zero cash. And, in order to operate as a business, it will have to acquire resources. If it's going to pay for those resources in cash, it will have to get that cash from somewhere. So, the first step in every firm's life is: **get financing**.

There are **two major kinds** of financing that companies can use: **equity from investors** and **loans from creditors**. Almost all companies use both kinds, and the distinction between the two is the core foundation of accounting.

1.4.1 Equity financing

Equity financing refers to when **investors** give cash to the business and receive, in return, **shares of ownership** (or 'stock' or 'equity')—these shares promise them a share of future profits and (usually) the right to vote. Equity investors are technically the **owners** of the business. The terms **'equity,' 'shares,' 'shareholders,' 'stock,' 'stockholders,' 'investors,' 'owners'** are all referring to this same basic thing. When an entrepreneur founds a company, she will invest some of her own money in the corporation and thus become the first owner. Eventually, she may seek other partners and financing and thus sell **shares of ownership** to them. A publicly-traded company (such as Microsoft) technically has millions of owners (including me).

One piece of jargon here requires a small 'mental rotation': when investors buy in, we say that the company **'sells common stock.'** When I first learned this, the word 'sells' confused me—it doesn't feel like selling in the everyday sense, since no product or service is going to any customer. Here's how to visualize it: imagine the corporation dividing its ownership, conceptually, into little slips of paper. Those slips are the **'shares.'** When the company 'sells common stock,' it is handing someone one of those ownership slips in exchange for cash. That's all the phrase means.

Why do investors buy shares/equity? They, naturally, hope to make a good return on their investment—to get more money back than they put in. When businesses distribute their profits to their shareholders, this is called a **'dividend.'** Shareholders naturally love it when businesses distribute these dividends—it is how they make their return, the whole reason they invested in the first place.[1] Companies are not contractually or legally obligated to pay dividends to investors—they are discretionary. But shareholders are the owners of the corporation and can vote on key matters. If the executives of the corporation **don't** prioritize their interests, the shareholders can **vote them out** of their jobs.[2]

There is also a deeper—and, I think, ultimately more satisfying—way to see why companies pay dividends at all. Ask yourself: why does a dental practice (a legal business entity) pay a dividend to the dentist (its owner)? Because the practice *is* the dentist's instrument. The corporation and its owners are not two strangers negotiating at arm's length; the corporation is a legal vehicle that the owners use to pursue their own ends. So when the company pays a dividend, it isn't one alien entity paying another—

[1] Those of you who have done finance or have a familiarity with it may know that companies can also directly **buy back** shares from shareholders and give them a return on investment that way. Additionally, shareholders can sell their shares to other investors in the **'secondary' market** (the stock market) and make a return that way—but, that return depends upon other investors' **expectations** about future payouts. For the purposes of this introductory book, I keep things simple. For now, let's work with this simple toy model of a business and accept that dividends are how investors get their return.

[2] In reality, there are a variety of **governance mechanisms** that shareholders use to get corporate managers to act in their interest. Perhaps the most direct one is just giving the corporate executives **stock-based compensation**—so that they do well if and only if common shareholders do well.

it's the owners paying themselves out of their own vehicle, moving cash over a line from the company's bank account into their own. Note, finally, the definitional boundary we have just drawn: equity carries *no contractual obligation of repayment.* If a claim entitles its holder to fixed payments that the company is contractually required to make, then—by definition—that claim is not equity. It is the other kind of financing, to which we now turn.

1.4.2 Credit financing

Most of us know that banks **lend** money to corporations. These **loans** are a form of **credit**, offered from the bank to the business. There is, annoyingly, a lot of different terminology for this one basic thing. We say that the banks or markets **'provide loans'** or **'supply credit'** and become **'creditors'** or **'debtholders'** or **'bondholders.'** Meanwhile, we say that the corporation **'receives loans'** or that it **'issued debt'** or **'sold bonds.'** All of these are just different terms for one basic thing: credit financing.

One of these terms genuinely tripped me up when I first studied finance: my professor kept saying that companies "sell bonds," and I kept thinking, 'Wait—isn't this just a loan? What exactly is being *sold*?' Here's the way to visualize it. The corporation writes up a contract—a **bond**—which says: "We promise to pay the holder of this document a fixed schedule of interest payments, plus the principal at maturity." That contract is itself a financial instrument—a thing of value. The company then sells *that piece of paper* to investors, in exchange for cash. Economically, the whole arrangement is a loan; the 'selling' refers to the contract that embodies it.

When creditors provide financing, they give cash to the business today, and receive a **contractual promise** that the business will pay them a fixed 'schedule' of **interest** and **principal** payments in the future.

Why do firms pay their debts? (More provocatively: Why do *you* pay your debts? Do you really *have to*?) We said above that managers of firms pay dividends to owners because those owners have ultimate control. But **creditors are not owners**. So why pay them back? Why don't companies just **take the money and run?**

The answer is that **creditors** also have various **enforcement mechanisms** specified in their **contracts**. If firms do not pay their creditors, those creditors can **seize collateral** from the firm—and since they are contractually within their rights, the **law** and local sheriff will **stand behind** the creditors as they do so. More generally, firms that do not pay their past debts will find it difficult to get new credit financing (just like a person with a **poor credit score**), and can be taken to **bankruptcy court**.[3]

[3] I recommend searching for **"airplane repo"** on YouTube to learn more about the ultimate mechanisms enforcing corporate credit. I also recommend watching the 2020 film "Uncut Gems," one of my favorite movies, to learn more about the enforcement of debt obligations in, ahem, more *informal* markets. What would happen if you, personally, didn't pay your debts? If you didn't pay off your credit card,

But otherwise, creditors do not have ownership, or voting rights over the business—and they are not entitled to any 'upside,' *beyond* their contracted interest and principal. (Your credit-card company, for instance, does not get to claim a share of your year-end bonus just because you happen to be flush.)

1.4.3 Going deeper: What exactly *is* 'ownership'?

Many undergraduate textbooks are content to simply assert that firms can finance themselves with two kinds of financing, credit and equity, as if this is an empirical fact or happenstance about the world. But, if we dig a little bit deeper, this is not a mere empirical happenstance of our world, but is, instead, a necessary **logical axiom** for any universe in which there are contracts. In other words, it's not clear that things **could be different**. A business has *resources* (assets), and does *activities* that generate cash flows (sales). Your *liabilities* are the things that you are *contractually obligated* to pay off. So, what happens *after* a company pays what it contractually owes? What happens with the *remainder*? As a matter of **corporate arithmetic** that remainder has to go *somewhere* and be controlled by *someone*. Put differently, there **must be a residual claimant** for things to **add up**. So, 'equity-like' relationships are a necessary consequence of any activity in which there are **contracts** and **uncertainty** about the future.

In 2016, **Oliver Hart** won the Nobel Prize in economics, largely for elucidating the meaning and implications of ownership. Here's a fundamental question: **What determines what businesses do**? Some of the things that businesses do, they do because of **legal contracts and obligations**—paying debts, obeying environmental laws, etc. But, what about **everything else**? In Oliver Hart's framework, ownership entails **"residual control rights."** Owners have the final say on everything that is **not contractually specified**. Once again, as a matter of "adding up," this must be the case—somebody has to make the decision when contracts are silent. So, in this framework, 'ownership'/'equity' is a necessary fact about any situation in which there are decisions to be made, contracts, and uncertainty about the future (and thus, "incomplete contracts"). This runs deep. E.g., **you are the 'residual claimant' in your own life**.

You can see a similar structure in your own day. You have 24 hours. Some of those hours are already spoken for: I have to teach my classes to keep my job, and I have to sleep and eat. Once those obligations are met, I decide how to use the

that would hurt your **credit score** going forward—not ideal at this phase of your life. If you financed the purchase of a Tesla, but then never made any payments, what would happen? Could the Tesla drive itself back to the dealership?

time that remains, and I receive the benefits of those choices. In that sense, I am the residual claimant in my own life.

Or consider a society on another planet. Suppose intelligent beings there form organizations, make contracts, and face uncertainty about the future. Their contracts will specify some obligations, but cannot settle every possible decision. Someone will have to decide what happens when the contracts are silent, and someone will have a claim on whatever remains after the obligations are met. The setting is unfamiliar, but the question is the same: who has the residual claim, and who has the residual control?

1.4.4 Comparing and contrasting

So, businesses **have to** pay off their **creditors** first. But once they've done that, they **can** distribute their remaining profits/value at will to their **equity investors**. This is why it's common to refer to the equity investors as **'residual claimants'**—they're entitled to **whatever is left over**, after the corporation has paid off its contractual obligations. Businesses pay off their creditors because they contractually, legally have to. And they pay dividends to their investors because the investors have ownership and control.

A familiar comparison might be **mortgages**: When people buy homes, they almost always finance the initial purchase with a mortgage from a bank. ('Mortgage' is just a term for 'credit financing provided for the purchase of a house.') As a part of that mortgage you agree to pay the bank back. And if you don't, the bank can seize the collateral—the house itself. But, if you sell the house for a huge profit, the bank doesn't get to claim anything beyond the amount of the loan that you are required to pay off. You, as homeowner, are entitled to all the additional 'upside'; and the **value of your home above the amount you still owe to the bank is your "home equity."**

The preceding paragraphs are just a very brief introduction to **corporate finance**, a deep, rich area of study worth many books of its own. Why this diversion in an introductory accounting book? The distinction between credit and equity financing is at the center of accounting. The core axiom of accounting, '**The Accounting Equation**' is about the **difference between creditors and owners**, and their respective rights.

1.5 Looping back: A simple model of a business

Let's go back to our simple model, and loop in what we've learned about finance.

A business is a legal entity that...:

1. ...uses financing from equity investors and creditors...

2. ...in order to acquire resources...

3. ...that it uses to produce goods and services...

4. ...for sale to customers.

5. ...The business *must* pay what it owes to its creditors...

6. ...and, beyond that, whatever is left over can be claimed by the equity owners.

We've only added a little bit to our initial model of a business—but these additions will turn out to be key to understanding accounting.

1.6 Financial Position & Performance: Core Identities

Accounting measures and summarizes both, (1.) a company's financial **position** (i.e., what resources, obligations, and thus remaining value it has) at a **point in time** and (2.) its financial **performance** (i.e., how it has added to or subtracted from that value) **over a period of time**.

Stocks and flows: An extremely general concept in accounting and other settings is the notion of "**'stock' measures vs. "flow" measures.**[4] "Stock" quantities are measures of *where things are at* at a point in time—the total, cumulative balance of some amount. "Flow" quantities are measured over some period of time, and, thus can also be the *change* in some related stock measure. So, in, e.g., the context of macroeconomics, the government's total outstanding debt would be a stock measure, whereas the Gross Domestic Product (GDP) produced by some economy in some year would be a flow measure. In the context of personal finance, your yearly salary or income would be a "flow" measure, and your total net worth or wealth would be the related "stock" measure. And in the context of accounting, the profit generated over some period is a flow, while the company's total assets or net worth would be stock measures.[5]

The bathtub metaphor: Almost every introductory accounting course illustrates the concept with an analogy to a **bathtub**. The **level** of the water in the bathtub is the **stock** measure, the total amount at some point in time. And the **rate** at which water is pouring in from the tap is the related **flow** measure. So, those are different measures and quantities. But, of course, the two are intimately and mechanically linked: The

[4] In this context, the word 'stock' is used in a fully general sense, and does not have to do with, e.g., share of common stock traded on stock markets.

[5] For those of you who enjoyed calculus: the flow is the *derivative* of the stock, and the stock is the *integral* of the flow. For those of you who did not enjoy calculus: the bathtub below will serve you just as well.

level is the **cumulation** of everything that has flowed in to date. And the rate of change of the level *is* the rate of flow, by definition. In accounting, you can (for now) think of the profits *flowing* into the company increasing the total *shareholders' equity*—the accounting measure of the total net worth of the company.

The core identities: The **core identities** of accounting are built around this concept: The '**Accounting Equation**' identifies the firm's financial position, based on the key "stock" measures; the '**Net Income definition**' identifies its performance over a time period, using the key "flow" measures; and the '**Retained Earnings Identity**' shows how the two are linked. These **identities are crucial, in that each one governs a key financial statement, and they show how everything fits together in accounting.** We will use them in every chapter of this book. The sections below discuss each one in detail.

1.6.1 The Accounting Equation / The Balance Sheet Equation

$$Assets = Liabilities + Shareholders' \ Equity \tag{1.1}$$

This is the equation that describes a business's position at a point in time.[6] **Assets** are defined as **economic benefits** that the company **owns**. The most basic, fundamental asset is **Cash**—the **OG** asset. But there are many other assets. These can include physical, tangible things like **factories**, or intangible things like **patents**—or, even, things like customers' **promises** to pay the firm in the future.

Liabilities are defined as **economic costs** that the company is **obligated** to incur. Liabilities include things like debt (**loans** from creditors), but also include the value of **obligations to pay suppliers, and employees**.

Shareholders' Equity is the accounting measure of the claims of the **owners/ shareholders** on the corporation, and includes what those shareholders have **put in** via financing (buying shares). And, according to the accounting equation above, Shareholders' Equity must be equal to whatever the **difference** between the two is.[7]

There are **two ways to visualize and understand the Accounting Equation.**

- **1. The "Sources of Financing" View**: The company has a certain amount of assets that it has acquired. And those assets had to be paid for somehow. And

[6] When you finish a PhD and become an accounting professor, you submit your dissertation and file the usual paperwork—and then there is one final, secret ritual: they hand you a ceremonial knife, you cut a small slice in your finger, and you sign in blood: "$A = L + E$." I am joking. But only barely: an accounting professor never, ever forgets this identity—it holds always and everywhere, for every firm, after every transaction—and by the end of this book, you will not forget it either.

[7] Some textbooks use the term 'Stockholders' Equity.' The two terms mean the exact same thing, but in the 21st century, it is more common to speak of 'shareholders.' So, to prepare you for your 21st-century careers, I will preferentially use the term 'shareholder' in lieu of 'stockholder.'

all of the company's financing had to come either from equity or credit. So the accounting equation tells you how much the company *has* (A), and how the purchase of those assets was *financed* (that is, L & E).

  - That is: $L+E=A$

- **2. The "What's Left Over" View**: The company's total economic **benefits** (that is, its **assets)** are equal to a certain amount (A), and its total **obligations** (that is, its **liabilities**) equal a certain amount (L). And so if the company's assets are greater than its liabilities, it can pay off everything that it owes, and whatever is **left over** is 'residual value' that can be claimed by the company's **owners**.

  - That is: $E=A-L$

Those are two different ways of **looking at** the equation. But it's one equation. So, the two are ultimately the same. Consider a simple example, and then try to visualize it both ways: I start my own business, and, as owner, paid in $600,000 of my own money, and then got a $400,000 loan from a bank, and then bought a store worth $1,000,000.

- **Sources of financing view:**

  - $L+E=A$
  - $\$400k+\$600k=\$1m$

- **"What's left over" view:**

  - $E=A-L$
  - $\$600k=\$1m-\$400k$

A natural question is: 'How do you know the store you bought is *truly worth* $1m?' That's a great question. And the answer is: We *don't* know it's 'truly worth' that. But accounting requires us to initially record the value of assets based on a **verifiable purchase price**, rather than a more subjective appraisal. There are a variety of reasons for that, which we'll discuss and debate later in this book. For *now*, let us accept it.

Now, you might wonder: what happens if the financing and the spending don't line up so neatly? Suppose I had paid in $800,000 of my own money instead of $600,000, kept the $400,000 bank loan, and still bought the $1,000,000 store. Does the extra $200,000 just wash out of the equation somehow? Not at all: it sits in the company's bank account, and Cash is also an asset. The company's assets would now be the store ($1m) plus cash ($0.2m); and the right-hand side would be the loan ($0.4m) plus equity ($0.8m). $\$1.2m=\$0.4m+\$0.8m$. It still balances. It always will.

Return to the original example, where the company spent the entire $1,000,000 on the store. It now has no cash, yet Shareholders' Equity is $600,000. That equity is not

another resource sitting somewhere, waiting to be spent. The store is the resource. Equity measures the owners' claim on the company's assets after its liabilities are accounted for. **Assets tell us what the company has; liabilities and equity tell us about the claims on those assets.**

1.6.2 Net Income definition

$$Revenues - Expenses = Net\ Income \tag{1.2}$$

This is the equation that identifies a business's performance **over a period/interval of time**. **Revenues** are the **value/benefits earned** when the company **sells** products or services to customers (often, but not always, for cash). **Expenses** are the **costs incurred in doing so**. The **difference** between those two over any period of time is the company's **net income** over that period. (If that value is negative, we might call it a 'net loss.') This is usually what people are talking about in everyday speech when they say '**profit**.' But, in accounting, we use the more precise term, 'net income.'

The net income equation and the accounting equation are **intimately linked**. Your revenues are the benefits you earned this period—and your assets are the total benefits you control right now. Your expenses are the costs you incurred and resources you expended—and your liabilities are the total obligations you have outstanding at a point in time. So, for now, it might make *vague* sense that the two are linked. In a few chapters, you'll understand *exactly* how.

1.6.3 Dividends: A special, important category

Dividends are conceptually important and potentially tricky for first-time learners. Recall that investors (or 'owners'/ 'stockholders'/ 'shareholders') put their money into a business in the hopes of getting a return on that investment. When a business **pays out cash to those investors**, it's called a 'dividend.' Investors love getting dividends (this is how they make their money). But it's a special kind of transaction.

If a company spends cash on rent, that's an expense that helps it stay in business and generate revenue. If a company spends cash on a major new investment, like a new plant, that's an asset that continues to provide value for the company. But when a business pays a dividend, it simply moves cash from its own bank account into the bank account of the owner. Thus why it's called a 'distribution.'

This is **not a bad thing**—far from it. The promise of getting paid is the major reason investors (and entrepreneurs) invest in the first place. Value is not being destroyed or lost—just **moved from the business's bank account to the investors'**. But it does mean that dividend distributions are a special kind of activity, that get some special treatment and can confuse first-time learners. In particular, many students

are tempted to think that dividends are an expense. They are **not an expense**: they are not a cost incurred in doing business. Net income is supposed to measure the value generated for shareholders. When firms move cash from their bank account to their shareholders', that *doesn't* decrease the value that was generated for them—just distributes it.[8]

1.6.4 Putting the identities to work

Let's pause and put these pieces to work on a small example—the kind of question you should now be able to answer in seconds. Eagle Corp operates MRI clinics throughout the Northeast. At the end of the current period, the company reports the following amounts: Assets of $40k; Liabilities of $30k; Dividends of $2k; Revenues of $13k; and Expenses of $6k. The company's management also reports that its long-term value is not being fully appreciated by the stock market, and that they will double their profits next year.

First question: What is Eagle Corp's Shareholders' Equity at the end of the period? My advice for problems like this: start from what you are being *asked*, and work backwards to the identity that governs it. We're asked for E; the Accounting Equation gives us $E = A - L = \$40k - \$30k = \$10k$. That is the entire solution.

Second question: What was Eagle Corp's Net Income over the period? Again: $Net\ Income = Revenues - Expenses = \$13k - \$6k = \$7k$. The single most common error here is to answer $5k—that is, to subtract the $2k of dividends as if they were an expense. By now, you know better: dividends are a *distribution* of profits, not a cost of generating them. Net income is always, and only, revenues minus expenses.

And what about management's stirring claims regarding the company's unappreciated long-term value and next year's doubled profits? They play no role whatsoever. That is not an accident of the example; it is the *point* of the example. *Of course* management believes the company is undervalued—we all want to believe we are underappreciated. Accounting has **rules** for what enters the accounts, precisely so that the numbers do not bend to whatever the people being measured would prefer them to say. Whatever the future may hold, $E = A - L$, always and everywhere. And part of learning accounting is learning to keep your eye on the ball amid self-serving narration—real financial reports come wrapped in plenty of it.

[8] A natural follow-up question: is there any limit? Could the owners pay themselves *too much* in dividends? From a pure accounting standpoint, no dividend—however large—is ever an expense. But think about who might object. If you were a bank that had lent the company $100k, would you enjoy watching the owners pay out all of the company's cash to themselves? Certainly not—you want to be sure your interest and principal get paid first. So, in practice, loan agreements typically include **'covenants'**: contractual terms that can restrict, among other things, how much the borrower is allowed to pay out to its shareholders. We'll meet covenants again much later, when we study long-term liabilities. For now, the accounting point stands: dividends are never expenses.

1.6.5 Retained Earnings Identity

This brings us to the final **core identity** of accounting, and one which links the previous two. We said that the **profits** earned by shareholders/owners are called 'Net Income,' and firms pay **dividends** to those shareholders. But corporations generally do not **pay out** the full amount of their profits **right away**. They may instead **retain** them in the corporation, in order to **invest** (that is, acquire more assets and resources) and earn more profits in the future.

In accounting, we call the **difference** between what shareholders have earned in total **(over all periods)**, and what has been paid out in total **'Retained Earnings.'**

$$Retained\ Earnings = Total\ N.I.\ Earned - Total\ Dividends\ Paid \tag{1.3}$$

This identity is **extremely important**. It is the **key link** between **financial performance** (over a period) and **financial position** (as of the end of the period). The firm's **Total Retained Earnings** are a **component of Shareholders' Equity**. One way to understand this is that the shareholders **could have claimed** all of those profits for themselves. So, whatever they **have not claimed**—it is **as if** they have contributed that in additional **financing**. And it crucially links together the **'flow measures'** in the Net Income Equation, to the **'stock measures'** in the Accounting Equation ($A = L + E$). Whatever net income (flow) has been earned and retained in a period **"lives on"** in the 'stock' measure of value (Shareholders' Equity) via Retained Earnings.

It's often useful to re-write this identity in terms of the *changes* between periods...:

$$\Delta Retained\ Earnings = Net\ Income - Dividends \tag{1.4}$$

So, e.g., if the firm had reported $100 in Retained Earnings at the **end of Fiscal Year 2019**, and earned $60 in Net Income and paid out $40 in Dividends **over Fiscal Year 2020**—then, it would report $120 in Retained Earnings at the end of **Fiscal Year 2020**. (Careful, here: the single most common error with this identity is to answer '$20'—the *change* in Retained Earnings—when asked for the ending *balance*. Retained Earnings is a 'stock' measure: it has a running balance, which the period's net income and dividends merely *update*. A distressing fraction of students still miss this on final exams; be the exception.)

In the **Bathtub Metaphor**, the 'flow' of water from the faucet is **Net Income**. Whatever is scooped out is **Dividends**. And so the 'stock' level of water (Shareholders' Equity) **increases by the difference.**

1.7 Periodic Reporting

There's one more concept we need in order to understand how these **core identities** and concepts link to the actual **Financial Statements**: Firms engage in activities **continuously in time** (of course!), but they necessarily report their Financial Statements on a **discrete**, **periodic** basis. You've likely heard the term 'Fiscal Year.' It's common for firms to report (that is, create and disclose) their Financial Statements on an annual basis. Such firms would report their financial statements **as of** the end of the fiscal year (e.g., December 31st for firms which align their **Fiscal Year** with the **Calendar Year**, and any other date for those which do not).[9]

So we have two facts, each individually obvious: (1.) firms *operate* continuously, in real time—a 24-hour CVS is delighted to sell you something at 3 a.m., because it wants your money; and (2.) firms *report* periodically, as of discrete dates. A remarkable amount of the structure of accounting comes from reconciling these two facts. Indeed, many of the subtler accounting questions we will meet in this book are really *timing* questions in disguise.

This may seem obvious (*of course* the company's accounts must be *as of some date*—how could it be otherwise?). But this is important for understanding some of the **"mechanics" of the financial reporting process**, and it can trip us up if we forget it. Here are some things to keep in mind on the **periodic nature of financial reporting**:

- Revenues and expenses are, we said above, measured **over a period**. If my company earned $50k in revenue in its first year, Fiscal Year (FY) 2020, then none of those revenues will "carry over" into FY 2021. They're set back to zero for the start of FY 2021. That doesn't mean they go away! They're just reported in FY 2020, and not in FY 2021. (And remember that they 'live on' in Retained Earnings.) Some textbooks refer to Revenue and Expense accounts as **"temporary accounts"**—because they are measured *over* a period, and zeroed out at the end—as opposed to the **"permanent accounts"** of assets and liabilities, e.g. *Land*, *Equipment*, etc.

- Consider the following: Suppose that you paid $100k for a 2-year real-estate rental contract (a lease) *in the middle of* FY 2020. What is the *expense* you would record for that rent in FY 2020 (given that you would only have "used up" one quarter of it as of that point)? We will answer this in Chapter 4. For now, the point is: The **discrete, periodic** nature of reporting matters for things like these.

[9] Firms can also report on a quarterly basis, and frequently have a fiscal year that *does not* align with the calendar year (i.e., ending on some date other than December 31). For the purpose of this introductory book, where I focus on communicating the main accounting ideas in the cleanest way possible, most of my example problems will assume annual reporting and a December 31 fiscal year end-date.

- Most generally, this means that companies don't really "*have*" Financial Statements *at a particular point in time*. At any **point in time**, they **track "accounts"** (e.g., Cash, Equipment, Revenue, etc.) And those accounts will be used to populate the financial statements at the end of the period, via the **mechanics of the reporting process**.

- In order for the financial statements to sync up, it's important we **update the Retained Earnings** balance (according to the core identity: $\Delta Retained\ Earnings = Net\ Income - Dividends$) right at the **end of the reporting period**, and **only** then. You'll eventually see how this keeps everything in sync.

1.8 Tour of the Financial Statements

Now it's time to get a bit more concrete. Let's talk about the actual financial statements that accountants produce and audit, and thus make it into the firm's financial reports. Unsurprisingly, there will be a close relationship between the core principles and identities we discussed above, and each statement.

There are **four financial statements**.

1. the Balance Sheet (B.S.),

2. the Income Statement (I.S.),

3. the Statement of Shareholders' Equity (S.S.E.) and

4. the Statement of Cash Flows (S.C.F.)

The Balance Sheet and Income Statement are the most important financial statements, in this book and in the outside world, and they illustrate the most important accounting concepts. Of the sixteen chapters of this book, we'll spend only one on the Statement of Shareholders' Equity, and one (plus a preview) on the Statement of Cash Flows. You'll have to know and understand the S.S.E and S.C.F. by the end, but I want to be clear about what to focus the most attention on now.

1.9 The Balance Sheet

The **Balance Sheet** is the statement of the company's financial **position** at a **point in time** (rather than its activities *over* a period). It's like the 'stock' measure (measuring the level of water) in the bathtub metaphor. It is governed by The Accounting Equation: $Assets = Liabilities + Shareholders' Equity$.

In a reductive sense, the Balance Sheet is just a **list**: A list of the Assets (resources), Liabilities (obligations), and, Shareholders' Equity accounts, that a company has, as of the end of the reporting period. The **left** side tabulates the **Assets** of the corporation (typically listed in order of **liquidity**—how quickly they could be converted into cash). **Assets** include things you would expect, like **Cash** in the bank, **Equipment**, and **Land**; but also some other less obvious kinds of benefits controlled by the business, such as '**Accounts Receivable**' (which are **promises** from customers to pay later).

The **right** side of the balance sheet shows how those resources are **financed and claimed**—i.e., the **Liabilities and Shareholders' Equity** of the corporation. **Liabilities** are obligations the company has to meet that will incur an economic cost—they include obvious things like **Debt** (provided by creditors/lender) but also things like **'Salaries Payable'** (money owed to employees for work they have completed), etc.

You might pause at that last one: why would a company owe its employees money *as of the reporting date*? Why wouldn't all salaries simply be paid up by then? Timing, once again: employees are typically paid every two weeks, or once a month, while the company reports as of one fixed date. If the reporting date is December 31, there will almost always be some late-December workdays that employees have already put in, but for which the paycheck won't go out until early January. As of December 31, the employees have *earned* that money, and the company *owes* it. So it is a liability—*Salaries Payable*—even though the cash will not leave the bank account until the next period.

The **difference** between the total Assets (the resources of the company) and its total Liabilities (the claims of creditors and other obligations it has) is the **Total Shareholders' Equity**. And Total Shareholders' Equity, we said above, can come from two sources: (1.) Amounts that shareholders/stockholders have directly *paid in* as investments (**Common Stock**); (2.) Profits that they *could have* paid out to themselves as dividends, but instead *retained* in the corporation (**Retained Earnings**).

A super-simple Balance Sheet could look something like this...:

Balance Sheet as of December 31, 2020

Assets	**Liabilities**
Cash: $20	Salaries payable: $30
Accounts receivable: $40	Long term debt: $120
Supplies: $80	**Total Liabilities: $150**
Equipment: $100	
	Shareholder's Equity:
	Common stock: $60
	Retained earnings: $30
	Total Shareholders' Equity: $90
Total Assets: $240	**Total Liabilities and S.E.: $240**

Here is a small exercise in reading a Balance Sheet the way an accountant does. Suppose a company reports Cash of $100 and Equipment of $100, along with an Accounts Receivable balance that has been smudged into illegibility; on the right side, it reports Total Liabilities and Shareholders' Equity of $250. What is the Accounts Receivable balance? Reason it out: the Balance Sheet must *balance*, so Total Assets must also be $250. And the Balance Sheet is a complete *list* of the asset accounts. Cash and Equipment account for $200 of that $250—so Accounts Receivable must be $50. Simple—but note that you had to consciously deploy two facts: the Accounting Equation, and that the statement lists *all* of the assets.

Remember the discussion above of the **two ways to see** The Accounting Equation ($A = L + E$): **(1.) How** the resources of the corporation were **financed** ($L + E = A$); **(2.)** What value is **left over** for the shareholders ($E = A - L$). These are important, key intuitions for *thinking about* what the Balance Sheet is telling us. But they aren't *perfectly* true, and this imprecision can be frustrating for first-time accounting students. The reason has to do with the second major general principle that governs the balance sheet. The balance sheet is intended to be **conservative**.

1.9.1 Balance Sheet: Key Principles

In accounting, **conservatism** is a general principle that constrains firms to **err on the side of understating the value of their assets** when that value is uncertain. The purpose is to restrain managers who may be tempted to 'cook the books' by overstating the value of their assets when they have discretion. There are two major implications of accounting conservatism:

- **First**, some **economic assets** are **not** recognized as **accounting assets**. E.g., part of the value of a company may come from **customers' brand loyalty**, or from the value of its Research & Development (**R&D**) in progress. But standard setters in the U.S. have decided that the value of these are too difficult to ascertain and verify. Therefore, firms are not allowed to measure and report these as assets on the Balance Sheet. (And this is controversial and debated.)

- **Second**, assets are by default recorded 'on the books' at their **historical cost** (purchase price) rather than their **true current economic value** (sometimes referred to as 'fair market value'). E.g., if a company buys Land for a price in 2019, and the value of Land in that region increased dramatically in 2020, it may not be able to recognize that increase in value on the Balance Sheet. However, if the value of that real-estate were to *decrease*, accounting standards could require it to recognize a loss.[10] Isn't this asymmetric, inconsistent? Yes, it is! And that

[10] Accounting standards are changing and evolving all the time, and, these days, some assets can be

makes it controversial. Or, as another example, if a firm buys a machine from another company for $100k, but manages the machine so well that it doubles its productivity—it cannot 'write the asset up' to $200k on the books. We generally defer to the **'verifiable' purchase price** in measuring the value of assets in accounting.

Why is accounting so distrustful? Because we all like to inflate our résumés. If managers could simply *declare* the value of their assets—"our brand is worth $50 billion; our customer relationships are priceless"—they would have every incentive to keep ratcheting the numbers up. So accounting anchors asset values to **verifiable** transaction prices, rather than to management's (inevitably sunny) estimates.

To feel what gets lost in the bargain, consider a story. Imagine a wonderful, zany little vintage clothing store in your neighborhood—a local institution for decades. The owner wants to retire, and you toy with the idea of buying the store. Now: how would you decide what it's worth? Would you walk in, appraise the musty old garments on the racks, sum up their value, and make that sum your offer? Obviously not. What you would really be buying is everything *else*: the store's reputation, its presence in the neighborhood, its loyal customer base built up over thirty years of chatting with customers and curating the vibe. That is the number-one source of the business's value—and *none* of it appears on its balance sheet, because there was never a transaction in which the owner 'bought' customer loyalty at a verifiable price.

The contrast is with an intangible resource the company *purchases*. If it buys a patent from another company, the purchase supplies a transaction price that can be recorded. The lesson is not that only physical things can be assets. It is that the value of a business and the amounts recognized in its accounts answer different questions. We will return to purchased and internally developed intangibles in Chapter 11.

For these reasons, the value of a company's 'real' 'economic' assets will **usually be understated** in the financial statements. This is why a **corporate acquirer** will typically pay more than the net '**book value**' (that is, the accounting value of Shareholders' Equity) of a firm to acquire ownership. And this is why the **market value of equity** (i.e., the value of all the shares of a company, traded on exchanges like the NYSE) is usually higher than the book value. This tension seems weird at first, but we all get used to it.

1.10 The Income Statement

The **Income Statement** seeks to measure and represent a business's **performance** over a particular **period**. It is governed by the 'Net Income Equation' discussed above.

recognized at 'fair value,' while others are still recorded at historical cost. These changes will continue during your career, and can make life challenging—i.e. interesting—for the financial analyst.

It starts with the proverbial '**top line**' of the **Revenues** earned in that period (often referred to more colloquially as 'sales'), subtracts all of the various categories of **Expenses** (such as rent expense, salaries expense, interest expense, etc.), and then arrives at the proverbial '**bottom line**' of **Net Income**. These revenues and expenses are **intimately linked** to the assets and liabilities on the Balance Sheet. You could define a revenue as an increase in the company's assets (or decrease in its liabilities) generated from its operations—selling goods or services to customers—and an expense as roughly the opposite.

A super-simple Income Statement could look like this...:

Income Statement for the Fiscal Year Ending Dec. 31 2020

Total Revenues	$100
Salaries Expense	($40)
Depreciation Expense	($30)
Interest Expense	($10)
Net Income	**$20**

(Note that, in accounting, by convention, parentheses indicate negative or subtracted amounts.) You might be tempted to assume that 'revenue' is just a fancy word for the total cash the company collected, that 'expense' is just a fancy word for the cash the company spent. But this is **not correct**. Such a system, which tracks inflows and outflows of cash, is called **cash accounting**.[11] But U.S. and global accounting systems have evolved over many centuries to use a system of **accrual accounting**.

The income statement's accrual accounting differs from cash accounting in two primary ways: **Revenue Recognition** and **Expense Matching**.

1.10.1 Accrual Accounting Revenue Recognition

Under accrual accounting, revenues are *recognized* not when cash is received from customers, but when it has been **earned**. So, e.g., when a **wholesaler** sells goods to its business customers of good reputation on **trade credit** (i.e., their promise—based on their *credibility*—to pay later), it would recognize those **sales** as **Revenues** after **delivering the goods**, even if the retailers would **not pay the cash** for several months.

Why do we do it this way? There are two reasons: **First**, financial-statement users (such as investors) may want the information they receive to be **timely**. **Second,** accounting tries to measure the profitability of the company's ongoing **activities and**

[11] Cash accounting is perfectly adequate for some organizations. Think of a small congregation or charity: members donate cash, and the trustees keep a simple ledger of cash in and cash out—mostly to demonstrate that nobody has absconded to Las Vegas with the collection plate. Such an organization has no need for a sophisticated measure of 'profit.' Apple does.

operations, irrespective of the **timing** of the arrival of **checks in the mail**.

- **Revenue recognition thought experiment**: Suppose that our company delivers a huge batch of product to WalMart on December 18. But, then, **two quantum universes diverge**. In one universe, the check from WalMart arrives at our headquarters on December 30. In the other universe, there's a huge **snowstorm**, that delays the arrival of the check until January 2 of the next calendar year. Should our company's profit for the year really be different in those two universes, depending on how a random weather event affects the mail, if we are trying to measure the profitability of the company's real *activities*?

If our company delivers a huge shipment of product to WalMart, the value of shareholders' ownership has already increased on that basis—our investors may wish to **know** about that fact **sooner**, during the next financial report, rather than **wait** until the end of the period in which the **cash is delivered**.

Let's sharpen the timeliness point, because the delays involved are longer than you might guess. Retail is highly seasonal: a retailer like WalMart stocks up on inventory from its suppliers in November, sells to holiday shoppers in December, and pays those suppliers in January or February. So suppose our young company lands its first huge contract, and ships the goods to WalMart in November 2022, with payment due in February 2023. Under cash accounting, that sale would land in our *2023* financial statements—which, after the usual months of preparation and audit, might not reach investors until early 2024. Our investors would learn about the transformative deal roughly *fifteen months* after we actually did the thing. If we are a growing startup, that is absurd: the whole story of our year would be missing from our own annual report. Accrual accounting records the revenue in 2022, when we *earned* it.[12]

To make the recognition rule mechanical, here is a simple worked example. Suppose our tutoring company has the following transactions with customers—and *only* these—in Fiscal Year 2021: (1.) we provide tutoring to Dane for $200 in cash; (2.) Alison pays us $300 in cash up front, for tutoring that we promise to provide next year, in FY 2022; (3.) we provide tutoring to Avery, who promises to pay us $100 next year, in FY 2022. What are our revenues for FY 2021? Take the rule at face value, and go transaction by transaction, asking one question each time: *did we provide the service in FY 2021?* Dane: yes—count the $200. Alison: no—we have her cash, but we have not *earned* it yet; count $0 this year (we will recognize that $300 as revenue next year, when we actually tutor her). Avery: yes—the work is done; count the $100, cash or no cash. Total

[12] 'But what if WalMart never pays?' An excellent question, and here is an honest answer: for now, we are assuming that promises from creditworthy customers are always kept. That is a deliberate simplification, one of several I will make early in this book and unwind later. What happens when customers *don't* pay—and, more interestingly, when your *expectations* and reality diverge—is the subject of Chapter 8. Things get delightfully gnarly there.

revenues for FY 2021: $300. Not $200 (only the cash-and-carry sale), and not $500 (all the cash that happened to arrive): $300.[13]

1.10.2 Going deeper: What is 'money,' anyway?

First-time students sometimes react to revenue recognition with healthy suspicion: "Wait—we're recording revenue even though we don't have the *money* yet?" It's a good instinct. But it invites a deeper question, one that rewards a moment of philosophical worrying: what exactly do you mean by *having the money*?

You probably have a checking account at some bank—say, Bank of America. When you log in, you see a number on the screen, and you say: "that's my money." But what does that mean, exactly? Does Bank of America keep a file drawer somewhere with your name on it, stuffed with dollar bills and coins equal to that exact amount, reserved just for you? Obviously not. The bank holds cash and other assets, including loans and investments; it does not set aside your particular dollars in a drawer. So what is that number?

Here is the answer: it is a **promise**. When Bank of America reports that you have $5,000 in your checking account, what that really means is that the bank has a legally enforceable obligation to pay you $5,000, on demand, whenever you show up and ask. In accounting terms, your 'money in the bank' is the bank's *liability* to you—and, from your side of the relationship, it is your *receivable* from the bank. Even the money you casually call "cash in the bank" is a claim on someone else.

Once you see this, accrual accounting's treatment of receivables looks far less alien. An $18k promise from creditworthy customers (like the one your shirt buyers made at the start of this chapter), and a $5,000 promise from Bank of America, are both legally enforceable claims to receive money. They differ in liquidity and credit risk, which is why accounting records them in different accounts. Accrual accounting simply takes this seriously. Wealth, for the most part, does not consist of green paper—it consists of enforceable claims.

1.10.3 Accrual Accounting Expense Matching

The second major difference is that, under accrual accounting, **expenses are, when practicable, matched to the corresponding revenues** that they helped to generate—that is, recorded as expenses **in the same reporting period**.

Before the formal mechanics, an everyday intuition. There are two very different

[13] You might wonder what Alison's $300 *is*, then, if it is not (yet) revenue. It is sitting in our bank account, after all—surely it must be accounted for *somehow*. It must, and it will be: hold that thought until the next chapter.

kinds of spending in your own life. If I stop at 7-Eleven on the drive home and buy a bag of Sour Patch Kids, I eat them immediately: the money is spent, the benefit is consumed, the episode is over. That is spending-as-*expense*. But when you pay tuition for your education, or renovate your house, you are spending money now in order to create benefits that will arrive over many future years. That is spending-as-**investment**. Accrual accounting's expense matching is the bookkeeping recognition of exactly this distinction.

An example of this is the **accounting "depreciation"** of investments. Suppose that we buy a new **Plant** in order to manufacture shirts that we will sell over the next ten years. In this case, we would not recognize the entire cost of purchasing the plant as an *expense* in this year—instead, we would attempt to match that expenditure to the revenues, by **apportioning the cost over the 10 years** over which we will use the plant. In accounting, that **apportionment** is called **'Depreciation expense.'**

Here's the flip side of that, though: *Because* we want to *apportion* that expense over those 10 years, we don't record the expenditure on the **Plant** as an *expense* right away. But, we must record that transaction, and *account for* what we spent that money on, and what we got in return. What gives? The answer is that when we *first* buy that **Plant**, we record it as an **Asset** (reported on the Balance Sheet). And then, later on, as we use it, over multiple periods, we'll record *Depreciation Expenses*. This is a core insight, and a deep link between the Balance Sheet and Income Statements. Generally, **Assets become expenses—get expensed out—*later*.**

- **Expense matching thought experiment**: Suppose our company spends $100k dollars to buy a plant that will produce goods over the next ten years. It doesn't seem correct to say that we 'lost' $100k that year—instead, it's more like we are **investing** now to gain more later. Similarly, if, over the next ten years, we were to earn only $5k per year in Revenues from the sale of goods made in that plant (after an initial investment of $100k), such that we lost $50k on net overall, it wouldn't seem correct to say it was 'profitable' in all those subsequent years. It seems intuitive to measure profitability in a way that **matches expenditures on resources to the periods in which those investments pay off.**

Here is one more worked example, using the world's simplest business—yet another shirt company, even simpler than the one you founded at the start of this chapter. Suppose I pay $100k in January 2019 to buy T-shirts (*Inventory*); I sell half of them in 2019 for $75k in cash, and the other half in 2020 for $75k in cash. If we simply charted the *cash flows*—no accrual accounting—we would get:

	2019	2020
Cash inflows	$75k	$75k
Cash outflows	($100k)	$0
Net cash flow	($25k)	$75k

Does it really make sense to say that I *lost* $25k in 2019, and then gained $75k in 2020—in two essentially identical years of the identical business? Of course not: the initial $100k payment was, conceptually, an investment that pays off across both periods. Under accrual accounting, we *apportion* that $100k to the periods in which the shirts are actually sold: a $50k *expense* (a different thing from the cash outflow) in each of 2019 and 2020, and thus $25k of *Net Income* (a different thing from net cash flow) in each year. Two identical years of selling shirts; two identical measures of performance. That is matching, doing its job.

1.10.4 Accrual Accounting Purpose

Why does accrual accounting do things this way? It's trying to measure the profitability of a company's real ***activities*** on an ongoing basis. The Income Statement gives a **'flow'** measure of the value generated for owners/shareholders from a company's real **activities/operations over the period**. And, it recognizes revenues and expenses in the **period** in which the **activities occur** rather than that in which the **cash changes hands / checks arrive in the mail**.

1.10.5 Returning to the bathtub

We can now complete the bathtub picture introduced earlier. The Income Statement measures the period's addition to value through revenues, less its expenses. The Balance Sheet reports the company's position at a date. Dividends move value out to the owners; they do not undo the profit that was earned. The link is Retained Earnings: its opening balance, plus Net Income, less Dividends, gives its closing balance. Keep the distinction clear: we are picturing the accumulation of value, not simply the movement of cash.

1.11 Statement of Shareholders' Equity

This statement is the simplest, has the least information, and is the one that investors pay the least attention to. Its distinctive focus is on tracking the business's transactions and standing with its own owners/shareholders. That is, it's all about **Share-**

holders' Equity (E). And there are fewer transactions—of less complexity—that a business makes with its owners/shareholders, than all the others tracked in the other statements. This statement is fairly intuitive, but presented in a different format than the others. Instead of listing **accounts** on the vertical axis (as in the Income Statement and Balance Sheet above), it instead lists particular transactions and items on the vertical axis, and tracks how those affected the Shareholders' Equity accounts on the horizontal axis. That is, it explicitly shows the *changes* in the accounts, and their drivers. A simple Statement of Shareholders' Equity (SSE) could look like this:

	Common Stock	Retained Earnings	Total Shareholders' Equity
Balance, Jan 1, 2020	$0	$0	$0
Issue Common Stock (Tr. #1)	15,000		15,000
Total Dividends Paid		(100)	(100)
Net Income		2,300	2,300
Balance, Dec 31, 2020	**$15,000**	**$2,200**	**$17,200**

Table 1.1: Statement of Shareholders' Equity

For present purposes, the important thing about this statement is how it helps to illustrate the **link** between the Income Statement and Balance Sheet. This statement is governed by two key equations:

$$Shareholders' \ Equity = Common \ Stock + Retained \ Earnings \tag{1.5}$$

This equation says that **E** comes from **two sources**: (1.) **'Common stock'** is the cash that investors directly paid in when they initially bought their shares of ownership (stock) from the business. It's often called **'paid-in capital.'** (2.) **'Retained earnings'** is the amount of profits that have been *earned*, but which the owners did *not* pay out to themselves, and instead *retained*. It's often called **'earned capital.'**

So, this third financial statement is based on the third **core identity**:

$$Retained \ Earnings = Total \ NI \ Earned - Total \ Dividends \ paid \ out \tag{1.6}$$

Retained earnings is cumulative sum of whatever profit the company has earned over its entire history, less whatever it has paid out as dividends—in other words, the total amount that it has *retained*. Combining these two equations, we can see that:

$$Shareholders' \ Equity = Common \ Stock + Total \ NI \ Earned - Total \ Dividends \ paid \tag{1.7}$$

- the world of the Balance Sheet:
  - $A = L + E$, **stock measures, permanent accounts, at a point in time**
- and the world of the Income Statement:
  - $R - E = NI$, **flow measures, temporary accounts, measured over a period**

The income-statement measures **'live on' ' in Retained Earnings**, which is a part of Shareholders' Equity, on the Balance Sheet. And the Statement of Shareholders' Equity illustrates this. It reports the Net Income earned, the Dividends paid out, and thus the *change* in Retained Earnings; it also reports all of the company's transactions with its owners/shareholders, including the Common Stock investments. Summing the two provides the updated Shareholders' Equity **(E)** reported on the Balance Sheet.

1.12 The Cash Flow Statement

The **Cash Flow Statement** reports the company's performance over the period, using (essentially) **cash-basis accounting** instead of **accrual-accounting** principles. It simply tracks how much **cash** the company **expended** and **received** on various different things. Importantly, it divides those cash flows into **three categories**: Cash flows from **Operating Activities**, **Investing Activities**, and **Financing Activities**. And it reports the net change in cash *within* each category of activity, and on net.

You might ask: don't we already know the company's cash position from the Balance Sheet? We do—Cash is right there, the first asset listed, and by comparing two consecutive Balance Sheets you could compute the net change in cash yourself. The value of the Cash Flow Statement is that it *explains* that change: it decomposes the net movement in cash into its underlying sources and uses—where the cash came from, and where it went.

So, they'll all have roughly this basic structure:

Operating cash flows...:		
…		$…
…	$(…)	
Net Change in Cash From Operations		**$X**
Investing cash flows..:		
…		$…
…	$(…)	
Net Change in Cash From Investing	**$(Y)**	
Financing cash flows..:		
…		$…
…	$(…)	
Net Change in Cash From Financing		**$Z**
Net Change in Cash Between Periods		**$X-Y+Z**

Table 1.2: Statement of Cash Flows, for FY ending 12/31/2020

We need to learn *a lot* more accounting before we can learn exactly how the Cash Flow Statement works, what goes into those ellipses (...) above. For now, let's just talk about two more general things, that will come up in the interim: (1.) **The categories** (operating vs. investing vs. financing activities), (2) **Why** financial-statement users would be interested in the cash-flow statement.

1.12.1 Categories of Transactions: Operating, Investing, Financing

- **Operating Activities** are transactions having to do with a firm's normal day-to-day activities and core business. These include things like **making sales of goods or services** to customers, **paying salaries and utilities**, etc.

- **Investing Activities** are transactions having to do with the purchase and sale of **long-term Assets**—resources that will pay off over multiple periods. This would include the purchase and sale of **Plant, Property, & Equipment**. And, if a company were to purchase or sell a *different* company's stocks or bonds, as an **Investment**, that would be an Investing activity as well.

- **Financing Activities** are transactions having to do with how the company **finances *itself*** —i.e., its transactions with its own **financial creditors and investors**. So, paying out **Dividends**, or getting extra cash from new investors (Common Stock) or from a new **loan**—these would all fall in the *financing* category.

One classification confusion is worth defusing right away. From the company's perspective, issuing common stock and receiving cash is a *financing* activity. But from the shareholder's perspective, buying that stock is an *investment*. The same transaction is 'investing' to the investor and 'financing' to the company—which is why the terminology gets flipped around in people's heads. The rule: when classifying cash flows on the Cash Flow Statement, we always speak from the *company's* point of view.

These will show up elsewhere in this book, everyday speech and business news.

1.12.2 Why the Cash Flow Statement?

What's the point of the Cash Flow Statement? Didn't we just make an argument for the principles of **accrual accounting**, as opposed to **cash basis accounting**, above? There are good, logical reasons for accrual accounting, and some real benefits. But there are also some downsides and potential problems. So, the Cash Flow Statement provides some incrementally useful information. In particular, it helps: (1.) As a **"check"** on potential abuses and manipulation of accrual accounting, (2.) To understand a company's **cash liquidity**, its ability to pay what it owes in cash.

- **(1.) Check on gimmicks:** Accrual accounting involves some unverifiable estimates and forecasts that can be abused. For example, we learned above that companies **record revenue** when they ship goods to **customers**, rather than when the check arrives in the mail. There's a reason for that. But what if a company started aggressively shipping product to customers who were never likely to actually pay, in order to boost reported revenues? Clearly, that doesn't sound like a good idea. As this book progresses, we'll see several other examples of cases in which accrual-accounting principles can allow firms to boost their *reported* profitability, even though the underlying economics is not improving.

  The cash flow statement doesn't depend on these accounting choices. It more closely tracks cash receipts. Thus, if you notice that a company's **Net Income** were increasing, while its **Cash Flows from Operations** were decreasing, that could be a **red flag**. It *could* indicate that the company's management is using accounting 'gimmicks' to make the company appear to be doing better than it really is.

- **(2.) Cash liquidity / 'Cash is King':** Companies have to **pay the bank**—that is, pay their contractually obligated **interest and principal** to their creditors—in **cash**. Otherwise, their creditors can seize collateral, take them to court, etc. And, in principle, a company *could be* highly profitable in terms of accrual **Net Income**, but **not** have enough **cash on hand and/or coming in** to pay its debts in cash. This could happen if, for example, the company was making all of its sales

to customers *on credit*—and not collecting cash from them. Or if the company was investing heavily in building up its *Inventory* (an asset), but not selling much of it. Such companies can be **profitable** but not **liquid**—i.e., not able to convert their value to cash to make payments when due. For these reasons, a company's **cash flow performance** is very important to creditors and managers.

1.13 Other information

The focus of this book is the four **Financial Statements**. But these are **not the totality** of the information that firms disclose. Firms are also required to, and voluntarily choose to, disclose plenty of other information as well—things that may be important to various stakeholders, but cannot be reliably quantified in the Financial Statements. Examples include: the management's *ideas* and *plans* for its **strategy**; and certain **long-term risks** that are **not quite quantifiable** as liabilities. These can be discussed verbally or qualitatively in the firm's annual reports (such as the **management discussion and analysis (MD&A)** section), or discussed on conference calls with analysts. While this 'other information' is, by definition, not part of the Financial Statements, it is an important part of the firm's "information environment"—the process by which it delivers useful information to the stakeholders who need it.

1.14 Says who? Standard setters, regulators, auditors

So, we've sketched out the major principles of accounting, and how they are used to produce the financial statements. But...**says who?** Who has the authority to say that firms must report those Financial Statements according to those principles, and why? There are two major standard-setting bodies and systems we'll talk about: U.S. GAAP standards, issued by the FASB, and international IFRS standards, issued by the IASB.

First, a bit of situating. 'Accounting,' in the broadest sense, just means *giving an account*—and people have been keeping accounts, in one form or another, for millennia. There is no single monolithic 'accounting'; there are different systems built for different purposes. *Managerial accounting* serves decision-makers *inside* the firm; *tax accounting* serves compliance with the tax code (and, let's be honest, the minimization of taxes); and *financial accounting*—the subject of this book—serves *external* users, like investors and creditors, who need standardized, comparable reports. When I say 'accounting' in this book, I mean financial accounting.

1.14.1 GAAP

In the U.S., **The Financial Accounting Standards Board (FASB)** sets financial accounting standards for entities that report under U.S. **Generally Accepted Accounting Principles (GAAP)**. U.S. public companies generally must use U.S. GAAP in their SEC filings. Accounting practices had evolved for centuries before either institution existed. In **1934**, after the stock market crash and the beginning of the Great Depression, Congress created the **Securities and Exchange Commission (SEC)** and gave it authority over public-company financial reporting. The SEC recognizes the FASB, established in 1973, as the designated private-sector accounting standard setter. The FASB is an independent, not-for-profit organization based in Norwalk, Connecticut; the SEC retains its statutory authority over reporting by companies within its jurisdiction.

1.14.2 IFRS

Globally, non-U.S. developed and developing countries have largely adopted a uniform set of accounting standards, called **International Financial Reporting Standards (IFRS)**, which are issued by an international NGO, the **International Accounting Standards Board (IASB)**. This body was developed in response to the rapid globalization of the late 20th and early 21st century. Financial-statement users increasingly wanted common, comparable standards if they wished to transact with and invest in companies from, e.g., Italy to Germany to Japan, etc. Prior to this, most large countries had their own country-specific accounting standards.

1.14.3 Convergence

Years ago, there was hope and some promises that the U.S. FASB and the international IASB would shortly "**converge**," and adopt a **common set of standards** for both U.S. and international companies. Such hopes have since fizzled out, and now there is little hope that we'll have one truly global set of accounting standards in the near future. As such, this book will mostly focus on U.S. GAAP. But most of the core accounting principles we'll cover are shared across the two systems. Of what we've discussed so far, there is only one exception: IFRS allows firms to record some of their R&D as an Asset (i.e., 'capitalize' it). More broadly, **IFRS** are often described as more **'principles based,'** while **U.S. GAAP** standards are seen as more **'rules based'** in the details of accounting for particular transactions.

1.14.4 Auditors

While the **standard setters** (FASB / IASB) are the ones who **make** the rules, **auditors** are the accounting professionals who independently check and **verify** that businesses are actually obeying those rules, and correctly reporting their Financial Statements. (The audit industry is dominated by the so-called **'Big 4'** firms—Deloitte, EY, KPMG, and PwC—names you will encounter constantly in your business career, and perhaps on your own résumé.) These auditors will do everything from verifying that the business used an acceptable **method** in calculating its depreciation, and calculated it correctly; to verifying in person, via random checks, that it actually has the amount of Inventory it claims. These auditors have an **extraordinarily valuable** role to play in ensuring that companies are not 'cooking the books'—and that investors can rely upon and trust the claims that companies make in their Financial Statements. But there have been some extraordinary failures (such as, e.g., the Enron-Arthur Andersen debacle).

Audit is a separate course of study, and we will not cover it in this book. But for now, just pause and appreciate what a profound question it is at its core: **How do we really know** that large institutions have actually done the things that they **claim** they have done? That the events they attest to have happened **really happened?** That they really **have** the resources they claim? Think about this in the context of the major institutional failures you have witnessed in your own lifetime—the moments when a large, official institution asked for the public's trust, and the curtain was then pulled away. (The pandemic era supplied several examples; every era supplies its own.) It's a deep question, and one we typically take for granted—often to our detriment.

2

Accounting for External Transactions

Chapter 1 was a high-level, conceptual overview. In this chapter, we'll take a more practical approach, and start doing some simple accounting. We'll **account for** 10 basic **transactions** using the **Accounting Equation** ($A = L + E$)—and then, once those ten have sunk in, go a bit beyond with two more.

Let me say something up front, so that you calibrate your effort accordingly: from my perspective, this is the most important chapter in the book. Later chapters will be more technically challenging and complicated—they have to be. But this is where we encounter the core principles of financial accounting *in action*, transaction by transaction. If you ask me what I most want you to still understand five years from now, it is the material in this chapter.

For now, we'll only account for a company's **"external transactions" during the reporting period**—i.e., the "real" exchanges of things or services between the company and some outside party, at the **exact moment of the exchange**. These are in contrast to "adjusting entries" or "internal transactions" at the *end of the reporting period*, which we will cover in subsequent chapters.

As a concrete example, suppose that our company pays for a 2-year lease on a building up front, halfway through the annual reporting period. The day the company pays for the lease, that is a "real" exchange—an external transaction. And the company receives, in return, a resource that it will use for multiple reporting periods—the ability to use the building for 2 years. So, in the moment of the exchange, it pays out cash, and receives an **asset**, which we happen to call **Prepaid Rent**.

Later on (Chapter 4), as the company "uses up" the lease, it will need to account for the **expense from using up the asset**—in this case, called *Rent Expense*. But there will be no "real" moment of exchange that will trigger that. Hence, we will account for that at the end of the reporting period, as an "adjusting entry" or "internal transaction." This distinction will become much clearer over time. For now, in this chapter, we are accounting only for what happens at the **exact moment of the transaction**. As we work through the transactions below, you will feel a strong temptation to 'think ahead'—to start mentally adjusting for how these resources will be used up over time. Resist that temptation. For now, be a punctilious accountant, and stay with the immediate exchange.

There are four things we need to keep in mind before we begin.

- **First**, remember that **the Accounting Equation (A = L + E) is an identity**—the two sides must always add up. Any transaction that affects one part of the equation must have a second effect that keeps the two sides in balance.

- **Second**, this is the equation that governs the Balance Sheet. But, as we discussed in Chapter 1, the Balance Sheet and Income Statement are intimately linked. So, we will see some Revenues and Expenses, and we'll account for them as eventual **contributors to** Shareholders' Equity, based on the **Retained Earnings Identity** from last chapter ($\Delta \textit{Retained Earnings} = \textit{Net Income} - \textit{Dividends}$).

- **Third**, we must define what we mean by an **'Account.'** An 'Account' has a commonsense meaning: It's a category for tracking changes in one type of resource or obligation. So, e.g., a company can have a **Cash** account (tracking the cash money it has in the bank) and an **Equipment** account (tabulating the value of the equipment it owns). Firms have some **discretion** over precisely how they **divide up the world**—i.e., what accounts they keep, which they call their '**Chart of Accounts.'** But there is also lots of similarity across firms, and strong conventions. We'll get familiar with the major categories in use over time. One more note on this: Some of what we learn in accounting is conceptually *deep*—the residual nature of equity, say, or the principle of matching. Other things, like precisely how firms name and group their accounts, are largely *conventions*. There is no profound reason that a construction company groups its cranes and bulldozers together in one account called *Equipment*, or that we call loans from banks 'Notes Payable.' We just get used to the standard categories, and use them consistently.

- **Fourth**, remember the definitions of assets and liabilities. In its conceptual framework, the Financial Accounting Standards Board (FASB) describes an asset as a present right to an economic benefit, and a liability as a present obligation

to transfer an economic benefit. The definitions below put those ideas in the terms we will use in our examples.[1]

- **Assets**: Rights the company currently has to economic benefits—for example, the right to use equipment or collect a customer's payment.
- **Liabilities**: Obligations the company currently has to provide economic benefits to others—for example, to repay a loan or deliver a service that a customer has prepaid.

The 10 transactions that we'll account for are:

1. Sell Shares of Common Stock for $15,000
2. Borrow $5,000 from the bank
3. Purchase equipment for $10,000
4. Pay $2,400 for two years of rent, up front
5. Purchase supplies 'on account' (that is, with promise to pay cash later) for $1,000
6. Provide soccer training to customers for $3,900 in cash
7. Provide soccer training to customers 'on account' (that is, with them promising to pay later) for $3,100
8. Receive cash payment in advance/up-front for future training, $800
9. Pay salaries to employees, $1,800
10. Pay cash dividends of $100 to shareholders.

2.1 Transaction 1: Sell Shares of Stock for $15k

This transaction is the original (**OG**) transaction in the life of the firm, in the "simple model of a business" discussed in Chapter 1. The first owner puts in their own money to finance the enterprise. And as additional investors put in more money (e.g., Shark Tank), they also become owners, receiving *shares*. In these cases, we say that the company has '**sold common stock**'—it has disbursed a legal contract (shares of ownership) and received cash in exchange.

[1] See FASB, Concepts Statement No. 8, Chapter 4, Elements of Financial Statements, issued December 2021. These definitions replaced the earlier Concepts Statement No. 6 definitions. The conceptual framework explains the elements; specific accounting standards govern recognition and measurement.

One note on perspective before we begin: As the owner, I am taking money *out* of my own bank account, and putting it into the corporation's. But in accounting, we always adopt *the company's* point of view. From the company's perspective, this transaction is simple: it receives cash, and disburses shares of ownership.

How to account for this? **Start with cash**. This is a mantra I will repeat throughout this book: *if there is cash in the transaction, start with cash.* Why? Because cash is simple and concrete. Some accounts (*Retained Earnings*, say) are abstract and conceptual; *Cash* is not. If we receive cash, that asset goes up; if we pay out cash, it goes down. It is a clean anchor, from which we can work out everything else. So: The company receives a cash inflow of $15k. **Cash** is obviously an economic resource controlled by the firm. So that is an **Asset** under the FASB definition. But the accounting equation must stay in balance. What else happens? Since this cash is a direct contribution from investors, **Shareholders' Equity (SE)** must increase ($A \uparrow$, $E \uparrow$). So, it would look like this:

Transaction	Assets	Liabilities	Shareholders' Equity
#1: Sell $15k of Common Stock	+$15k (Cash)		+$15k (Common Stock)

2.1.1 Takeaways

Even this simple transaction yields important, deep insights. The key **takeaway** from this transaction, the **weird thing** you need to wrap your mind around, is **the nature of double-entry bookkeeping**. Note that the company, at this point in time only **has one thing**—$15k in cash. (It *does not **have*** $30k of anything, even though the two accounts add up to that.) But we **account for that one thing in two ways**. This is what "double-entry bookkeeping" means. The left side of the accounting equation (A) tells you what the company *has*; and the right side ($L+E$) tells you how that thing was *financed*. This is a profound and deep insight, and even very senior, highly-remunerated financial professionals make mistakes because they haven't fully internalized this distinction. See, e.g., Prof. Anat Admati's "The Bankers' New Clothes."

One more piece of jargon-management: you will see the phrase **double-entry bookkeeping** in any serious discussion of accounting or its history, and it sounds like some special, fancy technique. But in this book, it will quickly fade into the background—because we will be doing it constantly, on every transaction, in every chapter. Like a fish in water, you soon won't even notice it.

2.2 Transaction 2: Borrow $5k cash from the bank

This transaction probably seems more familiar: A company getting a loan from a bank. How to account for this? **Start with Cash**. Clearly, Cash, an asset, goes up by $5k. How does the accounting equation stay in balance? The **bank is a creditor**, because the loan has to be paid back, so it is a **Liability** for the company ($A \uparrow$, $L \uparrow$). Now, here's a bit of **jargon** we just need to know: We use the somewhat antique-sounding term **"Notes Payable"** for this liability account. You can remember it this way: 'Notes Payable' are *bank notes* that you are *obligated to pay* off (a 'payable').[2] So, we account for it like so:

Transaction	Assets	Liabilities	Shareholders' Equity
#1: Sell Common Stock	+$15k (Cash)		+$15k (Common Stock)
#2: Borrow from bank	+$5k (Cash)	+$5k (Note payable)	

Note that, in the table above, we **separately account** for the **incremental effect** of each transaction on its own. The $15k received in Transaction #1 does not change the $5k increase in Cash from Transaction #2. We could also combine the two transactions to find the total balance in each account, as follows:

Transaction	Assets	Liabilities	Shareholders' Equity
#1 & #2 Aggregated	$20k (Cash)	$5k (Note Payable)	$15k (Common Stock)

Obviously, these are just two different ways of organizing the same information. But this distinction—between the ***incremental effect* of transactions vs. *aggregate balances***—can trip students up as things get more complex. (When you see a '+$5k' in the tables above, that plus sign denotes the *change* from that transaction—not the balance of the account. And when the company eventually reports *Cash* on its Balance Sheet, it will report the *balance*—everything that has accumulated to date—not the most recent change. The Balance Sheet is not 'the last transaction.')

2.2.1 Takeaways

The key **takeaway** from this transaction, the **weird thing** you need to wrap your mind around, is **how to see The Accounting Equation**. Recall that last chapter, we talked about the two ways to see it:

[2] In general, there will be several accounts referred to as "payables," and they will all be liabilities, since they represent amounts the firm *has to pay to others*. These will be in direct contrast to "receivables." Those are amounts the company is entitled to receive from others, which are thus assets.

- (1.) The **"Sources of Financing"** view ($L + E = A$): The company *has* \$20k in cash (A). And that cash had to come from somewhere. The L and E accounts tell us how much of it *was provided* by creditors vs. equity investors, respectively.
- (2.) The **"What's Left Over"** view ($E = A - L$): The company *has* \$20k in cash. But it has a legal obligation to pay the bank back the \$5k loan from the bank. So, if it does that, it will have \$15k left over, which can be claimed by the **equity owners**.

To make that second view fully concrete, try a little thought experiment. Suppose that, right after Transaction #2, I decide that entrepreneurship is not for me after all, and I shut the company down. The company has \$20k in cash. What happens? First, I *must* pay the bank back its \$5k—creditors get paid before owners; that is what a liability *is*. After paying the bank, \$15k remains, and I, the owner, can claim it. That is Shareholders' Equity: $E = A - L$. And note, once again, that there was never a separate 'pot' of equity sitting anywhere—just the \$20k of cash, and two kinds of claims upon it.

2.3 Transaction 3: Purchase Equipment for \$10k in Cash

So now, the firm has cash financing. Step 2 in the "simple model of a business" from Chapter 1 is to **acquire resources** that it will use to do its business.

The company buys **Equipment** for \$10k in **Cash**. How to account for this? Let's make it easy for ourselves and **start with cash**: That's an asset, and it goes down by \$10k. Next, how does the accounting equation stay in balance? In exchange for the cash, the company got **Equipment**—this is a *resource* that the company now *owns* and will use for its benefit for many periods. (It is not like a bag of candy, consumed on the spot and then gone; the equipment endures, and keeps serving the firm.) That means that it is an *asset*. So, this time, the change in one asset account is balanced by a change in another asset account ($A\uparrow$, $A\downarrow$). In other words, the transaction transfers value from one asset account to a different asset account.

Transaction	Assets	Liabilities	Sh.'s Equity
#1: Sell \$15k of Common Stock	+\$15k (Cash)		+\$15k (C.S.)
#2: Borrow \$5k from the bank	+\$5k (Cash)	+\$5k (N.P.)	
#3: Purchase equipment for \$10k	-\$10k (Cash) + \$10k (Equipment)		

And our **balance** (i.e., adding this transaction to the last balance) looks like this:

Transaction	Assets	Liabilities	Sh.'s Equity
#1, #2, & #3 aggregated	$10k (Cash), $10k (Equipment)	$5k (N.P.)	$15k (C.S.)

How do we know that the Equipment is **truly worth $10k**? That's a tricky and in some ways philosophical question. (Indeed, arguably, the firm should only pay $10k for the equipment if it thinks that it will be worth more than $10k to the firm. If you buy something for $100 in cash, then that means you preferred that thing to your $100 in cash—or in other words, that that thing was worth more than $100 to you.) So, the short answer is: **We don't** know what it's truly worth.

This is where accounting principles such as **verifiability, conservatism,** and the **monetary unit assumption** come into play. We need to **'convert'** all the company's various real resources into some **monetary unit**—dollars. And we can't just let the managers **say** that the resource is worth **whatever they want to say**. So, in accounting, we, by default, initially base asset measurements on the **purchase price** (often phrased as "historical cost"), based on a **verifiable receipt**, rather than more subjective estimates. The assumption is that if the firm bought that equipment in an **arms-length transaction** in a fairly **competitive market**, then the transaction price is the best available estimate of its value. Put differently, **"in accounting, you get what you pay for."**

2.3.1 Going deeper: Why does the purchase price deserve our confidence?

An auditor cannot inspect management's belief that a machine will be extraordinarily productive. But the auditor can inspect the invoice, the payment, and the evidence that the machine was delivered. A purchase price gives the accounts an anchor that another person can check.

There is a further condition hidden in the phrase **arm's-length transaction**. The parties should be independent, each bargaining in its own interest. If a manager sells something to a company that the same manager controls, a payment may be verifiable without being persuasive evidence of value. We need to ask who was on each side of the bargain.

Now test the distinction. Our company pays $15m for a factory and believes it can double the factory's productivity. What amount do we initially record? $15m. Management's belief may be right, and it may explain why the purchase was attractive. It does not turn the purchase into a $30m expenditure. The book value and the value management hopes to realize are different things.

2.3.2 Takeaways

The two **weird things** you need to wrap your mind around, are:

- In accrual accounting, when you **spend money on something you will use over multiple periods**, that thing is **initially an Asset**. Prior to starting this book, you probably thought that when companies spend money, that's an *expense*. So, this might seem counterintuitive. And, in Chapter 4, we will see that the Equipment will **"be expensed out"** as it is used up, over time. That expense will be factored in as a part of the "adjusting entries" at the end of the reporting period. This basic pattern will hold true for *most assets*. Assets are not expenses at the moment of the external transaction—but they come to be recognized as expenses later on.

- **"In accounting, you get what you pay for."** That is, we initially record assets "on the books" at their **purchase price**. There are reasons why we do this. But this means that "accounting value" and "true economic value" are not always going to be the same thing. This seems strange at first, but we all eventually get used to it. Remember: **"The map is not the territory."**

2.4 Transaction 4: Pay two years of rent in advance, $2.4k

How to account for this? **Start with cash**. That asset account obviously goes down by $2.4k. What else happens to keep the equation in balance? In exchange for the cash, the corporation gets **control** of a new economic **benefit/resource**: The ability to use the building for multiple periods. So that's an asset. We call this asset **"Prepaid Rent."**

Using the accounting equation, the transaction looks like this:

Transaction	Assets	Liabilities	Sh.'s Equity
#1: Sell $15k of Common Stock	+$15k (Cash)		+$15k (C.S.)
#2: Borrow $5k from the bank	+$5k (Cash)	+$5k (N.P.)	
#3: Purchase equipment for $10k	-$10k (Cash), +$10k (Equipment)		
#4: Pay rent up front, $2.4k	-$2.4k (Cash), +$2.4k (Prepaid Rent)		

And our **new balance** (i.e., adding this transaction to the last balance) is this:

Transaction	Assets	Liabilities	Sh.'s Equity
1 - 4 aggregated	$7.6k (Cash) $10k (Equipment) $2.4k (Prepaid Rent)	$5k (N.P.)	$15k (C.S.)

Now, it *must be true* that the Accounting Equation balances for the *aggregate account balances*, since we balanced it on each individual transaction, and the aggregate balances are just the cumulation of those. But, to reassure ourselves, we can verify:

$A = \$7.6k + \$10k + \$2.4k = \$20k$

$L + E = \$5k + \$15k = \$20k$

This verification contains a practical trick worth internalizing. Suppose someone asks: 'What are the company's *total assets* at this point?' Your first instinct might be to trudge through the whole asset column, transaction by transaction: $+\$15k + \$5k - \$10k + \$10k - \$2.4k + \$2.4k$... You can do that, and you will get $20k. But if you have the courage of your convictions that $A = L + E$ must *always* hold, you can take a shortcut: the right-hand side is often much cleaner. $5k of Notes Payable, plus $15k of Common Stock, is $20k—so total Assets *must* be $20k. Both routes are valid; one is just faster and less error-prone under pressure.

While we're here, note a related distinction that trips up more students than you would think: the company's total *Assets* are $20k, but its *Cash* is $7.6k. Those are different questions, with different answers. $A = L + E$ is an axiom, so we can always solve for total Assets from the right-hand side. But there is no axiom that says 'Cash equals something simple'—*Cash* is just one account *within* Assets, sitting alongside *Equipment* and *Prepaid Rent*. The arithmetic in this book is rarely hard; when people go wrong, it is usually because—under pressure, with tunnel vision—they lose track of *which quantity is being asked for*. The effect of a transaction, or a balance? Cash, or total assets? Slow down, and match your thinking to the question.

2.4.1 Takeaways

The two key **weird things** you need to wrap your mind around, are:

- **Abstract/intangible nature of assets:** The ***Prepaid Rent*** is not a real, physical thing like *Equipment*. Reductively, it is just **a contract**. (What do we 'have,' really? A signed PDF.) But that contract is a resource that will bring it benefits over multiple periods, and which it now controls. So it fits the definition of an asset. On this point: you will sometimes hear businesspeople assert that "accounting doesn't capture intangible assets." Stated that baldly, this is simply false. Real balance sheets are full of intangible assets: purchased trademarks, patents, lease

rights, and more. What matters for the definition is not whether you can *touch* the thing—it is whether it is a resource, obtained in a past transaction, that will provide future benefits under the firm's control.[3]

- **Assets and expenses again:** Just like last time, the company **spends money**, but it's **not an expense—*yet***. At the exact moment of the transaction, the company has acquired a resource, but not used it up. We will account for *Rent Expense* **later on**, as a part of the adjusting entries at the end of the period.

2.5 Transaction 5: Purchase Supplies for $1k "on Account"

Often, companies will buy things from their suppliers and promise to **pay cash later**. When they do this, we say that they are buying **"on account"** or **"on credit"** (that is, on the basis of the **credibility** of their promises, or their **creditworthiness**).

When I first learned this, buying 'on account' struck me as some abstruse accounting game, invented to torture students. It is, in fact, **the norm** in business-to-business relationships. Companies do not settle up in cash with their regular suppliers every single day, delivery by delivery. They receive goods and services throughout the month, keep records ('accounts') of what is owed, and settle the balance periodically. So this humble transaction type is one of the most common in all of commerce, and understanding how businesspeople think about it will pay off.

Unused *Supplies* are resources available for future use, so they are assets. They need not last for multiple reporting periods. By now, you surely know what *would* happen if we purchased supplies using cash. But **there is no cash** in this case. What do we do? One thing we may *not* do is nothing. You might be tempted to reason: 'No cash has moved yet, so there is nothing to record yet.' That is exactly how **cash-basis accounting** would work—recording things only when cash changes hands. But we are doing **accrual accounting**: we record transactions when the underlying *economic event* occurs, whether or not cash moves that day. And an economic event has plainly occurred here—we have more resources than we did this morning. (File this away as a general rule: "do nothing, because there's no cash yet" is *never* a valid argument in this book.) This is **still a "real" external transaction**, because there is a real **exchange** (the Supplies). How do we account for it? What balances the increase in Supplies ($A\uparrow$)?

Our company has *promised* to pay later. This means it has an *obligation*—a **Liability** ($A\uparrow$, $L\uparrow$) . When the company owed money to the bank, we called that *Notes*

[3]There *are* particular intangibles—like internally-developed R&D, and customers' brand loyalty—that accounting conservatism keeps off the books, as we discussed in Chapter 1. That is the defensible kernel inside the sloppy complaint.

Payable. And, in this case, when it **owes money to a *supplier*,** we call that ***Accounts Payable***. So, we would account for this transaction like so...:

Transaction	Assets	Liabilities	S.E.
#5: Buy supplies on account	+$1k (Supplies)	+$1k (Accounts Payable)	

And our **new balance** (adding this transaction to the last balance) looks like this:

Transaction	Assets	Liabilities	Sh.'s Equity
#1-#5 aggregated	$7.6k (Cash) $10k (Equipment) $2.4k (Prepaid Rent) $1k (Supplies)	$5k (N.P.) $1k (Accounts Payable)	$15k (C.S.)

2.5.1 Takeaways

There is one **weird thing** you need to wrap your mind around here:

- **Liabilities (L) are not just owed to the bank:** Up until this transaction, the only liability that we saw was *Note Payable*, the accounting term for a loan owed to a bank. And that fit with the intuition and corporate finance material we covered in Chapter 1, where we said that all of the resources of the company were financed by equity vs. credit financing. Here, we found out that *promises to pay suppliers* are also a liability.

  Does this contradict that early model, and our understanding of the meaning of the accounting equation? Not really. Most businesspeople would think of this as a **form of credit financing, provided by our suppliers**. When our suppliers give us a good today, and allow us to pay them back later, it is **as if** they are extending a loan. Just like a bank that gives us cash today, and requires us to pay them back later. And, we **have to pay our suppliers**—otherwise, they won't continue to do business with us. So they come **prior to our shareholders**. As such, this still fits into our framework for **how to see the accounting equation** ($E = A - L$).

You can also imagine the supplier doing this in two steps: first lending us $1k, and then selling us $1k of supplies for cash. The cash would come in and immediately go out again. What remains is exactly what we recorded: $1k of Supplies and a $1k obligation to pay the supplier. That is the sense in which buying on account provides financing.

2.6 A Quick Step Back: Our First Balance Sheet

Before moving on, let's step back and see where all of this is headed. In Chapter 1, we said two things about the **Balance Sheet**: (1.) it reports what a firm has, owes, and is worth, at a **point in time**; and (2.) reductively, it is just a *list* of the company's Asset, Liability, and Shareholders' Equity accounts, and their balances. Now look back at our aggregated balances after Transaction #5. We *have* such a list. So—perhaps surprisingly, this early in the book—we can already flip it into Balance Sheet format:

Balance Sheet, after Transactions #1–#5

Assets	**Liabilities**
Cash: $7.6k	Accounts Payable: $1k
Equipment: $10k	Notes Payable: $5k
Prepaid Rent: $2.4k	**Total Liabilities: $6k**
Supplies: $1k	
	Shareholders' Equity:
	Common Stock: $15k
	Retained Earnings: $0
	Total Shareholders' Equity: $15k
Total Assets: $21k	**Total Liabilities and S.E.: $21k**

Two things to notice. First, because we kept $A = L + E$ in balance on every individual transaction, the equation *must* balance in the aggregate, on the Balance Sheet—and it does: $21k on each side. Second, Retained Earnings is $0. Of course it is: the company has not yet *earned* anything. So far, it has only financed itself and acquired resources. That is about to change.

2.7 Transaction 6: Providing Services for $3.9k in Cash

How do we account for this? **Start with Cash**. That asset account goes up by $3.9k ($A \uparrow$). How does the accounting equation stay in balance? The company does not have any new liabilities here, and no other asset is affected. So it must be that **Shareholders' Equity** goes up by $3.9k, to keep things in balance ($A \uparrow$, $E \uparrow$). But this is a bit confusing. **No new equity investors have purchased Common Stock**. So, how does E go up?

Before turning to the formal identities, notice that the economics already makes sense of this. Think of the **"What's Left Over"** view of the equation ($E = A - L$). After this transaction, the company has $3.9k more in assets. And did it take on any new obligation by successfully doing its work and getting paid? No—there is no new liability. So if the owners shut the company down today, paid off the creditors, and claimed

whatever remained, there would be $3.9k *more* for them to claim than there was yesterday. Whatever we end up calling it, the owners' residual claim has genuinely grown. That is the first way to see why revenue must live in the Shareholders' Equity column. Now, the formal version.

We learned in Chapter 1 that when a company **earns** money from its operations, from providing goods or services to **customers**, we call that a **Revenue**. And further, we learned a couple of **core identities,** that we must **know cold**:

$$Net\ Income = Revenue\ - Expenses$$
$$\Delta Retained\ Earnings = Net\ Income - Dividends$$
$$Shareholders'\ Equity = Common\ Stock + Retained\ Earnings$$

Interpreting those identities verbally: **Revenues increase Net Income; Net Income increases Retained Earnings; and Retained Earnings is part of Shareholders' Equity.** That is why this transaction belongs in the equity column even though no owner has contributed new cash.

We also learned that **Retained Earnings** is sometimes called "Earned Capital"—amounts that shareholders have *earned* and retained—as opposed to "Paid-in Capital"—amounts that shareholders directly *Paid In* via purchasing Common Stock. So, when companies **earn Revenue**, we think of that as a **component of, or eventual contributor, to Shareholders' Equity**.

We need to be a bit careful here, though. Remember that **Revenue *per se* is never directly reported on the Balance Sheet** under Shareholders' Equity. Instead, it **eventually flows** into the **Retained Earnings** balance reported at the end of the period. As long as you keep that in mind, then, for now, you can just think of *Revenue* as a positive entry in the *Shareholders' Equity* column.

So, in short, when a company provides a service to customers (that is, **earns revenue**) in exchange for cash, we account for it like so...:

Transaction	Assets	Liabilities	Sh.'s Equity
#6: Provide training for cash	+$3.9k (Cash)		+$3.9k (Revenue)

And our **new balance** (adding this transaction to the last balance) looks like this:

Transaction	Assets	Liabilities	Sh.'s Equity
#1-#6 aggregated	$11.5k (Cash)	$5k (N.P.)	$15k (Common Stock)
	$10k (Equipment)	$1k (A.P.)	$3.9k (Revenue)
	$2.4k (Prepaid Rent)		
	$1k (Supplies)		

2.7.1 Takeaways

There are three key things you need to wrap your mind around:

- **Account for Revenues as positive entries in the Shareholders' Equity column:** Think through the **core identities**, and how **earning revenue** will eventually contribute to **Retained Earnings**, which is "earned capital." Or, just memorize.
- **Link between Income Statement and Balance Sheet:** When the company earns **Revenue**, this is balanced by an increase in **Cash**, which is an **asset** reported on the **Balance Sheet**. So, this is the first time we've seen, in concrete accounting for a transaction, what we talked about on a conceptual level in Chapter 1.[4]
- **Core concept—'Paid-in Capital vs. Earned Capital'**: This transaction helps to illustrate one of the core concepts of accounting, and business generally. Firms need *capital*—sources of cash or other resources—to do things. If our firm wants to buy an asset in cash, we need to get that cash from somewhere. Where do we get that cash? One way to get the cash is to raise it from investors, by issuing *Common Stock*, and getting **"paid-in capital."** But, **once we're up and running**, we'll also get cash and resources simply from running our business and **selling to customers**. *Revenue* is thus a source of **"earned capital."**

You now have three reinforcing perspectives on why Revenue belongs in the Shareholders' Equity column: the 'what's left over' logic (A up, L unchanged, so E must be up); the chain of core identities (Revenue $\rightarrow$ Net Income $\rightarrow$ Retained Earnings $\rightarrow$ Shareholders' Equity); and the notion of earned capital. These are not three separate rules to memorize—they are one fact, seen from three angles.

2.8 Transaction 7: Providing Services on Account, $3.1k

This transaction is just like the previous one except, this time, the customers don't pay in cash, and, instead, promise to **pay later**. How do we account for this? Recall from Chapter 1 that a core principle of accrual accounting is that we **recognize revenues when they have been earned**—that is, when the good or service has been provided—rather than when the check arrives in the mail. Since the service has been fully provided in this example, we recognize **Revenue** ($E \uparrow$).

How do we balance it out? The customers have promised to pay later. That is *their obligation*. From our company's perspective, we are **entitled to receive that payment**

[4] This will hold in general: Every way in which a company can earn Revenue will be linked to a change in an Asset account or a liability account.

in the future, and can reasonably expect we will. So that is an **Asset** for our company. In the **jargon** of accounting, we call this asset **'Accounts Receivable.'** (A 'receivable' is just the flipside of a 'payable'—amounts we are entitled to receive from others. We will see various kinds of both receivables and payables throughout this book.) This pattern is so common in business that it has a standard abbreviation: you will often see Accounts Receivable written simply as **'A/R.'**

Notice, too, the pleasing symmetry with Transaction #5. There, our *supplier* extended credit to *us*—we got the goods, and recorded *Accounts Payable*. Here, *we* are extending credit to our *customer*: we provide the service now, and let them pay later. On their books, they now have a payable; on ours, a receivable. Every credit sale is somebody else's credit purchase.

Transaction	Assets	Liabilities	Sh.'s Equity
#7: Provide training on account	+$3.1k (Acc. Receivable)		+$3.1k (Revenue)

And our **new balance** (adding this transaction to the last balance) looks like this:

Transaction	Assets	Liabilities	Sh.'s Equity
#1-#7	$11.5k (Cash)	$5k (N.P.)	$15k (Common Stock)
	$10k (Equipment)	$1k (Accounts Payable)	$7k (Revenue)
	$2.4k (Prepaid Rent)		
	$1k (Supplies)		
	$3.1k (Accounts Receivable)		

2.8.1 Takeaways

The two key, **weird things** you need to wrap your mind around, are:

- **Revenue Recognition:** The company here **recognizes revenue** even though it has **not yet received cash**, because it has provided the service. This is one of the two core principles of accrual accounting, which attempts to measure the profitability of the company's **activities**.

- **Link between Income Statement and Balance Sheet:** Last time, when the company earned **Revenue**, it was balanced by an increase in **Cash**, which is an **asset** reported on the **Balance Sheet**. This time, the **Revenue** is balanced by an increase in **Accounts Receivable**, which is usually the next asset listed on the Balance Sheet. So, either way that the company earns revenue that we've seen so far—whether selling for cash, or selling on account—is **linked to** the Balance

Sheet via an asset account. There is something deeper lurking here, too: a reliable promise, from a creditworthy customer, to pay us cash is, economically, *very close* to money. Accounting draws a line between the two—*Cash* is one asset, *Accounts Receivable* another—but both are assets, and either can serve as the balancing partner of Revenue.

The Three Timings of Cash

Let's pause and get organized, because one simple observation will make the whole topic of revenue recognition much less mysterious. When we make a sale, the only thing that can vary is the **timing of the cash**, relative to the moment we *earn* the revenue by providing the good or service. And since time moves in only one direction, there are exactly three possibilities: the customer pays us (1.) on the *same day* we provide the service; (2.) *after* we provide the service; or (3.) *before* we provide the service. That's it. That is the whole space of cases. Transaction #6 was case (1.); the transaction we just did was case (2.). The next transaction is case (3.)—and it is the trickiest of the three.

2.9 Transaction 8: Customers Pay Cash in Advance ($800)

This transaction is the oddest, hardest one so far. But it will help illustrate some very core, interesting accounting principles. In this case, **customers pay $800 cash in advance** for services that they will receive at a later period. To make the timing vivid: suppose the customers pay us in November of this year, for a training camp that we will run next February—and, as usual, our company reports annually on a calendar-year basis. So the cash arrives in one reporting period, and the service will be provided in the next. In which fiscal year will we report the *Revenue* from this sale? By now you know the answer: next year, when we actually provide the service. But something real has happened *today*, and we must account for it. How? **Start with cash**. (Don't overthink it: we did not earn Revenue today, but we certainly did get *Cash*.) That asset account goes up by $800 ($A \uparrow$).

What balances it? Is this a revenue? According to accrual-accounting principles, the answer is **no—our company has not yet provided the service, and so has not earned revenue**. Instead, because the customers have paid—honored their side of the **contract**—our company now has an **obligation** to provide the service in the future.[5]

[5] As a salient example of what Deferred Revenue is all about, consider the **Fyre Festival** debacle of 2017. Would-be festival attendees paid up front. But should the company be able to recognize those payments as profits before it actually put on the festival and provided the agreed service?

And we will have to expend resources in the future to satisfy the obligation. So it is a **liability** ($A \uparrow$, $L \uparrow$). In accounting, we call this liability **'Deferred Revenue.'** So this transaction looks like this:

Transaction	Assets	Liabilities	S.E.
#8: Receive cash for training in advance	+$800 (Cash)	+$800 (Deferred Revenue)	

And our **new balance** (adding this transaction to the last balance) looks like this:

Transaction	Assets	Liabilities	Sh.'s Equity
#1-#8	$12.3k (Cash)	$5k (Notes payable)	$15k (Common Stock)
	$10k (Equipment)	$1k (Accounts Payable)	$7k (Revenue)
	$2.4k (Prepaid Rent)	$0.8k (Deferred Revenue)	
	$1k (Supplies)		
	$3.1k (Accounts Receivable)		

'Deferred Revenue' is also sometimes called **'Unearned Revenue.'** The two terms are interchangeable, and mean the same thing. Both terms help convey the idea. The first term, 'Deferred Revenue,' denotes that we will **'defer' recognizing** the revenue until we actually provide the service. The second term, 'Unearned Revenue' denotes that we have **not yet earned** the revenue. When we provide the service, we will earn it, and it will be *Revenue* then. (Outside of accounting, where might you have heard the word 'defer'? College admissions: you get admitted, and *defer* matriculation for a year—you postpone actually starting. Same idea here: we have the cash in hand, but we postpone calling it revenue until we have earned it.)

Deferred Revenue vs. Accounts Payable: Non-Cash Liabilities

Here's a question you should be asking at this point: What, exactly, is the difference between *Deferred Revenue* and *Accounts Payable*? At a deep level, they are the same kind of thing: both are liabilities—obligations taken on in past transactions, which must be satisfied before our owners can claim anything. But they arise from opposite patterns, and are settled in different 'currencies.' With *Accounts Payable*, we received goods, and promised to pay **cash** later: a **cash liability**. With *Deferred Revenue*, we received cash, and promised to **perform** later—to deliver goods or provide services: a **non-cash liability**.

This is why the definition of a liability refers to transferring an **economic benefit**, rather than only paying cash. It must cover both kinds of promises. In our training-camp example, we settle the obligation by providing the promised training service.

2.9.1 Takeaways

There are three **weird things** you need to wrap your mind around here:

- **Revenue Recognition principles:** The form of this transaction is the opposite of Transaction #7, but the principle is the same. In this case, the company **does not recognize revenue** even though it has **received cash from customers**, because it has not **provided the service**. We record revenues when they are **earned.**

- **Deferred/Unearned Revenues are not Revenues, they are Liabilities:** Understand it. Memorize it. Don't let it trip you up. The terms have 'revenue' in them, but they are not *Revenue* accounts.

- **Advanced concept—*Deferred Revenue* as "customer-provided financing"**: In this transaction, we recorded **customers'** pre-payment as a liability (L). Does this contradict our initial model, in which L represents **financing from creditors**? Not quite. You can think of this as if **customers provided a loan** to the company, which the company is **obligated to settle at a later day**, by providing the service. If the company does not deliver on its promises to provide services to customers (a la Fyre Festival), it should pay them back—and if it can't do that, it will surely have trouble finding new customers and staying in business. So, this obligation comes **prior to shareholders**, and still fits with the **"what's left over"** intuition for the accounting equation ($E = A - L$).

  Business transactions and models like this are not uncommon. You have likely **bought tickets for concerts in advance.** Or, consider **Kickstarter.** On that platform, prospective **future customers** help **finance** the development of new products, in return for the right to get the product once it is produced. So, ***Deferred Revenues*** are, conceptually, a form of **credit financing from customers.** Innovative companies sometimes use this as one of their major sources of financing in their early stages.

  Customers may also understand a proposed product better than a bank does. Their willingness to pay in advance can help a young company fund production. For our accounting purposes, the key point is unchanged: receiving that cash creates an obligation to deliver, not revenue already earned.

Looking Ahead: What Happens When We Provide the Service?

We have recorded the November transaction. But let's peek ahead: what happens *next February*, when we actually run the camp, and finally earn the revenue that is currently deferred? Try to work it out with the tools you already have.

We are providing the service—and by now, that is always our cue to **record Revenue**, as a positive entry in the Shareholders' Equity column ($E \uparrow$). Now, what balances it? Here's where this case earns its title as the trickiest of the three timings. Do we get new cash? No—we were already paid, back in November. A new receivable? No—the customer owes us nothing further. What has changed is that we have **satisfied our obligation**: we no longer owe anyone a training camp. So the *Deferred Revenue* liability goes away ($L \downarrow$):

Transaction	Assets	Liabilities	Sh.'s Equity
Next Feb.: Provide the training		-$800 (Deferred Revenue)	+$800 (Revenue)

Assets do not change at all; a liability falls, equity rises, and the equation stays in balance. Notice what just happened: we **earned revenue by decreasing a liability**.

This completes a pattern worth savoring. Every way a company can earn revenue is linked to an increase in its **net assets**—that is, Assets minus Liabilities, the "what's left over" quantity. Sell for cash: an asset (*Cash*) rises. Sell on account: an asset (*Accounts Receivable*) rises. Provide a service that was prepaid: a liability (*Deferred Revenue*) falls. In every case, $A - L$ goes up. Indeed, this suggests a rather elegant definition: **revenue is an increase in net assets arising from the company's operations**—from doing things for customers. We flagged this link between the Income Statement and the Balance Sheet conceptually in Chapter 1; you have now seen every version of it in action.

Return to Dane, Alison, and Avery from Chapter 1. We already found $300 of revenue: $200 for tutoring Dane and $100 for tutoring Avery. What can we now add? Dane's payment increases **Cash**; Avery's promise increases **Accounts Receivable**; and Alison's $300 advance increases both **Cash** and **Deferred Revenue**, because we still owe her the tutoring. The answer about revenue has not changed. We can now explain where every other part of those transactions belongs.

2.10 Transaction 9: Paying Salaries to Employees, $1.8k

On payday, firms pay their employees in cash for the work they've done since the last paycheck, typically every two weeks. How do we account for this? **Start with cash.** That Asset account goes down by $1.8k ($A \downarrow$). What balances this? In previous transactions, when the company **spent money** on something, that *thing* was always an asset. But is there an asset here?

The payment is for work the employees have already performed. In this example, the company has consumed the benefit of that work; it has not bought an unused resource to carry forward. That is why we record an expense.

You will sometimes hear that "our employees are our greatest asset." In ordinary speech, that means they are valuable. Accounting uses *asset* more specifically: the company does not own the people it employs. Nor does calling an employee costly make the person a liability. An amount already owed for that employee's work, however, is a liability. We will distinguish Salaries Expense from Salaries Payable in Chapter 4.

So, in this case, the **money spent on salaries represents an *Expense*** rather than an asset. How do we account for this? **Expenses effectively just function as anti-Revenues**. So, we put expenses **in the Shareholders' Equity column, with a minus sign**, to denote how expenses **decrement** Shareholders' Equity. (And also recall the **core identities** of accounting, and how expenses decrement Net Income, which flows into Retained Earnings, which is a part of Shareholders' Equity.)

Transaction	Assets	Liabilities	Sh.'s Equity
#9: Pay salaries, $1.8k	-$1.8k (Cash)		-$1.8k (Expense)

And our **new balance** (adding this transaction to the last balance) looks like this:

Transaction	Assets	Liabilities	Shareholder's Equity
#1-#9	$10.5k (Cash)	$5k (Notes payable)	$15k (Common Stock)
	$10k (Equipment)	$1k (Accounts Payable)	$7k (Revenue)
	$2.4k (Prepaid Rent)	$0.8k (Deferred Revenue)	-$1.8k (Expense)
	$1k (Supplies)		
	$3.1k (Acc. Receivable)		

2.10.1 Takeaways

There are two **weird things** you need to wrap your mind around here:

- **"Immediately expensing" vs. "capitalizing as an Asset":** As we've seen, sometimes when a company spends money on something, that *thing* is an asset (and, as we've hinted, those assets will *become* expenses later). But sometimes, as in this case, that expenditure is an **expense right away**. What determines the difference? That question is what a lot of accounting is all about. Consistent with the FASB definition, if the company has obtained a right to **future benefits** that meets the applicable recognition requirements, it may record an asset. A resource need not last for multiple periods: supplies still on hand can be an asset even if they will be used tomorrow. If the benefits have already been consumed, the cost is generally expensed.

- **Expenses just function as 'anti-Revenues' and decrement S.E.:** We previously laid out the full logic behind why we put *Revenues* in the Shareholders' Equity column, using the core identities. Expenses are just the opposite of Revenues. So, symmetrically, we track them as negative entries in that column.

A Preview: Salaries *Payable*

What if the employees have done the work but have not yet been paid? Then the expense is accompanied by an obligation called **Salaries Payable**. That is an adjusting-entry problem, which we will work through in Chapter 4. Today's transaction assumes the work and its payment are accounted for together.

2.11 Transaction 10: Pay a $.1k Dividend

As we discussed in Chapter 1, companies *can* elect to **pay dividends** to their **investors (shareholders/stockholders)**. When they do so, it is as if they are just moving cash from the company's bank account, over a line, and into the owners' bank accounts. This is essentially how investors get paid back on their **Common Stock** investments.

How do we account for this? **Start with Cash**. The company disburses $100 in cash to its investors, so that Asset account goes down ($A \downarrow$). What balances this? Recall that dividends are **not expenses**, not a cost of doing business, or something that decrements the value *generated* for shareholders. But they do decrement the amount of that value that has been *retained* in the corporation—that is, the *Retained Earnings*. (**Core identity**: $\Delta Retained\ Earnings = Net\ Income - Dividends$.) So, we account for these as ***negative* entries in the Shareholders' Equity column**:

Transaction	Assets	Liabilities	Shareholders' Equity
#10: Pay a cash dividend $100	-$100 (Cash)		-$100 (Dividend)

And our **new balance** (adding this transaction to the last balance) looks like this:

Transaction	Assets	Liabilities	Shareholders' Equity
#1-#10	$10.4k (Cash)	$5k (N.P.)	$15k (Common Stock)
	$10k (Equipment)	$1k (Accounts Payable)	$7k (Revenue)
	$2.4k (Prepaid Rent)	$0.8k (Deferred Revenue)	-$1.8k (Expense)
	$1k (Supplies)		-$0.1k (Dividend)
	$3.1k (A/R)		

Something about this bothered me when I first learned it, and it may be bothering you. The whole reason owners invest is to *get paid*—dividends are the payoff, the good thing, the very point of the enterprise. Yet when we pay a dividend, we *reduce* Shareholders' Equity. And reducing equity *sounds* bad. How can the thing owners want most show up in the accounts as a decrease in their equity?

Here's an analogy that made peace between those two thoughts, for me. Suppose I own two properties, A and B, and at Property A I have installed a very nice $20k sauna. One day, I move the sauna over to Property B. The value of Property A—and thus my 'home equity' in it—goes down by $20k. Is that bad for me? Of course not: I own both properties; the sauna went from one of my pockets into another; and *I* chose to move it. A dividend is the same maneuver. Value moves from the corporation's bank account (where the accounting system measures it, as Shareholders' Equity) into the shareholders' own bank accounts (where the corporation's books no longer see it). The company's equity falls; the owners—who are the ones pulling the strings—are, if anything, pleased. The Balance Sheet is the *company's* map, and the sauna has simply been carried off the edge of that particular map.

2.11.1 Takeaways

There are two **weird things** you need to wrap your mind around here:

- **Dividends are not expenses:** Dividends are discretionary cash disbursements to shareholders. They are not a cost of doing business, or a cost incurred by shareholders. Think of them as **cash moving over a dividing line**, from the company's bank account into its owners' bank account.

- **But dividends do decrement S.E.:** You can think through the **core identities** again to fully understand the logic, of how dividends reduce Shareholders' Equity by reducing Retained Earnings ($\Delta Retained\ Earnings = Net\ Income - Dividends$). You can also think about it this way: When investors (shareholders/stockholders) **pay in cash**, we called that "Common Stock," which was a part of Shareholders' Equity. So, when the company turns around and **pays cash out** to them, that's just, in a sense, **reversing** that cash flow.

Anticipating a Question: Are Retained Earnings *Owed* to the Owners?

A question that often comes up at about this point: "If the company *retains* earnings—keeps the profits inside, and reinvests them in, say, a new factory—does it then *owe* that amount to the owners at some later date? Is Retained Earnings a sort of liability to the shareholders?" It is an excellent question, and the answer is **no**, for reasons that

run back to Chapter 1's foundations. A liability is a *present obligation* to transfer assets or provide services to someone—a contractual claim, with amounts and timing. Owners hold no such claim. If they were contractually owed fixed amounts on a fixed schedule, they would *be* creditors. Owners are **residual claimants**: they are entitled to whatever is left over after the actual liabilities are paid—which may turn out to be a lot, or nothing—and they get cash out when the company (which, recall, they control) chooses to pay dividends. Retained Earnings is not an IOU; it is simply the running measure of profits earned and not yet distributed.[6]

2.11.2 Going deeper: The owner who also does the work

To sharpen the line between salaries (an expense) and dividends (a distribution), consider a puzzle from the world of small-business valuation. Imagine someone who owns a hair salon—and also works in it, cutting the hair herself. Suppose you are thinking of buying her business, and she shows you her books: revenues, minus rent, utilities, supplies, products... arriving at a healthy profit. And then, being the sole owner, she pays that profit out to herself as a **dividend**. The dividend, as we have just learned, does not reduce her reported profit—it is a distribution of profit, not an expense.

Do you see the trap awaiting the would-be buyer? She is wearing *two hats*: owner, and stylist. Part of what she pays herself as 'owner' is really compensation for her *labor*. If you bought the salon, you could not step in and cut hair yourself (I certainly couldn't); you would have to *hire* a stylist—and pay a salary, which *is* an expense. So the salon's true profitability, to you, is its reported profit *minus* a fair market salary for the work she currently does for 'free.' In a solely-owned business, the return *to labor* and the return *to capital* arrive mingled in one bank account, and the analyst must disentangle them.[a] Same cash leaving the firm; very different accounting categories; and the difference matters enormously to anyone using the statements to value a business.

[a] Could she instead put herself on the payroll—paying herself a salary as an employee, *and* dividends as an owner? Absolutely, and many owner-operators do; how they split it is often driven by tax considerations, which I will spare you. The conceptual point stands: the buyer cares about what is left over *after* paying a market rate for all the labor the business requires.

[6] A related question: Can an individual shareholder of, say, Microsoft *decline* a declared dividend, and have 'their' portion put back into Retained Earnings? Essentially, no. When a dividend is declared, it is paid to all shareholders of record. What you *can* do is reinvest it—use the cash to buy more shares; many brokerages will even automate this. But note what that really means: you are typically buying shares from *other investors* in the secondary market, not from the company. Your 'reinvestment' does not touch the company's books at all.

2.12 Going Beyond: Transaction 11, Collecting on an Account

The ten transactions above are this chapter's core program. But let's go a bit beyond, with two more transactions that follow naturally—and that will test whether the principles have really sunk in.

Recall Transaction #7: we provided training to customers 'on account,' recording \$3.1k of *Accounts Receivable* and \$3.1k of *Revenue*. Now suppose those customers make good on their promise: they deliver the \$3.1k in cash that they owe us. (**Jargon**: they "pay off their account.") How do we account for that?

Start with cash—it's back! Cash comes in, so that asset account rises by \$3.1k ($A\uparrow$). Now, what balances it? Be careful here, because there is a tempting error lying in wait: is this new **Revenue**? *No.* Remember the mantra: we record revenue when we *provide the service*—and we provided this service, and recorded this revenue, back in Transaction #7. Recording it again now would be double-counting. Instead, think about the 'flip side': the customers have now satisfied their obligation to us, so our claim on them is extinguished. The *Accounts Receivable* must come back down—all the way to zero:

Transaction	Assets	Liabilities	Sh.'s Equity
#7: Provide training on account	+\$3.1k (Acc. Receivable)		+\$3.1k (Revenue)
#11: Customers pay off account	+\$3.1k (Cash)		
	-\$3.1k (Acc. Receivable)		

So the whole transaction happens inside the asset column ($A\uparrow$, $A\downarrow$), just like the equipment purchase in Transaction #3: we have simply *converted* one asset (a promise) into another (cash). Total assets are unchanged; there is no new liability; and there is no new revenue. The income was earned in Transaction #7. Transaction #11 is just the check arriving in the mail.

2.13 Going Beyond: Transaction 12, Paying the Electricity Bill

One more—and it is a subtle one. Our company pays its **monthly electricity** bill for its **headquarters**: \$300. (I have chosen each of those three bolded words carefully; each one is a clue.) How do we account for this?

Start with cash: it goes down by \$300 ($A\downarrow$). Now, the by-now-familiar question: is the thing we spent this money on an *asset*, or an *expense*? Run through the clues. *First*, it is a **monthly, recurring** bill—not something we pay once and then benefit from for

years, but something we will pay again next month, and the month after. *Second*, electric current is, for practical purposes, not storable: the current we paid for has already been consumed, keeping the lights on and the computers humming. There is no stock of it that we "control" going forward. So the expenditure fails the FASB asset definition, and is **immediately expensed**: we record **Utilities Expense**, a negative entry in the Shareholders' Equity column, just like *Salaries Expense*.

Transaction	Assets	Liabilities	Sh.'s Equity
#12: Pay HQ electricity bill	-$300 (Cash)		-$300 (Utilities Expense)

Why did I specify *headquarters*? Because a cost can be used up in producing something that we still hold as an asset. Electricity used to manufacture unsold goods may enter Inventory and become an expense when those goods are sold. We will develop that distinction in Chapter 10. For this transaction, the electricity was used in running the headquarters, so we record Utilities Expense now.

2.14 Putting It Together: A Worked Example

Before we close, try one integrative problem—the kind that tests whether this chapter's distinctions have really taken hold. Suppose a company has *only* the following transactions this year: it (1.) sells $50k of common stock; (2.) buys $30k of equipment, to be used for many years, in cash; (3.) provides $20k of services to customers for cash; (4.) pays $10k of salaries to employees; and (5.) pays $6k of dividends. **Question: What is the change in Retained Earnings for the year?**

Work from the core identity: $\Delta Retained\ Earnings = Net\ Income - Dividends$. So, first: what is Net Income? Revenues are $20k. Now, the expenses—and here are the traps. The common stock issuance is financing, not revenue (nothing was *earned*). The equipment purchase is **not an expense**: it is the acquisition of an asset, as in Transaction #3. The dividends are **not an expense** either, as we have just learned—they do not enter Net Income at all. The only expense on the list is the $10k of salaries. So Net Income $= \$20k - \$10k = \$10k$; and $\Delta Retained\ Earnings = \$10k - \$6k = \$4k$.

A nagging thought may have occurred to you: 'Shouldn't *some* of that equipment's cost count against this year's profit? The company did *use* the equipment this year...' Hold that thought. It is exactly right; the mechanism is called *depreciation*; and it is an 'adjusting entry'—the subject of Chapter 4. The problem above dodges it only by fiat, by stipulating that these are the *only* transactions.

2.15 What's next?

So, even at this early stage of this book, we've actually done a pretty good job **accounting for** a breadth of "real"/external transactions—that is, *exchanges* with outside parties. So, what are we missing? What's wrong with the picture so far?

- First, if you look at our aggregated table for Transactions #1–#10 above, **it's getting ugly and unwieldy**. This is despite the fact that we only had five asset accounts and three liability accounts! Most real corporations will have several dozen accounts—and thousands to millions of transactions. So, in **Chapter 3**, we won't overturn any of these accounting *concepts*, but we will learn some **new tools for organizing** all of this information better. Concretely, there are three things we will naturally find ourselves wanting: (a.) a compact record of each transaction's incremental effects, without redrawing the whole Accounting Equation every single time (these will be 'journal entries'); (b.) a running history, and current balance, of each individual account, like *Cash* (these will be 'T-accounts'); and (c.) one tidy list of all the accounts' balances at a moment in time (this will be a 'trial balance'). The problem will be how to keep the Accounting Equation in balance while we do all that. The solution: we'll learn to keep it in balance using **"debits"** and **"credits."**

- Second, as you've intuited, there are many *expenses* that we haven't accounted for yet. When the company first purchased Equipment, Supplies, and Prepaid Rent, those were all recorded as assets, initially. But, based on the accounting principle of **matching**, as the company **"uses those assets up"** we will recognize expenses. But there will be no particular moment of "real" exchange to trigger that—**no external transaction** during the reporting period. Instead, we recognize these expenses **at the end of the reporting period**, as **"adjusting entries"** (also called "**internal transactions**"). This is what we'll cover in **Chapter 4**.

- Third, we said at the beginning that accounting is ultimately about going *from* transactions *to* financial statements. We also have a couple of vague loose ends here. For example, in the above, I said that "Revenues and Expenses contribute to Shareholders' Equity, via Retained Earnings, but aren't reported as S.E. accounts." And we haven't yet aggregated the company's Revenues, Expenses, and Dividends into a **Retained Earnings** account. To get things just right—to avoid careless errors—we'll learn some fairly **rote** "**mechanics of the financial reporting process**." This will give us a reliable process for going *from* the transactions to the financial statements. This will be **Chapter 5**.

3

Debits and Credits and their uses: Journal Entries, T-accounts, and Trial Balances

In Chapter 2, we accounted for **10 external transactions** (plus two bonus ones at the end). We discussed the **logic** behind the accounting for each transaction and, each and every time, we verified that the core identity, **The Accounting Equation**, held. And at the end of this exercise, we ended up with a table that looked like this:

Transaction	Assets	Liabilities	Shareholders' Equity
#1: Sell Common Stock	+$15k (Cash)		+$15k (Common Stock)
#2: Borrow from the bank	+$5k (Cash)	+$5k (Note payable)	
#3: Purchase equipment for $10k	-$10k (Cash), +$10k (Equipment)		
#4: Pay rent up front, $2.4k	-$2.4k (Cash), +$2.4k (Prepaid Rent)		
#5: Purchase supplies on account, $1k	+$1k (Supplies)	+$1k (Accounts Payable)	
#6: Provide training for cash, $3.9k	+$3.9k (Cash)		+$3.9k (Revenue)
#7: Provide training on account, $3.1k	+$3.1k (Accounts Receivable)		+$3.1k (Revenue)
#8: Receive cash for training in advance $0.8k	+$0.8k (Cash)	+$0.8k (Deferred Revenue)	
#9: Pay salaries, $1.8k	-$1.8k (Cash)		-$1.8k (Expense)
#10: Pay dividend $0.1k	-$0.1k (Cash)		-$0.1k (Dividend)

Table 3.1: The 10 Transactions with the Accounting Equation, from Chapter 2

There's **nothing wrong** with this table, or missing from it. Every transaction the firm made was correctly and **fully accounted** for. The problem is just that it's getting

messy and busy, even though we have only tracked 10 transactions, affecting only 12 accounts. And real corporations would have **dozens of accounts**, and **many thousands of transactions** every reporting period. The remainder of this chapter is just about how to **organize and represent** this same accounting in more economical and useful ways, to be able to accommodate real-world accounting.

Remember, from Chapter 1, that the core problem of accounting is one of **"Too Much Information."** Accounting is ultimately about getting useful information into the hands of the people who need it—managers, investors, creditors, etc.—so they can make informed decisions. Now think about a company like Apple. At any given moment, Apple absolutely wants to know: "What is our total cash balance right now?" "How much do our customers owe us?" "How much do we owe our suppliers?" Its accountants cannot be working from one giant, raw $A = L + E$ table, scanning down thousands of rows and manually summing the Cash entries every time someone asks. They need a system that delivers clean, up-to-date account balances on demand. You can feel the beginnings of this problem even in our toy example: to answer "how much cash do we have?" from the table above, you have to pick out every Cash entry and add and subtract them all. That's annoying for 10 transactions. For Apple, it would be impossible.

Our examples have also been unusually simple. A sale of goods, for instance, involves both earning revenue and accounting for the goods that leave the company. Later transactions will affect several accounts at once. We need a way to keep those changes organized without redrawing the entire Accounting Equation each time.

These tools are the grammar we will use to express the accounting ideas in the rest of the book. Their purpose is practical: to help us keep a complicated transaction straight, preserve the Accounting Equation, and find the information we need. We will learn them by using the transactions whose economic meaning we already understand.

3.1 Organizing our accounting

Look at that big table again: In each **transaction**, we wrote out the **entire accounting equation** each and every time (with separate columns for **A, L, and E** throughout), even though we **only use two columns**. It seems a bit excessive. It seems more natural to just write out something like this:

"Transaction #1: ↑ $15k Cash (+A), ↑ $15k Common Stock (+S.E.)."

This is more straightforward and economical than what we've been doing (no extra empty column). And this is **pretty much** (with one change) what we're going to call the **Journal Entry** approach. We will use journal entries to track the **incremental effect** of

each transaction on the **accounts involved**, in a simple way.

Firms might additionally want to know **what has happened to one particular account**—that is, **all of the transactions** involving, e.g., Cash, or Supplies as well as the **final balance** of those accounts. While we do *have* all of that information preserved in Table 3.1 above, it would be a pain to extract the information for each account each time.

In short, it's natural for companies to want to track **all transactions affecting**, and the **final balances** of, **each account.** This is **pretty much** (with one change), what a **'T-account'** will be. Each individual account (like Cash, Supplies, Accounts Payable) will have its own tracking, with increases on one side of a **T**, and decreases on the other, and the *balance* of the account reported, like so:

Cash	
+$15k	-$10k
+$5k	-$2.4k
+$3.9k	-$1.8k
+$0.8k	-$0.1k
$10.4k	

If this format seems alien, notice that you already keep something like a T-account for yourself: your personal checking account. Money comes in—a paycheck, a transfer from your parents, sources better left unexamined—and money goes out, on rent, food, rideshares. What your banking app shows you, at any moment in time, is a running series of increases and decreases, and a current balance. A T-account for Cash is just a structured way of laying out that same information: increases on one side, decreases on the other, netted to a final balance.

Finally, a third thing that companies might naturally want is to know **where *all* their accounts stand** at a particular point in time—that is, the final **balance** of **all** of its accounts: Cash, Equipment, Revenue, etc. That is, it might like to have a **list** of balances, like this:

Account:	**Balance:**
Cash:	$10.4k
Equipment:	$10k
Revenue:	$7k
...Etc.	...Etc.

This is **pretty much** (again, with one change) what we're going to call the **Trial Balance**: a list of the final balances of each account. The Trial Balance will be especially

useful for the final step in the process of accounting—going **from transactions to the Financial Statements**. Remember that the Financial Statements are reported *as of* the end of the reporting period, and, reductively, tell you the **final balances** of the company's accounts as of that date. Those final balances are all lined up in a Trial Balance, by definition—so that's what we'll use to fill out the financial statements.

This might seem like a lot, right now, but bear in mind that there are **not any new accounting ideas** here. These are just three ways of organizing the information and accounting we already did. Summarizing:

- **Journal Entries** will track the **incremental effect** of each transaction on the accounts involved in the transaction—that is, the *change* in each account from it.
- **T-accounts** will track the **full sequence** of changes in **each** particular account, as well as the **balance**.
- **Trial Balances** consolidate and list the final balances of **all accounts**, in one list.

But, this raises a potential problem: The Accounting Equation is *the* core axiom of accounting. If it doesn't hold, we're not doing accounting correctly or coherently. So, if we stop using the format we used in Chapter 2 (with separate columns for A, L, and E for each transaction), how can we be sure to keep things in balance?

3.2 Debits and Credits: *The* Tools of Accounting

This is where **debits and credits** come in. These are simply the terms we use, for historical reasons, for a mnemonic tool for accounting. Keep the division of labor clear in your mind: **The Accounting Equation** is the *axiom*; debits and credits are the *procedure*—a little algorithm for recording transactions such that, if you follow the rules, the axiom automatically stays satisfied. They are not a separate theory of accounting; they are a technique for implementing the theory.

I still use these tools myself. When a transaction has several steps, a journal entry or a T-account can make the reasoning easier to hold in mind. Learn the procedure with its meaning attached: which resource or claim changed, and why?

Here's how it works:

3.2.1 The rules

- For practical purposes, all **debit** means is **'on the left'** and all credit means is **'on the right.'** That's it. That's really it.
  - **Debit: Left**

  - **Credit: Right**

- For **asset** accounts, when the account **increases**, that's a **debit** to the account (tracked on the left side of the account); meanwhile, for **liability and shareholders' equity** accounts, when the account **increases** that's a **credit**.
  - And, symmetrically, for asset accounts, decreases are credits, while for liability and shareholders' equity accounts, decreases are debits.
  - Summarizing visually:
    * **A: ↑ Debit, ↓ Credit**
    * **L, E: ↑ Credit, ↓ Debit**

- Finally, for any given transaction, and thus, across all transactions, the **total debit amount ($) must equal the total credit amount ($)**.
  - **Debits ($) = Credits ($)**

A caution before we go further: debits and credits carry **no moral content**. 'Debit' does not mean good, or increase, or money coming in; 'credit' does not mean bad. When I first learned this material, I tripped over exactly this. I had a little folk theory in my head: assets are good, liabilities are bad. So when an asset went up and that was called a 'debit,' I concluded that debits were the good ones; and when a liability went up and that was called a 'credit,' I concluded that credits were the bad ones. But now look at Shareholders' Equity. More equity in the corporation sounds like a good thing, right? Yet equity increases with a *credit*. So the moralized picture collapses. The correct picture is purely geographic: debit means the left side of the account, credit means the right side, and that is all.

The third rule deserves special emphasis, because it is about to become our daily mantra. Up until now, the thing we kept repeating was "Assets = Liabilities + Shareholders' Equity." Going forward, at the level of recording transactions, the mantra becomes: **Debits = Credits**. For every transaction, every time, with no exceptions. I am open-minded, tolerant, and forgiving about many things—questions, confusion, honest mistakes. But I draw the line here. If you ever hand me the accounting for a transaction in which the debits do not equal the credits, I call the dean. I call.[1]

What about **Revenues, Expenses, and Dividends**? Just like in Chapter 2, we will think of these as **contributors to** (or components of) **Shareholders' Equity**. Revenue increases Shareholders' Equity: Therefore, **revenues are credits**. **Expenses just function as anti-revenues, so they are debits**. **Dividends** also decrease Shareholders' Equity (by decreasing Retained Earnings), so they are also **debits**.

[1] I am, of course, joking about the dean. I am not joking about the importance. If we know nothing else, we must get this right: for every transaction, total debits equal total credits.

3.2.2 How to remember all this: memorize one thing

Presented as a list, these rules can look like an arbitrary table to be memorized, and you may be dreading that. Let me reassure you: I don't have a great memory, and I don't like memorizing lists of rules.[2] So I don't rely on memorizing the table. There is exactly *one* thing I consciously memorize: **when an asset goes up, that's a debit—and debit means left**. That's my one anchor. Everything else I reason out from first principles, on the spot, like so: Assets sit on the *left* side of the Accounting Equation, and increases in assets go on the *left* side of their accounts. Liabilities and Shareholders' Equity sit on the *opposite* side of the equation—so their increases must be recorded on the opposite side of their accounts: credits. And decreases must be the opposite of increases: an asset going down is a credit; a liability or equity account going down is a debit. One anchor, plus thirty seconds of reasoning, regenerates the entire rule set. I recommend this strategy to you.

3.2.3 The why

If you think about this carefully, you can see that this list of rules is just an **algorithm for keeping the Accounting Equation in balance** in any transaction.

It's easiest to see this by starting with simple concrete examples. For example, from Chapter 2, we know that if one asset account increases ($A\uparrow$), one of three things must happen to balance it: (i.) another asset must decrease ($A\downarrow$), (ii.) a liability must increase ($L\uparrow$), or (iii.) shareholders' equity must increase ($E\uparrow$).

In this **new world of debits and credits** we just entered, an increase in an asset account is a **debit**. And all **three** of the alternatives that could balance it out are **all credits**. If you keep thinking hard, you can see how this applies more generally. E.g., a decrease in an asset (such as paying out cash) is a credit and so it must be balanced out by a debit—an increase in another asset, or a decrease in a liability or shareholders' equity account. And so on.[3]

To make this fully concrete, replay the first three transactions in the life of our little company, all of which involved Cash. In Transaction #1, we got cash from investors: Cash up (a debit), balanced by Common Stock up ($E\uparrow$, a credit). In Transaction #2, we got cash from the bank: Cash up (a debit), balanced by Notes Payable up ($L\uparrow$, a credit). And in Transaction #3, we spent cash on equipment: Equipment up (a debit), balanced by Cash down ($A\downarrow$, a credit). Three different transactions, three different balancing moves—and every one of the balancing moves was a credit. Summarizing the equivalence we've been articulating, in compact form:

[2] I've got too much else going on in my life.

[3] Equivalently, write the equation as $A - L - E = 0$. A debit increases this expression and an equal credit decreases it, so equal total debits and credits preserve the equality. The worked transactions in the main text show what that means in particular accounts.

$$Debits\ (\$) = Credits\ (\$) \iff A = L + E$$

3.2.4 Why revenues are credits, and expenses are debits

Above, I asserted in a single line that revenues are credits. This is worth slowing down for, because it's the piece of the scheme that most often trips people up: the rules were stated in terms of **A, L, and E**—the permanent 'stock' accounts—and Revenues, Expenses, and Dividends are none of those. There are two routes to deriving their treatment, and you should have both.

The first route is deductive, through the **core identities** you learned in Chapter 1:

- $Revenues - Expenses = Net\ Income$
- $Net\ Income - Dividends = \Delta Retained\ Earnings$
- $Shareholders'\ Equity = Common\ Stock + Retained\ Earnings$

Follow the chain: higher revenues mean higher Net Income (holding expenses fixed); higher Net Income flows into higher Retained Earnings (holding dividends fixed); and Retained Earnings is a component of Shareholders' Equity. So, more revenue means more equity. Equity goes up with credits. Therefore, revenues are credits.

The second route is more inductive and intuitive. Suppose I'm a consultant, and I provide a service to a client, who pays me in cash. After that transaction, I clearly have more cash—my assets have gone up. Did I incur any new *liability* in the process? No: I don't owe anyone anything new merely because I earned revenue. So, if Assets went up and Liabilities did not, then, as a matter of arithmetic, Equity must have gone up. Earning revenue just *is* an increase in the owners' claim on the business. And increases in equity are credits.

Once revenues are anchored, expenses and dividends follow by symmetry. Expenses function as **anti-revenues**: whatever is true of revenues is true of expenses, in reverse. Higher expenses mean lower Net Income, which means lower Retained Earnings, which means lower equity—and a decrease in equity is a debit. Dividends work similarly through the second identity: more dividends, holding Net Income fixed, means lower Retained Earnings and thus lower equity. Debits again. Some textbooks formalize this by calling expenses and dividends **"contra-equity" accounts**—accounts that live on the equity side of the Accounting Equation but move equity in the opposite direction ('contra' just means 'against'). If the label helps you, use it; if not, you can safely ignore the jargon. What matters is the behavior: revenues increase equity and are credits; expenses and dividends decrease equity and are debits.

And here is a small consolation prize: in one respect, these accounts actually simplify your life. An account like Cash is sometimes debited (cash comes in) and sometimes credited (cash goes out), so you always have to think about the direction. For the ordinary recognition entries we are learning here, revenue increases are credits, while expense and dividend increases are debits. These accounts can also move in the opposite direction: corrections and reversals are possible, and in Chapter 5 we will debit Revenue to close its balance. So remember the direction of an **increase**, rather than a claim that an account can only ever move one way.

3.2.5 Jargon: 'debit accounts' and 'credit accounts'

It's very common in accounting to talk about the **"natural balance"** of particular accounts, or call accounts **"debit (vs. credit) accounts."** For example, **Cash** is an asset, so increases in Cash are **debits**. Obviously, corporations usually have *positive* (rather than negative) **balances** of cash. This means that the '**natural balance'** ' of the **Cash** account will be a **debit**. So, for shorthand, we sometimes call Cash (and other asset accounts) a **"debit account."** Similarly, **Revenue increases** are **credits**, and the account ordinarily has a credit balance before closing. So the "natural balance" of the revenue account is a credit, and we call it a **"credit account."** Put simply, if increases in an account are registered as a debit (vs. credit), we call that account a "debit account" (vs. "credit account").

While we're on jargon, note the grammar of these words: 'debit' and 'credit' function as both verbs and nouns. As verbs, we *debit* the Cash account, or *credit* Revenue—that is, we make entries on the left or right sides of those accounts. As nouns (or adjectives), Cash *is* a debit account, and Notes Payable a credit account, in the natural-balance sense above. Both usages are standard, and soon enough you'll toggle between them without noticing.

One more term you will encounter in the wild: the whole collection of a company's T-accounts—Cash, Equipment, Accounts Payable, Common Stock, and all the rest, with every debit and credit that has ever hit them—is called the **general ledger**. So when someone refers to 'the ledger,' they mean the master record of all of the company's accounts.[4]

At this point, you have all you need to understand the rest of this chapter: Everything else is just a matter of applying this new algorithm to track the transactions that we already solved previously, but using our new tools and formats. We'll use debits and credits—and the rule that total debits must always equal total credits—to track transactions across the three formats we discussed above: **Journal entries**, which track the

[4] In recent years, the word has been borrowed by cryptocurrencies—'distributed ledgers,' blockchains, and so on. Historically, though, 'ledger' is an accounting term. The crypto world borrowed it from us.

incremental effect (changes in accounts) from each transaction; **T-accounts**, which track the full sequence of transactions across **each separate account**; and **Trial Balances**, which list the balances of **all** of the company's accounts.

3.3 Journaling the 10 Transactions

Now, it's time to go back and do those same 10 transactions again, but this time using the algorithm we've learned for tracking transactions using debits and credits.

Before we begin, note the two **conventions** that govern how a journal entry is written; you will see them in every entry below. First, the account being *debited* is listed first, flush left. Second, the account being *credited* is listed second, and indented to the right.[5] I'll confess I don't care deeply about these stylistic details, but you might as well learn to do it the standard way, since that is how everyone else does it. I will also annotate each account with its economic role—(+A) for an asset increasing, (-A) for an asset decreasing, (+L), (+S.E.), and so forth. This is my own habit rather than a formal requirement of journal entries, but I recommend it: it keeps the economic meaning of the entry in view, and keeps you honest about the mapping back to the Accounting Equation.

Transaction 1: Sell Shares of Stock for $15,000

We learned last chapter that this increases Cash ($\uparrow A$) and Common Stock ($\uparrow S.E.$) by $15k. Now, we're just doing it with **debits and credits**. Since Cash is an asset account, that's a **debit** to cash. And the increase in Common Stock is the **balancing credit**.

Representing this in the journal entry format, we have…:

	Debit	**Credit**
Cash (+A)	$15k	
Common Stock (+S.E.)		$15k

And representing this in T-account format, we have:

[5] You may also encounter a notation in which each line is explicitly tagged 'Dr.' (debit) or 'Cr.' (credit). It means the same thing. In this book, the position—first and flush left, versus second and indented—carries that information.

Cash	
Debits	Credits
+$15k	

Common Stock	
Debits	Credits
	+$15k

And, at this very early stage, if we were to make a **Trial Balance** (a list of the final balances of accounts), it would look like this:

Account:	**Debit**	**Credit**
Cash (+A)	$15k	
Common Stock (+S.E.)		$15k

Transaction 2: Borrow cash of $5,000 from the bank

We know that Cash ($\uparrow A$) and Notes Payable ($\uparrow L$) increase by $5,000 as a result of this transaction. Since Cash is an **asset** account, that **increase is a debit**; since Notes Payable is a **liability**, that **increase is a credit**. So the journal entry for this transaction would be:

	Debit	**Credit**
Cash (+A)	$5k	
Notes payable (+L)		$5k

Remember again that the **journal entry** just measures the incremental effect of this transaction—the *changes* in the two accounts being affected. So, the fact that Transaction #1 had also incremented the company's cash account has no bearing on this journal entry. The journal entry isn't affected by what happened to the accounts previously. Instead, the effect of the whole previous **sequence of transactions 'lives on'** in the **T-account**.

And the updated T-accounts would look like this:

Cash	
Debits	Credits
+$15k	
+$5k	
$20k	

Common Stock

Debits	Credits
	+$15k

Notes Payable

Debits	Credits
	+$5k

And, the Trial Balance, at this very early stage, would look like this:

Account:	**Debit**	**Credit**
Cash:	$20k	
Common Stock:		$15k
Notes Payable:		$5k

Table 3.2: Trial Balance after Transactions 1-2

Note how, **across all three formats**, **total debits and credits balance**—meaning that the accounting equation is kept in balance as well.[6] Now, going forward in this book, I will **usually only display the journal entry**. The reason for this is that the T-accounts and Trial Balance take up **so much space**; and the journal entry, by definition, captures the *impact* on the company's accounts from the transaction, i.e., the *accounting idea*. Nonetheless, T-accounts and Trial Balances are useful tools for understanding accounting and for keeping track of complex series of transactions, and they will feature prominently in this book's practice problems. So you absolutely need to understand them, and should always try to think through the mapping *from* the journal entries I show, to the T-accounts.

[6]Though, obviously, within T-accounts, debits and credits *do not* balance *within* the particular T-account. If they did, this would just mean that the T-account had a balance of zero! Think about what that would imply for Cash: if every debit to Cash had to be offset by an equal credit *to Cash*, then every dollar that came in would have to go right back out, and the company would be condemned to a perpetual cash balance of zero. Not what we want. The equality of debits and credits holds at the level of each *transaction*: one half of a transaction hits one account, the other half hits another. And therefore, in the aggregate, debits and credits balance across all of the T-accounts as a whole—even as the individual accounts carry healthy, nonzero balances.

Transaction 3: Purchase Equipment for $10,000

We know that Cash decreases by $10k, but a different asset, Equipment, goes up by the same amount. Since both are assets, the decrease is a credit, and the increase is a debit.

	Debit	**Credit**
Equipment (+A)	$10k	
Cash (-A)		$10k

Note a small trap here, in the interplay between our old mantra and our new conventions. **Start with cash** is still an excellent way to *reason* through the transaction—Cash goes down, and an asset going down is a credit. But the writing convention says the debit line comes first, so *Equipment* leads the entry, even though Cash is where the reasoning started. Reasoning order and writing order are two different things.

Transaction 4: Pay for two years of rent in advance: $2,400

Cash decreases by $2,400, but the asset called Prepaid Rent increases by the same amount. Since these are assets, the increase is a debit, and the decrease is a credit.

	Debit	**Credit**
Prepaid rent (+A)	$2.4k	
Cash (-A)		$2.4k

Let's pause here for a quick self-test, to catch anyone dozing. Without looking back: what does the **Cash** T-account look like at this point, after Transactions 1–4? Sketch it before reading on. There are two classic wrong answers. The first is to write down only the most recent change (a lone $2.4k credit)—forgetting that a T-account is the *full history* of the account, not the latest journal entry. The second is to get the history right but the *sides* flipped. The correct answer is:

Cash

Debits	Credits
$15k	$10k
$5k	$2.4k
$7.6k	

And note that you can compute that $7.6k balance in two equivalent ways: arithmetically, by reasoning through the story ("we raised $20k in cash and have spent

$12.4k of it"), or mechanically, by netting out the T-account (total debits of $20k, less total credits of $12.4k). The two must always agree.

Transaction 5: Purchase Supplies for $1,000 on Account

Supplies goes up, and that's an asset, so that's a debit. Accounts payable goes up, and that's a liability, so that's a credit.

	Debit	**Credit**
Supplies (+A)	$1k	
Accounts Payable (+L)		$1k

Transaction 6: Provide Services for $3,900 in Cash:

Cash goes up and that's an asset, so that's a debit. And the company earns revenue here as well—since revenue increments shareholders' equity, that's a credit.

	Debit	**Credit**
Cash (+A)	$3.9k	
Revenue (+S.E.)		$3.9k

Notice the mantra from Chapter 2 doing its work here: we *provided the service*, and that is always the cue to record revenue—and a revenue increase is a credit. Then the algorithm does the rest: debits must balance credits, and we received cash, so we debit Cash. That cue will do quiet work for us later in this chapter, and beyond.

Transaction 7: Provide Services for $3,100 on Account

In a credit sale, both *Revenue* and *Accounts Receivable* increase. Since *Revenue* contributes positively to Shareholders' Equity, that increase is a credit. Since *Accounts Receivable* is an asset, that increase is a debit.

	Debit	**Credit**
Accounts Receivable (+A)	$3.1k	
Revenue (+S.E.)		$3.1k

Transaction 8: Receive Cash of $800 in Advance from Customer

Cash increases, so that's a debit. *Deferred Revenue* also increases, and that's a liability, so that's a credit.

	Debit	Credit
Cash (+A)	$0.8k	
Deferred revenue (+L)		$0.8k

Transaction 9: Pay Salaries of $1,800 to Employees

Cash decreases by $1,800, so that's a credit. *Salaries Expense* is an *expense*, an anti-revenue, so that is a debit.

	Debit	Credit
Salaries Expense (-S.E.)	$1.8k	
Cash (-A)		$1.8k

There are two equivalent ways to get here, and you can start from whichever end feels firmer. You can start with the debit: we know this is an expense, and expense increases are debits, so debit *Salaries Expense*—then find the balancing credit (the cash going out the door). Or you can start with Cash: cash is going out, an asset going down is a credit—then find the balancing debit. Either way, you land on the same entry.

Transaction 10: Pay Cash Dividends of $100

Cash decreases, so that's a credit. And Dividends decrement Shareholders' Equity, so that is a debit.

	Debit	Credit
Dividends (-S.E.)	$0.1k	
Cash (-A)		$0.1k

3.4 Revisiting T-accounts

As warned, we didn't track the T-accounts for every single account for every single transaction, because they take up too much space. But it is important to be able to understand and visualize how transactions affect T-accounts. So, let's just create the T-account for **Cash**. Cash was received (debited) in Transactions 1, 2, 6, and 8, and spent (credited) in Transactions 3, 4, 9, and 10. So, after all 10 transactions, the Cash T-account would be:

Cash

Debits	Credits
$15k	$10k
$5k	$2.4k
$3.9k	$1.8k
$.8k	$.1k
$10.4k	

Notice, by the way, that the plus and minus signs have quietly disappeared. In the earlier T-accounts, I wrote things like +$15k and -$10k—training wheels, while we were getting oriented. But in the world of debits and credits, the signs are redundant, because the *sides* carry that information: an entry on the left of the Cash account is an increase, and an entry on the right is a decrease, by construction. So we write plain, unsigned amounts, and let the geography encode the direction. That is the whole trick of the format.

Which other accounts require this kind of netting? Scan back through the journal entries and you'll find that only one other account was hit by more than one transaction: **Revenue**, credited $3.9k in Transaction 6 and $3.1k in Transaction 7. Its T-account looks like this (with the balance reported on the credit side, since Revenue is a credit account):

Revenue

Debits	Credits
	$3.9k
	$3.1k
	$7k

Every other account was touched exactly once, so its 'final balance' is just the amount from that single journal entry. And here's a satisfying forward link: if we ended the reporting period right now, that $7k Revenue balance would be the proverbial 'top line' of our Income Statement—total revenues for the period, inclusive of both the sale for cash and the sale on account.

3.5 The Trial Balance at this point in time

Remember that a 'Trial Balance' is literally just a **list of the balances** of the company's accounts. It is that simple. If you find it confusing, you are probably overthinking it. The reason it is useful is that accounting is all about going from transactions to the

financial statements. And the financial statements are a report of the company's accounts as of the end of the reporting period. So, a Trial Balance is a nice way to line things up, prior to filling out the financial statements. (And this is how we'll do it in the practice problems.) Please check my work, but I believe that, after our 10 external transactions, our Trial Balance would look like this:

Account:	**Debit**	**Credit**
Cash:	$10.4k	
Common Stock:		$15k
Notes Payable:		$5k
Equipment:	$10k	
Prepaid Rent:	$2.4k	
Supplies:	$1k	
Accounts Payable:		$1k
Accounts Receivable:	$3.1k	
Revenue:		$7k
Deferred Revenue:		$0.8k
Salaries Expense:	$1.8k	
Dividend:	$0.1k	
Retained Earnings:		$0
Total:	$28.8k	$28.8k

Table 3.3: Trial Balance after 10 external transactions:

There's one important detail here worth noting. Note how I added a row there for **Retained Earnings.** The balance of that account is **zero**. Why? Because this corporation **started its life in this reporting period**. So, it hasn't yet retained any earnings from previous periods.

Won't its Revenues, Expenses, and Dividends **flow in** to Retained Earnings? Yes, **they will**. But not yet. We still have balances of those Revenue, Expense, and Dividend accounts in this Trial Balance. We want those there, because we'll want to use this Trial Balance to fill out our Income Statement. So, if we kept those accounts there, but **also** add those quantities to Retained Earnings in the Trial Balance, we would be **double-counting** them, and throw the accounting equation (equivalently, our debits and credits) out of balance.

So, in a **Trial Balance**, the **Retained Earnings** is the Retained Earnings **from the end of the previous reporting period**. And we only update the Retained Earnings account right at the end of the reporting period, when we 'close out' the temporary revenue, expense, and dividend accounts. This is a detail—but one that will matter for getting things right.

3.6 Going beyond: two trickier transactions

I'm not letting you go quite yet. Recall that at the end of Chapter 2, we went 'beyond' the core 10 transactions: our credit customers paid off their account (Transaction #11), and we paid the electricity bill (Transaction #12). And, back in Transaction #8, we peeked ahead at what would happen when we finally provided the training that was paid for in advance. Let's now redo two of these with our new tools—because these are exactly the kinds of transactions where the tools earn their keep. They are also the kinds of transactions that confused me when I was in your shoes; I would not have gotten them right on the first try. The point is to see how journal entries, T-accounts, and the debit/credit rules help us reason our way through something that isn't immediately intuitive.[7]

3.6.1 Transaction 11: Our credit customer pays off their account

Recall that back in Transaction 7, we provided training to customers *on account*: they promised to pay the $3.1k later, and we recorded the journal entry:

	Debit	**Credit**
Accounts Receivable (+A)	$3.1k	
Revenue (+S.E.)		$3.1k

Now, as we supposed at the end of Chapter 2, those customers make good on their promise, and deliver the $3.1k in cash they owe us—they 'pay off their account.' We already worked out *what* happens, in Accounting Equation terms: Cash up $3.1k, Accounts Receivable down $3.1k, and—resisting the trap—*no* new Revenue. The exercise now is to get there in debits and credits.

Start with cash: we are receiving cash, so Cash goes up by $3.1k, and an asset going up is a debit. What balances it? Not Revenue—we recorded the $3.1k of *Revenue* back in Transaction 7, when we provided the service, and recording it again now would be counting the same sale twice. What's really happening is that the customers' outstanding promise to us—the *Accounts Receivable*—is being extinguished. That asset goes down, and an asset going down is a credit:

	Debit	**Credit**
Cash (+A)	$3.1k	
Accounts Receivable (-A)		$3.1k

[7] What about the electricity bill? I'll leave that one to you. In debits and credits, it works exactly like the salary payment in Transaction 9: debit *Utilities Expense*, credit *Cash*, $300.

Let me tell you why I think this one is deceptively tricky. Picture me in week 3 of my own introductory accounting course, years ago. I had just learned accrual accounting and revenue recognition the week before, and had just met *Accounts Receivable* for the first time in my life—this abstract thing, a customer's promise to pay later. Then I was told: the customer finally pays off their account. Intuitively, that sounds like *good news*. They're doing the right thing; they're paying us! And then the professor said: "and so your asset goes down." Why does a good event make my asset go down? That felt backwards.

If you feel that tension, you're not alone. Here's what dissolved it for me, and it illustrates the special usefulness of the T-account. *Accounts Receivable* means: 'the amount our customers still owe us.' After this transaction, these customers owe us nothing—there is no longer any outstanding promise. So the ending *balance* of the account must be zero. Now go to the T-account and work backward: before the payment, the account carried a $3.1k debit balance from the original sale. What entry takes it from a $3.1k debit balance down to zero? A $3.1k credit. And since we also know we received $3.1k in cash, the balancing debit is to Cash. That's the journal entry.

Accounts Receivable

Debits	Credits
$3.1k	$3.1k
$0	

Two general lessons lurk in this example. First: some of you will reason your way directly to the journal entry from first principles ("asset up must be balanced by asset down"), and that's great. But as transactions get more complex, there will be situations where the cleanest path is to ask, "what does the ending *balance* of this account need to be?"—and then work backward to the journal entry that gets it there. The T-account is built for exactly that kind of reasoning, and we will lean on it hard in Chapter 4. Second: a journal entry is always tied to a *specific event on a specific date*. A common mistake, when asked for the entry recording the customer's payment, is to answer with the *combined* effect of both dates—"debit Cash, credit Revenue"—as if the sale and the collection were one event. They are not. The revenue belongs to the date we provided the service; today's entry records only today's event, the swap of a receivable for cash.

3.6.2 Transaction 13: We provide the training that was paid for in advance

Finally, the transaction we could only preview in Chapter 2—call it Transaction 13. Think back to Transaction 8, in which customers paid us $800 in cash *up front* for train-

ing we promised to provide later (next February, on our timeline). We did not record revenue then—we hadn't earned it yet—and instead recorded a liability, *Deferred Revenue*, representing our obligation to provide the service in the future:

	Debit	**Credit**
Cash (+A)	$0.8k	
Deferred Revenue (+L)		$0.8k

Now February arrives, and we actually run the promised training sessions. The service is provided. How do we account for *that*? We previewed the answer in Chapter 2, in Accounting Equation terms; now let's get there with our new tools. This one is genuinely abstract, because no cash changes hands today—the cash moved back in Transaction 8. Here are two routes to the entry, and both are instructive.

Route one: start from the thing you know with absolute confidence. We provided the service, and 'provide the service' is the trigger for 'record revenue.' An increase in Revenue is a credit. So: credit *Revenue*, $0.8k. Now the algorithm goes to work for us: debits must equal credits, so *something* must be debited $0.8k—and asking 'what?' points us to the answer. Our obligation to these customers—the *Deferred Revenue* liability—has been discharged by providing the service. A liability going down is a debit.

Route two: start from the liability, using the work-backward-from-the-balance trick from Transaction 11. Before today, the *Deferred Revenue* T-account carried a $0.8k credit balance: the amount of service we still owed our customers. After today, we owe them nothing—so the balance must go back to zero. What takes a T-account from a $0.8k credit balance to zero? A $0.8k debit. And what is the balancing credit? The *Revenue* we have finally, in the accrual sense, *earned*.

	Debit	**Credit**
Deferred Revenue (-L)	$0.8k	
Revenue (+S.E.)		$0.8k

Deferred Revenue

Debits	Credits
$0.8k	$0.8k
	$0

Notice how this closes the loop on the "customer-provided financing" idea from Chapter 2. When the customers paid up front, they were effectively extending us a

loan—one we were obligated to settle not in cash, but in services. Today, we repaid it. The liability, like any loan that has been paid off, comes off the books. (Indeed, you now have a tidy little taxonomy of credit financing, organized by who is doing the financing: *Notes Payable* is financing from a bank; *Accounts Payable* is financing from our suppliers; and *Deferred Revenue* is financing from our customers.)

For practice, try updating the Trial Balance from the previous section for Transactions 11 and 13, and verify that it still balances. You should find that Cash rises to $13.5k, that Accounts Receivable and Deferred Revenue fall to zero, that Revenue rises to $7.8k—and that total debits and total credits are both still $28.8k. Can you see why neither total changed? Transaction 11 merely swapped one debit-balance asset (a receivable) for another (cash), and Transaction 13 merely swapped one credit balance (a liability) for another (a revenue).

And that's the hump. None of this becomes second nature without practice—that's what the practice problems are for—but there are no further secrets hiding here. In Chapter 4, we'll put these tools straight to work on the expenses we have been conspicuously postponing: the gradual 'using up' of the Equipment, the Prepaid Rent, and the Supplies. And in Chapter 5, we'll use them to 'close the books'—to move those temporary Revenue, Expense, and Dividend balances into Retained Earnings, and produce the financial statements themselves. From here on out, debits and credits are how we speak.

4

End of Period: Adjusting Entries ("Internal Transactions")

Last chapter, we showed how companies account for "real" external transactions, at the exact moment of the exchange (e.g., the sale of a good to a customer, the purchase of supplies, or the transfer of cash). We covered a broad set of transactions, and accounted for them correctly. But we're still missing a few final steps on the path to reporting the company's **financial statements**, which is ultimately what accounting is all about. There are two parts to this.

1. The first is the "adjusting entries" or "internal transactions" (in contrast to real "external transactions") the company will make at the end of the reporting period, before closing the books. This is how we'll account for various expenses that you intuited should be coming. We'll cover these in **this chapter**. We'll finally see how the expenditure on assets like Equipment, Prepaid Rent, and Supplies is factored in as an expense, as those assets are "used up" over time.

2. Second, we'll need to cover some of the rote '**mechanics**' of the financial-reporting **process**—i.e., how we dot our i's and cross our t's, to make sure we report everything correctly. In particular, we'll see precisely how to use a final trial balance to fill out the Balance Sheet and Income Statement. We'll cover these "mechanics" of the reporting process in the **next chapter**, Chapter 5.

4.1 Warming up: Reading T-accounts

Before the new material, a warm-up puzzle. Suppose that our company's *Supplies* T-account looks like the below. (Recall from last chapter that the netted line at the bottom represents the final **balance** of the account.)

Supplies (A)

Debits	Credits
$300	$X
$200	
$400	

Warm-up: Solve for X

Solve for X. The arithmetic here is trivial, but notice how much conceptual knowledge you have to deploy to get it right. You have to know that a T-account tracks the whole sequence of debits and credits to one account over time, with the netted balance at the bottom. You have to know that *Supplies* is an asset account, so increases are recorded on the left (debits) and decreases on the right (credits). And so, if total debits were $500, and the final balance is a $400 debit, the credit must have been $100.

Note also that the answer is $100—not 'negative $100.' Students are sometimes tempted to write a minus sign in front of it. But the "negative-ness" of that $100 is already encoded by the fact that it sits on the credit side. That is the whole trick of the format: in a T-account, the *side* carries the sign.

One more observation, which sets up this entire chapter: To this point in the book, we have never actually *credited* an asset like *Supplies* or *Equipment*. We have seen *Cash* go up and down plenty—we get cash, we spend cash. But the resources we bought with it—the *Supplies*, the *Equipment*, the *Prepaid Rent*—have only ever gone *up*. So what kind of event would produce a credit like that $X above? Nothing in Chapters 2 and 3 would. That is precisely what's new in this chapter: we are going to see how, why, and when these assets get credited.

4.2 "Adjusting Entries" / Internal Transactions

In Chapters 2 & 3 we accounted for "external transactions"—"real" exchanges—during the reporting period. At the end of the reporting period, we account for what we call, by contrast, **"internal transactions,"** or **"adjusting entries."** What are these? The best definition is that these adjusting entries are **everything else**: Everything else the com-

pany needs to do before closing the books.[1]

A quick note on the two terms: I use them interchangeably, and both are used in practice. It's a bit like 'deferred revenue' vs. 'unearned revenue'—two labels for one underlying thing, each highlighting a different aspect of it. "Adjusting entry" emphasizes the *when* and the *what*: the updates we make to our accounts right before reporting. "Internal transaction" emphasizes the contrast with external transactions: no outside party is involved.

To make this vivid, picture the scene: It is December 31, 11:59pm, and the poor accountant is at her desk, frantically doing everything that must be done before the stroke of midnight to "close the books" and report correct financial statements for the year. All of those last-minute updates are the adjusting entries.

It's worth dwelling on the contrast with Chapters 2 & 3 for a moment. In each of our 10 external transactions, there was a real *event*: cash changed hands, or a truck showed up with supplies. There was a specific *date* on which the exchange occurred, and that concrete event is what **"triggered"** us to update the books. Adjusting entries are everything we need to do to get the accounts right that is *not* triggered by an event like that. No one hands us cash; no truck shows up. But, economically, something has changed—and we need to reflect that change in our accounts before we report.

4.3 Prepaid Rent → Rent Expense

Let's make this concrete. Suppose that our company does annual financial reporting, aligned with the calendar year, January 1 through December 31. *During the reporting period*, our company had paid for **two years of rent in advance**. We have already accounted for that external transaction, like so:

Prepaid rent (+A)	$2.4k	
Cash (-A)		$2.4k

Get 2-year lease, July 1, 2020

I added one detail here: Let's make things simple and assume that our company got this contract exactly **halfway through** the reporting period, **July 1**. Now, it's December 31, the end of the fiscal year—time to "close the books" and report the Financial Statements. Now, the company has **used up** 6 months of the rental contract, **one quarter** of the value. So, it needs to account for that **expense**, from expending this resource, and,

[1] The practical requirement is that all necessary updates be included before the financial statements are prepared. The final section returns to borderline cases.

additionally, account for the reduction in the value of the rental contract still remaining. But there will be **no moment of "real" exchange** that will trigger that. This means that the company must account for this as an "internal transaction" or "adjusting entry" at the end of the reporting period.

Pause on how strange this is, relative to everything we did in Chapters 2 & 3. The company signed the lease on July 1—that was the external transaction. After that, it just *occupies the building*. From the perspective of the lease itself, December 31 is not a special date; nothing happens in the lease that day. December 31 is only special from the perspective of *financial reporting*: it's the date on which we stop, take stock, and prepare the financial statements. The company has "used up"—depleted, expended—one quarter of the resource, **simply through the passage of time**. So we must update our accounts ourselves, internally.

Intuitively, there are **two things** we need to do here. First, we have been consuming this resource, so we should recognize an *expense* for that consumption—that's the income-statement side of the story. Second, we no longer have the full two years of prepaid rent that we started with, only three quarters of it, so we should reduce the asset balance to its *remaining* value—that's the balance-sheet side. And here is the nice part: these two things turn out to be **two sides of the same journal entry**.

The reduction in the remaining value of the *Prepaid Rent (A)* is a credit. And the balancing debit is the *expense* from expending or using up that asset, which we call, unsurprisingly, *Rent expense*. So, the journal entry for this adjusting entry would be...:

Rent Expense (-S.E.)	$600	
Prepaid Rent (-A)		$600

End of period adjusting entry for Rent Expense, 12/31/2020

Since the company initially had Prepaid rent of $2,400, but has used up one quarter of that resource, the **remaining balance** would be $1,800. You can see this using the T-account for Prepaid Rent. Remember that Prepaid Rent is an asset account, so it has a **debit balance**.

Prepaid Rent

$2,400 (07/01/'20)	$600 (12/31/'20)
Balance: $1,800	

T-account after end-of-period adjustment, 12/31/2020

Think about what's happening here: The *initial purchase* of the rental contract (the

external transaction) was recognized as an asset, because it was a *resource* that the company was going to use over *multiple periods*. And then we **apportion** that expenditure as expenses over time, via adjusting entries/internal transactions, and **reduce** the remaining **balance** of the asset account. After two years, once the lease is up, that balance will be fully reduced to zero ($\$2,400 - \$600 * 4$). This will be the basic pattern for assets in general.

Note one more thing here—a theme that will recur throughout this book: the **link between the Income Statement and the Balance Sheet**. In this one adjusting entry, we have an expense on the Income Statement (*Rent Expense*), and that expense is directly linked to a reduction in an asset on the Balance Sheet (*Prepaid Rent*). Recall from last chapter that every way our company could earn a *Revenue* was linked either to an increase in an asset (*Cash*, *Accounts Receivable*) or a decrease in a liability (*Deferred Revenue*). Expenses, it will turn out, are the mirror image: every expense will be linked either to a decrease in an asset—as here—or an increase in a liability—as we'll see near the end of this chapter.

4.3.1 Asset vs. Expense: A Question of *When*

In earlier chapters, we saw that one of the most basic questions in accounting is whether some expenditure should be recorded as an **asset or an expense**. (Recall Transaction #9 from Chapter 2, where we had to reason through whether the money spent on salaries bought an 'asset'—and concluded that it did not.) An enormous amount of controversy in accounting—in standard-setting, in debates over earnings management—revolves around this question. But you can now see that the question is worded a bit misleadingly: it's not really a binary, 'asset vs. expense, full stop.' Most assets "get expensed out" over time, later on. The $2,400 we spent on the lease was first recorded as an asset, and is then gradually becoming *Rent Expense*. So, when you hear someone ask, "is this expenditure an asset or an expense?"—a better internal translation is: "should we recognize the expense *now*, or *later*—by first capitalizing the expenditure on the Balance Sheet, and then expensing it out over the periods of use?" It is (usually) a question of *timing*.

To see the contrast concretely, consider an expenditure that gets expensed immediately: the monthly utilities bill for our office. When we pay it, we would record...:

Utilities Expense (-S.E.)	$700	
Cash (-A)		$700

Paying the monthly utilities bill (an immediate expense)

There is no "Prepaid Utilities" asset sitting on the Balance Sheet for months—the cost goes straight through the Income Statement, because we consumed the electricity in the same period in which we paid for it. The rent, by contrast, followed the "capitalize now, expense later" pattern: first the debit to *Prepaid Rent (+A)* at signing, and then the *Rent Expense* recognized over time via adjusting entries. In both cases, the expenditure ends up as an expense; the difference is not *whether*, but *when*.

4.4 Supplies → Supplies Expense

Recall that *during the reporting period*, our company purchased **Supplies** for $1k, on account. *Supplies* are a resource that our company expected to use over multiple reporting periods—an asset. So, we recorded that initial external transaction as...:

Supplies (+A)	$1k	
Accounts Payable (+L)		$1k

Initial Purchase of Supplies, 7/1/2020

Now, it's December 31, the end of the reporting period. 6 months have gone by. Presumably, our company has **used** (expended) its supplies to operate over this period. But there has been no moment of "real" exchange—we have not yet recognized an *expense* from using that resource. So, we need to do so as a part of the "adjusting entries" at the end of the reporting period, before closing the books.

To make this concrete, suppose the supplies in question are printer paper. We are obviously not going to update our financial accounts every time somebody prints a page—so nothing during the period recorded our usage. And note that, unlike the lease, the supplies have no built-in clock: the mere passage of time doesn't tell us how much paper we've gone through. So what do we do? We go back to the supplies closet at year-end, take an **inventory**, and count what's left. Suppose we determine that 60% of our supplies are remaining, i.e., we have **used 40% of our supplies**. Then, intuitively, we need to reduce the remaining balance of the **Supplies** asset, and recognize the amount we used/expended as an expense. The reduction in the *Supplies* asset is a credit, and the corresponding debit is, unsurprisingly, *Supplies Expense*:

Supplies Expense (-S.E.)	$.4k	
Supplies (-A)		$.4k

End-of-period adjusting entry for Supplies Expense, 12/31/2020

Note that **Supplies** is an **Asset** account, and **Supplies Expense** is an **Expense** account. Since the company has "used up" 40% of its $1k in Supplies, it has a remaining **balance** of $600 in its Supplies asset account. You can see this using the T-account.

Supplies	
$1,000 (07/01/'20)	$400 (12/31/'20)
Balance: $600	

T-account after end-of-period adjustment, 12/31/2020

Using supplies does not settle the amount we owe their supplier. The $1k Accounts Payable stays outstanding until we pay it. That payment will be a separate external transaction, recorded on its own date:

Accounts Payable (-L)	$1k	
Cash (-A)		$1k

Future external transaction: Paying off the account, some later date

The adjusting entry records consumption of the supplies. The payment entry settles the obligation to the supplier. They concern the same purchase, but they record different events.

This raises a related question you might be wondering about: *are* adjusting entries always about the passage of time? The rent expense grew with the passage of time; depreciation, which we'll meet next, is calculated per year; and interest on a loan piles up with time (more on that in Chapter 7). So it's a decent heuristic—many adjusting entries are indeed closely tied to time passing. But it is not the *definition*, and *Supplies* is the counterexample: if we had bought our supplies and then, for whatever reason, never touched them all year, time would have passed, but there would be no *Supplies Expense* to record. What creates the expense is the *use* of the resource, not the calendar. This is why I stick with the broader, admittedly circular definition: adjusting entries are everything you need to do before closing the books that wasn't already captured by an external transaction.

4.4.1 Discussion: Income Statement vs. Balance Sheet Perspectives

Now that we've done two concrete adjusting entries, I want to introduce a concept and a bit of jargon that will show up more and more as this book proceeds. This is the notion

of "income-statement vs. balance-sheet *perspectives*" in accounting.

Have you ever seen the famous **"duckrabbit"** drawing? It's a simple line drawing that you can see as a duck (with its bill pointing one way), or as a rabbit (that same shape becomes the ears, pointing the other way)—one image, two ways of seeing it.[2] I try to keep the arithmetic in this book as simple as possible, because what I really care about is that you understand what's going on—and that usually means being able to see the same thing in more than one way. The adjusting entry we just did is a perfect little duckrabbit: one journal entry, two ways of seeing it.

The **"balance sheet perspective"** means that we focus our attention primarily on correctly measuring and updating the assets and liabilities of the firm, and reporting them correctly at the end of the period. That is, under this view, reporting the Balance Sheet, and thus the net assets of the firm, is the priority. And income-statement accounts (such as revenues and expenses) are simply "plugs," or ways of explaining the changes in the **net assets** reported on the Balance Sheet, between periods.

The **"income statement perspective"** means that we focus our attention on the 'flow' of profitability from the company's operations, and think of the balance sheet accounts as 'plugs' used to, e.g., store and apportion expenses over time.

This might seem abstract or tedious right now. But these perspectives can enrich our interpretation of many different transactions and help us understand the linkages in accounting. More concretely, this distinction will help us correctly understand and account for certain more advanced transactions and methods later in this book. For now, consider the adjusting entry we just did.

Supplies Expense (-S.E.)	$.4k	
Supplies (-A)		$.4k

End-of-period adjusting entry for Supplies Expense, 12/31/2020

Here are two ways to think about this transaction, from the balance-sheet vs. income-statement perspectives:

- **Balance-sheet perspective:** "It's the end of the period, so we need to report our company's assets and liabilities correctly. We check our supplies closet, and only 60% are remaining. So, we need to reduce (credit) the *Supplies* account, and correctly report that amount ($.6k) on our Balance Sheet. Oh, and to keep things

[2] If you haven't seen it, look up the "rabbit–duck illusion." The image became famous in early 20th-century philosophy—Wittgenstein discusses it, among others. In the older, Enlightenment picture of the mind (think Descartes), reality is objective, 'out there,' and our minds just passively reflect it, like a mirror. The duckrabbit is a nice little demonstration that our minds *actively structure and interpret* what we see. The flip from seeing the duck to seeing the rabbit—without a single line on the page changing—is called a **Gestalt shift**. We already performed one back in Chapter 2 (the "Advanced Gestalt Shift," on supplier financing), and we'll perform several more before this book is done.

balanced, we also need to debit *Supplies Expense*. With that, the Income Statement will explain *how and why* the net asset balance changed."

- **Income-statement perspective:** "When we first bought the Supplies, in July, we bought a resource that we planned to use over many periods. So, intuitively, it didn't make sense to account for that full $1k as an *Expense* against our profitability right away. Instead, we 'stored up' that amount in an asset account, to apportion out as we used it. Then, over this period, we ended up expending 40% of that resource—so, to get a sensible measure of our profitability, we **apportion** that $.4k as an expense. Oh, and to keep things in balance, we also need to credit the *Supplies* account, for the $.4k that we've just apportioned out."

Many more "old school" accounting professors and texts fall into the balance sheet perspective camp. For them, at the end of the day, accounting is all about correctly measuring and reporting our assets and liabilities according to GAAP, and the income-statement accounts are just "plugs," explaining how those changes arose.

Why is that? I think it's because many of those authors come from a professional **audit** background. Think about what an auditor does: they go into a company and say, "You claim you have this much inventory—let me check whether you actually have it. You claim these are your assets and liabilities—let me verify that they're measured and classified correctly." That work is genuinely important. (Consider the FTX debacle, which was, at its core, a case of an institution simply lying about the assets it had.) If your professional life is spent verifying balance sheets, it is natural to come to see the balance sheet as what accounting is *about*.

I personally (M. Shaffer) tend to lean more into income-statement perspectives. There's no right answer here, and I never wish to impose my perspectives on you. But I suspect that a lot of my explanations in my text, going forward, might have that flavor, in contrast to some other texts—so I wanted to note it, and *why*.

My own background is the mirror image of the auditor's: my professional experience and research interests run toward *equity research and valuation*—thinking about how to value companies, interpret earnings, and forecast future performance. That is the lens I most naturally reach for. And in that world—the world that most business students are headed toward, whether in finance, consulting, or general management—the focus is overwhelmingly on the income statement. When Apple releases its quarterly results, what does everyone talk about? Whether Apple met or beat its *earnings* estimates—income-statement language. Have you ever heard anyone say, "Wow, Apple's *balance sheet* really surprised the market this quarter—*Accounts Receivable* came in above expectations!"? That is not a thing.

My view is that it doesn't make sense to prioritize the Balance Sheet in our thinking. As we said in Chapter 1, the "book value" of a corporation will never be the same

as its "true" or market value. This is because there are many real economic assets (e.g., internally-developed R&D, customer loyalty, etc.) that are not recognized on the books under GAAP. And most assets that are recognized on the books are recorded "conservatively," asymmetrically. Wall Street analysts and investment bankers seeking to estimate the value of operating companies virtually never look at the book value of shareholders' equity as their metric—nor should they, given this.

Therefore, in my view, it doesn't make sense to focus our attention on measuring the Balance Sheet accounts *per se*, because *who cares*? The Balance Sheet, with its "stock" measures, doesn't tell us the real value of the company and never will—nor should it. Instead, the value of any company is a function of the expected future profits. So, in my opinion, the priority should be **measuring the flow of *profitability*** in a way that makes sense—that is persistent, captures the real, ongoing economics of the firm, and thus allows us to forecast *future profitability*. And the income statement can help us estimate that if it **matches expenses** in a rational, consistent, and predictable way over time. That's my opinion.

At this point in the book, these are just two ways of *looking at* this one same transaction. But, later on, we'll often have the ***choice of two different allowable methods*** for accounting for some transaction or issue, with one characterized as an income-statement-focused approach and the other a balance-sheet-focused approach. So, this conceptual discussion will also become practical. In fact, before this chapter is over, we'll see a real-world labor dispute—from Major League Baseball, of all places—that turned on exactly this difference in perspective.

4.5 Equipment → Depreciation Expense

During the reporting period, our company purchased *Equipment* for $10k. Equipment is a resource that we expect to use for multiple periods, so it was an asset. Suppose for the sake of this example that we bought this Equipment right at the beginning of the last reporting period, January 1, 2020. We accounted for that transaction like so:

Equipment (+A)	$10k	
Cash (-A)		$10k

Purchase of Equipment, 1/1/2020

Now it's December 31, 2020, time to close the books and report. We have **used** this resource over the year, and need to **apportion** the expenditure we made. But *Equipment* isn't really like *Supplies*—we can't just go and count how much of our Equipment is left over. It's not that kind of thing. Nor is it like the lease, which came with a built-in

clock—a fixed term that made the arithmetic unambiguous. How much of a *machine* have you "used up" by December 31? The question doesn't even quite compute. So, what do we do?

For Equipment, we **make an assumption**. Suppose we **assume** that we will use this Equipment for **10 years**, and then it will have reached the end of its **useful life** and we will dispose of it for $0 in cash. (The amount we expect to be able to get for an asset at the end of its useful life—here, $0—is called its **salvage value**, or 'residual value.') Then, under **straight-line depreciation**, we could **recognize** $1k in expense per year ($\frac{\$10k-\$0}{10\ years}$)—i.e., **apportion** the expenditure evenly over the *expected* useful life of the asset. And we **name** that expense *Depreciation*.

So, at the end of the reporting period, our adjusting entry would be...:[3]

Depreciation Expense (-S.E.)	$1k	
Equipment (-A)		$1k

End-of-period adjusting entry for Equipment, 12/31/2020

The remaining "book value" of the **Equipment** would then be $9k.[4]

Equipment	
$10k	$1k
$9k	

T-account after end-of-period adjustment, 12/31/2020

And, after 6 periods, as of 12/31/'25, the remaining "book value" would be $4k.

[3] N.B.: We will find out, next chapter, that many companies do this a little bit differently. But this is the basic concept.

[4] Note that this clean $1k figure depends on my assumption that we bought the Equipment right on January 1. Had we instead bought it halfway through the year, on July 1, we would recognize only half a year's worth of depreciation—$500—in the first year's adjusting entry, and then the full $1k in each subsequent full year. The 'straight line' is a constant rate *per unit of time*; the dollar amount recognized in any given entry depends on how much time has elapsed since the books were last updated.

Equipment	
$10k	$1k
	$1k
	$1k
	$1k
	$1k
	$1k
$4k	

T-account after end-of-period adjustment, 12/31/2025

In other words, the "book value" of the Equipment declines in a **straight line** over time. Hence the term "straight-line depreciation."

4.5.1 Why depreciation can feel strange

We have calculated the entry. Now consider two questions about what it means: does accounting depreciation measure a decline in market value, and how much does the expense depend on management's estimates?

- **First** we use the term 'depreciation' in everyday speech, but the **accounting meaning** of the term is **different** from both the **everyday meaning** and the **economic meaning.** In accounting, depreciation is defined as the way that the expenditure on a physical ('tangible') long-term asset is allocated over reporting periods. (When the asset in question is a nonphysical, i.e., intangible asset, we use the term 'amortization' for the same concept.) And this can very well **differ** from the **'true'**/'economic' depreciation of the asset, i.e., the actual decline in the functionality and/or market value of the asset.

  For example, it's a cliché that 'the day you drive a new car off the lot, it loses 25% of its value,' in terms of its *market* value—what you could resell it for.[5] But if a business were buying that vehicle, expected to use it for 15 years before disposal, and were using straight-line depreciation, it wouldn't recognize a depreciation expense of 25% in the first subsequent report. In other words, the **reduction in the market value of the asset** will **not** be the same thing as the **accounting depreciation.**

  Two related questions tend to come up here. The first: "What if the *market value* of our Equipment differs from its book value—don't we need to fix that?" My first

[5] Why a nearly new car may sell for less than a new one is an interesting question about information and resale markets. For our present purpose, the key point is simply that a resale price and an allocation of purchase cost are different measures.

response is: how would you even *know* the market value? A share of Apple stock has a price you can observe every second; a specialized crane, bolted to the floor of your factory, does not. But suppose you somehow did know it. The answer is still *no*: depreciation, under historical cost, is not in the business of tracking market values—it is in the business of allocating cost. The second question: "Okay, but what if we actually *sell* the Equipment, for more than its book value?" Then the sale must be recorded as a real, external transaction. A separate impairment rule may also require a write-down without a sale, as we will learn in Chapter 12. We would remove the Equipment at its book value, record the *Cash* received, and recognize the difference as a **'Gain'** (or, if we sold for less, a 'Loss').[6] The contrast is the point: ordinary upward market fluctuations do not change historical cost; a real sale does, and impairment rules can require recognition of a decline even before a sale.

- **Second,** remember that the *Depreciation Expense* we recognized was a function of how long the **management *expected*** to use the asset. In our example, we assumed **straight-line depreciation** and **zero salvage value**, which is the simplest possible case. And yet, even so, the annual *Depreciation Expense* ultimately depended on **management's proclaimed expectations about the future**. And the future is never perfectly knowable. That **forecast** is not **objective**, **verifiable**, or **certain.**

  So, accounting *depreciation* is a function of *unverifiable, subjective* forecasts and estimates. And yet—*Depreciation Expense* is *an expense*, so it directly impacts the "bottom line." A company could be profitable or unprofitable in one quarter depending upon its **depreciation *assumptions***, holding its **real activities** fixed. To make that concrete: suppose management had estimated a **40-year** useful life when it first put the Equipment into service, rather than 10 years. First-year *Depreciation Expense* would be $250 rather than $1,000, and first-year net income would be $750 higher—with not a single thing about the company's real operations changing. Unsettling? It should be, a little. But ask yourself: what's our alternative? Refusing to apportion the cost at all would give us the distortions we discussed in Chapter 1; and nobody, including management itself, truly *knows* how long the machine will last. In everyday speech, people use the term "bottom line" to mean something that is objective, unquestionable, or undeniable. But it turns out that, in accounting, the "bottom line" is **anything but**.

  This will be another theme of this book. **Accrual accounting often depends**

[6] E.g., if Equipment with a book value of $9k were sold for $10k in cash, we would debit *Cash* $10k, credit *Equipment* $9k, and credit *Gain on Sale of Equipment* $1k. Gains and losses get their full treatment in Chapter 11—including a technical wrinkle I am suppressing here—so consider this a preview.

upon unverifiable management *forecasts* and expectations — and those can sometimes be "gamed" or even abused.

Now, a good question you might be asking at this point: If depreciation depends on management's assumptions, what stops a company from just *choosing* whatever assumptions flatter it—depreciating everything over one or two years to look poorer (say, ahead of a labor negotiation), or over fifty years to look more profitable? The answer is that management has *some* degrees of freedom here, but not infinite ones. Remember the **auditors** from Chapter 1. The auditors who sign off on a company's financial statements are typically specialists in its industry; they know what 'normal' useful lives look like for that kind of asset, in that kind of business. There is usually no hard-and-fast FASB rule that says "this type of machine must be depreciated over exactly X years"—but if a company's assumptions are radically out of line with its peers', the auditors, who must put their own names on the statements, will push back. So the discipline here comes not from a rulebook, but from industry norms, the auditors, and, ultimately, the scrutiny of the capital markets.

You might also be wondering whether *time* is really the right basis for the apportionment in the first place. Wouldn't it make more sense, for some assets, to depreciate based on *usage*—miles driven for a delivery truck, or machine-hours for a lathe? That is a perfectly sensible idea, and there are indeed allowable "activity-based" depreciation methods that do exactly this; we'll cover them in Chapter 11. For now, we'll stick with straight-line depreciation—the simplest method, and the most common in practice.

4.5.2 Going deeper: A contract can run down while the player improves

Suppose a team pays an up-front amount for a five-year contractual right to a player's services. During those five years, the player improves. His services become more valuable. Should the team therefore stop allocating the cost of the contract as an expense?

It is tempting to say yes if we hear *amortization* as another word for deterioration. But look at the asset being accounted for. It is the finite contractual right the team purchased. After one year, one of the five years has been used. The player may be better than ever, while the remaining period of the contract is shorter.

The income-statement question is how to allocate the acquisition cost over the periods that benefit from it. Improvement may help the team earn more revenue. It does not eliminate the cost of obtaining those services. That is why an accounting expense can make sense even when the underlying talent is becoming more

valuable.

Now imagine the players and owners negotiating over compensation. Each side may prefer a different picture of the team's profitability. Before taking sides, ask whether they are disputing the economics or using the word *depreciation* in two different senses.

4.6 Generalizing: Asset → Related Expense

All of the cases we studied above had **one general form.** In each case, when we bought a resource that we expected to use over many periods, that was an **external transaction**, and we recorded the **Asset** at the **purchase price** ("you get what you pay for"). And then, at the end of the reporting period, we recognized an **Expense** based on how much of the asset we had "used up," as an **"internal transaction"** or **"adjusting entry." This basic pattern will hold for real assets in general.** You can think of the below as the "general form" for accounting for real[7] assets.

Asset (+A)	$(Purchase Price)	
Cash (-A)		$(Purchase Price)

Purchase of Asset, date during reporting period

Related Expense (-S.E.)	$(Amount Used Up)	
Asset (-A)		$(Amount Used Up)

End-of-period adjusting entry for expense of asset, 12/31/2020

Note also that, while all three of our assets shared this one general form, each had its *own method* for estimating the "amount used up." For the prepaid rent, it was purely the *passage of time* relative to the lease term—unambiguous arithmetic. For the supplies, it was a *physical count* at year-end. For the equipment, it was an *assumption* about useful life and salvage value. This is how we'll be able to move faster from here on: whenever we meet a new operating asset in this book, there are really only two questions to ask. **(1.)** What is the *name* of its related expense? **(2.)** What is the *method* for estimating the amount used up in a period? The pattern itself—capitalize first at cost, then expense out over time—will always be the same.

[7] 'Real' here is intended to indicate the caveat that, obviously, *Cash (A)* and other financial assets (e.g., investing in another company's stock) don't really behave this way, since we don't purchase cash, and we don't really 'use up' financial assets in the same way. Later on, we'll also learn about a few special 'real' assets that don't get "used up" over time.

Note how this is intimately related to one of the two core principles of **accrual accounting**: the **matching principle**. We want to "match" our expenses to the revenues they helped to generate, and/or the periods in which they were used. But if we want to apportion rent as an expense **over time**, we can't recognize it all as an expense **right away**. (If we did—and then also recognized rent expenses later on, as we used the space—we would be double-counting the cost.) So, instead, we first "capitalize" that expenditure as an asset, and then recognize that expenditure as an expense over time. In pure debits-and-credits terms: when we sign the lease, the credit to *Cash* needs *some* balancing debit—and the asset *is* that debit, a place to "store up" the cost until the periods in which it belongs. The **definition of an asset,** and the **principle of expense "matching,"** are thus **two sides of the same coin, in accounting**.

4.6.1 Takeaway: Asset Balances, Expenses, and Purchases

Another thing we can now do and understand, with the knowledge we've developed, is infer how much of an asset a company *purchased* in a particular reporting period, using just information from the Income Statement and Balance Sheet. This is a common exercise in this book's practice problems, because it's a simple quantitative application of a core concept. But it's also a common real-world application. Recall that as external users of financial statements, **we don't get to see each individual transaction in the company's journal entries. Instead, we only see their aggregated effects on the Financial Statements reported each period**. So, if we want to know how much a company spent on, e.g., *Supplies* in a period, this is the only way to do it.

When we purchase an asset, we write it up at the **purchase price** (i.e., amount we actually paid or agreed to pay); and, to date, the **only thing** that **reduces** the asset balance is the *expense* related to that asset. What this means is that **asset balances, purchases, and their related expenses**, are all linked—**"jointly determined"**—and so, if we know all but one, we can solve for the missing term.

In fact, based on the "general form" above, we know what *every* operating-asset T-account must look like:

Asset (A)

Debits	Credits
\$Purchase 1	\$Related Expense 1
\$Purchase 2	\$Related Expense 2
...	...
$\sum Purchases - \sum Related\ Expenses$	

The general form of an operating-asset T-account

(If you haven't seen it in a while: the big sigma, $\sum$, just means "the sum of all of

them.") Debits for the purchases, credits for the related expenses, and a balance that is, by definition, the sum of all the debits minus the sum of all the credits. That is all a T-account is.

For example, suppose that some company had reported a ***Supplies* balance** of \$100k at the end of FY2020, and a Supplies balance of \$120k at the end of FY2021, and recognized a **Supplies Expense** of \$40k for FY 2021. And we are asked, how much did the company **purchase** in *Supplies* over the period?

It may help to visualize with the T-account. We have:

Supplies	
FY2020 end bal.: \$100k	FY2021 expense: \$40k
FY2021 purchases: \$X	
FY2021 end bal.: \$120k	

T-account illustrating solving for purchases

Supplies Expense is, by definition, the thing that reduces the balance of *Supplies (A)*. And we had \$40k in *Supplies Expense* over the period. But **Supplies (A)** went up by \$20k. So, it must be the case that the company **purchased** \$60k of *Supplies*. That is something that we can *infer*, even though it is not directly *reported* anywhere. Something new we can learn using accounting concepts. **If we know w, y, and z, we can solve for x**.

Here is how I like to reason through it, purely verbally, without any formula: We started the year with \$100k of *Supplies*. If we had purchased *nothing* during the year, then, after expensing out \$40k, we would have ended the year with \$60k. But we didn't end with \$60k—we ended with \$120k. And the only thing that pushes the *Supplies* balance *up* is purchases. So we must have purchased \$60k. Assets go up with purchases and down with related expenses; everything else is arithmetic.

Or, as another example, if we were told only the beginning and ending balance of the asset account, and the purchases, we could infer the related expense. E.g., suppose the company ended the previous period with a balance of \$100k of *Supplies*, then purchased \$50k in additional supplies over this period, and then ended this period with \$120k in *Supplies*. Then we can know, from this information alone, that it must have had *Supplies Expense* of \$30k.

And one more, with a different asset, to show that none of this is special to *Supplies*: suppose a company reported *Prepaid Rent* of \$500k at the end of FY2021 and \$400k at the end of FY2022, and that, during FY2022, it signed one additional long-term lease, paying \$200k up front. What was its *Rent Expense* in FY2022? The 'purchase' here is the \$200k paid up front for the new lease. So: $\$500k + \$200k - \textit{Rent Expense} = \$400k$, and the *Rent Expense* must have been \$300k.

For these examples, assume that purchases and the related expense are the only changes in the asset account. Under that assumption, we have the following formula:

$$Asset\ Ending\ Balance = Asset\ Beginning\ Balance + Purchases - Related\ Expense \tag{4.1}$$

And we could rearrange and rewrite this in any number of ways, including, e.g.:

$$Purchases = Asset\ Ending\ Balance - Asset\ Beginning\ Balance + Related\ Expense \tag{4.2}$$

$$Purchases = \Delta Asset\ Balance + Related\ Expense \tag{4.3}$$

... etc. But, if you understand the "general form of an asset," you don't need to treat these as new formulas to memorize, or a crutch. These follow directly from the concept from **how assets work in accounting**.

4.7 Salaries Expense: A period expense

The pattern above is how (almost) every real asset works in accounting. But it's **not how every expense works.** Recall from last chapter, Transaction #9, that when a company **pays its employees**, that is fully recorded as an **expense** right away—not an asset. We accounted for that external transaction—**payday**—like so:

Salaries expense (-S.E.)	$1.8k	
Cash (-A)		$1.8k

Payday, during reporting period.

Now consider what happens at the **end of the reporting period**. There is no asset to "expense out," like in the cases above. But, we still need to make an adjusting entry. Why? You may know that, **typically, employees get paid every two weeks.** So, suppose that, as of the end of the reporting period, our employees have **worked** one week since their last **payday**, and they earn $900 per week. This means they have **earned** $900 in additional income in this period, which is our **Salaries Expense**, and we need to **account for** that before closing our books. But we have not paid them yet. This means that, when we close our books, we will have a **Liability**—the obligation to pay them in the future. As in previous cases, we call this obligation to pay cash later a **'Payable.'**

Salaries expense (-S.E.)	$.9k	
Salaries Payable (+L)		$.9k

End-of-period Adjusting Entry for unpaid Salaries, 12/31

Think of revenues and expenses as being about *activities*. Our employees have performed the activity—they've worked the week—so they have earned their pay, and we have incurred the expense. The fact that the cash won't actually go out the door until the first payday of January is a separate matter. (And when it does go out the door, *that* will be an external transaction.) As of December 31, they've earned it, and we owe it—so it belongs in *this* year's financial statements.

As this example shows, **not every end-of-period adjusting entry** is a matter of an asset being "expensed out" over time. Many are, but not all. In this case, we accounted for a liability that we *incurred*—or, to use accounting jargon, a ***liability that accrued*** over time—but which we have not yet paid. This is why I used the more general definition: "Internal transactions/adjusting entries, are everything you need to do before closing the books, but which *wasn't* accounted for as an external transaction."

4.7.1 Takeaway: Expenses are decreases in "net assets"

Let's put the whole chapter together. In the first part of this chapter, we saw expenses that were linked to **decreases in assets**: *Rent Expense* with *Prepaid Rent*, *Supplies Expense* with *Supplies*, *Depreciation Expense* with *Equipment*. Now we have seen an expense linked to an **increase in a liability**: *Salaries Expense* with *Salaries Payable*.[8]

Combining the two: expenses are **decreases in *net assets*** incurred through our operations. "Net assets" is a piece of jargon meaning assets minus liabilities. (You may have noticed the phrase sneaking into the "perspectives" discussion earlier; here is its official definition.) And you can decrease net assets either by decreasing an asset or by increasing a liability. And recall the mirror image from last chapter: every way we could earn a *Revenue* was linked to an increase in an asset or a decrease in a liability—i.e., an *increase* in net assets. This is the deepest version of the theme we've been tracking all chapter: the **link between the Income Statement and the Balance Sheet**. Every line of the income statement is tethered to the balance sheet. We'll develop this into fully general definitions of 'revenue' and 'expense' in Chapter 7.

[8] Interest that accrues on a loan works the same way—*Interest Expense* paired with an *Interest Payable (+L)*—as we'll see in detail in Chapter 7.

4.7.2 Discussion: Does the external-vs.-internal distinction really *matter*?

Some students get a bit hung up on—even stressed by—the somewhat philosophical question of what an external vs. internal transaction truly *is*. For instance: "Isn't every day our employees work an exchange? They give us labor; we incur an obligation. Why isn't that an external transaction, every single day?" My answer: sure—but then, why not every *hour*? Every minute? It's a fun question, but here is the liberating truth: **it doesn't really matter**. All that matters is that we report our financial statements correctly. Suppose our employees earn $180 per workday. There is no substantive difference between recording "debit *Salaries Expense* $180, credit *Salaries Payable* $180" at the end of each of the five workdays, and recording "debit *Salaries Expense* $900, credit *Salaries Payable* $900" once, at the end of the week. The accounts end up in exactly the same place, and the financial statements—the only thing anyone outside the company ever sees—are identical.

However frequently a company updates its books, it must include the work performed and resources consumed through the reporting date. The adjusting entries supply any updates still missing.

5

Closing the Books: The Mechanics of Financial Reporting

5.1 The "mechanics" of the financial reporting process

Back in Chapter 1, I offered a reductive definition of accounting: accounting is the function *f*, such that $f(Transactions) = Financial\ Statements$. For the last three chapters, we have been living inside that function—recording the external transactions during the period, and then the internal transactions/adjusting entries at the end of it. This chapter is the final step: actually *producing* the financial statements themselves. Fair warning: this is the most rote, 'procedural' material in this book. But it is also the *last* of the purely rote material—once we've gotten this process right, we can go back to ideas. And let me spoil the punchline now: in everything that follows, nearly every step is a matter of copying and pasting. The **only** thing you will genuinely have to *think* about is **Retained Earnings**.

We said that the "adjusting entries"/internal transactions are, by definition, "**everything else** we need to do before closing the books and reporting our financial statements." So, now that we have our adjusting entries, we are ready to do just that. Here is a simple summary of how we do it:

- (1.) We take the **final balances** of all of our accounts, after including the adjusting entries. By definition, a list of the balances of our accounts at a point in time

is a "Trial Balance." And we use the term **"Adjusted Trial Balance"** to denote where the Trial Balance stands *after* incorporating all of the internal transactions / adjusting entries at the end of the period.

- (2.) We just *list* all the Revenue and Expense accounts on the Income Statement, and calculate Net Income.

  - Since Revenues, Expenses, and Dividends are 'temporary accounts' that are measured over a period, and "live on" in Retained Earnings, we calculate the updated Retained Earnings. We could also explicitly go through the motions of setting those temporary accounts back to zero, which is sometimes called "The Closing Process."

- (3.) We just *list* all of the Asset, Liability, and Shareholders' Equity accounts on the Balance Sheet.

  - And, importantly, we use the *Retained Earnings Identity*, discussed several times, to calculate the new *Retained Earnings Balance*

5.1.1 Unadjusted Trial Balance

In order to report our Income Statement and Balance Sheet, we first need to get the **final balances** of all of our accounts, as of the end of the reporting period. We have a convenient tool for organizing all of those: **The Trial Balance**.

As of the end of Chapter 3, after completing the 10 "external transactions," our **unadjusted Trial Balance** looked like this...:

Account:	Debit	Credit
Cash:	$10.4k	
Common Stock:		$15k
Notes Payable:		$5k
Equipment:	$10k	
Prepaid Rent:	$2.4k	
Supplies:	$1k	
Accounts Payable:		$1k
Accounts Receivable:	$3.1k	
Revenue:		$7k
Deferred Revenue:		$0.8k
Salaries Expense:	$1.8k	
Dividend:	$0.1k	
Retained Earnings:		$0
Total:	$28.8k	$28.8k

Unadjusted Trial Balance, after 10 External Transactions

5.1.2 Adjusting Entries

And then, in Chapter 4, we did those four internal transactions/adjusting entries, at the end of the reporting period. Recapping, those were...:

Rent expense (-S.E.)	$600	
Prepaid rent (-A)		$600

End of period adjusting entry for Rent Expense, 12/31/2020

Supplies Expense (-S.E.)	$.4k	
Supplies (-A)		$.4k

End-of-period adjusting entry for Supplies Expense, 12/31/2020

Depreciation Expense (-S.E.)	$1k	
Equipment (-A)		$1k

End-of-period adjusting entry for Depreciation Expense on Equipment, 12/31/2020

Salaries expense (-S.E.)	$.9k	
Salaries Payable (+L)		$.9k

End-of-period adjusting entry for unpaid salaries, 12/31/2020

5.1.3 Adjusted Trial Balance

So, to get our **final**, adjusted Trial Balance—which we'll use to fill out the financial statements—we just need to add in those adjusting entries. This is tedious, but rote:

Account:	Debit	Credit
Cash:	$10.4k	
Common Stock:		$15k
Notes Payable:		$5k
Equipment:	$9k	
Prepaid Rent:	$1.8k	
Supplies:	$.6k	
Accounts Payable:		$1k
Accounts Receivable:	$3.1k	
Revenue:		$7k
Deferred Revenue:		$0.8k
Salaries Expense:	$2.7k	
Salaries Payable:		$.9k
Dividend:	$0.1k	
Rent Expense:	$0.6k	
Supplies Expense:	$0.4k	
Depreciation Expense:	$1k	
Retained Earnings:		$0
Total:	$29.7k	$29.7k

Adjusted Trial Balance, after 4 Internal Transactions

All I did was take the previous unadjusted Trial Balance—the list of where the account balances stood prior to the adjusting entries—and added extra rows for our new accounts (Salaries Payable, Rent Expense, Supplies Expense, and Depreciation Expense), and incorporated the *change* in each account indicated by each **adjusting journal entry.** Also note how *Retained Earnings* is still the balance from the beginning of the period (which happens to be $0 in this case, because the firm just started its life in this period), because the Revenue, Expense, and Dividend accounts for the period

haven't yet been zeroed out and 'flowed' in to that account. This is worth dwelling on for a second, because it is *the* subtlety of this chapter: the *Retained Earnings* number you see on any adjusted trial balance is really the balance from the *end of the previous period*. The current period's profits are all still sitting in the individual Revenue and Expense accounts, waiting. If we blindly copied that stale *Retained Earnings* number onto the Balance Sheet, we would be reporting a Balance Sheet that simply omits everything the company earned this period. Keep this in mind—it will be the one place, a few pages from now, where we have to stop copying and start thinking.

5.1.4 CTRL-C, CTRL-V to the Income Statement

Some accounting professors like to fill out the Balance Sheet first. I prefer to start with the Income Statement. (Given the 'income-statement perspective' I confessed to in Chapter 4, you probably could have guessed.)

So: we just **copy and paste** the **Revenue** and **Expense** accounts over to the **Income Statement**. This is easy. There's only one **Revenue** account (because **Deferred Revenue** is a liability, not a Revenue[1]), and all of the **Expense** accounts are explicitly named as such. Note that there is nothing new to compute here: that $7k Revenue figure is simply the ending balance of the *Revenue* T-account—the sum of the credits from every sale we recorded over the year—and reporting that total for the period is exactly the Income Statement's job. So, our Income Statement will look like this...:

Income Statement for the Fiscal Year Ending Dec. 31 2020

Revenue	$7k
Salaries Expense	($2.7k)
Supplies Expense	($.4k)
Rent Expense	($.6k)
Depreciation Expense	($1k)
Net Income	**$2.3k**

Two conceptual notes on that "bottom line." First, *Net Income* is not an account. We never once debited or credited an account called 'Net Income' during the year, and you won't find one on the trial balance. It is a **summary measure** that we compute at the end of the period: total revenues minus total expenses.[2] Second, notice one account that did *not* make the trip over: *Dividends*. As we discussed in Chapter 1,

[1] Despite its name. This is one of the few things in this book that you simply have to memorize: *Deferred Revenue* (a.k.a. *Unearned Revenue*) is a **liability**, and it lives on the Balance Sheet.

[2] And if that number comes out negative? Then, by convention, we present it in parentheses and/or re-label the bottom line "**Net Loss**." This is purely a matter of presentation—conceptually, it is still just revenues minus expenses; the difference just happens to be less than zero.

dividends are **not an expense**—they are a *distribution* of profits to the owners, not a cost of generating them. The Dividends account will enter the story momentarily, via Retained Earnings, but it never appears on the Income Statement.

Also, while we're here, we might as well calculate what our updated Retained Earnings balance will be, since that's reported on the Balance Sheet, which is up next.

$\Delta Retained\ Earnings = Net\ Income - Dividends = \$2.3k - \$.1k = \$2.2k$

5.1.5 Ordering the Income Statement

Before we move on to the Balance Sheet, one detail I glossed over: in what *order* do we list all those expenses? By definition, **Revenue** is always the "top line," and **Net Income** is always the "bottom line" (this is, in fact, where those everyday phrases come from). In between, there is no single hard-and-fast rule for the ordering of the expense accounts. But there is a convention you will see over and over: more **'operational'** expenses tend to be listed **higher up**, and less operational ones lower down.

Salaries, supplies, and rent are operating costs. **Depreciation on assets used in operations is also an operating cost**: it allocates the cost of those assets to the periods that use them. A company may show depreciation separately or include it within categories such as Cost of Goods Sold or selling and administrative expenses. **Interest Expense** reflects financing and is commonly presented after operating income for a nonfinancial company. The important distinction is between operating and financing costs, not between costs paid today and costs allocated from an earlier asset purchase.

Preview: 'EBIT' Separating financing costs also helps analysts compare businesses. **EBIT** means Earnings Before Interest and Taxes: it adds those two expenses back to Net Income. Chapter 6 will explain when that measure is useful. For now, we have what we need to finish our statements.

5.1.6 CTRL-C, CTRL-V to the Balance Sheet

And now, we once again just **copy and paste**, listing all of our asset, liability, and shareholders' equity accounts.[3] And the **only thing I need to think about is updating the Retained Earnings** balance—in this case, *from* 0 up to $2.2k.

[3] A natural question here: what about an account that was *used* during the year but ends the period with a balance of $0? Suppose, e.g., that we had fully delivered all the services our customers prepaid for, 'earning down' *Deferred Revenue* to zero by December 31. Would we list "Deferred Revenue: $0" on the Balance Sheet? In practice, no—companies don't report line items with zero balances on the face of their statements. Internally, the account would still exist in the company's bookkeeping system (its 'chart of accounts'), ready to be used again next period; it just happens to have a zero balance at the moment. As always, all that matters is that the reported statements are correct as of the reporting date.

Balance Sheet as of December 31, 2020

Assets	**Liabilities**
Cash: $10.4k	Accounts payable: $1k
Accounts receivable: $3.1k	Salaries payable: $.9k
Supplies: $.6k	Deferred Revenue: $.8k
Prepaid Rent: $1.8k	Notes Payable: $5k
Equipment: $9k	**Total Liabilities: $7.7k**
	Shareholder's Equity:
	Common stock: $15k
	Retained earnings: $2.2k
	Total Shareholders' Equity: $17.2k
Total Assets: $24.9k	**Total Liabilities and S.E.: $24.9k**

Pause and appreciate what just happened at the bottom of that statement: Total Assets came out to $24.9k, and Total Liabilities and Shareholders' Equity came out to $24.9k. $A = L + E$, all the way at the end. Why did that work? Because every journal entry we made, all year long, had total debits equal to total credits—and debits-and-credits, recall from Chapter 3, is nothing but an **algorithm** for preserving $A = L + E$. The Balance Sheet balancing here, at the finish line, is that algorithm paying off.

And this gives us a second way to see why *Retained Earnings* is the one number we must *compute* rather than copy. Suppose we had done the naive thing flagged earlier, and carried the stale $0 balance straight over from the adjusted trial balance. Then our assets would total $24.9k, but the right side would total only $22.7k—the Balance Sheet simply **would not balance**. The $2.2k of value the company generated for its owners this period is really *there*, sitting in the net assets on the books; the only question is where it gets *claimed* on the right side. *Retained Earnings* is that place. It is the account that ties the whole system together: the spot where the temporary 'flow' accounts of the Income Statement live on in the permanent 'stock' world of the Balance Sheet.

5.1.7 Note: We could have used the "closing process"

In the above, we just copied and pasted all of the revenue and expense accounts to the Income Statement, and all of the asset and liability accounts to the Balance Sheet, and used the **core identity** to update *Retained Earnings*. And we know that revenues, expenses, and dividends are 'temporary accounts' that are re-set to zero for the start of the next period. A more formal way of representing this is the "Closing Process." We could formally 'close out' all of the temporary accounts with journal entries, by fully **debiting** *Revenues* (to zero them out), and fully **crediting** *Expenses* and *Dividends* (to zero them out), and placing the **balance** in *Retained Earnings*, like so:

	Debit	**Credit**
Revenues:	\$7k	
Salaries Expense:		\$2.7k
Supplies Expense:		\$.4k
Depreciation Expense:		\$1k
Rent Expense:		\$.6k
Dividends:		\$.1k
Retained Earnings:		\$2.2k

'Closing process', update to Retained Earnings, 1/1/2021

This gets us to the exact same Retained Earnings entry as before (as it must, given that $\Delta Retained\ Earnings = Revenues - Expenses - Dividends$). And, in my view, this is a fairly baroque procedure for two simple ideas we should know conceptually: (1) that Revenues, Expenses, and Dividends are temporary accounts, measured *over a reporting period*, and, (2), the Retained Earnings identity. A legendary accounting instructor I once learned from, Merle Hopkins, said that students can simply remember that Revenues, Expenses, and Dividends "swim like a fish" into Retained Earnings. But some students find the Closing Process useful as a crutch, when first learning, and for **visualizing** the connection between the world of Income Statement "flow" accounts and Balance Sheet "stock" accounts. In the above you can *see*, in a journal entry, how all of the **temporary** flow accounts **'live on'** in balance of *Retained Earnings*.

Here's a way to see why the temporary accounts must be 'zeroed out' at all. Ask yourself: on January 1, 2021, one minute into the new reporting period, what is our company's Revenue *as of that moment*? The answer is zero—conceptually, necessarily. Revenue is a 'flow,' measured *over* a period, and the new period has only just begun. Contrast that with Cash: our company absolutely does have a Cash balance one minute into the new year (the same \$10.4k it had a minute earlier), because Cash is a 'stock,' measured *at* a point in time. The closing process is just the T-account implementation of this fact about flows and stocks: **flows reset, stocks accumulate**.

Now, you might wonder: does *Retained Earnings* itself get 'closed out' too, for the start of the next period? **No!** Retained Earnings is a *permanent* ('stock') account, reported on the Balance Sheet—it accumulates over the company's whole life, and is never re-set. Our company ends its first year with Retained Earnings of \$2.2k; at the end of its second year, the balance will be \$2.2k *plus* that year's net income, *less* that year's dividends; and so on, for as long as the company lives. It is the Revenues, Expenses, and Dividends—the flows—that get zeroed out, precisely *by* flowing into that permanent balance. And remember: investors never see any of this internal T-account housekeeping. They see only the reported financial statements. That is why I treat the

formal closing entry as optional: what matters is that you can compute the updated *Retained Earnings* correctly, and that you understand *why* the temporary accounts start each period at zero.

5.1.8 Reporting "mechanics" wrap

We have now followed the transactions all the way to the statements. The adjusted account balances supplied the reported amounts; the Retained Earnings identity supplied the closing equity balance. The reporting process keeps those pieces together and gives us a way to check that nothing has been omitted.

Summarizing, in this text, the "reporting cycle" involved these steps:

External Transactions → Internal Transactions / Adjusting Entries → Final Account Balances (organized in an Adjusted Trial Balance) → Financial Statements

Other standard textbooks (such as Spiceland et al.'s *Financial Accounting*) include some extra steps in this process: The "closing process" discussed above, and, then, a separate "Post-Closing Trial Balance" after the statements have been prepared and the temporary accounts closed. The concepts are all identical, and if you internalize the idea, you should have no trouble reconciling the two. But in this book, I seek to minimize the rote "procedural" steps and maximize our confidence in our ability to rely on concepts and ideas instead, so I won't show a separate post-closing trial balance.

5.1.9 Classification and Ordering

In the above, I wanted to get right into filling out the financial statements, so I skipped over explaining one detail about the *ordering* or **'classification'** of the financial statements. Reductively, the Balance Sheet is just a *list* of Asset, Liability, and Shareholders' Equity Accounts, and their final balances as of the end of the reporting period. The 'classification' of the Balance Sheet is just about the *order* in which we list things.

First, as we've seen, Balance Sheets always separately group each category (*A*, *L*, *and E*), and always list Assets first, followed by Liabilities, then followed by Shareholders' Equity accounts (A, then L, then E). Typically, the Balance Sheet is split into two columns, with Assets on the left side and Liabilities and Shareholders' Equity on the right.

Liquidity

Within the asset and liability sections, accounts are listed in order of **liquidity**. Liquidity is a term that is commonly used in accounting and finance. In this context, it means how quickly the asset or liability is expected to be converted to or discharged in cash—or, more generally, how soon it is expected to provide its benefit or be settled.

Put differently, liquidity is like the **time horizon** of the asset or liability. You could think of it as: how long until the *end of the story* for that asset or liability?

To get a feel for this, try a comparison: which is more liquid, *Accounts Receivable* or *Equipment*? Accounts Receivable, clearly. Receivables are amounts our customers have promised to pay us in the relatively near future—we are not going to wait ten years for our customers to pay us (and if we are, something has gone very wrong). Equipment, by contrast, is a resource we expect to use over many years; its 'story' plays out over a decade or more. Now flip to the liability side: which is more liquid, *Salaries Payable* or *Notes Payable* owed to a bank? Salaries Payable—it is just the slice of wages our employees have earned since the last payday, and we will pay it off within days of the year-end. The bank note, meanwhile, might not be fully repaid for five or ten years. And what about *Cash*? Cash is the most liquid asset *by definition*: its 'time to cash' is zero, because it already *is* cash. (It is, as we said in Chapter 1, the OG asset—all other assets are, in some sense, ranked by how readily they can be turned into it.)

"Current" vs. "Long-Term"

It is also common to bucket assets and liabilities into **two binary categories** based on **liquidity**: In our examples, **Current** assets are expected to be realized, sold, or consumed within one year, and **Current** liabilities are expected to be settled within one year. **Long-term** describes the remaining assets and liabilities. The normal operating cycle governs when it is longer than a year.[4]

So, 'current assets' are those that will or can be converted to cash or value within one year. 'Long-term assets' are assets with a longer horizon than that. 'Current liabilities' are those that will come due and need to be settled within one year or the normal operating cycle, if longer. And 'long-term liabilities' are those that will come due later. Note that the two categories are *disjoint* and exhaustive: every asset and every liability is either current or long-term, and the Balance Sheet is partitioned accordingly. There is, admittedly, something a little deflating about the terminology here: in accounting, 'long-term' just means *longer than one year*.[5]

Jargon: "Working Capital" While we're here, I want to introduce a bit of jargon that we will end up using *a lot* in later chapters. We use the term **"working capital accounts"** to refer to a company's **current asset** and **current liability** accounts. (And, similarly, we sometimes use the term **"net working capital"** to refer to the total *amount* of a company's **current assets minus current liabilities**.)

[4] For firms with operating cycles of greater than one year, the cutoff for the distinction is the length of their operating cycle, rather than one year. Imagine, e.g., a business built around the Olympics, operating on a two-year cycle—for it, 'current' would stretch to mean 'within two years.'

[5] When I think about the 'long term'—humanity's long-term problems, say—I think 2050, or 2100. Accounting thinks: next year. Adjust your expectations accordingly.

Ordering

Within the major groupings, accounts are listed in continuous order of liquidity—that is, how fast they can be converted into cash or otherwise used or discharged. So, in the Asset category, **Cash is always the first account.** Typically, **Accounts Receivable** will be next—those are accounts owed by customers, and will usually be paid within one year. And long-term assets such as **'Long-Term Investments,'** which are investments in another company's debt and stock, and **'Property Plant and Equipment'** will be further down the Asset side of the Balance Sheet.

Among *current liabilities*, **Accounts Payable** — amounts owed to suppliers for goods and services purchased on account — will typically be the **most liquid**. The various 'payables' are, in a sense, the flip side of receivables: short-term obligations that will be paid off soon, so they sit high in the liabilities section. (One could argue that *Salaries Payable*, which will be paid on the next payday, belongs above *Accounts Payable*—another marginal call.) And long-term loans from banks, etc., will be listed at the bottom of the liabilities section.

Do not lose sleep over the marginal cases. Know the pattern cold: Cash first, receivables high, long-term assets like Property, Plant, and Equipment down low; short-term payables high, long-term debt low. Whether *Supplies* goes above or below *Prepaid Rent* is a coin flip, and no reasonable reader of financial statements will fault you either way.

Finally, in the Shareholders' Equity section, the notion of liquidity doesn't really seem to apply (at least as far as I can tell), but it's conventional to list Common Stock before Retained Earnings. (I.e., **'paid-in capital' before 'earned capital.'**)

Playing it back: partitioning the Balance Sheet

These classifications aren't just formatting—they carve the Balance Sheet into quantities we can compute with. Here's a little problem that plays back everything in this section at once. Suppose ExpoCorp reports **Total Liabilities** of \$50m, **Common Stock** of \$20m, and **Retained Earnings** of \$20m on its Balance Sheet, and that its **current assets** total \$30m. What are its total **long-term assets**? Try it yourself before reading on.

Here's the reasoning: Shareholders' Equity has two components—paid-in capital (*Common Stock*) and earned capital (*Retained Earnings*)—so total Shareholders' Equity is $\$20m + \$20m = \$40m$. By the Accounting Equation, total assets must be $A = L + E = \$50m + \$40m = \$90m$. And since every asset is either current or long-term (the two categories are disjoint and exhaustive), long-term assets must be $\$90m - \$30m = \$60m$. Nothing here required any new formula—just three identities you already own: $A = L + E$; $E = Common\ Stock + Retained\ Earnings$; and the fact that current and long-term assets sum to total assets.

5.1.10 The end-product, out in the wild

With that, we have finally arrived at the ultimate end-product of accounting: the financial statements that companies actually report. Of course, the statements we just produced were for a tiny toy company with a dozen transactions. To see the real thing, go to the SEC's **"EDGAR"** website—the repository where all U.S. publicly-traded companies are required to file their mandatory reports—and look up the **10-K** (annual report) of any large corporation you're curious about. You will find the audited Balance Sheet and Income Statement, in exactly the structures we've been building, along with a frankly crazy amount of other audited information.

5.2 The Truth About Depreciation

5.2.1 The White Lie

In Chapter 4, I introduced *Accounting Depreciation*, which is accounting's way of **apportioning** the expenditure on long-term tangible assets, such as *Equipment*, over time. We accounted for the initial procurement of the *Equipment* at the **purchase price**, as we do for all assets, like so:

Equipment (+A)	$10k	
Cash (-A)		$10k

Purchase of Equipment, 1/1/2020

Then, at the end of the reporting period, as an adjusting entry, we **debited Depreciation Expense**, and, I *said*, **credited Equipment**, just as with other assets:

Depreciation Expense (-S.E.)	$1k	
Equipment (-A)		$1k

End-of-period adj. entry for Depreciation Expense on Equip, 12/31/2020

5.2.2 The More Complicated Truth: Introducing "contra-accounts"

But, it turns out that **I lied (or at least, elided) a little bit.** I did this because I wanted to focus on the **big-picture, unifying logic,** of how all the operating assets were "expensed out" via adjusting entries over time—the "general form" of assets—without getting bogged down in procedural details. And that big-picture concept still holds true. But many companies add an intermediate step in the process.

They do in fact **debit *Depreciation Expense*** as an adjusting entry. But, **instead of directly crediting Equipment**, many companies instead **credit a *separate account* called "Accumulated Depreciation."** As the name indicates, this account tracks the **accumulation** of *Depreciation* on the asset over time.

In particular, this account is called a **contra-Asset** account. This is a **new thing we haven't seen before**. It's called a contra-Asset account, because it "runs against," or **contra** to, an Asset account that it **references**, and **tracks its reductions separately**. Assets are debit accounts; **so contra-Assets are credit accounts.**

Making this more concrete, the **dirty truth** is that many companies instead account for *Depreciation Expenses* like the below:

Depreciation Expense (-S.E.)	$1k	
Accumulated Depreciation (+contra-A)		$1k

End-of-period adj. entry for Depreciation Expense on Equip, 12/31/2020

This means that, technically, the **Equipment** T-account *per se* is unaffected by depreciation expenses, since those reductions are all accounted for separately, in its **contra-Account**. So, after the first reporting period, the T-accounts would be:

Equipment (A)	
$10k	
$10k	

Accumulated Depreciation (-A)	
	$1k
	$1k

Equipment and related Accumulated Depreciation contra-account, 12/31/'20

5.2.3 Reporting at Book Value (or not)

In accounting, we use the jargon **"book value"** to indicate the **net** *accounting value* of an account—that is, net of reductions tracked in its contra-accounts. So, in the example above, that would be $9,000 (i.e., $10,000- $1,000).

Companies have **discretion** over how they **report this** on their Balance Sheet.

- (1.) they can **directly report Equipment at "book value"**, or
- (2.) they can **separately report** Equipment and Accumulated Depreciation right underneath it, with parentheses to indicate that it is a reduction.

In the former case the asset side of the Balance Sheet would look like what we saw earlier, before confessing to the "white lie":

Asset Side of Balance Sheet as of December 31, 2020
Equipment at Book Value

Assets
Cash: $10.4k
Accounts receivable: $3.1k
Supplies: $.6k
Prepaid Rent: $1.8k
Equipment (at Book Value): $9k
Total Assets: $24.9k

In the latter case (2.) it would look like this...:

Asset Side of Balance Sheet as of December 31, 2020
Accumulated Depreciation reported separately

Assets
Cash: $10.4k
Accounts receivable: $3.1k
Supplies: $.6k
Prepaid Rent: $1.8k
Equipment: $10k
Accum. Depreciation: ($1k)
Total Assets: $24.9k

As we noted, we **subtract Accumulated Depreciation** in calculating **total Assets**. So, we end up with the same total asset balance ($24.9k) either way.

5.2.4 Going deeper: What the separate balances can tell us

Why preserve the original Equipment balance and report Accumulated Depreciation separately if subtracting them gives the same net amount?

Consider two companies with the same small net Equipment balance. One may have bought relatively little equipment. The other may own a large stock of equipment that is nearly fully depreciated. The same net figure can therefore sit behind very different operating situations.

The separate balances let a reader begin to distinguish those cases. They do not tell us the machines' exact condition or replacement cost, but they preserve information that a net figure alone would hide. We will also find that keeping reductions in a separate account makes some later entries easier to organize.

5.2.5 Playing it back: book value, with a wrinkle

Let's close the chapter by playing back these new concepts with a worked example—one with a small wrinkle in it. Suppose that on **July 1, 2021**, a company buys *Equipment*, paying $24k in cash:

Equipment (+A)	$24k	
Cash (-A)		$24k

Purchase of Equipment, 7/1/2021

Management forecasts that it will use the equipment for **6 years**, with zero salvage value at the end. So, under straight-line depreciation, we apportion the cost evenly over the useful life: $\frac{\$24k-\$0}{6\ years} = \$4k$ per year. Now, the question: what is the **book value** of the *Equipment* on the Balance Sheet reported as of **December 31, 2022**? Work it out before reading on.

The tempting quick answer is $\$24k - \$4k - \$4k = \$16k$—two year-ends have passed, so two years of depreciation, right? But look again at the purchase date. We bought the equipment on *July 1*, 2021—halfway through that first reporting period. So, by December 31, 2021, we had only used the equipment for *six months*, and should only recognize *half* a year's worth of depreciation: $2k. (This is exactly the same apportionment logic we used for the rental contract in Chapter 4, when we expensed only 6 of the 24 months we had paid for. A footnote in that chapter even worked this exact half-year twist on a piece of equipment—if you caught it there, this wrinkle was no wrinkle at all.) Then fiscal year 2022 is a full year of use: $4k. So the *Accumulated Depreciation* contra-account stands at $6k as of December 31, 2022, and the T-accounts look like this:

Equipment (A)		**Accumulated Depreciation (-A)**	
$24k			$2k
			$4k
$24k			**$6k**

Equipment and related contra-account, after adjusting entries, 12/31/2022

The book value—the balance of the underlying account, *minus* the balance of its contra-account—is therefore $\$24k - \$6k = \$18k$. The company could report "Equipment (at book value): $18k" as a single line on its Balance Sheet, or report "Equipment: $24k" with "*less* Accumulated Depreciation: ($6k)" right underneath. Either way, the same $18k ends up in Total Assets. If you got $18k on your own—including the half-year wrinkle—then you have everything you need from this chapter: the mechanics are in place, and we can get back to ideas.

6

Introduction to Ratio Analysis

To date, the primary focus of this book has been understanding **how** a company's **transactions** are **recorded** in its accounts and, thus, produce its **Financial Statements**. But, in the vast majority of business careers, you will more likely **start with the Financial Statement numbers**, and then **use them** to **analyze, assess, judge, and understand** the company's operations, position, performance, and value. Understanding what goes in to a company's financial-statement numbers is a key prerequisite for this kind of analysis. So these are not separate endeavors. But this chapter, of everything we've done so far, will map most closely to what you would actually do in a career as a businessperson, analyst, consultant, or investor.

6.1 The basic idea

At its core, the basis for 'ratio analysis' or 'financial statement analysis'[1] is a very basic idea: **No one accounting number, by itself, will tell you whether a company is performing well or poorly**. Suppose that you are told that a company has earned **$1m in net income** this year. Is this **good or bad**? Obviously, **it depends.** If that company were a small, new startup, in its second year of operations, and only had $1m of capital invested in it, this would clearly be very good performance—the investors would already be earning 100% of their initial investment back in available profits already. If, however, the company were large and mature, with invested capital of $10bn, and had

[1] The terms are used essentially interchangeably.

previously had years where it had earned in excess of $2bn, then, clearly, $1m in net income would be disappointing in light of that.

In other words, in judging whether a number indicates good or bad performance, **it depends**. There are a couple of major categories of things that it *depends on*.[2]

- First, it depends on **how much its investors/owners have contributed in financing**. Owners invest in companies largely because they hope to get a good **return on** their investment. So, if you invest $1m in a company, you would clearly hope to accrue higher total profits than you would expect for a company in which you invested $10k.
- Second, it depends on the total **size** of the company, and the total assets it has employed in its operations. (Note that, by our core accounting identities, this is closely linked to the first item, since $A = L + E$. We'll say more about that later.)
  - So, some of the most basic **ratios** will simply involve **scaling** a performance measure (like NI), by measures of total investment or size.
- And, third, and most broadly, it depends on **what the question is**—or, put differently, what particular **kind of performance** we are interested in.
  - E.g., the **performance metrics** that **shareholders** are interested in are naturally **different** from the ones that **creditors** are interested in.
  - Or, even more specifically than that, we might be interested in analyzing, in greater detail, one specific **component of or contributor to** bottom-line performance. E.g., how your 'asset efficiency' or 'margin' **contributes** to your profitability. (We'll say more about this soon.)

These considerations will determine the **ratios themselves**, which we will discuss in greater detail, below. But even that will not be the end of a real ratio analysis / financial statement analysis project. Typically, we'll also want to compare those ratios to the company's **industry peers and competitors**, as relevant "benchmarks," and consider the company's relative performance and **growth** over time. (See Section 6.5.) But for now, we'll focus on understanding the ratios themselves.

Some students get intimidated when they reach this material and see so many new formulas. But all we're doing in ratio analysis is **throwing a division symbol between two financial-statement numbers that we have already discussed in depth, and interpreting the result**. So, these ratios really aren't new.

[2] Here's one more thing it can depend on—something we won't formalize into a ratio in this chapter, but which belongs in the back of your mind: *how* the company generated that net income. A company might report $1m in net income while burning through its cash and depleting its resources to do so—in which case, next year might look very different. Recall the discussion from Chapter 1 of using the **Cash Flow Statement** as a 'check' on accrual profitability.

We'll talk about the most well-known, conventional ratios used in practice. But these ratios were **not** handed down from Mt. Olympus, by law, or even from **GAAP**. These ratios were **invented by analysts** who were trying to understand companies. And if you truly understand the accounting we've studied so far, you could invent some ratios of your own to answer the questions you are interested in answering.

6.2 Profitability Ratios

6.2.1 Return on Equity:

A key, central performance measure is **Return on Equity (ROE)**. In fact, if you forced me to pick just *one* ratio for judging how well a company is doing—a request that would make me sad, since the whole point of this chapter is that no single number tells you everything—I would pick ROE.

$$Return\ on\ Equity = \frac{Net\ Income}{Shareholders'\ Equity} = \frac{NI}{E}$$

You might think that this is new material for this book. But is it really? All we are doing here is throwing a **division symbol in between** two of the **most central accounting numbers** in the book: **Net Income** and **Shareholders' Equity**.

Why is this simple measure so common and popular? Go back to our Chapter 1 foundations. **Net Income** is the proverbial 'bottom line' profit—the accounting system's way of estimating the amount of value (wealth) that has been realized for the company's **owners** over the reporting period. **Shareholders' Equity** is the total amount that shareholders have contributed to the company's financing, through both direct investment (Common Stock) and 'as-if' reinvestment (Retained Earnings). So we could interpret this ratio verbally as:

> Return on Equity: The profit generated for shareholders, scaled by the total financing they have contributed.

Before racing ahead, let's pause to interrogate both the numerator and the denominator of this ratio—each one attracts its own misconceptions. Start with the numerator. What *is* **Net Income**, conceptually? At a reductive level, it's just revenues minus expenses, measured according to the accounting rules—that's the mechanical definition you've been applying for several chapters now. But consider some candidate *conceptual* definitions. Is net income: (a.) the increase in the company's cash; (b.) accounting's way of measuring how much value has been realized for the owners, via the company's operations, over the period; (c.) the total wealth of the company's shareholders; or (d.) the increase in the company's stock price?

The answer is (b.). But it pays to see exactly why the others fail. Net income is certainly not the increase in the company's cash: we are doing accrual accounting, not cash accounting—and even under a pure cash basis, a company can increase its cash simply by issuing *Common Stock*. That brings in cash, but it isn't profit; it's new investment from owners. Nor is net income the total wealth of the shareholders: that would be a 'stock' measure, of how much they're worth at a point in time, whereas net income is a 'flow' over a period. And the stock-price increase is something else entirely: that's what you see on Yahoo Finance, and it reflects market expectations, risk, and sentiment—things well beyond what the income statement directly reports.

One more subtlety hides in that phrase 'for the owners.' Recall that **interest** paid to lenders is an **expense**: it reduces net income, because it's money that went to the creditors, not the owners. **Dividends**, on the other hand, are *not* an expense, and do not reduce net income: they are just a distribution of what was already earned for the owners—they don't change how much was earned in the first place. So net income is, by construction, what's left over *after* the lenders have been paid. It is measured from the owners' perspective, which is precisely what makes it the right numerator for ROE.

Now the denominator: what is ('book') **Shareholders' Equity**, conceptually? Here's a tempting answer: "how much value has been realized for the owners, via the company's operations, over the reporting period." It's a thoughtful answer, on the right conceptual wavelength—and it's wrong. Shareholders' Equity lives on the Balance Sheet, which means it is a stock measure at a point in time; anything measured "over the reporting period" is, by definition, a flow. In fact, that tempting answer is just our definition of *net income*, restated in slightly different words. Now, you might push back with the galaxy-brain version: "But net income flows into Retained Earnings at the end of each period, and Retained Earnings is part of equity—so isn't equity kind of the accumulated value realized for the owners?" And that's not crazy. Net income does indeed 'live on' in Retained Earnings. But the amount realized in any one reporting period is just one year's slice; the equity balance is the *cumulative* result of owner financing and reinvested profits over the entire life of the company. Nor is Shareholders' Equity "the amount of cash the company holds for the owners"—cash is an *asset*, on the other side of The Accounting Equation, and (as we'll see at the end of this chapter, in the context of bank regulation) confusing the two sides leads even Very Serious People into serious errors. The best conceptual definition is the one we gave above: the amount of financing that owners have contributed to the business, including their 'as-if' reinvestment via Retained Earnings.

Now it should be clear why ROE is such a central performance measure. For reasons we discussed in Chapter 1, the fields of finance and economics have traditionally assumed that companies are run as if for the benefit of the shareholders. The shareholders are the **owners** of the firm, and have the ultimate levers of control. If the company

does not perform well from their perspective, they can (indirectly) replace the board and/or executives. More generally, as owners, they use various **'governance mechanisms'** to protect their interests. ROE measures how well the company is doing with their money, from their perspective.[3]

ROE answers a question from the owners' perspective. That does not make it the answer to every question about a company. A creditor wants to know whether the company can meet its obligations; an employee or a community may care about other outcomes. Before choosing a performance measure, identify whose question you are trying to answer.

6.2.2 Important calculation detail: Use the *average* value of B/S scaling factors

For all of these ratios, when we **scale** an **income-statement (flow)** measure (like NI) by a **balance-sheet (stock)** number (such as shareholders' equity), we use the **average value** of the **B/S (stock) account** over the same period and the last.

The reason for this is that income-statement items measure performance *over a period*, but balance-sheet (stock) items measure accounts at a particular point in time—specifically, the end of the reporting period. With ROE, we are seeking to answer the question, "How well did the company do generating income over the last period, with the financing from shareholders that it tied up and employed over that period?" So, it would not make sense to use only *Shareholders' Equity* at the *end of the period*.

In theory, we would ideally want the average amount over the entire period (that is, the weighted average across every day). But, as external users of financial statements, we only observe Shareholders' Equity at the end of each reporting period. So we use the average of the value from the same reporting period and the previous.

So, to get the calculation technically correct, it would actually be:

$$Return\ on\ Equity_t = \frac{Net\ Income_t}{\frac{(Shareholders'\ Equity_t + Shareholders'\ Equity_{t-1})}{2}} \tag{6.1}$$

[3] Students often wonder why we use the **book value** of shareholders' equity rather than the **market value** of the shares in this ratio. Note that the market value of stocks fluctuates with investors' assessments of the performance of the company and their sentiment about its future prospects. So, for example, if a company's net income kept unexpectedly improving year to year, the company's market value of equity would likely increase in response. Thus, if we put the market value of equity in the denominator of this ratio, this company would be 'punished' for its excellent performance under this measure. Since the market value of equity fluctuates *in response to* companies' performance, it would usually be inappropriate to use it as a scaling factor in measuring the performance of companies. More fundamentally, book Shareholders' Equity *is what it is*—the **financing the owners have *contributed***. The market value of equity is something different. If the price of the stock trading in the "secondary market" increases, that's usually a good thing from the firm's perspective. But, stock trades *between* traders are not the same as *issuances* of *Common Stock* from the company. In the latter, the company receives cash financing. In the former, it does not. So, if our question is 'How well has the company performed, with the money its shareholders have put in?,' then we want **book** equity, not the market value of equity.

... where the 't' subscript would indicate the value from the Financial Statement for the same period, and 't-1' indicates the value from the FS from the previous period.

As you can see, this makes things a bit messier and more complicated-looking, **needlessly, given the simplicity of the idea**. In this book, we're focused on understanding and internalizing the concepts, in the most clean and straightforward way possible. So, throughout, I will ***represent*** it simply as $ROE = \frac{NI}{E}$ and expect you to **remember the rule** when working this book's practice problems. (This is consistent with practice in the outside world—most other texts and references will ***represent*** the ratios without additionally specifying the measurement periods each time.)

When you see a flow divided by a stock, ask whether the stock measure represents the resources or financing used over the period. That is the reason for averaging here.

6.2.3 Return on Assets:

Return on Assets (ROA) is another extremely common ratio.

$$Return\ on\ Assets = \frac{Net\ Income}{Total\ Assets} = \frac{NI}{A}$$

Once again, this 'new' ratio is just a matter of **throwing a division symbol between** two very familiar accounting numbers. As you can see, this is the same as the formula for ROE, except we replace Shareholders' Equity in the denominator with Total Assets.[4] Expressed verbally, this ratio is: "The total profitability of the company, scaled by the total assets the company uses to generate that profitability." At the simplest level, total assets are a rough proxy for the *size* of a company—so this ratio answers the most basic of the 'it depends' questions from the start of this chapter: given how big this company is, how much profit does it generate? In contrast to ROE, its **denominator** does not distinguish whether those assets (A) were financed by equity (E) or liabilities (L). Its numerator still includes interest expense, so this version of ROA is not independent of financing choices. A little algebra can be helpful for understanding the difference between ROA and ROE. Note the following identity:

$$ROE = \frac{NI}{E} = (\frac{NI}{A}) \times (\frac{A}{E}) = ROA \times (\frac{A}{E}) \tag{6.2}$$

As we can see, the multiplicative factor linking ROA to ROE is A/E, another extremely common, important ratio, called **'leverage.'** Given the fundamental identity of accounting, $A = L + E$, we can also write leverage as follows:

$$Leverage = \frac{A}{E} = \frac{L+E}{E} = \frac{L}{E} + 1$$

[4](As before, for calculating in practice use the average total assets, from the same period and last.)

This formula shows that leverage is essentially just a measure of how much the company has financed itself with **credit/liabilities relative to equity**. That notion is sometimes referred to as the company's "capital structure." The common term **"leverage"** denotes what we can see in Equation 6.2.3, above. Holding 'operational performance' (ROA) fixed, with higher credit financing, the shareholders of the company can "lever up" their return on their own investments.

Stated differently, the core difference is that *ROE* is a measure of performance that is **sensitive to leverage** (i.e., capital structure), while **ROA is less so**.[5] This can be an advantage or disadvantage, **depending on the question**. For example, if we simply wanted to compare how two different companies were performing operationally—how well the company was operating its asset base to generate revenues exceeding its costs—you might prefer to use return on assets (ROA). However, if you were yourself a shareholder in the company, interested in how well the management has performed with your money, on your behalf, ROE would be more relevant.

6.3 The Drivers of ROE: The DuPont Decomposition

The DuPont Decomposition is a famous and useful way of breaking ROE down into three underlying components to examine different **drivers** of the company's performance: **profit margin** (sometimes called "return on sales"), **asset turnover** (sometimes called "asset efficiency" or "asset utilization"), and **leverage** (discussed above). This decomposition is just an **algebraic identity**. We just take the definition of ROE (NI/E) and multiply it by various terms that cancel each other out, to get three separate **components**.

$$ROE = \frac{NI}{E} = (\frac{NI}{S}) \times (\frac{S}{A}) \times (\frac{A}{E})$$

Note how the 'S' (for sales) cancels out in the denominator and numerator of the first and second terms, respectively, and the 'A' (for Assets) cancels out in the second and third. That is, we have **decomposed** ROE into three separate ratios:

- $\frac{NI}{S}$: Profit Margin
- $\frac{S}{A}$: Asset Turnover
- $\frac{A}{E}$: Leverage

[5] Strictly speaking, if we wanted a measure of operating performance insensitive to capital structure, we would want to also take the debt-financing costs (interest expense) out of the profitability numerator—i.e., use **EBIT**: **E**arnings **B**efore **I**nterest and **T**axes—profit measured before the costs of financing, and before the taxman's cut. EBIT will quietly do real work for us in the leverage tables later in this chapter, precisely because it is the same number no matter how the firm happens to be financed. Beyond that, I defer the details for this introductory text.

Where did that middle expression come from? The trick is that we multiplied ROE by 1, twice: once written as $\frac{S}{S}$, and once as $\frac{A}{A}$. Multiplying by 1 changes nothing about the *value*—algebraically, we are still looking at plain old $\frac{NI}{E}$—but it changes everything about the *form*. Once you see it, it's obvious; the first time you encounter it, it can feel like a magic trick.

And what is the *point* of the exercise? It is **not** that we need all three terms to *calculate* ROE—if all you want is the number, $\frac{NI}{E}$ gets you there directly. The point is that we now have three distinct *drivers* of overall performance, each with its own economic meaning, which we can analyze separately in order to understand the company better. (You are not going to get paid well as a financial analyst if the only ratio you ever look at is ROE.) The subsections below take each one in turn.

6.3.1 Profit margin

NI/S is referred to as **"profit margin."** Here,'S' stands for Sales, which is just the word that is used in the context of this ratio for total Revenues. (I'm not changing the concept—just the letter. As we saw with 'credit financing' in Chapter 1, accounting has an annoying habit of using several different names for one basic thing.) 'Margin' is a term that is used in everyday speech, so you may already be familiar with this. In words, this ratio is the company's **bottom-line** profits relative to its **top-line total sales (i.e., revenues)**. Thus it is a measure of how much of the company's gross sales it manages to keep as 'take-home' profitability, after all expenses are accounted for. The higher a firm's profit margin, the more *net income* it gets per each dollar of sales.

Here's a quick worked example. Suppose that a company's Income Statement looks like this...:

Income Statement for the Fiscal Year Ending Dec. 31, 2021

Revenue	\$12.3k
Salaries Expense	(\$5.6k)
Supplies Expense	(\$0.8k)
Rent Expense	(\$1.5k)
Depreciation Expense	(\$2.0k)
Interest Expense	(\$0.3k)
Net Income	**\$2.1k**

Then its profit margin would be $\frac{\$2.1k}{\$12.3k} \approx 17\%$. Out of the top-line sales, about 17 cents on the dollar make it all the way down to the bottom line; the rest gets nibbled away by the expenses in between.

Another way to see the meaning of this ratio is with a little algebra. Net income is

revenues minus expenses—or, in our ratio-analysis notation, $NI = S - Expenses$. Divide both sides by S:

$$\frac{NI}{S} = \frac{S - Expenses}{S} = 1 - \frac{Expenses}{S}$$

So profit margin is essentially 1 minus the company's "expense ratio"—an inverse measure of how heavy its expenses are, relative to its sales.

It is often described as a measure of "how well the company is **managing the income statement**"—i.e., how well it is doing managing **all of the expenses that come between the top line and the bottom line** of the income statement. Profit margins famously vary across industries. For example, in highly **competitive "commodity" businesses** such as grocers, profit margins may be extremely **low**, perhaps 5% or less. In **luxury-goods** businesses and **subscription software**, profit margins are typically much higher.

6.3.2 Asset Turnover

S/A is referred to as **"asset turnover"** (sometimes also called 'asset efficiency' or just 'efficiency'). It is equal to total sales divided by total assets. Thus it is a measure of how effectively the company is **employing its resources (its assets) to produce marketable goods**—hence why it is sometimes also referred to as 'asset efficiency.'

You can interpret asset turnover as asking: for each dollar of assets the company holds, how many dollars of sales does it generate over the period? Think back to the simple model of a business from Chapter 1: a business uses financing to acquire **resources** (assets), which it uses to produce goods and services for **sale** to customers. Asset turnover measures how well the company is doing at exactly that step. It's a measure of how 'tight a ship' the company runs: how much saleable output it manages to produce, with how little in assets.

It is often described as a measure of how well the company is **managing the balance sheet**—i.e. how effectively it is employing the total assets it holds. Asset **turnover also varies widely across industries**. **Grocery stores and discount retailers** have thin margins (as referenced above), but **high turnover**. In contrast, a **luxury** jewelry retailer, or Tesla, likely has high margins but **relatively lower turnover**. This is not a coincidence. In general, industries with **high margins** tend to have **low turnovers**, and industries with **low margins** tend to have **high turnovers**.

To understand this fact note the following identities:

$$Margin \times Turnover = \frac{NI}{S} \times \frac{S}{A} = \frac{NI}{A} = ROA = \frac{NI}{(E+L)}$$

This shows that profit margin times asset turnover equals ROA. And, given the core

accounting identity, $A = L + E$, a company's total asset base is also equal to its total capital—the total financing it has received from both owners and creditors.

6.3.3 Going deeper: Why margin and turnover can move in opposite directions

The identity tells us that margin times asset turnover equals ROA. It does not, by itself, tell us why businesses choose different combinations.

Here is an economic intuition. If one business could reliably earn much more on its capital than another, new investment and competition might be drawn toward it. That pressure can make a low-margin business viable only if it generates a large volume of sales relative to its assets. A business with a higher margin may sustain itself with fewer sales per dollar of assets.

For a numerical illustration, imagine one company with a 2% margin and total-asset turnover of 12, and another with a 48% margin and turnover of 0.5. Each has an ROA of 24%. These are assumed ratios, not measurements of particular industries. They show how different combinations can produce the same result.

Whether competition actually brings returns together is a further question. Risk, barriers to entry, and differences between recorded assets and economic resources complicate the comparison. Keep the identity separate from the hypothesis: the multiplication is exact; the explanation of industry patterns requires judgment.

6.3.4 Leverage

A/E is referred to as "leverage." Of the three components of the DuPont decomposition, this one is the subtlest and the deepest—it is where much of the real-world drama lives, as the end of this chapter will show. Leverage can **increase the return on equity** when the return generated with borrowed funds exceeds their cost. It does not automatically improve shareholders' returns. But it also increases the riskiness of the company (since leverage multiplies both profits and losses) and **increases the risk that the company will go bankrupt**.

Before explaining why leverage means risk, let's distinguish between two categories of risk that a company faces. Some risks are purely **operational**: uncertainty about demand, technology, and competition—the risk that the new market you bet the company on (some speculative product line, say) never takes off, or that a rival eats your lunch. Those risks come from *what business you're in.* Leverage is about a different category, **financial risk**, which comes not from what you sell, but from how you *finance*

the business—your capital structure.

Here is the easiest way to understand this: By definition, from Chapter 1, companies **have to pay their debts** otherwise, they end up **in default**, and creditors can seize collateral, etc. All else equal, the more debt a company has, the higher its interest and principal payments, the higher the risk that it won't be able to make those payments in the future. In contrast, **dividends are discretionary**. A company without borrowing avoids default on loan payments. But it can still fail or become insolvent because of operating losses or other obligations, such as wages, supplier bills, or legal claims.

There's a complementary intuition on the upside. Suppose you're the sole owner of a company, and you want more financing. You have two broad options. If you issue more equity—sell additional shares of *Common Stock*—the share issuance itself adds no required interest or principal payments. But you are no longer the sole owner: you now have partners, with whom you must share all future profits and dividends. You have **diluted** your ownership stake. If, instead, you borrow, you avoid dilution—you keep a larger fraction of the upside for yourself—but you take on fixed interest and principal obligations that raise the downside risk. That's the tradeoff: equity financing means less financial risk, but more dilution; debt financing, the reverse.

You can see all of this compressed in the algebra from earlier: $ROE = ROA \times \frac{A}{E} = ROA \times Leverage$. Leverage is literally the scalar sitting between ROA and ROE—the factor that scales the company's operational performance up (or down) into the owners' return. For a fixed ROA and positive equity, higher leverage means higher ROE when ROA is positive, and more negative ROE when ROA is negative. But borrowing can itself change ROA through interest expense, so the identity does not by itself predict the effect of taking on more debt. Leverage 'levers up' the good *and* the bad—which, incidentally, is where the word comes from.

So, in short, leverage can mean **higher returns** for shareholders, but it also means **more financial risk**, especially risk of default and bankruptcy. Thus, on average, equity investors tend to like and to push for higher leverage levels, while debtholders tend to push for lower leverage levels.[6] Leverage ratios also tend to vary across **industries** and the **life cycles** of firms. **Early-stage** firms with uncertain prospects tend to be financed by **more equity** (that is, have lower leverage ratios), since their future cash flows are less certain and thus they would have a greater probability of bankruptcy if they were to rely heavily on debt financing. More **'mature' firms with more stable cash flows** are more likely to choose **higher leverage ratios** (that is, more debt relative to equity),

[6]You may have read in the financial press about "activist investors"—equity owners who buy up significant stakes in order to influence management to change policies in ways that they believe will increase the value of their shares. One of the major areas of focus for activist investors is leverage. Activist investors own the company's stock, and as such, they usually prefer higher leverage levels. They often encourage companies to issue more debt and pay out more cash to shareholders. Academic research has mostly suggested that activist investors are on average good for **shareholders**. But they remain controversial, partially because higher leverage levels increase riskiness.

because they have more certainty about their future cash flows and ability to service the debt. Historically, firms in the **financial industry** have had **very high leverage ratios**, and this became a topic of **controversy and new regulation** after the **financial crisis**, as leverage was seen to increase the **systemic riskiness** of the banks.

Finally, we can use The Accounting Equation to rewrite Leverage as:

$$Leverage = \frac{A}{E} = \frac{L+E}{E} = \frac{L}{E} + 1 = Debt\ to\ Equity + 1$$

The ratio $\frac{L}{E}$ is often given its own name, and called the "**Debt-to-Equity** Ratio." But note again that it encodes the exact same information and variation as **leverage**—just shifted by 1. So, everything that is true about leverage also applies to debt-to-equity. Other standard textbooks (such as Spiceland et al.'s *Financial Accounting*) give the debt-to-equity ratio some dedicated discussion. But I will not, because it is substantively the same thing as leverage, but leverage is the more conventional, commonly-used term in the outside world.

6.4 Credit Risk Ratios

In the above, we were mostly focused on understanding *performance* from a *shareholder's* perspective. But, the ratios and measures that are relevant also depend upon the question we are interested in. Financial *creditors* (that is, mostly debtholders) have a different set of concerns and focuses than shareholders. In particular, debtholders, by definition, stand to gain only their contracted interest and principal payments—they do not get any additional 'upside' beyond that. So, bluntly, they just want to know whether or not companies will make those payments.

There are a number of ratios that credit analysts have developed to measure companies' ability to pay their obligations. First a bit of **jargon**: We use the term **'solvency'** to indicate having assets that are worth more than liabilities—that is, positive **net worth**, over the **long run**. And we use the term **'liquidity,'** in this context, to mean the company's ability to **make specific cash payments coming due in the short term**. In business, **"cash is king"**—debtholders must be paid in cash, and so even companies that are highly profitable on *accrual measures* need to worry about converting their economic value into cold, hard cash. Since creditors do not share in profits beyond their contracted payments, **solvency and liquidity** are central to their analysis.

One of the measures we have already discussed, **leverage**, is somewhat relevant to assessing solvency. All else equal, the higher a company's *leverage*, the higher its **default and bankruptcy risk**. So, lenders pay attention to that ratio, and frequently include it in their **debt covenants (jargon)**. These are **terms in debt contracts** that stipulate certain penalties on firms that violate them. So, e.g., if a firm's **debt contract**

included a **covenant** around its **leverage ratio**, it might have to repay the principal early (before the original maturity date), if its leverage ratio grew to exceed the specified threshold.

For now, I'll cover just one ***liquidity measure***, the **current ratio**.

6.4.1 The current ratio

A simple measure of a company's ability to pay what it owes is the **current ratio**.

$$Current\ Ratio = \frac{Current\ Assets}{Current\ Liabilities}$$

Recall that **current** generally refers to the next year, or the normal operating cycle if longer. A current ratio of 1 means that the recorded amounts of current assets and current liabilities are equal. It does not guarantee that the company can pay every bill on time: some assets may be inventory or prepayments rather than readily available cash, and the timing of receipts and payments matters. A ratio below 1 deserves investigation, but can be sustainable for a business that collects cash quickly and pays suppliers later. A higher ratio can provide a larger cushion, all else equal; it can also reflect slow-moving inventory or uncollected receivables. Interpret the ratio alongside the business's operating cycle and cash flows.

6.5 Using Ratio Analysis: Benchmarking, and Comparative Analysis

In practice, even after we have calculated these scaled ratios, we're not done. In almost every analysis, we would start by calculating these ratios, but then do a bit more to **"benchmark"** them. Remember how we began this chapter discussing the factors that *it* (i.e., assessing a company's net income) **"depends on"**? Let's add a couple more:

- It depends on how well that company's **peers and competitors** are performing. This matters for two reasons: First, because the company will have to successfully **compete** with them in order to **survive**. And, second, because they provide another **benchmark** for the company, for how well, as a firm in the same line of business, it *could be doing* (i.e., its **'counterfactual'**).
  - So, a typical ratio analysis project would involve comparing peer firms.
- It depends on that company's **own history**, that is, how well it has, itself, done in previous years. This matters for two reasons: First, the company's own **past performance** provides another relevant **benchmark** for assessing its current perfor-

mance. And, second, because investors and managers want to see **growth** over time.

- So, a typical ratio analysis project would, even further, look at how those ratios and differences have *changed over time, and relative to those peer firms*. Is our firm growing, and, in particular, growing faster than our competitors? Are our margins staying flat while theirs are improving? Etc.

There is no hard science of ratio analysis. It is simply a matter of understanding what the ratios mean, and using your independent knowledge and judgment. The ratios are **tools for *us*** to use—not vice versa!

For example, suppose that you were comparing your own company, with an ROE of 20%, to a competitor with an ROE of 30%. If you were to discover that the two companies had the same asset turnover, but your company's profit margin were lower, this would suggest (a.) where the blame lay for your company's overall poorer performance, and (b.) a possible prescription. It might mean that your company should focus on *cutting out expenses* such as overhead (SG&A) to achieve a similar margin as your competitor. (This is where you 'bring in McKinsey,' as the joke goes—helping firms cut costs is the consulting industry's bread and butter.) After all, your company's competitor has shown that it is feasible to achieve that margin in your industry. The diagnosis suggests the prescription. In contrast, if you were to find that the two companies were matched on profit margin, but your company had a lower asset turnover, this might suggest you should work to improve your efficiency, such as by improving operations management, or upgrading your factories, etc.

This decomposition can also help you understand trends in a company's overall profitability **over time**. For example, if a company's profit **margin were increasing** over several years, this might suggest that the company's **competitive position** was improving, i.e., that the company was increasingly able to **command premium prices** on its brand. Or, if a company's asset turnover was decreasing over time, this might indicate that it was losing discipline and efficiency, that its **scope** was increasing too much (it had more assets than it could employ efficiently), or that its operations had become less well **organized**. This would inform an executive's decision about steps to take in response. For example, if **asset turnover** was the culprit in poor performance, the company might require consultation from "ops" (operations) experts.

6.6 Leverage and risk

In the text above, I offered a verbal explanation of the "easiest way to understand" why "leverage means risk": Our company *has to* pay its debts, even when times are bad—or end up in bankruptcy—while dividends paid to owners are discretionary. Therefore,

the more we finance ourselves through debt rather than equity, the greater the risk this imposes upon us, all else equal. So, why do companies issue any debt? Because, in good times, it helps owners "lever up" their overall rate of return (ROE).

In the below, I provide an example that illustrates in a more general and systematic way how **leverage** (A/E) can help a company **increase its Return on Equity** (ROE) on **average**, but comes at the cost of extra risk. (In particular, extra systematic risk, which is closely related to *beta*—a key concept you'll meet if you go on to study finance.) Here's the basic idea: We imagine a simple company and we hold its **operating performance fixed** (that is, its revenues and all of its expenses before its interest expense) and then **toggle its capital structure** (that is, its leverage, or how much it finances itself via liabilities vs. shareholders' equity), **between Columns 1 and 2**.

Then, we track how the differences between these capital structures play out, in terms of ROE, in two different states of the world: A **good** state of the world (**Scenario A**), in which the company earns **$100 in revenue**, vs. a **bad** state of the world (**Scenario B**, e.g., a macroeconomic recession), in which the company earns only **$90 in revenue**.

In order to illustrate the effect of **leverage** *per se* on ROE and risk, we **hold everything else fixed**. Suppose that all other expenses—that is, all operating / non-financial costs — are held fixed, at $75. Suppose that in both cases, the company's **interest rate on its debt is 20%.** (Note that this is a conservative assumption for our purposes—we would expect a company's interest rate to increase with its leverage, all else equal.) And suppose that the **tax expense** on its Earnings Before Taxes (EBT) is 21%—the assumed tax rate in this example.[7] (For this simplified example, assume no tax payment or tax benefit in the loss state.) Now watch how things play out:

	(1.) Balanced ($50 Debt / $50 Equity)	**(2.) Levered ($90 Debt / $10 Equity)**	**Explanation**
Revenue	$100	$100	By assumption
-(All other expenses)	-$75	-$75	By assumption
EBIT	$25	$25	By construction
-Interest expense	-$10	-$18	20% x Debt
EBT	$15	$7	EBIT - Interest
-Taxes	-$3.15	-$1.47	21% x EBT
NI	$11.85	$5.53	EBT - Taxes
ROE	23.7%	55.3%	ROE = (NI / Equity)

Scenario (A): The good state of the world:

[7] The federal corporate rate has stood at 21% since the 2017 tax reform took effect; for roughly three decades before that, it was 35%. The level matters for much of what follows: the higher the rate, the more valuable any deduction.

	($50 Debt / $50 Equity)	**($90 Debt / $10 Equity)**	
Revenue	$90	$90	By assumption
-(All other expenses)	-$75	-$75	By assumption
EBIT	$15	$15	By construction
-Interest expense	-$10	-$18	20% x Debt
EBT	$5	-$3	EBIT - Interest
-Taxes	-$1.05	-$0	21%. No tax if loss
NI	$3.95	-$3	EBT - Taxes
ROE	7.9%	-30%	ROE = (NI / Equity)

Scenario (B): The bad state of the world:

Note how in **Scenario A**, the highly-levered capital structure yields a far higher ROE—more than double that of the balanced capital structure—something that would be very pleasing to the company's shareholders. But notice how starkly things reverse in **Scenario B**. In that case, where the company's revenues decline by only 10%, the highly levered capital structure sends the company toward bankruptcy. It does not even earn enough to cover the interest expense on its debt, and its ROE is -30%. Meanwhile, the company with a balanced capital structure earns a healthy 7.9% ROE.

In general, **debt is good when times are good**: it allows equity investors (including homeowners, who have equity in their homes) to "lever up" their rate of return. But when **times are bad, debt makes them even worse**: Debt payments are fixed and non-discretionary, so companies have to pay them off even when it hurts them the most. In contrast, dividend payments to shareholders are discretionary, so when companies are in financial straits, they can simply cut their dividends. As such, equity financing does not impose the same risk on a company.

6.6.1 Leverage at home: houses and mortgages

Everything above was framed in terms of corporations. But the very same arithmetic governs what will likely be the biggest financial decision of your personal life: buying a home. Recall the mortgage example from Chapter 1: a **mortgage** is simply credit financing for the purchase of a house, collateralized by the house itself, and the value of your home above what you still owe the bank is your 'home equity.' Housing is a wonderful domain for understanding leverage, because the leverage ratios involved are large, the stakes are personal—and, as the next section will show, housing leverage sits at the center of financial crises.

Suppose you want to buy a house in Los Angeles, for $1 million. Is that the amount of cash you'll take out of your bank account? Almost certainly not (not least because

you almost certainly don't have it). Like nearly every homebuyer, you'll make a **down payment** and finance the rest with a mortgage. A common rule of thumb is that a 20% down payment gets you a good interest rate. So you put in $200,000 of your own money, and the bank lends you the remaining $800,000. Notice what we've just built: a little personal balance sheet. The house is the asset (A = \$1m); the mortgage is the liability (L = \$800k); and your down payment is your equity (E = \$200k). Your personal leverage ratio is $\frac{A}{E} = \frac{\$1m}{\$200k} = 5$. The Accounting Equation does personal finance just as well as corporate finance.

Now suppose the housing market has a great year, and your house appreciates by 30%, to $1.3 million. Here is a claim you will hear at practically every dinner party attended by thirty-somethings (homeowners in their thirties are obsessed with the housing market, and check Zillow constantly; enjoy your twenties): "My house went up 30%, so my investment went up 30%." Is that right? Let's do the arithmetic carefully. To make things fully concrete, suppose you actually sell the house for $1.3 million—no mere 'paper' gains. For this illustration, set aside interest, taxes, maintenance, transaction costs, and any principal payments. You receive $1.3 million in cash. But the sale doesn't make the mortgage disappear: you still owe the bank its $800,000. So you walk away with \$1.3m − \$800k = \$500k. You started with $200,000 of your own money; you end with $500,000. That's a gain of $300,000 on a $200,000 investment:

$$\textit{Return on Investment} = \frac{\textit{Ending Equity} - \textit{Beginning Equity}}{\textit{Beginning Equity}} = \frac{\$500k - \$200k}{\$200k} = 150\%$$

The *asset* went up by 30%—but your *equity* went up by 150%. That's because you were levered 5-to-1, and $5 \times 30\% = 150\%$. Leverage 'levered up' your return, exactly as in the corporate example above.[8]

Now run the film in reverse. Suppose instead that you bought at the top of the market, and the house *falls* by 30%, to $700,000. Is your return -30%? No—that's the return on the *asset*. If you have to sell at $700,000, you still owe the bank $800,000. Just to walk away clean, you'd have to come up with an extra $100,000 to plug the gap between the sale price and the remaining mortgage balance. Your equity has gone from $+\$200k$ to $-\$100k$: a return of -150%. This example assumes the borrower must cover the full shortfall; actual obligations depend on the loan and applicable law. Same 5-to-1 leverage; same multiplication; opposite sign.

And here's the part that should genuinely worry you: these risks are **correlated**.

[8] You might ask: if you *don't* sell, are the gains 'real'? That is partly a philosophical question. Economically, your net worth has changed when the market value of your house changes, whether or not you sell. Psychologically, people are more selective: at those dinner parties, gains are treated as very real when the Zillow estimates are up, and as "just paper" when they're down. We'll have much more to say about recognizing—or not recognizing—changes in the market values of assets later in this book.

Housing markets tend to fall in recessions. Recessions are also when people lose their jobs. So the scenario in which your house is underwater is disproportionately likely to be the very scenario in which you've lost your income, can't make the mortgage payments, and may need to move to find work—that is, to sell at the bottom. All the bad things tend to happen together. That is how leverage multiplies risk. And it is exactly the mechanism at the heart of the next section.

6.6.2 Going deeper: Bank equity is financing, not idle cash

Suppose a bank is asked to finance a larger share of its assets with equity. Does that mean it must keep a larger pile of cash idle?

Return to the two sides of the Accounting Equation. The asset side tells us what the bank holds: Cash, loans to customers, and other resources. The other side tells us about the claims on those assets and how the bank has been financed. Changing the mix of debt and equity is a question about that second side.

When the bank makes a loan, it exchanges Cash for a loan receivable. Both are assets. The distinction between those two assets is different from the distinction between debt and equity financing. An equity requirement is therefore not, by definition, a requirement to keep the same amount in cash.

This is why I care so much about the apparently elementary point from Chapter 1: **equity is a claim, not an extra resource**. It matters when we discuss leverage and financial regulation, where confusing the two sides can obscure the actual tradeoffs.

7

Looping Back: Accrual Accounting Deep Dive

By now, we've seen the complete process by which we go from transactions to periodic financial statements. We've learned the core tools of accounting: debits, credits, journal entries, T-accounts, trial balances. We accounted for a number of simple transactions and used them to illustrate some core principles of accounting. This chapter is about going deeper—we'll cover *harder* transactions, with more complex journal entries, and do a deeper dive on *accrual accounting principles*. Instead of just correctly journalizing transactions, we'll focus on *interpreting them*—the understanding we build from that will help us create new tools for financial analysis. In particular, we'll see how, by understanding accrual accounting principles, we can calculate and infer a company's actual **cash flows** from its Income Statement and Balance Sheet accounts.

Here's a way to see, right away, the flavor of what 'going deeper' means. To date, we have mostly gone *from* a description of a transaction *to* the journal entry: I tell you what happened, and you translate it into debits and credits. In this chapter, we'll frequently flip that around: I give you the journal entry, and *you* tell me what happened in the world. Suppose, for instance, that this is all you see:

Cash (+A)	\$3.5k	
Accounts Receivable (+A)	\$1.5k	
Revenue (+S.E.)		\$5k

Provide the soccer camp, 2/15/2022

Describe and interpret what has happened here. Take a moment and try to narrate it before reading on. Here's the short answer: we provided services worth \$5k—a soccer camp, in this case—and the customer paid \$3.5k of it in cash on the spot, promising to pay the remaining \$1.5k later. If you produced something like that sentence on your own, congratulations: you are no longer just *doing* accounting; you are starting to *read* it. And if you couldn't quite—don't worry. By the end of this chapter, you'll be able to say that, and quite a bit more, about this little three-line entry.

7.1 Timing Differences: (Revenues and Expenses) x (Deferrals and Accruals)

Recall from Chapter 1 that **accrual accounting** differs from cash-basis accounting in two primary ways: **Revenue Recognition** and **Expense Matching**.

- **Revenue Recognition**: Revenues are *recorded* when they have been **earned**—when we have **provided** the good or service—rather than when the cash is paid.
- **Expense Matching:** Expenses are, when practicable, *matched* to the corresponding revenues that they helped to generate—that is, recorded as expenses **in the same reporting period**—rather than when the cash is paid.

Before we organize anything, let's establish the principle that makes the organization possible. At any given point in time—in any given reporting period—accrual quantities and cash flows will *not* line up: $Revenue \neq Cash\ Inflows$, and $Expenses \neq Cash\ Outflows$. But now consider the **long term**. In the cash-settled transactions we are studying here, each revenue has a related cash flow: we sell on account, and the customer pays—eventually. (If we kept recording *Revenue* forever and were never actually paid, something would be absurdly wrong—a problem we'll take up in Chapter 8.) The same timing distinction holds for the cash-settled expenses in our examples. So, over a sufficiently long horizon, **accruals and cash flows must be "reconciled"**—they have to come back together. And this tells us something important about the accrual-vs.-cash distinction itself: it is not, ultimately, a difference of *substance*, but of ***timing***.

That insight lets us organize *everything*. Fix a revenue at a point on a timeline: today is the day we provide the good or service and *earn* the revenue. If the related cash must arrive at *some* point, then, relative to today, there are only three possibilities: the customer paid us *before* today (an advance payment); the customer pays us *today* (an ordinary cash sale); or the customer will pay us *after* today (a credit sale). There are no other options. The same exhaustive logic applies to any expense. We might pay the cash *before* recognizing the expense (we buy *Supplies* up front, and expense them later as we use them); *at the same time* (we receive the month's utilities bill and pay it on the spot); or *after* (we *accrue Interest Expense* over the year, and hand over the cash on the scheduled payment date).

Therefore, one way to organize all accrual transactions is to categorize them by whether the **revenue or expense** is recorded **before, at the same time, or after** the actual **cash** changes hands. We can put everything in a simple matrix, this way.

- We use the term **"deferral"** to indicate cases in which the revenue or expense is recorded *after* the cash changes hands (**jargon**). (I.e., we 'defer' recognizing the expense into the future, just like an admitted college student could defer matriculation for a year.)

- And we use the term **"accrual"** to indicate a case in which the revenue or expense is recorded *before* the cash changes hands. (E.g., we 'accrue' *Interest Expense* or *Salaries Expense* over time, before the date on which we actually pay.)

With that terminology, we can organize all transactions in this simple matrix.

	...before cash paid	**....same time**	**...after cash paid**
Revenue earned...	Accrued Revenue	Cash Sale	Deferred Revenue
Expense incurred...	Accrued Expense	Immediate Expense	Deferred Expense

So, we can place all of our revenue-related transactions in this scheme, like so:

- When we sold services to a customer on account (Transaction #7), that was ***accrued revenue***—earned before it was paid.

- When we sold services to a customer in cash (Transaction #6), that was a vanilla **cash sale, with no timing difference** between the cash flow and revenue.

- And when the customer paid up front (Transaction #8), that was ***deferred revenue***, paid before it was earned—we had to 'defer' recording revenue into the future, once we provided the service.

And, we can place all of our expense-related transactions in this scheme, like so:

- When we made an adjusting entry for *Salaries* at the end of the period, for the *Salaries Expense* that had been incurred but not paid since the last payday, that was an ***accrued expense***—incurred before it was paid.

- When we paid our headquarters' electricity bill, that full amount was "immediately expensed" (rather than capitalized), with no timing difference between the cash flow and expense.

- And, importantly, in this scheme, many of our **operating assets represent deferred expenses**—e.g., when we purchased *Prepaid Rent*, we initially capitalized that as an asset and *deferred* recording *Rent Expense* into the future, as we actually "used up" that resource over time.

For resources such as Prepaid Rent and Supplies, calling the balance a **deferred expense** draws attention to timing. We have recorded the acquisition, but we will recognize the expense as the resource is used. The question is not simply "asset or expense forever?" It is "what remains available now, and what has been consumed in this period?"

7.2 Accruals, Cash, and Working Capital

Now, let's cover a variety of accrual-accounting transactions, some of which we've seen previously, but place them in this framework, and do a deeper, more conceptual analysis, to get some new insights. The bottom line of what we'll learn is this: The differences between accrual revenues / expenses and their related operating cash flows are 'stored in' **working capital** accounts (current assets and current liabilities). And so, as a result, we can "reverse engineer" a company's actual cash flows *from* its income statement and balance sheet accounts. Understanding how and why we can do this is the main goal of this section. (I grant that this bottom-line sentence sounds Latinate and abstract on a first reading. In a way, this whole section is about unpacking that one sentence, transaction by transaction, until it feels concrete.)

7.2.1 Revenue Transactions

Let's first consider all the ways in which our company can earn *Revenue*.

Accrued Revenue: Revenue recognized *before* cash is paid

This will happen whenever a customer pays "on account" or "on credit"—that is, with a promise to deliver cash later. This was Transaction #7 in previous chapters. Recall

that, because there was a 'real' exchange (the good or service was provided), this is still an **external transaction**, even though no cash changes hands:

Accounts Receivable (+A)	$3.1k	
Revenue (+S.E.)		$3.1k

Service provided, customer pays on account, 10/1/2021

In the jargon of accounting, we say that the company has "accrued revenue"—it has earned it, and recognizes it, even though it has not collected the cash.

Later on, that customer will (we expect) eventually pay in cash. How do we account for that? **Start with cash.** Our company receives an **inflow** of Cash. And the customer will have then **fully discharged** their account. They will no longer owe us, so we will no longer have that **Receivable**. So we **credit A/R** to eliminate it.

Cash (+A)	$3.1k	
Accounts Receivable (-A)		$3.1k

Customer pays off their account in full, 2/1/2022

There are three **important takeaways** from this simple pair of transactions:

- **First**, note that **both** transactions here are **"external transactions,"** because there is a **real exchange** in both cases. When the company provides a good or service, that's an exchange. When the customer pays cash, cash changes hands. So there is an **accrual** here, but no internal transaction or adjusting entry. *Not every accrual* involves an adjusting entry. It depends on the specifics of the situation.

- **Second**, this simple pair of transactions illustrates a core accounting principle. **Over time, accruals should be 'reconciled' to actual cash flows.** Put differently, **differences between accruals and cash flows should 'revert' over time**. First the company *accrues* $3.1k in *Revenue*. Later it actually collects that amount in *Cash*. (If it doesn't, that's a problem, which we'll cover in Chapter 8.)

- **Third**, note how, in each part of the transaction, the change in *Accounts Receivable* is equal to the *difference* between the *Cash* received from the customer and the *Revenue* recorded. In the first part of the transaction, there's *Revenue*, but no *Cash*—and *Accounts Receivable* goes *up*. In the second part of the transaction, there's no *Revenue*, but the *Cash* is paid, and *Accounts Receivable* goes back

down. To make this totally explicit, I like to re-write these journal entries, with extra, extraneous zeroes, like so:

Cash (+A)	\$0	
Accounts Receivable (+A)	\$3.1k	
Revenue (+S.E.)		\$3.1k

Service provided, customer pays on account, 10/1/2021

Cash (+A)	\$3.1k	
Accounts Receivable (-A)		\$3.1k
Revenue (+S.E.)		\$0

Customer pays off their account in full, 2/1/2022

We can see here that *Accounts Receivable*, a working capital account, **"spans"** or **"bridges"** or **"plugs"** the difference between the accrual quantity—the *Revenue*—and the actual cash flow **across time**. When we earn Revenue, without cash, it goes up, to store the difference—and when we get cash, later, it goes back down. Across the pair of transactions, *Accounts Receivable* is like a switch that turns on and then off: it flips on when the revenue is earned, and off when the cash arrives. And while it's on, its balance is telling you something precise: exactly how much revenue has been **earned but not yet collected**.

One **key application** of this is that it shows how we can go **from** *accrual-accounting* revenue *to* actual *cash collected from customers*, or vice versa, using the change in the *Accounts Receivable* account. **If we assume the company has no Deferred Revenue and no other changes in receivables beyond credit sales and collections** (as we usually will for these quantitative problems, in this introductory text), then, on any given sale transaction, it must be that:

$$Revenue = Cash\ Collected\ from\ Customers + \Delta Accounts\ Receivable$$
$$Cash\ Collected\ from\ Customers = Revenue - \Delta Accounts\ Receivable$$

Since this identity holds on each transaction, it must hold across the sum of all transactions as well. And, the Financial Statements report the final balances of accounts, after all transactions. This means that we can infer how much cash a company actually received from customers over a period using its Income Statement and Balance Sheet accounts. Consider the example below:

Example problem: A company reported Revenues of $100m in FY2021, but its reported **Accounts Receivable** *increased* from $20m to $50m from FY2020 to FY2021 (and it had no Deferred Revenue). Calculate the cash it collected from its customers in FY2021.

$$Revenue = Cash\ Collected\ from\ Customers + \Delta Accounts\ Receivable$$
$$\$100m = Cash\ Collected\ from\ Customers + (\$50m - \$20m)$$
$$\$100m - \$30m = \$70m = Cash\ Collected\ from\ Customers$$

The company recognized $100m of revenue, while the amount still owed by customers increased by $30m. Under the assumptions of this example, collections were therefore $30m less than revenue: $70m. That is **cash collected from customers**, including payments on earlier accounts. It is not necessarily the amount of same-day cash sales.

Let's push on the intuition a bit further, because there are really **two scenarios** to keep straight, and it pays to think about what's happening in the world, not just in the T-accounts. What real-world events make *Accounts Receivable* go *up*? By definition: **credit sales**—we provide goods or services on account, and the customer hasn't paid yet. So if A/R rose over the period, some of this period's *Revenue* was not collected in cash—and to get from Revenue to cash collected, we *subtract* that increase. And what makes *Accounts Receivable* go *down*? **Customers paying off old accounts**. When that happens, cash comes in—but it is *not* current-period Revenue, because we recognized that revenue back when we made the sale. So a decrease in A/R means we collected cash *beyond* this period's revenues—and we *add* that decrease back. (And to be clear on terminology: when a customer pays off an old account, that absolutely counts as 'cash collected from customers.' It's cash, and it's from a customer. Don't overthink it.) Of course, for a real company, *both* things are happening constantly and simultaneously—every day it makes new credit sales *and* collects on old ones. That is why the formula runs on the *net change*, $\Delta Accounts\ Receivable$, which automatically handles both scenarios at once.

For those who want it fully airtight, here is the algebra. Over any period (still assuming no Deferred Revenue):

$$Revenue = Cash\ Sales + Credit\ Sales$$
$$\Delta Accounts\ Receivable = Credit\ Sales - Collections\ on\ past\ accounts$$
$$Cash\ Collected\ from\ Customers = Cash\ Sales + Collections\ on\ past\ accounts$$

Subtract the second equation from the first: the *Credit Sales* terms cancel, and we're left with $Revenue - \Delta Accounts\ Receivable = Cash\ Sales + Collections\ on\ past\ accounts = Cash\ Collected\ from\ Customers$. So the core formula is exact, on a net basis, no matter how the credit sales and the collections are jumbled together within the period.

While this may seem **esoteric** right now, I **promise it is not**. Many of the most im-

portant analyses in finance and general-management jobs are all about understanding a company's actual **cash flows** (because **"cash is king,"** and you have to pay the bank in cash, or else you can go out of business). History is littered with companies that were profitable, in accrual net income, right up until the day they ran out of cash. Indeed, one of the most common technical questions in investment-banking interviews is some variant of: "Here is a company's income statement—walk me to its cash flows."

And this example shows how, if you understand accounting concepts, you can unravel that relationship, just using the main financial statements, the income statement and balance sheet. More practically, this is (a.) a nice **setup for a quantitative problem** to test your mastery of the concepts—you'll meet it again in this book's practice problems—and (b.) the key to doing the **cash-flow statement**, later in this book. So it's definitely worth mastering.

Cash sale: Revenue recognized at the same time cash is paid

This one is easy, the most vanilla transaction. I'm just showing it for completeness. This was Transaction #6 in previous chapters:

Cash (+A)	$3.9k	
Revenue (+S.E.)		$3.9k

Cash Sale, 10/16/2021

In this case, because the customer pays in full, and the service is provided in full on the same day, there is no difference between the cash-basis sale and the accrual revenue. So this **one transaction** is the **full story** here. For a cash sale, there's no subsequent related adjusting entry or external transaction, because there's **no timing difference** between the activity and the cash flow. The date of the sale is the full story.

Deferred Revenue: Revenue recognized *after* cash is paid

In this case, customers pay cash *before* our company provides the actual good or service (Transaction #8). When the customer pays in advance, we do not recognize a revenue, because we have not yet earned it. Instead, we record a *liability* called **'Deferred Revenue,'** which represents our **obligation** to deliver on our side of the deal.

Cash (+A)	$0.8k	
Deferred revenue (+L)		$0.8k

Customer Pays in Advance, 10/17/2021

We have received cash, but the service remains to be provided. Revenue recognition waits until we earn it.

Later on, when we actually provide the service, how do we account for it? On that date, we will actually **earn the revenue—a credit**. And then, because we will have delivered on **our side of the deal** we will no longer be obligated, or liable, to this customer. So, we'll **eliminate the liability** (with a debit, since liabilities are credit accounts).

Deferred revenue (-L)	$0.8k	
Revenue (+S.E.)		$0.8k

We provide the agreed service to those customers, 1/5/2022

The **key takeaway** here is to note the **perfect symmetry** with the issues we discussed in the accrued revenue transactions in the section above. *Accounts Receivable* is a current asset in these examples, while *Deferred Revenue* is a current liability in these examples. *Accounts Receivable* represents amounts of Revenue we have **earned, but not been paid**. *Deferred Revenue* represents amounts we have **been paid, but have not earned.**

And, in perfect symmetry to the previous example, **Deferred Revenue**, a short-term liability account, **"spans"/"plugs"/"stores" the difference** between the cash and the accrual-accounting revenue in each transaction, **across time**. To make this fully explicit, I'll once again write the journal entries with the extra zeroes:

Cash (+A)	$0.8k	
Deferred revenue (+L)		$0.8k
Revenue (S.E.)		$0

Customer Pays in Advance, 10/17/2021

Cash (A)	$0	
Deferred revenue (-L)	$0.8k	
Revenue (+S.E.)		$0.8k

We provide the agreed service to those customers, 1/05/2022

Once again, a Balance Sheet working-capital account "spans" or "stores" the difference between the accrual revenue and the related cash flow.

Now that we have this symmetry in hand, we can loop back to the **assumption** I built into the "cash collected from customers" formula above: *no Deferred Revenue*. Why did I need that assumption? Because if a company *does* have Deferred Revenue activity, then some of the cash it receives from customers in a period will not correspond to current-period *Revenue* at all—it will be cash for services not yet provided. Here's a challenge question, then: what would the formula look like if we relaxed the assumption? Use the symmetry. If *Accounts Receivable* going *up* means 'more Revenue that was not collected in cash,' then *Deferred Revenue* going *up* means the mirror image: 'more cash collected that is not yet Revenue.' The two accounts push in opposite directions when reconciling Revenue to cash. For this expanded example, still assuming no writeoffs, sales returns, acquisitions, foreign-currency effects, or other changes in these accounts, the formula is:

$$Cash\ Collected\ from\ Customers = Revenue - \Delta A/R + \Delta Deferred\ Revenue$$

In many of this book's quantitative problems, we will assume no Deferred Revenue so that we can first isolate the role of Accounts Receivable. In an actual company, check both balances. Deferred Revenue can be central to subscription businesses and other companies that collect from customers in advance.[1]

7.2.2 Going deeper: Deferred revenue and the Zynga estimate

Back in Chapter 1, I promised that, as this book went on, we would see examples of how accrual accounting—for all its logic—can allow firms to boost their *reported* performance even when the underlying economics is not improving. Here is a real one, and it runs directly through *Deferred Revenue*.

Zynga was a gaming company whose flagship product, *FarmVille*, was an online multiplayer game accessed through Facebook. The game itself was free to play; the business model was selling **virtual goods** for real money. Some of those goods were consumed on the spot, but the interesting ones were **durable**—a virtual *tractor*, say, usable for as long as the player keeps playing. Suppose a player pays about $10 for one. By now, the phrase "the customer pays up front" should trigger two thoughts in your mind, automatically. First: *Cash*—Zynga receives $10. Second: *Deferred Revenue*—Zynga has not yet *earned* that $10, because it is obligated to keep that tractor available, in a running game, for as long as the player plays.[a]

But here's the wrinkle: over *what period* should Zynga 'earn down' that $10?

[1] A philosophical aside, courtesy of a question I could not immediately answer the first time it was posed to me: if someone has *paid* us but not yet *received* anything, in what sense are they already a 'customer'? The best answer I can give is that they have paid us with the expectation of receiving a good or service—unlike, say, an investor, who pays in with the expectation of dividends. But I concede the label is doing some quiet philosophical work.

Zynga does not assume each customer will play forever. It has to **estimate** how long the typical paying player will stick around—it simply has to, in order to apportion the payment across periods. Suppose Zynga estimates that the average player will stay for **five years**.[b] Assume the purchase occurs at the start of the year and service is provided evenly. Then each $10 purchase becomes $2 of *Revenue* per full year, with the remainder deferred. Combining the cash receipt and the first year's recognition:

Cash (+A)	$10	
Revenue (+S.E.)		$2
Deferred revenue (+L)		$8

Cash receipt and first full year of service (5-year estimated life)

Then, during 2011, Zynga's estimates changed: it now believed players were quitting its games faster. In our stylized version, say the estimate drops to **two years**. Pause on that for a moment, purely as a matter of *business*: is that good news or bad news? Clearly bad. If people are losing interest in your game faster, that's a problem—and *FarmVille* was a *platform* game, where much of the appeal was that your friends and family were on it too, so faster churn undermines the whole network effect.

But now watch the *accounting*. For a new purchase measured using the shorter estimate, each $10 is spread over two years instead of five: $5 of *Revenue* per year instead of $2. Reported revenue per customer purchase jumps by 150%—purely from a change in an **estimate** in management's heads, not because customers were paying more or staying longer. And remember that, for technology and platform companies, the 'top line'—revenue growth—is the number investors obsess over. So we get a very odd situation: customers quitting faster is *economically* bad news, dressed up by the accounting as spectacular 'revenue growth.'

Then, in December 2011, Zynga held its **IPO**, selling shares to the public. Its 2011 annual report disclosed that changes in estimated playing periods increased that year's revenue by **$53.9 million**, with an equal offsetting reduction expected in 2012. It also reported that the weighted-average life of durable virtual goods for bookings had fallen from 18 months in 2010 to 15 months in 2011.[c] Those are substantial changes in reported timing, without corresponding new cash receipts. A justified change in estimate is part of accrual accounting. But an analyst still needs to separate the effect of that change from growth in what customers are buying. The disclosure establishes the accounting effect; it does not, by itself, establish man-

agement's motive or a legal conclusion.

The takeaway: accrual accounting runs, in places, on **estimates**; estimates give management **discretion**; and discretion can be used to inform—or to mislead. This is one of the big reasons the analysis in this chapter matters. A company's *cash flows* are much harder to conjure with an estimate. If the accrual numbers look wonderful while the cash flows stagnate, that is a **red flag**—and, as we'll preview at the end of this section, it is a large part of why companies are required to report a Statement of Cash Flows at all.

[a] The accounting obligation is to provide the service promised in the arrangement. The customer's legal rights if the service ends depend on the contract and applicable law; a deferred-revenue balance does not by itself establish a right to a cash refund in bankruptcy.

[b] I have stretched the numbers for clean arithmetic. Zynga's actual disclosed estimates of the "average playing period of paying players" ran from roughly 8 to 25 months, depending on the game and the year; and the real estimate change, which we will get to in a moment, took the weighted-average life of these durable goods from about 18 months down to about 15. The mechanics—and the moral—are exactly the same.

[c] See Zynga's 2011 Form 10-K, "Revenue Recognition" and the discussion of online game revenue. The example above deliberately uses much longer lives and simpler amounts to make the timing mechanics visible.

7.2.3 Expense Transactions

Accrued Expense: Expense recognized *before* cash is paid

Recall that employees are typically paid every two weeks. In Chapter 4, we considered an example in which employees earned $.9k in salaries *per week*, and had their last **payday** one week **before** the **end of the reporting period.** As a result, as of the end of the reporting period, they had **earned**, but not been paid, $.9k in salaries. Our company had to account for that as an adjusting entry, like so (and note that I again include the **extraneous zero** line here, for **pedagogical** purposes):

Salaries expense (-S.E.)	$.9k	
Cash (A)		$0
Salaries Payable (+L)		$.9k

End-of-period Adjusting Entry for unpaid Salaries, 12/31/2021

In this case, we would say that the company has **"accrued"** a Salaries Expense. It has **incurred** the obligation, and owes its employees for salaries they have earned but has not paid them yet. Once again, a short-term Balance Sheet account, ***Salaries Payable,*** "plugs" the difference between the accrual quantity and the cash flow.

Now, consider the **next payday**. How do we account for it? This one is tougher, but doable. **Start with cash.** The employees earn $.9k per week, so they get paid $1.8k in

cash every two weeks. And, **since the last journal entry on 12/31**, they have *earned* another \$.9k. Moreover, on payday, the company **fully discharges** its lagging obligation to its employees. So, the **liability** from the previous adjusting entry is **discharged**.

Salaries expense (-S.E.)	\$.9k	
Salaries payable (-L)	\$.9k	
Cash (-A)		\$1.8k

Next payday, 01/07/2022

This is the hardest transaction we have done yet: it has three lines, and it's hard to process mentally, all at once. But it's doable if we just break it down. Cash is cash—on each payday, \$1.8k is paid. The **expense** is about the actual ***activity***, how much the employees have ***earned*** since the last time an expense was recorded, one week prior. And, after this payday, the company no longer has an outstanding *Salaries Payable* obligation—it has discharged it in full by paying—so that account has to go back to \$0. Alternatively, if we can just solve for two out of the three, we know that the third line must balance the difference—the beauty of debits and credits, making our lives easier.

Example problem: Since this transaction rewards practice, let's do a variant with a wrinkle. Suppose our employees earn \$700 per week and are paid on the **first Friday of each month**—so, depending on the calendar, sometimes four weeks elapse between paydays, and sometimes five. The last payday of 2021 fell on Friday, December 3. By Friday, December 31—the end of our reporting period—employees have worked four more weeks without being paid: they have earned $4 \times \$700 = \$2.8k$. Their earnings are our expense. So we make the adjusting entry:

Salaries expense (-S.E.)	\$2.8k	
Salaries Payable (+L)		\$2.8k

Adjusting Entry for accrued Salaries (variant example), 12/31/2021

Now it's Friday, January 7, 2022—the next payday, *five* weeks after the last one. What's the journal entry? **Start with cash**: cash is cash, and it's written right into the employment contract. Five weeks have gone by since the last payday, so we pay $5 \times \$700 = \$3.5k$—a credit to *Cash*. Next, the expense: since the last time we recorded a *Salaries Expense* (the 12/31 adjusting entry), the employees have earned one more week—a debit of \$0.7k to *Salaries Expense*. And, on payday, we discharge our lagging

obligation in full, so *Salaries Payable* must go back to zero—a debit of \$2.8k. (Check: $\$0.7k + \$2.8k = \$3.5k$. Debits equal credits.)

Salaries expense (-S.E.)	\$.7k	
Salaries payable (-L)	\$2.8k	
Cash (-A)		\$3.5k

Next payday (variant example), 1/7/2022

Now, here's an objection that sharp students raise about this entry, and it's worth taking seriously: "Wait—between December 31 and the January 7 payday, the employees worked another week. During that week, weren't we accruing *more* liability to them? Shouldn't *Salaries Payable* have gone *up* by another \$0.7k before being paid off—so that the payday entry clears \$3.5k of *Salaries Payable*, rather than \$2.8k?"

On economic—even *moral*—grounds, this is exactly right. With each passing day of that week, we really did owe our employees a little more. But here is the key distinction: did we actually *record* that growing obligation on our books? No. By convention, we only update the salaries-related accounts on **paydays** and at **end-of-period adjusting entries**—not continuously. (We'll say more about the logic of this convention later in this chapter.) So, going into payday, the only amount formally sitting in *Salaries Payable* is the \$2.8k from the 12/31 adjusting entry—and that is the balance that must be brought back to zero.

And here's the satisfying part: suppose we *had* practiced this kind of scrupulous 'moral accounting,' updating our books continuously. Then, during that first week of January, we would have recorded another accrual...:

Salaries expense (-S.E.)	\$.7k	
Salaries Payable (+L)		\$.7k

'Moral accounting' entry for one more week worked, early January

...and then, on payday, we would have simply paid off the full accumulated liability:

Salaries payable (-L)	\$3.5k	
Cash (-A)		\$3.5k

'Moral accounting' payday entry, 1/7/2022

Add up those two entries, and compare them to our single payday entry above: the net effect on every account is *identical*—$0.7k of new *Salaries Expense*, $3.5k of *Cash* out the door, and *Salaries Payable* ending at zero. The 'moral' books and the practical books arrive at exactly the same place. The only difference is how much bookkeeping labor was expended along the way—which is precisely why we adopt the practical convention.

The big takeaway here is to note how, once again, a working-capital account, *Salaries Payable* (a current liability), "plugs" the difference between the accrual-accounting expense and the cash flow. By definition, *Salaries Payable* is amounts of salaries that our employees have earned but not been paid—and therefore, for our company, amounts of *Salaries Expense* that we have *incurred* but not paid out. This means that, symmetric to the example problem above, we can infer the actual cash paid for salaries using the *Salaries Expense* and the *Salaries Payable*.

Example problem: ExpoCorp reported *Salaries Payable* of $120k on its FY2021 Balance Sheet, and $140k for FY2022. On its Income Statement, it reported *Salaries Expense* of $800k in FY2022. How much did ExpoCorp actually pay in salaries in FY2022?

Solution: ExpoCorp *incurred* $800k in *Salaries Expense* over FY2022—that's how much its employees earned. But it didn't necessarily pay them in full, yet. Notably, ExpoCorp's *Salaries Payable* went up by $20k over the period. This means that, on net, of the *Salaries Expense* it incurred, $20k went unpaid over the period. Therefore, it must be the case that it paid only $780k in cash for salaries over the period. Or, if you want a formula: $Cash\ Paid\ for\ Salaries = Salaries\ Expense - \Delta Salaries\ Payable$.

Accrued Interest

A transaction we haven't covered yet, but which illustrates a more commonsense use of the term *accrual*, is: **accrued interest**. Indeed, interest is the one place in everyday life where you already use this word: your savings *accrue* interest; your student loans, alas, *accrue* interest too. The amounts build up continuously with the passage of time, before any cash actually changes hands—the same slightly philosophical flavor we just encountered with salaries accruing between paydays. Consider a new example: our company receives a *Loan* of $10,000. You probably know that we are charged *interest* on loans. To make things simple, suppose that we got that loan **halfway through the reporting period, July 1, 2022**, and the loan requires **interest payments of 6% ($600) on an annual basis**, as long as the loan is **outstanding**.

Now, it's December 31, the end of our reporting period, and it's time to close the books. Because we have had this loan outstanding for half a year, we have **accrued**

interest—but the full **interest payment** will not be due for another half year. So, there has been no "real" exchange or payment related to the interest, yet. But we must account for the interest that has *accrued over time*, as an adjusting entry. Below is the adjusting journal entry (again with the extraneous zero, for pedagogical purposes).[2]

Interest expense (-S.E.):	$300	
Interest Payable (+L):		$300
Cash (A):		$0

End-of-period adjusting entry for accrued interest, 12/31/2022

Now, it's July 1 of the next year, and the first annual interest payment is due. The payment is an external transaction. How do we account for it? Start with Cash—we pay $600 in interest in cash, a credit. And, after making this payment, we have fully discharged the *Interest Payable* we recorded, so that liability must go back to a balance of zero, meaning we must debit it for $300. Finally, in the 6 months that have gone by since our last journal entry related to this loan, we incurred another half-year's-worth of *Interest Expense*—a debit of $300. (Alternatively, you can pin down just two of the lines, and solve for the third as the 'plug.') So, the journal entry would be:

Interest expense (-S.E.):	$300	
Interest Payable (-L):	$300	
Cash (-A):		$600

Paying annual interest payment, 7/1/2023

Now, you might wonder: we handed the bank one check for $600—a single interest payment. Why is it *split* on our books, half to *Interest Expense* and half to *Interest Payable*? Why not just debit the whole $600 to *Interest Expense*? Here's the answer: because we already recognized $300 of that interest as an expense—*last year*. That was the whole point of the 12/31 adjusting entry. If we debited the full $600 to *Interest Expense* now, we would be **double-counting** last year's interest: once in last year's income statement (via the adjusting entry), and again in this year's. The split ensures that each year's income statement reports exactly the half-year of interest incurred *in that year*—and that the liability we set up last year is cleared when it is finally paid.

You might also notice that, over the full cycle, the debits and credits to *Interest Payable* cancel out to zero. Correct—and that is exactly what *should* happen. But it

[2] If our company made a loan *to* another company, so that we were entitled to receive annual interest payments, the situation would be the "flip side." We would *accrue* **Interest Revenue** and an **Interest Receivable (A)** over time, even before the cash interest payments were paid to us.

doesn't mean the account was doing nothing. In the meantime, it was doing important work: separating *when the expense was incurred* from *when the cash was paid*—which is the entire enterprise of accrual accounting.

And, at the risk of repetition, note how, once again, a **working capital account** (*Interest Payable*, a current liability) is "spanning" or "bridging" or "plugging" the difference between the accrual expense and the related cash flow, over and across time.

Example problem: ExpoCorp reported *Interest Payable* of \$20k on its FY2021 Balance Sheet, and \$40k for FY2022. On its Income Statement, it reported *Interest Expense* of \$50k in FY2022. How much did ExpoCorp actually pay in interest in FY2022?

Solution: This one is worth working slowly, because there is a tempting wrong turn. Some students start adding and subtracting the *ending balances*: "We ended last year owing \$20k, so we must have paid that off; and we ended this year owing \$40k; so..."—and arrive at \$60k, or \$90k, or \$10k. The key is that we adjust the expense by the ***change*** in *Interest Payable*, not by its ending balance. *Interest Payable* is, by definition, interest we have *incurred but not paid*. It went *up* by \$20k over FY2022—so, on net, \$20k of the \$50k of *Interest Expense* incurred went unpaid. Therefore, ExpoCorp must have paid \$30k in cash: $Cash\ Paid\ for\ Interest = Interest\ Expense - \Delta Interest\ Payable = \$50k - \$20k = \$30k$.

You may notice that this example problem has exactly the same shape as the salaries problem a few pages back—swap 'Interest' for 'Salaries,' relabel the numbers, and it's the same problem.[3] That's because these are not two formulas, but **one formula in disguise**. For an expense X, assuming its only timing difference is captured in the related payable:

$$Cash\ Paid\ for\ X = X\ Expense - \Delta X\ Payable$$

Whether X is interest, salaries, utilities, or income taxes, the pattern—and the intuition—is the same: the increase in the payable is the portion of the expense that, on net, went unpaid.[4]

Expense recognized at the same time cash is paid

This will happen whenever the company pays for something in cash, and that thing is immediately recognized as an *expense* rather than an asset. When will this happen? Recall that we record expenditures as assets when they are resources that the company

[3] This is, in fact, roughly how I produce these problems: copy, paste, find-and-replace. I admit this not to confess laziness—the fact that it *works* is precisely the point.

[4] For now, every operating expense we've met has just *one* related working-capital account. Later in this book, when we get to **Inventory** (Chapter 10), we'll meet an expense with both a related *asset* and a related *liability*—and the formulas will get correspondingly richer.

will use over multiple reporting periods. So, if the company buys something that it will completely use up in the short-term, in the same reporting period, we would **"immediately and fully expense"** that. Consider, e.g., the **electricity bill for our corporate headquarters**. We cannot **store** it. We immediately use it, as it is delivered to us. It is considered too difficult, if not impossible, to reliably match the costs of the electricity bills at corporate headquarters to individual sales and revenues. So, we would fully expense this when the costs are incurred, rather than capitalize as an asset. So, we record each **Utilities** bill solely as an expense.

Utilities expense (-S.E.)	$.2k	
Cash (-A)		$.2k

Pay our utilities bill, 1/15/2022

As in the case of the cash sale, there is no timing difference here, between the recording of the accrual quantity and the cash payment. Because of that, this is the end of the story, and there is no subsequent related external transaction or adjusting entry.

Deferred Expense: Expense recognized *after* cash is paid

Note that this is what happened with **all of the assets we discussed in Chapter 4**: **Prepaid Rent**, **Supplies**, etc. Those assets had **one "general form."** Below, I show the "general form" of the journal entries for operating assets, once again with the extraneous zero lines for illustrative purposes.

When we first purchase an asset, we record the full **"purchase price"** to the **Asset** account, and **do not** recognize an expense. Since we will use it over many periods, we defer recognizing the expense into the future, as we do.

Asset (+A)	$(Purchase Price)	
Expense (S.E.)	$0	
Cash (-A)		$(Purchase Price)

Purchase of Asset, date during reporting period
(External Transaction)

Then, over time, we account for the **amount used up** as an expense, though there is no more cash exchange.

Related Expense (-S.E.)	$(Amount Used Up)	
Cash (A)		$0
Asset (-A)		$(Amount Used Up)

End-of-period adjusting entry for expense related to asset, 12/31/2020

Once again, a Balance Sheet account "stores" or "spans" or "plugs" the difference between the cash flow and the accrual expense. If the above is too abstract, to make it concrete, consider, e.g., the case of *Prepaid Rent* (Chapter 4).

Example problem: ExpoCorp reported Prepaid Rent of $700k on its FY2021 Balance Sheet, and Prepaid Rent of $600k on its FY2022 Balance Sheet, and Rent Expense of $400k for FY2022. How much did ExpoCorp actually purchase in rent (that is, spend to procure additional leases) in FY 2022?

Answer: *Prepaid Rent* represents amounts of money that the company has spent on rental contracts, but not yet expensed out. ExpoCorp's FY2022 *Rent Expense* was $400k. But, its *Prepaid Rent* balance only went down by $100k over the period. Therefore, it must have spent an additional *$300k* on rental contracts.

7.2.4 Going deeper: What does "investing in our future" mean?

A CEO explains a poor year by saying, "We are investing in our future." What should we ask next?

First, *what expenditure does the CEO mean?* If the company bought equipment, the purchase was recorded as an asset, and its cost is allocated over the periods of use. The cash outlay and this year's expense are therefore different amounts. Calling the whole purchase a reason for this year's low profit would need explanation.

But some spending intended to build future benefits is expensed immediately. R&D, advertising, and employee training can raise precisely that issue. In those cases, investment in the economic sense can reduce current accounting profit.

So the useful response is a follow-up question, not an automatic dismissal: which costs were incurred, how were they accounted for, and what future benefits does management expect from them? The distinction between an economic investment and a recognized accounting asset matters.

7.2.5 Generalizing: Cash, accruals, and "working capital"

So, in all of the examples we saw above, we saw how the difference between the **operating cash flow** and the **accrual revenue or expense** was "stored in" **changes** in current asset or current liability accounts—*Accounts Receivable (A)*, *Deferred Revenue (L)*, *Salaries Payable (L)*, *Interest Payable (L)*, *Prepaid Rent (A)*, etc. Recall from last chapter that we call current asset and current liability accounts "working capital accounts," and define "net working capital" as:

$$Net\ Working\ Capital = Current\ Assets - Current\ Liabilities$$

'Working Capital' is an absolutely ubiquitous, key term and concept in business and finance. And, with the work we just did above, we can now understand *why*. As users of financial statements, we don't see the individual transactions, we just see their aggregated effects on the financial statements: From that vantage point, we know that the difference between net income vs. related cash flows will be largely explained by changes in those accounts, and **working capital** as a whole. (This is largely what the **Cash Flow Statement** will be all about.) And many business and financial analysis applications center on understanding this—what drives the difference between net income and cash flows—to forecast future cash flows, to see if the company can pay its debts and stay in business. For a seasoned financial professional, the concept of working capital is as familiar, ubiquitous, and useful as "copy and paste."

There is one more way to see working capital that is worth planting now. The working-capital accounts represent an informal web of **financing** among the firm, its customers, its suppliers, and its employees. *Accounts Receivable* is, in effect, credit we have extended to our *customers*: we have delivered, and are floating them the money until they pay. *Deferred Revenue* is the reverse—financing our customers have extended to *us*, paying before we deliver (recall the Kickstarter discussion from Chapter 2). *Accounts Payable* is credit financing extended to us by our *suppliers* (also from Chapter 2). And *Salaries Payable* is, if you think about it, a short-term loan from our own *employees*, who work first and get paid later. None of these parties calls itself a bank, and no interest is charged—but real money is being provided by one party to another, across time, just as in any loan. This is another big reason finance professionals obsess over working capital: it tells you *who is financing whom* in the company's day-to-day operations. A company that manages this well can, in effect, get much of its operations financed by its suppliers and customers—for free.

7.2.6 A Preview: The Statement of Cash Flows

Everything we've done in this chapter has a payoff at the level of the financial statements themselves. Recall from Chapter 1 that there are four financial statements, and

that one of them—the **Statement of Cash Flows**—reports the company's performance on (essentially) a *cash* basis. We will learn to actually construct that statement at the end of this book (Chapter 16), and you should *not* worry about doing so yet. But you've now earned a preview—and, having done the work of this chapter, you will understand at a glance something that mystifies most first-time readers of financial statements.

First, a bit of history. For most of the twentieth century, U.S. companies were not required to report a cash-flow statement at all. The FASB began requiring it in the late 1980s, for precisely the reasons we've been discussing: accrual accounting is useful, but it runs on estimates and judgments, and some companies were making sketchy maneuvers with it (recall Zynga). Standard setters wanted financial-statement users to be able to check the accrual story against the cash reality. Here is what the top of a typical Statement of Cash Flows looks like, under the common 'indirect method':

Cash flows from operating activities:	
Net Income	**$25k**
...adjustments to reconcile N.I. to cash from operations:	
$-\Delta$*(Accounts Receivable)*	($20k)
$-\Delta$*(Prepaid Rent)*	$20k
$+\Delta$*(Accounts Payable)*	$15k
$+\Delta$*(Salaries Payable)*	$10k
$+\Delta$*(Interest Payable)*	($5k)
$+\Delta$*(Income Tax Payable)*	$1k
Net cash flows from operations:	**$46k**

Statement of Cash Flows (operating section), indirect method

Notice two things. First, the statement *starts with Net Income*—an odd choice, on its face: a cash-basis statement that begins from the flagship *accrual*-basis number! Second—and here is where this chapter pays off—look at the adjustment lines that carry us from *Net Income* down to *Cash Flows from Operations*. Every single one is the **change in a working-capital account**: receivables, payables, prepaids. Why? You already know why. As we've now seen, transaction by transaction, the differences between accrual revenues and expenses and their related cash flows are stored in exactly these accounts. The Statement of Cash Flows is simply aggregating, at the whole-statement level, the same relationships we've been deriving at the level of individual journal entries: take accrual net income, undo the timing differences parked in working capital, and you arrive at operating cash flow.

Again, don't worry yet about *doing* this—about which changes get added and which get subtracted, and why. That's Chapter 16's job. For now, just appreciate the shape of the thing: the mysterious middle of the cash-flow statement is nothing but this chapter,

written in aggregate.

7.3 Accrual Accounting x Internal Transactions

Let's talk about the relationship between **accrual accounting** and the **"internal transactions" / "adjusting entries"** companies make at the end of the reporting period. The two concepts are *related* in important ways. But some students **overthink it**, and this can lead to confusion.

7.3.1 The Relationship Between the Two

External vs. Internal: Earlier, we defined **'external transactions'** as **"real" exchanges** of things or services, between our company and some external party. And we defined **"internal transactions" / "adjusting entries"** as **"everything else"** a company has to do in order to update its accounts before closing the books. The adjusting entries we've seen so far had to do with how certain assets are "used up" over time, or how certain liabilities (e.g., salaries or interest) *accrue* over time, before they are paid.

Accrual accounting: Accrual accounting is defined by way of contrast to **cash-basis accounting**, in which we only account for cash transactions—payments and receipts. And all **cash transactions** are, necessarily, **external transactions**—real exchanges.

Relationship: Therefore, since **all internal transactions / adjusting entries** are non-cash, they all ultimately have to do with **accrual accounting**—with the fact that we do not *only* account for cash payments and receipts.

In short, **all adjusting entries have to do with accrual accounting**. But that relationship **doesn't really go both ways**. Companies can have ***accruals* that don't involve any internal transactions or adjusting entries**. For example, as we saw, when a company makes a **credit sale**, it *accrues revenue* before the cash is paid. That sale is an **external transaction.** Later on, when the **customer pays off their account**, that is **also** an external transaction, since there's a real exchange of cash.

So, it's a **rectangle-square** situation. Every internal transaction has to do with accrual accounting. But not every accrual or deferral involves an internal transaction.

More generally, we don't need to make any end-of-period internal transactions / adjustments of accounts related to transactions where there's nothing "left over"—if the external transactions were a full accounting. This will happen whenever:

1. a cash transaction and the related accrual revenue or expense recognition happen at the same time (e.g., cash sale, or an immediately-expensed item);

2. they don't happen at the same time *but* the difference between them is fully accounted for in external transactions (e.g., credit sale, which is later paid off).

I emphasize this to anticipate confusion, because many students **overthink it**. They see some accrued quantity—**Accounts Receivable**, or **Accounts Payable**—on the books at the end of the reporting period, and think they're supposed to **do something**.

Nope. We **adjust** some assets and liabilities that are **used up or discharged over time**. Others we don't. E.g., unlike *Interest Payable*, *Accounts Payable* doesn't change *over time per se*—that account gets updated if and when we actually incur and pay our bills, both of which would be external transactions. In cases like those, at the end of the reporting period: "Don't just do something—stand there."

7.3.2 External vs. Internal: What really *is* the difference?

Some first-time students get a bit **hung up** on "philosophical" questions about what really is an external or internal transaction. The truth is, sometimes, the boundary between the two is fuzzy. But in these **cases, it doesn't really matter**. All that matters is that we **report the financial statements correctly** at the end of the reporting period.

Remember that the company doesn't really "have" financial statements until then. So, all that matters, is that if we *don't* account for something **as an external transaction,** we *do* account for it as an **adjusting entry,** before closing the books.

Consider the **Deferred Revenue** example we referenced above. There, a customer **paid in advance** (debit Cash, credit Deferred Revenue) and later on, we provided the service (debit Deferred Revenue, credit Revenue). In the case of that example, we accounted for **both** of those as "real" exchanges, external transactions.

Now, suppose you were a lawyer, and, on July 1st, a customer pays $10k up front to have you **"on retainer"** for a year. A "retainer contract" means that the client pays for the **right to call upon you** and **prevent you from representing any of his or her enemies** over the agreed period of time.

How do we account for this contract? Well, the *initial transaction* is surely an external transaction, because $10k in cash is paid:

Cash (+A)	$10k	
Deferred revenue (+L)		$10k

Customer Pays for 1-year retainer contract, 7/1/2022

What next? **When** do you, as the lawyer, **recognize revenue** from this contract? Your obligation was to be available to your client, to be on call. There's not necessarily

going to be any *particular moment* of "real exchange"—or, in more reductive accounting terms, anything that will 'trigger' you to update your books.

But, at the end of the year, December 31, you will have to close your books. And, with half a year gone by, you would have earned half of your retainer contract—you would have satisfied half of your **'performance obligation' (jargon)** to be available and loyal to that client for one year. So, in this case, it seems logical to account for half of that retainer contract as *Revenue* as an **internal transaction** / adjusting entry at the end of the reporting period.

Deferred revenue (-L)	$5k	
Revenue (+S.E.)		$5k

Adjusting entry for half year of Retainer Contract, 12/31/2022

As this pair of contrasting examples shows, the account itself or general transaction type (in this case, *Deferred Revenue*) won't tell you whether a transaction is external or internal. You have to think about the specifics of the situation, and what makes sense.

The payday example earlier in this chapter showed that recording each intervening day separately would lead to the same balances. In our examples, we record salaries on paydays and supply any remaining accrual at the reporting date. The essential requirement is to include the work performed through that date.

7.4 Connecting the Income Statement and Balance Sheet

With the transactions that we have covered so far, we're now in a better position to understand the deep links between the two primary financial statements, the Income Statement and the Balance Sheet, which is one of the core themes of this book.

7.4.1 Earning Revenue from increasing assets

Consider the first two revenue-generating Transactions our company made, #6 and 7. In the first, the company sold services for cash, and we accounted for it like so:

Cash (+A)	$3.9k	
Revenue (+S.E.)		$3.9k

Cash Sale

In the second, the company sold services on account (in exchange for the customer's promise to pay later), and we accounted for it like so:

Accounts Receivable (+A)	\$3.1k	
Revenue (+S.E.)		\$3.1k

Credit Sale

In both cases, the Revenue, is balanced by / linked to a change in an asset account—either Cash or A/R. Of course, we know these journal entries well by now. The important thing here is how this provides a deep way to *think about* accounting. Instead of thinking about A/R as a 'plug' account, you could instead look at it this way: "An increase in Assets generated from our operations/sales ***is*** a Revenue. Revenues *are* increases in net assets created by our operations, whether the asset is *Cash* or *A/R*."

Or, going deeper, consider the following example, in which we get a **combination of *Cash* and *A/R* in a sale**: Suppose that a customer buys \$1k in goods from us, but only pays \$600 now, and promises to pay the other \$400 later. Since we have provided the good, we have earned revenue. But we have only received \$600 in cash. We account for this like so:

Cash (+A):	\$600	
Accounts Receivable (+A):	\$400	
Revenue (+S.E.):		\$1000

Sale Partly in Cash, Partly on Credit
Customer Pays in Part

The first thing we learned in this example is that we can have a sale transaction with a combination of Cash and A/R, and therefore three lines in the journal entry. (The rule is that the total **dollar value** of **debits equal** that of credits.) You may also recognize the shape: this is the soccer-camp entry that opened this chapter, in miniature. But there's also a more general insight here. There are two useful ways to *see* this one transaction:

- $Revenue\ Earned - Cash\ Collected = \Delta Accounts\ Receivable$: "We have earned \$1,000 in Revenue, but the customer has only paid \$600 to us. So we have an entitlement to receive the difference later. Accounts Receivable 'plugs' the difference over time."

- $Revenue\ Earned = \Delta Accounts\ Receivable + Cash\ Collected$: "From our sale, we have **obtained two assets**: (1.) The \$600 in **Cash, an asset**. (2.) The customer's

promise to pay \$400 later. That **promise is an asset**. So, to **calculate our revenue, we add up the two assets we have gained**: $\$600 + \$400 = \$1,000$."

These are the two perspectives from Chapter 4, applied to a sale. Starting with the revenue, we ask where the uncollected amount went. Starting with the assets received, we add the Cash and the receivable to measure what the sale brought in. Both routes explain the same entry.

Notice, too, that this second view falls straight out of the mechanics. In the journal entry, the credit to *Revenue* must equal the sum of the two debits—one to *Cash*, one to *Accounts Receivable*. So the 'formula' $Revenue = Cash\,Collected + \Delta Accounts\,Receivable$ is nothing more than the journal entry itself, re-written as an equation. This is worth savoring: the reconciliation formulas we've been using all chapter are not extra machinery bolted onto the accounting—they *are* the accounting, read sideways.

Keep both routes available. In Chapter 8, the distinction will help us compare methods that begin by estimating an expense with methods that begin by estimating the receivable left on the Balance Sheet.

7.4.2 Earning Revenue from decreasing liabilities

But the transactions immediately above, in which the *Revenue* was linked to increases in assets, are not the only ways a company can **generate revenue**. The other way is to 'earn out' a ***Deferred Revenue*** liability, as we saw earlier:

Deferred revenue (-L)	\$0.8k	
Revenue (+S.E.)		\$0.8k

We provide the agreed service to those customers, 1/5/2022

That is, earning a *Revenue* can also be linked to a **decrease** in a *Liability*. Now, with this, we've described **all possible ways** in which a company can earn *Revenue*—since the cash has to be paid either before, at the same time as, or after the sale. We sometimes use the term **"net assets"** in accounting to denote the total **Asset minus Liabilities** (**jargon**). So a useful definition of Revenue for our examples is:

> **Revenue:** "An increase in *net assets* (i.e., an increase in assets and/or a decrease in liabilities) earned from our operations (sales)."

This can also help us better understand the connection between the worlds of the Income Statement and the Balance Sheet. Here's one way to visualize this:

$\uparrow$ **Revenue** $\leftarrow$ **Transactions** $\rightarrow$ ($\uparrow$ **Assets** or $\downarrow$ **Liabilities**)

$\uparrow$ **Revenue** $\rightarrow$ $\uparrow$ **Net Income** $\rightarrow$ $\uparrow$ **Retained Earnings** $\rightarrow$ $\uparrow$ **Shareholders' Equity**
($\uparrow$ **Assets** or $\downarrow$ **Liabilities**) $\rightarrow$ $\uparrow$ **(Assets-Liabilities)** $\rightarrow$ $\uparrow$ **Shareholders' Equity**

What this visual represents is that the *same underlying transactions* (sales) that generate **Revenues** also either increase **Assets** or decrease **Liabilities**. **Revenues** positively contribute to Shareholders' Equity via the **Net Income definition** and **Retained Earnings** identity. And increasing your Assets, or decreasing Liabilities, also do, according to the **Accounting Equation** ($E = A - L$).

7.4.3 Expenses from decreasing assets or increasing liabilities

Pushing even further, recall that **Expenses** are just "**anti-revenues**." And everything is **beautifully symmetric**. So far, we've seen **two ways in which a company can incur an expense**:

1. By **reducing an Asset**: e.g., spending **Cash** on **Utilities**, or "using up" **Prepaid Rent** (and crediting it as an adjusting entry).

2. By **increasing a Liability**: e.g., *accruing* a liability for employee salaries, or paying for an expense *on account* (Accounts Payable).

So, the **fully general definition** of an *Expense* is:

> **Expense:** "A decrease in *net assets* (i.e., a decrease in assets and/or an increase in liabilities) incurred from our operations."

7.4.4 Linking the Income Statement and Balance Sheet

And so, finally, the most general way of visualizing the connection between the Income Statement and the Balance Sheet is like so:

$\uparrow$ **(Revenues-Expenses)** $\leftarrow$ **Transactions** $\rightarrow$ $\uparrow$ **(Assets-Liabilities)**
$\uparrow$ **(Revenues-Expenses)** $\rightarrow$ $\uparrow$ **N.I.** $\rightarrow$ $\uparrow$ **Retained Earnings** $\rightarrow$ $\uparrow$ **S.E.**
$\uparrow$ **(Assets-Liabilities)** $\rightarrow$ $\uparrow$ **S.E.**

Every transaction that affects Revenues or Expenses also affects Assets or Liabilities in balancing, opposite ways. And the differences between the two both **flow into** Shareholders' Equity equally. Everything stays in sync. **It's a beautiful system**.

8

Receivables Part 1: Accounting for Uncollectible Accounts

So far, we've always assumed that when companies make credit sales, they will be paid back in full. And we've accounted for those transactions like so:

Accounts Receivable (+A):	$6,000	
Revenue (+S.E.):		$6,000

Credit Sale

Cash (+A):	$6,000	
Accounts Receivable (-A):		$6,000

Collecting on the Account

But, a truth that we've avoided confronting, so far, is that the **future is never knowable with certainty**. There's always **risk**. And there's always **credit risk**—that we won't be able to collect what we are owed. Some customers might be truly **unable** to pay—unexpected contingencies might send them into bankruptcy. And some customers might simply be **unwilling** to pay. They might decide they don't need to do business

with us anymore, refuse to pay their last bill, and ask, essentially, 'What you gonna do about it?'

Selling on account means relying on a customer's future payment. A contractual obligation is not a guarantee that we will collect. Some losses come from inability to pay; others come from disputes or failures of enforcement. For the examples in this chapter, we will state when an account has become uncollectible, and focus on how to reflect that information in the accounts. Chapter 9 returns to bankruptcy.

It's time to account for that. And we need to. A core accounting principle is that **accrual quantities** (such as *accrued revenue and receivables* from credit sales) and **cash flows** must eventually be **reconciled**. So, if we sell $6k of services *on account*, and book that full amount towards our bottom-line profit, but then 15% of our customers never actually pay, that's a **problem**, a violation of this core accounting principle. So, we need a way to **account for "Uncollectible Accounts."** That's what this chapter is about.

In the previous chapters, for the introductory phase of this book, the text tried to lay out one simple, clean **"right answer"** for each issue we encountered—*the* way to account for some transactions. We are now in its intermediate phase. That means covering precisely those areas where things are **not so simple**.

To study the tension between matching expenses and updating asset values, we will compare three approaches. One of them—the deliberately simplified percentage-of-credit-sales model—holds the expense to a fixed fraction of current sales and does not revise it for later news. That makes the contrast clear, but it must not be mistaken for a complete current reporting method. **Under current U.S. GAAP, expected credit losses on receivables must be estimated and the allowance updated as relevant information changes; balances deemed uncollectible are written off.** A sales-based estimate can inform that process, but cannot justify leaving a materially wrong net receivable on the Balance Sheet. The debate below asks what each approach reveals and obscures, while preserving that distinction between a teaching model and a reporting requirement.[1]

8.0.1 Going deeper: Why spend a chapter on this one problem?

Uncollectible accounts are one example of a broader problem: we do something today, and we know it carries a risk of future cost, but we cannot yet know the exact outcome.

A warranty makes the structure easy to see. A manufacturer sells a product to-

[1] The expected-credit-loss framework is in ASC Topic 326. FASB's ASU 2025-05 permits certain simplifications for current receivables and contract assets; it does not remove the obligation to estimate expected losses. Our fixed-percentage examples isolate the mechanics, rather than reproduce every requirement of that framework.

day and promises to fix certain defects later. It cannot identify every future claim, but it can form an expectation about their cost. Waiting for the claims and estimating them at the sale date tell different stories about this year's performance.

Receivables give us a manageable setting in which to examine that tension carefully. What should we record for expected losses? What should we do when reality differs from the estimate? If we work through those questions here, we will have a way to think about other uncertain costs without treating each as an unrelated rule.

8.1 Goals:

The exact 'mechanics' of the methods will be a bit gnarly. So, let's start reasoning at a higher level, about how we *should* account for this. Logically, we want two things:

- **Goal #1, Account for *expected* credit losses on sales each period:** First, it seems logical that, when we sell **on account**, and expect some losses, we should not count the full amount toward bottom-line net-income. Instead, before reporting our financial statements, we should **estimate** our **expected losses**—and that amount should **not** contribute to net income.

- **Goal #2, Account for 'surprise' realized collections:** The future is never perfectly predictable. This means that we will almost certainly **end up** collecting either **more or less** on our credit sales than we expected. Call that difference the 'surprise' or the 'news.'[2] So, e.g., if we had **expected** to collect \$4k of our credit sales, but **actually** collect \$5k, then the 'surprise' would be \$1k (good news). But if we **actually** collect \$3k, the 'surprise' would be –\$1k (bad news). It seems logical that this 'news' should also be factored in somehow.

8.2 Overview of Methods

In the sections below, we'll cover three different methods for accounting for uncollectible accounts, in detail, describing the rules for each method, and accounting for the same set of transactions using each. For now, a brief preview and verbal description of each, before we get into the dirty technical details. The three methods are:

[2] This use of 'news' is not accounting jargon—it's how finance people use the word, in a slightly technical sense: you start with an *expectation* about the future, and whatever actually happens, *relative to that expectation*, is the 'news.' If you follow the stock market, you've seen the concept in action. When a company announces its earnings, the stock price doesn't react to the raw earnings number—the price already reflected investors' expectations—it reacts to earnings *relative to* expectations. Did they 'beat' or 'miss'? *That's* the news.

- **1) The "Direct Writeoff Method"**: Under this method, we don't make any allowance for uncollectible accounts when we make a credit sale *per se*. Instead, if and when we determine that a customer will not pay, due to, e.g, bankruptcy—then and only then we "write off" their account, and record a "Bad Debt Expense." Thus, I like to call this the **"wait and see method."**

- **2) The "Percentage of Credit Sales Method"**: Under this method, every period, we "make an allowance" for *expected* losses on credit sales we made in that period and record a "Bad Debt Expense." That's it. That's all. Really. It's often called the **"Income Statement method,"** because the *Bad Debt Expense* is based on the 'flow' of credit sales in that same period. Later on, in future periods, if and when particular customers declare bankruptcy, and we learn that we'll actually collect less than expected, what do we do? Under this method, *nothing*. We already "allowed for" that, factoring in expected credit losses as best as we could. Thus, I call this the **"make our best guess and keep moving method."**

- **3) The "Allowance Method"**: Under this method, during the period, if a customer declares bankruptcy, we "write off" their account at that time; and then, at the end of every reporting period, we focus our attention on correctly estimating the *balance* of an *"Allowance for Doubtful Accounts."* This *Allowance* is a contra-account to *Accounts Receivable*, and, under this method, tracks what portion of our still-outstanding accounts we expect we will *not* collect. This method is often called the **"Balance Sheet method,"** because the focus is on correctly estimating the "stock" / Balance Sheet measure of ***Net*** *Accounts Receivable*—the total amount we expect to eventually collect.

In the next section, I'll lay out a very simple example scenario and set of transactions involving credit sales, collections, and uncollectible accounts, across two reporting periods. Then, in the sections that follow, I'll show how to account for these transactions under each of the three methods, sequentially. Finally, for each method, I'll also calculate and discuss the key ***summary measures*** related to credit losses, the ***Net*** *Accounts Receivable* reported on the Balance Sheet, and the *Bad Debt Expense* reported on the income statement, to assess the implications of these methods.

8.3 Scenario

Below is the scenario we will use. MattCo is a simple company that starts its life in 2021, does annual, calendar-year reporting, and has the following events in FY '21 and '22:

- **11/10/'21**: MattCo starts up & sells $2m of textbooks to **StudentCo** on account.

- **12/01/'21**: MattCo sells $8m of textbooks to misc **other customers** on account.
- **03/03/'22**: All of MattCo's **other customers pay** off their accounts in full—$8m.
- **03/17/'22**: **StudentCo** collapses and declares **bankruptcy**.

In each case, we'll assume that MattCo has information to indicate that $9m of the credit sales / outstanding accounts made in 2021 will be collected.

Therefore, **an even more distilled summary** of this scenario is this: In the first year, MattCo makes $10m of credit sales, and has reason to believe $9m will be collected. In the second year, MattCo actually collects $8m (a 'surprise' of -$1m).

8.4 Direct Writeoff Method

As the name indicates, under this method, instead of forming an initial forecast / expectation of what portion of accounts receivable will ultimately be uncollectible, we just **"write them off" at the time** when we decide that those **particular accounts** are uncollectible. (For simplicity, in this text, the trigger for this will be the customer filing for bankruptcy.) In accounting, "writeoff" is an informal but common term for "setting an account back to zero" and/or "acknowledging that some asset no longer has value."

When we determine that some particular *Accounts Receivable* are in fact uncollectible, that means that asset must be eliminated from our books, with a *credit* to A/R. What's the balancing debit? It's a new one. We call it **"Bad Debt Expense,"** for self-explanatory reasons. So, under this method, we only account for the **sale** itself, the collections, and the **day of the writeoff**, on the days on which they actually happen—all external transactions.

Two clarifications before we formalize the rules. First, on the mechanics: 'writing off' an account does **not** mean going back and erasing the original entry. The original debit to *A/R* stands. In T-account land, the way we 'erase' something is by posting an equal and opposite entry. So a writeoff is one *more* entry—a credit to *A/R* that cancels out, going forward, the debit that's already sitting there.

Second, a question you might reasonably ask: if we recorded *Revenue* that we're never going to collect, why don't we just *reverse the revenue*? Why treat the loss as an *expense* ('Bad Debt Expense') rather than as a kind of 'anti-revenue'? There would be nothing logically impossible about a system designed that way. But here's the thinking behind the actual convention: under the **revenue recognition** principle, we record revenue when it is *earned*—and we really did earn it. We delivered the goods; we performed our side of the contract. The fact that the customer later fails to pay is treated as a **cost of doing business on credit**—an expense—rather than as evidence that the

original revenue was illegitimate. And remember that, conceptually, and in their effect on bottom-line net income, **expenses *are* just anti-revenues**—so the distinction is ultimately a bit semantic.[3]

8.4.1 Direct Writeoff Method Rules

Summarizing, we can reduce the *Direct Writeoff Method* to these **Rules**:

- **Rule #1**: Account for both *credit sales* and *collections* like normal, like we've done so far. (That is, on the date of the credit sale, debit *A/R* and credit *Revenue*. On the date of the collection, debit *Cash*, credit *A/R*.)
- **Rule #2**: If and when we determine that some customer will not pay (triggered by the customer declaring bankruptcy), "write off" that account by crediting *A/R*, and debiting *Bad Debt Expense* for that amount.
- **Rule #3**: Since, under this method, we "wait and see" and just account for collections and writeoffs when they happen, there is **no related adjusting entry**.

8.4.2 Direct Writeoff Method Accounting

Therefore, the journal entries for the scenario above would simply be:

Accounts Receivable (+A):	$2m	
Revenue (+S.E.):		$2m

11/10/'21 Credit Sale to Student Co.

Accounts Receivable (+A):	$8m	
Revenue (+S.E.):		$8m

12/01/'21 Credit Sale to Other Customers

Cash (+A):	$8m	
Accounts Receivable (-A):		$8m

3/03/'22 Collection on Other Customers' Accounts

[3] Spiceland's *Financial Accounting*, a standard textbook, offers a framing worth thinking about: your *revenue* is the full amount you have *earned* and are entitled to; your *expenses* are all the things you expect to come between that amount and your bottom-line *net income*. Credit losses fit the latter.

Bad Debt Expense (-S.E.):	\$2m	
Accounts Receivable (-A):		\$2m

3/17/'22 Write Off StudentCo

Now, a subtle but important point about **timing**. That \$2m *Bad Debt Expense* hits the books on 3/17/2022—so it reduces **2022's** net income. You might wonder: shouldn't it, somehow, go back and reduce the income we reported in *2021*, when we made the sale? It's a natural thought, but no. This is how periodic financial reporting works: we issued our 2021 financial statements, and then they were *done*. Barring fraud or some extraordinary restatement, companies do not go back and rewrite prior years' statements when new information arrives. StudentCo's bankruptcy is a 2022 *event*, and, under this method, its full effect lands on the 2022 income statement. Hold onto this point—it is precisely what's wrong with this method, as we'll now see.

8.4.3 Evaluating the Direct Writeoff Method

The direct writeoff method is not allowed under GAAP for financial reporting purposes (though can be used for tax reporting). Why not? As the simple example above shows, **this method fails on Goal #1, above.** We should *expect* that some of our credit sales will not be collected. And this system **does not account** for that, in the period in which those sales are made. Therefore, it is a **violation of the matching principle**.

To see how this could be abused, imagine you're a manager who wants this year's reported profits to look as good as possible. Under the direct writeoff method, one way to do it would be to **ship a lot of goods to customers who are terrible credit risks**. You ship the goods, send the invoices, and recognize revenue in full—with no expense recognized for the risk you're taking on. This period's income looks great. Next period, when those customers inevitably fail to pay, all of the bad news arrives. You've front-loaded the good news and back-loaded the bad. (We previewed this very gimmick back in Chapter 1, when motivating the Cash Flow Statement.)

Quantifying, the key measures related to uncollectible accounts—*Net Accounts Receivable* and the *Bad Debt Expense*—using this method would be:

	Bad Debt Expense	*Net Accounts Receivable*
FY 2021	\$0m	\$10m
FY 2022	\$2m	\$0m

In **FY '21**, under this method, the ***Net Accounts Receivable*** we would report would be the full **\$10m**—because we debited *A/R* for all of those sales in full, without making

an allowance for expected credit losses. Relatedly, the ***Bad Debt Expense*** in FY '21 would be **$0**, so all $10m from those credit sales would directly count toward bottom-line net income in that year. Therefore, because we did not account for expected credit losses, we arguably overstated our net assets and net income in FY '21.

8.5 I/S Teaching Model: Percentage-of-credit sales

Under this method, in each period, the company records a *Bad Debt Expense* as a simple **percentage** of its total **credit sales in that period**. The matching credit is to a new account called the ***"Allowance for Doubtful Accounts."*** This account is a **contra-account** to *Accounts Receivable*. That is, it is a separate account that tracks a *"negation"* in the value of the *A/R*, "allowing for" the fact that not all of those accounts will be paid.

Now, we need a bit of **jargon**: We use the term **Net Accounts Receivable** to indicate the **total Accounts Receivable minus the total Allowance**. (This is a common pattern in accounting: **'Net'** indicates the balance of an account *less* the balance of some related reduction, such as its contra-account.) Formally:

$$Net\ Accounts\ Receivable = Accounts\ Receivable\ (balance) - Allowance\ (balance) \tag{8.1}$$

And, in fact, it is this number, the ***Net*** *Accounts Receivable*, that is actually **reported** on the Balance Sheet—a detail we omitted from the earliest chapters, to simplify.

If this contra-account setup feels familiar, it should: the *Allowance* is our **second contra-asset**, after *Accumulated Depreciation*. The parallel is exact. *Equipment* tracks the asset at historical cost, *Accumulated Depreciation* separately tracks the reductions, and the difference is the 'book value.' Here, *Accounts Receivable* tracks the gross amounts customers owe us, the *Allowance* separately tracks the portion we expect never to see, and the difference—*Net Accounts Receivable*—is the amount we actually expect to collect. One difference in reporting practice, though: whereas companies often present *Equipment* and *Accumulated Depreciation* as separate lines on the Balance Sheet, receivables typically appear only as the single, netted number.

8.5.1 I/S Method Rules

Now, let's lay out the rules formally, and then apply them to the scenario above. Here are the rules for the deliberately simplified *Percentage-of-Credit Sales* model used in

this book:[4]

- **Rule #1**: During the reporting period, account for both *credit sales* and *collections* just like we've done so far. Don't overthink it and worry about expected credit losses just now—we'll deal with that as an adjusting entry.

- **Rule #2**: At the end of the reporting period, **as an adjusting entry**, account for expected credit losses as a **given percentage of the credit sales made during that period**, debiting *Bad Debt Expense* and crediting the *Allowance*.

- **Rule #3**: During the period, if and when our customers declare bankruptcy, **do nothing**. We do not update our *Net Accounts Receivable* or *Bad Debt Expense*. We already "allowed for" expected losses. We did our best (shrug), and keep moving.[5]

So, distilling those rules down even further, **in short, we just record a *Bad Debt Expense* and add to the *Allowance* based on the flow of credit sales each period.**

It really is that simple. And yet, every year, students get these problems wrong—not because the method is hard, but because they don't *believe* it can be this simple. A problem will mention beginning *A/R* balances, collections on past accounts, customers declaring bankruptcy—and their confidence gets shaken, and they start 'adjusting' for things this method tells them to ignore. Under this method, computing the *Bad Debt Expense* requires exactly **two** pieces of information: the **credit sales made this period**, and the **assumed percentage**. Everything else is a red herring. To test yourself: under the percentage-of-credit-sales method, all else equal, the greater our *collections on past periods' accounts* during this period, will our *Bad Debt Expense* be higher, lower, or unchanged? The answer: **unchanged**. Economically, collecting more of last year's accounts is good news, and your instinct may be that it should reduce this period's expense somehow. But under this method, we made our best guess and kept moving—collections on old accounts enter the calculation nowhere.

[4] Also, test your knowledge of the rules in general with this practice problem: Our company accounts for bad debts / receivables using the **percentage-of credit-sales method**, assuming 15% of credit sales will be uncollectible. We began FY2020 with a balance of $150k in Accounts Receivable. During FY2020, we made an additional $100k in credit sales, and collected $60k on accounts from past credit sales. Customers with accounts totaling $20k declared bankruptcy. What would be our FY2020 Bad Debt Expense?

[5] Note: Some other texts teach that, under the percentage-of-credit-sales method companies do account for writeoffs by reducing both *A/R (A)* and the *Allowance (contra-A)*. However, given that companies only report *Net Accounts Receivable* (total *A/R* minus the *Allowance*), and, under, the I/S method, the end-of-period adjusting entry for the *Bad Debt Expense* and update to the *Allowance* is only based on the same period's incremental credit sales, this adjustment is nominal, with no substantive impact on the key reported measures, the *NAR* or *Bad Debt Expense*. Therefore, this question is purely procedural, and all the key substantive ideas here—e.g., showing the income-statement method does not factor "surprise" (unexpected) collection into the next period's BDE or NAR—hold regardless of which procedure is used.

8.5.2 Where does the percentage come from?

In this text, the percentage of credit sales assumed uncollectible will simply be *given* to you—it's a parameter, call it *p*, and your job is to apply the method correctly and interpret the results. But you might reasonably ask the deeper, real-world question: how do companies actually come up with that number?

This is a forecasting problem. A company can use its historical collections, information about its customers, and relevant industry experience. The estimate is not certain, but it should have a basis that can be explained and compared with subsequent results. In the examples below, we will give the percentage so that we can concentrate on what the method does with it.

A natural follow-up: given that this is an estimate, couldn't managers just pick whatever number flatters this year's income? There is genuine **discretion** here—just as there was in the useful-life assumptions behind depreciation—and this is one of the inherent features (and risks) of accrual accounting. But the discretion is not unlimited. Neither a 0% nor a 100% loss estimate should be assumed without evidence; either can be appropriate for particular receivables when the facts support it. And a company's **auditors**, who typically have deep experience with the industry, must sign off on the assumption—if a firm's actual losses keep running at 30% while it keeps assuming 10%, the auditors will, eventually, force a change.

8.5.3 I/S Method Accounting

Now, let's revisit our scenario above, and account for it under the **percentage-of-credit-sales method**, with the **assumption that 10% of credit sales** will be uncollectible. As a refresher, here's our scenario:

- **11/10/'21**: MattCo starts up & sells $2m of textbooks to **StudentCo** on account.
- **12/01/'21**: MattCo sells $8m to misc **other customers** on account.
- **03/03/'22**: MattCo's **other customers pay** their accounts in full—$8m.
- **03/17/'22**: **StudentCo** collapses and declares **bankruptcy**.

So, the journal entries during the period in 2021 would be:

Accounts Receivable (+A):	$2m	
Revenue (+S.E.):		$2m

11/10/'21 Credit Sale to Student Co.

Accounts Receivable (+A):	$8m	
Revenue (+S.E.):		$8m

12/01/'21 Credit Sale to Other Customers

Now, it's time for our December 31 adjusting entry, before we close the books. We made $10m of credit sales this period, and we're told to assume that 10% will be uncollectible. So the adjusting entry would just be...:

Bad Debt Expense (-S.E.):	$1m	
Allowance for Doubtful Accounts (+contra-A):		$1m

12/31/'21 Adjusting Entry for Expected Bad Debts

Now, move on to 2022. First, we collect on the other customers' accounts:

Cash (+A):	$8m	
Accounts Receivable (-A):		$8m

3/03/'22 Collection on Other Customers' Accounts

Now, on March 17, StudentCo declares bankruptcy. What do we do? Under this method: Nothing. Under this deliberately simplified teaching model, we omit write-offs. That is not the rule for current financial reporting. We *already* allowed for the *expected loss*. So, we don't update our *Net Accounts Receivable* or *Bad Debt Expense* now. We already **made our best guess, so we keep moving**. So, after March 17, 2022, our *Accounts Receivable* T-account and its related *Allowance* will still look like this:

Accounts Receivable (A)

Debits	Credits
11/10/'21: $2m	03/03/'22: $8m
12/01/'21: $8m	
Bal: $2m	

Allowance (contra-A)

Debits	Credits
	12/31/'21: $1m
	Bal: $1m

That's all for this method. There's no FY22 adjusting entry required, because there were no new credit sales during 2022—and, at the risk of repetition, under the *percentage-of-credit-sales* method, the *Bad Debt Expense* and the credit to the *Allowance* are only and solely determined by the credit sales made during that period. Therefore, those are the final account balances for FY22, and *Net Accounts Receivable* is $1m.

8.5.4 Evaluating the Percentage of Credit Sales Method

We already verbally previewed the problems with this method. Now let's add rigor by quantifying the key measures related to uncollectible accounts that would be reported in each period, under this method: the *Bad Debt Expense* (the 'flow' / Income Statement measure) and the *Net Accounts Receivable* (the 'stock' / Balance Sheet measure).

As a reminder, the scenario here was:

- **11/10/'21**: MattCo starts up & sells $2m of textbooks to **StudentCo** on account.
- **12/01/'21**: MattCo sells $8m to misc **other customers** on account.
- **03/03/'22**: MattCo's **other customers pay** their accounts in full—$8m.
- **03/17/'22**: **StudentCo** collapses and declares **bankruptcy**.

And, gathering from the above, the key measures under this method, with 10% of credit sales assumed uncollectible, would be:

	Bad Debt Expense	*Net Accounts Receivable*
FY 2021	$1m	$9m
FY 2022	$0m	$1m

So, what do we think? Does this method make sense?

Clearly, this method initially **succeeds on Goal #1**. We account for expected credit losses in FY21, recording a $1m *Bad Debt Expense*. But we clearly **fail on Goal #2**. We expected only $1m in credit losses. But in FY22, our customers with accounts worth $2m went bankrupt. So, it now looks like our actual realized credit losses will, in fact,

ultimately be $2m, a "surprise" of -$1m, but we don't make any adjustment or update for that. Across **both periods combined**, FY21 and FY22, our ***total*** *Bad Debt Expense* is $1m.

Turning to the Balance Sheet, as of the end of FY22, the *Net Accounts Receivable* on our books is still $1m—even though all of our customers have either paid or declared bankruptcy. Because we don't make adjustments, under the *Percentage-of-Credit Sales* method, *Net Accounts Receivable* became obviously misstated by the end.

This stylized example shows what happens if we never revise the original expense estimates. It is precisely why current reporting requires the allowance to be updated. The percentage-of-credit sales method is often called the **"Income Statement method"** because it gives us a reasonable measure of the *expense* from expected credit losses each period, matched to those same period's sales. But, precisely because it does not factor in subsequent "surprise" collections and losses, the "stock" measure of *Net Accounts Receivable* on the Balance Sheet—the amount of our outstanding accounts that we expect to collect—becomes misstated over time.

This brings us to our next method, the *Allowance Method*.

8.6 The B/S Method / The Allowance Method

This method is harder to summarize verbally. Broadly, this approach, in contrast to the previous, is often referred to as the **"Balance Sheet Method,"** because we focus our attention on correctly estimating, updating, and reporting *Net Accounts Receivable*—the Balance Sheet / "stock" measure. But, as we know, Balance Sheet and Income Statement are **intimately linked**. So, under this method, the *update* to the *Allowance* that we make at the end of the period will determine our *Bad Debt Expense*. And that number, it turns out, will *simultaneously* incorporate the effects of (i.) expected losses on new accounts and (ii.) "surprise" collections on past credit sales.

Understanding exactly *how* is hard. So, let's first lay out the **rote rules**.

8.6.1 B/S Method Rules

- **Rule #1**: During the reporting period, account for both *credit sales* and *collections* just like we've done so far. Don't overthink it—we'll factor in expected uncollectible accounts as an adjusting entry, at the end of the period.

- **Rule #2**: During the period, when our customers declare bankruptcy, account for this by **extinguishing the account** (that is, crediting *Accounts Receivable*) and "using up" the *Allowance* we made (as the matching debit).

- **Rule #3**: At the end of the reporting period, estimate what *remaining outstanding accounts* we expect not to collect. **Adjust** the *Allowance for Doubtful Accounts* **balance** to that number. The matching debit is the *Bad Debt Expense*, by definition.

- **Rule #4**: Obey rules #1 - #3. Resist the temptation to stray. The most common mistake I see from students comes from "overthinking it"—in particular, students try to account for the *concepts* of expected losses on each credit sale, i.e., during the reporting period. Don't. Trust the process. The "mechanics" of the adjusting process will take care of that for us, if we just follow Rules #1-#3.[6]

8.6.2 Scenario and Assumption: Percentage-of-Receivables

Now, let's account for our scenario using this method. As a refresher, it is:

- **11/10/'21**: MattCo starts up & sells $2m of textbooks to **StudentCo** on account.
- **12/01/'21**: MattCo sells $8m to misc **other customers** on account.
- **03/03/'22**: MattCo's **other customers pay** their accounts in full—$8m.
- **03/17/'22**: **StudentCo** collapses and declares **bankruptcy**.

Now, let us suppose our company **assumes that 10% of outstanding accounts** at the end of each period will be **uncollectible**. That is, it estimates the allowance based on a simple **percentage-of-receivables**. (Later on, we'll find that companies can use other, more complicated methods for estimating the uncollectible amounts.)[7]

8.6.3 B/S method accounting

The external transactions in 2021 would be:

Accounts Receivable (+A):	$2m	
Revenue (+S.E.):		$2m

11/10/'21 Credit Sale to Student Co.

[6] The first rule of Fight Club is: don't talk about Fight Club. The second rule of Fight Club is: *don't talk about Fight Club.* Rule #4 is in that spirit.

[7] Note that, in this simple toy example, of a firm that starts its life and just makes two credit sales (and has no collections) in 2021, the amount of *outstanding receivables* at the end of the period ($10m) is the same as the amount of credit sales made during that period. But, this equality will not hold in general—usually, the total ***outstanding*** *Accounts Receivable* (a stock amount) will not be the same as the flow of credit sales made in that period. And so, assuming 10% of the stock of outstanding receivables will prove uncollectible (under the B/S method) is a substantively different assumption than assuming that 10% of the flow of credit sales will be (under the I/S method).

Accounts Receivable (+A):	$8m	
Revenue (+S.E.):		$8m

12/01/'21 Credit Sale to Other Customers

Now, it's time for our December 31 adjusting entry, before we close the books on 2021. We know that we are assuming that 10% of outstanding accounts will be uncollectible. And, under this method, we simply need to adjust our *Allowance* to that balance. And the current balance of the *Allowance*, before the adjusting entry, is $0m, since the company just started its life, and none of the transactions to date involved the *Allowance*. Our current balance of *Accounts Receivable* is $10m, meaning that the **new ending balance** of our *Allowance* must be $1m.

So, in sum, we need to **update the *Allowance*** from $0 to $1m. And, the matching debit is the *Bad Debt Expense*, by definition. So, the adjusting entry would be:

Bad Debt Expense (-S.E.):	$1m	
Allowance for Doubtful Accounts (+contra-A):		$1m

12/31/'21 Adjusting Entry for Allowance for Uncollectible Accounts

So, for 2021, so far, things are the same as they were under the percentage-of-credit-sales method. But they won't stay that way. Now, move on to 2022.

First, we collect on the other customers' accounts:

Cash (+A):	$8m	
Accounts Receivable (-A):		$8m

3/03/'22 Collection on Other Customers' Accounts

Then, StudentCo declares bankruptcy, so we now can be reasonably confident that they will not actually pay their $2m account. Under the **percentage-of-credit sales** method, we did nothing. But under the *Allowance* method, we do something. We "write off" the account by (i.) extinguishing (crediting) *Accounts Receivable*, and (ii.) "using up" the *Allowance*, as the matching debit. (Recall that the *Allowance* is a contra-asset account, so it is a credit account, so reducing the *Allowance* is a debit.)[8]

[8] A major customer default can require material disclosure, but Form 8-K is triggered by specified reporting items, not by every event investors might consider important. Whether a particular default requires an 8-K depends on the facts and the applicable item; companies can also report other events voluntarily.

Allowance for Doubtful Accounts (-contra-A):	$2m	
Accounts Receivable (-A):		$2m

3/17/'22 Write Off StudentCo

Before moving on: *why* does it make sense to debit the *Allowance* at the moment of the writeoff? Here's the intuition. Last year, when we set up the *Allowance*, we were saying, "we expect about $1m of these receivables to eventually go bad." The *Allowance* is a **provision**—a bucket of expected losses, set aside in advance. Now a specific loss has actually materialized; we had (in part) anticipated exactly this kind of event. So we **draw down** the provision we already made—we take this particular loss out of the bucket we had set aside for it. That's what the debit to the *Allowance* represents: 'using up' expected losses that we had already allowed for.[9]

Now, finally, it's December 31, 2022, time to make the adjusting entries and close the books for the year. Under the *Allowance* method, we need to make an adjusting entry to update the *Allowance* to whatever it needs to be (even though we haven't made new credit sales). It's best to be careful with these problems, and draw out the T-accounts, to make sure we're not forgetting anything. Here is what the T-account for *A/R* and the *Allowance* would look like at the end of the reporting period ***before*** making the required adjusting entries:

Accounts Receivable (A)

Debits	Credits
11/10/'21: $2m	03/03/'22: $8m
12/01/'21: $8m	03/17/'22: $2m
Bal: $0	

Allowance (contra-A)

Debits	Credits
03/17/'22: $2m	12/31/'21: $1m
Bal: $1m	

Since the balance of *Accounts Receivable* is now 0 (all customers have either paid or been written off), the new balance of the *Allowance* must be zero as well. It currently

[9] An analogy: a professor who builds one 'freebie' absence into the attendance policy has, in effect, made an *allowance*—a provision—for the fact that something will inevitably come up for everyone during a semester. When a student uses that freebie, nothing new gets 'charged' against their grade; they're just drawing down a provision that was already made.

has a debit balance of $1m, after the previous period's adjusting entry and this period's writeoff. So, for our adjusting entry, we need to get it to a balance of $0 *via* a $1m credit. And the matching debit is the *Bad Debt Expense*, by definition. So, coming to our conclusion, our adjusting entry would be...:

Bad Debt Expense (-S.E.):	$1m	
Allowance for Doubtful Accounts (+contra-A):		$1m

12/31/'22 Adjusting Entry to update the Allowance

Now, you might ask: where did that $1m *Bad Debt Expense* 'come from'? We didn't make a single credit sale in 2022. Mechanically, it came from the adjusting entry—the credit required to bring the *Allowance* back to where it needed to be, after the writeoff had 'overdrawn' it. Conceptually, it is the **bad news**: we provisioned for $1m of write-offs, reality delivered $2m, and the difference forced its way onto the income statement in the period in which the news became clear. (More on this in the evaluation below.)

If the *Bad Debt Expense* feels abstract under this method, that's because it *is* abstract. With *Supplies*, the expense is easy to visualize: you can watch the stack of printer paper shrink, and the *Supplies Expense* is the amount used up. Under the *Allowance* method, the only reliable way to think about the *Bad Debt Expense* is from the **Balance Sheet perspective** we developed in Chapter 7: first pin down where the balance-sheet accounts must *end up* (here, an *Allowance* of $0—with no outstanding accounts left, there is nothing left to be doubtful about), and then the expense is, by definition, **whatever adjustment gets you there**. It's the same logic as *Supplies*—if *Supplies* must end at $60k, the *Supplies Expense* is whatever reduction gets you from the beginning balance to $60k—just one level more abstract, because the account being adjusted is an expectation about the future rather than a stack of paper.

8.6.4 Evaluating the B/S Method

In order to understand what happened here, and interpret this method, let's once again quantify the key measures related to uncollectible accounts in each period, under this method: the *Bad Debt Expense* (the 'flow' / Income Statement measure) and the *Net Accounts Receivable* (the 'stock' / Balance Sheet measure).

As a reminder, the scenario here was:

- **11/10/'21**: MattCo starts up & sells $2m of textbooks to **StudentCo** on account.
- **12/01/'21**: MattCo sells $8m to misc **other customers** on account.
- **03/03/'22**: MattCo's **other customers pay** their accounts in full—$8m.

- **03/17/'22**: **StudentCo** collapses and declares **bankruptcy**.

And, gathering from the above, the key measures under the **Allowance Method, assuming 10% of outstanding accounts are uncollectible,** would be:

	Bad Debt Expense	*Net Accounts Receivable*
FY 2021	\$1m	\$9m
FY 2022	\$1m	\$0m

So, what do we think? Does this method make sense?

The pros: There are some really cool, really good things about this method. First, this method definitely achieves **Goal #1** in FY2021: We account for *expected* credit losses on the credit sales we made. Second, in FY2022, this method achieves **Goal #2**—we make an adjustment for the fact that, contrary to our initial expectations, the **total uncollectible accounts** for the periods turned out to be **\$2m**. And, consistent with that, the ***total*** *Bad Debt Expense* over the two periods, combined, is **\$2m**.

This is, in my view, the coolest thing and deepest insight about the *Allowance Method*. Under the *Allowance Method*, the *Bad Debt Expense* **simultaneously** factors in both (i.) expected losses on new accounts, and (ii.) "surprise" writeoffs, contrary to our previous periods' expectations, on past credit sales. In FY 2021, the \$1m *Bad Debt Expense* is driven by the former. In FY 2022, it's driven by the latter.

That's the concept. But how precisely did that work out, in the accounting mechanics? Under the *Allowance Method*, "news" about actual realized collections, worms its way into the *Bad Debt Expense* indirectly. When we have a writeoff, that "uses up" the *Allowance* (reduces it with a debit). And, under the Allowance method, we need to get the balance of the Allowance to where it needs to be at the end of the period—and that number determines the *Bad Debt Expense*, by definition. That is, the greater our **write-offs** during the period, the greater the reductions in the *Allowance* during the period (per **Rule #2)**, the bigger the credit to the *Allowance* in the end.

The cons: But there's a dark side, too. As the table above noted, the *Bad Debt Expense* for FY 2022 was \$1m. This is despite the fact that the company did not make any credit sales—or, indeed, any *Revenues* at all—in this period. That \$1m *Bad Debt Expense* came from the "surprise" writeoffs during the period. Is that a good thing? Some people think it is, that it means that this method achieves **Goal #2**, above.

But I am not sure it is, for the following reason: That expense is not well "matched" to the related revenues—the expense is recorded in 2022, even though there are *no sales at all* in FY2022. So, that seems economically illogical. It also has practical implications: If each year's *Net Income* is being affected not just by that same year's sales,

but also by the *"surprise" collections and writeoffs of past years' sales as well*, that can make the bottom-line profit more **volatile and opaque—harder for us, as financial-statement users, to understand and forecast**.

Indeed, under the *Allowance* method, we can even have a ***negative** Bad Debt Expense*—that is, boosting *Net Income*—if the positive "surprise" collections on past periods' accounts outweigh the expected losses on new credit sales.

For example, consider the example scenario we used above, but now suppose that, instead, StudentCo paid off all $2m of its account on 3/17/'22. Then, in that case, there would have been no $2m debit to the *Allowance* during the period, so its balance, prior to adjustment, would be a *credit* of $1m. But, since all accounts had been paid off, the ending balance of the *Allowance* would still need to be $0. So, the required adjusting entry would be a *debit* of $1m to the *Allowance*, meaning a *credit* of $1m to the *Bad Debt Expense*—a negative expense.

This does not seem to make economic sense, but it can happen in companies' accounting, and has happened in reality, under the allowance method.

8.6.5 The general form of the Allowance

One way to see all of this more formally (and get some nice shortcut formulas for solving quantitative problems) is with the "general form" of the T-account for the *Allowance*, under the assumption that a percentage (p) of outstanding receivables (AR) will prove uncollectible.

Allowance T-account

Allowance for Doubtful Accounts (contra-A)	
$Writeoffs_t$	Beg. Bal: $(p * AR_{t-1})$
	12/31: $Bad\ Debt\ Expense_t$
	End. Bal.: $(p * (AR_t))$

This nicely summarizes what drives the *Bad Debt Expense* under the *Allowance Method*. By definition, the 12/31 adjusting entry is what determines the *Bad Debt Expense*. And, as this T-account shows, that number, in turn, will be determined by three other factors:

1. The outstanding receivables at the end of this period, AR_t, and the percentage (p) that we expect not to collect.

2. Vs. the beginning balance of the allowance, which is last year's ending balance ($p * AR_{t-1}$).

    1. That is, the *change* in outstanding accounts receivable, from last period.

3. The writeoffs during the period, which "use up / reduce the *Allowance*, and therefore influence the required adjusting entry. The higher those quantities, the higher the required adjusting entry on 12/31, the higher the *Bad Debt Expense*.[10]

Equations

Or, generalizing a bit further, we can express the same idea in algebraic form. First, note from the T-account above that the beginning balance of the Allowance ($p * AR_{t-1}$) minus the $Writeoffs_t$ plus the ($Bad\ Debt\ Expense_t$) equal the ending balance. Or, written algebraically:

$$p * AR_{t-1} - (Writeoffs_t) + (Bad\ Debt\ Expense_t) = p * AR_t \tag{8.2}$$

Next, isolating the *Bad Debt Expense* on the left side, we have:

$$Bad\ Debt\ Expense_t = p * AR_t - p * AR_{t-1} + Writeoffs_t \tag{8.3}$$

Next, we can rewrite that in terms of the *change* in outstanding *Accounts Receivable* between the two periods:

$$Bad\ Debt\ Expense_t = p * (\Delta AR) + Writeoffs_t \tag{8.4}$$

The formula above may be the simplest quantitative formula to memorize for solving quantitative problems. But we can build better *understanding* by further ***decomposing*** the change in *Accounts Receivable*, into the underlying events that drive it. A/R is a "stock" / Balance Sheet measure, and it *increases* with new credit sales, and decreases with collections and writeoffs (under the Allowance method). Therefore:

$$\Delta AR = Credit\ Sales_t - Collections_t - Writeoffs_t \tag{8.5}$$

So, with that decomposition, we can plug in for ΔAR and rewrite our previous for-

[10] To test your confidence and understanding of that T-account above, try the following **practice problem**: Our company accounts for bad debts / receivables using the **allowance method**, assuming 10% of outstanding accounts will be uncollectible. We began FY2020 with a balance of $50k in Accounts Receivable and $5k in our Allowance for Doubtful Accounts. During FY2020, we made an additional $30k in credit sales, collected $20k on accounts from past credit sales, and wrote off $3k in accounts. What would be our FY2020 Bad Debt Expense?

mula like so:

$$Bad\ Debt\ Expense_t = p * (Credit\ Sales_t - Collections_t - Writeoffs_t) + Writeoffs_t \tag{8.6}$$

And, finally, gather terms and simplify **to this formula for a constant loss percentage, with no recoveries or other adjustments**:

$$Bad\ Debt\ Expense_t = p * (Credit\ Sales_t - Collections_t) + (1-p)Writeoffs_t \tag{8.7}$$

That formula above, if you can memorize it, provides a very handy shortcut for solving quantitative problems (instead of making multiple T-accounts). And, it also beautifully **distills** how different events affect the *Bad Debt Expense* under the allowance method. As the formula above shows, **all else equal**, the higher our new credit sales during the period, the higher our *Bad Debt Expense* will be. The higher our collections during the period, the *lower* it will be. And the greater our writeoffs during the period, the higher it will be. So, this equation shows how, under the Balance Sheet method, the *Bad Debt Expense* simultaneously factors in both (a.) the effect of new credit sales during this period and (b.) "news" about collections vs. writeoffs from past accounts.

8.6.6 Checking the formula against the mechanics

Formulas are only trustworthy if you can see them agree with the underlying accounting. So let's do one problem both ways—once with the full T-account mechanics, and once with the formula—and confirm that they land in the same place.

Problem: Our company uses the Allowance method, assuming 10% of outstanding accounts will be uncollectible. We begin the period with \$100k in *Accounts Receivable* (and thus, per our assumption, a \$10k credit balance in the *Allowance*). During the period, we make \$50k in new credit sales, collect \$40k on past accounts, and write off \$5k in accounts. What is the *Bad Debt Expense*?

The mechanics: First, *Accounts Receivable*: $\$100k + \$50k - \$40k - \$5k = \$105k$ outstanding at the end of the period. Under our assumption, the *Allowance* must therefore end at 10% of \$105k—**\$10.5k**. Where does the *Allowance* stand *before* adjustment? It began at \$10k (credit), and the \$5k writeoff debited it down to \$5k. So the adjusting entry must be a **\$5.5k credit** to the *Allowance*, taking it from \$5k to \$10.5k—and the matching debit is the *Bad Debt Expense*: **\$5.5k**.

The formula: Recall that $Bad\ Debt\ Expense_t = p * (Credit\ Sales_t - Collections_t) + (1-p)Writeoffs_t$. Plugging in: $.1 * (\$50k - \$40k) + .9 * (\$5k) = \$1k + \$4.5k = \$5.5k$. The two

paths agree, as they must.

A variation: Now change the facts: same starting position (\$100k in *A/R*, \$10k *Allowance*), same \$50k of new credit sales—but this period, miraculously, we collect the *entire* \$100k of past accounts, with no writeoffs at all. Ending *A/R* is $\$100k - \$100k + \$50k = \$50k$, so the *Allowance* must end at \$5k. But nothing has touched the *Allowance* during the period (no writeoffs), so it still sits at \$10k. The adjusting entry must therefore be a \$5k *debit* to the *Allowance*—and the matching *credit* goes to *Bad Debt Expense*. There it is again: a **negative** *Bad Debt Expense*, which we first met in the StudentCo variation above—but this time it emerges even while the company keeps making new credit sales, because the good news about past accounts outweighs the expected losses on the new ones. (Check it against the formula: $.1 * (\$50k - \$100k) + .9 * (\$0) = -\$5k$.)

One warning, though: not every problem can be solved with the formula. Sometimes you'll be given only, say, the pre-adjustment balance of the *Allowance* and the ending *Accounts Receivable*. In that case, go straight to the definition: compute the required ending *Allowance* from the ending *A/R*, compare it to the pre-adjustment balance, and the difference *is* the *Bad Debt Expense*. The formula is a shortcut; the mechanics are the method.

8.7 Debating the methods: The Balance Sheet vs. Income Statement Approach

While the above were toy examples, they illustrate the general problems and tradeoffs, which are a source of real debate and consternation, even among non-accountants. Current U.S. reporting requires an updated estimate of expected credit losses, and it is common for Wall Street investors to complain about how "opaque" the *Bad Debt Expense* is as a result. This gets at a major theme in accounting, though a bit advanced: the tradeoff in **Income Statement vs. Balance Sheet principles and emphases**.[11]

Broadly, under "balance sheet perspectives," we focus on measuring and reporting the net assets of the firm "correctly" *per se*. But, as we know, via the core linkages and identities of accounting, changes in asset and liability accounts "flow through" to revenues and expenses, and *vice versa*. (This is especially palpable in the Allowance

[11] This debate is a favorite essay prompt of mine. Two questions worth writing out full answers to, in complete sentences: (1.) Compare and contrast the percentage-of-credit-sales method and the Allowance method. What are the strengths and limitations of each? Why is one called the 'Income Statement method' and the other the 'Balance Sheet method'? Which do you think is better overall? (2.) Under the Allowance method, we never debit *Bad Debt Expense* at the moment of a writeoff—and yet, all else equal, the higher our writeoffs during a period, the higher our *Bad Debt Expense* ends up. Explain precisely how that works.

method—the *Bad Debt Expense* **is**, by definition, the **revision in the Allowance**.) However, given those linkages, if we prioritize measuring assets and liabilities "correctly" each period, then the *revisions* in estimates of those assets and liabilities will *flow through* to revenues and expenses, and impact *Net Income*, even when they do **not** have to do with any changes in the company's real **operations and activities during that period.**

This is a general tradeoff in accounting, and it shows up in a lot of places, not just the methods for accounting for *Uncollectible Accounts*. (Indeed, the real reason I give such a thorough discussion of these methods is that they serve, in this book, as a *synecdoche* for this broader, more general tradeoff and debate.)

If we prioritize measuring the net assets that companies acquired in previous periods "correctly," then we will likely make *Net Income* less *persistent and predictable* (due to the downstream effects of those asset and liability re-valuations) and less informative about the company's current and recurring operations and activities.

But, if we eschew such revisions, in order to keep Net Income smoother and better "matched" to the same period's activities, then our net assets on the Balance Sheet will become misstated over time. It's an inevitable tradeoff, at least as far as I can tell. **We can't have it all.** The universe is tragic.

I lean toward income-statement perspectives. In my view, it's not a big problem if *Net Accounts Receivable* gets a bit misstated over time, because, who really cares about the Balance Sheet "stock" accounts? What matters, in my view, is having a rational, predictable, understandable measure of the profitability of the company's current and ongoing activities. And I think the *percentage-of-credit sales method* does better on that. Current U.S. standards require the allowance to reflect updated loss expectations. My argument here concerns the information an analyst would like to see, not permission to disregard that requirement. And I've changed my mind on this question each time I've taught it. I'm curious whether a reader will change it again.

8.7.1 Going deeper: Why an analyst cares about the components of expense

My sympathy for the income-statement perspective comes from valuation. To estimate what a business may earn in the future, an analyst begins with what it earns now and asks which parts are likely to persist.

A reported Bad Debt Expense can combine expected losses associated with current sales and revisions of expectations about older accounts. Both pieces can matter. But they answer different questions, and a reader who sees only the total may have trouble separating them.

That is the attraction, to me, of a charge whose relationship to this period's sales is easy to interpret. It is also why I want you to understand the allowance mechanics rather than merely calculate the expense: you can ask what drove the number. Was it new business, a revised estimate, or news about older customers?

My preference does not eliminate the other demand on the accounts: receivables should tell readers something meaningful about what remains collectible. The question is how best to convey both kinds of information. I have changed my mind about the balance before; the argument is here for you to examine, not to memorize.

8.7.2 The case for 'best guess and keep moving'

To understand my position, and the argument in favor of the income-statement method, consider a somewhat more realistic scenario (as opposed to the toy example above, with only two periods, one in which all the credit sales were made). Instead, consider this new scenario, in which a company keeps making new credit sales and new collections each period, and its estimates are off by small amounts each time.

8.7.3 A more realistic scenario

Assumption: MarshallCo uses the percentage-of-credit sales method, and assumes 10% of new credit sales each period will be uncollectible.

- **2019**: MarshallCo starts up its life and sells $1m on account.
- **2020**: MarshallCo (i.) collects all $1m from 2019, and (ii.) makes an additional $1m in new credit sales
- **2021**: MarshallCo (i.) collects $800k from 2020, while the customers with the other $200k in accounts go bankrupt, (ii.) makes $1m in new credit sales.

Distilling and summarizing verbally: In each period, MarshallCo makes $1m in new credit sales and assumes 10% will be uncollectible. And in 2020 it collects a bit more than expected on the previous period's credit sales, while in 2021 it ends up collecting a bit less than expected.

The rule under the percentage-of-credit sales method is that we just record *Bad Debt Expense* and add to the *Allowance* based only on "flow" of new credit sales each period. Therefore, quantifying the key measures, under this method.

	Bad Debt Expense	*Net Accounts Receivable*
FY 2019	$100k	$900k
FY 2020	$100k	$800k
FY 2021	$100k	$900k

This, more realistic, example illustrates the argument in favor of the income statement method. Reality will always play out a bit differently than we expect. But, if we're forming our expectations well, things will probably roughly even out over time—sometimes we'll collect a bit more than expected, and sometimes a bit less.

Arguably, using this income-statement method, our *Net Accounts Receivable* is a bit misstated in 2020. As of the end of 2020, we had collected all $1m in accounts from sales made in 2019, and just made another $1m in credit sales in 2020—so, arguably, *NAR* should really be $900k. Instead, applying the rules of the *percentage-of-credit-sales* method mechanically, the number is $800k (since total A/R will be $1m and the total *Allowance* will be $200k). But I argue that that's not a big deal, since (1.) who cares about the Balance Sheet measures? and (2.) it will "wash out" over time.

Let's dwell on that 2020 misstatement, because an objection can sound persuasive at first: "We collected the extra $100k, so Cash is higher than expected. Aren't total assets fine?" Cash is correctly recorded, but that does not make the receivable correct. Both methods record the same actual cash collection. At the end of 2020, the remaining $1m of receivables is expected to yield $900k. Reporting it at $800k therefore understates **both Net Accounts Receivable and total assets by $100k**. It also leaves cumulative income and equity $100k lower than under an updated allowance. The debate is about whether and when to recognize that better-than-expected collection, not about an offsetting error in Cash.

The second confusion is mechanical: since the *Bad Debt Expense* is the same $100k every year, students sometimes expect *Net Accounts Receivable* to be the same every year too—or, alternatively, to sink steadily downward as the *Allowance* piles up. Neither. *A/R* and the *Allowance* are **stock** accounts: they accumulate everything ever posted to them. By the end of 2021, *A/R* has accumulated to $1.2m (three years of $1m credit sales, less the $1m and $800k of collections), and the *Allowance* to $300k (three annual $100k credits)—so *Net A/R* is back up to $900k. And notice: as of the end of 2021, that $900k once again makes perfect economic sense. We have $1m of freshly outstanding accounts from 2021's sales, and we expect 10% of them to go bad. The 2020 wobble has already washed out.

Why did it wash out? Because reality erred in *both* directions, and the errors offset: 2020's collections came in $100k better than expected, and 2021's came in $100k worse. If our percentage assumption is well-calibrated, that is roughly what we should expect

in general. Think of a fund manager picking stocks: on any individual stock, luck dominates; across a whole portfolio, the idiosyncratic noise washes out, and what remains is the underlying quality of the picks. Likewise here: on any given cohort of receivables, our 10% will be wrong—sometimes too gloomy, sometimes too rosy—but across millions of transactions and many years, a well-calibrated *p* will produce a *Net A/R* that gravitates around the right number.[12]

And, on the positive side, we now have a measure of the *Bad Debt Expense* that is more *understandable, predictable,* and driven by *this period's activities*. In my view, having an *income statement*—and, thus, measure of *profitability*—with these desirable properties is worth the cost of slightly "misstating" the value of our Balance Sheet accounts from time to time. But, there is an inevitable tradeoff, and current U.S. reporting requirements give priority to updating expected credit losses.

8.8 Methods for estimating the allowance

In the example above, I was focusing on the big new ideas: The nature and interpretation of the three methods, the tradeoffs, the intuition for how "expectations" and "news" about bad debts get factored in, etc. So, therefore, when covering the *Allowance Method*, I made a very **simple assumption**, that the company always assumed that it would fail to collect a certain **fixed percentage of all outstanding accounts**. This is the simplest possible method for estimating the *Allowance,* which we can now call the "percentage-of-receivables method" (for self-explanatory reasons).

But in reality, this is only **one possible method of estimating the Allowance**, and a fairly **crude** one at that. In principle, companies could use big data, machine learning, fancy statistical methods, and **all of the information they have about their customers** to get a better estimate of the *Allowance*. After all, under the Balance Sheet method, the goal is just to get a good estimate of *Net Accounts Receivable*—the Accounts Receivable we will actually collect. So, the question here is really just **a general forecasting problem,** and, in principle, there's nothing stopping us from bringing all the information and statistical tools we have to that problem.

However, since this is not a statistics or data science text, we compromise, and just cover **one particular conventional method** that factors in just ***a bit*** **more information** to get a more precise estimate of the *Allowance*: The Aging of Receivables Method. Note that this method is like a sub-category of the *Allowance Method*, not some alternative. The *Allowance Method* means that we focus our attention on updating the

[12] And what if the errors *don't* offset—if we are persistently wrong in the same direction, assuming 10% while actually losing 30%, year after year? Then *Net A/R* would become systematically overstated. But that is not a flaw of the *method*; it's a flaw of the *forecast*—we're simply using a bad *p*. And in the real world, the auditors will eventually refuse to keep signing off on an assumption that the company's own experience keeps contradicting.

Allowance at the end of the period, based on the outstanding accounts (rather than simply accruing a *Bad Debt Expense* based on credit sales in that period). We previously covered one way **to do that**: The percentage-of-receivables method. Now, we'll cover another approach for estimating the allowance: The aging of receivables.

8.8.1 Aging of Receivables

The conventional alternative method that we'll cover is the **Aging of Receivables Method**. All this really means is that, instead of crudely applying a constant percentage to all of its receivables, the company **uses information about how long each account has been past due, to estimate** what portion will be collected. Generally, the assumption is that the **longer an account has been outstanding, the lower the probability** that it will in fact be **collected**.

So, concretely, we would put different outstanding accounts into different "buckets," depending on how long they have been outstanding. And, for each bucket, we would estimate what portion of those accounts in that age range will prove uncollectible. And so our estimate of the total accounts uncollectible will just be the **"dot product"** of the two: The amount of each bucket, multiplied by its estimated percent uncollectible, all added together. Implementing this is just a matter of arithmetic. It requires no genius, just a little care and diligence.

For example, suppose that our company, based on our historical experience and data, estimates that 10% of accounts that have been outstanding less than 60 days will be uncollectible; and 30% of accounts outstanding longer than that will be uncollectible. These are the only two "buckets" we use. Suppose we currently have \$2m in accounts outstanding less than 60 days, and \$1m in accounts outstanding more than that. Then, this implies that our updated *Allowance for Doubtful Accounts* should be: $(\$2m * .1) + (\$1m * .3) = \$500k$. That's all! The more buckets we use, the more arithmetic terms we will have. But the concept will always be that simple.

We can also organize this in a matrix or table structure, like the below:

	Bucket #1: Accounts $<$ 60 days	**Bucket #2: Accounts $\geq$ 60 days**
Outstanding Amount	\$2m	\$1m
% Uncollectible Assumption	10%	30%
Est. Amount Uncollectible (\$)	\$200k	\$300k

Total Est. Amount Uncollectible (\$): \$500k

Matrix for visualizing/calculating *Allowance* based on *age* of accounts

Once we have that number—our updated estimate of how much will prove uncol-

lectible, and, hence, the updated balance of the *Allowance for Doubtful Accounts*—we're back on familiar terrain. Always and everywhere, using the allowance method, at the end of the period, we update the *Allowance* by whatever amount is required to get it to its new balance, and the balancing debit is the *Bad Debt Expense*. All that we have added here is a different method for getting that balance of the *Allowance*. Everything else remains unchanged.

Practice problem: Our company uses the Allowance method, and estimates the Allowance using the *aging-of-receivables* method. We began FY2022 with a balance in the *Allowance* of $20k. Over the period, we wrote off accounts totaling $10k, and make additional credit sales as well. We end FY2022 with $100k in Accounts outstanding less than 30 days, and $80k in Accounts outstanding more than 30 days. We assume 10% of the former category and 15% of the latter category will be uncollectible. What will be our *Net Accounts Receivable* and *Bad Debt Expense* for FY2022?

Solution: First, *Net Accounts Receivable* is trivial: We are given that we expect 10% of $100k in accounts and 15% of $80k in accounts to be uncollectible, or $22k uncollectible out of $180k total. Therefore, *NAR* is $158k.

Solving for the *Bad Debt Expense* under the Balance Sheet method is always harder, and I recommend writing out the T-accounts. Here is a compressed verbal summary of how to do that: We start out the period with an Allowance of $20k (a credit balance in that T-account). During the period, we have writeoffs of $10k—writeoffs 'use up' / reduce the allowance in the B/S method. So, before adjustment, the Allowance will have a balance of $10k (i.e., $20k - $10k). We assume 10% of $100k and 15% of $80k will prove uncollectible. So, the ending balance of the Allowance must be $22k (i.e., .1*100k + .15*80k). So, the adjusting entry must get our Allowance from $10k to $22k. That means the Bad Debt Expense will be $12k.

9

Receivables Part 2: The Remainder

In Part 1 on *Receivables*, we covered the biggest, most challenging, most novel idea: accounting for **uncollectible Accounts Receivable**. But, our company might have other kinds of **receivables**—most significantly, *Notes Receivable*, when we make a loan to another company. We've talked about the case where our company receives a loan (which is a *Note Payable* for us), but not this. And, there are a few steps in the life of a loan that we haven't accounted for so far. We'll learn how to account for it all.

There are some more interesting, conceptual issues we'll cover as well. We've talked about the fact that companies and customers can't always pay their debts, and mentioned 'bankruptcy' in a couple of different contexts—as a dreaded outcome for our company if we can't pay, or a potential reason why our customers might not pay. But we haven't explained *exactly* what bankruptcy is, and what happens. Finally, we'll cover two different ways in which companies can put interest-like terms into their contracts with customers who pay on account, though with different accounting treatments. These examples will show that, sometimes, the distinction between 'operating' vs. 'financing' can be a bit 'fuzzy'—the two can be tangled up in interesting ways.

9.1 Notes Receivable vs. Accounts Receivable

9.1.1 What's the difference?

Both *Notes Receivable* and *Accounts Receivable* are, obviously, **receivables**—amounts we are entitled to receive from others, and thus assets. To date, we've drawn a simple, clean line, and always said that an **Account** is an amount owed from a **customer** from a credit sale, while a **Note** is a purely financial instrument, a **loan**. But in reality, the categories can blend together a little bit. Sometimes, our company might require our **customer** to **sign a more formal contract** for a **credit sale**, with specified **interest terms**. In this case, it might be categorized as a **Note Receivable**.

Why would our company do one vs. the other? Very often, if we are selling to a **regular customer** with **good credit** (that is, enough cash and cash flows to pay us back, and a track record of doing so), we could be easily satisfied with a **more informal understanding**—we ship the goods and invoice, and trust that they will pay us later, in order to continue our business relationship. That would be an **'Account.'**

But, in other circumstances, we might be worried about the **ability or intention** of our customer to **actually pay**. So, in that case, we might demand a more **formal contract**, including **specified terms of interest** designed to **incentivize** early payment, and **compensate** us for risks and delays in payment. In that case, it would be a '**Note**.'

A written note documents the amount, timing, and other terms of the debt. But being called a **Note** does not, by itself, give a claim higher priority than an **Account**. **Seniority** depends on such things as collateral, contractual subordination, and statutory priority. An unsecured note and an ordinary unsecured trade payable may rank together. To assess which creditor is likely to recover more in bankruptcy, we need to read the actual terms and understand the applicable priority rules.

9.1.2 Going deeper: Bankruptcy does not necessarily mean that operations stop

We've referenced "bankruptcy" a number of times in this book. And it is a key business concept. But most people outside of the corporate law, finance, and accounting worlds don't understand what it *really* means. In everyday speech, when people use the term 'bankruptcy,' they seem to think it means **corporate death**. But that's a very **inapt and misleading metaphor**, which causes a lot of the confusion.

In reality, in the United States, bankruptcy is mostly a **technical legal proceeding** for determining **which creditors** get paid **how much** when a company is unable to meet *all* of its legal obligations. Contrary to popular belief, **companies may**

continue operating, especially under Chapter 11, while they are in bankruptcy court—they continue to sell their goods and services, run their factories, employ their employees, etc. American Airlines was in bankruptcy court from November 2011 until December 2013; and if you flew American in those years, you probably didn't know it.

Go back to our Chapter 1 foundations, on the nature of *credit vs. equity* financing. We said that companies **"have to" pay their debts.** But what if they *just can't*, because they simply don't have the cash on hand or incoming cash flows (liquidity) to do so? If they just can't, then they just can't! If they *don't*, then the company is said to be **"in default"** (or "to have defaulted on its loans"), and the company's creditors have various forms of **recourse** specified in their **debt contracts**, including, for example, **seizing collateral** (see, e.g., "Airplane repo" on YouTube). But that **collateral** is the assets—the resources—that companies use to run their operations. So, when creditors seize collateral, that significantly **disrupts** their operations, which, in turn, disrupts their ability to generate revenues and cash flow going forward. That, in turn, could just make things **worse** for all of the **other creditors.**

As a result, U.S. corporate law has evolved a rich, sophisticated, and valuable body of **bankruptcy law**. When companies see that they will be unable to pay their debts, they **"file for bankruptcy protection."** Under bankruptcy **protection**, an automatic stay generally pauses creditors' collection and foreclosure actions, subject to exceptions and court-approved relief. That way, the bankrupt company can **continue operating**. However, in exchange, the company has to go to **bankruptcy court**, with its creditors as the other parties to the case. And they litigate and negotiate, and the court eventually determines **who gets paid what, and how**.

There are a couple different forms of bankruptcy, but, most significantly, under **Chapter 11** bankruptcies, the parties will negotiate **"The Plan"**[a] for **reorganization**—that is, how the company will change its policies and strategies in order to pay back its creditors as much as feasible.

[a] This is, in fact, the formal legal term.

9.1.3 Accounting for Notes Receivable: Interest and Principal

To date, we have only shown how to account for *Notes Payable*, when our company receives a loan from somebody else. And, even then, we didn't show the *final step*—paying off the loan. Let's now cover *Notes Receivable*, when we make a loan to another company, including the date on which the loan is paid off.

Example: Suppose that our company does annual, calendar-year reporting. On July 1, 2022 (exactly **halfway through the year**), we make a $10,000 loan, with a 5% stated

interest rate (paid annually), and a one-year **maturity** (jargon).

That means that the $10,000 "face value" of the loan will be paid off and the loan will be discharged in full in exactly one year. And, since the interest rate is 5%, paid annually, we will also be paid the $500 interest payment on that date as well. Since this loan is so simple, with only one payment date, we need to account for just three dates:

- (1.) the day we make the loan,
- (2.) the Dec. 31 adjusting entry for accrued interest, before closing the books,
- (3.) the date on which the loan is paid off in full.

Those journal entries would be as follows:

Note receivable (+A):	$10,000	
Cash (-A):		$10,000

Jul. 1, 2022: We make the initial $10,000 loan

Interest Receivable (+A):	$250	
Interest Revenue (+S.E.):		$250

Dec. 31, 2022: adjusting entry for **half year of *accrued* interest**

Cash (+A):	$10,500	
Note Receivable (-A):		$10,000
Interest Revenue (+S.E.):		$250
Interest Receivable (-A):		$250

Jul. 1, 2023: Paying off interest and principal

The final transaction is the only really new one. And it's our **first four-part journal entry**. So it looks big. But it's simple when we break it down.

Here is how I think of it: **Start with Cash**. We receive the principal ($10k) and the interest ($500) on that date, and they are both just cash, so we **debit the sum** in one line. What else? Now that the borrower has paid off their loan in full, they no longer have an obligation to us, so we no longer have a receivable from them. So we need to **extinguish** *both* the initial *Note Receivable (A, $10k)* as well as the *Interest Receivable (A,*

$250) we accrued in the previous journal entry. Those are reductions in assets, so both credits. Finally, in the **half year since the last journal entry**, we have **accrued**/earned another $250 in interest. So, we credit another $250 to **Interest Revenue**.

Why credit Interest Receivable by $250 when another half-year of interest has accrued since December? Because the payment entry starts from the balance actually on the books. We recorded $250 receivable at December 31, and the final payment settles it. The additional $250 earned since then is recorded directly as Interest Revenue in the same entry. This is the pattern from Chapter 7's payday example: settle the previously recorded obligation or claim, and recognize the amount earned since the last update.

9.2 'Sales discounts': Incentives for quick payment

The last concrete thing to cover is "Sales discounts."[1] A Sales Discount is when a company **sells to a customer on credit**, sends an invoice with a **stated price**, but offers a **discount for "early" payment**. (You'll see why I put "early" in scare quotes, by the end.) Companies do this to **incentivize earlier**, more secure **payment** of their accounts. I'll show how this is accounted for, and then raise some discussion topics.

Suppose our company sells to a customer "for" $10,500, but states that if the customer pays **before** a certain date (say, December 1, 2021), the customer will get a **Sales Discount** of $500 off. For this example, assume that the company initially expects to collect $10.5k and that recognizing that amount meets the revenue standard's constraint on variable consideration. Under that assumption, it records the full amount initially. If it instead expects the customer to take the discount, it must reflect that expectation when measuring revenue; it cannot simply use the higher list price to inflate earnings.

Accounts Receivable (+A):	$10,500	
Revenue (+S.E.):		$10,500

Initial Credit Sale of a Service, October 1, 2021

And, *if* the customer paid *after* December 1, and thus had to pay $10.5k, we would *also* account for it as usual—debiting cash and crediting A/R for that full amount.

But what if the customer pays early, say, on November 20th, and is thus able to claim the Sales Discount? **Start with the Cash (debit)**. Per the stated terms of the contract,

[1](Note that these are different from **Trade Discounts,** which are just 'discounts' from the quoted/stated 'list price,' and, so, have no significance in accounting. We already know from earlier chapters that 'the list price is just marketing material, the transaction price is what matters'—so there is no real accounting for 'trade discounts.' We just use the actual purchase price as usual.)

that customer will only actually pay $10,000. And the customer's **Accounts Receivable** will be fully **extinguished**—a credit. But that *Account* was recorded, above, at the full $10,500. How do we **plug the difference**?

The answer is we use an account called **Sales Discounts (contra-Rev)**. This is another **contra-Account**, specifically our first **contra-Revenue**.

Cash (+A):	$10,000	
Sales discount (contra-Rev):	$500	
Accounts receivable (-A):		$10,500

Customer pays early, Nov. 20, 2021, claiming discount

You can think of this account as just a 'plug' for this transaction. But it also should make intuitive sense to think of it as a contra-Revenue. Given the **core principle** that **accruals and cash flows must be 'reconciled' over the long run**, and given that the customer is only paying $10,000, the Revenue recognized should be reduced to that amount.[2]

Now, one last bit of **jargon**: Within the context of a company's own **internal accounting**, we use the term **'Net Sales'** to mean the total Revenues **net of** contra-Revenue accounts. It's like the 'book value' of Equipment in that respect. (And **'net'** often has this meaning in accounting.) The **Revenue** that companies report on the top line of their income-statement *is* this netted number. That is, we do not report contra-Revenues as separate lines on the Income Statement.

9.2.1 Going deeper: When does a price difference become financing?

In accounting, as in life, we need to **draw lines** and make distinctions. But the underlying **reality** of the world is always a bit **fuzzier**. One thing that this chapter brings home is how our 'financing' and 'operating' categories can blend together quite a bit. Allowing your customers to buy 'on account' is like extending credit financing *to* them. Take a closer look at the last two transactions we considered, the *Notes Receivable* transaction and the *Sales Discount* transaction.

- In the **Notes Receivable** transaction, our customer paid $10,500 in cash (interest plus principal) **later** instead of paying $10,000 in cash **sooner**.

[2] There are also a variety of other **contra-Revenue** accounts used to account for things like customer returns. Because learning those would be somewhat rote, and not add much conceptually, I do not cover them in this text.

- In the **Sales Discount** transaction, our customer paid $10,000 in cash (list price minus Sales Discount) **sooner**, rather than pay $10,500 in cash (list price) **later**.

Summarizing visually, the **Notes Receivable** contract looks like this:

Condition/timing:	pay earlier	pay later
Jargon:	list price	charged 'interest'
Cash payment:	$10,000	$10,500

Credit Sale Financed with Note Receivable

And the **Sales Discount** contract looks like this:

Condition/timing:	pay earlier	pay later
Jargon:	'sales discount'	list price
Cash payment:	$10,000	$10,500

Credit Sale with Sales Discount

The earlier worked examples used different dates. For this comparison, imagine the same earlier and later payment dates in both contracts, with the same $10,000 and $10,500 alternatives. On that common timeline, the cash-flow choices are identical. The **jargon** we use is **different**, and **so is the accounting**. Admittedly, this pair of examples is a bit stylized. But here are a couple of challenging questions to chew on, for those who like to go deeper:

1. First, what exactly is the difference between offering a **'discount'** to customers who pay **earlier** rather than later, as opposed to imposing **interest** terms on those who pay **later** rather than earlier?

2. Second, could managers use this to 'game' their accounting at all? Suppose that we want to report higher revenues in 2021, and it's now December. Instead of selling on account with interest, and reporting $10k in revenue from the sale, could we instead sell with a list price of $12k, and offer our customers a $2k 'discount' if they pay before June—thereby allowing us to report 20% higher revenues as of the end of the period (Dec. 31)? Would our auditors stop us from doing that?

3. Third, this issue raises a philosophical question: What do we **mean** by a **'transaction price'** when that **price is in fact conditional upon the timing of the payment**?

4. Fourth, and most generally of all, this suggests that, perhaps, for complex modern businesses, with rich and complex contracts, we really *cannot* always perfectly cleanly separate 'operating' vs. 'financing' activities and cash flows.

10

Inventory and COGS

Most companies sell **physical goods—products**. But, to date in this book, we have always used examples of service companies selling **services**. Why did we do that? The reason is that when companies sell a physical good, there are extra steps involved in accounting for the sale. When we sell services, the accounting was simple, involving a credit to Revenue and one balancing debit. But when companies sell **physical goods**, they also account for the costs attributed to the goods they just sold as an expense. This chapter is mostly about how to do that.

When a company first acquires and/or produces physical goods that it intends to eventually sell, we account for that as an **asset** called **Inventory**. We've talked about the "general form of an asset." When we purchase a resource that we expect to use over multiple periods, we initially write that asset up at the **purchase price**, and then expense it out over time, over the period of its use. For each asset so far, the only other questions have been:

- What do we *call* the associated expense?
- How do we measure the 'amount used up'?

For *Inventory (A)*, the associated expense that is recognized as it is 'reduced' or 'used up' is called **Cost Of Goods Sold**. And the general pattern for assets we've seen so far *largely* applies, but with a couple of major caveats. For example, **manufacturing companies** produce their own products—that is, make their own **Inventory**. So, for

these companies, the Inventory does not have a **purchase price**. Instead, these manufacturing companies have a complex process for estimating and **allocating** all the **costs of production** *to* their products in **Inventory**. These techniques are complex, and are covered in intermediate-level financial and managerial accounting courses. In this introductory text, we will consider only **merchandising companies** (such as retailers you are familiar with), which **buy finished products** from manufacturers, for resale to customers. As ever, for these companies, we write the **Inventory (A)** up at the **purchase price**.

It's worth lingering on this business background for a moment, to know the lay of the land. **Manufacturing companies** are the firms that actually *produce* physical products: they buy raw materials and components, run factories and plants, and transform those inputs into finished goods. **Merchandisers**, again, do not produce anything themselves—in effect, they outsource the production function to their suppliers—and they come in two flavors: **wholesalers**, who buy in bulk from manufacturers and re-sell to other businesses ("B2B"), and **retailers**, who sell to end consumers like you and me ("B2C"). If you sit down and try to list companies off the top of your head, the ones that come to mind most readily will mostly be retailers—because those are the ones you personally transact with. But behind those familiar consumer-facing brands, all across the S&P 500, sits a whole ecosystem of manufacturers, wholesalers, and distributors, quietly running the supply chain in the background.

And here I should come clean about some deliberately careful wording from earlier in this book. When I first introduced *Utilities Expense*, I was oddly specific: I said, consider the electricity bill for our corporate *headquarters*, and I told you that we immediately and fully expense it—since, as I argued there, there is no sensible way to match such a cost to any particular future sales. I chose those words carefully. If, instead, the electricity is powering a manufacturing company's *factory*, the story is different. Imagine we are a furniture manufacturer building tables. Each table required a certain number of kilowatt-hours of electricity in the plant, plus, say, 40 hours of labor at $20 per hour, plus some quantity of wood, glue, and hardware. Manufacturers use their cost-accounting systems to **trace or allocate** these production costs—materials, labor, plant utilities, overhead—*into* the cost of each unit produced, and those traced costs are **capitalized** into the *Inventory (A)* asset. Then, when the table is finally sold, the COGS recognized on that sale is, in principle, the sum of all the costs that were attributed to that particular good. Designing and running these costing systems is the bread and butter of *managerial accounting*—and, depending on your tastes, it is either good news or bad news that we set all of that aside in this book.[1]

[1] I have had a few students who were disappointed to hear this, because they found the cost-tracing problem intellectually interesting. Most are relieved. If you are in the first camp, managerial accounting

So, for our concrete purposes, there are really only **two** major deviations from the general pattern of assets we've seen to date:

- First, under the **Perpetual Inventory System**, we recognize the expense from using up **Inventory** (that is, COGS), **on the date of the sale transaction, as an external transaction**, rather than as an adjusting entry at the end of the period.[2]
- Second, companies have a larger number of allowable **methods** for calculating the amount of Inventory **"used up"** (that is, the regular COGS expense). And these different methods, in turn, have different implications for the bottom-line net income a company will report, and the value of the *Inventory* on its books.[3]

I'll discuss these in order now.

10.1 Perpetual Inventory System

To date, we've had one "general form of an asset." When we first purchase an asset, we record the full **purchase price** to the **Asset** account. Then, over time, as we **use the resource up**, we account for the **amount used up** as an expense, via an *adjusting entry* at the end of the reporting period. We record the *related Expense* and reduce the balance of the **Asset** by the same amount. (In the case of *Equipment*, we did this *indirectly* via the *Accumulated Depreciation* contra-Account.)

However, in accounting for **Inventory (A)** and **COGS (the related expense)** under the **Perpetual Inventory System**, things are a bit different. We still write the Inventory *up* at the purchase price, on the date of its purchase:

Inventory (+A)	$(Purchase Price)	
Cash (-A)		$(Purchase Price)

Purchase of Inventory, date during reporting period

However, under the **Perpetual Inventory System**, we do not wait for the end of the reporting period. Instead, we recognize the **COGS** expense, and reduce the **Inventory**

awaits you.

[2]Technically, this is specifically true of companies using the **perpetual inventory system**, as opposed to a **periodic inventory system**—in the latter, **Inventory** accounts are updated only at the end of the period, as was the pattern with previous assets. But, given that nearly all modern retailers use the **perpetual system**, and this is what is interesting and novel in this chapter, it is the default in this book. Unless otherwise specified, in this book, a **perpetual inventory system** is assumed.

[3](Note that these methods for estimating the amount of Inventory used up—the expense—do not affect or influence the initial **debits** to **Inventory** from purchases. Inventory **purchases** are always and everywhere written up at the **purchase price**.)

balance accordingly, **on the date of the sale**, as an **external transaction**. That is, we expense the Inventory out on the **date it leaves our store**.

The sale gives us a specific event to record: the customer receives the goods, and those goods leave our Inventory. Under the perpetual system, we record that reduction on the sale date, alongside the revenue. This differs from the rent and supplies adjustments in Chapter 4, where we updated the amount consumed at the reporting date.

To make things a bit more palpable, suppose that our company is a **Liquid Yeet** retailer. On May 13, 2021, we sell 10 liters of **Liquid Yeet** to students for a **sale price** of $300 in cash. And suppose, for the sake of the example, that we estimate the **cost** of those 10 liters of our **Inventory**, which have now departed our store, was $100. (We'll talk about where that estimate could come from, next.) Then, on the date of the transaction, we would account for **both parts**:

Cash (+A):	$300	
Revenue (+S.E.)		$300
Cost of Goods Sold (-S.E.):	$100	
Inventory (-A):		$100

Sell Liquid Yeet from inventory for $300, 5/13/2021

While this still fits the general theory of assets, and is a small change, there are some cool things about this. First, a bit of **jargon**: We use the term '**gross profit**' to mean **Revenue minus Cost of Goods Sold**. I.e., it is your 'profit' after subtracting only your most "basic," "direct" product costs (without subtracting expenses from things like "overhead," rent, etc.). We have to wait until the **end of the reporting period** to see our **bottom-line profit, Net Income** because it is affected by all of the expenses that are recognized as *adjusting entries*.

But now, because all of the most 'direct' **product costs** are allocated to the **Inventory** asset, and we expense that out on the date of the sale, a store owner or manager can **visualize the gross profit on a particular sale**. That is novel, and valuable for many entrepreneurs, who wouldn't want to wait until the end of the year to know if they are generating any value.

Second, that four-part journal entry vividly illustrates how "value-generation" works in accrual accounting, in one single step. Note that, **total debits and credits balance**, as they must. However, the **Asset gained** (Cash (+A), $300) surpasses the **Asset expended** (Inventory (-A), $100); just as the **Revenue** (+$300) surpasses the **COGS expense** (-$100). In other words, while **debits and credits balance**, the effects on the total Assets, Shareholders' Equity, and Net Income **do not** net out to **zero.** So, in

one transaction, we can see the link between the Income Statement and the Balance Sheet, "value creation" in one step, but everything kept in balance.

The two sides of a journal entry balance, but the company can still come out ahead. Here, the $200 increase in net assets is also the $200 gross profit on the sale.

10.2 Inventory Cost Methods: Calculating COGS

How do we calculate the **COGS expense**? Companies replenish their *Inventory* with fresh purchases on a regular basis. (Inventory is a short-term asset.) This means that different units in inventory will come from different purchase 'batches,' bought at different prices. Per **core accounting theory**, we should want to **allocate the cost of purchasing Inventory** as we use it up **over time** and **match** that expense to the revenues it helped to generate. And, given that different inventory units are purchased at different prices over time, there's **no one obviously-superior way to do this**. Companies have the choice of four major methods:

- Specific Identification,
- First-In First-Out (FIFO),
- Last-In First-Out (LIFO), and
- Weighted-Average Cost.

In the real world, each company picks one of these methods and **discloses** its choice in the notes to its financial statements. (In this book's problems, you will always be *told* which method to apply—the interesting question of *why* a company might pick one over another comes later in this chapter.) I'll show how each of these methods works, and then back up to discuss. It will be easiest to understand these methods with a concrete but simple example.

Example Suppose that our company first **buys three batches of 100 units each** of *Inventory*, on January 1st, January 8th, and January 15th, each at different prices, tabulated below; and then it **sells 150 units** on January 22nd, and then **another 75 units on January 25th**.

	Batch 1	**Batch 2**	**Batch 3**
Date	January 1	January 8	January 15
Units purchased	100	100	100
Price per unit	$10	$20	$30
Total cost	**$1,000**	**$2,000**	**$3,000**

Inventory Purchases

	Sale 1	**Sale 2**
Date	January 22	January 25
Units sold	150	75
Sale price per unit	$40	$40
Total Revenue	**$6,000**	**$3,000**

Sales

No matter what Inventory cost method the company uses, as of January 15th, before the company makes any sales, the company's **Inventory account/balance** would simply be written up with the purchase prices, and look like the below:

Inventory	
Jan 1: $1,000	
Jan 8: $2,000	
Jan 15: $3,000	
$6,000	

The **costing methods** only determine what **quantity** will be recorded as a **COGS expense** (and Inventory credit) on January 22nd.

One assumption I want to make explicit before we begin, because it will matter for the discussion afterwards: in all of these examples, assume that the company *could* know exactly which unit came from which batch. With modern inventory-management technology, that is a perfectly realistic assumption for a large corporation today. And yet—as you're about to see—GAAP still allows the company to choose among these methods. Hold onto that: the choice of costing method is **not** about what information is technologically available.

The four methods are below:

10.2.1 Specific identification

As the name indicates, under this method, the company would **specifically identify** which exact units had been sold, and which batch each unit came from. In the modern era, this could be possible if the full physical flow of inventory goods was tracked with barcodes or RFID tags. Suppose we determine that, of the 150 units we sold on January 22nd, **80** came from the **first batch**, **60** came from the **second**, and **10** came from the **third**. Then the Cost of Goods Sold (and matching Inventory credit) for the January 22

sale would be:[4]

$$COGS,\ Jan\ 22,\ Specific\ ID = (80 \times \$10 + 60 \times \$20 + 10 \times \$30) = \$2,300$$

And, therefore, the journal entry for the January 22 sale, including both components, would be:

Cash (+A):	\$6,000	
Revenue (+S.E.)		\$6,000
Cost of Goods Sold (-S.E.):	\$2,300	
Inventory (-A):		\$2,300

January 22

And, the balance of **Inventory after this sale** would be \$3,700 (i.e., \$6,000 - \$2,300).

Inventory	
Jan 01: \$1,000	Jan 22: \$2,300
Jan 08: \$2,000	
Jan 15: \$3,000	
\$3,700	

Then, suppose we are told that the 75 units we sold on January 25 all came from the third batch. Then, the Cost of Goods Sold (and matching Inventory credit) for that sale would be:

$$COGS,\ Jan\ 25,\ Specific\ ID = (75 \times \$30) = \$2,250$$

And, the balance of **Inventory after this sale** would be \$1,450 (i.e., \$3,700 - \$2,250).

Inventory	
Jan 01: \$1,000	Jan 22: \$2,300
Jan 08: \$2,000	Jan 25: \$2,250
Jan 15: \$3,000	
\$1,450	

[4]For those of you with a mathematical bent: the COGS here is just the *dot product* of the quantity vector and the unit-cost vector. That observation holds for all of these methods—they differ only in *which quantities* get matched to which costs.

One practical note: specific identification only works if we are, in fact, told (or can determine) which exact units were sold. If a problem gives you the purchase batches and a quantity sold, but no information about which particular units went out the door, then, under specific identification, the honest answer is 'not enough information'—you would need one of the methods below.

While specific identification may seem like the most natural, obviously 'correct' inventory costing method, in practice, **companies rarely use it**. Sometimes it may be too costly and complex to track inventory units individually. But there are also arguments to be made that other methods better capture the *economics* of the company. This is something that first-time learners find odd and counterintuitive, so I'll first show those other methods, and then back up and discuss.

10.2.2 First In, First Out (FIFO)

As the name indicates, under this inventory costing method, the company **recognizes expenses *as if*** the **first** item **added** to the company's inventory (and still remaining in its balance) was the first **one 'out'**—the first one to be **sold**. Note the phrase 'as if.' We are "assuming" this in the same sense in which economists "assume" things in their models—as a modeling convention, not as a factual claim. FIFO is an assumption about the flow of *costs*, not a claim about the flow of *physical goods*; nobody is asserting that the warehouse staff actually moves the oldest boxes out first. (More on this when we compare the methods below.)

In the example above, the first batch was 100 units purchased at $10 each. But the company sells **150** units on January 22nd. So, under this method, we would calculate the cost as if it took **100 units from the first batch and 50 units from the second** batch. So:

$$COGS,\ Jan\ 22,\ FIFO = (100 \times \$10) + (50 \times \$20) = \mathbf{\$2{,}000}$$

And the balance of *Inventory* after the Jan 22 sale would be $4k (i.e., $6k - $2k).

Next up: the January 25 sale of 75 units. We're still using FIFO, but as of now, we've already "used up" or "expensed out" all 100 units from the first batch, and 50 of the units from the second batch. (I like to imagine a little **'pac-man'** shuttling along, 'consuming' the batches in order.) So, we would "take" the remaining 50 units from the second batch, and move on to take 25 units from the third batch.

$$COGS,\ Jan\ 25,\ FIFO = (50 \times \$20) + (25 \times \$30) = \mathbf{\$1{,}750}$$

And the balance of *Inventory* after the Jan 25 sale would then be $2,250.

Inventory	
Jan 01: $1,000	Jan 22: $2,000
Jan 08: $2,000	Jan 25: $1,750
Jan 15: $3,000	
$2,250	

Another way to calculate that quantity would be to just note that, under FIFO, as of January 25, we only have 75 units of inventory left over, which must all be in the **last** batch (since we expensed out the **first** ones first). So, that's: 75 × $30 = $2,250.

10.2.3 Last In, First Out (LIFO)

As the name indicates, this method is just the opposite. We calculate the inventory cost as if the **last** item that we **purchased** and added to Inventory is the **first** one **'out.'** So, since the company sold 150 units, and each batch it purchased has 100, it would 'take' 100 units from the last batch, and 50 units from the second batch. So:

$$COGS,\ Jan\ 22,\ LIFO = (100 \times \$30) + (50 \times \$20) = \mathbf{\$4{,}000}$$

And the balance of *Inventory* after the Jan 22 sale would be $2k (i.e., $6k - $4k).

Next up: the January 25 sale of 75 units. (Under LIFO, our little pac-man shuttles from *right to left*—starting with the most recent batch and working backwards—and, of course, he cannot eat the same units twice.) We're still using LIFO, but as of now, we've already "used up" or "expensed out" all 100 units from the third batch, and 50 of the units from the second batch. So, we "take" the remaining 50 units from the second batch, and the final 25 units from the first batch.

$$COGS,\ Jan\ 25,\ LIFO = (50 \times \$20) + (25 \times \$10) = \mathbf{\$1{,}250}$$

And the balance of *Inventory* after the Jan 25 sale would be $750 (i.e., $2,000 - $1,250). Another way to calculate this is just to note that, under LIFO, as of January 25, we only have 75 units of inventory left over, which must all be in the **first** batch (since we expensed out the **last** ones first). So, that's: 75 × $10 = $750.

10.2.4 Weighted-Average Cost

As the name indicates, in this method, we simply **pool all the units remaining** in the inventory account together and recognize COGS at their **average cost**, calculated as the **total Inventory balance divided by the number of units remaining**. In this example, the inventory account balance is $6,000, and there are 300 units, indicating an

average cost of $20 per unit (i.e., $6,000/300 units). 150 units are sold. So:

$$COGS,\ Jan\ 22,\ Weighted\ Average\ Cost = 150 \times \$20 = \mathbf{\$3{,}000}$$

And the balance of Inventory after the Jan 22 sale would be $3k (i.e., $6k - $3k).

Next up: the January 25 sale of 75 units. As of right now, after that January 22nd sale, we have 150 units of inventory remaining on our books at a total Inventory cost of $3,000.

Inventory	
Jan 01: $1,000	Jan 22: $3,000
Jan 08: $2,000	
Jan 15: $3,000	
$3,000	

That means the average cost is still $20 per unit, $3,000/150 (but note that the average cost would have changed if we had had another purchase in the interim, at a different price). So, the COGS for the January 25 sale would be:

$$COGS,\ Jan\ 25,\ Weighted\ Average\ Cost = 75 \times \$20 = \mathbf{\$1{,}500}$$

And the balance of Inventory after the Jan 25 sale would be $1,500.

Inventory	
Jan 01: $1,000	Jan 22: $3,000
Jan 08: $2,000	Jan 25: $1,500
Jan 15: $3,000	
$1,500	

In practice, the Weighted-Average Cost method gets far less attention than FIFO and LIFO—though plenty of large firms do use it. Which is a bit of a shame, in a way: when I teach this material, some student usually *invents* this method on their own, before I've presented it—it seems so natural. And it fits nicely with an instinct we've seen throughout accrual accounting: the instinct to **smooth** big, lumpy costs over time, just as depreciation smooths the cost of a machine over its useful life. Weighted-average cost is the smoothest possible way to do inventory costing: rather than worrying about which batch a unit came from, you blend everything into one average. It is perfectly allowable under GAAP. But it has never had LIFO's devoted constituency, for a reason that will become clear when we get to taxes below: when costs are rising, blending old and new prices captures only *part* of the tax benefit that LIFO delivers in full.

10.2.5 Inventory Costing Method Problems with Timing Tricks

The example above was simple, designed to illustrate the idea. And, for the most part, problems involving inventory costing methods will be quite straightforward—a matter of checking your work and doing the arithmetic, **diligence, not genius**. But the problems can get slightly **more complex** when you have **intervening inventory purchases**. For these problems, you need to remember that you cannot **expense out items that have already been expensed out**. For example, suppose that, after the sale on January 22nd, the company purchased another batch of inventory, 100 units for $40 each. (Denote this as: **Batch 4—Jan. 23, 100 units, $40 each**.) And *then*, the company sold **another 150 units afterwards, on January 24**.

Under LIFO, to calculate the COGS on this January 24 sale, in this counterfactual scenario, the company would **start** with the **last purchases in**—the 100 units purchased on January 23rd in Batch 4. But, since there were only 100 units in that batch, it would need to find 50 more units to "expense." But **we had already** "used up" (expensed out) all of the units in Batch 3, and 50 of the units in Batch 2, on our **original January 22nd sale**. So, we would need to 'skip over' Batch 3, and 'expense out' the 50 remaining units in Batch 2.

The mirror-image complication is when a **sale** occurs **before** a later batch has even been purchased. Under LIFO, "last in" means the last batch *in our Inventory as of the date of the sale*—a batch that hasn't been purchased yet obviously cannot be expensed out. Since this is about as tricky as these problems get, let's do one as a full worked example. Keep our same three purchases (Batch 1: Jan. 1, 100 units at $10; Batch 2: Jan. 8, 100 units at $20; Batch 3: Jan. 15, 100 units at $30), but now suppose that the company instead **sells 75 units on January 10**, and then **sells 140 units on January 20**. Calculate the COGS for each sale, under LIFO.

For the January 10 sale: as of that date, only Batches 1 and 2 exist—Batch 3 won't be purchased until January 15. So the "last in" is Batch 2, and we take all 75 units from it:

$$COGS,\ Jan\ 10,\ LIFO = 75 \times \$20 = \mathbf{\$1{,}500}$$

For the January 20 sale: by now, Batch 3 has arrived, and our remaining inventory consists of 100 units from Batch 1 (at $10), 25 units from Batch 2 (at $20—remember, we already expensed out 75 of them), and 100 units from Batch 3 (at $30). Working backwards from the most recent batch: we take all 100 units of Batch 3, then the 25 remaining units of Batch 2; and we still need 15 more units, so we reach all the way back to Batch 1. So:

$$COGS,\ Jan\ 20,\ LIFO = 100 \times \$30 + 25 \times \$20 + 15 \times \$10 = \mathbf{\$3{,}650}$$

(Picture our pac-man going back and forth *through time*: he eats from the latest batch; then a new batch is appended on the right; then he eats from that—always keeping track of what he has already eaten.)

Or, similarly, using the Average Cost Method, if **new inventory is purchased at a different price** than the previous average, we need to **recalculate the average cost** as the total remaining Inventory balance divided by the number of units remaining, in order to calculate COGS on the **next sale**. So, it can get a bit more complex than the simple example I used above. But again, it requires just diligence, no genius.

10.2.6 Why do these problems?

Admittedly, you will never have to hand-calculate COGS expenses in your careers. So, why do I make you do these 'rote' arithmetic problems in this book's practice sets?

- First, doing these problems helps to **bring home** what an expense really is in **accrual accounting**. An expense is often an **allocation or apportioning** of a past expenditure on an asset. By making you calculate expenses for the **same transaction**, using **different costing methods**, we bring this home.
- Second, it illustrates a core theme and takeaway, of how the 'bottom line' is not such an objective, unquestionable thing. It depends on your **method** for calculating certain expenses. And different companies might use different methods.
- Third, the harder problems force you to think carefully about the intimate relationship between **assets and their related expenses**. E.g., to get hard **LIFO**- and **Weighted Average Cost**-method problems right, you have to think carefully about which units are remaining in your Inventory balance, after past expenses, and how that affects the expense allocated to the next sale, etc.

10.3 Comparing Inventory Costing Methods

Before we compare, let's put all four methods side by side. For the very same January 22nd sale of 150 units, the four allowable methods gave us four different COGS expenses:

Method	**COGS**
FIFO	$2,000
Specific Identification	$2,300
Weighted-Average Cost	$3,000
LIFO	$4,000

COGS for the January 22nd sale of 150 units, by method

Same transaction, same physical goods out the door—and an expense that differs by a factor of two, depending on the **method**. Before reading on, ask yourself: which of these four numbers do you think 'makes the most sense'? Be warned that this is not a question like "what is 2 + 2?"—it is more like asking which *lens* is most useful, for the purposes we care about. When I poll my students, most gravitate toward FIFO and specific identification, with a couple of bold souls voting for LIFO. Here is something worth pondering as you read this section: companies that choose **LIFO** when costs are rising can be *voluntarily reporting lower earnings*. The tax link later in this section helps explain why they would make that choice.

10.3.1 Why not specific identification?

Probably the most frequent question that first-time learners ask when learning about inventory costing methods is why don't companies always just use specific identification? This seems like the most natural, obvious, correct, and direct way to recognize costs—to use the purchase cost of the actual unit that you literally sold.

There are a couple of answers to this question. An easy answer is that, in many businesses, it's costly and complex to track the actual flow of individual units of inventory. But, in the modern, digital era, there are plenty of firms that can track each individual unit with RFID codes (note that companies know when you do or do not open their promotional emails), but that still don't use specific identification for inventory accounting.

A **deeper answer** is that, **based on core accounting principles**, specific identification isn't necessarily a better way of **representing** the **economics** of the firm.

In general, the expenses associated with 'using up' assets are the way a company **allocates** the cost of its initial investment over time, **matched** to sales, to give a measure of whether the company is selling at above its costs on an ongoing basis. And, from that perspective, the dollar amounts that the company paid for individual inventory units are **sunk costs**—those purchases are in the past. So, when the company sells a particular unit of inventory, it's not obviously important that we expense the exact cost of that *particular* unit, rather than an **identical unit from a different batch**. Consider the following **thought experiment**, which illustrates this more vividly.

Winter storm shovels: Suppose our company is a local **general store** in the Northeast, and it is late fall. We currently have 15 **shovels** on our shelves, all of which we purchased from a single batch, for $10 each. Now, suddenly, a surprising **early winter storm** is approaching. We know that there will be **demand for shovels**, so we want to stock up. But, because of the surge in demand, and treacherous roads, our **supplier** now charges us **$20 each**. Even so, we buy another 15 at this price.

Now, **fast forward to the late-April thaw**. There are still four shovels left over on the shelves—two from each batch. A **customer** walks in, sees four identical shovels on the shelves, and picks one of them **at random** to purchase.

Discussion Question: Is it important in any way, economically, or for the **users** of our company's **Financial Statements**, that our COGS expense for that sale reflect the particular batch it happened to be purchased in ($10 or $20), depending upon that customer's random choice? That cost **variance**, from the past winter storm, is a **sunk cost**, a one-off event. Of course, we do need to allocate the **total expenditure** in some way, at some point, over time. But should this customer's random decision decide **exactly how or when**? This thought experiment shows that specific identification isn't really 'better,' if we're focused on the **big-picture goal of accounting**, measuring and communicating the financial performance and position of a company in a way that is **useful** to the **users** of the **Financial Statements**.

If that abstract argument doesn't persuade you, consider the empirical fact: essentially no company of any size uses specific identification for interchangeable goods. They use FIFO, LIFO, or weighted-average cost.

10.3.2 Comparing FIFO and LIFO

Companies need to consider a couple of different factors when they choose whether to use FIFO or LIFO for inventory costing—and, in turn, investors need to think about those same factors when interpreting those companies' financial statements. We can understand the tradeoffs here by just thinking carefully about the *meaning* of the terms. And if the framing below feels familiar, it should: just as with the methods for uncollectible accounts in the receivables chapters, we can think of one method as embodying an "income-statement" perspective and the other a "balance-sheet" perspective. This is a recurring motif in accounting.

- **LIFO pros and cons:** By definition, LIFO starts with recognizing/allocating costs from the later batches of inventory purchases.
  - **LIFO pros:** You could argue that LIFO is normatively better—gives you a better picture of the 'true' economics of the firm—since it uses the **most up-to-date measure of the costs of acquiring the company's Inventory**. If **net income** is supposed to be a measure of whether the company is **selling at above its ongoing cost of production**, the most recent inventory acquisition costs are the most relevant for that. For this reason, **LIFO** is sometimes characterized as the **"income-statement method**' ' in this context.

  - **LIFO cons:** One obvious problem with LIFO is that, because it starts with the latest inventory purchases, if the company maintains a positive inventory balance, its earliest inventory purchases will remain on the books at their original, now-stale costs. The costs of those earliest purchases are never allocated to sales. So, in general, under LIFO, the Inventory *balance* reported on the Balance Sheet will be understated.

- **FIFO pros and cons:** By definition, FIFO starts with recognizing/allocating costs from the earlier batches of inventory purchases.

  - **FIFO pros**: So, you could then argue that FIFO is normatively better: It doesn't have the same problem LIFO does, of the earliest purchases remaining on the company's books un-allocated as costs. So the *Inventory* balance reported on the Balance Sheet will more closely approximate its current market value . For this reason, FIFO is sometimes called the **"balance-sheet method"** in this context.

  - **FIFO cons**: But, the tradeoff is just the flipside. As a result, the revenues from sales are not **matched** to the most up-to-date measure of its Inventory costs. So, FIFO arguably does a **worse job** providing timely information, via the income statement, about whether the company is selling at above its cost of production on an ongoing basis.

Let me make the 'income-statement' argument for LIFO vivid, because it is more than a debating point. Consider a period in which inflation suddenly surges—and suppose our company's input prices rise by 30% in a year.[5] Now consider a company in one of those industries that uses **FIFO**. It records its revenues at *today's* selling prices, but computes its COGS from inventory batches purchased *last year*, before the price surge. On paper, it can look wonderfully profitable: today's high prices matched against yesterday's low costs. But economically, it may now be selling its product for *less* than the current cost of replacing that inventory—its margin, at today's input prices, may have vanished. The FIFO income statement will not tell you this until the cheap old batches run out. LIFO forces the company to confront current costs in COGS *right away*, giving a more realistic answer to the question an analyst really cares about: *can this firm still earn a margin at today's prices?*

On the other side, the 'stale Balance Sheet' problem with LIFO can be quite extreme. Imagine a company that has been operating for forty years and has always maintained a positive inventory balance. Under LIFO, our pac-man may *never* have eaten all the way back to the earliest batches. In principle, some of the inventory on that company's

[5] A historical illustration: Kroger reported a LIFO charge of $197 million in fiscal 2021 and $626 million in fiscal 2022. See its fiscal 2022 Form 10-K.

Balance Sheet could still be carried at the cost of the first inventory it ever purchased, decades ago—a number bearing little resemblance to any current economic reality.[6]

Two more arguments are commonly made for FIFO. One is simple orderliness: under FIFO, pac-man only ever moves left to right, eating from the oldest batch forward, no matter how purchases and sales interleave—mechanically tidier than LIFO's back-and-forth through time. The other is the claim that FIFO "more closely tracks the real physical flow of goods," since many businesses do, operationally, try to move their oldest stock first (think of perishables, or goods that go out of style). I don't find this second argument very compelling, and here's why: **it proves too much**. If your goal is really to track the physical flow of goods, then nothing will ever beat specific identification—*literally* tracking the physical flow. Taken seriously, the 'physical flow' argument doesn't point to FIFO at all; it points all the way back to specific identification, which we have already found wanting.

One clarification, before we go on. When we say LIFO is the "income-statement method" and FIFO the "balance-sheet method," we do **not** mean that a company gets to use one method for one statement and another for the other. You pick *one* method, and it flows through *both* statements: the method determines the COGS on the Income Statement, and—because whatever hasn't been expensed remains in the asset—it thereby determines the *Inventory* balance on the Balance Sheet. The labels just describe which statement each method arguably serves *best*.

And a second clarification, on the slippery words 'better' and 'accurate.' When we say a method makes a statement 'better,' we could mean two very different things: (1.) *bigger*—higher reported net income, higher assets, higher equity; or (2.) *more informative*—a more faithful picture for the users of the financial statements. In an inflationary environment, FIFO makes both statements 'better' in the first sense: lower COGS means higher net income, and, via the linkages, a higher Inventory balance and higher equity. But the interesting normative debate is entirely about the second sense. Similarly for 'accuracy': both methods are fully GAAP-compliant, so both are 'accurate' in the formal sense. When we argue about which is 'more accurate,' we are implicitly comparing the accounting numbers to some benchmark of *true economic performance*—which is a concept we hold in our heads, not a GAAP quantity. For what it's worth, my own sympathies lie with the income-statement perspective, and thus with LIFO. My background is in valuation: what I want from an income statement is a good measure of the profitability of the firm's *ongoing* activities, so that I can forecast the future and figure out what the firm is worth. For that purpose, the most recent costs

[6] A sharp reader might notice a flip side: if, in some unusual year, that company ran its inventory way down, pac-man would finally reach those decades-old batches—and then LIFO's *COGS* would suddenly be based on wildly stale costs, too. Fair point. But that is an edge case; in the normal course of business, firms continuously replenish their inventory, and the oldest batches just sit there, untouched.

are the most relevant—whether they happen to be rising or falling.[7]

Empirically, the choice of method has **financial implications** for the company's **reported** bottom-line net income. Usually, **in most industries, costs tend to rise over time**. Most advanced economies experience **inflation** rather than deflation, which is, by definition, a rise in average prices. And since **LIFO** 'picks out' the **most recent** inventory purchases, while **FIFO** picks the **earliest** inventory purchases, this means that the **LIFO** method usually generates **higher COGS expense** recognition. By core accounting identities, that means that LIFO will also entail lower net income, and a lower Inventory (and net Asset and Shareholders' Equity) balance. And **just the opposite for FIFO**.[8] And, of course, these effects don't stop at the bottom line: because *Net Income* is the numerator of *ROA*, *ROE*, and *profit margin*, the choice of inventory costing method ripples through all the profitability ratios we learned in the ratio-analysis chapter. Given this, **why would companies ever choose to use the LIFO method**? Here, it's time to introduce a new accounting theme: **tax reporting vs. financial reporting**.

10.3.3 FIFO, LIFO, and Taxes

10.3.4 Going deeper: Why reported profit and taxable profit differ

A company can report one profit number to investors and a different taxable-income number under the tax rules. The two systems serve different purposes and apply different recognition and measurement requirements.

That distinction matters when we ask why a company chooses an accounting method. A method used in its financial statements does not automatically determine its tax deduction. We have to ask what the tax rules require, and whether those rules link the two choices.

Inventory supplies a useful case because the LIFO conformity requirement creates such a link. Keep the general distinction in mind as we turn to that particular requirement. We will need it again when discussing depreciation.

[7] That last clause matters: if prices are *falling* in your industry, the numerical comparisons between FIFO and LIFO all flip (see below)—but the normative argument for LIFO is unchanged, because the most recent costs are still the most relevant for assessing ongoing profitability. And if prices are perfectly flat, the methods converge, and the whole debate is moot.

[8] In the minority of times and industries in which inventory purchase prices are decreasing over time, the situation will be just the opposite.

Relevance of *tax* to FIFO/LIFO

So far, we've assumed that the methods that these two accounting systems, **financial vs. tax reporting**, use are **independent**—that a particular accounting choice that the company makes for financial accounting purposes has no effect on its tax profit. And that is still the **general rule—the two systems are independent, by default**. This point is worth committing to memory now, because it will save you from a tempting fallacy later. Whenever you hear someone claim that a company reports some way *for financial accounting purposes* "because of the tax benefit," your default reaction should be suspicion: the tax benefit comes from the *tax* method, which is generally free to differ. (When we cover accelerated depreciation in the next chapter, remember this—there is no 'depreciation conformity rule.')

But *Inventory* accounting is a **special case**, the one exception to this rule in this book. Under the **"LIFO conformity rule,"** a company choosing **LIFO for U.S. tax purposes** generally must also use LIFO in the relevant financial reports to owners and creditors. This is an asymmetric requirement, not a rule that every tax and financial-reporting inventory method must match.[9] As above, since costs tend to rise over time, **LIFO** will usually be the method that will show higher COGS expenses and thus **lower profit**. Lower tax profit means **lower taxes owed**—a good thing for companies' owners.

For a company using LIFO for tax purposes, the conformity rule generally means accepting LIFO in its financial reports as well. When inventory costs are rising, this can mean **lower reported Net Income in exchange for a cash-tax benefit**. That is a real tradeoff. Investors who understand the accounting choice can take the tax benefit into account rather than treating lower reported profit as evidence of worse operations. But this does **not** establish that most companies choose LIFO, or that tax savings always outweigh the other considerations.

The choice also warns us against assuming that managers always prefer the highest reported profit. If a disclosed accounting choice reduces a real cash tax payment, investors can consider that saving while adjusting their comparison of reported earnings. The lesson is to distinguish a change in the number reported from a change in the cash available to the business.

Two natural questions remain. First: *if LIFO reduces tax revenue, why does Congress allow it at all? And why does the FASB allow multiple methods, when mandating a single one would make companies so much easier to compare?* I don't have a definitive historical answer, but here is my honest one: path dependence and politics. Both methods were established as allowable long ago; companies have since built systems and tax

[9] Strictly speaking, the rule is asymmetric: *if* you use LIFO for tax reporting, you must use LIFO for your financial reporting. There is no reverse "FIFO conformity rule"—nothing legally stops a company from using FIFO for tax and LIFO for its financial statements (though it's hard to think of a reason to want to). We call it the *LIFO* conformity rule because that is the direction in which it has bite.

strategies around them; and any attempt to eliminate LIFO now would face intense lobbying from the firms that benefit. Bills to change the treatment of LIFO surface in Congress with some regularity, and they tend to die there. Standard-setting, like much else in the U.S., is path-dependent and political.

Second: *why might a company decline a possible LIFO tax benefit?* The size of that benefit depends on changes in its inventory costs and the applicable tax rate. Keeping track of LIFO layers also costs money, and a company reporting under IFRS cannot use LIFO. In an industry where costs fall, the direction of the cost comparison can reverse. These factors make the choice more than a contest to minimize one year's reported profit or tax bill.[10]

How common is LIFO in practice?

The conceptual comparison should not be mistaken for a description of which method most companies use. **LIFO has become a minority method among U.S. public companies.** Its use declined over the decades after the high-inflation period of the 1970s and early 1980s. For one dated benchmark, Tinkelman and Ling report that **6.8% of Compustat companies with inventory used LIFO for some inventory in 2019**, while **4.8% used it as their primary method**.[11] Those are 2019 observations, not a current census.

More recent evidence points in the same direction. PwC's March 2025 report counted **121 public companies using LIFO as their primary method in 2023**, compared with **220 in 2013**.[12] Be careful with the unit being counted: a company can use different methods for different inventories. Using LIFO for some inventory, using it as the primary method, and reporting a LIFO reserve are different measures. And a small share of companies need not imply a small economic role; the remaining users include large firms in inventory-intensive industries.

The alternatives are not limited to FIFO. **Twin Disc switched its U.S. inventories from LIFO to FIFO in fiscal 2026**, citing comparability and consistency among its reasons. **Kaiser Aluminum switched from LIFO to weighted-average cost in 2025**, cit-

[10] People make analogous tradeoffs in their personal tax planning all the time—consider everyone who could, in principle, cut their tax bill by establishing residency in a lower-tax state, and stays put anyway, because they like where they live. Firms, too, are allowed to decide that tax minimization is not the only thing that matters.

[11] Daniel P. Tinkelman and Qianhua Ling, "The Rise and Decline of LIFO", *Accounting Historians Journal* 49(2), 2022, pp. 103–120. The percentages refer to 2019 and to companies with inventory in their Compustat sample. The article is also reproduced in the authors' September 2024 submission to the FASB.

[12] PwC, *Economic and Fiscal Analysis of the LIFO Method of Inventory Valuation*, March 2025, p. 7, prepared for the LIFO Coalition. These are counts in its Compustat tabulations, not directly comparable percentages from the earlier study. Its measure of firms reporting a reserve is broader than its primary-method measure.

ing comparability and a simpler financial close.[13] These are companies' stated reasons for particular changes, not proof of the cause of the aggregate decline.

So what should we take from the evidence? The tax benefit matters, but the size of that benefit, administrative costs, comparability, and a company's inventory practices can all matter too. The choice to *retain* LIFO also differs from the choice to *adopt* it: a long-time user may face a tax cost from changing methods and bringing accumulated differences into taxable income. The comparison in this chapter gives us a way to understand those choices; it does not tell us that every company, or even most companies, will choose the same side of the tradeoff.

10.4 Inventory, COGS, and their relationship

We just learned a couple different allowable **methods** for estimating COGS. But the **core axioms of asset accounting** still apply to **Inventory** regardless. This means that there are a lot of things we can know, and trick-question problems we can solve, without knowing which **costing method** is being used—just knowing how assets work.

Recall that, with Inventory (like assets in general), we write it up at the purchase price, and *credit* it (reduce it) with each corresponding expense. Based off of what we've seen so far, these are the only things that affect the Inventory T-account.[14] So, the balance of the Inventory account will be the cumulation of purchases that have incremented it and the COGS expenses that have decremented it:

Inventory	
$(Purchase i.)	$(COGS from sale 1)
$(Purchase ii.)	$(COGS from sale 2)
$(Purchase iii.)	$(COGS from sale 3)
	$(COGS from sale 4)
	$(COGS from sale 5)
...	...
$Inventory Balance	

Or, put differently:

$$Inventory\ Balance = \sum \$(Inventory\ Purchases) - \sum \$(COGS)$$

[13] Twin Disc, fiscal 2026 Form 10-K, Note A; Kaiser Aluminum, first-quarter 2025 Form 10-Q, Note 1. These examples illustrate continuing changes in method; they do not establish a 2026 population frequency.

[14] The rare exception to this is when Inventory is **impaired**—written down because its value has fallen below its cost, a topic for our later discussion of asset impairments. In general, impairments are rare events, and so if a problem does not specifically specify one, you can assume them away.

You can see that this is just a case of the general asset identities we discussed in Chapter 4:

$$Asset\ Ending\ Balance = Asset\ Beginning\ Balance + Purchases - Related\ Expense$$

$$Purchases = \Delta Asset\ Balance + Related\ Expense$$

So, we can often **calculate a COGS expense without knowing the specific method used,** just by understanding these relationships. For example, suppose we have a beginning balance of Inventory (x, e.g., \$100), and ending balance (y, e.g., \$120), and are told the COGS expense (z, e.g., \$40)—then we can infer how much Inventory the company must have purchased over that period (v). Specifically, it must be that $v = (y-x)+z = \$60$. More generally, **given any three of the four variables** we can **solve** for the fourth.

In fact, when I teach this material, I open the very first lecture on inventory with a problem of exactly this kind—before I've said a single word about costing methods, or perpetual systems, or anything else in this chapter. Try it: at the end of FY2020, our company reported \$100k in *Inventory*. Over FY2021, it spent another \$80k purchasing *Inventory*. At the end of FY2021, it reported \$70k in *Inventory*. What was the company's COGS for FY2021? You already have every tool you need. *Inventory* is an asset; purchases are debited to it at the purchase price; and whatever reduced the balance from \$180k (i.e., \$100k + \$80k) down to \$70k must have been the 'using up' of the asset—the related expense. So COGS must have been **\$110k**. The point of opening with this quiz is the point of this whole section: the new labels (*Inventory, COGS*) come with no new logic. Everything you already know about the general form of an asset carries over.

10.5 'Levels of profitability' / Multi-Step Income Statement

The income statement starts with top-line revenues, subtracts out all of the company's expenses, and gets us to the proverbial bottom-line net income. Now that we've talked about COGS, we've talked about most types of expenses that are in the income statement. A common practice in business and financial analysis is to analyze different 'levels' of profitability, as we go down the income statement, netting out various categories of expenses.

Based on what we've seen so far, a simple Income Statement could look like this:

Income Statement for the Fiscal Year Ending Dec. 31 2020

Total Revenues	$120
Cost of Goods Sold	($40)
Salaries Expense	($20)
Rent Expense	($10)
Interest Expense	($10)
Tax Expense	($9)
Net Income	**$31**

A **multi-step income statement** is a presentation of the income statement that shows intermediate subtotals, such as **Gross Profit** and **Operating Income**, before arriving at Net Income. These can be GAAP amounts. Analysts may also calculate additional, non-GAAP measures, such as EBIT or EBITDA. Keep those categories distinct: a useful subtotal is not automatically a non-GAAP measure.

10.5.1 Gross Profit

After starting out with **top-line revenue**, if we **only subtract** out the first expense line item, **Cost of Goods Sold (COGS)**, the difference is called **Gross Profit**.

$$Gross\ profit = Revenues - COGS$$

(A rule of thumb worth knowing: on a typical income statement, the expenses are listed roughly in order of how *operational* they are—and the first expense listed is almost always COGS, the expense most tightly tied to the core act of selling the product. So the gross profit 'level' sits right near the top.)

One way to understand it is that gross profit is what the company's profit *would be* if its only expenses were its most 'basic' costs involved in the direct physical production or acquisition of its product (i.e., its *Inventory*). A company's **Cost of Goods Sold**—that is, the cost of its product Inventory—is often considered its most **'basic' or 'inflexible'** cost. Analysts, private equity investors, or consultants might believe that companies could **"cut out fat"** in terms of **other expenses**—e.g., the salaries of middle managers, or headquarters' rent, or advertising—but, COGS is harder to 'budge.' You can negotiate a bit with your suppliers, and improve your processes at the margin, but you can't usually cut your raw product costs in half overnight. By contrast, painful as it is, a company *can* make much bigger, faster changes to the other operating expenses—the layers of overpaid middle management, the luxurious headquarters—which are often lumped together under the label **SG&A** (Selling, General, and Administrative expenses). For this reason, you can think of gross profit as a kind of **ceiling** on

the company's *potential* profitability: roughly what the bottom line could approach if all the slack and fat in the middle were cut out. If gross profit is healthy but net income is weak, examine the costs between those two subtotals. They may include avoidable overhead, but also necessary R&D, selling costs, interest, or other economically meaningful costs. If even the gross profit is weak, cutting overhead can only do so much: the core product economics are the problem. The **classic private-equity business model** is to try to identify companies with decent **Gross Profit** margins, but with plenty of "slack" in other expenses. Private-equity investors would buy control of these firms, and cut out those other costs, in an attempt to increase the total profitability of the company and thus the value of the equity. So, the **Gross Profit Margin** is an important input in **understanding companies, as well as shaping and transforming them.**

10.5.2 Operating Income (and EBIT)

The next 'level' of profitability in our framework is **"operating income."** Here, it's easier to start with the **idea**, and then talk about the **calculation**. As the name indicates, operating income seeks to measure the value the company is generating from its actual normal business **operations**. That is, it is supposed to be exclusive of items classified as **nonoperating**, such as interest expense for many nonfinancial companies. An expense does not become nonoperating merely because it is unusual or occurs once. So, the best definition is, admittedly a bit circular:

$$Operating\ Income = Revenue - All\ Operating\ Expenses \tag{10.1}$$

In the context of what we've done so far in this book, we haven't seen any of these 'special items.' So, for our purposes, it's probably easiest to just **start at the bottom of the income statement**, and then **work our way up, adding back every expense that is *not* operational.** In the context of what we've done so far, that's a short list, of only two items, **Interest Expense** and **Tax Expense**. So, at this point in the book, *Operating Income* is identical to another **very common measure**, **Earnings Before Interest and Taxes (EBIT)**. The name indicates that it is the profit *before* taking out expenses for interest and taxes—that is, *after* adding them back.

$$EBIT = Net\ Income + Interest\ Expense + Tax\ Expense \tag{10.2}$$

EBIT is commonly used by financial analysts, because it allows us to compare the performance of two firms that have different **capital structure (leverage ratios)**, in an "apples-to-apples" comparison. The firm with more debt-financing would have higher interest expenses, holding all else equal. Therefore, if we want to compare the *operational* performance of the two companies, we would want to adjust for that by

'excluding' *Interest*, using a measure like *EBIT*.

Which of these 'levels' people care about most depends on who they are and what they are doing. For ordinary public companies, the headline Wall Street metric is usually bottom-line earnings—think of the ubiquitous Price-to-Earnings (P/E) ratio, and the financial press's obsession with 'earnings per share.' The classic private-equity investor, as above, fixates on *Gross Profit*. And in the world of **mergers and acquisitions (M&A)**, the metric of choice is **EBIT**. Why? Think about what happens in an acquisition—especially a **leveraged buyout (LBO)**, in which the acquirer borrows heavily and uses that debt to buy up the target's equity. The acquirer is going to *redesign the target's capital structure*: how much debt, of what kind, at what rates. So the target's *current* interest expense is close to irrelevant—it is about to be replaced. The target's current tax situation may change too, with the new structure and jurisdiction. If you are comparing potential acquisition targets, then, you want a measure of what the *underlying operating business* earns, before the financing and tax arrangements that you, the acquirer, are about to rip out and rebuild. That is exactly what EBIT gives you. (This, by the way, is a classic finance-interview question.)

In the simple income statement above, **Operating Income** equals **EBIT**. That need not hold generally. EBIT, defined as Net Income plus interest expense and income tax expense, can include nonoperating gains or income that Operating Income excludes. Conversely, gains or losses on assets used in operations may be included in Operating Income even though the sale proceeds appear in the investing section of the cash flow statement. Income-statement classification and cash-flow classification answer different questions.

A further distinction matters: **depreciation and amortization of operating assets are operating costs**. They are included when calculating GAAP Operating Income, whether shown separately or included in other expense lines. A measure that adds them back is a different measure and needs a different label. For example, **EBITDA** adds Depreciation and Amortization to EBIT. When reading a report, check both the name and the calculation rather than assuming that all these profit measures are interchangeable.

In this book's problems, include depreciation and amortization of operating assets in Operating Income unless the question explicitly asks for a different measure. If a question supplies no such expense, there is simply nothing to subtract on that account.

10.5.3 Multi-Step Income Statement

So, adding in these extra 'steps', our previous Income Statement would look like the below. Gross Profit comes right **below** COGS; **Operating Income** comes right **above** all the **'non-operating'** items, such as interest expense and tax expense—in this particular

case, that's equivalent to **EBIT**.

Multi-Step Income Statement for the Fiscal Year Ending Dec. 31, 2020

Total Revenues	$120	
Cost of Goods Sold	($40)	
Gross Profit		$80
Salaries Expense	($20)	
Rent Expense	($10)	
Operating Income / EBIT		$50
Interest Expense	($10)	
Tax Expense	($9)	
Net Income	**$31**	

10.6 New Levels, New Ratios

These new 'levels of profitability' also give us some new **ratios**—new tools to add to the ratio-analysis toolkit from Chapter 6. As ever with ratios, the point of dividing is to **scale**: to convert raw dollar amounts into measures that are comparable across companies of different sizes, and across time.

10.6.1 Gross Profit Margin

In the ratio-analysis chapter, we defined the **profit margin**:

$$Profit\ Margin = \frac{NI}{S} = \frac{Revenue - All\ Expenses}{Revenue} = 1 - \frac{All\ Expenses}{Revenue}$$

Now that we have gross profit, we can define the **Gross Profit Margin** in exactly the same spirit:

$$Gross\ Profit\ Margin = \frac{Gross\ Profit}{S} = \frac{Revenue - COGS}{Revenue} = 1 - \frac{COGS}{Revenue}$$

The interpretation follows from everything above: this is your margin with respect to your most **direct, basic, inflexible** product costs, before accounting for any 'overhead.' It is, in ratio form, the *ceiling* on your potential overall margin. A strong gross margin with a weak net margin tells us where to investigate, but does not prove that the intervening costs are wasteful. A weak gross margin directs attention toward pricing, product mix, and product costs as well as overhead.

10.6.2 The Common-Size Income Statement

In fact, for financial analysis, we can put *all* of the 'levels' in percentage terms, by scaling every line of the multi-step income statement by top-line Revenue. The result is called a **"common-size" income statement**:

A Multi-Step and "Common-Size" Income Statement				
Total Revenues	$200			
Cost of Goods Sold	($80)		(40%)	
Gross Profit		$120		60%
Salaries Expense	($40)		(20%)	
Rent Expense	($20)		(10%)	
Operating Income / EBIT		$60		30%
Interest Expense	($20)		(10%)	
Tax Expense	($10)		(5%)	
Net Income	**$30**			15%

Now we can separately analyze the **gross margin** (60%), the **operating margin** (30%), and the plain **profit margin** (15%)—and compare each, line by line, against a competitor's, or against the same company's own history. If our gross margin matches our competitor's but our operating margin lags, we know exactly *where* in the income statement to look for the problem.

10.6.3 Inventory Turnover

In the ratio-analysis chapter, we also covered **Asset Turnover** ($\frac{S}{A}$, sometimes called 'asset efficiency' or 'asset utilization')—a measure of how efficiently a company employs its assets to generate sales. Now that we have this chapter's new asset and its new expense, we can define a more focused cousin, the **Inventory Turnover Ratio**:

$$Inventory\ Turnover = \frac{COGS}{Inventory}$$

Note that it follows the same general pattern as our other turnover ratio: an Income-Statement 'flow' measure in the numerator, scaled by a related Balance-Sheet 'stock' measure in the denominator.[15] The interpretation: how fast is the company **'turning**

[15] In practice, analysts often use the *average* of the beginning and ending Inventory balances in the denominator, since the Balance Sheet number is a point-in-time snapshot, while COGS accrues over the whole period. (Recall the rule of thumb from the ratio-analysis chapter: whenever you see a flow-over-stock ratio, you probably want an average in the denominator.) For the simple examples in this book, I will just use the ending Inventory balance.

over' its inventory into sales—how many times per period does it sell through the stock of inventory it keeps on hand?

A frequently-asked question: why is the numerator *COGS*, rather than *Sales (Revenue)*? Because we sell our inventory at a **markup** to its acquisition cost—that markup is the very source of our margin. If we want a *turnover rate*—a measure of how many times we cycle through our stock—we should measure the numerator and denominator in the same "units": acquisition cost, not sale price. Otherwise, a fat markup would masquerade as fast turnover.

Why do we care about this ratio? Two reasons. First, it is not costless to hold inventory: shelf space in a desirable retail location, warehouse space, capital tied up in goods sitting unsold—these all cost money, and better-managed companies run a 'tighter ship.' Second, and more interestingly, inventory turnover can be a **leading indicator** of trouble. Remember: under the perpetual system, a company records COGS—often its largest expense—*only when sales actually happen.* So consider a company that keeps buying inventory but is failing to sell it. Its *Net Income* won't look so bad yet, because the unsold inventory just sits on the Balance Sheet, unexpensed. The first place the problem shows up is in a **declining inventory turnover ratio** and a **ballooning Inventory balance**—often well *before* it shows up as GAAP losses.

10.6.4 Average Days in Inventory

Finally, it is common to restate inventory turnover in more intuitive units—days:

$$Average\ Days\ in\ Inventory = \frac{365}{Inventory\ Turnover}$$

Two stylized examples will build intuition for these quantities. First, imagine a maximally nimble company: it buys \$x of inventory each day, and fully sells off and replenishes that stock the very next day. Its inventory balance at any moment is \$x, and its total COGS for the year is \$365x. So:

$$Inventory\ Turnover = \frac{365x}{x} = 365; \qquad Days\ in\ Inventory = \frac{365}{365} = 1\ day$$

Now imagine a maximally sluggish company: it always holds \$365x of inventory, and sells through that amount just once per year. Then:

$$Inventory\ Turnover = \frac{365x}{365x} = 1; \qquad Days\ in\ Inventory = \frac{365}{1} = 365\ days$$

So 'days in inventory' answers, roughly: *how long does the typical unit sit on our shelves before it is sold?* One day for the nimble firm; a full year for the sluggish one.

10.7 Capitalizing Shipping Costs

If you pay for **shipping** for your inventory (**'freight-in'** in the jargon of the business), how do you account for it? Here, the **rule is simple**, but the discussion around it is more complicated and interesting. First, **the rule**: We **add the shipping cost to that batch of Inventory**—and, by core accounting identities, that will, in turn, affect the COGS.

Example: Suppose our company is a retailer, and we buy **100 units of Inventory** from a wholesaler at a price of **$10,000**, but we also separately pay **$500** to ship the goods to our store. We account for **both** as part of the **Inventory asset**.

Inventory (+A):	$10,500	
Cash (-A):		$10,500

Purchase of 100 units of Inventory ($10k) and $500 of shipping

So, as a part of this batch purchase, we now have **100 units**, at a total **Inventory cost** of $10,500. This means that when it is time to **'expense out'** this batch as COGS, the inventory **cost per unit** will be: $\$10,500/100\ units = \$105\ per\ unit$.

This is something that would have seemed strikingly odd to you in the opening chapters of this book, for understandable reasons. How does paying out money for shipping increase the accounting value of the Inventory asset? Isn't cash payment to the shipping company, a bad thing, a cost, something that decreases value? By this phase of the book, it may seem less shocking. By now, we are accustomed to the core principles of accounting, and how expenditures are frequently 'capitalized' as assets.

There are a couple of principles we can appeal to, in order to understand this **rule**.

- First, most generally, recall that **Assets** *are* 'deferred expenses'/'prepaid expenses.' So, by adding the **Shipping Cost** *to* the **Inventory asset**, we're not *denying* that it's an expense — we're just expensing it out *later* (via COGS), rather than *now*. So, by adding the **shipping** to Inventory, to be 'baked in' to the COGS on the sale date, we're achieving better **matching of expenses to their corresponding revenues**. To make this concrete, if we were to then sell and expense out 10 units from the batch above, which we had acquired at $100 plus $5 per shipping *per unit*, the COGS from that sale would be:

COGS (-S.E.):	$1,050	
Inventory (-A):		$1,050

COGS from Selling 10 units from the batch above

- Second, suppose that, instead of **charging for shipping separately**, our wholesaler simply **baked it into the invoice for the inventory itself**—much like the psychologically-manipulative fiction of **"free shipping"** from **Amazon prime**.[16] Should the accounting be *different* in that case, based off of the way the invoice is written? Given the **rule** above—that we add shipping costs to Inventory—these different 'forms' will be accounted for identically.

10.7.1 Going deeper: Should rewriting an invoice change profit?

Suppose we pay $10,500 in total for goods delivered to our store. The invoice could call that $10,000 of goods and $500 of shipping, or $2,000 of goods and $8,500 of shipping. The goods and the cash payment would be the same.

If shipping were expensed immediately while the goods were capitalized, rewriting the invoice could move a large expense from the period of sale into the period of purchase. A new manager might welcome the resulting pattern: a bad first year followed by apparently improved margins.

Including the costs of getting the inventory ready for sale prevents that particular change in labels from determining the timing of expense. The cost follows the goods into Inventory and then into COGS.

Next chapter, we'll find out that this is a **pattern** for a **variety of assets**. In the early chapters of this book, we wanted to keep things simple, so we said that assets are written up at their 'purchase price.' But the reality is slightly more complicated than this. In fact, many assets are written up at their **purchase price plus all of the costs necessary to get them ready for use**. Shipping is the first example, but not the last.

[16] A related pet peeve of mine: vacation-rental listings advertised at "$200 per night" that, by checkout, have accreted a $250 cleaning fee and sundry 'service' charges. At the end of the day, *the price is what you pay*; everything else is marketing and presentation. Accounting, in its own quiet way, refuses to let this kind of presentation game distort the financial statements.

11

Long-Term Assets Part 1: Complications

11.1 What do we know about assets so far?

We've been learning the broad theory of assets since the beginning of this book. In Chapter 2, we described an asset as a present right to an economic benefit.

Initially we learned that companies record the value of their assets on their books at their actual **purchase price** (rather than the company's *opinions* about what those assets are worth). We appealed to the accounting principles of **verifiability and conservatism**. Later on, as we learned about accrual accounting, we characterized assets as 'deferred expenses': If you are going to use some resource over future periods, you 'write it up' or 'capitalize' it as an **asset now**, so that you can **expense it out later** on, **matching** the expense to the period in which it is used. We've talked about the **"general form of an asset"**:

Asset (+A)	$(Purchase Price)	
Expense (S.E.)	$0	
Cash (-A)		$(Purchase Price)

Purchase of Asset, date during reporting period
(External Transaction)

Related Expense (-S.E.)	$(Amount Used Up)	
Cash (A)		$0
Asset (-A)		$(Amount Used Up)

End-of-period adjusting entry for expense related to asset, 12/31

And that is **still the basic theory**, at a high-level. What we will learn in this chapter is a bunch of **caveats, details**, and a couple of things that might *look like* exceptions to that rule. And we'll talk about how to **interpret the financial statements** that are produced by these accounting rules, given these complications.

11.1.1 The plan for this chapter and the next

This chapter and the next are both about long-term assets, and both are about complicating that simple picture—but in two different ways. Everything in *this* chapter stays within the world of **historical cost**: we will still write assets up based on what was actually paid, and expense them out over time; we are just adding the extra steps, wrinkles, and interpretive caveats that come with real long-term assets. The *next* chapter covers something more radical: the special cases in which accounting *abandons* historical cost, and instead uses "**fair value**"—for special reasons we will discuss when we get there.

11.1.2 Full circle: 'What is profit?'

Before we start complicating, let's pause and zoom out, to appreciate how far we've come. On the very first page of this book, I posed a deceptively simple question: What *is* 'profit'? And I gave you a scenario: You incorporate a company and put in $50k of your own money; you get another $20k investment from your cousin, and a $100k loan from a bank; you buy a batch of shirts (inventory) for $90k; you sell 50% of them for $80k in cash, and another 10% to customers who promise to pay $18k later; you pay the bank $6k in interest; and you pay proportional dividends of $5k and $2k to yourself and your cousin. What was your profit?

At the time, you could only gesture at an answer. Now—and this is the payoff of ten chapters of machinery—you can answer it rigorously, item by item:

- The $50k and $20k that you and your cousin put in are **owner contributions** (*Common Stock*)—financing, not profit.
- The $100k loan is not profit either. You gain an asset (*Cash*) and incur a liability (the loan) of exactly the same size. Nothing has been earned; the two net out,

and Shareholders' Equity is untouched.

- Buying the shirts is not an expense. It is an exchange of one asset (*Cash*) for another (*Inventory*); under accrual accounting, that cost gets expensed out later, as the shirts are sold.

- Selling 50% of the shirts for $80k in cash: the $80k is *Revenue*, recognized on the sale date—and, on that same date, under the perpetual system, we recognize the *COGS* on the shirts that left the store: $45k (that is, 50% of the $90k—and note that, with only one purchase 'batch,' every costing method from last chapter agrees).

- The sales on credit still count: $18k of *Revenue* (debit *Accounts Receivable*), along with the corresponding $9k of COGS. (As we learned in Chapter 8, we might also need to estimate an allowance for the customers who won't pay—set that aside here.)

- The $6k of interest is *Interest Expense*: a genuine cost of doing business, which reduces profit.

- The dividends are *not* an expense. They are distributions *of* profit, not determinants of it—they reduce *Retained Earnings*, not *Net Income*.

Total it up: Revenues of $98k, minus $54k of COGS, minus $6k of Interest Expense. Your profit—your *Net Income*—was $38k (setting aside taxes). It took ten chapters to fully answer a question I posed on page one. That's not because accountants enjoy making you wait; it's because 'profit' is a genuinely intricate construction, and each piece of the machinery had to be built in order. With that morale boost in hand, let's get complicating.

11.2 Complicating and caveating 'purchase price'

11.2.1 Capitalizing costs for 'getting ready for use'

Earlier in this book, we repeated the mantra that 'assets are initially written up at their purchase price.' The reality is slightly different: assets are written up at the **purchase price paid to the seller plus the costs the company incurs to get the asset ready for use**. The mantra we repeated earlier wasn't really a lie, but, instead, an *elision*: It remains the case that accounting **primarily relies on 'historical costs'** to determine the initial value of the asset (i.e., what the company actually paid, rather than what it *wants to believe*). It's just that, now, we learn that this includes capitalizing certain **additional costs involved in the preparation** of the asset.

This relates to the discussion from the last chapter, of why **Shipping** costs are capitalized in **Inventory**. Since retailers must pay shipping in order to get their *Inventory* in a place where they can sell it, under the **matching principle**, the expense should be allocated to the period in which the goods are actually sold. But if those costs will be expensed out later, they can't be expensed out now—so they are first capitalized in *Inventory*, and thus to the *COGS* recorded on the date of the eventual sale. (Consider the alternative: if we expensed the shipping immediately, and later recorded COGS based only on the sticker price of the inventory, we would recognize part of the cost of the goods *before* any revenue arrives, and the rest upon the sale—mis-matched on both ends.) The same logic applies to other assets as well.

That's the accounting **principle**. But note that managers could potentially **use** that accounting principle **opportunistically**. Capitalizing something as an asset rather than immediately expensing it means that the company will report a higher asset balance and net income in that period. As such, U.S. **GAAP** provides a decently high level of **detail on what specific expenditures can be capitalized.**

11.2.2 Going deeper: Where should capitalization stop?

"Necessary to get the asset ready" sounds like a simple test until we begin following connections. A factory needs workers; hiring workers requires administration; administration needs offices and computer systems. In a business, almost everything helps something else happen.

If that indirect connection were enough, a manager could argue that almost every expenditure should be capitalized. Current expenses would shrink, and current profit would rise, without any change in the business itself.

That is why the broad principle needs boundaries. The question is not merely whether a cost helps the business. We must identify the asset to which it belongs and the applicable rules for including it. The examples in the main text give us a first set of distinctions; they are not a substitute for every detailed rule.

Those specifics vary by assets. For example, for *Land*, the costs of "attorney, real estate commissions, title, title search, and recording fees" will be included. For buildings (i.e., *Plant*), "realtor commissions and legal fees" are included. And for Equipment, "sales tax, delivery insurance, assembly, installation, testing, and even legal fees" count. For example, if a buyer assumes delinquent property taxes owed by the seller as part of acquiring land, that amount can be part of the acquisition cost. Paying the buyer's own old tax obligation is different: making that payment a condition of a new purchase does not turn it into the cost of the new asset.[1] However, **recurring costs**

[1] The distinction is whose obligation is being paid and why. Taxes assumed from the seller as part of

such as annual property insurance premiums and taxes are **fully expensed as they recur**, rather than capitalized (Spiceland et al, *Financial Accounting*, 5 ed., pp. 345-347).

For this introductory text, ask whether the cost is directly attributable to acquiring the asset and getting it into its intended location and condition for use. Being one-time or necessary in a broad business sense is not sufficient by itself. The examples specify the acquisition and preparation costs that belong in the asset.

There is also a deeper, conceptual reason why this particular line makes sense—why *recurring* costs, specifically, are the ones we can safely leave out. The whole point of capitalizing a cost into an asset is **matching**: we decline to expense the cost *now* precisely so that we can expense it *later*, in the periods when the asset is generating revenue. But a recurring, periodic cost—a monthly insurance premium, a monthly utilities bill—is *already* being matched, in a rough-and-ready way, to the periods of its benefit, simply by being paid and expensed as it recurs. There is nothing to be gained by routing such a cost through an asset account and then dripping it back out again; the drip is already happening on its own.

11.2.3 Subtracting out salvaged materials

The flip side of the above is that we also must **subtract** the value of materials or benefits that are **salvaged** and then **sold** off from the original asset. This is something that probably would have seemed strikingly **odd** back in Chapter 1 of this book. How does getting and selling something of value **decrease** the value of the asset?

Note that if a company salvages some material from purchased land, and then sells it for *Cash (A)*, and deducts that amount from the value of the *Land (A)*, the **total asset balance** of the company will remain **unchanged**. This will just **re-allocate** that value from *Land* to *Cash*. To make sense of this, consider the following thought experiment.

Suppose we buy land for $2 million, knowing that some material can be salvaged and sold during preparation. The expected salvage value is part of what we and the seller considered in agreeing to the price. We then sell that material for $200,000. If we left the full purchase price in Land as well as recording the cash received, we would fail to reflect that recovery of acquisition cost. The entries show the reallocation:

Land (+A):	$2,000,000	
Cash (-A):		$2,000,000

Purchase of Land, at the negotiated price

an asset purchase can be acquisition cost. Taxes the buyer already owed for its own prior operations are a separate obligation; paying them does not create a new asset cost.

Then, as we clear and grade the land for the solar installation, we recover the material we expected to salvage and sell it for $200,000 in cash. We know we must debit *Cash* for $200,000. But what is the balancing credit? We never recorded a 'Pot of Gold' asset; all we have on the books is *Land*, at $2 million. Per the rule above, the credit comes out of *Land*:

Cash (+A):	$200,000	
Land (-A):		$200,000

Sale of salvaged material

Notice that this second transaction is just a change in the *form* of an asset—like the very earliest transactions in this book. The value was embedded in the parcel when we bought it; we have now extracted that value in the form of salvageable material, and converted it to *Cash*. One asset down, another asset up; total assets unchanged; no *Revenue*, and no 'Gain.'

Here is a second intuition, in terms of cash flows: the *net* cash cost of ending up with this parcel of land, in usable condition, was $1.8 million—$2 million out to the seller, $200,000 back in from the salvaged material. Looked at that way, it makes perfect sense that the *Land* account ends up at $1.8 million. The salvage proceeds are, in effect, a reduction of the acquisition cost of the land.

This example assumes ordinary salvage associated with preparing the acquired asset for use. An unexpected discovery worth more than the land's purchase price would raise a different accounting question. It should not be treated as a mechanical invitation to make the Land balance negative.

11.2.4 Practice Problem: Capitalizing and Subtracting

Problem: MattCorp purchases Equipment from a seller for a price of $10m. MattCorp pays legal fees of $500k to secure title, as a part of the transaction; pays a nonrefundable purchase tax of $250k on the equipment; pays a recurring annual $1m insurance premium to insure the Equipment; and salvages scrap materials from the equipment that it then sells for $1m in Cash. What would be the initial carrying value of the Equipment on MattCorp's books (i.e., after all these events but before any subsequent depreciation expense)?

Solution: This is all new content—but, as with most things in this book, if you break the problem down and simply ask 'in or out?' for each item, it becomes very manageable. Walk down the list:

- The $10m purchase price: obviously in.
- The $500k of legal fees to secure title: in. It is a one-time cost, and a condition of getting the asset into a legally usable state, so it is capitalized.
- The $250k nonrefundable purchase tax: in, as part of the acquisition cost.
- The recurring annual $1m insurance premium: out. It is a periodic cost, expensed year by year as it recurs.
- The $1m of salvaged scrap: it counts, but negatively—it reduces the net cost of the Equipment.

So the initial carrying value of the Equipment is $\$10m + \$0.5m + \$0.25m - \$1m = \$9.75m$.

Follow-up problem: Suppose MattCorp then assumes a useful life of 10 years, zero residual value, and straight-line depreciation. What is the annual *Depreciation Expense* on this Equipment?

Solution: We have already done the hard part. Once we have the correctly capitalized initial value, straight-line depreciation works exactly as before: $\$9.75m/10 = \$975{,}000$ per year. Note what just happened: if we had lazily used the $10m 'sticker price,' we would have calculated $1m per year. The point of the follow-up is that these capitalization (and salvage) rules are not just about the initial balance-sheet entry. Whatever we put into the asset flows through, via depreciation, into the income statement of *every subsequent year* of the asset's use.

11.3 Selling off or disposing of an asset: Gains and Losses

One thing that's missing from the 'general form of an asset' we've discussed to date (purchasing an asset and then expensing it out as it is used up), is the **final step**: Either selling the asset off or disposing of it. (One of those two will have to happen *eventually*.)[2] The **accounting** for this is relatively **simple**, but the implications are quite **interesting**—and they introduce a new category of accounts.

[2]When I was in college, a mentor consoled me after a breakup with the observation that every relationship ends in exactly one of two ways: you break up, or you die married. Not the most comforting thing to hear at the time, but admirably precise. Assets are like that: either you get rid of the asset at some point, or you hold it until the heat death of the universe—and the accounting for the latter is beyond the scope of this book.

Before the mechanics, let's set up the **problem** from first principles. Recall a point we've emphasized before, and will keep emphasizing: *accounting* depreciation does not equal 'real,' *economic* depreciation. Our depreciation schedule is a smooth allocation of a past cost, based on assumptions management made years ago—and the market does not consult our depreciation schedule. Moreover, if we are reasonably competent managers acting in our shareholders' interests, we will not sell an asset *at* its book value just because that is the number on our books; we will sell it for as much as we can get. (More money is better than less money—a fairly safe economic principle.) So the sale price will almost never exactly equal the book value. And that creates a bookkeeping puzzle: when we extinguish an asset from the books at its book value, and receive cash in some *different* amount, our total assets change by the difference—and something must be recorded to account for that.

11.3.1 Gain on the Sale of a Long-Term Asset

Consider the following example: Suppose that on Jan. 1, 2020, our company bought a piece of **Equipment** for $10 million, and assumed a useful life of 10 years, zero salvage value, and straight-line depreciation—i.e., $1m of depreciation each year. Now, fast forward to January 1, 2025. The Equipment **now has a book value** of $5m (i.e., $\$10m - 5 * \$1m$). Suppose we now decide to sell the Equipment, and fetch $6m in Cash upon the sale. How do we account for this?

The *intuitive* way to think about this is like so: We are getting **$6m in Cash** (an asset), but we are extinguishing the *Equipment*, an asset that has a net / **book value of $5m**. And so we need a **$1m 'plug'** to balance things out. And we can infer that this 'plug' **must increment *Shareholders' Equity***: The value of the *Cash (A)* we gain from the sale is greater than the book value of the real asset we discharged, **so our *net assets*—another way of saying Shareholders' Equity—increased.** We call this plug a **'Gain,'** and treat it as a contributor to Shareholders' Equity (like *Revenue*) accordingly. So, the way *I think about* this is like so:

Cash (+A):	$6m	
Equipment at book value (-A):		$5m
Gain (+S.E.):		$1m

Gain on Sale of Equipment: Intuition / logic / 'white lie'
(NOT technically correct)

But, in reality, there's **one extra step here**. As we learned in Chapter 5 many companies keep separate accounts for **Equipment (A)** and its **Accumulated Depreciation**

(contra-A). So in that case we would need to separately extinguish them. So, the **technically correct journal entry** is:

Cash (+A):	\$6m	
Accumulated Depreciation (-contra-A):	\$5m	
Equipment (-A):		\$10m
Gain (+S.E.):		\$1m

Gain on Sale of Equipment: Dirty truth (technically correct)

Written that way, it looks intimidating and unwieldy. That's why I showed the more intuitive, technically incorrect, version first, where a 'Gain' is just a plug between the book value and the sale price. But, as you can see, in the 'technically correct' version, the Gain is still that same number and plug. To solve these problems, I recommend first **thinking about** that **intuition**, and then **reminding yourself** to deal with the **extra step** for the Accumulated Depreciation contra-account.[3]

11.3.2 What is a 'Gain'?

So what exactly is a 'Gain'? It's something that *behaves* like a revenue—contributes positively to net income, on the income statement—but isn't one. We need to account for this gain, for things to balance. Our **net assets** have increased as a result of the sale, since the *Cash (A)* we got was greater than the **book value** of the real asset we just got rid of. But the sale of long-term assets is not part of the 'core,' typical **operations** of the firm. We generally don't expect Gains to be 'persistent' (repeatable) like revenues from sales of goods and services. So, we want something that increments Net Income, but isn't a Revenue—we call that thing a 'Gain.' (A gain's classification within the Income Statement depends on its nature; gains on assets used in operations can be included in Operating Income.)

Going deeper: notice that it *has to* work this way—we can *derive* the Gain's place in the accounting system, rather than memorize it. Our net assets increased as a result of the sale (the *Cash* received exceeds the book value of the asset extinguished), and no new liability was incurred. Therefore, by the accounting equation ($A = L + E$), Shareholders' Equity must increase. Within Shareholders' Equity, there are only two sources. This was not a transaction with our own shareholders, so it is not new **paid-in capital** (*Common Stock*); therefore, it must be **earned capital** (*Retained Earnings*). And the

[3] Companies can also **dispose of** assets. As you would logically expect, the basic concept is identical. If the company disposes of the asset for nothing, it is as if \$0 is the 'sale price' and everything else is the same. If the company salvages materials and sells them for \$X, it is as if \$X is the 'sale price.'

only route into Retained Earnings is through *Net Income*. So the plug *must* be something that flows through the Income Statement and increments Net Income—and, as argued above, we decline to call it 'Revenue.' Hence: a new category.

How do we interpret a Gain? Is it good news? It'll be easiest to answer this question once we look at a slightly different scenario, with a different result.

11.3.3 Loss on the Sale of a Long-Term Asset

Suppose everything 'real' above was held constant (i.e., the real purchase price, use and sale of the asset were all the same), but the management of the company had initially decided to make a **more aggressive depreciation assumption.** E.g., suppose the company had initially assumed the **useful life of the asset would be 20 years**. This, in turn, would mean the annual **Depreciation Expense** was **$500k**, and the accumulated depreciation as of the sale on January 1, 2025 would be $2.5m, leaving a **book value of $7.5m**. So, when the Equipment is sold for **$6m in Cash**, we once again need to **plug the difference**—but this time, it needs to go **in the opposite direction**.

When an **asset sells for less than its book value**, the 'plug' is called a **'Loss.'** And, *Losses* are just the opposite of *Gains*, so everything in this case is symmetric to that case. Our ***net assets*** have decreased as a result of the sale. So, the 'plug' must **decrement** *Shareholders' Equity*, like an expense. But, we do not wish to *call* this an expense, since it is not part of the company's core, recurring operations. So, instead, we create a new category, and call this a '*Loss*.' And, to get things **technically correct**, we once again need to remember that **Equipment** and **Accumulated Depreciation** are tracked in **separate accounts**, and need to be extinguished separately. So, the technically correct journal entry is:

Cash (+A):	$6m	
Loss (-S.E.):	$1.5m	
Accumulated Depreciation (- contra-A):	$2.5m	
Equipment (-A):		$10m

Loss on Sale of Equipment, January 1, 2025

Now, you might have a clever objection here: If the problem is that our book value ($7.5m) turned out to be 'too high' relative to reality, why record a 'Loss' at all? Why not simply *fix* the *Accumulated Depreciation* account—catch it up to what it 'should have been'—and be done with it? Let's follow that suggestion through, because it's more right than wrong. To bring Accumulated Depreciation up by $1.5m, we would

credit it for $1.5m. And every credit needs a balancing debit. What would that debit be, conceptually? We would be recognizing an additional $1.5m of 'using up' of the asset—that is, the debit would be an expense-like item, decrementing Shareholders' Equity. But that is *precisely* what the Loss is. Your suggested fix and the official treatment are, at bottom, the same entry; the difference is only the label. And this reveals the right way to *think about* a Loss: it is the difference between the sale price and the remaining book value. In this example, changing the depreciation assumption changes that difference. In general, a loss can also reflect market changes or an unexpected event; it does not establish that past depreciation was an accounting error.

A second question that occurs to careful readers: "Wait—I thought depreciation was recorded via end-of-period *adjusting entries*, $500k at a time. Why is $2.5m of Accumulated Depreciation being debited all at once?" Because we are not *recording* depreciation in this entry—we are *removing* it. Over the five years, we recorded five separate year-end adjusting entries, each crediting *Accumulated Depreciation* by $500k; by December 31, 2024, the account holds a $2.5m credit balance. Then, on January 1, 2025, the sale is an **external transaction**—a real exchange with an outside party—so it is recorded on its own date, and it must clear the *whole accumulated balance* off the books, along with the full $10m in *Equipment*. The $2.5m debit extinguishes what those five adjusting entries had built up.

One more small confession, while we are at it. Early in this book, I told you that the income statement was 'revenues, minus expenses, equals net income.' Technically, the complete statement is: revenues *and gains*, minus expenses *and losses*, equal net income. Gains and Losses are 'temporary'/'flow' accounts—reset to zero at the start of each period, so last year's gains do not carry over—reported on the Income Statement, and flowing into Retained Earnings via Net Income, just like their operational cousins. This is a refinement of what you learned, not a contradiction of it: the separate labels exist to signal, to the users of the financial statements, which items are recurring and operational, and which are not.

Now, note that we've **gone from a 'Gain' to a 'Loss,' holding all the 'real' transactions fixed, just by changing the initial *estimate*, the depreciation *assumption***. If you find this weird, and arguably problematic, you should!

11.3.4 Are gains (vs. losses) good (vs. bad)?

So, does a 'Gain' vs. 'Loss' mean 'good' vs. 'bad' news? Not really. Or at least a 'gain' (vs. loss) does not mean that the company did a good (vs. bad) job using or selling the asset, or made a wise (vs. unwise) choice to sell at that price. Gains and Losses simply **are what they are**, by definition: The difference between the book value of the asset and its realized sale price. And the **book value** of the asset is determined by the

accumulated depreciation charged to the asset, which, in turn, is a function of **past, subjective managerial estimates.**

A Gain means that the sale price exceeds the remaining book value. In our paired examples, the depreciation assumption explains the difference between a gain and a loss. More generally, market prices, scarcity, inflation, or other changes can also affect the sale price. So a gain does not prove that the past depreciation estimate was wrong, or that the asset was managed well. It tells us the result of comparing a realized sale price with an accounting carrying amount.[4]

This has a direct implication for financial analysis. Suppose you are building a model of a company—forecasting its future earnings from its past financial statements. You notice that net income jumped sharply in the most recent year. Don't take that at face value; look at *why*. If the jump came from growing operating revenues at stable margins, that tells you something about the underlying business. If it came largely from a one-time Gain on the sale of a building, that is a very different story: you should not pencil in 'sells another factory' as a line item for every future year. This is analogous to a subtlety we saw with the allowance method in Chapter 8: some items that flow through net income are perfectly 'real,' but reflect one-time events, or the delayed consequences of past periods' estimates, and tell you little about the ongoing operations you are trying to forecast. Analysts therefore typically separate such items out.

Gains and Losses are also common suspects in "earnings management" — techniques that managers tend to use to 'massage' their reported net income over time. For example, managers might **intentionally over-depreciate assets** on their books (i.e., in accounting terms) in order to 'store up' the ability to sell the asset and realize a 'gain' in the future. That way, if, e.g., the company has a bad year in terms of sales, it can use the gain to **buffer** its earnings. Savvy analysts are on the lookout for these gimmicks, and know how to adjust for them. Check the classification of the particular gain or loss. A nonoperating item is excluded from Operating Income; a gain or loss on an operating asset may be included.

11.3.5 A useful formula: backing out the sale price

We now have two ways of seeing a Gain or Loss: the **intuition** (the plug between the sale price and the book value), and the **journal entry** (the mechanical implementation). For working problems—and for real financial analysis—it is worth making a third layer explicit, the **formula**. By definition:

$$Gain\ (or\ Loss) = Sale\ Price - Book\ Value$$

[4]The same caution applies to losses: a sale below book value does not, by itself, establish that the original depreciation estimate was unreasonable. Accounting depreciation allocates cost; it does not continually measure market value.

...where 'Book Value' is the carrying amount of the asset at the moment of the sale. If that difference is positive, we call it a Gain; if negative, a Loss. (A Loss is just a negative Gain.) And, of course, we can rearrange:

$$Sale\ Price = Book\ Value + Gain$$

(subtracting a Loss, if applicable, instead of adding a Gain). Why bother with the rearrangement? Because it corresponds to an inference that analysts actually make. If you are not the bookkeeper inside the firm, the sale price of a particular asset may not be disclosed separately. Sale proceeds can appear in the cash flow statement or notes, but sometimes only as an aggregate. What you *do* observe is the book value of the assets (from the Balance Sheet), the depreciation expense (from the Income Statement), and any Gain or Loss upon sale (also from the Income Statement). From those pieces, you can reconstruct the book value at the time of the sale, and *back out* the sale price. This is not an artificial textbook trick; it is exactly the kind of detective work that reading financial statements consists of.

Worked example: On January 1, 2017, Joe's Inc. bought a machine for $100,000. The company assumed a useful life of 20 years, a salvage/residual value of $20,000, and straight-line depreciation. On January 1, 2022, Joe's Inc. sold the machine and recorded a $10,000 Gain upon the sale. What was the sale price?

First, reconstruct the book value at the time of the sale. The depreciable value is $\$100,000 - \$20,000 = \$80,000$, allocated over 20 years: $4,000 of depreciation per year. Five years elapsed between purchase and sale, so Accumulated Depreciation stood at $20,000, and the book value was $\$100,000 - \$20,000 = \$80,000$. Then apply the formula: $Sale\ Price = \$80,000 + \$10,000 = \$90,000$.

Another, with a Loss: On January 1, 2020, our company bought Equipment for $20m, assuming a useful life of 10 years, zero salvage value, and straight-line depreciation. On January 1, 2025, we sold the Equipment, and recorded a $2m *Loss* upon the sale. What was the sale price? The book value on the sale date was $\$20m - 5 \times \$2m = \$10m$; a $2m Loss means we sold for $2m *below* book value; so the sale price was $8m.

11.3.6 Gains, Operating Income, and EBIT

In Chapter 10, when we introduced the multi-step income statement, I promised that *Operating Income* and *EBIT*—identical at that point in the book—would eventually come apart. Gains and Losses are what pry them apart. Consider a company that sells land held as a nonoperating investment and reports the following Income Statement:

Total Revenues	\$120k
Cost of Goods Sold	(\$40k)
Salaries Expense	(\$20k)
Gain (upon Sale of Land)	\$5k
Interest Expense	(\$10k)
Tax Expense	(\$5k)
Net Income	**\$50k**

Income Statement, for the Fiscal Year ending 12/31

Notice the presentation: the expenses are in parentheses, because they subtract from net income; the Gain is shown as a positive number, just like Revenue, because it adds—mechanically, it behaves exactly like a revenue, even though we refuse to call it one.

Now: What is this company's *Operating Income*? Operating Income means income from core operations—so it excludes Interest Expense (a financing item), Tax Expense, and, in this example, the Gain on land held as a nonoperating investment. That is: $\$120k - \$40k - \$20k = \$60k$. The \$5k Gain is real, and it does increase Net Income; but it is not operating performance, so the Operating Income line is drawn *above* it.

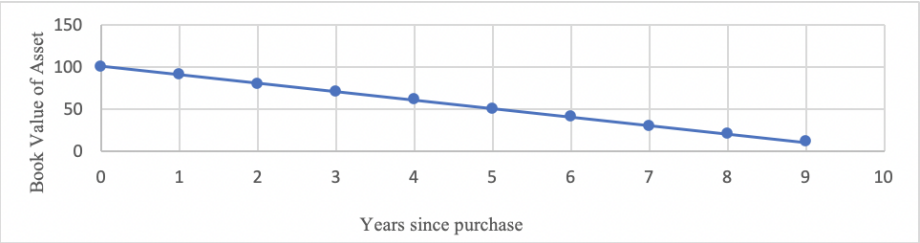
What about *EBIT*? If we hold ourselves to the strict definition—Net Income, plus Interest Expense, plus Tax Expense—then $EBIT = \$50k + \$10k + \$5k = \$65k$. EBIT adds back interest and taxes, but it makes no judgment about operating vs. non-operating items, so the Gain stays in. So, here, Operating Income is \$60k, while EBIT is \$65k. As promised in Chapter 10: closely related concepts, sometimes identical, but not the same thing.

11.4 Other depreciation 'patterns' and methods

Depreciation expense is one of the largest expense items on companies' income statements. Here are the things we've learned about depreciation to date:

1. Accounting depreciation is accounting's way of **apportioning** the **net cost** (that is, purchase price minus salvage value) of assets across the period of their use.
2. It does not necessarily equal the 'real'/economic depreciation of the asset.
3. It is justified by the 'matching' principle: Since investments in assets will be used to generate revenues over many periods, we first write those assets up, then expense them out over time as depreciation.
4. Accounting depreciation is based on managerial *estimates* about the future and, thus, unverifiable and potentially manipulable for "earnings management."

Those are still the **high-level ideas**. But, to date, since we've been focusing on the concept, we've only talked about one kind of depreciation **method**—and, in particular, the very simplest kind, **straight-line depreciation**. In straight-line depreciation, we apportion the **depreciable value** (purchase price minus estimated salvage value) **evenly** over **time**. So, concretely, suppose we bought an asset for $100 on January 1st of some year, and expected to sell it for $10 9 years later. Then the **depreciable value** would be $90, and we would record $10 in *Depreciation Expense* each year. So, the **book value** of the asset would decline as the **eponymous straight line**, over time:



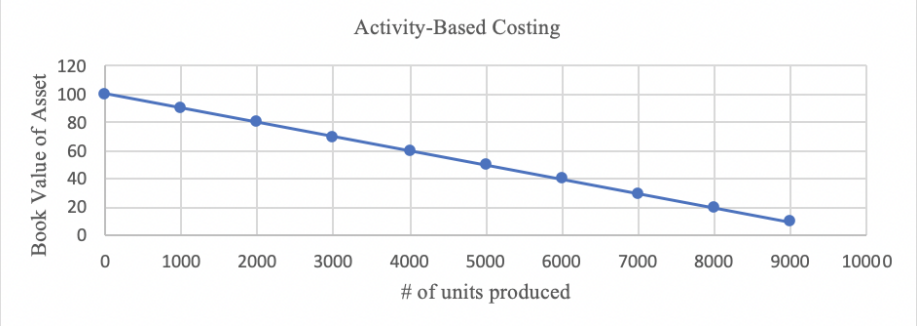
But, in principle, there are **any number of other ways in which we could apportion the expected net cost (depreciable value)** of the Equipment over the period of its use.

11.4.1 Activity-Based Depreciation

Instead of apportioning the expense **evenly per year**, we could instead **apportion it evenly per unit produced**. This is what we call **activity-based depreciation**. (Units produced are the most common measure of 'activity,' but companies can use others—hours of machine operation, miles driven by a delivery truck, etc. Whatever best measures how much of the asset's productive capacity is being consumed.) And it's quite intuitive. As before, we estimate what the salvage/residual value of the asset will be at the end of its useful life. The difference between that value and the purchase price is, once again, the *depreciable value*. But, then we **estimate** how many **units** we expect the Equipment to produce over that time period. And then we evenly allocate that depreciable value on a **per-unit** basis.

$$Depreciation\ exp.\ per\ unit = \frac{Purchase\ Price - Salvage\ Value}{Expected\ no.\ of\ units\ to\ be\ produced}$$

Note that with activity-based depreciation, the book value of the asset will once again be a 'straight line,' just with the **numbers of units produced on the X-axis,** instead of years.



However, in any given **reporting period**, the depreciation expense will **vary** depending on **how many units were actually produced** in that period. So the pattern of depreciation expenses and the Accumulated Depreciation T-account will be **uneven**, on a **periodic basis**, even though the expense is **evenly** allocated on a **per-unit basis**. So, for example, if we purchased some Equipment at the beginning of 2021, and did annual reporting, then our *Accumulated Depreciation* account at the end of 2023 would be like so:

Accumulated Depreciation (contra-A)	
	$(Depreciation per unit * units produced in 2021) $(Depreciation per unit * units produced in 2022) $(Depreciation per unit * units produced in 2023) ...

Worked example: Our company purchases a piece of *Equipment* for $25m in FY2020. We expect the equipment to produce 500,000 units over its useful life, and to have zero salvage/residual value at the end; we use activity-based depreciation. During FY2020, it produces 100,000 units. What is the *Depreciation Expense* for FY2020?

The depreciable value is the full $25m (zero salvage). Allocated evenly across the 500,000 expected units, that is $50 of depreciation per unit. The equipment produced 100,000 units this year, so the Depreciation Expense is $100,000 \times \$50 = \$5m$. (Or, as a mental shortcut: 100,000 units is one-fifth of the expected lifetime total, and one-fifth of $25m is $5m.)

Worked example, with a different measure of 'activity': Glacier Company purchases Equipment for $30,000. It expects to operate the equipment for a total of 20,000 *hours* over the next 5 years, at which point it expects a residual/salvage value of $10,000. The company uses the equipment for 4,600 hours in its first year. What is the first year's Depreciation Expense? The depreciable value is

$\$30,000 - \$10,000 = \$20,000$; allocated over 20,000 expected hours of operation, that is \$1 per hour. So the first year's Depreciation Expense is $4,600 \times \$1 = \$4,600$.

Why do companies choose to use activity-based depreciation, or not? There is a quite persuasive **argument in favor** of this method: By allocating expenses on the basis of the actual units produced, it seems it could do a **better job of matching the expenditure on the equipment to the revenues** it generated, a major principle and goal of accounting. And, indeed, many academics and consultants advocate for it on that basis, arguing that it gives a better picture of a company's profitability.

Why don't all companies use it, then? It can be **difficult to forecast** how productive a new piece of equipment will be. And so, in turn, when companies use activity-based depreciation, it may be difficult to forecast annual depreciation expenses; or, they could produce many more units than expected, and need to make an awkward adjustment to keep its book value above zero. Another reason is simple **convention**: People and firms are more accustomed to time-based straight-line depreciation, and **norms** inside companies are **sticky.**

There is a sharper way to frame this tradeoff: ask *who the number is for*. For *internal* decision-making—the province of managerial and cost accounting, and the daily bread of consultants—activity-based depreciation is often exactly what you want. If you are trying to work out which divisions or factories are truly profitable, tying the equipment's cost to its actual usage gives you a very sharp per-unit tool. For *external* financial reporting, however, a CFO may prize *predictability*: she would like a reasonably stable sense of what the income statement will look like over the next ten years. Straight-line depreciation delivers that by construction; activity-based depreciation requires forecasting not just how long you will keep the factory, but how much it will produce in each individual year. So it is common for firms to use activity-based measures in their internal analyses, while sticking with time-based methods in their GAAP statements. (All of the methods in this section are allowable under GAAP.)

11.4.2 Accelerated Depreciation Methods

Another thing companies can do is use a **time-based** pattern of depreciation, but **vary the rate per time-period**. By definition, **'accelerated depreciation'** is when companies charge **higher depreciation expenses earlier on**, and, in turn, lower expenses later on in the asset's life.[5] Many companies use accelerated depreciation schedules for most of their long-term tangible assets.

[5]This can confuse first-time learners, because 'accelerated' depreciation is, as it were, decelerat*ing*: the expense gets smaller every year. The name makes sense if you think of how we use 'accelerated' in everyday language. An 'accelerated math program' does not mean you do more and more math every year, forever—it means you get to the harder material *earlier* than usual. Likewise, accelerated depreciation recognizes more of the total depreciation *earlier* in the asset's life.

Why do companies do this? A normative argument is that this may, in some cases, provide better **matching**, for assets that are expected to be **more productive in the earlier years** of their use. A more pessimistic reason is that companies may want to better 'manage' earnings. They may want to record higher depreciation earlier on so that they can then show more 'growth' in the future, or store up the ability to recognize a Gain upon the sale.

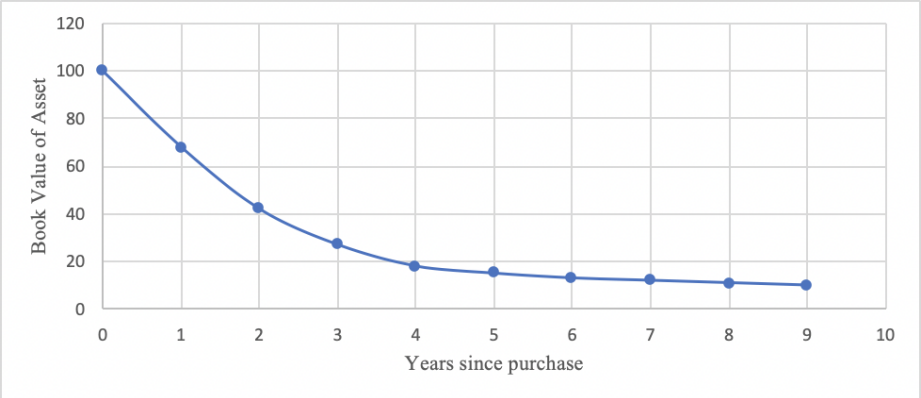
Formalizing, by definition, *accelerated depreciation* is when the depreciation expenses are higher earlier on. So, the T-account for Accumulated Depreciation will be some instantiation of this general form:

Accumulated Depreciation T-Account (Accelerated Depreciation)

	$X
	$Y
	$Z
	...

...where $Z < Y < X$.

And the pattern of the asset's book value over time will look something like this:



That's the definition of accelerated depreciation. And, in principle, there are infinite ways in which a company could depreciate their assets that satisfy this definition. For those of you with calculus background, you may note that there are *infinite convex curves* that could take us *from* the purchase price down to the estimated salvage value. In many introductory accounting courses and textbooks, for historical reasons, they teach **one particular method** for doing this, the **double-declining balance method**. However, this method has **no special logic** to it, and a number of **illogical flaws**. Therefore, I do not teach it—in my classroom or in this book—and instead just expect you to understand the **concept** of accelerated depreciation. (If you're desperate to know the method, you can Google it.)

Worked example: Notice that we can work real problems using only the *concept*. Suppose our company has only one item in its *Equipment* account, and uses an accelerated depreciation method. At the end of FY2017, the book value of the Equipment was $900k; at the end of FY2018, it was $600k. Based on this information alone, which of the following is a *possible* Depreciation Expense for FY2019: $200k, $400k, or $500k?

Reason it out by remixing a few core ideas. What makes the book value of Equipment fall? Its related expense—depreciation. (If it helps, picture the T-accounts: *Equipment* sits untouched at historical cost, while *Accumulated Depreciation* takes the credits that reduce net book value.) So the Depreciation Expense for FY2018 must have been $300k. And, by the definition of accelerated depreciation, the expenses shrink each year—so FY2019's expense must be *less than* $300k. Of the choices, only $200k qualifies. (A variation, to test yourself: if the question had instead asked for a possible ending *book value* for FY2019, then any value strictly between $300k and $600k would do—$600k, minus some depreciation expense between zero and $300k.)

11.4.3 Why (and why not) accelerated depreciation?

Let's linger on the 'why,' because it's a perfect case study in one of the recurring themes of this book: how, and why, companies 'massage' their reported earnings over time. And let's also dispatch a tempting non-reason.

The **normative story** we have already seen: matching. Many assets genuinely are more productive when new—they run more efficiently, break down less, and support a product line that is being pushed hard in its early phase. If the revenues attributable to an asset are front-loaded, then front-loading its depreciation lines the expenses up with those revenues, which is exactly what matching asks for.

There can also be incentives to shift reported income across periods. Higher depreciation now leaves less cost to allocate later and may leave a lower book value against which a future sale is compared. We saw the latter effect in the gain-and-loss examples above. An analyst should therefore ask whether the depreciation pattern is supported by the asset's expected use, rather than interpreting a later increase in profit as operating improvement automatically.

Finally, the **non-reason**: taxes. Reasoning by analogy to LIFO from last chapter, you might guess that companies choose accelerated depreciation in their financial reporting in order to report higher expenses now, and thereby pay lower taxes. Careful! Recall the general rule from that discussion: tax reporting and financial reporting are *separate systems*, and the "LIFO conformity rule" was the one-off exception. There is **no conformity rule for depreciation**: the depreciation method a company uses in its GAAP financial statements does not constrain, and is not constrained by, its tax accounting. So 'tax advantages' is *not* a valid reason for choosing accelerated depreciation for

financial-reporting purposes—the tax game is played on a different board.

11.5 Do not depreciate Land: Why?

Here's a new **rule**, worth memorizing:

- You do not depreciate Land.[6]

If you just want to know the rule, you can stop here. But why is this the rule? And does it contradict the theories we have learned previously, in particular, the core axiom that 'expenditures on assets are apportioned out as expenses over time'?

First, notice why the question is trickier than it looks. From a *balance-sheet* perspective, the rule may not seem counterintuitive at all: a factory rusts, wears out, and gets used up over time; a parcel of land does not. But recall what this book has been at pains to emphasize: accounting depreciation was never about tracking declines in *market value*. It is about *allocating a past cash expenditure* over the periods of an asset's use, per the matching principle. We paid good cash for the land, and we will use it for many years to help generate revenues. So 'land doesn't wear out' cannot, by itself, be the whole answer.

The answer is that there is **no strict contradiction**, but understanding why requires some precision. Recall that when we **depreciate** Equipment, we **don't** always expense the **full purchase price** out as depreciation. Instead we expense out the **depreciable value**—the purchase price minus the expected salvage value. Put differently, we expensed out the **net cash outflow** from acquiring and then disposing of the asset.

The accounting reason is that ordinary **Land has an indefinite useful life**: its service potential is not used up over a finite, estimable period in the way a machine's is. That does not require us to predict that its resale price will equal its purchase price. Market prices can rise or fall. The absence of a finite useful life explains the absence of periodic depreciation; impairment and eventual sale are separate questions.[7]

You can connect that to the straight-line formula as an intuition: without a finite period of use, there is no finite number of years over which to allocate the cost. But "indefinite" is an accounting assessment, not a literal forecast that the land will exist forever, and it is not a claim that its depreciable amount must be zero.[8]

[6] If you ever go searching for accounting memes online—an activity I recommend only in moderation—you will find that roughly half of them are variations on 'you do not depreciate Land.' By accounting standards, this is S-tier humor.

[7] Market efficiency does not require an asset's expected future price to equal its current price. The rule against depreciating ordinary land rests on its indefinite useful life, not on a zero expected price change.

[8] What about land with oil, gas, timber, or minerals on it, which we very much *do* use up? In that case, the extractable resource is tracked in a *separate* account from the *Land* itself—and that account *is* expensed out as the resource is extracted and sold, via an expense called **depletion**. The Land underneath still is not depreciated.

Does this mean that Land is **"free"**? Can companies just buy Land extravagantly, with no sensitivity to their cost, and never incur an expense? Not quite. Companies **still need to test these assets for impairment** (next chapter), as with other long-term assets. So, *Land* balances can be reduced if "fair value" (which we will cover in the next chapter) falls below the purchase price. Also, purchases of *Land* need to be paid for somehow—that is, they must be financed by debt or equity. If they are financed by debt, there will be associated interest expenses. If they are financed by equity, shareholders will pay attention to that dilution, and expect an appropriate return on their investment. So purchasing these assets is not 'free' in accounting terms.

You might also ask about the opposite direction: What if the Land *increases* in value? (For readers in Southern California, this may feel less like a hypothetical than like a law of nature.) If we know—from appraisers, from brokers, from every real-estate app on our phones—that the parcel is now worth more than we paid, do we write the *Land* account *up*? The answer is no. This is the same core principle from the very first chapter: assets are recorded at their **verifiable** transaction prices, not at what we *believe* they are worth—and the belief that our land has appreciated, however well-founded, is not a transaction. The *Land* account stays at its historical cost until there is an actual transaction (such as a sale, at which point—as we learned earlier in this chapter—any difference from book value is recognized as a Gain or Loss). Book value is not market value, and does not try to be.[9]

11.6 Intangible assets: interpretation and amortization

The universe of assets can be divided into two mutually exclusive and collectively exhaustive categories: **tangible vs. intangible assets**. Tangible assets are assets that have real 'physical substance,' like Land, Equipment, etc. And **intangible** assets are those that **do not have physical substance**—things like Trademarks, Patents, etc.

In terms of accounting rules and journal entries, there is nothing new when companies *purchase* intangible assets. As usual, we write them up at the purchase price, and then expense them out over the time period of their use. For most tangible long-term assets, as we know by now, that expense is called *Depreciation*. For **intangible assets, the analogue is called *Amortization***.

[9] And a word of caution about extrapolating the recent past: the decades-long run-up in land values that once looked like a law of nature reflected a particular historical period—falling interest rates, strong demand, constrained supply—and, sure enough, when interest rates rose sharply in 2022, that seemingly permanent ascent leveled off. Whether any such trend resumes is exactly the kind of question markets price in. Recall the earlier footnote on market efficiency: if you could be *certain* a parcel would keep appreciating rapidly, why would its current owner sell it to you at today's price?

Consider *Patents*, a familiar example. A U.S. utility patent generally expires **20 years from the relevant filing date**, subject to adjustments and extensions. That is not necessarily 20 years from grant or purchase. For accounting, its useful life may be **shorter** than the remaining legal protection—for example, if a competing technology makes it obsolete. Amortization follows the expected useful life, constrained by the rights the company actually holds.

For our purposes, we will always assume **straight-line amortization**. So, if we buy a **Patent** for **$2m** on January 1, 2021, and assume a useful life of **20** years with **zero residual value** (once the patent expires), then we would recognize annual amortization expenses like so:

Amortization Expense (-S.E.):	$100,000	
Patent (-A):		$100,000

Annual Amortization of a Patent, Dec 31, 2021:

(One small wrinkle to watch for in problems: patents are often purchased *mid-life*. If we buy a patent with only 10 years remaining before it expires, we amortize its purchase price over no more than those *remaining* 10 years, and less if its expected economic usefulness is shorter. So a $1m patent with 10 years left would also carry a $100,000 annual amortization expense.)

Trademarks: the 'Land' of intangibles. What about *Trademarks*—"Coca-Cola," the Nike swoosh, "Just Do It"? Here it is worth a short detour into intellectual-property law, because legal rights can constrain these assets' useful lives. Why does the law grant patents at all? A patent gives its holder the exclusive right to profit from an innovation, and the policy goal is to motivate people to invest in inventing things: if, the moment you invented something, everyone could freely copy it and compete away your profits, you would have little incentive to innovate in the first place. But the law balances this against the interests of consumers, who benefit when innovations eventually 'go generic' and prices fall. Congress has drawn that line at 20 years.

A trademark serves a different function. The Nike swoosh is not an 'innovation' in the way a new cancer drug is. A trademark's job is to *identify the source* of goods: when you buy a soda, you want to know who made it, that it is what the label says it is—and whom to sue if it turns out to be poisonous. That identification function should not expire; indeed, it would be actively harmful to consumers if it did. So trademark law is designed differently: trademarks can be **renewed indefinitely**, for as long as they are used and defended. Coca-Cola's trademark is not going to lapse in 20 years; it will be protected *in perpetuity*, for as long as the company maintains it.

Now, here is the accounting question: do we expect a finite period of benefit from this particular acquired trademark? **If its useful life is indefinite, we do not amortize it**, although it remains subject to impairment testing. If the expected benefit is finite—perhaps the company plans to retire the brand—we amortize it over that estimated useful life. The legal ability to renew a trademark indefinitely does not automatically give every trademark an indefinite accounting life.

The more **interesting issue** is the **debate** over the initial recognition of intangible assets. People commonly claim, in the press and public conversations, that 'accounting numbers do not recognize the value of a company's intangible assets.' This is not really true. The truth is that accounting does recognize the value of intangibles that meet the FASB asset criteria, just as it recognizes tangible assets that meet those criteria. If an **intangible asset is acquired via a purchase**—such as Trademarks or Patents purchased on the open market—it will be recognized. So the operative distinction in accounting is not really *tangible vs. intangible* at all—it is **purchased vs. internally developed**. It just so happens that many of the most economically important intangibles (brands, technology, organizational know-how) are of the internally-developed kind. That coincidence is what gives the popular claim its surface plausibility.

The **controversial** issue has to do with intangible assets that are **developed internally**, rather than purchased—e.g., the patents and intellectual property that the company creates as a part of its Research and Development (R&D). Under **U.S. GAAP**, ordinary **internal R&D** is generally **expensed as incurred**. There are specific exceptions, including some software costs, and certain legal and registration costs of securing a patent can be capitalized. Our examples set those exceptions aside.

What does 'internally developed' mean, exactly, and why does it change everything? Imagine that, instead of buying a patent from someone else, we hire a team of scientists and pay them for ten years, at the end of which the Patent Office grants us a patent. Economically, we have created a valuable patent. But there was never a purchase—never an external transaction with a verifiable price tag. Even the grant itself is not a transaction: no cash or consideration changes hands on the grant date. So those R&D expenditures do not become a *Patents* asset. This does not rule out capitalizing qualifying legal and registration costs. All of the costs that went into creating it—the scientists' salaries, the lab costs—were expensed, as *R&D Expense*, in the years they were incurred. The same logic applies to brand-building: spend heavily on advertising to make your brand beloved, and the *Advertising Expense* is recognized as incurred; no 'Brand' or 'Trademark' asset appears on the books, no matter how effective the campaign.

It is worth seeing the contrast in journal-entry form. 'Building' a patent, year after year:

Research and Development Expense (-S.E.):	$1m	
Cash (-A):		$1m

'Build': pay scientists to develop technology internally

...versus 'buying' a ready-made patent on the open market:

Patents (+A):	$1m	
Cash (-A):		$1m

'Buy': purchase a patent on the open market

The same cash goes out the door; the financial statements could hardly look more different. The first entry hits net income immediately and leaves no trace on the balance sheet. The second parks the cost on the balance sheet, and releases it into net income gradually, as *Amortization Expense*, over the years of the patent's use.

This is controversial for two reasons.

- First, according to core accounting principles, if that R&D will help create valuable products, those expenditures should **ideally be matched** to their future sales.
- Second, many people fear that this has **socially undesirable** consequences. They fear it makes companies less inclined to invest in **R&D** that may have positive **externalities** that could benefit society as a whole.

But the FASB believes that expenditures on R&D cannot be **verifiably matched** to future sales with **sufficient certainty**. As such, it could be very problematic to allow companies to capitalize those expenditures as assets as they please.

So, **the FASB has a point. But so do the critics**. There is no doubt that many companies have intangibles that are real 'economic' assets, but which cannot be recognized on their books—e.g., Apple with its trademarks, and pharmaceutical companies with their internally-developed patents. And this has implications. It means that companies that are **investing in internally developing** R&D and other intangibles may have low or negative net income, because those **'economic' investments** are fully recorded as **expenses** in the company's **accounting**.

The flip side, however—which the CEOs of these companies are less inclined to note—is that once those investments start to pay off (e.g., a pharmaceutical development company passing a clinical trial, and then selling its drug), those companies will have *overstated* net income (and ROA and ROE), because the economic investments in developing those intangibles were **already fully expensed** in past years.

11.6.1 'Build vs. Buy': A tale of two companies

Real firms differ, strategically, on exactly this dimension. Apple famously *builds*: it develops its operating systems, chips, industrial design, and brand in-house. Cisco, by contrast, is famous for *buying*: it watches the startup ecosystem, waits for a technology to prove itself in the market, and then acquires the company. (Famously, in the dot-com era, whole cohorts of startups had, as their entire business plan, 'build something cool and get bought by Cisco.') Two firms can compete in the same technology markets while acquiring their intangible assets by completely different routes—and, given the accounting above, their financial statements will look very different *even if their underlying economics are identical.*

Let's make that precise, with a deliberately stylized thought experiment: two companies, in parallel universes, with identical later revenue streams, but different costs and timing of acquiring the technology.

Company A ('build') hires scientists at $t = 0$, and spends \$1m per year for ten years developing a technology; during those years, it earns no revenue from it. All \$10m is expensed as *R&D Expense*, as incurred. At the end of year 10, the Patent Office grants the patent, and, from year 10 to year 30, the company earns \$2m per year in revenue from it—with **no associated expense**: the R&D was fully expensed years ago, and there is no asset on the books to amortize.

Company B ('buy') spends nothing for ten years. At the end of year 10, it buys the (identical) patent on the open market for \$20m: debit *Patents*, credit *Cash*. From year 10 to year 30, it earns the same \$2m per year in revenue—but now with a \$1m annual *Amortization Expense* (\$20m, over the 20 remaining years). So it reports \$1m per year of income from the patent, where Company A reports \$2m.

	Company A ('build')	**Company B ('buy')**
Years 1–10	R&D Expense \$1m/yr; Revenue \$0	Nothing
Year 10	Patent granted (no entry: not a transaction)	Buys patent for \$20m (*Patents (+A)*)
Years 10–30	Revenue \$2m/yr; no related expense	Revenue \$2m/yr; Amortization \$1m/yr
Income from patent, yrs 10–30	\$2m/yr	\$1m/yr

Build vs. Buy: same later revenues, different cost histories

From year 10 onward, these two companies are doing *exactly the same thing*: each owns the same patent, and earns the same \$2m per year from it. But Company A looked terrible for a decade (all expense, no revenue), and then looks fabulously profitable ever after; Company B never looked bad, and now looks merely decent. An analyst

comparing the two at, say, year 15—as 'comps,' in the argot of investment banking and equity research—would see wildly different net incomes, margins, ROA, and ROE, despite identical later revenues. Their overall economics are not identical: one spent $10m gradually, and the other paid $20m later. The accounting comparison must keep those different cost histories in view. If you don't understand the build-vs.-buy asymmetry in intangibles accounting, you will misread that comparison badly. If you do, you can adjust for it.

Here is a real-world number that this framework helps us interpret. Apple's fiscal 2024 Net Income was $93.736 billion. Its beginning and ending book equity were $62.146 billion and $56.950 billion, respectively. Using the **average equity denominator** from Chapter 6 gives an ROE of approximately **157%**.[10] That is an accounting return on book equity, not a 157% return earned by someone buying Apple shares that year. Internally developed resources can help generate income without appearing as separate assets, and share repurchases reduce book equity. Both can contribute to a high ratio. The number alone does not tell us how much of the result is attributable to either factor; the analyst must examine the business and its accounting.

11.7 Contemporary issues: Discussion Questions

11.7.1 Assets, investments, and intangibles, and implications

Return to the claim from Chapter 7: "We are not profitable because we are investing in our future." We can now make the follow-up more precise. Identify the expenditure, decide whether it is capitalized or immediately expensed in the example, and then explain how it affects current and future profit. Buying equipment and funding internal research can create very different paths through the statements.

Finally, it's common to hear claims, in the popular press and on CNBC, that "Accounting does not recognize the value of intangible assets." And, as we've seen, that's *not quite correct*. In fact, accounting recognizes acquired (purchased) intangibles on the Balance Sheet, but not *internally-developed* intangibles. Those internally-developed intangibles are significant, and do matter, increasingly in recent decades. (Consider, e.g., the value of Apple's brand value, and its proprietary hardware design, and software.) In fact, in recent decades, by various measures, firms' 'book value' has **increasingly diverged** from their market value. And this is often attributed to the growing importance of these internally-developed intangibles.

[10] Apple's fiscal 2024 Form 10-K, Consolidated Statements of Operations and Shareholders' Equity. The calculation is $93.736/[(62.146+56.950)/2]$, approximately 1.574. Using ending equity alone would give about 165%, but would not follow this book's averaging convention.

11.7.2 Going deeper: Where might unrecorded value show up?

An internally developed brand may not appear as a separate asset on the Balance Sheet. Does that make the brand invisible to someone reading the accounts?

Not necessarily. If customers are willing to pay more because of the brand, that willingness may affect the company's sales and margins. We can observe consequences of an economic resource even when the accounts do not assign the resource a separate carrying amount.

That is the sense in which I want to say its value is reflected in performance. It is not the same as recognizing a Brand asset. The two statements provide different kinds of evidence, and an analyst can use both.

12

Long-Term Assets Part 2: Fair Value

12.1 Introducing 'Fair Value'

Early in this book, we had to wean ourselves off the temptation to think in terms of 'true value.' Most people arrive at accounting with the opposite intuition—that book values and earnings are, or at least should be, true economic values—so I have pushed hard against it: in accounting, we anchor on **historical cost**, the verifiable price we actually paid, and then apportion that cost over time via depreciation, amortization, and so on. That is the default, and the 'general form of an asset' remains most of the story.

But there are some special situations where that simple historical-cost story breaks down, and accounting is forced to bring in a market-based estimate of an asset's current exit price. The accounting term for this measurement is **"fair value."** This chapter collects those situations. And I want to emphasize one thing at the outset: each time fair value comes into play, it is *not* because we have suddenly abandoned our principles and decided to chase 'true value' wherever it leads. In each case, there is a specific reason why our hand is forced—the purchase price is *unavailable*, or it is *unreliable*—and fair value becomes the lesser evil. So keep this question in mind throughout the chapter: in each situation, *why, exactly, can't we rely on historical cost here?*

12.1.1 What if purchase price is unavailable?

Sometimes, a company gets a thing that counts as an **accounting asset** under the FASB definition we first met back in Chapter 2, but there's no **purchase price** that we can appeal to. How can this happen? There are several different situations in which this might happen, but I'll start with one example: a **basket purchase**. In a basket purchase, the company buys several different things that fall into different **asset accounts**, for one purchase price from one seller. This might sound odd, but is actually quite common.

For example, suppose our company buys a **factory and the property** it is on, for $1 million. This—buying the property together with the factory—would be *the rule, not the exception.* (Think about why: if you bought the factory but *not* the land underneath it, you would be at the mercy of whoever did own the land—they could, in principle, block access and shut down your operations. So real-world transactions of this type are almost always basket purchases: multiple assets acquired together, for one lump-sum price.) But, for our **accounting purposes**, it's a challenge, because the *factory* itself should be in the **Plant** account (or, aggregated with **Property, Plant, and Equipment (PP&E)**), while the *property* itself should be separately tracked in the **Land** account.

And why *must* the two be tracked separately? Why not just lump the whole purchase into one combined 'Plant and Land' account and be done with it? Because the subsequent accounting treatment is different: we will **depreciate** the Plant over its useful life, but—per the rule from last chapter—we do **not depreciate Land**. To apply the correct rules going forward, we need a separate dollar amount for each.

So, in this situation, our company has two different assets, that must be allocated to **two different accounts**, but only **one purchase price**. We know that the total **Cash** credit is $1m; and **total credits must equal total debits**; so that **total** amount must be balanced by debits to those asset accounts. But how much to each? Here's a journal entry representing what we know so far, and the **state of our ignorance**:

Plant (+A):	$???	
Land (+A):	$???	
Cash (-A):		$1 million

Basket purchase, 4/4/2021

What do we do? Here, the accounting rule is that companies must **apportion** the **total purchase price** based on the assets' **relative** estimated **"fair value."**[1]

[1] These days, companies often get specialized **appraisal/valuation firms** to do these fair-value estimates.

Example: Suppose that our company, in consultation with our appraisal/valuation experts and advisers, estimates the "fair value" of the **Plant** and **Land** to be **$660k** and **$440k**, **respectively**. Those two **appraisal estimates** add up to $1.1 million. But the verifiable total purchase price, and Cash credit, was $1 million. So, we would apportion that total purchase price $1m, to the two assets proportionally, with the *proportions* determined by the appraisal estimates. $660k is 60% of $1.1 million and $440k is 40%. Visualizing:

Appraisal Estimate

Account	Amount ($)	(%)
Plant	$660k	60%
Land	$440k	40%
Total	$1.1m	100%

Accounting Allocation of Total $1m Purchase Price

Account	Estimated / Appraised Fair Value (%)	Accounting Allocation ($)
Plant	60%	$600k
Land	40%	$400k
Total	100%	$1m

And so, applying this rule, the journal entry for this basket purchase would be:

Plant (+A):	$600k	
Land (+A):	$400k	
Cash (-A):		$1m

Basket purchase, 4/4/2021, w/ relative fair value estimates:

Before moving on, let me flag the most common conceptual mistake with basket purchases. Students see the appraised total of $1.1 million and are tempted to record the assets at $1.1 million—and then, needing something to balance the extra $100k, to book it as some kind of 'gain.' That is **not** what we do. We remain **anchored to the actual purchase price**: the $1 million credit to *Cash* is real—verifiable, historical cost. We are relying on the fair-value estimates only because we have to, and only for one job: to tell us how to *split* the $1 million between Plant and Land. The appraisals supply the *proportions* (the 60% and 40%); the purchase price supplies the *total*.

12.1.2 Why don't the appraisals match the purchase price?

Now, you might wonder: If the experts think the assets are worth $1.1 million in total, how did we manage to buy them for $1 million? Shouldn't those numbers agree? This little discrepancy actually illustrates something important about **valuation**: it is inherently somewhat subjective. There is real science to it—data, models, comparable sales—but there is also a lot of art, and reasonable people can disagree about what something is 'really' worth. Think about Zillow. Every house in America now has a Zillow estimate; and then there is the actual price the house sells for, when a buyer and seller sit down and negotiate. Those two numbers are almost never exactly the same. They're usually in the same ballpark, but they differ because of all the messy, real-world factors that go into a negotiation: how motivated the seller is, how many bidders there are, timing, emotions, and so on. Our situation is the same: the appraisers' combined estimate is $1.1 million, the negotiated transaction price was $1 million, and a 10% difference like that is entirely realistic and not at all alarming. Appraisals are estimates, not oracles.

A large gap should prompt us to investigate the assumptions behind the appraisal and the circumstances of the sale. For the basket-purchase problem, however, the accounting task remains specific: allocate the price actually paid using the relative values supplied.

12.2 What exactly is "fair value"?

Now we have a reason to ask: what was the appraiser trying to estimate? **Fair value is a market-based measurement.** For an asset, it estimates the price that market participants would pay to acquire it in an orderly sale at the measurement date. For a liability, it estimates the price to transfer the obligation to another market participant. This is an **exit-price** idea: what the holder could receive for the asset, or pay to transfer the liability. It is not a declaration of a uniquely knowable "true value."[2]

How can we estimate that price? There are two useful routes to keep in mind.

1. If there is an active market for the same asset, observed prices may supply the most direct evidence. For an asset without a quoted price, comparable market transactions can help.

2. A valuation model can estimate the **present value** of expected future cash flows. As we will learn next chapter, discounting accounts for the timing and risk of those cash flows. To estimate fair value, the model must use assumptions that

[2] See FASB, Conceptual Framework, Chapter 6, and the fair-value measurement objective in ASC Topic 820.

market participants would use. A value based only on what this particular company expects to accomplish with the asset may be a different number.

The two routes are connected: buyers consider the benefits they expect an asset to produce when deciding what they will pay. But the connection does not guarantee that every model gives the same number, or that an asset has the same value to every owner. Estimates require judgment about the relevant market, the cash flows, and the risks.[3]

For this chapter's problems, the valuation calculation will remain a **black box**. We will give you a fair-value estimate, and ask what the accounting rules require you to do with it. The next chapter opens that box far enough to explain present-value calculations. For now, keep the target clear: **an estimated market-participant exit price at a particular date**.

12.3 Where else is 'fair value' used?

12.3.1 When there is no cash involved

Let's make this more concrete. When **investors invest** cash in a company they get shares of **common stock**. Here is the same type of transaction as Transaction #1, with a new amount:

Cash (+A):	$50k	
Common Stock (+S.E.):		$50k

Stock-for-cash (Transaction #1)

But, there's nothing that stops companies from **striking other kinds of deals with current and potential investors**. Suppose that our company wants to **buy equipment** from another company, but **doesn't have cash** available. Our company could **pay** for the asset **with shares** of our common stock. (This is in fact common practice among cash-constrained startups, which could pay part of their rent by issuing equity to their landlord.) In other words, we could **swap shares for equipment**. This would have the same form as Transaction #1, with only one change: we get **Equipment instead of Cash**. No other asset or liability is affected by this transaction. So the journal entry must be some form of:

[3] One of the hard things about learning business—and, honestly, about being a human—is that everything is interconnected. You can't fully understand accounting without understanding some finance, and you can't fully understand finance without understanding some economics, and so on. But you have to start somewhere. Hence the 'black box' strategy adopted below.

Equipment (+A):	$???	
Common Stock (+S.E.):		$???

Stock-for-equipment swap (our company's perspective):

But what is the precise **amount**? There is **no purchase price** that we can appeal to or rely upon. So here, accounting must **rely instead upon an estimate of the fair value** of the Equipment and/or shares. For a publicly-traded company, you could easily use the **price the shares trade at in the stock market** as an estimate of that fair value. (For a smaller, private company, you might need to rely on a valuation model of the company's shares or the equipment. But, for our purposes whatever the 'black box' spits out as fair value is the number we use in this situation.) Suppose that the company is a large, publicly-traded company, and on the date of the deal, its shares are trading for $50 per share, and it issues 1,000 shares in exchange for the Equipment. Then, the entry would be:

Equipment (+A):	$50,000	
Common Stock (+S.E.):		$50,000

1,000 shares-for-equipment swap (our company's perspective):

There is a bit of jargon worth knowing here. **'Consideration'** is the business-law term for what each party gives up in a transaction—anything of value that is exchanged. Cash is the most common form of consideration, but it is not the only one: parties can also pay with goods, services, or equity. So deals like the one above are said to involve **"non-cash consideration."**

Another worked example: The same reasoning applies if a company issues shares in exchange for a prepaid lease: debit Prepaid Rent and credit Common Stock. The new question is how to measure the exchange when the company's own shares do not trade in a public market.

Where could the number come from? At the end of the day, we need an estimate of *either* what the shares we're issuing are worth, *or* what the lease we're receiving is worth—either route is acceptable, so long as the fair-value estimate is reasonable. If we're publicly traded, the natural starting point is our **stock price**: suppose we issue the landlord 10,000 shares, and on the lease-signing date our stock is trading at $80 per share. Then the fair value of what we've given up is $10{,}000 \times \$80 = \$800{,}000$, and the entry is:

Prepaid Rent (+A):	$800,000	
Common Stock (+S.E.):		$800,000

Stock-for-lease swap: 10,000 shares, trading at $80 per share on the transaction date:

But suppose instead we're privately held. Our shares don't trade on an exchange, so there is no clean market price per share—perhaps the last venture-capital financing round was long ago, and its price is stale or unrepresentative. In that case, it can be more reliable to start from the *lease* side: What are **comparable leases**—similar office space, in the same building or neighborhood, on similar terms—currently going for? That market evidence gives us an estimate of what the lease is worth, and we use *that* as our number. Or, finally, we could commission a full **financial valuation** of our own shares: project the company's future cash flows, discount them, and back out an implied share value. (That's the 'black box' again—we open it up next chapter.)

This is a bigger deal in practice than the toy example may suggest. The classic startups everyone loves to follow—your Ubers and Airbnbs, in their startup years—were famous for paying their employees with **stock-based compensation**. Under U.S. GAAP, reported salaries expense must include the *fair value* of that stock-based compensation: giving employees stock or options is part of the cost of compensating and retaining them, so it must be measured and run through the income statement. Which means that privately held startups preparing to go public have to bring in valuation experts to estimate the fair value of their shares, and of all that stock-based compensation—often going back several years. This comes as a shock to some founders: they've been running a company of salespeople and engineers, and then discover that, to go public, they need a small army of finance and accounting people to get their GAAP financial statements in shape.[4]

12.3.2 When a 'purchase price' can't be relied upon

Previously we've said that when a company buys something in an **arms-length transaction in a competitive market**, the **purchase price** is **reliable**. This statement may become more palpable when we contrast it to the **opposite situation**:

Related Party Transaction:

Suppose that there are two distinct legal corporate entities, Corporation A and Corporation B. But, although they are legally separate entities, **Corporation A controls**

[4] A practical career tip, free of charge: if you can land a job in the financial planning & analysis (FP&A) or valuation group of a company that's preparing to go public, that can be a very nice gig—lots of responsibility, lots of visibility, and often very good pay.

Corporation B, by owning more than 50% of its voting shares. That is, **Corporation B is a subsidiary** of Corporation A. Now, suppose we observe the following behavior: First, **Corporation A** buys a piece of real estate from an **outside party** for **$1m**. Then, a month later, Corporation A sells the real estate to **Corporation B for $5m**. Should Company A be able to recognize a 'gain' on the real-estate, and increase its net income and shareholders' equity by $4m?

Instinctively, this situation seems corrupt, or at least somewhat **unreal**. The 'purchase price' in this case does not seem to be 'real' or reliable, because it is not *competitive*. It doesn't seem like Corporation B bought that piece of real estate for $5m because it believed that was the **best deal** available to it. It seems like it bought it because it is controlled by Corporation A, and was **told to** do so. This is one example of the general notion of a **related-party transaction**. This is a highly technical area, and so we won't cover all the details in this introductory text. For now, what we need to know is that, in such situations, a quoted 'purchase price' may not be considered reliable (and why)—and the applicable accounting may require measures or consolidation adjustments that do not simply accept that quoted price.

Adam Neumann WeWork Example:

Or consider another high-profile example, (featured, like many of the best case studies in this book, in a streaming documentary). Adam Neumann was **founder, controlling shareholder, and CEO** of *WeWork*, a business that *subleased* hip working spaces to startups and provided co-working spaces for remote workers.

In July 2019, WE Holdings LLC, in which Neumann was a managing member, assigned rights relating to the "we" family of trademarks to the company. In exchange, the company issued partnership interests valued at approximately **$5.9m**. That issuance was later unwound and the interests returned, while the company retained the trademark rights.[5]

At the time, WeWork was planning to go public in an IPO, which would require filing public financial statements according to GAAP.

Discussion question: For the discussion, round the amount to $6m. Suppose WeWork had gone public as planned in 2019. Should it have valued the *Trademark* on its books at the $6m purchase price?

Intuitively, this situation seems fairly *corrupt*. Economically: This "purchase price" does not seem real / competitive. Adam Neumann was, himself the controlling shareholder and CEO of WeWork, the corporate entity. So, WeWork didn't "shop around" for the best deal in purchasing this trademark. So, that $6m transaction price doesn't seem

[5] See the amended We Company IPO filing, "Unwinding of Receipt of Partnership Interests Relating to 'We' Family Trademarks."

like a "real" / reliable / "market" valuation. The conflict calls for scrutiny of the transaction, its measurement, and its disclosure. It does not, by itself, establish the carrying amount required by GAAP. (And, separate from the accounting issues here, there's also obviously a *moral* issue as well. The WeWork IPO eventually fell through, and, under public scrutiny and pressure, the partnership-interest issuance was unwound; this was not simply a cash payment returned.)

This example repays some further dwelling. First, consider Neumann's best available defense. The natural first reaction to this story is that it is simply *embezzlement*. But when we think of embezzlement, we picture someone taking money out of the corporate till and running off with it—no pretense of a transaction, just theft. Neumann's defense, if we try to be charitable, would be: "I didn't just take the money and run. I *sold* something. There was a transaction: a seller, a buyer, a price." What should we make of that argument? Here's the problem: it is a 'transaction' in scare quotes. Formally, there is a contract. But economically, Neumann was pulling the strings on *both* sides—as the seller personally, and as founder, CEO, and controlling shareholder of the buyer. He was, in effect, **negotiating with himself**. The important governance question is whether independent decision-makers on the corporate side adequately challenged the price: "Is this really worth $6 million?" And notice that the problem is especially acute for an asset like a trademark, for which there is no deep, transparent market price (the way there is for, say, publicly traded stock, or a commodity). With no market to discipline the number, and no independent party empowered to say 'no,' the price is essentially arbitrary—he could just as well have called it $60 million, and the same mechanics would have gone through.

Second, be careful not to over-generalize from this scandal. There is no rule that says the owners of a company can never transact with that company—that would be far too strong, given how complex modern ownership structures are. Suppose a big bank's venture-capital arm takes an equity stake in a young fintech startup, and the bank also happens to have proprietary technology that would genuinely help that startup. Is it inherently corrupt for the bank both to own a stake *and* to license its technology to the company? Not necessarily; that can be a perfectly legitimate related-party transaction. The point is narrower: when the parties are related, the *price* was not set by competition—so we don't automatically *trust* it.

Third—and this is the accounting bottom line—a related-party price is not automatically evidence of an arm's-length value. But GAAP does **not** impose a blanket rule to replace every related-party transaction price with fair value. The accounting depends on the type of transaction and the applicable standard, along with required related-party disclosures. In the parent-and-subsidiary example above, the intercompany sale and its profit are eliminated in the consolidated financial statements. The group cannot create a $4m gain simply by selling property to itself.

Another example: Transfer pricing. Related-party issues are not only about colorful founders. Consider a multinational company with subsidiaries in countries that tax corporate profits differently. If one subsidiary licenses patents or software to another, the **transfer price** affects how much profit each reports. The group therefore has an incentive to choose prices that move profit toward the lower-tax location. Tax authorities examine those prices under their transfer-pricing rules, often asking what independent parties would have agreed. That is related to the concern about arm's-length measurement, but tax transfer pricing is not simply a universal GAAP fair-value rule. In consolidated financial statements, the group's internal charges are eliminated. The same arrangement raises distinct questions for tax reporting and for financial reporting.

12.3.3 In a variety of asset impairments and writedowns

In Chapter 1, we introduced the core accounting principle of "conservatism," that accounting rules force managers and companies to err on the side of *understating* the value of their assets, and briefly mentioned the idea of "writing down" assets, when their value falls below historical cost, but, asymmetrically, *not* writing them *up*. Now, it's time to fill in the technical details.

Under U.S. GAAP, the impairment test depends on the asset. **Long-lived assets held and used**, including Equipment, ordinary Land, and finite-lived Patents, are tested for recoverability when events or changes in circumstances indicate a possible problem. The test ordinarily applies to an **asset group** that generates largely independent cash flows. **Indefinite-lived intangibles** and **Goodwill** have different tests, generally performed at least annually and when relevant indicators arise. A new competitor, technological obsolescence, physical damage, or an adverse regulatory change may prompt a review. We will start with the mechanics of a required writedown, then distinguish the tests below.

What is a **'Loss (-S.E.)'**? This is the *same account* we used in the case in which we sold a long-term asset for less than book value. As before, for practical purposes, a **Loss 'behaves' like an expense**—it is subtracted from *Revenue* on the Income Statement, and thus *decrements* Net Income—but **is not called an expense**.[6]

Why not call it an expense? Given accounting **conservatism**, we want to write the impaired asset **down**. And, as we know, the Income Statement and the Balance Sheet

[6] In practice, companies have some discretion in how they label and aggregate these accounts. Internally, a company might keep a specific account called *Impairment Loss*, or lump all its non-operating losses together; on the face of the income statement, some firms show a line item explicitly labeled 'Impairment Loss,' while others aggregate it with other non-recurring items. For our purposes, the label matters less than what the thing *is*: a loss that may be included in operating income, depending on its nature, and that reduces the asset and, passing through Net Income to Retained Earnings, reduces Shareholders' Equity.

need to be kept in sync. So, this must 'pass through' the Income Statement as well. But **impairments are not** part of the **'core'** operations of the firm—they are not expected to be **'repeatable'** or **persistent** like Salaries Expense or Depreciation. So, we want something that is like an expense, but **classified separately**. That's what a *Loss* is.

For an indefinite-lived intangible asset under the quantitative fair-value test, the basic comparison is:

- **Estimate the fair value** of the asset (with assistance of valuation experts).
- If 'Fair Value' $\geqslant$ Book Value: No change. Do nothing (accounting conservatism).
- If 'Fair Value' $<$ Book Value, recognize an impairment.
  - Write the asset down to the estimated fair value, balanced by a 'Loss,' with a journal entry like the below.
    - (Note that to bring the book value *from* the former value to the fair value, we must credit it for *the difference*.)

Loss (-S.E.):	$(Book – Fair Value)	
Asset (-A):		$(Book – Fair Value)

Impairment of Long-Term Asset, after testing, 7/19/2021:

Concretely, suppose this test applies and the Book Value of the asset *was* $100. Then, today, we suspect that it might be impaired, so we 'test for impairment,' and estimate that its Fair Value is now $80. Then we would need to credit (reduce) the asset by $20, and recognize that as a *Loss*. And the *balance* of the asset account after this impairment would be $80. It can be useful to see this in T-account form: the ending balance, after the impairment, *is* the estimated fair value—by construction.

Asset

Debits	Credits
$(Prior Book Value)	
	$(Book − Fair Value)
$(Fair Value)	

Note: Impairments disrupt our simple formula for asset balances!

In earlier chapters of this book, the only things that *increased* (debited) operating assets were *purchases*, and the only things that decreased them were *expenses*. This means that we were able to use this simple formula for many problems:

$$Asset\ End.\ Balance = Asset\ Beg.\ Balance + Purchases - Related\ Expense$$

Impairments throw a wrench in this. How so? They obviously reduce the asset balance, without being registered as an expense. This is one reason why it's better to focus on understanding the accounting principles and their *meaning*. If you understand impairments and losses, you should still be able to solve similar problems, without relying on a formula. Hopefully, you can now understand this more general formula:

$$Asset\ End.\ Bal. = Asset\ Beg.\ Bal. + Purchases - Related\ Expense - Related\ Impairment\ Loss$$

For practical purposes, if a problem like this—on the relationship between asset balances, expenses, and purchases—does not specifically indicate that an impairment occurred, you can assume it did not. But when it does, you should be able to handle it.

Worked example: Our company owns several different patents. At the end of FY2019, we reported *Patents* (at book value) of $900k. Over FY2020, we recorded total *Amortization Expense* of $200k, recorded a $300k impairment *Loss* on one of our patents, and purchased an additional patent for $100k. What is the book value of *Patents* on our FY2020 Balance Sheet?

This should feel like a very familiar setup—it's the same 'beginning balance, plus purchases, minus expenses' pattern you have been solving since early in this book—with the one new twist that impairment losses *also* reduce the asset. So: start with $900k; add the $100k purchase; subtract the $200k of amortization; and subtract the $300k impairment. The ending book value is **$500k**. (I am implicitly assuming here that we aggregate all of our patents into a single *Patents* account. In reality, the $300k of impairments might be one patent written down by $300k, or three patents written down by $100k each; for our purposes, the aggregate amounts are what matter.) And, of course, this kind of problem can be flipped around: given the ending balance, the purchases, and the impairment loss, you should be able to solve for the amortization expense.

The white lie and the more complicated truth

As is often the case, for the purpose of this introductory book, I first focused on providing a simple and coherent mental picture, of the main **overarching principle**, the idea of **"book value" vs. "fair value,"** and the **asymmetry** of impairment testing (that we write assets down to fair value, but not up). And that is the main important idea.

But, unfortunately the truth is that the FASB makes things **more complicated for some assets**. For long-lived assets held and used, including finite-lived Patents and Equipment, there is instead a **two-step process**, where the company first compares the carrying amount of the asset group with the **undiscounted cash flows** expected from its use and eventual disposition. That is, in reality, for these assets, the real, two-step process is:

1. If the (undiscounted) net cash flows are determined to be less than the book value the asset is said to be "impaired" (Step 1)

2. and the asset is then re-valued and written down to "fair value" (Step 2)

What is the difference between those two numbers? **Undiscounted expected cash flows** add the amounts expected from use and disposition without adjusting for their timing. **Fair value** estimates a market-participant exit price. A discounted-cash-flow model may be used to estimate it, but its assumptions must reflect market participants; it is not necessarily the company's own cash-flow forecast with a discount applied.

The Money Machine: To make the distinction concrete, suppose we own a magical asset—the **Money Machine**—which sits in the corner like a safe and will spit out $1,000 per year for the next 5 years:

Time	**+1 Year**	**+2 Years**	**+3 Years**	**+4 Years**	**+5 Years**
Expected Cash Flow	$1,000	$1,000	$1,000	$1,000	$1,000

The expected net cash flows are easy—don't overthink this part: $\$1{,}000 \times 5 = \$5{,}000$. But is the machine *worth* $5,000 today? Would you *pay* $5,000 for it, today? No. A dollar five years from now is not the same as a dollar today: there is inflation; there is the interest you could have been earning elsewhere in the meantime; and there is the risk that the promised cash never materializes. So the amount you would rationally pay today—the present value of these positive future receipts—is less than their $5,000 total when the discount rate is positive.[7] For now, the one relationship worth committing to memory is: for the same positive future cash flows discounted at a nonnegative rate, **present value** $\leqslant$ **undiscounted total**. This is a feature of those assumptions, not a universal ordering of every fair-value estimate and every entity-specific forecast.

[7] For the curious, here is a preview of next chapter's machinery. If we assume a 7% interest rate (a 'discount rate'), the fair value is $\frac{\$1{,}000}{(1.07)} + \frac{\$1{,}000}{(1.07)^2} + \frac{\$1{,}000}{(1.07)^3} + \frac{\$1{,}000}{(1.07)^4} + \frac{\$1{,}000}{(1.07)^5} \approx \$4{,}100$. Don't worry yet about where that formula comes from—that is Chapter 13's job. And in this chapter, you will not have to compute present values; the fair-value number will be handed to you by the 'black box.'

This is another area where the rules from the FASB have changed in recent years and seem likely to change in the future. Hence my desire to first focus on the main overarching concept, book value vs. fair value, rather than the details of which assets have a "two-step impairment test." Suffice to say that, in this book's problems, I don't plan to trip you up with minor procedural details—it'll be unambiguous with either *both* net cash flows and fair value being less than book value (indicating impairment and a writedown to fair value) or *both* of them being greater than book value (indicating that we do nothing). But I wanted to reveal this truth now, so that you are not misled, or surprised when you see other sources revealing the more complex truth.

Worked example: Suppose our company purchased a patent for a hypothetical drug, **Drug A**, whose current book value is $10m. A competing drug is introduced, and new evidence leads management to expect lower sales of Drug A. For this exercise, assume the patent's cash flows can be evaluated separately and that the long-lived-asset recoverability test applies. Our experts estimate undiscounted net cash flows of $9.5m and a market-participant fair value of $8m. What journal entry do we make?

Both numbers are below the $10m book value, so the asset is unambiguously impaired. And despite all the two-step testing complexity above, once we have concluded *that*, the mechanics are straightforward: we write the asset down from its $10m book value to its $8m **fair value**. (The undiscounted $9.5m plays no role in the *measurement* of the writedown; it matters only for the initial 'test.') So we credit *Patents* for the $2m difference, and the balancing debit is the *Loss*:

Loss (-S.E.):	$2m	
Patents (-A):		$2m

Impairment of the Drug A patent:

Worked example, flipped: Same setup: the Drug A patent is on the books at $10m, a competitor appears, and our experts run the analysis. This time, they determine that the expected net cash flows are $11m, and the fair value is $10.2m. What journal entry do we make? *None.* We do nothing. And this is where the asymmetry really bites: even though the fair value ($10.2m) is now *above* the book value ($10m), we do **not** write the asset up and record a $0.2m gain. Conservatism only requires (and only permits) us to recognize revisions in the *downward* direction.[8]

[8]The alert reader will now ask about the case in between: What if the two numbers *straddle* the book value—say, expected net cash flows of $11m, but a fair value of $9.5m? Strictly speaking, under the two-step model, the asset then *passes* the first test (its undiscounted cash flows exceed book value), so it is not considered impaired, and no writedown is recorded—even though its fair value is below its book

Two more notes round out the mechanics. **First**, an impairment need not take the asset all the way down to zero (and usually doesn't). If a patent on the books at $10m is impaired, and its estimated fair value is $5m, then the writedown is $5m; and afterward, we would amortize the *remaining* $5m of book value over the remaining useful life—so the annual amortization expense going forward would be about half what it was before. The general principle stands: after the impairment, the asset's new book value *is* its estimated fair value, and ordinary accounting resumes from there.

Second, you might wonder how all of this applies to *internally developed* intangibles. Suppose we had internally developed a drug patent (rather than purchasing it), and then the crushing news arrives that a rival drug has rendered ours worthless. What do we write down? *Nothing!* Recall from last chapter that the R&D costs in this example were expensed as incurred, and we assume no capitalized legal or registration costs, so the patent's carrying amount is zero. There is nothing to impair. (And note the flip side, which makes for a nice test of your grasp of last chapter's *Gains*: if we instead *sold* that internally developed patent to another company for $5m, the entry would be a $5m debit to *Cash* and a $5m credit to *Gain*—the entire sale price is a Gain, because the asset's book value was zero.)

Interpreting impairments: 'pulling' the future into the present

So much for the mechanics. What do impairments *mean*? Here is the interpretation I find most useful: an impairment is what happens when accounting takes some bad news about the *future*, and pulls it *all* into the current period, *at once*.

This is genuinely new, and it actually cuts against some of the earliest intuitions we built in this book. Think back to depreciation and the matching principle. The story there was one of *smoothing*: we make a one-time expenditure on a long-term asset, and then we smoothly allocate that cost over the periods that benefit from it, in order to get a stable, representative measure of the profitability of our ongoing operations. Impairments are the opposite. Instead of letting bad news dribble out over time, we bunch all of the expected future losses into a single period. Impairments are, as it were, **anti-smoothing**.

Illustrative example: Suppose we have a *Patent* for a drug on our books at $10m, with 10 years of useful life remaining—so, ordinarily, $1m of *Amortization Expense* per year. Now a dramatically superior rival drug is approved, and completely replaces ours in the marketplace: no doctor will ever prescribe our drug again, so we will never make another dollar of sales on it.

value. This is precisely the kind of procedural edge case I promised not to trip you up with. But if you go on to more advanced accounting coursework, or the CPA exam, you will meet it there.

First, imagine a world *without* the impairment rule. We would simply keep doing what we were doing: recording $1m of amortization per year for the next 10 years—against zero revenues:

Year	**+1**	**+2**	**+3**	**+4**	**+5**	**+6**	**+7**	**+8**	**+9**	**+10**
Revenues from *Patent*	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Amortization Expense	$1m	$1m	$1m	$1m	$1m	$1m	$1m	$1m	$1m	$1m
Net Loss	$1m	$1m	$1m	$1m	$1m	$1m	$1m	$1m	$1m	$1m

A worthless *Patent*, in a world with *no* impairment rule:

We would be dribbling the $10m loss out as ten separate $1m hits, over a decade. Now bring in the FASB's actual rule. The moment the news arrives, we have strong evidence that the fair value of our patent is (approximately) zero. We are required to test for impairment and, in this case, to write the patent down immediately from $10m to $0—recording the full $10m loss in this period alone. Going forward, there is no more amortization (there is nothing left on the books to amortize), and no revenues either:

Year	**0**	**+1**	**+2**	**+3**	**+4**	**+5**	**+6**	**+7**	**+8**	**+9**	**+10**
Revenues from *Patent*	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Amortization Expense	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Impairment Loss	$10m	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Net Loss	$10m	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0

The same worthless *Patent*, under the FASB's impairment rule:

Same real events; a very different pattern of reported profits. Instead of ten years of $1m losses, a single $10m loss, now. That is what I mean by 'anti-smoothing': the impairment rule pulls all of the expected future losses into the present period.

Impairments in the wild: "adjusted" and "non-GAAP" earnings

Once you know what an impairment is, you will start seeing them everywhere. Public companies report their full GAAP financial statements in their annual 10-K filings, and they also put out a press release each quarter announcing their earnings—announcements that Wall Street watches intensely (will the company meet, beat, or miss analysts' expectations?). And in those press releases, an extremely common pattern is: (1.) "Here is our GAAP net income, as required"—including, say, a large impairment loss taken that quarter—followed by (2.) "Here is our **'adjusted'** (or **'non-GAAP'**) earnings measure, excluding certain items"—most commonly, impairments and other non-recurring charges. In other words, the company reports the GAAP

number it is obligated to report, and then also shows you what earnings *would have been* without the one-time hit.

An obvious effect of excluding a loss is that the 'adjusted' number is higher. So the easy, cynical reaction is: "How devious—they are deviating from GAAP just to make themselves look better." And, to be sure, there are some genuinely sketchy things that companies do under the banner of 'adjusted' earnings. But in my judgment—and in the judgment of most sophisticated investors—this particular adjustment is quite legitimate. Indeed, investors don't merely tolerate this treatment; they *like* it, and they make the same exclusion in their own analyses. Why? Recall the interpretation above. An impairment loss is one-time. It is not smooth, not recurring, not persistent, and not related to the ongoing, repeatable activities of the business. Including it in this year's earnings makes that number *less* informative about what next year's earnings will be. So if your goal is to forecast future profitability, or to value the company based on its sustainable earnings power (investors commonly value companies as some multiple of their earnings), you might quite reasonably want to strip it out.

Now, you might raise a sharp objection here: If investors simply strip impairment losses out of earnings, and never do anything else about them, aren't they systematically *ignoring* real economic losses? In the no-impairment-rule world of the tables above, the analyst would have been forced to confront the loss, $1m at a time, every single year; in the actual world, if she excludes the one-time $10m hit and moves on, she never confronts it at all. That objection is exactly right, and dealing with it is what separates sophisticated analysts from naive ones. The sophisticated analyst treats the impairment as a one-time *accounting recognition* of a long-run *economic* loss—and then accounts for that bad news somewhere in her own forecasts and valuation model, for instance by spreading it over time, rather than dropping it entirely. An analyst who always excludes impairments and never compensates elsewhere will end up with measures of profitability—and valuations—that are systematically too rosy.

A pattern worth knowing: Here is a piece of practical financial-statement detective work that follows from all this. If you compute profitability ratios for a real company over several years, you will sometimes find a year in which the bottom line collapses—return on assets and net profit margin fall off a cliff—while the *gross* profit margin sails along smoothly, and everything bounces back the following year. Revenues and COGS look normal; whatever is going on is happening *below* gross profit. When you see that pattern, your first move should be to open the company's 10-K and search for the word "impairment." Very often, that is the culprit. Procter & Gamble's fiscal 2019 is a classic example: its bottom-line profitability collapsed in that one year, while its gross margin was stable—and, sure enough, its income statement shows a line item for a large impairment loss in 2019 only, which explains much of the difference. (And,

true to the pattern above, P&G's own preferred adjusted measure—which it calls "Core" earnings—excludes that charge.)

Whose rule is this, anyway? A closing debate

The tables and the adjusted-earnings discussion leave us with a real tension. Recognizing the impairment now updates the asset's carrying amount promptly, but it also concentrates a loss in one period. Spreading the original cost over the remaining years would produce a different pattern of reported performance. My instinct often favors a measure that is easier to use in forecasting ongoing operations. The counterargument is that readers also need a timely account of what the company's assets can support.

Different users of the financial statements will genuinely disagree here. An equity investor doing valuation tends to want smoothing, for the forecasting reasons above. A creditor may feel very differently. Go back to Chapter 1 first principles: creditors do not share in the upside; they bear only the downside. A bank that has lent the company money doesn't much care whether earnings are smooth and forecastable; it cares whether it will be repaid. So creditors tend to *like* conservative accounting: if there is credible evidence of a big drop in asset values, recognize it *immediately*—don't hide it from us by smoothing it over ten years.

Managers have incentives too. A new CEO may prefer to recognize a large set of losses early, attribute them to the preceding management, and then show stronger earnings in later years. This is the **"big bath"** concern. A writedown reduces the remaining asset balance and can reduce later depreciation or amortization, so some apparent improvement may reflect the timing of expense rather than an improvement in operations.

Finally: who enforces all of this? Publicly traded U.S. companies do not get to opt out of impairment testing because it is inconvenient; GAAP compliance is mandatory, and it is literally the auditor's job to push back if a company ignores an undeniable impairment trigger (a clearly superior rival drug, say). But most cases are not undeniable. Sales are soft, but not catastrophic; a competitor has appeared, but its eventual market share is uncertain. In those gray areas, management has real discretion over whether an impairment 'indicator' even exists—and, within a reasonable range, auditors may accept management's judgment. So, as in so many areas of accounting: the rule is mandatory, but its application runs on judgment—and judgment runs on incentives.

12.4 Corporate Mergers and Acquisitions and Goodwill

One of the most contentious, complex, and important areas of accounting is **Goodwill,** which is an intangible asset that is created in corporate **mergers and acquisitions**.

Understanding *Goodwill* and M&A accounting is important for any career in banking, law, accounting, valuation/appraisal, and private equity. In this introductory text, we will just learn the **conceptual bare bones**. But even that will take some effort.

Let's set up the problem in the framework of the discussion we've been having recently. In general, in this book, we've talked about how and why the **'true value'** of the corporation (or, more concretely, its **'market value'**) will differ from its accounting or **'book value.'** Those values are different, everyone knows they are, and that's okay. But in a corporate acquisition, the two different values collide, and this creates a challenging accounting problem. Here's a very simple definition and model of an acquisition:

Acquisition: In an acquisition, Company A (with 'A' standing for 'acquirer') pays a purchase price, $PP, to buy the entire target corporation, Company T, from T's now-former shareholders, and assumes all of its assets and liabilities.

Before doing any accounting, it is worth slowing down on what this definition actually says, because everyday language obscures it. In the press and in ordinary conversation, people say things like "Company A bought Company T for $10 billion," and the mental picture is of big Company A simply swallowing little Company T, amoeba-style. But ask: *who receives that $10 billion?* A natural first guess—and a genuinely common confusion—is that A pays the cash 'to the target,' and then, since the target is now part of A, the cash ends up back inside the combined company... making the acquisition somehow 'free.' That is not what happens. You do not buy a corporation *from* the corporation; you buy it from its **owners**—its shareholders. It is just like buying a used car: if I buy your car, I pay *you*, not the car. The cash leaves the acquirer and lands in the bank accounts of the target's now-former shareholders, who hand over their shares and exit the scene. The corporate entity is the thing being *transferred*; it is not the recipient of the purchase price.[9]

And what does the acquirer get? All of the target's assets—and, just as importantly, all of its **liabilities**. The target's debts and obligations do not vanish in the deal. (They couldn't: if liabilities evaporated upon acquisition, every struggling company would arrange to get itself acquired as a way of escaping its debts.) The acquirer assumes them.

So, in accounting terms, **A disburses Cash to T's now-former Shareholders** and, in return, **gets all of T's assets and liabilities**. This poses an accounting problem: All of Company T's assets and liabilities now need to be on Company A's books. If we simply moved all of those assets and liabilities over, without doing anything else, the **net debit** to A's books would be Company T's book value of Shareholders' Equity ($E_T = A_T - L_T$).

[9] It is thus mildly maddening to read, in the business press, that "the acquirer paid $10 billion *to the target*." No: to the target's *shareholders*.

But, the Acquirer just paid the purchase price, **$PP**, in cash—a **credit** of $PP. [10] So, if the negotiated purchase price ($PP) does not precisely equal the book value of the erstwhile Target (E_T), which it never will, we have an accounting problem.

And, indeed, not only will $PP not *equal* E_T—it will typically be far *greater*. Why? By now, this is a recurring theme of ours: book value is not true value. The target's assets sit on its books conservatively, at historical cost or below; and its internally developed intangibles—often the very things the acquirer covets—are not on its books at all. The purchase price, by contrast, is whatever it takes to convince the target's shareholders to sell: a negotiated, *market* value, reflecting the economic worth of all of it. So the gap between $PP and E_T is not an anomaly to be embarrassed about; it is the book-value-versus-true-value gap made flesh. And M&A accounting is the special set of rules for bridging it.

How does accounting deal with this? In two steps:

- **First**, for this special situation, the acquirer generally recognizes the target's **identifiable assets and liabilities at acquisition-date fair values**, subject to specific exceptions. The adjustment can be upward or downward.
  - 'Identifiable' means pretty much what you would expect: Assets that can be separately 'identified.' So, the assets that the target has recorded on its books (e.g., Equipment, etc.) would definitely count, and so would the target's *internally developed patents and trademarks* (which previously would not have been recognized on the target's books). However, the benefit of positive social-media posts from customers is, while a real source of economic value, not really separably 'identifiable'—so that would not count.
  - We call the difference **Net Identifiable Assets** (NIA).
    - $NIA = Net\ Fair\ Value\ of\ Target's\ Identifiable\ Assets\ and\ Liabilities$
- **Second**, we *plug* the *difference* between the **purchase price** ($PP) and the **Net Identifiable Assets** with **'Goodwill (A)'**.

Pause on that first step, because it is a very special moment in this book: this is the *only* situation we will encounter in which assets get written *up* to fair value. For the historical-cost operating assets we have studied, upward revaluation is generally not permitted. Other accounting topics, including some financial investments, do use recurring fair-value measurements that can move either way. In an acquisition, for this one purpose, the target's identifiable assets and liabilities are reset to fair value before coming over to the acquirer's books.

[10] Companies can also pay for acquisitions using their own stock (either partially or entirely). I'm assuming cash for now to keep things simple. The basic concept is the same in stock-financed acquisitions.

For an acquired **intangible** asset, "identifiable" has a specific meaning: it arises from contractual or legal rights, or it is separable—for example, capable of being sold, transferred, or licensed, alone or with a related item. Patents and Trademarks meet that test. Some customer relationships also qualify. Merely having seen a label on a Balance Sheet is not the test. In our simplified examples, "positive buzz" means a general reputational benefit that does not meet either criterion.[11]

In accounting, 'Goodwill' is an **intangible asset** account that only arises in acquisitions, and is a 'plug' that spans the gap between the price the target was purchased for and the net fair value of the target's identifiable assets and liabilities.[12] As such, it represents **everything and anything** that can explain the **difference** between the **identifiable accounting** (net) assets of the Target and the **'real'** value that the Acquirer saw and paid for. The term 'Goodwill' seems to *connote* things like customer and employee loyalty. And these would be implicitly included in Goodwill in an acquisition. But the technical accounting definition encompasses much more as well. For example, the value from positive social-media buzz about the Target's products would be *implicitly* baked into Goodwill—because that is real value, but would not be considered an identifiable accounting asset.

12.4.1 Acquisition example

Suppose our company acquires a very simple Target, paying **$10m in cash** to its now-former shareholders. Suppose the Target has the following identifiable assets and liabilities, and our valuation experts have provided the following estimated fair values:

Target's Prior Book Assets & Liabilities and Estimated Fair Values

Account	Prior Book Value ($)	Estimated Fair Value ($)
Patents	$1m	$8m
Notes Payable	$2m	$2m

Then the Net Identifiable Assets would be...:

$$Net\ Identifiable\ Assets = \$8m - \$2m = \$6m$$

And the Goodwill would be:

$$Goodwill = Purchase\ Price - Net\ Identifiable\ Assets = \$10m - \$6m = \$4m$$

[11] The contractual-legal and separability criteria are explained in FASB's Statement 141(R), the predecessor to the business-combination guidance now in ASC Topic 805.

[12] A natural question: What if the target itself carries *Goodwill* on its books, from its own past acquisitions? Do we haul that old Goodwill over, and stack the new Goodwill on top? No—old Goodwill is not an 'identifiable' asset. The acquirer effectively re-does the acquisition accounting fresh, at the new purchase price: fair-value the target's identifiable assets and liabilities as of *this* deal, and plug *this* deal's difference as the new Goodwill. The target's old Goodwill is subsumed in that process.

And the journal entry for the acquisition would be:

Patents (+A):	\$8m	
Goodwill (+A):	\$4m	
Cash (-A):		\$10m
Notes Payable (+L):		\$2m

Acquiring Simple Target, July 4, 2021

Why these numbers are realistic: It is worth pausing on the *shape* of our experts' estimates—a huge write-up for the *Patents* (from \$1m to \$8m), and none at all for the *Notes Payable*. The Patents write-up is exactly what you should expect, given last chapter. If the target developed its patents largely internally, most of that R&D was *expensed* as incurred, so the patents' book value dramatically understates their economic value. (Indeed, it is common for a target's most valuable patents to sit on its books at *zero*.) A patent under development is like an option on a technology: the company takes a big risk, invests heavily, and the outcome is uncertain—a flop, or a blockbuster. When it turns out to be a blockbuster, book value and fair value part ways completely. And note what the \$8m estimate is based on: the expected *future cash flows* from applying the patent—not the amount that was historically spent on the R&D. Historical cost is precisely what we are leaving behind in this step.

The *Notes Payable*, by contrast, gets no adjustment, and that too is realistic. A liability is, by definition, a contractual obligation to pay specified amounts on specified terms—and that contract does not change just because the company is being acquired. Absent big moves in interest rates or credit risk, the fair value of plain-vanilla debt sits close to its book value. In general, in acquisition accounting, expect the fair-value 'action' to be concentrated in intangibles (especially internally developed ones), while short-term contractual items (like *Prepaid Rent*, whose book value is essentially the remaining contractual payments on a recently signed lease) and plain liabilities barely move.[13]

12.4.2 Goodwill discussion

There is *a lot* to talk about and debate with Goodwill. First, what *is* Goodwill? It's hard to say. It *is what it is*. A necessary plug for this special situation, representing *anything and everything* that accounts for the *difference* between what an acquirer is willing to

[13] 'Barely' is not 'never.' For some liabilities—think of an environmental obligation to clean up a polluted river at some future date—the acquirer's reassessment could produce a fair value meaningfully different from book value.

pay, and the estimated fair value of the assets and liabilities that are considered 'identifiable' by the accounting system. So, we know we need to account for it, but we can't say what's really 'in' that account—so many different things are 'in' it. And yet, Goodwill is large. It can account for a substantial share of the assets of companies that have made large acquisitions. That means a significant part of an acquisitive company's Balance Sheet may be this **residual** or 'catch-all' account. Many accounting professors and standard-setters see this as problematic.

Goodwill can be a substantial part of an acquisitive company's assets. Its size tells us to examine the acquisitions behind it: what was purchased, what identifiable assets were recognized, and what expectations supported the price. The balance alone does not answer whether those expectations were well founded.

How should you *interpret* a big Goodwill balance, then? Carefully. Management does not simply pick the number—Goodwill is mechanically determined by the purchase price and the fair values of the identifiable net assets—but there is discretion in those fair-value estimates and, more fundamentally, in the strategic decision to pay a given price in the first place. A large Goodwill balance tells you one thing for sure: the company has done acquisitions, and paid substantially more than the fair value of the identifiable net assets it acquired. Whether that is good news or bad depends entirely on whether those deals were wise. If you believe in the CEO's strategy—the synergies are real, the integration will work—then the Goodwill reflects genuine economic value the company will realize. If you think the CEO overpaid (perhaps hoodwinked by eager investment bankers), then that same balance is a ticking time bomb: 'value' that isn't really there, waiting to be written off.

Written off how? This brings up a question you may have been itching to ask: Goodwill is an intangible asset—so do we *amortize* it, the way we amortize a *Patent*? Under the U.S. GAAP model for public companies, Goodwill is not amortized. Certain private companies and not-for-profit entities can elect an accounting alternative that amortizes it. An indefinite-lived Trademark is also not amortized, but a finite-lived Trademark is. But—linking this chapter's topics together—companies *are* required to test their Goodwill for **impairment**. And think about what that testing entails: since Goodwill is not a separable thing you can appraise on its own, testing it requires estimating the fair value of the entire operating unit it is attached to—one of the most advanced (and contested) exercises in all of accounting. When Goodwill impairments come, they tend to be enormous. Indeed, the large 2019 impairment at Procter & Gamble that we met earlier in this chapter included impairments of both *Goodwill* and the Gillette indefinite-lived intangible assets—the delayed reckoning for an acquisition (of Gillette, over a decade earlier) that had not delivered the value paid for it.

A common trap: Here is a question that trips up many students. After an acquisition, which of the following is implicitly 'baked into' the *Goodwill* on the acquirer's balance sheet: (a) the value of the target's internally developed trademarks; (b) the value of ordinary assets such as *Prepaid Rent*; or (c) the value of positive social-media buzz about the target's products? The answer is (c) *only*. The buzz is real value, but not an identifiable asset (by the stated assumption, it does not meet the contractual-rights or separability criteria), so it gets swept into Goodwill. Option (b) shouldn't tempt anyone: *Prepaid Rent* is about as identifiable as assets get, so it is counted in Net Identifiable Assets, not Goodwill. Option (a) is the trap. The tempting-but-wrong reasoning goes: "internally developed trademarks aren't recognized on the balance sheet—we learned that last chapter—so their value must be hiding in Goodwill." This overlooks **Step 1** of the two-step process: upon the acquisition, we get to—indeed, must—write those internally developed trademarks up to fair value as *identifiable* intangible assets. Once written up, they are part of Net Identifiable Assets; and, by definition, anything in Net Identifiable Assets is *not* in Goodwill. Goodwill is the residual: the plug for whatever is left over *after* the identifiable items have been counted.[14]

Challenge question: negative Goodwill? Look back at the formula: Goodwill is the purchase price *minus* the Net Identifiable Assets. Arithmetically, nothing stops that number from being *negative*: the purchase price could, in principle, be less than the net identifiable assets. Could that realistically happen? Think it through from the target's side. The target's board and shareholders have to *agree* to the deal. Why would they ever accept a price below the fair value of the identifiable net assets they know they have—especially given that, on top of those, they also possess all the unidentifiable sources of value (such as synergies and other benefits not separately identifiable) that normally make Goodwill *positive*? So, in ordinary arm's-length M&A, negative Goodwill essentially does not occur; I have never seen it in a normal transaction. You can construct exotic scenarios—fire sales, forced liquidations, distressed banks shoved into a rescuer's arms by regulators over a weekend—where a buyer pays less than net identifiable asset value, and accounting standards do have provisions for such 'bargain purchases.' After reassessing the identification and measurement of the acquired items and the consideration, a remaining excess of net identifiable assets over the purchase price is recognized as a gain, not a negative Goodwill asset. But in the standard M&A world we are modeling here, do not expect to see it.

Finally, I want to add some clarifying notes to **anticipate confusion**, for readers who follow the business press or have had business or finance internships. You may

[14]Timing matters in that statement. *Before* the acquisition, the target's internally developed trademarks are in neither place: not on its balance sheet (last chapter's rule), and not in any Goodwill either—because Goodwill does not yet exist. Goodwill comes into existence only at the moment an acquirer pays a purchase price that can be compared to net identifiable assets.

have heard that during an acquisition, if the acquirer is purchasing a publicly-traded target company, the acquirer will have to pay a **"market premium"** to the pre-deal stock market price of the target. And this should seem logical: If the target's current stock price were greater than what it was being offered by the acquirer, why would it accept that offer? You may have also heard talk of **"synergy"** in acquisitions—a term for the idea that the Target will be worth more as a part of the Acquirer than it was worth on a standalone basis (i.e., Value (A&T together) > (Value A + Value T)), which provides the logical basis for the acquirer's willingness to pay that premium.

These facts and concepts are somewhat **related** to Goodwill accounting, but also **distinct and technically separate**. Some students are tempted to think that the Goodwill the acquirer will recognize is equal to market premium it pays. But this is **not correct**. The two are related in that, all else equal, the more the acquirer pays, the higher the market premium and the higher the Goodwill. But they would not be the same number unless the net fair value of the Target's identifiable Assets and Liabilities exactly equaled the Target's pre-deal standalone market price—which it never will. So, if you know a little about M&A and know about market premiums and synergies, just don't get distracted by them and forget the precise **accounting definition** of Goodwill.

12.4.3 Going deeper: Can an imperfect measure still be useful?

Suppose, just for a thought experiment, that every reported asset and liability were exactly half its economic value, and that every reader knew this. The reported numbers would be biased, but a reader could recover the economic amounts by **multiplying by two**. A known difference need not make a measure useless.

Real accounting is less tidy. Different accounts use different measurement rules, and no single adjustment translates the entire Balance Sheet into market value. That is why knowing how a number was produced matters.

The point of learning the rules is not to mistake the map for the territory. It is to understand what the map shows, what it leaves out, and which additional questions to ask. After this chapter, those questions include acquisition prices, valuation assumptions, impairment losses, and the source of Goodwill.

12.5 Contemporary issues: Discussion Questions

12.5.1 Fair-value accounting is a 'live' and controversial area

The use of *fair value* estimates is one of the most significant and fast-changing issues in accounting. This chapter has only scratched the surface. And fair-value account-

ing standards will likely change over the first years of your career. So, rather than go into ever more details about where fair-value is used, just think. Where do you think it *should be* used, and why?

Suppose that our company has invested in the stock of another company. If we continue to hold that stock, but the stock price gyrates, with macroeconomic and company news, should we recognize 'Gains' and 'Losses' on a quarterly basis? If not, why not? Or, would your answer change depending on whether our company was in the financial sector, vs. otherwise?

Or, suppose that our company has an office in Culver City, in a building that we have owned for the last 20 years. Should we write that real estate (*Property*) up to its current estimated fair value, and book profit? If so, how frequently? What if our office were in a less well-populated area, where it would be harder to get a reliable current appraisal of the value of commercial real estate? What do you think?

13

Intro to Finance

Our next *accounting deliverable*, in the next chapter, is accounting for **corporate bonds**. Bonds are the major way in which large, mature companies *finance themselves*—that is, get the cash they need to make investments.

A bond is essentially like a *Loan*, but is just traded in a *financial market* (like a stock) instead of directly, bilaterally contracted with one bank. When a company **'issues a bond,'** it **sells a contract**: It **promises** to make a fixed schedule of **interest and principal** payments in the future. And in exchange for this promise, **bond buyers**—such as mutual **funds, banks, corporate pension plans**, insurance companies, and even individual **savers**—give the company **cash today.**[1]

It turns out that the ***Interest expense*** that companies recognize on their bonds will be **determined** by this **financial market**. So, to understand the accounting for bonds, we need to understand the basics of **finance**. Since most readers of this book will not yet have taken an introductory finance course, we cover those basics now.

There is a **second motivation** as well. Last chapter, I made you a promise. We agreed that, for the time being, **'fair value'** would be a number that gets **spit out** of a **'black box'**—and we agreed to postpone the question of what is going on inside the box. Recall that one way of estimating fair value is to calculate the **present value of expected future cash flows**, using assumptions that market participants would use. If, at the time, that read like a phrase to memorize rather than something you could

[1] Here's some jargon you'll encounter in the financial world: because these contracts promise a *fixed schedule* of payments, bonds and loans are collectively referred to as **'fixed income'** assets.

really *see*, that is about to change. By the end of this chapter, the black box will be open. More generally, accounting and finance are tightly intertwined—each depends on the other, so you have to learn both, in one order or the other. This chapter is where we pay finance its due.

Finance is the theory of **value, risk,** and **time**—that is, it is about the question of **'How much are uncertain (that is, risky) future cash flows worth today, and why?'** As we will see, this is **equivalent** to asking, **'How much can we expect to get in the future, in return for investing some of our cash today?'** Put differently, finance is just about the relationship between cash/value *today*, and expected *future* cash/value. The theory is **elegant and deep.** It's challenging at first, but, once you internalize it, you'll see that everything you need to do boils down to just **one formula**.

13.1 Bonds, loans, and installments are just Cash Flows

A **bond** is essentially the same as a **loan**—both are forms of **credit**, in which the corporation receiving the credit promises to pay a **fixed stream** of **interest and principal** payments. The major distinction that is conventionally made between bonds and loans is that **loans** are **directly** negotiated with and provided by **one bank** (so interest and principal is paid directly to that bank); while **bonds** are sold to and traded in the **financial markets** more broadly (so, interest and principal is paid to whoever **purchased that bond contract in the financial market**).[2]

By now in this book, we've seen a lot of simple examples of accounting for loans, so I think that we know the basic form. Just as with loans, for normal bonds, the interest and principal payments are defined by the **face value**, the **maturity date**, the **stated interest rate** (or 'coupon rate'), and the **frequency of interest payments**. These terms mean pretty much what they sound like, but let's pin them down. The **face value** (or 'principal') is, by definition, the total amount you owe and pay back at the very *end* of the loan or bond. The **maturity date** is the date on which you pay off that face value in full and the contract ends. The **stated interest rate**—the **'coupon rate'**—determines the size of the periodic cash interest payments you make along the way, before you finally repay the principal: each period, you pay that percentage of the face value. (We'll see, soon enough, why I keep emphasizing the word 'stated.') And the **payment frequency** is just how often those interest payments are made—annually, semiannually, etc.[3] You should be able to write down the payments based on those terms. For ex-

[2] As usual, the reality in the 21st century is a bit fuzzier and more complex than this stylized simplification. In principle, there's nothing stopping banks from selling the loans they have extended to other parties, i.e., 'the market.' And the market for corporate bonds is in fact famously **illiquid**—there is very little trading, relative to the market for corporate stocks.

[3] In the real world, most corporate bonds actually pay interest *semiannually*—twice a year. But since this book is about concepts rather than messy institutional details, I'll assume annual payments

ample, consider a bond with a face value of **$10,000**, a stated **6%** coupon rate, with interest paid **annually**, and maturing in **5 years**. Tabulating that, this means that this bond promises the following sequence of payments:

Time	**+1 Year**	**+2 Years**	**+3 Years**	**+4 Years**	**+5 Years**
Interest Paid	$600	$600	$600	$600	$600
Principal Paid					$10,000

Contract we call 'loan or bond'

Once you see the actual dollar amounts laid out, the terminology becomes much less mysterious. The $600 is just the 6% coupon rate applied to the $10,000 face value, paid every year up to and including the fifth—and, in that fifth year, on the maturity date, the $10,000 principal comes due as well.

Now, back up. This is **not the only form of credit financing**. Another thing that both you and a corporation can do is to finance the purchase of a good by promising to make a series of future **installment payments**. For example, suppose that the corporation (or you) gets **$10,000 in cash** (or a car worth $10,000) today, and you promise to pay **5 annual installment payments** of **$2,400** in return.[4] This, by the way, is how a typical home **mortgage**—the credit financing we met back in Chapter 1—actually works. You hear people talk about the 'mortgage interest rate,' but the cash flows are not structured as small interest payments followed by one giant repayment of the whole house price at the end. Instead, you pay one fixed installment every month (or year), for the life of the mortgage—inside each payment there is an interest component and a principal component, but the contract you *experience* is 'pay this fixed amount every period.'[5] Then, tabulating, the future promised, fixed cash flows would look like this:

Time	**+1 Year**	**+2 Years**	**+3 Years**	**+4 Years**	**+5 Years**
Install Payment	$2,400	$2,400	$2,400	$2,400	$2,400

Contract we call 'installment payment'

throughout, to keep the arithmetic and the logic clean.

[4]For our purposes, an 'installment' contract means, by definition, the *same* payment every period. Could a contract be written otherwise—say, with payments that adjust when market interest rates move? Sure. At the end of the day, these are all just contracts written between people, and there is nothing in nature that forces them to be written one way or another. But in this book, 'installment payment' will always mean fixed, equal payments per period.

[5]You might also wonder: what if you pay a loan off *early*—can you escape the interest for the remaining periods? The answer is entirely contract-specific. A typical residential mortgage allows early repayment (say, when the owner sells the house or refinances). But corporate lenders sometimes write contracts that explicitly *prohibit* early repayment—because losing that contracted stream of future interest is itself a risk to the lender. There is no law of nature here; just terms in contracts.

There are two important things to note here, comparing the two tables above. **First**, while we call these contracts different **names** ('loan' vs. 'installment payment'), and have different **words** for the *types* of payments (installment vs. interest / principal), they have the **same basic form**. In both cases, you **get something today**, and in return, **promise** to make a set schedule of **future payments**. The only difference is the precise **amounts due on each date.**

To make this vivid, imagine that you are the *lender* on the loan contract above, receiving the payments. The money arrives in your bank account: some of it is labeled 'interest' in the contract, some of it is labeled 'principal.' Once it hits your account, is any of it literally color-coded? Do you have blue dollars for interest and red dollars for principal? Of course not. It's just money. **Money is fungible**—a dollar is a dollar, regardless of whether the contract calls it 'interest' or 'principal' or 'installment.'

And, indeed, this is the view that the field of **finance** takes. From the perspective of finance, there's just an exchange of **cash today, for promised future cash,** and the **names don't matter**. So, finance just uses the general term **'cash flow.'** These contracts are all just specified **cash flows**, in and out. So, we can represent the two tables in a more simple and general way, like this:

Time	**+1 Year**	**+2 Years**	**+3 Years**	**+4 Years**	**+5 Years**
Cash flow promised	$600	$600	$600	$600	$10,600

Contract we call a 'loan' or 'bond'

Time	**+1 Year**	**+2 Years**	**+3 Years**	**+4 Years**	**+5 Years**
Cash flow promised	$2,400	$2,400	$2,400	$2,400	$2,400

Contract we call 'installment payment'

The **second** important thing to notice is that, in the installment payment contract, there is an **implicit cost of the financing**. The **total installment payments** add up to $12,000. By now, we know that loans and bonds entail interest payment—and, conceptually, interest is the cost of the loan for the borrower, the extra amount they have to pay in order to get cash financing today. What we learn here is that there is an **implicit cost** of borrowing in that **installment payment**, even though nothing there is **named** 'interest' vs. principal. Soon, we'll find out that the installment contract *does have* an implicit, true interest rate, and we'll learn how to calculate it.

13.2 Linking present and future value: 'discount rates'

So, now we understand that all forms of credit financing are, despite their **different names**, really just one thing: **Cash flows, in and out.** Next, we need to back up to some deep, high-level theory. The field of finance has a systematic, elegant, deep framework for understanding the link between **cash today** and **future promised cash.**

In finance, the **future value** of any asset is linked to the **present value (PV)** by what we call the **'discount rate'**—that is, how much you **'discount'** a promised future payment in terms of your **willingness to pay for it today**. We 'discount' the cash flows that are promised by financial assets—that is, we **do not value them 1-for-1** in present-value terms—because we usually **do not like delay or risk**. That's the **core idea of finance**. Almost **every formula** you will ever see in finance is just a way of re-expressing this idea in a mathematical form. **I am dead serious**. If you understand that idea, stated verbally, you can absolutely master finance.

For a payment one year from now, let i denote the **annual discount rate**, expressed as a decimal. A 25% rate is $i = 0.25$. The denominator is $1 + i$, so at 25% we divide by 1.25. The rate is 0.25; 1.25 is the gross factor. The **discount factor** multiplying the future payment is its reciprocal, $1/(1+i)$. With that distinction clear, we can write the idea as an equation:

$$Present\ Value\ of\ future\ cash\ flow = \frac{Promised\ future\ cash\ flow\ amount}{1+i}$$

Some textbooks present this as a *formula* that allows you to *calculate* things. But I think **it's important to understand that this is a mathematical identity**—we are just writing, in equation form, what we **meant** by **'present value' and 'discount rate.'**

Let me pause on this, because I know that some readers' minds go a bit blank when an equation appears on the page. (I sympathize—I did not come out of high school loving mathematics, either.) If that's you, just re-read the idea in *words*—'we don't value future promises 1-for-1; we discount them'—until you can see that the equation above is literally that sentence in symbolic form: a compression of the idea, nothing more. And here is something that should be reassuring: even in the highest-level PhD finance courses, *this* is the anchor relationship that everything else is built on.

"Discount rate" is a general term. For a bond, we often infer a **yield** from its market price and promised payments. The bond's **coupon rate** is different: it specifies the cash interest payments in the contract. The two rates can differ, as we will see below.

13.3 What determines discount rates?

So, **what determines** 'discount rates' that we—and thus, in aggregate, **'the market'**—apply to risky future cash flows? Simply put, people do, according to their own **beliefs and preferences**. Discount rates are not handed down by **law or from Mt. Olympus**. They are a function of people's choices, **beliefs, preferences, and constraints** as expressed in a **financial marketplace**—just like the **price of jeans** in a **retail market.**

This point is worth dwelling on, because it corrects a common misconception. People often hear that the Federal Reserve 'sets interest rates.' The Fed certainly *influences* interest rates—it has various tools and levers that affect the overall level of rates in the economy—but it does not decree, 'The interest rate on this particular bond shall be X.' The actual interest rate on any specific loan or bond **emerges** from the interaction of borrowers and lenders in the marketplace—from what people are willing to pay or accept for particular streams of future cash, given their beliefs about risk and time. How much is someone willing to pay for a pair of jeans? The government doesn't tell you that; the price emerges from supply, demand, and people's preferences. Financial assets work the same way.

Before reading the numbers below, imagine lending to someone you know. How much would you be willing to pay today for that person's promise of $1,000 in one year? Keep your answer in mind as we work through Taylor's example.

Consider a **toy example.** Suppose that you have a friend named Taylor, who **promises** to pay you **$1,000 in one year**, if you front them some **cash today**. How much is that promise worth to you? Taylor's promise is **not a sure thing**, and even if Taylor *does* deliver on the promise, that cash will be delivered **in the future.** So, that promise **shouldn't** be worth a full $1,000. That is, it should be *discounted* relative to that. But, by **how much**? That's up to you.

Suppose that, after considering everything, including what you know about Taylor's future earnings potential, you determine that the maximum amount that you would be willing to give Taylor today in exchange for that promise is **$800**. So, restating that **verbally**, that means that in valuing Taylor's promise in *present value* terms, you 'discounted' $1,000 down to $800. We can write this idea in **math** as:

$$\textit{Present value of Taylor's promise} = \$800 = \frac{\$1,000}{1+i}$$

Now, we have only **one free variable** in the equation on the right side, above, and so we can mathematically **solve for the discount rate**, which previously we only discussed at a conceptual level. Rearranging terms, this implies that the gross factor for Taylor's promise is $1+i = \$1,000/\$800 = 1.25$, so the annual rate is $i = 0.25$, or 25%. That is:

$$\$800 = \frac{\$1,000}{1+i}$$

$$1+i = \frac{\$1,000}{\$800} = 1.25$$

Now suppose that your friend **Matt** promises to pay you $1,000 in one year's time. But Matt is known to be a **volatile and shifty** figure. You suspect he might lose his job next year, due to his repeated failure to balance his debits and his credits. So **you don't trust Matt to pay you back** as much as you trusted Taylor. So, put simply, **Matt's promise is not worth as much** to you as Taylor's.

So, after factoring in all considerations, suppose you decide that the maximum you are willing to pay for Matt's promise is $600. Restating that in arithmetic:

$$\textit{Present value } (PV) \textit{ of Matt's promise} = \$600 = \frac{\$1,000}{1+i}$$

Once again, we have only one free variable, so we can solve for the rate implicit in Matt's promise: $1+i = \$1,000/\$600 \approx 1.667$, so $i \approx 0.667$, or 66.7%.

So, from this very simple pair of examples, we can extract a **deep insight** that is at the **core of finance**, and that will hold true generally: For these two promises of the same amount on the same date, greater concern about nonpayment leads you to offer a **lower price**, which implies a **higher promised yield**. This is not a guarantee of a higher realized or expected return: the borrower may not pay in full. By now, this pattern should feel familiar: in this subject, the deepest, most far-reaching implications tend to fall out of the simplest identities. (Recall how the humble equation $A = L+E$ eventually handed us the whole logic of *leverage*.) This chapter's little identity will do comparable work for us.

How exactly do you **define and estimate 'risk'**? Isn't this 'risk' inherently about the **future**, which we cannot observe, and thus inevitably somewhat **subjective and uncertain**? Great question. And, yes. These gnarly issues are the **subject of finance** classes. We can't go into too much depth here, about the particular *methods* for defining and estimating risk. For now, don't overthink it: **All we need to know, for now, is that some companies are riskier than others, and thus there is variation in the discount rate** that market participants will apply to them. Clearly, lending to a company that is on the verge of bankruptcy is riskier than lending to one that is not.

13.4 Discount rate and interest rates

The great thing about accounting and finance (as opposed to, say, math) is that you have both some quite **deep theory**, but you also have **real-world** institutional prac-

tices that have **evolved for thousands of years**, and the two converge (and sometimes clash) in subtle ways. 'Discount rates' are the general, theoretical idea, and that term is applied to *all assets* (including stocks, which you'll meet in depth if you go on to study finance). But, when we are speaking of **forms of credit**, it is more conventional in practice to use the term **'interest rate.'**

Now, you might wonder: why the distinction? When we value a *stock*, aren't we doing the very same thing—discounting expected future cash flows? Yes. Exactly the same thing. The difference is purely one of convention, and it tracks the credit-vs.-equity distinction we've known since Chapter 1. A loan or bond has a *contracted* rate written into its terms ("6% coupon"), so we naturally speak of 'interest rates' there. A stock's cash flows—discretionary dividends, an uncertain future sale price—are not contractually guaranteed; the shareholder is, as ever, the residual claimant. With no contractual rate to point to, when we discount a stock's expected cash flows, we instead call the discount rate the **'required return'** or the **'cost of equity.'** Same math, different name. I know—I wish the terminology were simpler, too. But I didn't design the world; the conventions were here before I arrived. My job is just to show you how they all map back onto one simple idea.

The percentage convention is the same whether we say "discount rate" or "interest rate." In either case, i is the percentage expressed as a decimal, and the denominator for one year is $1+i$. Rewriting Taylor's example makes the connection explicit:

$$PV\ of\ Taylor's\ promise = \$800 = \frac{\$1,000}{1+i} = \frac{\$1,000}{1+i} \rightarrow i = .25 = 25\%$$

This implies that i = .25 = 25%. So now, bringing this together, **we can say something that sounds a bit more conventional: 'The annual interest rate you charge to Taylor, based off of your assessment of Taylor's credit and ability to repay, is 25%.'** (Similarly, the i for Matt's promise is .67—so you are charging an annual interest rate of 67% to Matt.)

So, the **big takeaways** from what we just learned are as follows:

1. In financial markets, people trade **cash today** for promises of **future cash** flows.

2. People generally don't value those future promises of cash 1-to-1. They **discount** them, because we generally **do not like** risk or delay.

3. The **degree** to which you discount them (i.e. the discount rate) is primarily determined by the perceived **risk**.

4. When we are talking about forms of **credit** financing—as we are in this part of the book—we re-express 'discount rate' as the **'interest rate.'**

5. That interest rate is the **i** in the formulas above.[6]

13.4.1 Present values and future values: it goes both ways

The equation we started with is an **identity**—it is **true** by the nature of **what the terms mean**. And there are **three terms** in it: **present value, future value,** and the **interest/discount rate**. So, this means that we can do **other calculations** as well—if we know any **two** of those variables, we can **solve for the third**. Above, we had:

$$Present\ Value = \frac{Future\ Value\ (cash)}{(1+i)}$$

But, trivially, we can rewrite this as:

$$\rightarrow\ Future\ Value\ (cash) = Present\ Value \times (1+i)$$

So, as a concrete example, if we are told that somebody is paying **$100** to invest in a bond that will pay interest and principal in one year, and we are told that the interest rate is **15%**, then its promised payment consistent with that yield is **$115** in total (principal and interest both included), since $\$100 * (1+.15) = \$115$.

And we can also rearrange terms and solve for i, which is basically where we started:

$$\rightarrow\ i = \frac{Future\ Value\ (cash)}{Present\ Value} - 1$$

13.4.2 Valuing a stream of future cash flows:

There's only one more thing that we need to do: Understand how this relationship works when we have a **stream or sequence of several future cash payments**, not just one lump-sum payment in one year. And note that this is the *normal* case, not an edge case. In accounting terms, an asset is something that brings *future benefits*—and those benefits typically arrive over *multiple* future periods. In finance terms, that means most real assets pay a **stream** of cash flows: a bond pays interest every year and principal at maturity; a factory generates sales year after year. If our framework could only handle a single lump sum, it wouldn't be worth much. Let's go back to the example we used at the start of this chapter, of an **installment payment**. The borrower in this relationship gets cash (or a car) today, and, in return, promises to make annual payments of $2,400 for the next 5 years. We represented this as:

[6] And generally that interest rate should be somewhere in the order of 1% to 40%, depending on the company and situation. (If you solve for an **i** of 130%, outside of extreme circumstances, that's probably a calculation error.)

Time	**+1 Year**	**+2 Years**	**+3 Years**	**+4 Years**	**+5 Years**
Cash flow promised	$2,400	$2,400	$2,400	$2,400	$2,400

Installment Payment Contract

How much is this whole **future stream worth, in present value**? Let's **start with what we do know**. The **first** installment payment is due in one year, **just like** in the examples **above**. So let's just talk about that **first payment alone** for now.

$$Present\ value\ of\ the\ first\ cash\ payment = \frac{\$2,400}{1+i}$$

Suppose that, based on our assessment of the credit of this borrower, we decide to apply a **12% annual interest rate** to this promise. Then, that becomes:

$$Present\ value\ of\ the\ first\ cash\ payment = \frac{\$2,400}{1+.12} = \frac{\$2,400}{1.12} \approx \$2,142.86$$

Okay, now, what to do with the **remaining cash payments**—the second, third, fourth, and fifth? Here, we make a **simplifying assumption** for this book's calculations. We assume that we apply the **same discounting/interest rate *per each time period***. Or, in other words, to value the **second cash payment** in present value terms, you apply the same discount/interest rate you used for the first cash payment **twice**:

$$Present\ value\ of\ the\ second\ cash\ payment = \frac{\frac{\$2,400}{1.12}}{1.12} = \frac{\$2,400}{1.12^2} \approx \$1,913.27$$

There is *some intuition* for this: You'll have to **wait** for the second cash payment for **twice as long**, and the borrower will be **exposed to the same risks** (that will possibly prevent them from paying you) for twice as long. So that's an intuition. (But admittedly, this logic is not totally **rock solid**—and if you do a graduate-level program in finance, you might consider *time-varying discount rates*. Indeed, if you've ever heard finance people talk about the **'yield curve'**—the fact that market interest rates differ across different maturities—that is exactly this relaxation. In any case, we use a constant per-period rate in this book. In practice, analysts may use rates that differ across maturities.)

Generalizing that, we would **apply that discounting** a **third time** to the **third payment**:

$$Present\ value\ of\ the\ third\ cash\ payment = \frac{\$2,400}{1.12^3} \approx \$1,708.27$$

And finally, the present value (PV) of the whole **stream** of future cash payments is

just the **sum** of the present values of the **individual** cash payments. This means that:

$$PV\ of\ instllmnt\ pymnts = \frac{\$2,400}{1.12} + \frac{\$2,400}{1.12^2} + \frac{\$2,400}{1.12^3} + \frac{\$2,400}{1.12^4} + \frac{\$2,400}{1.12^5} \approx \$8,651$$

So, that's the answer. The arithmetic looks sort of ugly at first. But, it really is **really, really simple.** We're just saying that people don't like risk or delay, so they discount future risky cash flows, and as the time horizon increases, the **'discount' multiplies in turn.** The above is just putting numbers to that idea.

Before moving on, look back at what we just did, and describe it in a phrase: we took the *expected future cash flows* of an asset, *discounted* each one, and summed them up. **'Discounted expected future cash flows.'** That is the discounted-cash-flow technique discussed last chapter, which can estimate **fair value** when its assumptions reflect market participants—the one I asked you to accept, at the time, as the output of a **black box**. The black box is now open. This calculation is also what Wall Street calls a **'DCF'**—a 'discounted cash flow' valuation. If you've been doing investment-banking interview prep, you've seen that acronym thrown around as though it names something fancy and forbidding. It doesn't. It names exactly what we just did: forecast the cash flows, apply the appropriate discount factor to each, add them up. There is no magic beyond that.

13.4.3 Value as of a particular date

The present value of the third payment is what that year-three cash flow is worth **today**, not what it will be worth in year three. The date from which we are valuing is part of the question.

13.4.4 Going deeper: A single payment can itself be an asset

Suppose the holder of our installment contract sells someone the right to receive only the third payment. The other payments remain with the original holder. We have already calculated the present value of that separate right: $2,400 discounted for three years at the assumed rate. The terms in our sum therefore have a concrete interpretation; each values a particular payment from today's vantage point.

Now ask a different question: what would the whole contract be worth *as of year 3*? We can absolutely answer this, with nothing but the tools we already have. Fast-forward in time: we are now standing in year 3, just after the third payment has been made. What is left of the contract? Reductively: one year from our new vantage point,

we'll get $2,400 (the year-4 payment), and two years hence, another $2,400 (the year-5 payment). That's all that's left. So, if our interest rate is still 12% at that time:

$$PV\ as\ of\ year\ 3 = \frac{\$2,400}{1.12} + \frac{\$2,400}{1.12^2} \approx \$4,056$$

It's the same identity we've been using; we've just shifted our 'present' from today to year 3. And this reveals a general lesson worth internalizing: **'present value' always means present *as of a specific date***, given what is known and believed on that date. There is no free-floating, date-less 'value.'

Third, and relatedly: can the discount rate itself *change* over time—say, if events change the riskiness of the borrower while we're waiting for the cash flows? Yes—and this is not a violation of the formula, but a feature of it. The discount rate we apply is always conditional on what we know at the moment of valuation. Take a corporate bond. Today, given the company's financial health and the state of the market, investors might apply an 8% rate, and that pins down today's present value. Two years from now, suppose the company has stumbled, or the economy has soured: investors will no longer pay as much for the very same remaining promised cash flows—which is just to say, they are now applying a *higher* discount rate. The identity is always local in time: at any given date, present value equals the discounted expected *remaining* cash flows, from that date forward, at whatever rate is appropriate *then*.

Finally, here's an implication you can now derive yourself: all else equal, a 20-year promise is riskier than a 1-year promise. There is simply more time for things to go wrong—interest rates can move, inflation can surprise you, the borrower's credit can deteriorate, the whole macro environment can shift. So, in general, longer-term bonds are riskier.[7]

13.5 Doing the same for the bond

Now, let's **apply** what we've learned to the **simple bond** we used as our starting example, above: That bond had a 5-year maturity, and paid annual interest payments, with a stated coupon rate of 6% and face value/principal of $10,000. And, as we learned, despite the different names, these are all just **cash flows**. So, we can represent this bond like so:

[7] A related question you might ask: as a bond approaches its maturity date, does its present value drift upward toward the final payoff, as the remaining delay and uncertainty melt away? For a bond that pays a single lump sum at maturity—a so-called **'zero-coupon' bond**—and with interest rates held constant, yes, exactly: its value rises smoothly toward its face value. For coupon-paying bonds like ours, the path is more complicated, because cash is also being paid out along the way. More on bond prices in the next chapter.

Time	**+1 Year**	**+2 Years**	**+3 Years**	**+4 Years**	**+5 Years**
Cash flow	\$600	\$600	\$600	\$600	\$10,600

Now, in even doing this calculation, we're already encountering something new and **seemingly paradoxical**. We already said that this bond has a **'stated' interest rate**—called the **'coupon rate'**—of 6% annually. But now, we are **not** taking it for granted that this is the **'real'/ 'true'/ or 'market' interest rate**. The bond's **stated coupon rate** is just **text in a contract**—it's the way of specifying the **size** of the **cash interest payments** relative to the **face value**. The company issuing the bond **can't force us**—that is, 'the market'—to accept that 6% stated interest rate as the 'true' discount rate. We—or, more generally, the financial marketplace—**get to decide** how much we're **willing to pay** for that bond.

How do we do that? The amount that we'll pay is, **by definition, the present value.** We **already specified the promised future cash flows**. So, to calculate the present value, we need to decide on our discount/interest rate. Suppose we determine that, based off of this company's riskiness, we require an **8%** interest rate. Then, the present value will be: [8]

$$PV\ of\ bond = \frac{\$600}{1.08} + \frac{\$600}{1.08^2} + \frac{\$600}{1.08^3} + \frac{\$600}{1.08^4} + \frac{\$10,600}{1.08^5} \approx \$9,201.46$$

[9]

It's worth previewing how this plays out in the real world, from the perspective of the *company issuing the bond*—because this is precisely the situation we'll be accounting for next chapter. The company drafts the bond contract *first*, and the contract must spell out, in advance, exactly what will be paid and when: say, a \$10,000 face value with a 6% annual coupon. Once that contract is drafted, approved, and the compliance boxes are all checked, the company takes it to the market. At that point, the promised cash flows are **locked in**—they cannot be changed. But when the bond actually goes on sale, the market may look the company over and say, in effect, "No—we think you're more of an *8%-interest-rate kind of company.*" Once the terms are fixed for the pricing example, the coupon stays unchanged. An issuer can revise a planned offering before

[8] If it helps, you can paste this formula into Google to verify the calculation: (600/1.08) + ($600/1.08^2$) + ($600/1.08^3$) +($600/1.08^4$) +(10600/1.08^5).

[9] Notice how because our **actual interest rate is higher** than the bond's **stated** coupon rate, the present value of the bond—and thus, the cash we would actually pay to the corporation issuing this bond—is **lower than its face value**. This will hold generally: If the 'real'/ 'true' market interest rate on a bond is higher (vs. lower) than its stated or 'coupon' rate, then its present value and price upon its issuance will be lower (vs. higher) than the face value. And, if the market interest rate is exactly equal to the stated coupon rate, the present value and issuance price will be the same as the face value. This is *not* a coincidence, but the proof of this fact takes up a bit too much space. To 'prove' it to yourself, you can just paste calculations like this into Google: $600/1.06 + 600/1.06^2 + 600/1.06^3 + 600/1.06^4 + 10600/1.06^5 = 10000$.

issuance, but that would be a different set of promised payments. So what adjusts? **The price.** The bond still promises $600 a year and $10,600 at maturity, but investors, requiring an 8% return, will only pay about $9,201 for it—and *that*, not the $10,000 face value, is the amount of cash the company actually receives (the debit to *Cash*, in our terms). The stated rate is text in a contract; the market rate is expressed through the price. Hold on to this picture: it is the key to the bond accounting we'll do next chapter.

13.6 Generalizing: *The* identity of finance

Let's back up and restate what we've learned in a more general form. This will be useful for two reasons:

- The general theory is very important, for the rest of this book, and any business career.
- As we'll see below, some calculations we'll do when we apply this theory will *look* very big and unwieldy and force us to use Excel. As a result, if you just jump into those calculations, you might get the impression there are a lot of new hard formulas and ideas to master. But there aren't. There's just **one idea**, but a bunch of Excel shortcuts for calculation. So, mastering *the idea* will spare you anxiety.

What we've learned so far is that the **present value** of any asset is, **by definition**, equal to the **future cash flows it promises, discounted** by some discount/interest rate. Assuming, as we do, a **constant interest rate** per time period, this means that the following is always true:

$$Present\ Value = \frac{[Cash\ flow_1]}{1+i} + \frac{[Cash\ flow_2]}{(1+i)^2} + \frac{[Cash\ flow_3]}{(1+i)^3} + ... + \frac{[Cash\ flow_n]}{(1+i)^n}$$

Or, we can put this in **even more general terms** as...:

$$Present\ Value = \sum_{t=1}^{\infty} \frac{Cashflow_t}{(1+i)^t}$$

In this constant-rate framework, present value is the **sum of the discounted future cash flows**. A finite-lived asset has no payments beyond its final date; an infinite stream also requires the sum to converge. More advanced valuations can use different discount rates for different dates or risks. The organizing idea remains the same: identify the payoffs and value them from a specified date.

The structure is now in place: identify the cash flows, choose the appropriate discounting, and add the present values. Forecasting the cash flows and assessing their

risk can be difficult. The formula organizes those judgments; it does not make them for us.

13.6.1 An arbitrary example: the Crazy Money Machine

Test the same method on an irregular stream: $420 in year 1, $1 in year 2, $1,000 in year 3, and $1,000,000 in year 100, with nothing between. At an assumed 15% discount rate, the present value is:

$$PV\ of\ machine = \frac{\$420}{1.15} + \frac{\$1}{1.15^2} + \frac{\$1,000}{1.15^3} + \frac{\$1,000,000}{1.15^{100}}$$

Plug that into a calculator or Excel, and out comes a number.[10] However bizarre the pattern of cash flows, the logic is identical: What are the cash flows? What discount factor applies to each? Done.

13.6.2 What this framework applies to (and what it doesn't)

One caveat about scope, before we go on. This framework is about **financial assets**—things you hold *because of the cash flows they generate*. It does not directly apply to, say, a painting you buy because you love looking at it, or because you want to impress your dinner guests. Nor does it fully capture why billionaires buy professional sports teams—very often, that is about status, vanity, and courtside seats, with the financial returns a secondary consideration. Those are real forms of value; they're just not *cash-flow* value, so they call for a different kind of analysis. But stocks, bonds, and loans are not like this. Nobody hangs a bond certificate over the fireplace to impress their guests. You buy these assets for one reason: the money they are expected to pay you. For such assets, 'discounted expected future cash flows' is not merely *a* way to think about their value—it is what their value *is*.

That said, I'll confess that I think the underlying *way of thinking* is far more general, and is one of the more valuable things you can take from this chapter. Present versus future, under uncertainty: that is the structure of a huge share of life's decisions—investing in your education, taking a risky job, starting a company, deciding how much to save versus enjoy now. You can't literally plug your life into the formula. But the discipline of asking 'What am I giving up now? What do I expect to get later? And how should delay and uncertainty weigh on that?' pays off far beyond finance.[11]

[10] About $1,024. Notice, by the way, how little of it comes from the $1,000,000: a hundred years of compounded discounting at 15% crushes even a million dollars down to *under one dollar* of present value. (1.15^{100} is a staggeringly large number.)

[11] One place I put this to use: personal-finance gurus love to exhort young people to save every possible penny in their twenties, because compounding is powerful. Compounding *is* real and powerful. But applied rigidly, that advice can be a bad analysis of *value*. A year of cheap backpacking may be feasible and glorious at 22, and impossible at 45—in which case the 'cash flows' (experiences, in this case)

13.7 Reversing: solving for the discount/interest rate

Everything we did above was **correct**, but also a bit *backwards*. Usually, *in reality*, we are not *given* a discount rate by a professor. Instead, usually we can directly **observe the stated terms of the debt contract** (that is, the future cash payments promised), and we can directly observe a **market price for the bond**—that is, how much people are actually paying for it in the market. The **'real'/market interest rate is not written down** and directly observable in the same way. (The universe does not helpfully put little 'interest rate' labels on things.) And so a more realistic calculation in the real world would be to use a bond's market price (that is, its present value in the market) and its future payments to **calculate its actual interest rate.**

Suppose that we observe that the same bond we specified above ($10k face value, 6% coupon rate paid annually, maturing in 5 years) was issued to the market and traded at a price of $9,500. Then, putting this in math terms, this implies that:

$$Market\ PV\ of\ bond = \$9,500 = \frac{\$600}{1+i} + \frac{\$600}{(1+i)^2} + \frac{\$600}{(1+i)^3} + \frac{\$600}{(1+i)^4} + \frac{\$10,600}{(1+i)^5}$$

There is only one free variable in this equation—i, the interest rate. So, there **is a solution** here, an **implicit market interest rate**. But, even for this simple 5-year bond, the solution involves a **polynomial of order five**—I may be rusty on my high-school algebra, but I don't remember how to solve this by hand. So, here, we would **just rely on** a financial calculator or **Excel shortcut** to solve this for us.

Or, suppose that our bond instead had a 20-year maturity. In this case, the formula for calculating its present value would be:

$$PV = \frac{\$600}{1+i} + \frac{\$600}{(1+i)^2} + \frac{\$600}{(1+i)^3} \ldots + \frac{\$600}{(1+i)^{19}} + \frac{\$10,600}{(1+i)^{20}}$$

The concept is still very simple. It's still that **one core identity of finance**. But this is now a very **long calculation**—involving twenty terms, and polynomials of order 20—and we would likely make errors if we tried to add up all those terms on a scientific calculator. So, even though the concept is really simple, the calculation gets gnarly, and so it's best to save some time and trouble and use Excel. **Online resources** can walk you through how to use these **Excel shortcuts**. I won't add value by repeating them here.

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are worth radically different amounts at different dates, and the discounting cuts the *other* way. I am not advising you to max out your credit cards. I am advising you to actually think about present versus future value, rather than outsourcing the thinking to a slogan.

13.8 Net present value (NPV)

An extremely important concept in finance, business, and econ—a term that you will hear ubiquitously in the business world—is **'Net Present Value' (NPV)**. What is it? We've just discussed **present value** at length. *Net* Present Value is just... (drumroll)... the *net* present value. It is the **present value of the benefits/payoffs (cash inflows)** you get from some asset or investments, **net of** the **present value of the costs (cash outflows).** This gross-versus-net move should feel familiar by now—it's the same one accounting makes when it goes from *Accounts Receivable* to *Net Accounts Receivable*, or from assets to net assets. 'Net' just means: after subtracting the offsetting side.

If you've followed this chapter so far, you can see that this is ***the metric for rational investment decisions***. Present Value is just, by definition, how much future risky cash flows are worth to us in present-value terms.

13.8.1 NPV examples

In most investments, you have to **pay out something now** in order to get the future benefits. So, even if those future benefits are large, that investment might be a bad idea, if the **price you have to pay today is too high**. To determine whether an investment is worth it, you need to decide whether the benefits you expect to get in the future are worth what you have to pay out today to get them. That is, you need to decide whether the present value of the benefits is greater than the present value of the cost. That is, you need to know whether the *Net* Present Value is positive.

Recall the example of the **installment payment contract**, where we were promised 5 annual payments of $2,400 each. When we applied an interest rate of 12% to that asset, its present value was:

$$PV\ of\ instllmnt\ pymnts = \frac{\$2,400}{1.12} + \frac{\$2,400}{1.12^2} + \frac{\$2,400}{1.12^3} + \frac{\$2,400}{1.12^4} + \frac{\$2,400}{1.12^5} \approx \$8,651.46$$

Now, suppose that we were considering buying this asset, the entitlement to **receive those future cash flows**. Suppose that the price to buy this asset was $8,000 (or, put differently, we would need to loan someone $8k to get this installment contract). How would we factor that price in?

In finance terms, purchasing this asset would just be a **negative cash flow of $8,000 today**. What's the appropriate **discount/interest rate on that cash outflow**? **Nada.** Go back to our core theory. The cash purchase would be a **sure thing, paid today**, so it would **not be discounted.** There's no interest on an immediate, certain payment. So, the **Net Present Value (NPV)** of this investment opportunity would be:

$$NPV = -\$8,000 + \frac{\$2,400}{1.12} + \frac{\$2,400}{1.12^2} + \frac{\$2,400}{1.12^3} + \frac{\$2,400}{1.12^4} + \frac{\$2,400}{1.12^5} \approx \$651.46$$

(Of course, you could shortcut to this answer by subtracting the purchase price from the present value of the future payments, which we had already calculated above.)

Based on this analysis, this is a **good deal**. You *should* be willing to buy this installment contract, at that price. Why? Because, the price of buying it is *lower* than the *present value of the benefits*. If we claim that the *present value of the benefits* is higher than the cost, but we *don't* want to make that trade, we're **contradicting** ourselves. So, **calculating NPV** is really just a matter of **expressing our beliefs and preferences formally**.

Now, suppose that the price of this asset were instead $9,000. In that case, the NPV of this opportunity would be -$9,000 + $8,651.46 = -$348.54. In other words, the net present value would be negative, because the value of what you have to pay is **greater** than the present value of what you would receive.[12]

Beyond the typical case: In the examples above, all of the cash *outflow* came at the start, as the purchase price, and everything afterward was inflow. That is the most typical shape of an investment, but NPV is more general than that. Suppose we are considering buying a factory: we pay a big amount today, and the factory will generate future cash *inflows* (sales)—but it will also require future cash *outflows*: maintenance, repairs, property taxes. No problem. The 'cash flows' in our identity are simply the **net** cash flows of each period—inflows minus outflows—each discounted back to today in the usual way. Don't overthink it: NPV really is just the net-net value, in today's dollars.

Here's a fuller worked example in that spirit. Suppose I have the opportunity to buy a restaurant for $1 million. I estimate that it would bring in $200k per year in cash sales for the next 20 years, but would also incur annual cash outflows of $120k on ingredients, maintenance, and salaries. Based on my assessment of the riskiness of the restaurant business, I determine that the appropriate discount/interest rate is 15%. **Should I buy the restaurant?** The net cash flow in each of the next twenty years is $\$200k - \$120k = \$80k$. So:

$$NPV = -\$1,000,000 + \sum_{t=1}^{20} \frac{\$80,000}{1.15^t} \approx -\$499,000$$

Deeply negative—I should walk away. And notice how badly the *undiscounted* arithmetic would have misled me: twenty years of $80k net cash flows total $1.6

[12] Note that this is true even though the total promised cash flow from this investment is $12,000—meaning that the expected **net cash flow** (undiscounted) is actually +$3,000 in this scenario.

million, which appears to beat the $1 million price handily. But at a 15% discount rate, that whole future stream is worth only about $501k in present-value terms—roughly half of what I'd have to pay today to get it.

13.8.2 Present Value Discussion

Fair-value definitions: Now that we have covered the calculation, return to Chapter 12's fair-value question. A present-value model can help estimate what market participants would pay. If our estimate differs from the quoted market price, that may signal an opportunity—or a difference in assumptions, information, risk, or the benefits available to this particular owner. Market competition connects prices and valuations; it does not turn every company-specific forecast into the accounting fair value.

13.8.3 Going deeper: Present value already looks ahead

A decision can reduce this year's accounting profit while increasing the value of the business. Spending on a project may hurt current earnings, for example, while creating benefits expected in later years.

Present value gives us a way to ask whether those benefits justify the cost. The word *present* describes the date to which we translate the benefits; it does not restrict our attention to benefits received soon. The calculation can include cash flows far into the future.

This makes conversations about "short-term versus long-term value" worth unpacking. Does the speaker mean current profit, today's market price, or the present value of the full expected cash-flow stream? Those are different quantities. Clarifying the quantity helps us evaluate the actual tradeoff, rather than arguing over a phrase.

The calculation still depends on judgments about what will happen and how to value the risk. A strategy of accepting losses now is not justified merely by calling it long-term. The future benefits have to support the case.

13.9 Other finance topics

13.9.1 Risk and return

While we're on the topic of finance, I want to talk about another idea that comes up a lot, which we now have the tools to understand. You may have heard phrases of the form like **'high risk, high return.'** I think that many people think that this is a kind of moralistic exhortation, indicating that companies need to make big, *bold* risky moves

in order to *generate* high profits. But in the context of finance, that is *not what this really means*. Instead, the real meaning is something we've already discovered in the first pages of this chapter, and is, frankly, more **boring**.

In finance, the useful version of "high risk, high return" concerns **expected compensation for bearing risk**, not a promise that a risky investment will pay off. Holding expected payoffs fixed, investors may require a lower price for risks they dislike bearing. That lower price raises the expected percentage return. For one future payoff, with no intervening distributions, the relation is:

$$Present\ Value = \frac{Expected\ Future\ Value}{1+i}$$

$$Expected\ Return = \frac{Expected\ Future\ Value}{Present\ Value} - 1 = i$$

The distinction between **promised** and **expected** payoffs matters here. A distressed bond can promise an enormous yield because its price is low, while investors expect to recover far less than its contractual payments. And not every kind of risk earns an expected premium: diversification and the way an asset's payoff covaries with investors' other risks matter. Our earlier examples show the price-and-yield arithmetic; they do not establish that taking more risk guarantees a better outcome.

Now, a careful reader might raise an objection here—one that trips up a lot of people. 'If risky assets are cheap, and I therefore put in *less money*, haven't I taken on *less* risk? Doesn't that undo the whole story?' This objection rests on a confusion between two different notions of risk, and pulling them apart is clarifying. When finance speaks of the **risk of an asset**, it means the uncertainty in that asset's own future payoffs—the spread of possible outcomes—*not* how much of your money you happen to put into it. A wildly volatile stock is a high-risk *asset* even if you only hold $10 worth of it. Your *personal* risk—the risk of your overall portfolio—depends on both the riskiness of the assets you hold *and* how much you have allocated to each. When we say 'higher risk means a higher discount rate,' we are talking about the payoff-uncertainty of the asset, which determines its price *per unit*. How much of it you then buy merely scales your own exposure. Keep those two notions—**asset risk** and **portfolio risk**—separate in your mind, and a lot of finance gets easier.

13.9.2 'Market efficiency'

Finally, I want to discuss one last very important finance term: **market efficiency.** I think that when much of the general public hears the term 'market efficiency,' they think that heartless economists are claiming that markets are morally ideal. But this is **not** what this term means, in the **context of finance.** 'Market efficiency' simply means

that, if you want to buy or sell an asset in a financial market, you'll be entering a very **competitive marketplace**, where the other people you need to buy from or sell to are pretty smart and also looking for good deals. So, you can't expect anyone to sell you a financial asset (whether a bond or stock or anything else) at a price at which it is an obvious, no-brainer, good investment—near-certain to sharply increase in value, with little risk. Why not? Consider this **'proof by contradiction'**: If it *were* certain that the asset would drastically increase in value, with little downside risk, **why would the seller sell it at that price** right now?

Generalizing this, in financial markets, the **prices** that assets trade at are set in a way that **already takes into account** the available information about the company **and its future prospects**. That is, the current prices of financial assets *already reflect* predictions about their future performance. A more ***trippy*** way of putting it is this:

> "The **current price** of an asset in a financial market ***already is***, essentially, a ***prediction*** about its future price. In a sports betting market, you might be able to predict an ***event***, e.g., that the Lakers will beat the Knicks in the next game, with some success, better than a coin flip. But you can't really predict how the **prevailing prediction itself**—that is, the betting odds—will change in the interim. Similarly, you can't predict whether a stock will outperform.[13]"

Less wordy professors thus sometimes define market efficiency as:

"Prices reflect all available public information."

This notion has two practical implications for our purposes:

1. First, if an asset (such as a bond) promises a **higher** rate of **return** (which is the market interest rate in the case of a bond), that return must come with a **catch**—presumably, more **risk**. So, a higher interest rate means higher risk and vice-versa.

2. Second, given market efficiency, we will often assume that market prices and interest rates are basically **'correct.'** If we want to calculate the net present value of an investment, we'll often take for granted some relevant **market discount rate** as our 'true' discount rate for doing so. And, next chapter, we'll see that the market interest rate (rather than the company's 'stated' coupon rate on the bond) is what determines the actual ***Interest Expense*** a company will recognize in its financial statements.

[13] That is, yield returns beyond the *expected return* entailed by its risk.

14

Long-Term Liabilities

This chapter is about **accounting for bonds.** The main challenging work in this chapter is what we call **'bond amortization.'**[1] And it is among the **hardest**, most complicated, and most technical parts of this book. But, as we will see, it is a **necessary process** for accounting for some interesting and **inevitable complications** in how companies **finance themselves with bonds**. And we can reduce the accounting to a fairly **rote algorithm**. So, while this material will look tough and gnarly at first, it is absolutely doable.

14.1 What is a bond, and what is the problem?

Bonds are the major way in which **mature** companies finance themselves. They are a form of **credit**. The company makes a contractual promise to pay out a specified stream of payments—interest (or 'coupon') payments and the face value on the maturity date. They are **like loans** in that respect, but with **one major difference**: Companies **issue (that is, sell) their bonds to the financial marketplace**, where financial market participants (such as pension plans, hedge funds, or even individual investors) bid on those bonds and **trade** them. So, the interest payments are paid to whoever has

[1] (This is quite distinct from the *Amortization expense* we covered in Chapter 11—the allocation of the purchase price of an *intangible asset*. Same word, quite different meaning. Though the two uses do share a root idea: taking some lump amount and spreading—apportioning—it over multiple periods. People even use the word metaphorically: 'I put a lot of effort into learning accounting, but I'll amortize that investment over the rest of my career.')

bought that bond in the market. Whereas with a loan, the company negotiates and contracts privately with a **bank**, and the bank holds onto the loan instead of trading it.

The **jargon** here is a bit confusing to first-time learners, but important to grasp. We say that a company **'issues (or sells)** a bond,' because it is as if it is **selling a piece of paper** to the financial markets (the contract specifying the future payments it will make), in exchange for getting **cash today**. In the jargon of the financial world, bonds are often referred to as **'fixed income'**—because the payments (income) they offer to the owners of the bond are **fixed** in advance (in direct contrast to the dividend payments on *shares*).

The basic **accounting challenge** is this: The company issues its bonds to the **financial free market**. The company **cannot sell** a bond contract to the financial market **until it has set the terms** of the contract. So, the company has to specify a **'face value'** of the bond (the amount that it will pay off at the end) as well as its **cash interest payments**. By definition, the **cash interest payments** are determined by the **face value and the 'stated' interest rate**. Going forward, we'll refer to the 'stated' interest rate as the '**coupon rate**,' to more clearly distinguish it from the 'true market rate.'[2]

So, in short, the company has to specify its **stated coupon rate** prior to selling the bond. But the financial market is a free market. And so, as we learned in the last chapter, the participants in that market get to **make up their own minds** about what those future promises are worth—the **present value**—which is what the market will actually pay for the bond on its issuance date. And, by the **core identity of finance**, that is equivalent to determining the **real interest rate** on that bond.[3] We could also say it this way: 'The market gets to decide the real interest rate it will assign to the company's bond, which determines the present value.' Given our core finance identities, those are **equivalent statements**.

But this creates an **accounting problem**: The **cash received** on the bond (a debit to Cash for the issuer), may **differ from the face value** that the company is obligated to pay on the maturity date. And the **stated coupon rate may differ from the market interest rate**. So we need a method that handles any such difference. **'Bond amortization'** is accounting's way of dealing with that. But before we can get to that, we need to be clear on a few more mechanics.

[2] This language has to do with the history of bonds. Back in ye olde days, pre-digital era, bonds were **literal paper contracts** printed and sold by corporations to investors—those investors would literally clip a coupon off of that bond on the specified interest payment dates, take that physical coupon to the bank, and redeem it for cash there.

[3] ...since $i = \frac{future\ value}{present\ value} - 1$, and related identities.

14.1.1 Going deeper: Fruit as an analogy for a financial contract

Before we descend into those mechanics, I want to establish one big idea that will pay off repeatedly in this chapter: **a deal has to be worth it to both sides**. When our company issues a bond, we get to propose the terms—the face value, the coupon rate, the maturity date. But we do *not* get to dictate the price. The market decides what our promises are worth. If we want the financing, we have to make the deal attractive enough, given the risk, that the other side willingly participates.

Here's an analogy I am fond of.[a] Think about plants and animals as two 'parties' in an ecosystem, each facing a different problem. Plants can do photosynthesis—they can use sunlight to turn water and carbon dioxide into sugars, and draw minerals from the soil, which is an astonishing superpower. But a plant cannot walk its seeds across the landscape. Some plants rely on animals to carry them. Animals cannot photosynthesize, but they can move.

So, what is **fruit**? Fruit can be understood through an analogy to a **contract between these two kingdoms of life**. It is not literally negotiated, but each side benefits. The plant packages together, in one bundle: (1.) sugar and nutrients—the incentive, the 'payment' offered to the animal—and (2.) its seeds. The animal eats the fruit because it is sweet and energy-dense, and, in doing so, swallows the seeds and transports them across the landscape, dispersing them via a digestive process that I will not elaborate on further. The plant gets what it cannot do for itself (distribution); the animal gets what *it* cannot do for itself (food). And notice: the plant cannot dictate terms. If the fruit is not sweet enough, the animal simply will not participate, and the deal is off.

That is exactly the situation of a company issuing a bond. We can propose whatever contract we like. But we have to sell it at a price that makes it worth it to the actual investors on the other side, given the risks they are taking on. Keep this principle in mind—we will call back to it when we discuss 'callable' bonds later in this chapter.

[a] Here's a little perspective shift I had one time: I was backpacking in Peru, licked a frog I found in a stream, and suddenly saw the whole theory of bond accounting laid out before me. In reality, it was just a long hike, and an animal eating a piece of fruit. But the vision held up.

14.1.2 Bond specification:

Now, let's start making this more concrete: The first step in basically any bond problem is to be able to go from a **verbal description of a bond's terms** to understand the **cash payments it promises**. The key parameters in a bond contract, for our purposes, are:

- the coupon rate,

- the maturity date,
- the face value,
- the interest payment frequency.[4]

For example, suppose we have the following **verbal description**: 'A $10,000 face value bond, with a 5-year maturity, and a 6% coupon rate (stated interest rate), with annual payments.' This means that over the next 5 years, the company pays once-annual payments of 6% x $10,000 = $600; and on the maturity date (5 years from now) it additionally pays the **'face value.'** In other words, that verbal description of the bond indicates the following **schedule of cash payments**:

Time	**+1 year**	**+2 year**	**+3 year**	**+4 year**	**+5 year**
Cash flow promised	$600	$600	$600	$600	$10,600

Table 14.1: 5-year bond, $10k face value, 6% coupon paid annually:

You should be able to generalize from that example to be able to **specify the cash payments on any bond**, given any combination of face value, maturity, and coupon rate. For example: 'A $100,000 face value bond, with a 3-year maturity, and a 12% coupon rate, with annual payments.' For this bond, the schedule of cash flows is:

Time	**+1 year**	**+2 year**	**+3 year**
Cash flow promised	$12,000	$12,000	$112,000

3-year bond, $100k face value, 12% coupon paid annually:

14.1.3 Bond valuation:

The next step is to be able to understand and calculate how the **'real' / market interest rate** will determine the **present value** of the bond, and thus, the price it will actually sell for in the financial market. That, in turn, will be how much **Cash** the company issuing the bond will receive upon the bond's issuance. Given that we have **already specified the cash flows**, this is just an application of the **core identity of finance—the one formula—that we drilled on last chapter.** All we need to go from a schedule of cash flows to a present value is the **discount / interest rate**.

How do investors assess and price the 'riskiness' of corporate bonds? That's largely a question for dedicated finance courses. For our purposes, we will be *given* an interest rate, and need to know how to plug it in.

[4] I will always assume annual interest payments to keep the exposition of the ideas clean and simple. In reality, it's more common for bonds to pay interest *semiannually*, twice yearly.

As Chapter 13 showed, we can also work in the other direction: observe the price and promised payments, then solve for the implied yield. In the examples below, the rate will be supplied so that we can focus on the accounting.

Consider our first bond above: 'A $10,000 face value bond, with a 5-year maturity, and a 6% coupon rate (stated interest rate), with annual payments.' We already laid out the cash flows it pays (Table 14.1). So, if the real / market interest rate on that bond is 8%, then the PV and issuance price will be:

$$P.V. of\ bond = \frac{\$600}{1.08} + \frac{\$600}{1.08^2} + \frac{\$600}{1.08^3} + \frac{\$600}{1.08^4} + \frac{\$10,600}{1.08^5} = \mathbf{\$9,201.46}$$

Now, what if, **instead**, the market had assigned a real/market interest rate of **5%** to a bond with the same contractual terms? Then the PV and issuance price would be:

$$P.V. of\ bond = \frac{\$600}{1.05} + \frac{\$600}{1.05^2} + \frac{\$600}{1.05^3} + \frac{\$600}{1.05^4} + \frac{\$10,600}{1.05^5} = \mathbf{\$10,432.95}$$

Note how, in this simple pair of examples, when the **real** / market interest rate (8%) is **greater** than the **coupon rate**, the present value and issuance price is **less** than the face value. And when the real / market interest rate (5%) is **less** than the coupon rate, the present value and issuance price is **greater** than the face value. It also turns out that, if the real interest rate is **the same** as the coupon rate, the present value will **equal** the face value.

This is **not a coincidence**, and will **hold generally**. The formal proof of this is too long and gnarly for this book. But, to 'prove' the last one to yourself—that if the coupon rate equals the true interest rate, present value equals face value—do the following calculation and similar ones: $\frac{\$600}{1.06} + \frac{\$600}{1.06^2} + \frac{\$600}{1.06^3} + \frac{\$600}{1.06^4} + \frac{\$10,600}{1.06^5}$. If the coupon rate in the numerators is equal to the true interest rate in the denominators, then the present value will always equal the face value, **every time**. Then, once you've convinced yourself of *that*, you can remember that, **holding the cash flows fixed** (coupon payments and face value), a higher (vs. lower) **interest rate** means a lower (vs. higher) **present value.** So, you should be able to know those *directional differences* without having to do a calculation. (This is, honestly, how I handle these questions myself. I do not have a great memory, so instead of memorizing a table of cases, I memorize exactly *one* fact—coupon rate equals market rate means price equals face value—and reason everything else out from it. A higher market rate means bigger denominators, means heavier discounting, means a lower present value—interest rates *are* discount rates—and vice-versa.)

Summarizing these *directional* relationships:

- **market interest rate = coupon rate $\longleftrightarrow$ present value = face value**
  - Jargon: Bond 'issued at par.'
- **market interest rate $>$ coupon rate $\longleftrightarrow$ present value $<$ face value**
  - Jargon: Bond 'issued at a discount'
- **market interest rate $<$ coupon rate $\longleftrightarrow$ present value $>$ face value**
  - Jargon: Bond 'issued at a premium'

14.1.4 A challenge question: the net cash cost of financing

Before we get to the accounting, here's a challenge question that will pay dividends later in the chapter: **What is the total *net cash cost*, to our firm, of each of the bonds above?** Forget discounting and present value for a moment. Just count the raw, undiscounted dollars: over the whole life of the bond, how much cash comes in, and how much goes out?

Let's start with a benchmark where the answer is obvious: a simple $10,000 loan at 6% interest, for 5 years—which is equivalent to our bond issued 'at par.' We receive $10,000 today. We pay $600 per year for five years, and repay the $10,000 principal at the end. The $10,000 in and the $10,000 out cancel each other. So the net cash cost of the financing—what we paid, in total, for the privilege of using that $10,000 for five years—is just the interest: $3,000.

Now consider our bond issued at a **discount**. We receive only $9,201.46 today, but we pay out the same $13,000 in total ($600 $\times$ 5 years of coupons, plus the $10,000 face value). So the net cash cost is $13,000 $-$ $9,201.46 = $3,798.54. That is, we are really paying **two costs** for this financing: the $3,000 of coupon payments, *plus* the $798.54 gap between the face value we must repay and the cash we actually received.

And for the bond issued at a **premium**, symmetrically: we receive $10,432.95 today and pay out $13,000, for a net cash cost of $2,567.05. The $432.95 'extra' that we collected up front—money we got, but never have to repay—partially **offsets** our coupon payments.

Here's the punchline, and really the moral of this whole chapter: whenever a bond is *not* issued at par, **the coupon payments alone are not the true cost of the financing**. The initial discount is an *additional* cost, on top of the coupons; the initial premium is an *offset* to them. The whole 'bond amortization' machinery we are about to learn is just accounting's way of taking that extra $798.54 cost (or that $432.95 benefit) and **spreading it out**—apportioning it, *amortizing* it—over the life of the bond, so that each year's *Interest Expense* reflects the true economic cost of the debt. If you hold onto that idea, the rules below will feel much less arbitrary.

14.2 Accounting for a bond issued 'at par'

Now, to segue to our **accounting** content, let's start with the **simplest** case of a bond issuance. As we saw above, if and only if the 'real' / market interest rate is equal to the stated / coupon rate on the bond, then the present value and issuance price of the bond will equal its face value.[5] In this case, we say that the bond is issued **'at par' (jargon)**.

14.2.1 Bond issuance

Consider again the bond contract we used as an example above, with a 6% annual coupon rate and a face value of $10,000. If and only if the real / market interest rate upon issuance were also 6%, the present value would be:

$$P.V.of\ bond = \frac{\$600}{1.06} + \frac{\$600}{1.06^2} + \frac{\$600}{1.06^3} + \frac{\$600}{1.06^4} + \frac{\$10,600}{1.06^5} = \textbf{\$10,000.00}$$

New bonds are often priced at or near par when their terms are set close to prevailing market conditions. We use an exact par issue as the simplest **benchmark** for bond accounting. But, hypothetically, if this were to happen, the **accounting** for the bond issuance would be very **simple**: The company would get a **$10k Cash** inflow (a debit), and the face value that it is liable to pay off at maturity will also be **$10k**, and recorded in the *Bonds Payable* liability account. The journal entry would be:

Cash (+A):	$10,000	
Bonds Payable (+L):		$10,000

Bond issued at par, 1/1/2021

This is essentially how we accounted for loans (**Notes Payable**) earlier in this book, just with *Bond* in lieu of *Note*.

14.2.2 Subsequent Interest Expenses / Payments

Additionally, in this special case of a bond issued at par, accounting for subsequent interest expenses is also easy. Let's keep things simple and assume our company does annual reporting on January 1st, and also issued its bond paying annual interest payments on January 1st. That way, **we don't need to worry about interest accruals** (because the interest cash payment and expense recognition happen in full and on the

[5] Appendix 2 gives the algebraic proof that present value equals face value when the coupon rate equals the market rate. The numerical examples here illustrate the result.

same date). We'll make this simplifying assumption throughout this chapter. Since the coupon rate and true market interest rate upon issuance were both 6%, the annual **interest payment and interest expense are both $600**. So, we would account for each annual interest payment like so:

Interest expense (-S.E.):	$600	
Cash (-A) :		$600

Annual interest payment and expense, 1/1/2022

So, here the accounting is easy. When the bond does not issue at par—that is, when the market rate is not equal to the coupon rate—things get more challenging.

14.2.3 Why work with both premiums and discounts?

The coupon specifies the promised cash payments; the market price determines the rate investors obtain on those payments. Issuers generally try to choose terms close to current market conditions, so a new bond may be priced close to its face value. Our examples use a larger gap to make the accounting visible. Whether the gap is large or small, the task is the same: connect the proceeds received at issuance with the payments the company will make over the bond's life.

14.3 Bond issued at a discount:

Consider again our example of a bond with a $10,000 face value, a 6% annual coupon rate, and a 5-year maturity. But now, suppose the **real / market interest rate** upon its issuance is **8%**. Then the implied present value and, thus, the issuance price, would be:

$$\textit{Issuance Price} = \textit{P.V.} = \frac{\$600}{1.08} + \frac{\$600}{1.08^2} + \frac{\$600}{1.08^3} + \frac{\$600}{1.08^4} + \frac{\$10,600}{1.08^5} = \mathbf{\$9,201.46}$$

Because the $9,201 issuance price is lower than the face value, we say the bond is issued "at a discount" **(jargon).**

14.3.1 Issuance

Now, consider the **accounting problem**. Our company gets a $9,201 **Cash (+A)** infusion (debit); but the face value we are liable to pay in 5 years is $10,000. How does

accounting deal with this difference? The **rule** is we credit **Bonds Payable for the full face value** ($10,000), and **'plug' the difference** with a new account called **'Discount on Bonds Payable.'** So, obeying that rule, the journal entry upon issuance would be:

Cash (+A):	$9,201.46	
Discount on Bonds Payable (+contra-L):	$798.54	
Bonds Payable (+L):		$10,000

Bond issued at a discount, Jan 1 2021:

What exactly *is* this **Discount on Bonds Payable**? As the journal entry above indicates, it is a **contra-liability account**. This is the first contra-liability account we've seen in this book. But it fits previous patterns we've seen, of how a contra-account will plug a difference, and separately track some kind of a reduction in another account. In this case, upon issuance, the **Discount on Bonds Payable** tracks the difference between the **face value** (the payment due on the maturity date) and the **present value** of the bond. And we want them tracked separately, because the company *will eventually pay* that face value. Setting up the T-accounts, those would look like so, at this point:

Bonds Payable (L)

Debits	Credits
	$10k

Discount on B/P (contra-L)

Debits	Credits
$798.54	

Another way to think about this is to **forget about the 'meaning,'** and simply note that we need to **'plug'** the difference between the **cash flow** and the related **accrual account** (the Bonds Payable liability), a common accounting move.

14.3.2 Introducing 'Carrying Value'

Now, we need to introduce one vital bit of terminology (**jargon**). We use the term **'Carrying Value'** to indicate the 'net' value of the Bonds Payable on the company's books—that is, net of its *Discount* contra-account. (This is quite similar to 'book value,' the

value of some asset less its related contra-account, e.g. *Equipment* and *Accumulated Depreciation*.) Formally:

$$Carrying\ Value\ =\ Bonds\ Payable\ -\ Discount\ on\ Bonds\ Payable \tag{14.1}$$

(Or, you can just think of it in terms of those 'stacked' T-accounts above—the *Bonds Payable (L)* account minus its contra-account amount.) So, after the issuance, per the journal entry above, the **initial carrying value** will be $10,000 - $798.54 = $9,201.46. Note that, per our **rule**, *Bonds Payable* is always credited for the full **face value on the issuance date**. So, on the issuance date, the carrying value will also equal the *Face Value* minus the *Discount on BP*, which is just the issuance price ($9,201.46). But, as we'll see, this will not hold going forward—-because the *Discount on BP* (and thus the carrying value) will be *updated* with each interest payment.

One more helpful parallel here: just as the *Equipment* account itself stays at the original purchase price forever, while all of the updating action happens in *Accumulated Depreciation*, the *Bonds Payable* account will stay put at the $10,000 face value for the entire life of the bond, while all of the updating action happens in the *Discount* contra-account.

14.3.3 FAQ: Why aren't the coupon payments a liability?

Now, you might raise a deep question here: The company is contractually obligated to pay the $10,000 face value—and we duly record a $10,000 liability in *Bonds Payable*. But the company is *equally* contractually obligated to pay those five $600 coupon payments. So why is there no 'Coupon Payments Payable' on the balance sheet for the $3,000 of promised interest? Aren't those obligations too?

Here's the answer: the coupons *are* in there—implicitly. The **Carrying Value** of the bond on the issuance date is the issuance price, which is precisely the **present value, at the market interest rate, of *all* of the promised future payments**—every coupon *and* the final face value. So the Carrying Value is the single number that summarizes, in present-value terms, the entirety of the obligation. (This is also consistent with how we've treated interest on plain loans throughout this book: interest is the cost of borrowing *over time*, so it accrues, and gets recognized as *Interest Expense*, period by period as time passes—not all up front, before we've done the borrowing.)

14.3.4 Subsequent expenses:

So we just solved the accounting problem of the **difference** between the **face value** and the **issuance price** of the bond on the *issuance date*—just the old accounting trick of creating a contra-account as a **plug**.

So, now, we need to account for subsequent interest payments and expenses. (Note that, per our simplifying assumption that the bond is issued on Jan 1st, the reporting date, and pays annual interest, both the expense and the payment will be recorded at the same time.) Let's start with what we know for sure. By definition, the **annual cash interest payment** on this bond is fixed by the terms of the bond itself. So, we know that that will be a credit of $600. So, here's what we know right now:

Interest Expense (-S.E.):	???	
???	???	
Cash (-A):		$600

First interest payment, Jan 1 2022 **(State of our Ignorance)**:

But what is the *Interest Expense*? The **next accounting problem** we face is that the **real / market interest rate** will be different from the stated coupon rate. And, per our core finance theory, the real / market interest rate determines the real / **'economic' cost** to the firm of getting its debt financing. So it would be problematic to allow companies to base their reported *Interest Expense* on their *stated* coupon rate.[6]

So, how exactly does accounting deal with this issue? By now, you are probably primed to recognize that we **use the real/market interest rate upon issuance** (8% in our current example) rather than the stated/coupon rate (6% in our current example) to calculate the Interest expense per period. So, do we multiply that 8% times the $10,000 face value, and recognize $800 in annual *Interest Expense*? **No.** Remember that on the initial issuance date, the market actually paid **$9,201.46** (which is also the initial carrying value of the bond). So that is like the real initial **loan** the company got.

It turns out that the **rule** is that we multiply the real / market interest rate by the **Carrying Value** (CV), not face value, to calculate the *Interest Expense* each period. That is:

$$Interest\ Expense\ =\ CV\ \times Market\ Interest\ Rate\ upon\ issuance \tag{14.2}$$

So, after the first year, that would be a debit of $9,201.46 x .08 = $736.12. So, our **next and final accounting problem** here is that our Cash credit ($600) and our *Interest Expense* debit ($736.12) do not match. How do we deal with this? Once again, the

[6] As an edge case, suppose that a company issued a $10,000 face-value, zero-coupon bond—in other words, had an annual coupon payment of $0—but this company was *actually* quite *risky*. In that case, the market would assign a high *real interest rate* to the bond, and its *issuance price* would thus be far below $10,000—say, e.g., $6,000. So, that case, there would be a **real cost of that debt financing,** even though there was **$0 in annual coupon payments**. Intuitively, we shouldn't allow that company to falsely report that it has $0 in *Interest Expense*. We need some way to account for the real cost of the debt. (You can think of the zero-coupon bond as the *limit case* of a bond issued at a discount.)

Discount on Bonds Payable (-contra-L) account will *plug* the difference. So, in fact, our full journal entry for the first annual interest expense would be:

Interest Expense (-S.E.):	\$736.12	
Discount on Bonds Payable (-contra-L):		\$136.12
Cash (-A):		\$600

First annual interest payment/expense, Jan 1, 2022, in full:

This means that, after this journal entry, the balance of the *Discount on Bonds Payable* will be \$798.54 -\$136.12 = \$662.42. (Remember that the Discount on BP is a debit account so credits to it are *reductions*.) Visualizing with the T-accounts:

Bonds Payable (L)

Debits	Credits
	\$10k

Discount on BP

Debits	Credits
\$798.54	\$136.12
\$662.42	

And, per its definition, the updated *Carrying Value* of the bond would be:

$$Carrying\ Value\ =\ Bonds\ Payable\ -\ Discount\ on\ B.P.\ (balance) \tag{14.3}$$

$$Carrying\ Value\ = \$10,000 - \$662.42 = \$9,337.58 \tag{14.4}$$

(Note that we also could have updated the Carrying Value more directly by just adding the \$136.12 change in the *Discount* to the prior Carrying Value.)

We'll discuss some *intuitions* for what's going on in the next section. But for now, let's just keep applying our **rote rules**. What do we do about the *next interest payment* (in 2023)? As always, the **cash interest payment** is fixed at \$600. The *Interest Expense* is, by definition, $Carrying\ Value \times Market\ Interest\ Rate\ upon\ issuance$. Calculating, that is: $\$9,337.58 \times .08 = \$747.00$. And we 'plug' the difference between the two by reducing the *Discount on BP*. So, the journal entry for the next interest expense and payment would be:

Interest Expense (-S.E.):	$747.00	
Discount on Bonds Payable (-contra-L):		$147.00
Cash (-A):		$600

Second annual interest payment/expense, Jan 1, 2023

And then, after that interest payment and expense, our updated *Discount on Bonds Payable* balance will be $662.42 - $147.00 = $515.42.

Discount on BP

Debits	Credits
$798.54	$136.12
	$147.00
$515.42	

Discount on BP (contra-L) after first Interest

And so our updated Carrying Value will be: $\$10,000 - \$515.42 = \$9,484.58$. And so, in turn, our *Interest Expense* for the *next* year (2024) will be: $\$9,484.58 \times .08 = \$758.77$. And then we just keep *plugging and chugging*, as long as our question requires!

14.3.5 Backing Up: Summarizing The Rules

So, we just worked through an example in linear order, and introduced a lot of new **jargon/terminology**. So, if your head is spinning a bit, mine is too. But everything we did can be boiled down to a couple of **rote rules**, which we can memorize and **cling to** as we develop our understanding.

Rules for Bonds Issued at a Discount (market interest rate $>$ coupon rate)

- **Rule #1:** On the **issuance date**, credit the full **face value** ($10,000) to ***Bonds Payable***; debit the **issuance price** (present value) to **Cash**; and 'plug' the difference with a debit to ***Discount on Bonds Payable (contra-L)***.
- **Rule #2:** Define: $Carrying\ Value = Bonds\ Payable - Discount\ on\ BP\ (balance)$
- **Rule #3:** In each period, the *Interest Expense* is calculated as: $Interest\ Expense = Beginning\ Carrying\ Value \times Issuance\ Yield$.
- **Rule #4:** 'Plug' the difference between the **cash interest payment** and that *Interest Expense* with a credit to *Discount on Bonds Payable*.

- **Rule #5:** And remember to *update* the Carrying Value, based on the updated *Discount*, before calculating the subsequent year's *Interest Expense*.

Or, we can show the **journal entries** in general form. The journal entry for the initial *issuance* of a bond at a discount will always be:

Cash (+A):	$(Issuance Price [PV])	
Discount on BP (+contra-L):	$(Face Value - Issuance Price [PV])	
Bonds Payable (+L):		$(Face Value)

Bond issued at a discount, Jan 1 2021:

The Carrying Value (CV) will always be as defined above. And the subsequent *Interest Expenses* and cash interest payments will always be:

Interest Expense (-S.E.):	$(CV * Market Interest R)	
Discount on BP (-contra-L):		$(Difference)
Cash (-A):		$(Coupon Rate * Face Value)

Subsequent annual interest payment/expense, Jan 1, 202x

14.3.6 Intuition: What's going on here?

It is very challenging to get intuition for what's going on here. For solving problems, it's best to cling to the rules. But here are some things that could make some intuitive sense:

- **First**, the actual **issuance price** of the bond is like the amount that the market actually 'loaned' to the company. So it should be intuitive that, the *first* interest expense is the market interest rate times the initial carrying value, which is that number.

- **Second**, note that with each *Interest Expense* and interest cash payment, we **credit** the *Discount on BP (contra-L)* account. Since that is a *debit account*, the credits are **reductions.** So, we are **"using up"** the 'Discount' over time, allocating it across periods. And this means, in turn, that the **Carrying Value** increases up to the **Face Value** as of the **maturity date**. This is why this process is called **bond "amortization"**—it's a way of *apportioning* an abstract B/S account. (See the bond-amortization figure a few pages below.)

- **Third**, note that for bonds issued at a **discount**, the market interest rate is **higher** than the pre-set **coupon payments**. So, one way to think about this is that it is *as if* the company is 'falling behind' on paying its real cost of debt. So, the *Carrying Value* of the bond—and, thus, in turn, the *Interest Expenses*—**increase** over time. This is just like what would happen if you fell behind on credit-card interest payments. Suppose you have a $10,000 balance on a card charging 5%. Just to tread water, you need to pay $500 this year. If you only pay $200, the credit-card company does not shrug and forgive the other $300—it gets added to your balance. Your liability grows, and next year's interest is charged on that bigger balance: your interest *compounds*. Our discount bond is the same idea in a more structured setting: the market requires 8%, we are only paying out 6% in cash, and the shortfall gets capitalized into the carrying value of the liability.

- **Fourth**, in simple cash terms, the company issuing this bond is **paying two forms of cash costs** for its bond financing. It's paying the cash **interest payments**, obviously. But it's **also** paying the **difference between the face value and the issuance price**. I.e., when a company issues a bond at a discount, the cash it gets at the beginning ($9,201.46) is less than the face value it will have to pay at the end ($10,000). The **difference** between them is, by definition, the **initial Discount**. And, what this whole algorithm is doing is **apportioning** that Discount out as expenses over time, **on top of** the company's cash interest payments. So, this looks like a **familiar tactic** in accrual accounting—apportioning a net cash outflow over time at a constant effective rate, to reconcile cash to accruals.

If none of these intuitions make sense to you yet, don't panic. The **rules work** whether or not we think they should. If you just want the right answer, the algorithm has your back.

14.4 Bonds issued at a premium:

Now, let's cover the final possible case, when the **market** interest rate on the bond is **lower** than the pre-set **coupon** rate, and it issues at a **premium**.

14.4.1 Differences between a bond issued at a discount

After all the work we did in the previous section, this will be relatively quick and easy, because virtually everything above applies *symmetrically*, with only two minor, and predictable, semantic differences:

- **First**, because the bond is issued at a *premium*, instead of having a contra-liability account (*Discount on BP*) as a plug, we use ***'Premium on Bonds Payable (+L).'*** This is a normal liability account, a credit account. So, we write it up with a *credit* on the issuance date, and will reduce it via *debits* in the later amortization.

- **Second**, predictably and symmetrically, the Carrying Value (CV) is now defined as:

$$CV = Bonds\ Payable + Premium\ on\ Bonds\ Payable. \tag{14.5}$$

14.4.2 The Rules

And so our **rules** for bonds issued at a premium are **substantively identical** to those for bonds issued at a discount, just with those changes.

- **Rule #1:** On the **issuance date**, credit the full **face value** to ***Bonds Payable***; debit the **issuance price** (present value) to **Cash**; and 'plug' the difference with a credit to ***Premium on Bonds Payable (L)***.

- **Rule #2:** Define: $Carrying\ Value = Bonds\ Payable + Premium\ on\ BP\ (balance)$

- **Rule #3:** In each period, the *Interest Expense* is calculated as:
$Interest\ Expense = Beginning\ Carrying\ Value \times Issuance\ Yield$.

- **Rule #4:** 'Plug' the difference between the **cash interest payment** and that *Interest Expense* with a debit to *Premium on Bonds Payable*.

- **Rule #5:** And remember to *update* the Carrying Value, based on the updated *Premium*, before calculating the subsequent year's *Interest Expense*.

And, as before, we can show the **journal entries** in general form. The journal entry for the initial *issuance* of a bond at a premium will always be:

Cash (+A):	$(Issuance Price [PV])	
Premium on BP (+L):		$(Issuance Price [PV] - Face Value)
Bonds Payable (+L):		$(Face Value)

Bond issued at a premium, Jan 1 2021:

The Carrying Value (CV) will always be as defined above. And the subsequent *Interest Expenses* and cash interest payments will always be:

Interest Expense (-S.E.):	$(CV * Market Interest R)	
Premium on BP (-L):	$(Difference)	
Cash (-A):		$(Coupon Rate * Face Value)

Subsequent annual interest payment/expense, Jan 1, 202x

14.4.3 Example

Go back to our primary bond example, with a **5-year maturity**, **$10k** face value, and a **6% coupon** paid annually. Now let's suppose that the **market** interest rate on the bond upon issuance was **5%**. Then the present value and issuance price of that bond would be:

$$\textit{Issuance Price} = \textit{P.V.} = \frac{\$600}{1.05} + \frac{\$600}{1.05^2} + \frac{\$600}{1.05^3} + \frac{\$600}{1.05^4} + \frac{\$10,600}{1.05^5} = \textbf{\$10,432.95}$$

And, applying our **rules**, the journal entries for the issuance and subsequent interest payments would be:

Cash (+A):	$10,432.95	
Premium on Bonds Payable (+L) :		$432.95
Bonds Payable (+L):		$10,000.00

Bond issuance at a premium, Jan 1 2021

The **initial Carrying Value** would be $10,432.95. So the first annual *Interest Expense* would be $10,432.95 x .05 = $521.65. Applying our **rules**, the journal entry for the first interest payment date would be:

Interest Expense (-S.E.):	$521.65	
Premium on Bonds Payable (-L):	$78.35	
Cash (-A):		$600

First annual interest payment/expense, Jan 1, 2022:

And so, the updated balance of the Premium would be: $432.95 - $78.35 = $354.60. Or, if you prefer to see this with the T-account:

Premium on BP	
Debits	Credits
$78.35	$432.95
	$354.60

Premium on BP (L) after first Interest

So the updated Carrying Value would be: $10,000 + $354.60 = $10,354.60. So the subsequent Interest expense would then be: $10,354.60 x .05 = $517.73...and we would keep applying that algorithm, **updating the Carrying Value** after each interest payment, and **plugging and chugging**, as long as our question requires.

14.4.4 Intuition

It's very hard to get intuition here. Let's try to just 'flip' the intuitions we worked on for bonds issued at a discount.

- **First**, note that for bonds issued at a **premium**, the pre-set coupon rate is **higher** than the real **market interest rate**. So, one way to think about this is that it is *as if* the company is paying **more** than its real interest cost of debt, so it is as if it is **'paying down' the principal of the initial 'loan' of $10.4k** as well. This is just like how **installment payments** have an *implicit interest rate*, but fixed payments over time, so each payment represents interest but **also a reduction in the outstanding principal**. (This is also just how home mortgages work. Mortgages have an implicit interest rate, but homeowners pay essentially fixed monthly mortgage payments. Homeowners increase their **home equity** with each payment, because each one represents both *interest* and paying down the principal.)

  And, since the *Carrying Value* is reduced over time, the *Interest Expense* recorded in each period decreases as well. This should also make intuitive sense. It's like if you paid off *more* than your minimum interest payment on your credit card. That would reduce your outstanding *balance*, which would reduce the interest you were charged.

- **Second**, in simple cash terms, the company issuing this bond is incurring the **cash cost** of paying **coupon payments** ($600 $\times$ 5 years), but that is partially **offset** by the **positive net cash flow** from selling the bond at a premium to its final **face-value** payment. That is, it gets $10.4k on issuance, but will only have to pay $10k of face-value on the maturity date. So, that approximately $433 difference **"offsets"** the cash interest payments, reducing the *Interest Expense* over time.

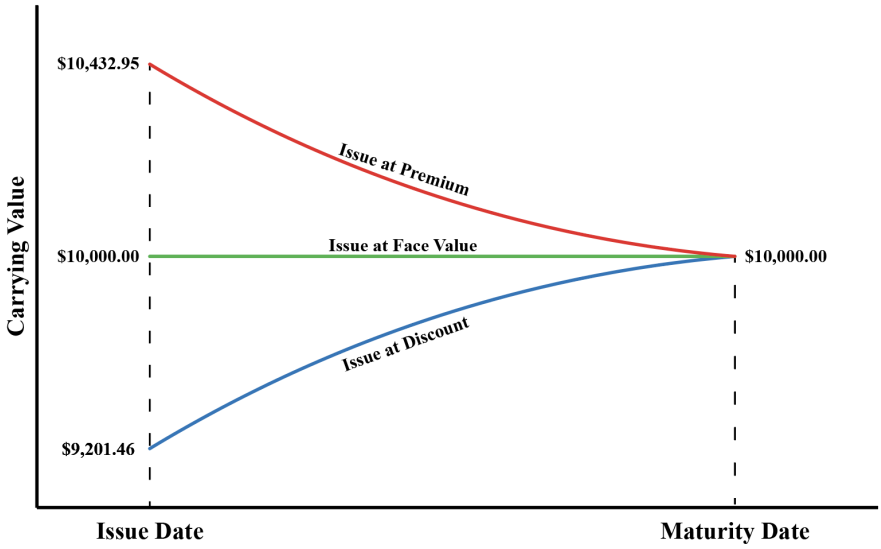
- **Third**, note that with each *Interest Expense* and interest cash payment, we **debit** the *Premium on BP (L)* account. Since that is a *credit account*, the debits are **reductions.** So, we are **"using up"** the 'Premium' over time, allocating it across periods. And this means, in turn, that the **Carrying Value** decreases down to the **Face Value** as of the **maturity date**. (See the figure just below.)

14.4.5 FAQ: Why would anyone buy a bond at a 'premium'?

One question that students often have when we discuss bonds issued at a premium, is 'Why would the market pay more than the face value of the bond—the cash payment it will actually get at the end?' And while this does seem odd to first-time learners, there is an **easy answer** to this question: Bond buyers in these cases are willingly paying more than face value, because they are getting **compensated** by the **relatively high coupon payments** in the **interim period**. Since the coupon rate is higher than the real/market interest rate, they are still getting the **return** they require **overall**, even though the final face-value payment they will receive at the end is less than their initial investment.

14.4.6 Visualizing premium/discount amortization

It can be very helpful to visualize the 'amortization' process for bonds issued at a premium vs. discount together. In both cases, the *face value* is different from the *issuance price*, which is the *initial carrying value*. And then, as we reduce the *Discount* or *Premium*, amortizing it with each interest payment and expense, the Carrying Value converges to the *Face Value*. See the figure below for a visual representation:



Bond Amortization

14.4.7 A deeper intuition: Carrying Value tracks fair value

Here is one final intuition—the deepest of the lot. It turns out that this whole plug-and-chug algorithm has a beautiful property: **the Carrying Value tracks the *fair value* of the bond over time**—so long as the market interest rate on the bond stays constant.

Recall Chapter 12: **fair value** is a market-participant exit price, which can be estimated using present-value techniques. Now let's pin down two easy anchor points for our bond issued at a discount. **What was the fair value of the bond on the issuance date?** Easy: $9,201.46. It satisfies both definitions—that is what the market actually paid for it, and that is the present value of the promised cash flows at the 8% market rate. **And what will the fair value be a microsecond before the maturity date** (assuming the final coupon has just been paid separately)? Also easy: $10,000. At that instant, the bond is simply an entitlement to receive $10,000, essentially immediately—there is nothing left to discount.

Notice: those two anchor points are exactly where the Carrying Value *starts* and *ends* (see the figure above). And it turns out—the algebraic proof is too gnarly for this book—that the amortization algorithm connects those two endpoints in exactly the right way: immediately after each scheduled coupon payment, the Carrying Value equals the present value of the *remaining* promised cash flows, discounted at the original 8% market rate. In other words, our humble rote rules are quietly performing a continuous, principled valuation of the liability, the whole time.

One honest caveat: this works *if and only if* the market interest rate on the bond holds constant. In reality, the bond's yield will wander around with market conditions and sentiment, and then the Carrying Value will no longer match the true market value. Do we then re-mark the bond on our books? **No.** Remember what accounting records: *transactions*. We issued the bond and received cash (a transaction); we will pay the face value at maturity (a transaction); and unless we actually repurchase or extinguish the bond in between, we do not constantly re-mark it to market. This is the same posture we take with long-lived assets—we don't revalue a factory every time the market for used factories wiggles. Bonds payable are carried at this **'amortized cost,'** rather than at fair value. As we're about to see, this is not just a bookkeeping nicety—it was at the center of one of the most consequential bank failures of the 2020s.

14.4.8 Real-world application: Interest-rate risk and Silicon Valley Bank

With the tools we've now built, we can understand one of the most significant financial events of the early 2020s: the March 2023 collapse of **Silicon Valley Bank**

(SVB). It's a perfect application of this chapter's bond math, and of a deep tension we've been wrestling with throughout this book: **fair value versus historical cost**.

First, how does a bank work, at a very basic level? A bank has **depositors**. Those depositors can be individuals—you deposit your paycheck because you need somewhere for that cash to sit—or companies. (SVB's depositors were famously concentrated among tech startups, parking the cash they had just raised from venture capitalists until they needed it for payroll and rent.) When you deposit your cash, the bank shows you a number on a screen. What *is* that number, in accounting terms? It is **not** a label on a pile of cash sitting in a vault with your name on it. It is the bank's **liability** to you: the amount you are entitled to withdraw, on demand. The bank does not leave the corresponding cash sitting idle—it lends it out and invests it.

For our purposes, distinguish two risks. **Credit risk** is the risk that the bond issuer will fail to make its promised payments. **Interest-rate risk** is the risk that the market value of those payments will change as market rates move. U.S. Treasury securities have very low credit risk, but that does not make a long-term Treasury bond immune to interest-rate risk. Banks can therefore hold securities with little credit risk while still exposing themselves to large changes in market value.

To isolate that mechanism, take a **hypothetical** 30-year bond with a $10,000 face value and a 2% annual coupon. If the market yield is 2%, it sells at par, $10,000. It promises $200 each year, plus $10,000 at maturity. This deliberately simplified bond illustrates the duration risk in long-term fixed-rate securities; it is not a description of SVB's exact portfolio.

Now suppose the required market yield rises from 2% to **6%**, while 30 years remain. The bond's promised cash flows have not changed. But every payment is discounted more heavily, and the final principal is divided by 1.06^{30} rather than 1.02^{30}. Applying our formula gives a new value of about **$4,494.07**. The large decline comes entirely from the change in the discount rate. These are illustrative rates; the Federal Reserve does not directly set a 30-year Treasury yield.

Now, here's the genuinely interesting question, and it's not as obvious as it first appears: **Is this drop actually a problem for the bank?** There are two sides to argue. On one side: the bank's liabilities to its depositors are unchanged—they can still demand every dollar back, any day—while its assets have lost a huge chunk of their market value. Its equity cushion, its net worth, looks dramatically thinner. On the other side: the *contractual cash flows haven't changed at all.* If the bank simply holds those Treasuries to maturity, it will collect every dollar it always expected to collect, and can use those cash flows to pay depositors as they come and go. In normal times, deposits roll over—some people withdraw, others deposit—and a

bank with sufficiently stable funding may not need to sell those securities before maturity. On that view, the bank can avoid realizing a loss in a sale if it can hold the securities. The economic decline still matters: its funds remain tied up at a below-market return, and funding costs or withdrawals can change. Whether the decline in market value is a real problem depends on what the bank has to do, and *when*.

That timing mismatch was central to SVB's failure. On March 8, 2023, the bank announced that it had sold $21 billion of available-for-sale securities, realizing a $1.8 billion after-tax loss, and planned to raise capital. Depositors then withdrew more than $40 billion on March 9, and the bank failed on March 10. Its concentrated reliance on uninsured deposits made a rapid run especially dangerous.[a] The security sale, the announcement, and the run are parts of a sequence; the story is not simply that every bond was sold after the run began.

The accounting question sitting underneath all of this is one you are now equipped to appreciate: should a bank be required to write its bonds down from their amortized cost to their current fair value as interest rates rise, even if it intends (and expects) to hold them to maturity? Reasonable people disagree—this is the fair-value-versus-historical-cost debate in its sharpest modern form. Actual accounting distinguishes held-to-maturity securities, generally measured at amortized cost, from available-for-sale securities, measured at fair value with most unrealized changes outside net income. My point here is not to settle the debate, but to show how the pieces connect: it's nothing but our one formula of finance, plus the tension between recording verifiable transactions and reporting current market values.

[a] Federal Reserve, Review of Silicon Valley Bank: Evolution of Silicon Valley Bank, April 2023.

14.5 Repurchasing or calling a bond:

A bond, like a piece of equipment, has a **life cycle**. With equipment: (1.) we purchase it, (2.) we depreciate it over time, and (3.) eventually we sell or scrap it. With a bond: (1.) we issue it, (2.) we amortize the premium or discount over time, and (3.) eventually the bond is extinguished—at maturity or, sometimes, *early*. This section is about that third step, when it comes early.

14.5.1 Callable bonds

So far in this chapter, we've talked about bonds in a fairly **general way**. But, as always, **reality** can be incredibly **complex**. At the end of the day, **bonds are just contracts**—agreements by the company to make certain payments at certain dates, and specifying other rights, conditions, and obligations for the company and bondholder. A typical

credit agreement can run to almost 100 pages, or more. (This is one reason why there are so many white-collar jobs in and around the financial sector.)[7] And so, the **complex reality** is that these contracts can be written in **any number of ways**.

One of the most significant parameters in a bond contract, worth knowing about, is whether or not the bond is **'callable.'** A 'callable' bond is one in which the bond contract stipulates that the **issuer *can* 'call'** the bond—that is, essentially, pay to *extinguish it* and cancel all the future payments—at a pre-specified **price**. **Economically,** this is essentially the **same** as the issuer having the right to **repurchase** the bond at the specified **call price**. But, the legal structure, mechanics, and **semantics** are a bit different. For callable bonds, it's conventional to refer to this as **'redeeming'** the bond rather than 'repurchasing.'

Why would our company 'redeem' one of its callable bonds? The most common and significant reason would be a change in **overall market interest rates**. For example, suppose that when we originally issued our bond, the interest rate for bonds for companies with our credit rating was 7%. But now, thanks to Federal Reserve policy and other factors, that prevailing market interest rate is 3%. Clearly, we would rather pay a 3% interest rate than 7%! So, in such a condition, we could issue **new bonds** at the 3% interest rate, and use the **cash proceeds** to **redeem** our old bond, and thereby **reduce** our **interest payments**.

Let's make the mechanics concrete, because **refinancing** usually involves two linked transactions. Suppose our old bond has a 7% coupon, a $10,000 face value, and is currently callable at $10,500. Market rates fall to 3%. The remaining cash flows of an otherwise identical **noncallable** bond might be worth $11,500. But our callable bond does not offer that unrestricted stream: investors know we can redeem it. Its market price therefore reflects the call right and would generally be limited near the amount obtainable on an immediate call, allowing for accrued interest and notice terms. To refinance, suppose we issue $10,500 of new debt at par and 3%, and use those proceeds to redeem the old debt for $10,500. The new annual coupon is $315, compared with the old $700. The initial call premium and other costs also belong in the decision. The issuer's option is valuable because it limits what must be paid to retire the old contract; it was part of the bargain investors priced at issuance.

If your parents ever 'refinanced the house,' by the way, this is exactly what happened, in personal-finance form: they had an old mortgage owed to Bank A; interest rates fell; they took out a *new*, cheaper mortgage from Bank B; and they used the cash from Bank B to pay off Bank A. Old loan gone, new cheaper loan in its place.

[7] I'll confess a mildly cynical view here—don't hate the player, hate the game: the finance industry *loves* designing very complicated contracts. Partly that is to tailor them to genuine client needs; but partly it is so that the industry can charge higher fees, and make clients feel they are buying deep sophistication. In my view, much of the fine print doesn't change the fundamental economics very much. The complexity is often part of the sales strategy.

14.5.2 Repurchasing bonds

Even if the bond **contract does not** make it 'callable' at a **pre-specified price**, bonds are *traded* in free **financial markets**. So, companies can *repurchase* their own bonds, directly from sellers, or via brokers. In repurchasing a bond, the company essentially pays the **present value** (market price) today, in order to extinguish all of its future payments. So, this is, in essence, just **reversing** the bond **issuance**. Why would companies do this?[8] There could be any number of reasons. Corporate finance is about companies' needs for **cash today** vs. **cash in the future**. Perhaps the company issued the bond in the past, because it needed a cash infusion at that time. But then, perhaps its subsequent sales in the interim were better than expected; so, now it has extra cash, and wishes to pay off its debts early. Or, perhaps the company has a new CFO who has decided to change its **capital structure**. It could issue more Common Stock and use the cash proceeds from that sale to **pay off debts** thereby decreasing its **leverage**.

14.5.3 FAQ: Why not only issue callable bonds?

So, based off of what we learned above, companies can repurchase 'normal' (non-callable) bonds in the bond market, at the prevailing **market price.** However, if they issue 'callable' bonds, then they have the *right* but no obligation (that is, to use a core finance term, the ***option***), to repurchase it at a pre-specified **'call price.'** Thus, 'callable bonds' give the **issuing company** additional **options** or **optionality**—most typically, the ability to profitably **refinance** if interest rates decrease.

In that case, why don't firms *only* issue **callable** bonds? Go back to our **core finance theory**. For the corporation **issuing the bond**, the call provision is an **option** that gives it **potential upside**, if interest rates decline. But, the **flip side** is that the **issuing corporation's optionality is the bond buyer's risk**. For the bond buyer, that is the risk of being **deprived** of the interest payments, if market rates decrease in the future. So, if the **bond buyer faces more risk** from a callable bond, then, given our core finance theory, that means they will charge a **higher interest rate**. There's **no free lunch** in competitive markets—just **tradeoffs**.

This is the two-sided-deal question from the earlier box: if the issuer gains an option, how does that change what investors are willing to pay? The call feature belongs

[8] Warning: advanced material. If you understand our core finance theory and present-value relations very deeply, you may be able to understand that, for non-callable bonds, our company would not benefit from refinancing—e.g., repurchasing a bond issued at 7%, with the purchase financed by issuing a new bond at 3%—in quite the same way. Why? The reason is that, based on present-value identities, the decline in the market interest rate from 7% to 3% would increase the present value, and thus the market price, of our bond. So we'd have to pay much more than face value to repurchase it for the market price. And so it turns out that repurchasing a bond at *the market price* doesn't, by itself, make us any better off. There's no free lunch in a competitive marketplace. The reason why *refinancing* a callable bond does help us is that the 'call price' is fixed in advance. If this makes your head spin, don't worry. Finance courses spend weeks on it.

in the terms of the bargain; it is not a free benefit added afterward.

14.5.4 Accounting for repurchasing or calling a bond

In **either case**—whether a company 'redeems' a callable bond at a pre-specified price, or repurchases its bond in the bond market—the accounting **journal entry** is essentially the **same**. Why? Because in both cases, the company is paying out **Cash** and **extinguishing** all of the bond-related accounts. The same **accounts** are updated, so the **journal entries** are updated, regardless of the underlying bond's **type**. The journal entry will typically have **four rows**, but is essentially an extremely simple algorithm:

The rules:

- **Rule #1:** Credit **Cash** for the repurchase / call price
- **Rule #2: Extinguish** that bond from all of its accounts (*Bonds Payable* and the *Discount* or *Premium*)
- **Rule #3: Plug** the remainder as a *Gain* (if credit) or *Loss* (if debit)

That's really it. Let's show a concrete example, and then talk about the intuition. Let's pick up where we left off in the example of a bond issued at a premium, right after its first interest payment on January 1, 2022 (the journal entry we recorded above). Suppose that the company either calls or repurchases this bond for **$10,000** immediately after that. What do we do?

Following the rules, we **credit cash** for the $10k disbursement. Then we **extinguish** the bond-related accounts: *Bonds Payable* and, in this case, the *Premium on Bonds Payable*. *Bonds Payable* is on the books at $10k, and the *Premium* now has a balance of $354.60 (see the T-accounts above). Those are *both* credit accounts, so we **extinguish** them both with **debits**. This means that the required **plug** is a **credit** of $354.60—a **Gain**.

Bonds Payable (-L):	$10,000.00	
Premium on Bonds Payable (-L):	$354.60	
Cash (-A):		$10,000.00
Gain (+S.E.):		$354.60

Repurchasing or calling our bond-at-premium for $10k, Jan 2, 2022:

To be safe, I just wrote out the journal entry step by step, then added the Gain as a plug at the end. But you can also think through this **intuitively**, as being like the 'flip side' of the sale or disposal of an asset.

Intuition

The **Carrying Value** of this bond (a liability) was **$10,354.60**. The carrying value is essentially like the 'book value' of a bond. The company **paid** out **$10k** in cash in order to extinguish that liability from its books. That is, it *paid less* than 'book value' to get rid of the liability. So, it should make intuitive sense that that is a 'Gain' of $354.60. As you might be able to see, extinguishing a bond liability is beautifully **symmetric to** the Gains and Losses from selling (extinguishing) an asset, as covered in Chapter 11.

- If you sell an asset at more (vs. less) than book value, that's a gain (vs. loss).
- If you repurchase a bond liability at more (vs. less) than CV, that's a loss (vs. gain).

If you would rather *derive* the sign than memorize it, reason it through the accounting equation. A *Gain* increases Net Income, which increases Shareholders' Equity—and Equity is **net assets**, $A - L$. Suppose a bond's Carrying Value is $9,400 and we pay $9,500 to extinguish it. Then our liabilities fall by $9,400 (good), but our assets (Cash) fall by $9,500 (bad). Net assets fall by $100: a **Loss** of $100. Had we instead paid only $9,300, net assets would *rise* by $100: a $100 **Gain**. And, comfortingly, the journal-entry 'plug' will deliver exactly this answer automatically—losses are debits and gains are credits, so whatever plug balances the entry is guaranteed to come out with the right sign.

14.6 Installment Loans

Another way in which companies can finance themselves and their purchases is with **Installment Loan** contracts. In this case, instead of paying back small interest payments in the interim, and a large face-value payment on the maturity date, the debtor pays **the same amount each period**.

Time	**+1 Year**	**+2 Years**	**+3 Years**	**+4 Years**	**+5 Years**
Install Payment	$2,400	$2,400	$2,400	$2,400	$2,400

Contract we call 'installment payment'

You may recall that we discussed this at the very beginning of the *Introduction to Finance* chapter. We noted that, although nothing was explicitly called 'interest' in this contract, there was an *implicit* cost of financing. And so, despite the name, an installment contract is essentially the same as a loan / bond, just with a different **pattern** of payments. We should understand that now given our mastery of the **core identity of finance**.

Now, suppose that our company's **interest rate** is 8%, so we get $9582.50 in cash today, in exchange for signing the installment-payment contract above. That is:

$$PV\ of\ Installment = \frac{\$2.4k}{1.08} + \frac{\$2.4k}{1.08^2} + \frac{\$2.4k}{1.08^3} + \frac{\$2.4k}{1.08^4} + \frac{\$2.4k}{1.08^5} = \mathbf{\$9582.50}$$

14.6.1 Issuance date

How do we account for this? Let's start with the initial **contract date**. That's easy enough. Our company gets a **cash infusion** (a debit) equal to the installment loan amount, which is the present value, $9,582.50. And we balance that with a **simple single credit to *Notes Payable***. Easy enough:

Cash (+A):	$9,582.50	
Notes Payable (+L):		$9,582.50

Getting $9,582.50 Installment Loan, 1/1/2021

14.6.2 Subsequent installment payments and interest expenses

Now, the harder part: **How do we account for the subsequent installment payments?** Recall that, as we said in Chapter 13, there is a 'real' cost of this debt financing here, a real interest rate. So we do need to account for an *Interest Expense* here, even though nothing is called *Interest*.

It turns out that we use essentially the same algorithm we used for *Bonds Issued at a Premium*, with only some minor semantic differences. Note that you can **conceptualize this installment payment** as a **bond with a $2.4k coupon and zero face value**. The Interest Expense we would recognize each period would be the same. This will become clearer once we do an example. The **rules** for Installment Payments, with their slightly-revised semantics, are:

- **Rule #1:** As before, the issuance amount ($9,582.50 in this case) is the **initial Carrying Value**.
- **Rule #2:** As before, the *Interest Expense* recognized is the Carrying Value times the 'real' market interest rate upon issuance.
- **Rule #3:** But, in the case of an installment contract, we **plug the difference directly in *Notes Payable*** via a debit, rather than a contra-account.
- **Rule #4:** And, the updated **Carrying Value** is just the updated balance of *Notes Payable*, since we *directly* debited it for the principal reduction.

14.6.3 Installment Payment Examples

Let's make this concrete, and just apply these rules, before backing up. We already covered the issuance of this installment note on Jan. 1, 2021 above. So, next, let's cover the **first annual installment payment** and interest expense recognition, on Jan. 1, 2022. **Per the contract terms**, the **Cash** credit is $2,400. Per **Rule #2**, the *Interest Expense* we will recognize is the Carrying Value times the real market interest rate upon issuance: $\$9,582.50 \times .08 = \$766.60$. And per **Rule #3**, we directly plug the difference in *Notes Payable*. So, our journal entry will be:

Interest Expense (-S.E.):	$766.60	
Notes Payable (-L):	$1,633.4	
Cash (-A):		$2,400

First Installment Payment and Interest Expense, 1/1/2022

And, per **Rule #4,** the updated **Carrying Value** will be the new balance of *Notes Payable*, which will be: $9,582.50-$1,633.40 = $7,949.10. Or, showing this via the T-account:

Notes Payable

Debits	Credits
$1,633.40	$9,582.50
	$7,949.10

Notes Payable (after first annual installment payment)

So, then, the *next* Interest Expense, recognized on January 1, 2023, will be $\$7,949.10 \times .08 = \$635.93$. And, the journal entry on that date will be:

Interest Expense (-S.E.):	$635.93	
Notes Payable (-L):	$1,764.07	
Cash (-A):		$2,400

Second Installment Payment and Interest Expense, 1/1/2023

And, the new balance of *Notes Payable*, which is the **Carrying Value** in this case, will be: $7,949.10 -$1,764.07 = $6,185.03. Or, showing this via the T-account:

Notes Payable

Debits	Credits
$1,633.40	$9,582.50
$1,764.07	
	$6,185.03

Notes Payable (after second annual installment payment)

14.6.4 Intuition: Each payment represents interest and principal

Despite slightly different **mechanics**, the idea here is exactly the same as in the case of a *Bond Issued at a Premium*. Indeed, the intuition might be somewhat clearer in the case of an installment payment. In an **installment payment**, there is no large 'face value' payment at the end. So each and **every** payment clearly represents **economic interest**—the cost of debt financing—as well as **paying down the initial principal**. It is just like **paying off a mortgage** and thereby building **home equity**. This is why the *Interest Expense* is **smaller** than the **installment payment** in each year. Also note that the **Carrying Value** decreases over time, reflecting this. Because each Installment Payment represents *interest* as well as principal, the book value of the liability decreases over time.

This is exactly the same as the *Bond Issued at a Premium*, with only one difference: In that case, we had a separate account for *Bonds Payable* and the *Premium*, and only updated the latter account with each interest payment. For the installment contract, we use just one account, *Note Payable*, and update it *directly* with each payment. So, working through one should help you build intuition for the other.

14.6.5 Application: Mortgages, home equity, and the rent-vs.-buy decision

Why have I dwelt on installment contracts? Because the same calculation governs a familiar and important financial decision: **buying a home with a mortgage**.

Consider a standard, fully amortizing, fixed-rate mortgage. The contract specifies the interest rate and a schedule of monthly payments. The monthly payment for **principal and interest** is fixed. But the split between those two parts changes over time. The outstanding mortgage balance plays the role of the Carrying Value in our installment-loan example. Multiply it by the monthly interest rate to calculate that month's interest. Subtract that interest from the payment, and the remainder reduces the loan principal. Next month, interest is calculated on the new, lower balance. A payment collected by the lender may also include property taxes or insurance; those amounts are separate from principal and interest.

Now consider a piece of personal-finance folklore: **"If you're renting, you're just throwing money away."** Is that right? When you buy food, you consume it. You do not retain an asset afterward, but that does not mean you got nothing for your money. Rent buys a flow of **housing services**: a place to live for an agreed period. The useful question is not whether rent buys anything. It is which way of obtaining those housing services costs less in your circumstances.

Take a hypothetical house priced at $1,000,000. You make a 20% **down payment**, $200,000, and borrow the remaining $800,000. Assume a fixed annual interest rate of 6%, with monthly payments over 30 years. The monthly principal-and-interest payment is about $4,796.40. In the first month, the interest is $\$800{,}000 \times (0.06/12) = \$4{,}000$. The remaining $796.40 pays down principal. The next month begins with a balance of about $799,203.60, so the interest portion falls slightly, and the principal portion rises. The total payment stays the same.

Suppose a comparable home rents for $3,500 a month. Comparing $4,796.40 with $3,500 does not, by itself, settle the decision. Part of the mortgage payment is interest, a cost of financing. The rest reduces a liability. If the house's value is unchanged, that principal repayment increases your **home equity** by the same amount. It uses cash, but it is not an expense in the same sense as interest or rent.

To see that distinction over a longer period, suppose you have eventually reduced the mortgage balance from $800,000 to $700,000. If you sell the house for the same $1,000,000 purchase price, you would receive $300,000 after repaying the lender, before selling costs. That consists of your original $200,000 down payment plus the $100,000 of principal repaid along the way. Paying down principal did not make that money disappear; it reduced what you owed against the house.

The full rent-versus-buy comparison must also account for property taxes, insurance, maintenance, transaction costs, and the return you forgo by tying up the down payment and later principal repayments in the house. The house's value can rise or fall, too. Our accounting does not decide the choice for us. It gives us an essential first step: distinguish the **cash payment**, the **financing expense**, and the **reduction in the liability**.

14.7 Lease accounting:

The last topic we'll cover in this chapter is something that *sounds boring*, but is actually a surprisingly live, controversial, and *interesting issue*: lease accounting.

Let me set this up with a **thought experiment**. Suppose that a small **commercial airline** wants to **use a plane** that costs **$8m** to purchase, but it **doesn't have the cash** on hand. For this thought experiment, assume a ten-year useful life, no residual value, and otherwise comparable rights to use the plane. There are two different ways

in which the **airline** and the manufacturer could **structure a transaction.**

- **Option #1**: The manufacturer could finance the airline's purchase of the plane with an installment note—giving it the $8m plane in exchange for promised installment payments of $1m per year for each of the next 10 years. Then, the airline would get the plane today, but have to pay back $1m per year over the next 10 years. Here's a way of visualizing this contract, from our airline's perspective:

Time:	**At signing**	**Year 1**	**Year 2**	**...**	**Year 10**
Exchange:	+1 Airplane	-$1m	-$1m	-$1m	-$1m

Installment-note financed purchase of plane (airline's perspective)

- **Option #2**: An alternative is that the airline could lease/rent the plane from the manufacturer, agreeing to pay annual rental/lease payments of $1m per year over the next 10 years. Here's a way of visualizing this contract, from our airline's perspective:

Time:	**At signing**	**Year 1**	**Year 2**	**...**	**Year 10**
Exchange:	+1 Airplane	-$1m	-$1m	-$1m	-$1m

10-year Leasing agreement for plane (airline's perspective):

Do you see what this thought experiment is getting at? We *call* **Option #1** a **'purchase' financed by an installment note**, and we *call* **Option #2** a **lease**. But, the **real economics**—the actual exchanges of resources—are **exactly identical** in the two cases.

Airlines do, in fact, use both owned and leased aircraft. A lease can offer flexibility and change who bears residual-value risk, so real contracts need not be economically identical. The simplifying assumptions above isolate the recognition question.

This is a general challenge for accounting. Despite the identical *real economics* of the two examples above, until recently, the two were accounted for differently. Many companies realized that they could make themselves look less risky, by reducing their reported liabilities, by choosing to **legally structure** transactions as "leases," instead of debt-financed purchases.

To understand the implication more clearly, recall that one of the **primary metrics** that creditors use to assess **credit risk** is $\textit{Leverage} = \frac{A}{E} = \frac{L}{E} + 1$ (since $A = L + E$). If a company were to structure this transaction as **Option #1**, above—a purchase financed by a *Note*—that would increase both its Assets and its Liabilities by $8m. So, its Shareholders' Equity would be unchanged, and *Liabilities* would increase, increasing its reported **Leverage**. Under prior accounting standards, if the company were to,

instead, use **Option #2**, structuring it as a **lease**, then the company's reported **Liabilities** and **leverage** would be **unchanged**. Instead, the lease payments would have been recorded as *periodic expenses*.

Notice which ratios move under the old off-balance-sheet operating-lease treatment. At inception, recognizing the asset and liability increases leverage while leaving equity unchanged. Later, the expense pattern can also differ: depreciation plus interest on a financed purchase need not equal a level rental expense in each year. So ROA and ROE comparisons require attention to both the numerator and denominator. The broad concern remains: substantial contractual payments could sit outside the recognized liability total.[9]

In short, companies could **conceal** their **real, economic liabilities** and **riskiness** by choosing to **structure** transactions as **leases** rather than **loan-financed purchases**.

The aircraft example illustrates a broader measurement problem. Once a reported ratio affects lending terms or managerial rewards, people have a reason to change the ratio. Sometimes they improve the underlying business; sometimes they seek a different contractual form. To interpret the number, we need to ask which has happened.

The current U.S. lease standard, **ASC Topic 842**, addresses that recognition gap. A lessee generally records a **Right-of-Use Asset** and a **Lease Liability** for both **operating leases** and **finance leases**. A policy election can exempt qualifying short-term leases, generally those with a term of 12 months or less and no purchase option reasonably certain to be exercised.[10]

The distinction between an operating lease and a finance lease still matters, chiefly for the expense pattern and cash-flow presentation. Finance-lease classification depends on whether the arrangement effectively transfers the economic benefits of ownership. Indicators include a transfer of ownership, a purchase option reasonably certain to be exercised, a lease term covering a major part of the asset's remaining economic life, payments representing substantially all of the asset's fair value, or an asset so specialized that it is expected to have no alternative use to the lessor at the end. These are **classification criteria**, not a list of conditions that an operating lease must pass to appear on the Balance Sheet.

For a simple lease beginning now, with no advance payments, incentives, or initial direct costs, the initial entry is:

[9] If you want to see the ultimate enforcement mechanism in action, I once again refer you to the television genre of *Airplane Repo*: people literally repossessing aircraft from operators who can't pay their debts. Very on-the-nose for this section.

[10] See FASB's Leases project and ASU 2016-02. The older 75% and 90% thresholds were associated with the former capital-lease model; they are not conditions for recognizing an operating-lease liability under Topic 842.

Right-of-Use Asset (+A):	$(PV of unpaid lease payments)	
Lease Liability (+L):		$(PV of unpaid lease payments)

Initial recognition of a simple lease, 10/31/2021

The liability measures the present value of the required unpaid lease payments. The asset represents the right to use the underlying property; it does not mean that the lessee owns that property. In more complete examples, advance payments, incentives, and certain direct costs can make the asset's initial amount differ from the liability.[11]

14.7.1 The view from practice

A few postscripts to this story, from the world of practice. First, sophisticated analysts never really fell for the lease game. For years, one of the classic 'advanced' moves taught in financial-analysis courses and investment-banking interview guides was to *undo* it by hand: estimate the present value of a company's disclosed lease commitments, and add it back to its liabilities, to get a truer picture of its leverage. In effect, the new standard made GAAP do what good analysts were already doing on their own—and made a lot of interview-prep guides obsolete overnight.

Second, you can see the rule change with your own eyes in the data. When the new standard kicked in, leverage ratios computed from GAAP financial statements jumped, discontinuously. Companies did not suddenly binge on debt that year; the accounting simply began *recognizing* obligations that had been there all along. (I stumbled over this spike myself in my own research data, having briefly forgotten about the rule change—a good reminder that when a time series of accounting numbers jumps, the first suspect should often be an accounting rule, not the underlying economics.)

And third, a more recent, less catastrophic, but still instructive example of the underlying game: **WeWork**. WeWork's business model is to sign long-term leases on commercial real estate, renovate and furnish the space, and then sublease it—by the desk, by the office, month to month—to startups, freelancers, and established companies. Where does the demand for that service come from? Partly from genuine value: the space is flexible, already built out, and pleasant to work in. But part of the demand was, arguably, our lease game at one remove. A company that signs its *own* long-term lease (let alone buys a building) now has to show a large lease liability on its balance sheet. A company that buys flexible, short-term space from WeWork does not—the long-term lease obligations sit on *WeWork's* balance sheet instead. Some customers were paying WeWork not just for desks and coffee, but for tidier-looking financial statements.[12]

[11] In companies' actual financial statements, you'll often see this asset labeled a *'Right-of-Use Asset'*—which is an admirably honest name for what you're really getting.

[12] For entertaining contemporaneous coverage, see Chris Bryant's aptly titled "WeWork's Balance

Sheet Looks Ugly So Yours Doesn't," Bloomberg, September 2019; and Matt Levine's Bloomberg newsletter from the same week. Epilogue: WeWork itself filed for bankruptcy in 2023 and restructured—the long-term lease obligations it had absorbed from everyone else turned out to be quite real.

15

Statement of Shareholders' Equity

This chapter is about the Statement of Shareholders' Equity (SSE). The unique focus of this statement is tracking a company's **transactions** and **standing** with its **own shareholders**. That's why we can cover this entire financial statement in just one chapter. There are simply far **fewer** and less varied transactions that companies have with their own shareholders than with their business partners, suppliers and customers. So, this financial statement will typically take up the least space in most companies' financial reports—and, frankly, it is probably the one that investors spend the *least* time analyzing. Every quarter, traders and financial journalists sit glued to their screens awaiting Apple's earnings announcement—"What was Apple's earnings per share this quarter?" Nobody has ever waited breathlessly to see Apple's Statement of Shareholders' Equity. That's just not how the world works. But, this brief chapter is still important, for at least three reasons.

- **First**, you could argue that a company's transactions with its own shareholders are what the corporation is all about. One view of a corporation is that it is a legal entity which investors create and found by putting in their own money, in the hope and expectation of making a return on their investment. In that view, the corporation's operations are just a mechanism for transforming the shareholders' *Common Stock* investment into future cash distributions (e.g. *Dividends*) to shareholders. And the SSE fully illustrates the beginning and end of that process.

- **Second**, there are a couple of data points that are important for investors that we

can **only** get from this statement.

- **Third**, there are a number of current topics of **public conversation and controversy** relating to corporations' relationships with their own shareholders. The technical knowledge we'll cover in this chapter will allow you to understand those debates.

15.1 Where we stand

15.1.1 What do we know about shareholders' equity so far?

To date, we've learned that Shareholders' Equity (**E**) represents the shareholders' claims on and financing contributions to the corporation, and consists of two components:

1. **Paid-in capital**: amounts that shareholders have literally and directly paid in to the corporation, via the purchase of ***Common Stock*** from the corporation.

2. **Earned capital**: amounts that the shareholders have *earned* (as net income), but *not* paid out to themselves as dividends, represented by ***Retained Earnings***.

In Chapter 1, we expressed that with this **core identity**:

$$Shareholders' \ Equity = Common \ Stock + Retained \ Earnings \tag{15.1}$$

The key journal entry for **paid-in capital**, the issuance of *Common Stock*, was Transaction #1:

Cash (+A):	$15,000	
Common Stock (+S.E.):		$15,000

Issuance of Common Stock, Jan. 2, 2020

And we know the **core identity** for updating *Retained Earnings* between periods:

$$\Delta Retained \ Earnings = Net \ Income - Dividends \tag{15.2}$$

Chapter 5 showed how closing moves the period's revenues, expenses, and dividends into Retained Earnings. Here we will use the same identity to explain the change in equity, alongside transactions directly with shareholders.

15.1.2 The Statement Itself

One thing we haven't discussed much to date is how the **Statement of Shareholders' Equity** is **presented** in the financial statements. It's fairly self-explanatory, but also a bit different than the Income Statement and Balance Sheet, which can make it a bit jarring on a first encounter. The I/S and B/S list account totals *vertically*. But the Statement of Shareholders' Equity instead lists **individual transactions** with shareholders, as well as Net Income earned and Dividends paid out, on the y-axis, and shows the affected Shareholders' Equity accounts *horizontally* (that is, along the x-axis). For example, based on the Adjusted Trial Balance at the Chapter 5 (after the 10 external transactions and 4 adjusting entries), the Statement of Shareholders' Equity would look like this:

	Common Stock	Retained Earnings	Total Shareholder's Equity
Balance, Jan 1, 2020	$0	$0	$0
Issue Common Stock (Tr. #1)	15,000		15,000
Total Dividends Paid		(100)	(100)
Net Income		2,300	2,300
Balance, Dec 31, 2020	**$15,000**	**$2,200**	**$17,200**

Statement of Shareholder's Equity

This is still the **big picture**, the key concepts. But, as usual, that **high-level** overview of Shareholders' Equity left out some details and complications.

15.1.3 What are we missing?

Let's first briefly cover **what** we're missing, before filling in those gaps and details (in subsequent sections).

1. **Legal and institutional background on shares**: So far, we've learned that shareholders invest cash in the company, and receive, in return, *Common Stock* which gives them *ownership*, specifically the right to vote their shares and receive dividends. But we also know that corporations **can issue more shares** at various times, to raise additional financing. So, **where do those shares come from**, what are they, why do companies do that, and are there any restrictions on a company's ability to do so? We need to learn some new terminology and legal details to understand this.

2. **Par value:** In reality, the accounting for **Transaction #1** (the issuance of common stock) is slightly **more complicated** than the above. For historical reasons, shares of stock have something called **'par value'**, which has **no meaningful economic content** today, but which is still accounted for, due to historical and institutional reasons.

3. **Repurchases**: To date, the way in which our corporation has returned cash to shareholders has always been through dividends. But in reality, corporations can **also return cash to their shareholders** via an arguably more direct route: **repurchasing shares**. In fact, today, this is the **preferred way** for companies to get cash to their shareholders, and also one that has become a source of political and social **controversy**. We'll discuss why, and also the accounting for repurchases.

4. **Preferred Stock**: To date, we've always talked about *"common"* stock. But there is another category of stock, called 'preferred stock,' that is like a **hybrid of equity and debt**. We'll define what preferred stock is, discuss why it's interesting and controversial, and cover the accounting for it.

5. **Stock Splits**: It is very common for companies to do stock *"splits"*—that is, declare that every 1 share is now 2 shares. After a split, every shareholder's *percentage* ownership is unchanged, and each individual share is essentially worth half as much. So, why do companies do these, and what are the implications?

Perhaps the best motivation for all of this detail is simply to pull up a real company's balance sheet. Try it with essentially any large public company—Nvidia, Pfizer, Ross Stores. The Shareholders' Equity section in the wild looks considerably more complicated than the toy "Common Stock + Retained Earnings" world we've worked in so far. You'll see a line for *Preferred Stock*. You'll see *Common Stock*, but with a dollar amount that is bizarrely tiny relative to the size of the company, accompanied by a cryptic note about 'par value.' Then a much larger line called *Additional Paid-in Capital*. And, further down, *Treasury Stock*, sitting there in parentheses, *reducing* the total equity. By the end of this chapter, you will know exactly what every one of those lines means.

There are some issues in accounting for shareholders' equity that became very torturous and arguably illogical, often due to the arcana of **par value** (e.g., certain aspects of stock splits and 'stock dividends'). And those items are not very important for broader understanding. I encourage my introductory and business-major students not to go beyond the topics I cover here.

15.2 Legal and institutional details on shares

If you and I **both have 50 shares** of a corporation, and are the only outstanding shareholders, we usually think that means that we both have **50% ownership**. But we also know that the corporation can **issue more** shares and **repurchase** shares from shareholders. So, how do we make sense of all of this, and understand who has what percentage ownership at any time? This suggests we need some **legal background and terminology**.

1. **'Authorized Stock'**: Corporations have **articles of incorporation**, which are sort of like their corporate **constitution**. A corporation's articles will explicitly state the **total number of shares** that the corporation can issue. We call these the '**authorized shares**.' So, ultimately, all shares are a subcategory of the authorized shares.

2. **'Issued Stock'**: These are the shares that companies have issued and have not subsequently retired (that is, sold to external investors), out of their authorized stock.

3. **'Treasury Stock'**: These are all the shares which the company *had issued*, but which it has **since repurchased** from shareholders. (I discuss *why* companies repurchase shares, below.) If the company holds the repurchased share rather than retiring it, you can think of it as being 'held by the corporation' (as in its treasury). But, since repurchased shares, by definition, no longer belong to outside investors, they **cannot vote and cannot receive dividends**. (It wouldn't make much sense for the corporate legal entity to receive dividends from itself, or vote).

   Many students ask, how does treasury stock—that is, shares, the company issued and then repurchased—*differ* from shares that *were authorized* but never issued? That's a great question. Philosophically, it's not clear how those shares really *are* different in their essence—both are 'held' by the corporation, and therefore do not receive dividends or votes, and can be issued in the future. But ***creating* this distinction** between the two will make some of our accounting more tractable (see below).

4. **'Outstanding Stock'**: These are the shares that have been issued and are *currently* held by outside shareholders—that is, those that are not currently in Treasury Stock. So, these are the only shares that can actually be voted and receive dividends. This is where the real action is. By arithmetic identity, the number of outstanding shares must equal the number of issued shares minus the treasury stock.

So, to make this concrete, suppose the following:

- Our company's articles authorize 10 shares, numbered #1-10.
- We first sell Common Stock, issuing shares #1-6.
- And then later we repurchase shares #4-6.

Then, after those three stock transactions:

- The **authorized** shares are all shares, #1-10.
- The **issued** shares are shares #1-6.
- The **treasury** stock is shares #4-6.
- And so the **outstanding** shares are #1-3.
- (And shares #7-10, which were authorized but never sold to anyone, we call the **'unissued'** shares.)

Now let's add one more move, to cement the trickiest of these categories. Suppose that, after all of the above, our company **re-sells** shares #4-5 out of its Treasury Stock. (We'll cover the *accounting* for such re-sales later in this chapter.) What are the categories now? The authorized shares are still #1-10—that never changes without amending the articles. The **issued** shares are *still* #1-6. This is the point that trips people up: 'issued' means the share has *left the shelf at some point*, not that it is currently in investors' hands. Repurchasing a share does not un-issue it, in this terminology. The **treasury** stock is now just share #6 (repurchased, and not yet re-sold). And so the **outstanding** shares are #1-5. At all times, per the identity above: outstanding shares = issued shares − treasury shares.

Now, here's a question you might be chewing on: what *are* those unissued shares #7-10, really? Do they have value? Do they show up on the balance sheet as some kind of asset? Here's the short answer: No. Earlier, we said that repurchased shares can be thought of as 'held by the corporation,' and you will sometimes hear the same said of unissued shares. But I find that phrasing unsatisfying here. Unissued shares are better thought of as pure **capacity**: a legal permission, written into the articles, to issue up to a certain number of shares in the future. Nobody has paid the company anything for them, they collect no dividends, they carry no votes, and they have no market value as separate objects. They are potential, not property. So there is nothing to record.

Discussion question: Now, let's go back to our initial question. Suppose that you currently own **40 shares**, there are a total of **100 outstanding shares**, and a total of **200 authorized shares**. Then, what is your percentage ownership—40% or 20%?

That would usually be considered **40%** ownership. If a vote or dividend distribution were held today, you would get 40% of either. That is, conventionally, when we talk

about shareholders' percentage ownership, we mean of the **outstanding** stock. But, as always, the real world is **fuzzy**. Suppose that the **other 60** outstanding shares are all owned by **your enemies**. Then, if your enemies ally against you, they could use their **control**—from their 60% majority ownership—to get the corporation to issue the other **100 un-issued shares** to **their friends**. And, presumably, the friends of your enemies are also your enemies. After that dilution, you would then have 20% ownership. So, the reality of *control* is more complicated and *contingent*—it depends on who the other shareholders are, and what constraints they are working with.

15.2.1 Going deeper: Takeover battles and poison pills

If the distinctions between *authorized*, *issued*, and *outstanding* shares still feel like legal trivia, let me show you where they become the decisive weapon in billion-dollar corporate battles: **hostile takeovers**.

Recall the simple model of an acquisition from Chapter 12: an acquirer, Company A, buys a target, Company T, by paying cash to Company T's shareholders in exchange for their shares. Now suppose the *management and board* of Company T do not want the company to be acquired. Can the acquirer simply go around them? In principle, yes: if Company T's shares trade for $50 on the stock exchange, Company A can offer $60 per share directly to the shareholders, and keep buying until it holds more than 50% of the **outstanding** shares—which can confer voting control, subject to the company's voting structure, governance provisions, and applicable law. That is a **hostile takeover**, and it is roughly how takeovers worked through the 1980s—the heyday of the corporate 'raiders,' when acquirers really could buy companies out from under reluctant boards.

Takeover defenses can make that route much harder. During that era, some very clever lawyers, working for managers who did not want their companies taken over, invented a defense that exploits exactly the distinction we've been drawing: the **'poison pill'** (or, by its polite name, a 'shareholder rights plan'). Here is a deliberately simplified illustration of the dilution effect. Actual plans generally use rights to acquire shares at a discount, with detailed triggering and exercise provisions. Suppose our target company has 1,000 authorized shares, of which 100 are outstanding. A would-be acquirer quietly buys up 15 of those 100 shares—a 15% stake—and is plainly still buying. The board responds by adopting a plan that says: if any single shareholder's stake reaches a threshold (say, 20%), then every *other* shareholder's shares automatically convert into, say, 10 shares each—but the triggering shareholder's shares do *not*.[a]

Run the arithmetic. Suppose the acquirer pushes on and buys up to 20 of the

100 shares—20%, reaching the trigger. The other 80 shares become 800, so the acquirer now holds 20 out of 820 shares: about 2.4%. The acquirer paid prices reflecting a 20% stake, and is left holding a sliver. And where did those 720 new shares come from? From the pool of **authorized but unissued** shares. That pool is the ammunition. (Which also tells you the limit of the maneuver: a board can only keep doing this until it exhausts its authorized shares, at which point it would need a charter amendment to authorize more.)

Note that the pill almost never actually 'goes off.' It works like any credible threat: because crossing the threshold would be financially catastrophic, no sane acquirer crosses it. Instead, they stop just short of the line and negotiate with the board. This is precisely what happened in the most famous recent example: when Elon Musk moved to acquire Twitter in 2022, he accumulated a stake of just over 9%; Twitter's board promptly adopted a poison pill, with a 15% trigger; and Musk negotiated a deal that the board approved. The board amended the plan in connection with the merger agreement. Nordstrom's board adopted a pill that same year, when an activist investor began building a large stake. So this is a live tool of corporate governance, not a historical curiosity.

Is all this legitimate? Boards frame these plans as protecting small shareholders from being steamrolled by a powerful acquirer. But notice what the mechanism literally does: it lets the board pick and choose *among its own shareholders*—massively diluting one of them while enriching the others, and, not incidentally, protecting the incumbent management from being replaced. If you ask corporate-law and business-school professors, critics argue that poison pills are problematic for exactly this reason: they can entrench managers against owners who are willing to sell the company at a premium. But whatever your view, you now understand the machinery—and you can see why the lawyerly distinction between *outstanding* shares (who controls the company today) and *authorized but unissued* shares (the board's reserve ammunition) is anything but trivia.

[a] Real-world rights plans accomplish this through more convoluted means—rights to purchase newly issued shares at a deep discount—but the economic effect is what I've described. As usual, I simplify the legal machinery to expose the logic.

15.3 Par value

15.3.1 Background and meaning

The **'Par Value'** of shares is a weird, and somewhat frustrating, **historical artifact**. I call it the **'appendix of the accounting system,'** because it had a role in our evolutionary history, but **now** causes only **irritation** and inflammation.

Par value comes from the legal-capital rules governing share issuance. In jurisdic-

tions such as Delaware, shares with a stated par value generally cannot be issued for less than that amount. The amount classified as legal capital can also matter to distributions. Those rules do not make par value a measure of market value, nor a floor under the price at which investors trade the shares. A company can specify a very small par amount, which is why the accounting number can seem so detached from the economic value of the business.

For the ordinary transactions in this introductory chapter, par value is mostly a bookkeeping detail. A par amount of $0.01 per share does not mean that the share is worth one cent: investors may pay much more, as our example below shows. It still has a legal role, so it is better to say that it is **not a valuation measure** than that it has no significance at all.

Some companies issue no-par stock; others specify a small par amount. For the entries in this chapter, the important distinction is how the proceeds are divided between Common Stock and Additional Paid-in Capital. Neither the par amount alone nor the Common Stock balance tells us the market value of the company.

15.3.2 Accounting for par value

And, **accounting is sticky, too**. It turns out that we **elided** a bit when we first covered **Transaction #1**, when we debited *Cash* and credited *Common Stock*. In reality, that is **only how we do it for companies that issue 'no par stock.'**

In **reality**, for companies that issue stock with a stated **'par value,'** we only credit *Common Stock* for the *par value* of the shares, and plug the **remainder** to a new account called **Additional Paid-in Capital (APIC)**. This account is a Shareholders' Equity account, and, unsurprisingly, a part of paid-in capital. But it's tracked and reported separately from the *Common Stock* account.

Example: If our company sells 1,000 shares of common stock with a par value of $.01 per share, for an actual market/sale price of $30 per share, we would account for it like so:

Cash (+A):	$30,000	
Common Stock (+S.E.):		$10
Additional Paid-In Capital (+S.E.):		$29,990

Issuing 1k shares of $.01 par value stock for $30 per share, Jan 2, 2021

Since the corporation sells the 1,000 shares for $30 per share, it receives $30,000 in Cash (+A). The total par value of the shares issued gets credited to Common Stock

($.01/share x 1000 shares); and the **remainder is plugged to APIC.** Note that here, we refer to APIC as a **'plug,'** but, for most modern share issuances, the credit to APIC will **actually be much larger** than the credit to the Common Stock account, since modern corporations' stated par value is typically so low, for the reasons discussed above.

You can verify all of this on essentially any real balance sheet. Pull up Ross Stores' annual report, for example, and look at the Shareholders' Equity section. The first line reads "Common stock, par value $.01 per share," and—as of its report for the fiscal year ended January 31, 2026—shows a balance of about $3.2 million: a comically small number for a company of that size. Divide by the one-cent par value, and you recover roughly 322 million shares issued. And immediately below it sits *Additional Paid-in Capital*, at over $2.2 billion. The 'plug' dwarfs the account it plugs. Once you know the story of par value, these otherwise-baffling lines read naturally. The figures are from Ross Stores' Form 10-K for the year ended January 31, 2026.

There are two reasons why it's worth learning this detail:

- **First**, while 'par value' is not a market-value measure, the **distinction between paid-in capital and earned capital** is important, conceptually, and for various practical applications (there are some trading strategies that depend on decomposing shareholders' equity into those two parts). And, so, it's important to know and internalize that, unless a company issues no-par stock, its paid-in capital will not, in fact, be fully tracked in the *Common Stock* account. A significant fraction of the paid-in capital will be tracked in Additional Paid-in Capital. Modifying our core identity from Chapter 1, we now have:

$$Shareholders' \ Equity = Paid \ in \ Capital + Earned \ Capital \tag{15.3}$$

$$Paid \ in \ Capital = Common \ Stock + Additional \ Paid \ in \ Capital \tag{15.4}$$

$$Earned \ Capital = Retained \ Earnings \tag{15.5}$$

- **Second**, we do get *some* **new information** from this. You may know that stock market analysts and traders are very focused on **'earnings per share' (EPS)** and other **per-share** numbers, because they are ultimately trying to calculate the value of the individual shares that they trade in the stock market. In all of the accounting we've seen to date, we've only talked about the total accounting numbers for the corporation as a whole. For **basic EPS**, we divide earnings available to common shareholders by the **weighted-average common shares outstanding during the period**. Preferred dividends can affect the numerator; diluted EPS

has additional rules. How do we get this number? Well, if we know the *par value* of the shares, we can now use the Common Stock account to calculate the total number of shares **issued**. (In the example above, it would be \$10/(\$.01/share) = 1000 shares.) Now, **outstanding shares** and **issued shares** are **not the same thing**. So, once the company repurchases shares into its Treasury Stock, the math for calculating outstanding shares won't be so simple anymore. So, calculating EPS is not quite *that* simple, but *par value* gives us *some information* that is relevant toward that.

15.4 Repurchasing shares

15.4.1 Background on repurchases

Companies can and **increasingly** do **repurchase shares** from their shareholders. Why do they do this? This is one way of **getting cash back to their investors**, which, from the investors' / owners' perspective, is largely why they founded and/or invested in the firm in the first place. So, in that respect, they are **just like dividends**—moving cash over a line, from the company's bank account into the shareholders' bank accounts. So, the next question is, why would corporations return cash to shareholders via **repurchases *rather than* dividends**?

There is a **famous theorem in finance** that implies that if a corporation has a choice between returning \$X in cash to shareholders via repurchases vs. returning \$X in cash via dividends, the two have **identical effects on overall shareholder wealth**.[1] How to make sense of this? Here's an intuition:

Assume a company is worth \$1,000, including \$100 of cash it will distribute. There are 100 shares, worth \$10 each, and you own one. Set aside taxes, transaction costs, and any new information conveyed by the payout.

- **Dividend:** The company pays \$1 on each share, distributing \$100 in total. The business is then worth \$900, so each of the 100 shares is worth \$9. You hold a \$9 share plus \$1 in cash: total wealth of \$10.

- **Repurchase:** The company instead spends \$100 buying ten shares at \$10 each from other holders. It is again worth \$900, now divided among 90 outstanding shares. Each remaining share is worth \$10. If you do not sell, your share remains worth \$10. Your ownership percentage rose, but the cash leaving the company reduced the value being divided.

[1] This is one of the celebrated 'irrelevance' results of Franco Modigliani and Merton Miller, both Nobel laureates, whose names you will encounter constantly if you go on to study finance.

Thus, under these assumptions, neither payout creates extra wealth simply by changing its form. A continuing shareholder can sell a fraction of a share to obtain cash; a dividend recipient can reinvest. Taxes, information, and trading costs can make the real-world choices differ.[2]

A dividend is paid across the eligible shares; a repurchase pays the shareholders who sell. That difference matters to the people deciding whether to take cash now or continue holding shares. The simplified comparison above is a way to examine the payout choice, not a claim that every shareholder faces identical circumstances.

So, from the perspective of high-level finance theory, dividends and repurchases are really just the **same thing**—cash flows from the corporate entity, to its owners' bank accounts. So why are companies increasingly choosing repurchases? First, repurchases are often effectively **tax-advantaged** (for the shareholders themselves). The full tax treatment is a course unto itself, but the heart of the advantage is a matter of **timing**. When a company pays a dividend, that cash lands in your brokerage account whether you like it or not—and for a taxable investor, it generally creates a tax consequence in that year; you cannot politely decline to receive it. When a company instead repurchases shares, continuing shareholders generally receive no cash at that time and need not realize an existing gain merely because other shareholders sell. Any later capital gain is generally taxed when they sell and realize it. So *you* control the timing: you can let the gains compound untaxed for years, and realize them in a year when your tax rate is low (say, in retirement). Even when the statutory tax *rates* on dividends and capital gains are identical—as they now are for many investors—the option to defer is worth real money.

An interesting benefit for the **firms** is that repurchases effectively give them more **flexibility**, for psychological/sociological reasons. Historically, the stock market has been somewhat **'fixated'** on stocks' **dividend** levels. If a company has been paying a dividend of $1 per share for many years, and then the company **suddenly cuts** that dividend, the markets interpret that as a **bad signal** and sell off steeply (or so it is feared). For psychological/sociological reasons that are beyond the scope of this book, stock markets and traders **don't seem to get so fixated on total annual repurchases** in the same way. So, if a company has excess cash on hand this year, it may be more comfortable **disbursing** that cash **via repurchases**—that way, it doesn't feel **locked in** to paying out that same amount in **future** years as well.

Repurchases, you may know, have become a source of **political controversy** in recent years, with many politicians fearing or arguing that corporations are doing wrong by (some say) 'manipulating' their stock prices through repurchases instead of rein-

[2] In reality, there are lots of complications here about expectations, surprises, how we are defining and measuring returns, etc., which would be a great topic for a **PhD** seminar. For now, I am just trying to convey the intuition for the purposes of this introductory text. So, I apologize to my PhD-endowed readers for the technical elisions.

vesting that cash in their growth, their employees, etc. (Interestingly, this is not a simple left-versus-right issue—you can find politicians across the spectrum proposing to tax or restrict buybacks.) There are a number of tricky issues, and we don't do politics in this book. But, as a **technical matter**, repurchases and dividends are really the same thing, two forms of **shareholder payouts**, just structured differently. That common payout effect is important, although differences in taxes, timing, information, and who sells can also matter to policy arguments. My opinion is that there are **specific scenarios** in which we would want certain companies and industries to **reduce shareholder payouts as a whole**—e.g., to reduce *leverage* in the financial sector, per the discussion in Chapter 6. (Note that reducing payouts increases the accumulation of earned capital.) That said, because dividends and repurchases are identical in that respect, it is puzzling why some public figures are specifically focused on *repurchases per se* (and not equally on dividends)—and I tend to view that as a signal that the speaker has not thought very clearly about their policy proposal.

15.4.2 Accounting for repurchases

How do we account for repurchases? *Intuitively*, a repurchase is **'reversing' the prior issuance** of stock. So, it's **tempting** to just **'extinguish'** the Common Stock and APIC credits from that initial issuance. But, that **can't quite work**. Why not? Because the company can—and usually will—repurchase that stock at a different price than what it initially received when it issued it. So, if we just 'extinguished' the initial credits, that wouldn't balance with our cash repurchase cost.

For shares held in treasury under the **cost method used in this chapter**, the rule is: We debit *Treasury Stock* and credit cash for the full amount of the *repurchase*. So, if a company repurchases 500 shares for $20 each, the journal entry would be:

Treasury Stock (+contra-S.E.):	$10,000	
Cash (-A)		$10,000

Repurchasing 500 shares for $20 each, August 1, 2021

Treasury Stock is a new *account*. It is a contra-Shareholders' Equity account (the first one we have seen). This should make intuitive sense. By repurchasing shares from investors, paying out cash to them, the company is reducing the amount of financing contributed by shareholders. As a contra-Shareholders' Equity account, it has a debit balance, and it decrements the total amount of Shareholders' Equity. In the Shareholders' Equity section of the balance sheet, it is reported as a negative number—indicated with a minus sign or parentheses—reducing the total S.E. One small contrast with the

contra-accounts you've seen before: *Accumulated Depreciation* is 'contra' to one specific account (*Equipment*), whereas *Treasury Stock* is not paired against *Common Stock* or *APIC* in particular. It simply stands on its own, as a kind of 'anti-equity.'

There are three things worth noting about this accounting rule:

- **First,** the fact that we *previously issued* the stock for $30 does not matter—the accounting for the repurchase would be the **same** whatever the **initial issuance price** was. This is counterintuitive and can trip students up.
- **Second**, the **par value**, which had already been credited to *Common Stock*, also does not factor in here. We do not 'reverse' the *Common Stock* credit from previously-issued, now-repurchased stock *at all*. This is counterintuitive. It also means that, after a repurchase, the *Common Stock* account will not *directly* tell us how many shares are *outstanding*—only the total number that have been *issued*. There is a flip side, too: the *Treasury Stock* account accumulates the *dollars* spent on repurchases, not a count of shares. Without knowing the repurchase prices, you cannot convert its balance into a number of treasury shares. (Tuck away both of these facts—we will put them to work in a puzzle shortly.)
- **Third**, this means that a company's *repurchases* over a reporting period will be tracked in changes in this account, *Treasury Stock*. So, that's where you would start your research if you wanted to know about a company's activities in this area of controversy. (And, recalling the transaction categories from Chapter 1: a repurchase is a *financing* cash outflow—a transaction between the company and its own investors.)

15.4.3 Going deeper: Which part of equity did the repurchase reduce?

There are two tempting ways to describe Treasury Stock. A repurchase reverses the direction of cash in a share issuance, suggesting a reduction of paid-in capital. But a repurchase also distributes cash to owners, as a dividend does, suggesting a connection to earned capital.

Under the cost-method entries we use here, Treasury Stock is shown separately as a reduction of total equity. We can record that effect without forcing the transaction into either of the two simple categories from Chapter 1.

What does that tell us about our original model? It was useful for getting started, but additional transactions require additional accounts. Keep the distinction between paid-in and earned capital, while recognizing what the separate Treasury

Stock account contributes to the fuller presentation.

15.5 Re-selling repurchased stock

In the previous section, we learned that companies can return cash to their shareholders via repurchases, and those repurchased shares are then categorized as part of its *Treasury Stock*. Companies can also **reverse things again, and re-sell those shares**. Why would a company do this? For the **same reason it issued shares originally**—to get cash financing for its growth, investment, etc.

If a company re-sells treasury shares, the **accounting is slightly different** from the initial issuance. In this case, the accounting depends upon *which particular shares* the company is re-selling, and the **price it had repurchased them for** previously. (Obviously, for concrete problems, this information will be provided.)

Before I state the rule, notice how far we can get from first principles, with no new rules at all. As always, don't overthink it—start with *Cash*: the debit to Cash is just shares sold times sale price. And this transaction creates no new liability. So, for the accounting equation to hold, the credits must all land in Shareholders' Equity, and must total exactly the cash received. The only genuine question is *which* equity accounts get those credits, and in what amounts. That is all the rule below decides.

The rule for accounting for the **re-sale** of Treasury Stock is this: (i.) debit Cash for the proceeds, (ii.) credit *Treasury Stock* for the shares at their **prior repurchase price** (that is, *reverse* those shares' debit to *Treasury Stock*), and (iii.) credit any excess proceeds over cost to *APIC from Treasury Stock*. If proceeds are below cost, debit that APIC balance only to the extent available from prior treasury-stock transactions; debit any remaining shortfall to *Retained Earnings*. Neither case creates a gain or loss in Net Income.

15.5.1 Example

Let's continue with the example we used above for the initial repurchase of 500 shares for $20 each. So, now, our company has 500 shares in Treasury Stock. If we were to re-sell 100 of those shares for $30 each, the journal entry would be:

Cash (+A)	$3,000	
Treasury Stock (-contra-S.E.)		$2,000
Additional Paid-in Capital (+S.E.)		$1,000

Re-selling 100 Treasury Shares (purchased for $20) for $30, 12/17/2021

And note that if we **resold** the shares for **less** than we had initially paid for them, the **plug** to *APIC* would have to be a **debit** instead. Test yourself: Suppose those 100 Treasury Shares were instead sold for $10 in this transaction, and assume at least $1,000 is available in APIC from prior treasury-stock transactions. How would you account for it?

Here's the answer. The Cash debit is $1,000 (100 shares $\times$ $10). The credit to *Treasury Stock* is *still* $2,000—this leg is fixed by the **repurchase** price of the shares ($20 each), and doesn't care what we re-sell them for. So the entry can only balance with a $1,000 **debit** to *APIC*:

Cash (+A)	$1,000	
Additional Paid-in Capital (-S.E.)	$1,000	
Treasury Stock (-contra-S.E.)		$2,000

Re-selling 100 Treasury Shares (purchased for $20) for $10, 12/17/2021

And what does a *debit* to APIC mean? APIC is a normal, credit-balance equity account, so a debit shrinks it: we are reducing the recorded paid-in capital. This gives you a handy self-check for any treasury re-sale: re-sell *above* the repurchase price, and the APIC plug is a credit; re-sell *below* it, and the shortfall is a debit to available treasury-stock APIC, then to Retained Earnings if needed. Either way, the *Treasury Stock* leg is the same.

So, that's the rule. There are two important discussion points here.

15.5.2 Why APIC and not a 'gain'/'loss'?

First, note how we *do not* account for this transaction. The company *had* repurchased those shares for $20, and now *re-sells* them for $30. So, many students are *tempted* to think this should be a 'Gain.' If you had purchased an **asset** like *Equipment* for $2k, and re-sold it for $3k later in the day, it **would be**. Further, a 'Gain' would be a credit satisfying the 'plug' we needed in the transaction above, and balancing our accounting. So, **why do we put it in APIC** instead?

Accounting **standard-setters** have declared that companies **cannot book gains by trading in their own stocks**. *Gains* contribute to net income, the bottom line measure of how much value the company has **generated for its shareholders** via its operations. Company **management** has lots of **private information** about next quarter's expected earnings, etc., which could make it **easy** for them to **'buy low and sell high'** from their own shareholders. (Remember, too, that management holds the levers on the *timing* of information: it decides when to issue the press release, when to update the market

on how the drug trial is going, when to pre-announce good or bad news.)

But if companies are **buying low and selling high to their shareholders**, the **shareholders** who participate in those transactions are actually **getting hurt**. In any trade, one side's 'buying low' is the other side's 'selling low': if the company buys low from, and sells high to, its own shareholders, then those shareholders are buying high and selling low—the exact opposite of the first mantra of finance. So, if we allowed companies to book net income on those trades, that could have very problematic implications. Companies that were desperate to meet or beat certain earnings targets would then be tempted to **manipulate information** to exploit their own shareholders—driving down their stock price before buying it, and driving up their stock price before selling it.

And yet, the company has received the extra cash financing, from shareholders, and so that amount must go into Shareholders' Equity, somehow, in order for everything to balance. APIC fits the bill. The **bigger reason** it's valuable to **learn** about this is that there are **in fact** a variety of **other transactions**, more advanced than we will cover in this book, that have a similar logic. You will likely see some of these in your future careers. So, this transaction is really about this **general idea**: Transactions that *must* end up on the Balance Sheet, but where we must **'bypass' Net Income**.[3]

15.5.3 Why separate accounts for these three 'reversals'?

At a high level, all the transactions above did was: (i.) issue common stock, (ii.) reverse that issuance by repurchasing the common stock, and (iii.) reverse that repurchase by re-issuing the common stock.

So it *is* frustrating and counter-intuitive that we have three separate rules, three separate sets of accounts, for them. Why do it this way? Given that the stock price varies between those three transactions, they could never be simple, total reversals. There would always have to be some kind of 'plug' involved, at some point. There's no way around that. And per the logic above, we can't use 'Gain / Loss.' The particular rules are **not the only imaginable way of doing it**—and let me know if you see a better approach that satisfies all of our accounting goals and constraints. But they are GAAP, and that's what we need to know.

[3] If you go poking through real annual reports, the biggest example you'll encounter is **'Other Comprehensive Income'**—a whole category of value changes that are recorded in Shareholders' Equity on the Balance Sheet while deliberately bypassing Net Income. The specific items differ, but the mental template is exactly the one we just built: something economically real happened, it must be recorded somewhere in equity for things to balance, and we do not want it distorting the bottom-line performance measure. We won't cover comprehensive income in this introductory text—but when you meet it, you'll recognize the move.

15.6 Putting the pieces together: reverse-engineering share counts

Let's consolidate everything we've learned about par value, *Common Stock*, and *Treasury Stock* with a puzzle. Recall that analysts care intensely about per-share numbers, above all earnings per share (EPS)—and to compute any per-share number, you need to know how many shares are **outstanding**. So: how much can we recover about a company's share counts, just from the equity accounts on its balance sheet?

Suppose that MarshallCo begins its life in 2018, and issues Common Stock with a stated par value of $1 per share. Assume a single class of common stock, unchanged par value, no share retirements, and no other changes to Common Stock. Below are selected amounts reported on its balance sheets:

	2019	**2018**
Common Stock	$10,000	$10,000
Additional Paid-in Capital	$900,000	$900,000
Treasury Stock	($200,000)	$0

MarshallCo, selected Shareholders' Equity accounts

Based on this information alone, which of the following can we determine: (a.) the number of shares *outstanding* at the end of 2018; (b.) the number of shares *outstanding* at the end of 2019; (c.) the number of shares *issued* at the end of 2019? Try it yourself before reading on. This little problem integrates everything in this chapter so far.

Start with (a.). When shares are issued, *Common Stock* is credited for the total *par value* of the shares issued. So a $10,000 balance, at $1 par per share, means 10,000 shares have been **issued**. And at the end of 2018, *Treasury Stock* is zero: no shares remain in treasury under our assumptions, so every issued share is outstanding. So, yes: we can determine that there were **10,000 shares outstanding**.

Now (b.). During 2019, *Treasury Stock* grew from $0 to $200,000, so we know the company repurchased shares. But recall: the *Treasury Stock* account records the *dollars* spent, not the number of shares. Were those repurchases made at $20 per share (10,000 shares—every share outstanding!) or at $400 per share (a mere 500 shares)? We have no idea. Without the repurchase price, we cannot convert dollars into a share count—and so we **cannot** determine the shares outstanding at the end of 2019.

Finally, (c.). Here's where the seemingly-pedantic definition of 'issued' pays off. When a company repurchases shares, we debit *Treasury Stock* and credit *Cash*—and we do *not* touch *Common Stock* at all. So the *Common Stock* balance tracks the par value of issued shares under our stated assumptions, including shares held in treasury. That balance is still $10,000, and we can also tell that the company issued no new shares

during 2019 (because *Common Stock* and *APIC* are both unchanged). So, yes: **10,000 shares issued**, same as in 2018.

The general morals: the number of issued shares always equals the *Common Stock* balance divided by the par value. And outstanding shares equal issued shares minus treasury shares—but the treasury share *count* is exactly the quantity that the balance sheet won't hand you directly.[4]

15.7 Preferred Stock

From Chapter 1, we've said that companies can finance themselves via either **equity** or **credit financing**. Equity financing, so far, has been represented by *Common Stock*, and we have seen several liability accounts representing credit financing. Our high-level definition has been:

- 'Credit' financing is when our company gets cash today, and is *contractually obligated* to pay it back in the future. We **have to** pay our creditors back, but they do not get any additional 'upside' beyond our contractually specified payments.
- 'Equity' financing is when our company gets cash today from those investors, who have no contractual entitlements to cash flows, but become 'owners' (usually with voting rights). As 'residual claimants,' they can claim **whatever is left over**, after credit obligations have been met.

That seemed like a clean distinction. But, as we've seen many times before, the real world is more complicated and fuzzy. **Reductively**, both equity and credit financing are ultimately just **financial contracts**—contracts in which people provide financing to corporations in exchange for *something*. And lawyers and investment bankers and corporations can write up contracts in ever-more complicated ways. One kind of financial contract we haven't discussed yet is ***preferred stock***. Let's first define how it *works*:

Preferred Stock:

- Outside investors give the corporation cash in exchange for *shares* of *Preferred Stock*.
- The preferred stock **contract pre-specifies certain cash distributions** called *preferred dividends*. They are *called* 'dividends,' just as with *Common* Stock, but work a bit differently.

[4] So how do analysts actually get shares outstanding? Companies disclose the number—on the face of the balance sheet, on the cover page of the 10-K, and elsewhere. The point of this puzzle isn't that the number is a secret; it's to test whether you really understand what each of these equity accounts does and does not track.

- The corporation is ***not*** **unconditionally obligated** to pay those preferred dividends. But, *before* the company can pay regular dividends to Common Stockholders, it must **first pay out** the dividends specified for **preferred** stockholders. (In many preferred stock transactions, the company must also first pay off all preferred dividends "in arrears"—the cumulative sum of what it missed in previous years.)

- Preferred stockholders also typically do **not** get **voting rights**. So, they lack the element of *control* that we have associated with ownership since Chapter 1.

Restating, our company is not **unconditionally obligated** to pay dividends to *Preferred Stockholders*, but ***conditional on*** wanting to pay out common stockholders, it is.

How do we classify this contract? The answer depends on its **terms**, under both U.S. GAAP and IFRS. A nonredeemable preferred share whose distributions remain discretionary may qualify as equity. A share with a mandatory cash-redemption obligation can instead be a liability; under U.S. SEC reporting, some redeemable instruments are presented outside permanent equity. So there is no blanket rule that "GAAP calls preferred stock equity while IFRS calls it debt." Read the contractual obligations first.[5]

It's worth knowing that many financial analysts and investment bankers split the difference in practice: when they compute leverage ratios and assess risk from GAAP financial statements, they will often mentally *reclassify* preferred stock as debt—"GAAP calls this equity, but it behaves like a liability, so I'll model it as one." There is a nice lesson in that. Classification in accounting is sometimes a judgment call, not a law of nature: the legal name "stock" does not settle the financial substance. The standards can differ in details, but the instrument's obligations matter in both systems.[6]

15.7.1 Accounting for preferred stock

For the equity-classified preferred shares assumed in this example, we account for issuance much like *Common Stock*, just changing the specific name of the stock account. It's that simple. E.g.:

[5] See IAS 32 and the U.S. distinction between liabilities and equity in ASC Topic 480. FASB's Statement 150 summary explains the mandatory-redemption principle that preceded the codification.

[6] Nor is preferred stock the only hybrid lurking out there. Financial engineers keep inventing securities that straddle the equity–debt line—for instance, the 'AT1' bonds issued by large banks, which are designed to convert into equity or be written off in a crisis, and which made global headlines in 2023 when Credit Suisse's AT1 holders were wiped out in its emergency sale. Wherever accounting draws a line between two categories, someone will engineer a contract that sits exactly on top of it.

Cash (+A):	$30,000	
Preferred Stock (+S.E.):		$10
Additional Paid-In Capital (+S.E.):		$29,990

Issuing 1k shares of preferred stock, $.01 par value, for $30 per, Jan 3, '21

15.8 Stock Splits and etc.

15.8.1 What?

It's common for companies to engage in **2-for-1 'stock splits.'** This is **just what it sounds like**. Every 1 share is turned into 2 shares. But, since there are now **twice as many** shares, **each individual** share only has **half** as many rights and **value**, in terms of voting power and entitlement to future share of dividends. So, the share price after a split is essentially half as much as what it was immediately beforehand,[7] and each shareholder is left with the same overall voting power.

15.8.2 Why?

As such, stock splits are essentially just **'paper transactions'** with **no real economics**—no real change in value, power, or resources. So, why do companies do them? Even in the 21st century, exchange trading is generally organized around whole shares, although many brokerages now offer fractional-share purchases. So, many companies fear that if their stock price goes to a quite high-level—more than, say, $100 each—ordinary 'retail' investors ("mom and pop") will be unable to buy them. This could reduce investor demand for the shares, and thus, in theory, hurt its stock price. Or at least seem undemocratic. So, typically, companies will do a stock split after their stock price has experienced a sharp rise, in order to keep the per-share price in a 'palatable' range. Companies can also do 3-for-1, or 5-for-1 stock splits, which are just what you would expect. For example, Tesla did a 5-for-1 stock split in the summer of 2020, after years of extraordinary stock performance (and then split again, 3-for-1, in 2022).

15.8.3 Reverse splits and stock dividends

Companies can also do a **reverse stock split**, which is just the opposite, as you would expect, consolidating every 2 shares into 1. Symmetrically, after a 1-for-2 reverse stock

[7] As always, things can get a bit more complicated at the graduate level. If a split is interpreted as a *signal* of managerial optimism—for reasons we'll discuss soon—that signaling effect could increase the aggregate valuation.

split, each stock would be worth essentially twice as much, all else equal. Why do companies do this? As before, sometimes this is about keeping the stock price in a "psychologically palatable" range, after years of rough decline. Investors are used to seeing per-share prices in the $10-$100 range. So, if your stock is trading for $.50 per share, it **looks bad**. Another reason is that some U.S. **stock exchanges** have **listing requirements** related to the stock price. E.g., a company can fall out of compliance with an exchange's minimum-price requirements and eventually face delisting. The rules use specified measurement periods and procedures; a single trade below $1 does not automatically cause immediate delisting. This factor adds an **extra element** to reverse stock splits however. For example, suppose that a company is currently trading at $1.50 per share on the NYSE, and the management suddenly decides to do a reverse split. You might reasonably suspect that the management has information that might cause the stock price to drop further in the future, and is trying to get ahead of de-listing.

These listing requirements also help explain why companies may prefer a buffer above the minimum share price. But there is no fundamental "correct" dollar price for an individual share, and companies choose very different denominations. The number of shares and the value per share must be read together; splitting the same ownership into more pieces does not, by itself, change the total value.

Finally, companies can also do something called 'stock dividends,' which are essentially just *fractional stock splits*, though with slightly different legal and accounting mechanics. I.e., after a 20% "stock dividend," every 1 share becomes 1.2 shares.

15.8.4 Accounting for these: We don't, in this book

At this point in the chapter, we've covered all the important concepts I want to convey. It turns out that the accounting for stock splits, reverse splits, and 'stock dividends' is quite arcane, gnarly, and, in some cases, seemingly illogical. And, very often, the difficulties here are created by the mechanics of adjusting 'par value,' the appendix of the accounting system. Since these are largely 'paper transactions,' with limited economic significance, and the accounting is so challenging, I do not cover them in this book, and discourage my students from going further in this topic at this phase.

But there is one conceptual point about the accounting that you should carry with you, because it follows directly from first principles: **a stock split does not change total Shareholders' Equity**. Why not? Ask what the company actually *received* or *gave up* in the split. Nothing. No cash came in, no asset was acquired, no liability was incurred—there was no transaction with any outside party at all. The company simply re-denominated its existing equity into more, smaller slices. Even if the split nudges the *market* value of the equity up a percent or two (via the liquidity and psychology discussed above), remember that we do not write book equity up to market value—that

has been true throughout this text. So, whatever fiddly reclassifications the formal split entries make *within* the equity accounts, the total is untouched. A practical corollary: if you are ever given a list of transactions and asked to compute ending total Shareholders' Equity, and a stock split appears somewhere in the middle of the list—skip right over it. Its contribution is zero.

15.9 The Statement Itself

Now that we've learned about these other accounts and transactions with shareholders, we can also show a more realistic version of the Statement of Shareholders' Equity, relative to the simple one we presented above. We would simply include the **additional Shareholders' Equity accounts** we just learned about (*Preferred Stock*, *APIC*, and *Treasury Stock*), as additions on the 'horizontal' axis, like so:

	Preferred Stock	Common Stock	Additional Paid-in Capital	Retained Earnings	Treasury Stock	Total Shareholder's Equity
Balance, Jan 1st	$0	$0	$0	$0	$0	$0
Issue Common Stock		15,000				15,000
Dividends				(100)		(100)
Net Income				2,300		2,300
Balance, Dec 31st	$0	$15,000	$0	$2,200	$0	**$17,200**

Statement of Shareholder's Equity, FY2020

Note that, in the example we were using above, the company after its first 10 transactions and adjusting entries, we had no-par stock, and no repurchases or preferred. So, those new columns are all zeroes. But triangulating between what we've learned, and this template, you can surely fill in the rest.

16

Statement of Cash Flows

In Chapter 1, we introduced the concept of *accrual accounting*, which we contrasted to simple *cash-basis accounting*. By now, we're accustomed to accrual accounting—we probably just think of it as *accounting, full stop*. But in this chapter, we cover the Statement of Cash Flows, which reports a company's financial performance on a **cash basis**.

16.1 Overview

16.1.1 Why?

Why are we doing this? There are some **good reasons for accrual accounting**. By recognizing revenues when they are **earned** and **matching** expenses, it attempts to measure whether the company is actually **selling above cost and generating value** for its shareholders on an **ongoing basis**, regardless of the **timing of checks in the mail**.

But there's also been a **dark side to accrual accounting**. Managers can affect their **bottom-line** net income simply by changing their **depreciation assumptions and estimates**. Or, think back to Chapter 8, and all the complex mechanics and unverifiable estimates that go into the *Allowance for Uncollectible Accounts*, and how revisions in that balance affect the *Bad Debt Expense* and, thus, bottom-line *Net Income*. In general, accrual accounting allows some ***scope*** for managers to use their ***discretion and***

estimates to **"massage"** their earnings a bit. And even when managers have **no ill intention**, some of the more **baroque** aspects of accrual accounting (e.g., the *Bad Debt Expense*) may make it an **opaque** or distortionary window into the company's operations.

The earlier cases gave us several reasons to compare reported profit with cash: an estimate can change, a loss can be recognized without a new payment, and revenue can precede collection. A growing gap between Net Income and operating cash flow is therefore a question to investigate. It is not, by itself, proof that anything is wrong.

Another reason for reporting performance on a cash basis is that **"cash is king"**—you'll hear this mantra repeated in business settings and used to mean many different things. In this context, what this means is that companies must make their **interest and principal payments to their creditors in cash**. **You cannot pay the bank in receivables.** Only cash. A company could be **highly profitable on an accrual basis** (in terms of net income), but default on its debt payments and go into bankruptcy, if it is **unable to convert its *Accounts Receivable* into cash** in time to pay the bank, or because its expenditures on *Inventory* or other assets (which are not immediately expensed in accrual accounting) **consumed too much cash**. Put differently, if assets like *Accounts Receivable* and *Inventory* are increasing rapidly, a company might report **high net income**, but be **bleeding cash**.

Let that *Inventory* case sink in, because it is the more insidious of the two. Picture a company that keeps buying and buying inventory—warehouses steadily filling—while customers, it turns out, do not actually want the product. Economically, this is terrible news: cash is being tied up in goods that may never sell. The purchase itself does not immediately reduce Net Income. But inventory that has lost value may require a write-down, even before it is sold. We recognize the expense—*COGS*—only when the goods are *sold*; until then, the unsold inventory just sits on the Balance Sheet, as an asset, at cost. So *Net Income* can sail along unbothered while the company quietly bleeds cash into a warehouse full of things nobody wants. This is exactly the kind of disconnect the Statement of Cash Flows is designed to expose.

Historically, then, the classic constituency for cash-flow reporting has been **creditors**: a banker making a five-year loan wants to know whether the borrower will generate the *cash* to service it over that horizon—full stop. Goodwill, trademarks, receivables: there are good reasons those things are on the balance sheet, but none of them is legal tender. Increasingly, though, **equity investors** demand cash-flow information too, and for a somewhat different reason. They are not merely asking 'can you pay me back over the next five years?'—as residual claimants, they are trying to evaluate the entire business model and its long-term upside. And a nontrivial fraction of them have concluded that, even for that purpose, they cannot take the accrual story on faith: they want to check it against the cash.

16.1.2 How?

In this chapter's exercises, we use the **Income Statement**, **comparative Balance Sheets**, and **stated transaction assumptions** to reconstruct cash flows. In practice, companies also use transaction records and supporting schedules. Two Balance Sheets show net changes; they do not, by themselves, identify all the gross receipts and payments that produced those changes.

This may remind you of the more theoretical-seeming stuff we covered in Chapter 7. E.g., we showed how differences between *Revenues* and *Cash Collections from Customers* related to **changes** in *Accounts Receivable*. We wrote certain journal entries using an extra zero row, to show how '**differences** between **accrual** revenues / expenses vs. their **related cash** flows are stored in or **plugged** by **changes** in **balance sheet** accounts.' Almost everything we do in this chapter, in filling out the Statement of Cash Flows, will be an extension of that idea.

To give you an early taste of this detective work, consider the following puzzle. Suppose our company bought a parcel of *Land* fifteen years ago, for $1m. What is that *Land* carried at on the books today? Still $1m—we do not depreciate *Land*, and (absent an impairment) we do not revalue it. Now suppose that this year, the *Land* account drops from $1m to $0, and the Income Statement reports a *Gain on Sale of Land* of $0.5m. Here's the puzzle: suppose the information supplied in this exercise does not directly state the *cash* received for that land. Can we figure it out anyway? Yes. We know the book value that came off the books ($1m), and we know that the *Gain* 'plugs' the difference between the sale price and the book value. So the sale price—the actual cash inflow—must have been $1.5m. We just reverse-engineered a cash flow from the finished statements, using nothing but our accounting knowledge. That is the spirit of this entire chapter—and I think you will be pleasantly surprised by how much you can figure out simply by staring at an Income Statement and a Balance Sheet with your accounting brain switched on.

One more piece of setup. You might reasonably ask: if this statement is about tracking cash, why do we need a whole *statement* at all? *Cash* is literally the first line of the Balance Sheet—anyone can see that it was $220k last year and $230k this year, and subtract. Here's the answer: what we want from the Statement of Cash Flows is not the *change* in cash, but an understanding of *why* cash changed—which activities drove it. A company whose cash went up because it issued a pile of new bonds is in a very different position from a company whose cash went up because customers are happily paying for its products. The first company borrowed; the second company *performed*. Lumping the two together into a single number would throw away exactly the information we are after. Hence, the statement's defining structure.

In practice, the Statement of Cash Flows **decomposes** the **total change in cash** into

three different components, each reported in their own section of the statement:

- (1.) Cash Flows from **Operations**,
- (2.) Cash Flows from **Investing**,
- (3.) Cash Flows from **Financing**.

For most purposes and analyses, Cash Flows from Operations is the most important number—the major alternative, competitor, or complement to *net income* as a measure of the company's performance. And there are **two different methods** for filling out the **operating** section of the cash flow statement:

- The **"direct method,"** in which we **adjust each** individual income-statement line item to its **cash-basis analogue**, one-by-one.
- The **"indirect method,"** in which we instead **start with** *Net Income* and make those adjustments required to go to *Cash Flows from Operations* at a more **summary level**, instead of line by line.

These two methods get you to the same place (**cash flows from operations**), using the same concepts, just with the *adjustments* presented in a different way. The large **majority of companies** report under the **indirect method**. But the **direct method** is better, in my view, at **helping you understand** what's really going on in the cash flow statement. So, I cover it first. Once we do that, the indirect method will be relatively easy. This distinction—direct vs. indirect method—only applies to the **operating section**. For the investing and financing sections, we only use one approach.

So, the remainder of this chapter will proceed as follows.

- **First**, we'll cover the **distinction** between operating, financing, and investing cash flows, the **three sections** of the statement, in greater depth.
- **Second**, we'll cover the **operating section** in detail, showing **both** the direct and indirect methods.
- **Third**, we'll fill out the remaining sections, investing and financing.

16.2 Operating vs. Investing vs. Financing

All cash flow statements decompose the net change in cash into cash flows from operating vs. investing vs. financing activities. So, they'll all have this basic structure:

Operating cash flows:		
...		$...
...	$(...)	
Net Change in Cash From Operations		**$X**
Investing cash flows:		
...		$...
...	$(...)	
Net Change in Cash From Investing	**$(Y)**	
Financing cash flows:		
...		$...
...	$(...)	
Net Change in Cash From Financing		**$Z**
Net Change in Cash Between Periods		**$X-Y+Z**

Statement of Cash Flows, for FY ending 12/31/2021

We use **U.S. GAAP cash-flow classifications** in this chapter. IFRS can differ. For clarity, our examples also set aside foreign-currency effects and changes in restricted cash. Our first order of business is to understand what transactions go into which categories. These are **mostly** pretty **intuitive**, but there are a couple of **notable exceptions**, counterintuitive cases, and especially tricky ones that I'll highlight.

- **Cash flows from operations (CFO)**
  - **Big idea**: These are cash flows having to do with the company's normal short-term business operations, the ongoing sale of goods and services to customers.
    - So, **cash collections from customers** count as operational, as do cash **payments of salaries** to workers, cash payments for short-term **supplies, rent**, and other miscellaneous operating expenses, etc.
  - **Tricky and counterintuitive cases**.
    - For the purpose of the **Cash Flow Statement**, cash spent on **interest payments** is categorized as an operating cash flow, even though it has to do with the company's debt financing. This is done to make the operating section of the cash flow statement **more comparable to the income statement** and thus more **useful as a check** on it. Since *Interest Expenses* are on the income statement, we include cash interest payments as operating cash flows. (Symmetrically, cash interest *received* on investments is also categorized as operating—*Interest Revenue* is on the income statement, too.)

    * **Cash spent on *Inventory*** is considered operating, not investing, because *Inventory* is a short-term (current) asset. This is potentially confusing for first-time learners (and your author), because *Inventory* is capitalized as an asset, just like purchases of PP&E, etc., which might seem to connote that it is an investment. The classification follows the activity, not just the balance-sheet maturity. Inventory purchases are operating; a purchase of another company's securities may be investing even if the securities are classified as current assets. Put differently, there is **not** a 1-to-1 mapping from capitalizing vs. expensing an expenditure to the investing vs. operating categories.

- **Cash flows from investing**
  - **Big idea:** These are cash flows having to do with the company's **purchase and sale of long-term assets**.
    * So the cash purchase of **Equipment or Land**, or ***another company's* stock or bonds**, would be investing cash **outflows**. And, the **sale** of those same assets would be an investing cash **inflow**. If you think back to Chapter 11, you may recall that we can calculate the **cash sale price** on an asset by triangulating between its **prior book value** and the **gain or loss** upon sale.
  - **Tricky cases:**
    * As above, remember that if a company spends money on an asset, but that asset is a **short-term**, current asset, like *Inventory*, that's considered operational, not investing. And, if a company buys or issues *its own* stock or bonds, that has to do with how it *finances itself*, so that's financing, not investing.

- **Cash flows from financing**
  - **Big idea:** These are cash flows having to do with **how the company finances itself**, its **transactions with its shareholders and financial creditors**.
    * So, the **issuance or repurchase of bonds or common stock**, and the payment of **dividends**, all count as financing cash flows.
  - **Tricky cases:**
    * As we discussed in the **operational** section, we categorize **cash interest payments** as operational, not financial, to make the operating section more comparable to the income statement. Additionally, note that if

a company buys *another company's* stocks or bonds, that would be an **investing** cash outflow. Financing cash flows have to do with how our company finances *itself*—i.e., its *own* stocks and bonds.

Before moving on, play this back to yourself, because this categorization is the skeleton that everything else in the chapter hangs on. Test yourself: If our company repurchases its own shares into *Treasury Stock*—which category, and inflow or outflow? (Financing; outflow. Transactions in our *own* stock are about how we finance ourselves.) If we purchase *Inventory* for cash? (Operating; outflow. Inventory is an operating asset, even though the expenditure is capitalized.) If we sell some used *Equipment*? (Investing; inflow.) If we pay cash interest to our bondholders? (Operating; outflow—the counterintuitive exception we flagged above.)

16.3 Operating Section: Direct method

Under the direct method, we convert each line item in the income statement (that is, each revenue and expense) to its cash-basis analogue, line-by-line. That's it. And, doing this is absolutely feasible, based on what we've learned so far. Why? Because, the differences between accrual-basis revenues and expenses, and their related cash flows, are 'stored in' changes in the related asset and liability accounts (remember Chapter 7, Section 7.2.5). Let's start by diving into an example, and then back up to the theory and rules, afterwards.

16.3.1 Example

Suppose we are given the Income Statement and Balance Sheets below, and asked to prepare the Operating section under the Direct method, for FY2021. In order to make this tractable at an introductory level, we need to make a **simplifying assumption**, that will be our default throughout this book: Assume that the *Accounts Payable*, below, are all **related to the purchase of *Inventory***.[1]

For the main example, also assume: no bad-debt writeoffs, returns, acquisitions, foreign-currency effects, or other noncash changes in working capital; no deferred-tax amounts or bond discount/premium amortization; Investments increased only through cash purchases; the Land decrease is one cash sale, with no other land transactions; Equipment increased only through cash purchases, with no disposals or impair-

[1] Why do we need this assumption? To convert each expense to its cash analogue, we have to know which income-statement item each *Payable* 'belongs to.' If our company had both *Inventory* and, say, *Supplies* feeding into a single *Accounts Payable* balance, we could not uniquely decompose that payable between them without further disclosure—and the problem would be unsolvable at our level of detail. Real filings provide more granular disclosures. In this introductory text, we deal with one complication at a time.

ments; bonds were issued at par for cash, with no repayments; and no-par Common Stock increased only through cash issuances, with no repurchases or retirements. All dividends declared were paid in cash, and Net Income and those dividends were the only changes in Retained Earnings. These facts make the reconstruction determinate; they cannot be inferred from the balances alone.

Revenue:	**$200k**
COGS:	**$80k**
Salaries expenses:	**$40k**
Depreciation expenses:	**$30k**
Loss on sale of land:	**$5k**
Interest expense:	**$10k**
Income tax expense:	**$10k**
Net Income:	**$25k**

MattCorp Income Statement, For the Year Ended December 31, 2021:

	2021	2020
Assets:		
Cash	**$230k**	**$220k**
Accounts Receivable	**$90k**	**$70k**
Inventory	**$120k**	**$140k**
Investments	**$130k**	**$0**
Land	**$220k**	**$280k**
Equipment	**$250k**	**$200k**
Accumulated Depreciation	**$(80k)**	**$(50k)**
Total assets:	**$960k**	**$860k**
Liabilities:		
Accounts payable	**$80k**	**$90k**
Salaries payable	**$30k**	**$20k**
Interest payable	**$7k**	**$12k**
Income Tax payable	**$16k**	**$15k**
Bonds payable	**$260k**	**$220k**
Shareholders' Equity:		
Common Stock	**$432k**	**$383k**
Retained Earnings	**$135k**	**$120k**
Total Liabilities and S.E.:	**$960k**	**$860k**

MattCorp Balance Sheets, Year Ended December 31:

Where do we start? Under the direct method, we convert each line item of the income statement to its cash-basis analogue. So, **start at the top** of the I/S, with *Revenue*.

16.3.2 Adjusting Revenue to Cash Collections from Customers

The **cash analogue** to accrual-basis *Revenue* is the *"Cash Collected from Customers"*. Our company has reported $200k in *Revenue* from this period. But we can't be sure it actually collected all of that in cash. This company has **no** *Deferred Revenue*, so that doesn't factor into this analysis. So, we only need to worry about *credit sales* and collections, which are ***stored in*** changes in *Accounts Receivable*. So, to go from Revenue to Cash Collected from Customers, we'll use the change in *Accounts Receivable (A/R)*. To understand this, consider the following:

- Suppose our company sold $50k in goods *on account*—then that would count toward *Revenue*, but not *Cash Collected*, and would **increase** *A/R* by $50k.
- Similarly, suppose our company collected $40k on **past accounts**, that is, from *Revenue* earned in **previous periods**. Then that would be *Cash Collected*, but wouldn't count toward *Revenue* in this period, and would **reduce** *A/R* by $40k.

Netting out, and generalizing, we have the following core identity (which holds **under our stated assumptions, including no Deferred Revenue or other noncash receivable changes**):

$$Cash\,Collected\ from\,Customers = Revenue - \Delta\,Accounts\,Receivable$$

In the case of our example above, this means that:

$$Cash\,Collected\ from\,Customers\ (FY2021) = \$200k - (\$90k - \$70k)$$

$$Cash\,Collected\ from\,Customers\ (FY2021) = \$200k - (\$20k) = \$180k$$

So, the **top line** of our **Operating Section**, under the Direct Method, will be just that.

In this simple example, we only had one kind of *Revenue*. But suppose that we had *Interest Revenue* as well. In that case, the same general pattern would hold, but we would use the *Interest Receivable* in lieu of A/R. Additionally, as will *usually* be the case in this book, we had **no** *Deferred Revenue*. But what if the company did have some? Recall that *Deferred Revenues* represent customers' pre-payments—so they *are* cash collected from customers, but have not yet been recorded as accrual-basis *Revenues* (since they have not yet been earned). In this sense, *Deferred Revenue* is the **mirror**

image of *Accounts Receivable*: *A/R* is revenue that has been *earned but not yet collected*, while *Deferred Revenue* is cash that has been *collected but not yet earned*. So, it should make sense that we would make just the opposite adjustment, *adding* changes in *Deferred Revenue*. We can state this as a general rule.

Rule: Under the Direct method, we adjust each *Revenue* to its cash-basis analogue by subtracting the change in the related asset account (the *Receivable*), and adding the change in the related liability account (*Deferred Revenue*) ***if any.***

16.3.3 Adjusting COGS to Cash Paid to Suppliers

Next, going down the Income Statement line-by-line, is *COGS*, the accrual expense from using up *Inventory*. The **cash-basis analogue** to this is *"Cash Paid to Suppliers for Inventory."* Remember that, even though *Inventory* is an asset, it is short-term (**current)**, so purchases of *Inventory* are categorized as *operational*.

Our company reported $80k in *COGS*. And yet, *Inventory* only decreases by $20k over this period ($140k down to $120k). How can this be? By our core identities around **purchases and expenses**, it must be that it **purchased an additional $60k** in *Inventory*. Put more generally, we know that:

$$Asset\ Ending\ Balance = Asset\ Beginning\ Balance + Purchases - Related\ Expense$$

Applied to *Inventory*, we could rewrite this as:

$$Inventory\ Purchases = COGS + \Delta Inventory \tag{16.1}$$

So, is that *Cash Paid to Suppliers for Inventory*? **No, not yet**. While we now know the company *purchased* $60k in *Inventory*, we can't be sure that was all in cash. It could have purchased some *on account*. How do we calculate that? As specified above, in this book, we make a **simplifying assumption** that all *Accounts Payable* are related to the purchase of *Inventory*. This means that the difference between the *Inventory* **purchases** and the **cash payments** to those suppliers will be fully tracked in *Accounts Payable*. Consider the following examples:

- If we purchased $60k of *Inventory*, but only paid for $40k of it in Cash, then those $20k of credit purchases would increase *Accounts Payable*.
- If we paid off **past periods'** *Accounts Payable*, that would be a cash payment to suppliers, but not an expense in this period, and would decrease our *A/P*.

Netting out, and generalizing, we have the following core identity (assuming all *Accounts Payable* are related to the purchase of *Inventory* from suppliers):

$$Cash\ Paid\ For\ Inventory = Inventory\ Purchases - \Delta\, Accounts\ Payable \tag{16.2}$$

Or, consolidating Equations 16.1 and 16.2, we then have:

$$Cash\ Paid\ For\ Inventory = COGS + \Delta Inventory - \Delta\, Accounts\ Payable$$

In the case of our example above, that would be:

$$Cash\ Paid\ For\ Inventory\ FY2021 = \$80k + (\$120k - \$140k) - (\$80k - \$90k) = \$70k$$

These formulas may seem like a lot. But they're just helping us keep track of and formalize a simple idea. If your *Inventory* went up, it must be that your inventory *purchases* exceeded your COGS by that amount. If your *Accounts Payable* went up, it must be that those purchases were not fully paid in cash. That's the idea. The formulas just help us keep track of things on a ***net*** basis. We can also restate what we learned with a general rule:

Rule: Under the Direct method, we adjust each *Expense* to its cash-basis analogue by **adding the change in the related asset and subtracting the change in the related liability** (typically a payable).

16.3.4 Adjusting Salaries Expense to Cash Paid for Salaries

The next line item on the I/S is *Salaries Expenses* of $40k. This one will be easy after the work we did on *COGS*. Our general rule is that we *adjust accrual expenses to their cash-basis analogue* using the changes in the related asset and liabilities. Unlike for COGS (Inventory), **there is no related asset account** for *Salaries Expense*. So, we can do the full adjustment using only the liability account, *Salaries Payable*.

- If we incurred $100k of Salaries Expense over the reporting period, but only paid them $80k, that would increase our *Salaries Payable* by $20k.
- If we paid off **past periods'** *Salaries Payable* in cash (that is, paid back our employees what we had owed them), that would be a cash payment to our employees, and would decrease our *Salaries Payable*.

Netting out, and generalizing, we have the following:

$$Cash\ Paid\ for\ Salaries = Salaries\ Expense - \Delta(Salaries\ Payable)$$

In our particular example, that would be:

$$Cash\ Paid\ for\ Salaries\ FY2021 = \$40k - (\$30k - \$20k) = \$40k - \$10k = \$30k$$

For many expenses (not COGS, though), it will be this simple. If there is not a related asset account (that is, if the expenditures from this expense are ***not* capitalized**), then the difference between the expense and the cash is fully tracked in the related *Payable*.

16.3.5 Aggregating: Where are we going?

We have converted the first three income-statement lines into cash amounts. The table below keeps the original amount, the adjustment, and the resulting cash flow together.

Or, it could be most helpful to visualize the Income Statement line items, the adjustments, and the cash-basis analogues, side-by-side, like so:

I/S Item	I/S Amount	Direct Method Adjustment	Cash Flow Analogue	Cash Flow Amount
Revenue:	\$200k	- Δ Accounts Receivable	Cash Coll. from Customers	\$180k
COGS:	-\$80k	- Δ Inventory + Δ A/P	Cash Paid to Suppliers	-\$70k
Salaries Expenses:	-\$40k	+ Δ Salaries Payable	Cash Paid for Salaries	-\$30k

Showing Income Statement, Direct Adjustments, and Cash-Basis Analogues

Because this table begins with expenses shown as **negative** amounts, its adjustment signs are the opposite of the formulas for positive "cash paid" amounts. For example, $-80-(-20)+(-10)=-70$ for the supplier-payment row. The remaining work is to continue down the Income Statement.

16.3.6 Skip over Depreciation Expense: Why?

The next item on the Income Statement is *Depreciation Expense*. What do we do with it? **Nothing: We skip over this one.** Our goal is to adjust each line item of the I/S to its operating cash flow equivalent. There **is no operating cash flow related to *Depreciation***. It is, by definition, an ***allocation*** of the net cash outflow from investments in **long-term tangible assets**. The actual cash flows from the purchase and sale / disposal of long-term assets will all show up in the **investing section** of the cash flow statement.

There is a one-line explanation you will hear everywhere: 'we skip *Depreciation* because it's a *non-cash expense*.' Be careful with that phrase—it explains less than it seems to. *Every* line item we've adjusted so far is 'non-cash': *Salaries Expense* is not cash; *Revenue* is not cash; the entire project of this statement is to start from non-cash accrual figures and adjust them to cash. What actually distinguishes *Depreciation* is not that it is non-cash, but that its related cash flows are classified as **investing**, not operating—so there is no operating cash flow for it to be adjusted *to*.

16.3.7 Skip over Gains / Losses on Sales: Why?

Next on the I/S is *Loss on Sale of Land*. What do we do? Once again: **We skip over it.** Why? If you consult **YouTube accounting instructors**, many of them will claim that this is because 'Gains' and 'Losses' are simply accounting plugs, not 'real' cash flows. But this is not correct. Indeed, selling *Land* for cash obviously *does* yield a real cash flow—and the *Gain or Loss* is a part of that overall purchase price. (Since $Gain\ (Loss)\ upon\ Sale = Sale\ Price - Book\ Value$.) The **actual reason** we skip over it for now is that that cash flow, from selling the asset, is part of the **investing section** of the cash flow statement, not operating. It will *implicitly* show up there.[2]

16.3.8 Adjusting Interest Expense to Cash Paid for Interest

The last two items are easy: In the case of both *Interest Expense* and *Tax Expense*, there is no related asset account, and so we can adjust both of them back to their cash payment analogue using only the change in their related liabilities, their *Payables*. Consider *Interest* first:

- If we incurred \$100k of *Interest Expense* over the reporting period, but only paid \$80k, that would increase our *Interest Payable* by \$20k.
- If we paid off **past periods'** accrued interest in cash, that would be a cash payment, but **not** an *Interest Expense* in **this period**, and would decrease our *Interest Payable*.

Netting out, and generalizing, we have the following:

$$Cash\ Paid\ for\ Interest = Interest\ Expense - \Delta(Interest\ Payable)$$

In this case, that's:

$$Cash\ Paid\ for\ Interest\ = \$10k - (\$7k - \$12k) = \$10k + \$5k = \$15k \tag{16.3}$$

[2] But, note that there is some dissonance here. Earlier, we learned that companies count **cash interest payments** as operating cash flows to make the operating section more **consistent** with the income statement. But here, gains and losses affect the income statement, but are excluded from the operating section of the cash flow statement. In the case of *Gains* and *Losses*, these are usually "one-time," "special items." We **do not** expect *Gains* and *Losses* from large asset sales and impairments to be *persistent and repeatable*. So, this inconsistency doesn't make the two measures, **CFO vs. NI**, incomparable on a regular basis. However, *Depreciation* and *Amortization* **are** persistent and repeated. Put differently, the *Income Statement* allocates the investing cash outflow as an expense over time, but we don't do the same for *Operating Cash Flows*. So there's some inconsistency there; and, this creates a tendency for CFO to exceed NI, other things equal; it is not a universal inequality. (We'll return to this asymmetry, and what it implies, at the end of the chapter.) It's not clear how else to deal with this issue, however. What this means is that, as always, analysts can't just use one measure to tell them everything they need to know. They need to use the full mosaic of information, from all the firms' disclosures, and analyze them holistically.

16.3.9 Adjusting *Income Tax Expense* to Cash Paid for Taxes

Once again, for *Income Tax Expense*, there is no related asset account, and so we can adjust it back to the related cash payment using only the change in the related liability, *Income Tax Payable*.

- If we incurred \$100k of *Tax Expense* over the reporting period, but only paid \$80k, that would increase our *Tax Payable* by \$20k.

- If we paid off **past periods'** *Tax Payable* in cash (back taxes), that would be a cash payment, but not a *Tax Expense* in this period, and would decrease our *Tax Payable*.

Netting out, and generalizing, we have the following:

$$Cash\ Paid\ for\ Income\ Tax = Income\ Tax\ Expense - \Delta(Tax\ Payable)$$

In this case, that's:

$$Cash\ Paid\ for\ Income\ Tax = \$10k - (\$16k - \$15k) = \$9k \tag{16.4}$$

16.3.10 Entire Operating Section under Direct Method

So now, we've adjusted each and every income-statement line item. Next up is to just consolidate all of that work, and net out the cash flows, to calculate and report the **Operating Section**, and its **'bottom line,'** ***Net Cash Flows from Operation***.

Cash Collected from Customers:	**\$180k**
Cash paid to suppliers for Inv.:	**\$70k**
Cash paid for salaries:	**\$30k**
Cash paid for interest:	**\$15k**
Cash paid for income taxes:	**\$9k**
Net Cash Flows from Operations:	**\$56k**

MattCo Operating Section (Direct Method), FY ending Dec. 31, 2021

Finally, it could be useful to once again show this in a wide table like this, with the I/S items, the adjustments, and the cash-flow analogue all lined up:

I/S Item	Amount	Direct Method Adjustment	Cash Flow Analogue	Amount
Revenue:	$200k	- Δ Accounts Receivable	Cash Coll. from Customers	$180k
COGS:	-$80k	- Δ Inventory + Δ A/P	Cash Paid to Suppliers	-$70k
Salaries Expenses:	-$40k	+ Δ Salaries Payable	Cash Paid for Salaries	-$30k
Depreciation Expenses:	-$30k	Skip Over	None	None
Loss on sale of land:	-$5k	Skip Over	None	None
Interest Expense:	-$10k	+ Δ Interest Payable	Cash Paid for Interest	-$15k
Income Tax Expense:	-$10k	+ Δ Tax Payable	Cash Paid for Tax	- $9k
Net Income:	$25k	(All of the above)	**Net Cash from Operations**	$56k

Showing Income Statement, Direct Adjustment, and Cash-Basis Analogue

16.3.11 Summarizing: Three Rules

In the above, I discussed the intuition and logic of each adjustment in some detail. But, it turns out that we can boil all of that work down to just **three rules**. You may have noticed this pattern yourself:

- **Rule #1:** Adjust each **Revenue** to its cash-basis equivalent by *subtracting* the change in its related non-cash asset account (*Receivables*), *adding* the change in its related liability account (e.g., *Deferred Revenue*), if any.
- **Rule #2:** Adjust each **Expense** to its cash-basis equivalent, by doing **the opposite**, ***adding*** the change in the related asset account, if any (e.g., *Inventory* for COGS), and subtracting the change in the related liability account (typically a *Payable*).
- **Rule #3: Skip** over *Depreciation* and *Amortization*, as well as *Gains* and *Losses*, because the related cash flows are in the **Investing** section, not **Operating**.

You could boil that down even further, if you liked, given the symmetry in Rules #1 and #2. Expenses are just anti-Revenues, so it's intuitive that they have opposite rules for adjustments with their related asset and liability accounts.

Notice, also, *what kinds of accounts* did all of this adjusting for us: every single one was a **working-capital** account—*Accounts Receivable*, *Inventory*, and the various *Payables*. This is the "informal web of financing" from Chapter 7 making its scheduled reappearance. *Receivables* are, in effect, short-term loans we have extended to our customers; *Payables* are short-term loans our suppliers and employees have extended to us. Those in-kind loans are precisely what pry a company's accrual performance apart from its cash performance—and so the working-capital accounts are precisely the dials we turn to convert one into the other. When a finance professor someday tells you that 'working capital represents customer and supplier financing,' you will know exactly what they mean—and you will have personally filled out the financial statement that proves it.

After all the work we did understanding each adjustment in the direct method in detail, the *Indirect Method* will just require some **mental rotation**, and then it will be a cinch.

16.4 Operating Section: The Indirect Method

The Indirect Method is just a **different method** for filling out the **operating section** of the cash flow statement. It gets us to the exact same 'bottom line' of that section—Net Cash Flows from Operations. We use **the exact same concepts**, and, moreover, **we make the exact same adjustments** as we did in the Direct Method above. But we simply make those adjustments in a **different** way and **presentation**.

Per the name, under the Indirect Method, instead of adjusting each I/S line-item directly, one-by-one, we **start with** *Net Income*, and **apply the adjustments** required to go to *Net Cash Flows from Operations* at a more summary level. That is, we *apply the adjustments* to go between those numbers, instead of **directly** adjusting each line item. So, in broad outline, it will look something like this:

Net Income (NI):	**$25k**
...adjustments to reconcile NI to CFO	
...	$...
...	$(...)
Net change in cash from operations (CFO):	$56k

MattCo Operating Section (Indirect Method), FY ending Dec. 31, 2021

Two remarks before we dive in. **First**, that second line—"adjustments to reconcile NI to CFO"—is not just pedagogical scaffolding of mine. Many companies' actual filings print that phrase, or something very close to it, verbatim. So when you open a real 10-K and see the operating section begin with *Net Income*, followed by a line about "adjustments to reconcile," you have your tell: this company reports under the indirect method.

Second, a question that sharp students often raise at this point: if we are taking *Net Income* and stripping things out of it, aren't we just backing our way into one of the intermediate subtotals of the Income Statement—'operating income,' or 'EBIT' (earnings before interest and taxes), which you may have heard tossed around? No. Those are still **accrual**-basis earnings measures: revenues minus some subset of expenses, just with the tape cut at a different line. What we are doing here is categorically different—converting an accrual measure into a **cash** measure. The destination is not a different flavor of earnings; it is cash.

Using what we learned from the Direct Method, we can do the Indirect Method just by making **two key mental rotations**:

- **Mental Rotation #1:** Because the Indirect Method **starts with** *Net Income*, and since *Depreciation* and *Amortization* and *Gains* and *Losses* have **already** been included in calculating *Net Income*, we have to **back them out** (instead of skipping over them).

  – I imagine the "ghost of the Income Statement" above.

- **Mental Rotation #2:** In the direct method, we first calculate **Cash Paid for Inventory**. An increase in Inventory makes that payment larger. We then subtract the cash payment when calculating operating cash flow.

  The indirect method begins with Net Income, where COGS has already been deducted. To make the same adjustment from that starting point, we subtract the increase in Inventory directly. The two routes have the same final effect; the direct method performs the calculation in two steps. The related payable works the other way: an increase means some cost has not yet been paid, so it increases cash flow relative to income.

16.4.1 Example

This will become clearer once we dive in to an example. Let's fill out a Cash Flow Statement using the Indirect Method, using the same example we used last time. I reproduce the same Income statement and Balance Sheet below:

Revenue:	**$200k**
COGS:	**$80k**
Salaries expenses:	**$40k**
Depreciation expenses:	**$30k**
Loss on sale of land:	**$5k**
Interest expense:	**$10k**
Income tax expense:	**$10k**
Net Income:	**$25k**

MattCorp Income Statement, For the Year Ended December 31, 2021:

	2021	2020
Assets:		
Cash	**$230k**	**$220k**
Accounts Receivable	**$90k**	**$70k**
Inventory	**$120k**	**$140k**
Investments	**$130k**	**$0**
Land	**$220k**	**$280k**
Equipment	**$250k**	**$200k**
Accumulated Depreciation	**$(80k)**	**$(50k)**
Total assets:	**$960k**	**$860k**
Liabilities:		
Accounts payable	**$80k**	**$90k**
Salaries payable	**$30k**	**$20k**
Interest payable	**$7k**	**$12k**
Income Tax payable	**$16k**	**$15k**
Bonds payable	**$260k**	**$220k**
Shareholders' Equity:		
Common Stock	**$432k**	**$383k**
Retained Earnings	**$135k**	**$120k**
Total Liabilities and S.E.:	**$960k**	**$860k**

MattCorp Balance Sheets, Year Ended December 31:

Let me shortcut to **the answer**, and then explain what's going on. Under the indirect method, the **Cash Flow Statement** would look like this:

Net Income (NI):	**$25k**
...adjustments to reconcile NI to CFO	
+ Depreciation Expense	$30k
+ Loss on Sale of Land	$5k
- Δ Accounts Receivable	($20k)
- Δ Inventory	$20k
+ Δ Accounts Payable	($10k)
+ Δ Salaries Payable	$10k
+ Δ Interest Payable	($5k)
+ Δ Income Tax Payable	$1k
Net change in cash from operations (CFO):	$56k

MattCorp Operating Section (Indirect Method), FY ending Dec. 31, 2021

If you look and think carefully, you'll be able to see that these are the **exact same adjustments** we made under the *Direct Method*, just with those two **mental rotations**.

- **First**, we **added back** *Depreciation* and the *Loss*. We add these back because they are not operating cash flows, but they were *already* taken out of *Net Income*. (Symmetrically, we would subtract out a *Gain*.) This is the analogue to "skipping over" these items in the **direct method**. **(Mental Rotation #1.)**.

- **Second**, **everything else** is the same adjustments we made in the Direct Method, just with the **expense adjustments 'flipped'** up front. **(Mental rotation #2.)** (Note that we don't need to 'flip' the adjustment we made to *Revenue*, since *Cash Collections from Customers* **was not** subtracted in the Direct Method.)

  This means that for the **Indirect Method**, we have **just one rule** for all of the current asset and liability adjustments: We **subtract changes in current operating assets** (other than cash itself) and **add** changes in **current operating liabilities**, after excluding noncash changes. Borrowings and other financing balances are not operating adjustments. You may recall from Chapter 5 that a company's current asset and liability accounts are called its **"working capital."**

16.4.2 Comparing Direct and Indirect Methods

To make this ultra clear, let me show the Indirect Method, and the table tracking the adjustments we made under the Direct Method again, so you can compare them side-by-side:

Net Income (NI):	**$25k**
...adjustments to reconcile NI to CFO	
+ Depreciation Expense	$30k
+ Loss on Sale of Land	$5k
- Δ Accounts Receivable	($20k)
- Δ Inventory	$20k
+ Δ Accounts Payable	($10k)
+ Δ Salaries Payable	$10k
+ Δ Interest Payable	($5k)
+ Δ Income Tax Payable	$1k
Net change in cash from operations (CFO):	$56k

MattCo Operating Section (Indirect Method), FY ending Dec. 31, 2021

I/S Item	I/S Amount	Direct Method Adjustment	Cash Flow Analogue	Cash Flow Amount
Revenue:	$200k	- Δ Accounts Receivable	Cash Coll. from Customers	$180k
COGS:	-$80k	- Δ Inventory + Δ A/P	Cash Paid to Suppliers	-$70k
Salaries Expenses:	-$40k	+ Δ Salaries Payable	Cash Paid for Salaries	-$30k
Depreciation Expenses:	-$30k	Skip Over	None	None
Loss on sale of land:	-$5k	Skip Over	None	None
Interest Expense:	-$10k	+ Δ Interest Payable	Cash Paid for Interest	-$15k
Income Tax Expense:	-$10k	+ Δ Income Tax Payable	Cash Paid for Tax	- $9k
Net Income:	$25k	(All of the above)	**Net Cash from Operations**	$56k

Showing Income Statement, Direct Adjustments, and Cash-Basis Analogues

As you can see, these are the **same adjustments**, just factored in in a different way, which we can **understand** with those **two mental rotations**.

16.4.3 Boiling The Indirect Method Down to Just Two Rules:

So, that's how to **understand** the Indirect Method. To implement it, you can essentially boil it all down to **just two rules**.

Net Income (NI):	**$25k**
...adjustments to reconcile NI to CFO	
+ Depreciation Expense	$30k
+ Loss on Sale of Land	$5k
- Δ Accounts Receivable	($20k)
- Δ Inventory	$20k
+ Δ Accounts Payable	($10k)
+ Δ Salaries Payable	$10k
+ Δ Interest Payable	($5k)
+ Δ Income Tax Payable	$1k
Net change in cash from operations (CFO):	$56k

MattCorp Operating Section (Indirect Method), FY ending Dec. 31, 2021

- **Rule #1: "Back out"** the non-cash / non-operating items from the Income Statement: *Depreciation* and *Amortization* and *Losses* and *Gains*.
  - This means *adding* D&A and *Losses*, and subtracting *Gains*.
- **Rule #2:** Everything else is a **working capital adjustment**:
  - **Subtract** the operating changes in non-cash **current operating assets** (like *Inventory* and *A/R*)
  - **Add** the operating changes in **current operating liabilities** (such as trade and payroll payables).

That's it. That algorithm will allow you to fill out the operating section correctly under the indirect method.

The worked reconciliation and the two rules above are enough to continue to the investing section. If you would like another route to the intuition, the following box and explanations revisit the add-back and working-capital adjustments.

16.4.4 Going deeper: Does adding back depreciation create cash?

Suppose Depreciation Expense were $40k instead of $30k, with the cash transactions and other assumptions unchanged. Would the larger add-back increase operating cash flow?

No. Net Income would begin $10k lower, because the larger depreciation expense had already been deducted. The reconciliation would then add back $10k more. The changes cancel.

The add-back is part of the route from income to cash; it is not itself a receipt. The expenditure for the long-term asset belongs in the investing section when the cash is paid. Keeping the starting point, adjustment, and destination separate resolves the apparent paradox.

16.4.5 Why gains and losses are removed in the operating cash-flow reconciliation

The gain or loss affected Net Income, but the full proceeds from selling the long-term asset belong in the **investing** section. We therefore remove the gain or loss from the operating reconciliation. Later, we use book value and the gain or loss to reconstruct those proceeds. This is the same classification we used when we skipped the item under the direct method.

16.4.6 Explaining everything else: working capital adjustments

In our simplified example, the remaining adjustments are changes in **operating working capital**, and follows just one rule: **Subtract** the operating change in current non-cash **operating assets**, and **add** the operating change in current **operating liabilities**. We've already built and stated that rule a couple of times now. But this is hard stuff to grasp the first time around. So here are some concrete examples to help you visualize and build intuition:

- If *Accounts Receivable* **increased**, that means that the company made a credit

sale, which means it recognized *Revenue* (which goes into *Net Income*) that wasn't collected in cash. So to go back **from** *Net Income* (NI) toward *Cash Flows From Operations* (CFO), we subtract out that increase.

- If *Accounts Receivable* **decreased**, it's just the opposite: That means we collected cash on prior accounts, but that wasn't recognized as revenue in this period. So, to go from NI to CFO, we add that decrease back.
  - Generalizing and netting: To subtract the increases and add the decreases in A/R, we just **subtract the net change** between periods.
- If a current asset like *Inventory* (or *Supplies*) increased, that means the company **purchased** that asset, **but capitalized** that expenditure as an asset, so it was **not** recognized as an **expense**. But purchases of *Inventory* and *Supplies* **are** considered operating cash flows. So, to move from NI toward CFO, we subtract out that increase, just like with A/R. (If the company made that purchase on account, that will be taken care of with our adjustments for payables, below.)
- If a current asset like *Inventory* decreased, it's just the opposite: That means that the asset was 'expensed out' (via *COGS* in the case of *Inventory*), even though the actual cash purchase had been made in the past. So, to go from NI toward CFO, we add that decrease back.
  - Generalizing and netting: To subtract the increases and add the decreases in these **current assets**, we just **subtract the net change** between periods.

For **current operating liabilities**, the logic is symmetric. The payables in our example all relate to operating activities; a current portion of a bank loan would instead be a financing item.

- If *Salaries Payable* increased, that would mean that we incurred *Salaries Expense* that we haven't yet paid in cash. Since *Salaries Expense* is already deducted from NI, to go from NI to CFO, we need to **add back the increase** in *Salaries Payable*.
- Symmetrically, if *Salaries Payable* decreased, then that would mean we paid down past payables in cash, related to expenses that had been recognized in previous periods (that is, not deducted from this period's NI). So, to go from NI to CFO, we need to **subtract that decrease**.
  - Generalizing and netting: To add the increases and subtract the decreases in *Salaries Payable*, we just **add the net change** between periods.
- The same logic applies to the operating payables in this example: *Interest Payable, Income Tax Payable, Accounts Payable*, etc. If a *Payable* increases, that

indicates an accrual *Expense* that **was not paid in cash**—so we need to add it back in going from NI to CFO.

- If a *Payable* decreases, that indicates it was **paid down**—but the payment of **past periods' expenses** is not an expense in this period. So that payment was a cash outflow, but had not been deducted from NI. So, to go from NI to CFO, we need to **subtract that decrease**.

  – Generalizing and netting: To add the increases and subtract the decreases in these *operating Payables* and other **current operating liabilities**, we just **add the net change**.

16.4.7 Reading the reconciliation as a diagnostic

One of the running themes of this book has been the tension between **accrual profitability** and **cash flows**. So suppose an analyst confronts the situation we flagged at the start of this chapter: a company with solid *Net Income* but weak *Cash Flows from Operations*. What could account for the difference? Here is the beautiful thing about the indirect method: it answers that question *explicitly*. Whatever is driving the wedge between NI and CFO, it is—literally—the list of adjustments in the operating section. That is what the reconciliation *is*.

So you can read the "usual suspects" right off the statement. Is the company recognizing lots of revenue on credit sales without collecting the cash (a ballooning *Accounts Receivable*—i.e., financing its customers)? Is it spending heavily on *Inventory* that isn't selling (a ballooning *Inventory* balance)? Is it being propped up, cash-wise, by stretching out its suppliers (a ballooning *Accounts Payable*)? Are unusually large *Depreciation* charges dragging down NI while leaving cash untouched? None of these, by itself, proves anything sinister—growing companies legitimately build up receivables and inventory. But these lines are where a serious analyst starts digging. The indirect-method reconciliation, which may have looked like an arbitrary pile of pluses and minuses a few pages ago, is in fact a diagnostic instrument: a line-by-line decomposition of the gap between the two 'bottom lines.'

16.5 Investing section:

The **operating section** is the main meat of the cash flow statement. But the other sections are still non-trivial, both in difficulty and importance. We'll now proceed to the **investing** section. (Remember that the distinction between the indirect and direct method does not apply here—that distinction is only for the operating section.)

Investing activities are, by definition, about the **purchase and sale of long-term assets**. So, let's start by **literally looking at** the long-term assets on the company's Balance Sheet, and using our accounting knowledge to figure out what must have happened. In the interest of space, I include just *that section* of the Balance Sheet below:

	FY2021	**FY2020**
Assets:		
Cash	$230k	$220k
Accounts Receivable	$90k	$70k
Inventory	$120k	$140k
Investments	$130k	$0
Land	$220k	$280k
Equipment	$250k	$200k
Accumulated Depreciation	($80k)	($50k)
Total Assets:	**$960k**	**$860k**

MattCorp Balance Sheets, Asset Section

MattCorp only has three long-term assets—**Investments, Land, and Equipment**—so our work will be fairly easy. A **major risk** in filling out this section is **overthinking it**.

16.5.1 Investments

Start with **Investments**. Its balance rose from $0 to $130k. The problem states that this change came entirely from cash purchases, so the investing cash outflow is $130k. Without that assumption, the increase alone would not be enough: purchases and sales, noncash acquisitions, and fair-value changes could combine to produce the same ending balance.

16.5.2 Land

Between 2020 and 2021, the company's *Land* account went down by $60k. What happened? We know there are **three things that make assets go down**:

- (i.) the expense from 'using them up'
- (ii.) sale / disposal
- (iii.) impairments.

We learned in Chapter 11 that we **do not depreciate *Land***. The problem also states that there were no impairments or other land transactions.

So, by process of elimination, this means that we now know that this company **sold off *Land* that had a book value of $60k**. But was that the **sale price, the cash flow we're interested in? No.** $60k is the prior book value of the Land, and the sale price is a negotiated market price that can be higher or lower. The Income Statement indicates a ***Loss* from the sale of *Land*** of $5k. If the company sold land with a book value of $60k, and incurred a loss of $5k, that means the actual **sale price** was $55k.

Earlier in this book we learned that Gains and Losses on sales 'plug' the difference between the book value of the asset and the sale price.

$$Gain\ (Loss) = Sale\ Price - Book\ Value$$

Rearranging, this means that we can calculate the **Sale Price** using the prior book value of the asset sold, and the gain (loss) upon sale.

$$Sale\ Price = Book\ Value + Gain\ (Loss) = \$60k - \$5k = \$55k$$

Note also that here, in the **investing** section, is where *Gains and Losses* **ultimately end up**, after they were **backed out** of the **operating** section. This is why it was important to get the logic there correct.

16.5.3 Equipment

Finally, *Equipment* went up from $200k to $250k. As before, if a long-term asset account goes up, that means there was **purchase** debited to that account—and, unless we are told otherwise, we can default to the assumption that it was in cash. Do we need to worry about anything else here? Backing up, we know that, overall, there are **four things** that can change a long-term asset balance:

- (i.) purchases,
- (ii.) the expense from 'using them up',
- (iii.) sale / disposal,
- (iv.) impairments.

Do we need to **worry** about any of the **others** here? In this case, **no**. The reason is the stated assumption that Equipment had no disposals, impairments, or noncash acquisitions. A sale at book value would generate no Gain or Loss, so the absence of those lines would not, by itself, prove that no sale occurred. And, since this company reports **separate accounts for Equipment vs. its Accumulated Depreciation** (rather

than reporting *Equipment* at book value), this means that the *Equipment* account *per se* is unaffected by the *Depreciation Expenses* charged. (Much earlier in this book, I told you that contra-accounts would, despite their initial tortuousness, end up making our lives easier. This is one such case.) So, this means that we don't need to overthink this one, and can **directly infer** that the company **purchased $50k in *Equipment***.

Now, an anticipated question: this inference was easy *because* MattCorp reports *Equipment* and *Accumulated Depreciation* as separate line items. What if it didn't—what if the Balance Sheet showed only a single net line, '*Equipment, net*'? Then we would have to be a slightly more energetic detective, triangulating with the Income Statement. In this example, the net book value went from $150k ($200k − $50k) to $170k ($250k − $80k), while the Income Statement reports $30k of *Depreciation Expense*. Run the familiar logic: had we purchased nothing, depreciation would have carried the book value from $150k down to $120k. Instead, it ended at $170k. So we must have purchased $50k of *Equipment*—the same answer, reached via the same 'beginning balance, plus purchases, minus expense' identity we have leaned on throughout this book.

16.5.4 Consolidating

There are no other long-term asset accounts, so nothing to indicate any other investing cash flows. So, the final step is just to pull it together, and net out the total investing cash flows:

Investing Cash Flows	
Purchase of Investments	$(130k)
Cash from Sale of Land	$55k
Purchase of Equipment	$(50k)
Net Cash Flows from Investing	**$(125k)**

MattCorp Cash Flow Statement, Investing Section, FY 2021

16.5.5 Boiling Down to Rules: Investing Section

Under the simplified assumptions used in these exercises, the **Investing** section can be approached this way. In real filings, use transaction details and notes to separate gross purchases, sales, and noncash changes:

- **#1**: *Look at* long-term asset accounts on the Balance Sheet.
- **#2**: If a long-term asset has **increased**, that indicates a **purchase**. Assets are **written up** at their purchase price, and we can assume it was a **cash transaction, by default**.

- **#3**: If a long-term asset has **decreased** and that is not accounted for by the associated expense (e.g., *Equipment* decreasing for a company that tracks *Accumulated Depreciation* separately) or an impairment, that indicates a **sale**.
- **#4**: We can infer the **cash sale price** of long-term assets by triangulating between their **prior book value** and the *Gain* or *Loss* reported on the Income Statement.
- **#5**: When it's **complicated** (i.e., there is an *Impairment*, or *Equipment* is reported at book value), **slow down** and use your knowledge, like a **detective**.

16.6 Financing section

The next and **final section** of the cash flow statement is the **financing section**. Broadly, this section tracks and nets the cash flows related to **how the company finances itself**—transactions in its own stock and debt (with the notable exception that interest payments on debt are categorized as operational, to make the operating section more comparable to the Income Statement).

The company's major transactions in its bonds and stocks will be reflected in the **long-term liabilities** and **Shareholders' Equity** sections of the **Balance Sheet**. So, once again, let's start by **literally looking there**. There will be one very **important thing to remember at the end**—but *looking* will get us most of the way there. See the *Liabilities* and *Shareholders' Equity* sections of the Balance Sheet from the original problem below:

	FY2021	**FY2020**
Liabilities:		
Accounts payable	$80k	$90k
Salaries payable	$30k	$20k
Interest payable	$7k	$12k
Income tax payable	$16k	$15k
Bonds payable	$260k	$220k
Stockholders Equity:		
Common Stock	$432k	$383k
Retained Earnings	$135k	$120k
Total Liabilities and S.E.:	**$960k**	**$860k**

MattCorp Balance Sheets, FYs ending Dec 31, LT Liabilities and Shareholders' Equity Sections excerpted

The **first four** liability accounts are all **current** liability accounts related to short-term, operating cash flows. We already accounted for these in the operating-cash flow

section. So, the only liability we need to worry about in this section of the problem is *Bonds Payable*.

16.6.1 Bonds Payable

The problem states that the bonds were issued *at par*. The absence of a separately displayed Discount or Premium would not establish that in an actual company's statements, because debt may be reported net. (And we will always **make that simplifying assumption** in this section, in this introductory text.) Since the *Bonds Payable (L)* account went up by $40k, and is *at par*, that means that our company must have received $40k in cash from the issuance of a bond. We have no reason or information to believe any old Bonds were called or repurchased. So, we can infer that this company issued $40k in bonds at par, a **financing cash inflow of $40k**, in this year.

16.6.2 Common Stock

Next, the problem specifies **no-par Common Stock** and only cash issuances during the year, and the cash proceeds from issuing shares is fully credited to the *Common Stock* account. From 2020 to 2021, the *Common Stock* account went up by $49k. This directly indicates that the company got a **$49k cash inflow** from the sale of *Common Stock*. Do we need to worry about *repurchases*? The problem explicitly rules out repurchases and retirements. A zero or absent Treasury Stock balance alone would not rule them out.

16.6.3 The Key Final Financing Cash Flow: Remember Dividends

Are we **done** here? **Not quite.** We've ***looked at*** all the long-term liabilities and Shareholders' Equity accounts on the Balance Sheet. But there is one key **financing cash flow** that is **not *directly*** tracked here: ***Dividends***. We know that companies pay *Dividends*, that it's an important financing cash flow. But *Dividends*, as we learned from Chapter 1, are not *Expenses*, so they are not reported in the Income Statement. And they are a temporary account, so they are not reported on the Balance Sheet. So, we won't **see *Dividends* anywhere**. But we do have **enough information to infer** what our company paid. Remember that:

$$\Delta \textit{Retained Earnings} = \textit{Net income earned in period} - \textit{Dividend paid out in period}$$

On our two Balance Sheets, we are told that *Retained Earnings* increased by $15k (from $120k in FY2020 to $135k in FY2021). And, as reported on the Income Statement,

the company earned $25k in Net Income during the period. This means that the company **must have paid out $10k in *Dividends***. Or, if you want a **general formula**, we can rearrange the identity above like so:

$$Dividends\ Paid\ out\ in\ Period = Net\ Income - \Delta\ Retained\ Earnings$$

Personally, I prefer the version in words to the formula: we *earned* $25k for our shareholders this period, but *Retained Earnings*—the account that accumulates whatever we earn and don't pay out—only went up by $15k. The missing $10k must have gone somewhere. It walked out the door, to the shareholders, as *Dividends*.

So, pulling that all together, our financing section would look like this:

Financing Cash Flows	
Issuance of Bonds	$40k
Issuance of Common Stock	$49k
Dividends Paid Out	$(10k)
Net Cash Flows from Financing	**$79k**

MattCorp Cash Flow Statement, FY '21, Financing Section

16.6.4 Boiling Down to Rules: Financing Section

Under this exercise's assumptions, we can approach the **Financing** section as follows. Outside those assumptions, use the supporting disclosures to identify cash issuances, repayments, repurchases, and noncash changes separately:

- **#1**: *Look at* the long-term liability and Shareholders' Equity accounts on the B/S.
- **#2**: If *Bonds Payable* or *Common Stock*, etc., has *increased*, that indicates a financing **cash inflow** for the company.
- **#3**: If *Bonds Payable* has decreased that indicates a repurchase or call, or paying off face value at maturity. If *Treasury Stock*, has *increased*, that indicates a stock repurchase. Both are financing **cash outflows.**
- **#4**: **Remember to think about *Dividends*,** a key financing **cash outflow**. They are not *reported* anywhere on the B/S or I/S. But you can infer them using the **Retained Earnings identity**.
- **#5**: When it's more **complicated**, **slow down** and be a **detective**.

16.7 Consolidating the Three Sections:

The last thing we do is consolidate the three sections together, ***net*** the **net change in cash** in each category, and check that it equals the change in cash on the Balance Sheet. The change in cash from this calculation is $10k, which is the same change we see inspecting the Balance Sheets. So, I can sigh with relief: I probably got this problem right.

Operating cash flows:	
Net Income:	**$25k**
... adjustments to reconcile NI to CFO	
+ Depreciation Expense	$30k
+ Loss on Sale of Land	$5k
- Δ Accounts Receivable	($20k)
- Δ Inventory	$20k
+ Δ Accounts Payable	($10k)
+ Δ Salaries Payable	$10k
+ Δ Interest Payable	($5k)
+ Δ Income Tax Payable	$1k
Net change in cash from operations:	**$56k**
Investing Cash Flows:	
Purchase of Investments	($130k)
Cash from Sale of Land	$55k
Purchase of Equipment	($50k)
Net Cash Flows from Investing	**($125k)**
Financing Cash Flows	
Issuance of Bonds	$40k
Issuance of Common Stock	$49k
Dividends Paid Out	($10k)
Net Cash Flows from Financing	**$79k**
Net change in cash:	**$10k**

MattCorp Cash Flow Statement, For the Year Ended December 31, 2021 (indirect method)

16.8 Some 'tricks': e.g., non-cash transactions

In the above, I wanted to lay out the main ideas in a clean example, and then try to provide simple rules, to make things easier as you learn. But there are a couple of things that can get thrown into problems, which **you have the knowledge** to solve, but which require going **beyond the rules above**. For example, suppose that our company's *Common Stock* was ***not*** no-par. Then you should be able to apply your knowledge from Chapter 15 to still infer the **cash proceeds** from the issuance of *Common Stock* (i.e., by looking at *APIC*.)

Let's make that concrete, since it also gives us a chance to consolidate the machinery from the previous chapter. Suppose that a company's stock carries a par value of $1 per share, and that it reports the following Shareholders' Equity accounts:

	FY2021	**FY2020**
Common Stock	$1.5m	$1m
APIC	$40m	$10m
Treasury Stock	$0	$0

Shareholders' Equity accounts, FYs ending Dec 31

What was the financing cash inflow from issuing *Common Stock*? For this worked example, assume the only changes in Common Stock and APIC were cash issuances, with no repurchases, retirements, stock-based compensation, or other noncash changes. Zero Treasury Stock balances alone would not establish those facts. Now recall the journal entry for issuing par-value stock: debit *Cash* for the total proceeds, credit *Common Stock* for the par value of the shares issued, and credit *APIC* for the remainder. Which means: *Common Stock* and *APIC*, *together*, track the total proceeds. *Common Stock* rose by $0.5m and *APIC* rose by $30m—so the company collected **$30.5m** in cash from issuing stock. (As a bonus inference: at $1 par, the $0.5m increase in *Common Stock* tells us the company issued 500,000 new shares—so it issued them at an average price of $30.5m / 500,000 = $61 per share.)

Treasury Stock, for its part, works even more simply, because par value plays no role in repurchases at all: we debit *Treasury Stock* and credit *Cash* for the full repurchase amount. If Treasury Stock rose from $0 to $200m solely through cash repurchases, with no reissuances, retirements, or other changes, the repurchase outflow would be $200m. Without those facts, the net balance change would not identify gross repurchases.

Another common trick in these problems is to include **non-cash purchases of long-term assets**. As we saw in Chapter 12, companies can finance the purchase of long-term assets by signing a *Note Payable*, etc. By default, we would normally

assume the purchase of a long-term asset was in cash. But suppose we are explicitly told that the company financed the purchase of new *Equipment* with a *Note Payable*. That means that the company *directly* swapped *Equipment* for a *Note*—without cash changing hands (yet). That is, the purchase of the *Equipment* was ***not* made in cash**, and the issuance of the *Note Payable* was not a **cash inflow**. If the transaction was **not in cash**, it's **not a cash flow**, full stop. And the Statement of Cash Flows is, **by definition**, about tracking the actual flow of cash. So, in that case, we would **not** put that amount ($) in either the investing section (purchase) or the financing section (*Note*). Instead, we would just **add a footnote** at the bottom of the cash flow statement, saying something like, **"Note: $50k in Equipment purchased via issuance of Note payable."**

Here is another worked complication, of the 'long-term assets at book value' variety. Suppose our company owns several patents, which it reports at book value (i.e., with no separate accumulated-amortization line). At the end of FY2019, it reported *Patents* of $900k. Over FY2020, it recorded $200k of *Amortization Expense*, recorded a $300k impairment *Loss* on one of its patents, and ended the year with *Patents* of $800k. What was the investing cash outflow from purchasing patents? Assume all purchases were paid in cash and there were no other changes in the account. Be a detective. Had the company purchased nothing, the amortization and the impairment together would have carried the balance from $900k down to $400k. It ended at $800k. So the company must have spent **$400k** purchasing patents. In identity form:

$$Ending\ BV = Beginning\ BV + Purchases - Amortization - Impairment$$

$$\$800k = \$900k + Purchases - \$200k - \$300k \;\Rightarrow\; Purchases = \$400k$$

One more. Throughout this chapter, our example company sold products, so the two-step expense adjustment ran through *COGS* and *Inventory*. But suppose we were a *services* company instead—no products, no *Inventory*—whose only short-term operating asset was *Supplies* (printer paper, toner, and the like), with an *Accounts Payable* owed to the suppliers of those supplies. What then? Nothing new, really: *Supplies* plays exactly the role that *Inventory* played. Under the Direct Method:

$$Cash\ Paid\ for\ Supplies = Supplies\ Expense + \Delta\ Supplies - \Delta\ Accounts\ Payable$$

And under the Indirect Method, you don't even need to think that hard: *Supplies* is a current asset, so you subtract its change, like any other. The rules were never really about *Inventory* per se—they were about current operating assets and their related payables.

You may see other miscellaneous complications thrown into cash-flow statement

problems. There's no way to boil all of them down to one set of rules. The cash flow statement is ultimately about using your **understanding** of what the Balance Sheet and Income Statement *mean* to infer what the underlying transactions were. With everything you have learned over the course of this book, that's a lot to go on! So, when you see a wrench thrown into these problems, slow down, have courage, and show yourself what you can do.

16.8.1 Going deeper: The statement over a firm's life cycle

We have now completed the statement. Let's step back and do some first-principles thinking about what it should look like for different kinds of companies. Denote the three section totals X (net operating cash flows), Y (net investing cash flows), and Z (net financing cash flows). For a given company, which of these should we expect to be positive, and which negative?[a]

Consider, first, an **early-stage startup**. It is spending on product development, engineers, marketing, perhaps building up inventory—but it is not yet generating much in sales. So X, its operating cash flow, will typically be **negative**: it is burning cash. And that is precisely *why* it must raise outside financing, making Z **positive**. Indeed, you can run the logic in reverse: if a young company *were* already generating healthy positive operating cash flows, why would it sell off chunks of its ownership to a venture capitalist? The very fact that a company is out raising capital is usually a sign that its operations do not yet pay for themselves.[b]

Now consider a **mature, successful, household-name company**—the kind that populates the S&P 500. Its situation is the mirror image. It is successful precisely because it makes a great deal of money selling products and services to customers, so X is **strongly positive**. And because its customers are, in effect, funding it, it has little need to keep raising capital from the markets. What should it do with the excess cash? Corporate finance gives the classic answer: return it to the providers of capital—pay dividends, repurchase shares, pay down debt. All of those are financing *outflows*, so for mature companies, Z is typically **negative** on net. Pull up the cash flow statement in the 10-K of practically any household name and check: the large majority report negative net financing cash flows, year after year.

This gives you a lens on a complaint you may have heard in public conversation: that big public companies "aren't really using the stock market to raise capital anymore"—that "all they do is pay dividends and buy back stock." Whatever your moral or political read on that, notice that, from a purely financial perspective, it is hard to see how it could be otherwise. A company gets to be a household name by

succeeding; succeeding means generating cash from operations; and a company generating ample cash from operations has no reason to keep diluting its shareholders for more. The stock market's capital-*raising* function is mostly for young firms; for mature ones, the pipe naturally runs the other direction. That is not a scandal—it is the life cycle completing itself. Investors put cash in at the beginning precisely in order to take more cash out at the end.

What about Y, the investing cash flows? Here there is no neat life-cycle reversal: for a normal operating company, Y is typically **negative** at *every* stage. Think about what the investing section contains: mostly, purchases of long-term assets, punctuated by occasional disposals. A company continually buys equipment at full price and, years later, sells it used—almost always for far less than it paid, since it has been using the thing up. Early on, you build capacity; later, you replace and upgrade it. Either way, the outflows to acquire long-term assets systematically dwarf the dribble of inflows from disposing of old ones. (A company with persistently *positive* investing cash flows is a company selling off its asset base—occasionally a strategy, but more often a distress signal.)

So: the canonical startup has $X < 0$, $Y < 0$, and $Z > 0$—burning cash on operations and investment, funded by financiers betting on the future. The canonical mature company has $X > 0$, $Y < 0$, and $Z < 0$—cash pouring in from customers, continually reinvested in the asset base, with the remainder handed back to investors and creditors. These patterns are useful starting points, not rules for classifying a company from three signs alone. A profitable company can raise financing for expansion, and positive investing cash flow can reflect a planned asset sale or sales of investment securities. Read the transactions behind the totals.

[a] Look back at the schematic statement at the start of this chapter: even there, in a generic template, the investing total was written as (Y)—parenthesized, negative—while X and Z were left unadorned. That was not an accident, as we are about to see.

[b] You might wonder whether startups do this fundraising by borrowing from banks or by selling equity. Overwhelmingly, the latter. Recall the logic of leverage from Chapter 6: debt multiplies risk, and early-stage companies have plenty of that already. They also hold little tangible collateral—a banker cannot repossess an idea. But they are natural candidates for equity, which offers financiers the upside needed to compensate for the risk. Hence the venture-*capital* industry, rather than a venture-lending industry.

16.9 Applications: What Do Analysts Do With It?

We have now built the entire statement—and the operating section twice over. Let's close the loop on *why*—what do people actually *do* with a Statement of Cash Flows, out in the world?

Cash-basis performance ratios. Recall the ratio-analysis toolkit from Chapter 6. Every ratio we built there with *Net Income* in the numerator has a natural **cash-**

basis cousin: simply substitute *Cash Flows from Operations*. Return on Assets ($NI/Avg.\ Assets$) becomes a 'cash return on assets' ($CFO/Avg.\ Assets$); profit margin ($NI/Revenue$) becomes a 'cash margin' ($CFO/Revenue$); and so on. For all the reasons this chapter has covered, running the two side-by-side can be illuminating: when an accrual-based ratio is doing something you cannot explain, the trajectory of its cash-based cousin is often the clue.

Valuation. Recall, from Chapter 13, what the value of a financial asset *is*: the present value of its expected future cash flows. For a *bond*, those cash flows are specified in the contract—you can read them off and discount them. But for *equity*, no contract specifies anything: to value shares, you must *forecast* the cash the firm will generate that is available, after necessary reinvestment, to be paid out to its owners. Where does such a forecast begin? With the firm's historical financial statements—and with the Statement of Cash Flows above all. This is the essence of the "free cash flow" analysis you will meet if you go on to study finance, and it is why serious valuation work is impossible without the material in this chapter.

Credit and liquidity analysis. Finally, the original constituency: creditors. A banker deciding whether to extend a five-year loan is, at bottom, forecasting whether the company's cash generation will cover its interest and principal payments over that horizon. And the same analysis runs *inside* the company, too: a Chief Financial Officer planning next year is asking how much cash the operations will throw off, how much the investment plans will consume, and therefore how much financing will need to be raised—which is to say, she is drafting next year's Statement of Cash Flows before it happens.

16.10 Coda: Which Is Better—Net Income or Cash Flows?

We opened this chapter with the case *against* accrual accounting: the estimates, the discretion, the gimmicks, the investors pleading 'just show me the cash.' And we have now spent a chapter building the statement that answers that plea. So let me pose the question directly: if you had to evaluate a company's performance using *either* its Income Statement *or* its Cash Flow Statement—but not both—which would you choose?

When I pose this question to my students at this point in the course, the verdict is nearly unanimous: the Cash Flow Statement. And I understand why—the freshest material in their minds is a chapter's worth of net income's defects. But this is a genuine, live debate, in practice and in the academic literature, and I want to close the chapter

by making sure you can argue both sides. Because the case for cash is *not* as one-sided as it looks from here.

16.10.1 Cash measures can be gamed, too

A thought experiment. Suppose our company adopts a bold new strategy: we will become systematically evasive about paying our employees and suppliers. 'Give me another month,' we say. 'Give me another quarter.' We delay, we stall, we stiff. Set aside the ethics (and the long-run wisdom) of this strategy, and just ask: what happens to our financial statements?

Our *Cash Flow Statement* looks *better*. Cash is cash: if we are not paying it out, our Cash Flows from Operations improve, mechanically. But our *Income Statement* does not improve at all. Our employees are still working; our suppliers are still delivering; so we are still *incurring* those expenses, and the unpaid amounts simply pile up as *Payables*. Accrual accounting sees straight through the maneuver—indeed, this is precisely what accrual accounting was *built* to see through: it measures the activities we actually undertook, not the checks we happened to write. So the manipulability argument cuts both ways. Just as there are accrual gimmicks that cash flows expose, there are cash gimmicks that accruals expose.

16.10.2 The missing charge for long-term assets

Here is a deeper, structural point. Ask yourself: for a normal, steady-state company doing nothing sketchy at all, which should be higher on average—*Net Income*, or *Cash Flows from Operations*?

Think about the structure of the indirect-method reconciliation. The working-capital adjustments are **timing differences**: the check hasn't arrived, and then it arrives; the bill hasn't been paid, and then it is paid. Receivables and payables cannot drift off to infinity—over time, these adjustments reverse and wash out. But the *Depreciation* add-back is different. It never reverses. It recurs every single period, always with the same sign, for as long as the company holds long-term assets—which, for a normal operating company, is forever. For a steady-state company with roughly stable operating working capital and no offsetting items, the depreciation add-back tends to make **CFO exceed NI**. Growth in receivables or inventory, gains, and other adjustments can reverse that comparison.

Now interpret that fact. It means that *Cash Flows from Operations* is a performance measure that makes **no charge whatsoever for the cost of the company's long-term assets**. Those factories and machines were paid for with real cash—cash that went out through the investing section—and the company must keep spending to replace them,

forever, before a dollar can go to shareholders. Are those costs relevant to judging performance? Obviously. The Income Statement charges them, smoothly, as *Depreciation* and *Amortization*; CFO simply never does.[3] If you evaluate companies on CFO alone, you are grading them on a curve that ignores their single largest category of investment.

16.10.3 The verdict

So: net income runs on estimates that can be massaged, but it charges companies for all the resources they consume. Operating cash flow is hard to fake in the ways accruals can be faked, but it can be gamed through payment timing, and it gives investment costs a free pass. If you truly forced me to pick one, I would—perhaps unfashionably—take old-school accrual *Net Income*. In my reading, the empirical research generally supports this: accrual earnings, for all their warts, tend to contain more information about a company's future performance and value than raw cash flows do.[4]

But of course, nobody is forcing you to pick one. The financial statements were designed as a *system*: the Balance Sheet, the Income Statement, the Statement of Shareholders' Equity, and the Statement of Cash Flows, each viewing the same underlying reality from a different angle, each checking the others. A serious analyst uses the whole mosaic. And you, now, can read every panel of it—and you know how each one is made, down to the journal entries.

[3]The same critique applies to **EBITDA** ('earnings before interest, taxes, depreciation, and amortization'), a non-GAAP measure beloved on Wall Street, which by construction also strips out the D&A charge. Warren Buffett has spent decades ridiculing EBITDA on exactly these grounds—search for "Buffett EBITDA" for some entertaining reading; his argument is precisely the one in this section. Wall Street remains unmoved: in mergers and acquisitions, practitioners still overwhelmingly center their valuations on EBITDA multiples. Whether that reflects wisdom that Buffett and I are missing, or the industry's taste for measures that flatter the deals it wants to sell, I leave to you.

[4]Which is, if you think about it, a vindication of those medieval Venetians: the elaborate accrual machinery earns its keep. But the research also shows that the *gap* between earnings and cash flows carries information of its own—recall the gap between accrual earnings and cash flows discussed at the start of this chapter. The two measures are more informative together than either is alone.

A

Practice Problems and Essay Questions

A.1 Chapter 1

A.1.1 Practice Problems: MCQ's

1. Which of these is one of the four financial statements that the firm's financial accounting system ultimately produces?

   (A) The Chart of Stock Price Performance

   (B) The Statement of Long-Term Value

   (C) The Balance Sheet

   (D) The Social Impact Statement

   (E) None of the other answers

   Solution. There are exactly four financial statements: the Balance Sheet, the Income Statement, the Statement of Shareholders' Equity, and the Statement of Cash Flows. The other three options are inventions—and note what they all have in common. Each one gestures at something a company might *like* to tell you about itself: how its stock has done, how much long-term value management believes is lurking unappreciated, how good it is for the world. Those are exactly the kinds of claims the financial statements are built *not* to be. So the answer is **(C)**.

2. Which of the following indicates *equity* financing?

   (A) Interest payments.

   (B) Owners.

   (C) Contractual obligation to pay.

   (D) Collateral.

   (E) None of the other answers

   Solution. Equity financing is when investors hand over cash and receive **shares of ownership** in return. The one word in this list that means 'ownership' is *owners*. Every other option is a marker of the *other* kind of financing: interest is what you pay a creditor for the use of their money; a contractual obligation to pay is the defining feature of a liability (equity carries no such obligation); and collateral is what the creditor seizes when you don't. So the answer is **(B)**.

3. Which of the following indicates *credit* financing?

   (A) Ownership.

   (B) Shares.

   (C) Stock.

   (D) Debt.

   (E) None of the other answers

   Solution. This is the mirror image of the previous question, and it is worth doing both, because the terminology is where people get lost. 'Ownership,' 'shares,' and 'stock' are three different words for the same thing—equity. 'Debt' is credit financing: cash today in exchange for a contractual promise of interest and principal tomorrow. (And recall that credit financing has its own thicket of synonyms: the company 'receives a loan,' 'issues debt,' or 'sells bonds'; the counterparty is a 'creditor,' 'lender,' 'debtholder,' or 'bondholder.' One thing, many names.) So the answer is **(D)**.

4. Eagle Corp operates MRI clinics throughout the Northeast. At the end of the current period, the company reports the following amounts: Assets = \$40k, Liabilities = \$30k, Dividends = \$2k, Revenues = \$13k, Expenses = \$6k. The company management asserts that its long-term value is not being fully appreciated by the stock market, and that they will double their profits next year. **Calculate Shareholders' Equity at the end of the period**.

   (A) \$10k

(B) \$23k

(C) \$70k

(D) \$17k

(E) None of the other answers

Solution. Start from what you are asked for, and work back to the identity that governs it. You are asked for E, and E lives in the Accounting Equation:

$E = A - L = \$40k - \$30k = \$10k$

That is the entire computation. Note what you did *not* need: the revenues, the expenses, the dividends, and—especially—management's confidence about next year. Option (C), \$70k, is what you get by adding A and L, which balances nothing. Option (D), \$17k, is \$10k plus the \$7k of net income, i.e., double-counting income that is already inside that \$40k of assets. So the answer is **(A)**.

5. Eagle Corp operates MRI clinics throughout the Northeast. At the end of the current period, the company reports the following amounts: Assets = \$40k, Liabilities = \$30k, Dividends = \$2k, Revenues = \$13k, Expenses = \$6k. The company management asserts that its long-term value is not being fully appreciated by the stock market, and that they will double their profits next year. **Calculate Net Income at the end of the Period**.

   (A) \$5k

   (B) \$7k

   (C) \$9k

   (D) \$10k

   (E) None of the other answers

   Solution. Same drill, different identity:

   $Net\ Income = Revenues - Expenses = \$13k - \$6k = \$7k$

   Option (A), \$5k, is the single most common wrong answer, and it is wrong for an interesting reason: it subtracts the \$2k of dividends as though dividends were an expense. They are not. An expense is a cost incurred in *generating* value for the owners; a dividend is a *distribution* of value already generated—cash moving over a line from the company's bank account into the owners'. Subtracting it from net income would be like docking your salary because you spent some of it. So the answer is **(B)**.

6. At the end of Fiscal Year (FY) 2020, ExpoCorp had Assets of $200m, Liabilities of $120m, and Common Stock of $30m. Over FY 2021, ExpoCorp earned Net Income of $20m, and paid out Dividends of $10m. What would its *Retained Earnings* be at the end of FY 2021?

   (A) $10m

   (B) $20m

   (C) $50m

   (D) $60m

   (E) None of the other answers

   Solution. Two steps, and the trick is noticing that there *are* two.

   First, find where Retained Earnings *started*. We are handed A, L, and Common Stock as of the end of FY 2020, so:

   $E_{2020} = A - L = \$200m - \$120m = \$80m$

   $Retained\ Earnings_{2020} = E - Common\ Stock = \$80m - \$30m = \$50m$

   Second, roll it forward one year using the Retained Earnings identity:

   $\Delta Retained\ Earnings = Net\ Income - Dividends = \$20m - \$10m = \$10m$

   $Retained\ Earnings_{2021} = \$50m + \$10m = \$60m$

   Now look at the distractors, because they are a menu of the ways to stop early. $10m is the *change* in Retained Earnings—the answer to a question you weren't asked. $50m is the *opening* balance, right before you rolled it forward. $20m is just the net income. Retained Earnings is a **stock** measure with a running balance; net income and dividends merely *update* it. Keep the bathtub in mind: you were asked for the water level, not the flow. So the answer is **(D)**.

7. What is this company's *Accounts Receivable* balance?

Balance Sheet as of December 31, 2020

Assets	**Liabilities**
Cash: $100	Salaries payable: $100
Accounts receivable: $???	Long term debt: $100
Equipment: $100	**Total Liabilities: $200**
	Shareholder's Equity:
	Common stock: $50
	Retained earnings: $0
	Total Shareholders' Equity: $50
Total Assets: $???	**Total Liabilities and S.E.: $250**

(A) $0

(B) $50

(C) $250

(D) Not enough information.

(E) None of the other answers

Solution. You have to consciously deploy two facts here, and the second one is the one people forget.

Fact one: the Balance Sheet *balances*. The right-hand side totals $250, so Total Assets must also be $250.

Fact two: the Balance Sheet is a complete **list** of the company's asset accounts. Cash and Equipment are listed at $100 each, accounting for $200 of that $250. There is nowhere else for the remaining $50 to hide. So Accounts Receivable = $250 - $100 - $100 = **$50**.

Option (D), 'not enough information,' is the tempting one, and it is tempting precisely because you have to *believe* fact two. If the statement were merely a sample of the company's assets, (D) would be right. It isn't; it's the whole list. So the answer is **(B)**.

8. If we issue *Common Stock*, is that an operating, financing, or investing activity?

(A) Operating

(B) Investing

(C) Financing

(D) None of the above

Solution. Financing activities are the company's transactions with its own creditors and investors—how it **finances itself**. Issuing common stock is the archetype.

The trap here is a genuine one, and it's worth defusing now: from the *shareholder's* side of the table, buying that stock is unmistakably an *investment*. Same transaction, two names. The rule is that on the Cash Flow Statement we always speak from the *company's* point of view, so it is financing. ('Investing,' in the company's vocabulary, means buying and selling long-term assets—factories, equipment, other firms' securities.) So the answer is **(C)**.

9. From our company's perspective, which of the following would be an **operating cash inflow**?

(A) Paying salaries to employees.

(B) Selling Common Stock to investors.

(C) Making a Cash sale to customers.

(D) Purchasing a factory.

(E) None of the other answers

Solution. Two conditions have to be satisfied at once—*operating*, and *inflow*—and each distractor fails exactly one of them. Paying salaries is operating, but it is an outflow. Selling common stock is an inflow, but it is financing. Purchasing a factory fails both: outflow, and investing. Selling to customers for cash is the company's core day-to-day business (operating), and the cash comes in (inflow). So the answer is **(C)**.

10. Suppose our company has the following transactions with our customers, and *only these*, in Fiscal Year 2021:

    - We provide tutoring to Dane for $200 in cash.
    - Alison pays us $300 in cash up front, for tutoring that we promise to give her next year, in Fiscal Year 2022.
    - We provide tutoring to Avery, who promises/agrees to pay us $100 next year, in Fiscal Year 2022.

    What are our (accrual) revenues for FY 2021?

    (A) $200

    (B) $300

    (C) $500

    (D) $600

    (E) None of the other answers

    Solution. Take the revenue-recognition rule at face value and go customer by customer, asking one question each time: *did we provide the service in FY 2021?*

    Dane: yes. Count the $200. (The cash arriving the same day is a coincidence, not the reason.)

    Alison: no. We have her $300, but we have not earned a penny of it—we owe her tutoring. Count $0 this year; that $300 becomes revenue in FY 2022, when we actually teach her.

    Avery: yes. The work is done. Count the $100, check or no check.

Total: \$200 + \$100 = **\$300**.

The two distractors are the two ways of not believing the rule. \$200 is cash-basis-with-cold-feet: only the sale where the money and the service arrived together. \$500 is pure cash accounting: everything that landed in the bank this year. Neither is what we mean by revenue. So the answer is **(B)**.

11. MattCorp does annual reporting, with the fiscal year ending December 31. In FY2020 (January 1, 2020 to December 31, 2020), MattCorp earns a total of \$100 in *Revenue*, and incurs \$60 in Expenses over that year. **How much *Net Income* would MattCorp report on its Income Statement as of July 1 (exactly halfway through the year)?**

    (A) \$40

    (B) \$20

    (C) \$0

    (D) Trick question / question makes no sense.

    (E) None of the other answers

    Solution. Yes, it's a trick question—but a trick that teaches something, so let me say exactly *which* thing is broken.

    Three separate problems, actually. First, 'Net Income *as of* July 1' is a category error. Net income is a **flow** measure: it is only ever defined *over* a period, never *as of* a date. Asking for net income as of a moment is like asking how fast the tap was running at 3 p.m. without specifying an interval—er, well, ask a calculus student. Balances (Cash, Retained Earnings) are 'as of'; income is 'over.'

    Second, even if we generously rewrote the question as 'net income for January 1 through June 30,' we could not answer it. We are told the *annual* totals, and nothing tells us they accrued evenly. A ski resort does not earn half its revenue by July. Option (B), \$20, is the answer you get by splitting \$40 in half and assuming otherwise.

    And third, as a matter of plumbing: MattCorp reports annually. It does not *have* an Income Statement on July 1. At any point in time a company merely tracks its accounts; the statements get assembled at period end. So the answer is **(D)**.

12. Which of the following is a good way of describing what *Net Income* (profit) ***is***, conceptually, in accounting?

    (A) The increase in the company's cash

(B) Accounting's way of measuring how much value has been realized for the owners, via the company's operations, over the period.

(C) The total wealth of the shareholders of the company.

(D) The stock price increase.

(E) The amount of money made for the bank lenders of the company.

Solution. Option (B) has all four of the load-bearing pieces: it is a *measure* (accounting's attempt, not metaphysical truth), of value realized *for the owners* (not for lenders, not for society), *via operations* (not via financing or dividends), *over the period* (a flow, not a level).

The distractors each break one piece. (A) is cash accounting—but revenue can be earned before the cash arrives, and cash can arrive without any revenue being earned. (C) confuses a flow with a stock: total shareholder wealth is a level, and in any case accounting's stab at it is Shareholders' Equity, not net income. (D) is the market's opinion, which is not something the accounting system produces. (E) points at the wrong claimant entirely—interest paid to lenders is *subtracted* on the way to net income. So the answer is **(B)**.

13. Which of the following is a good definition of what ("book") Shareholders' Equity ***is***, conceptually, in accounting?

(A) The amount of financing that owners have contributed (including reinvestment) to the business.

(B) The amount of cash that is held separately for the owners.

(C) How much value has been realized for the owners, via the company's operations, over the reporting period.

(D) None of the other answers

(E) The amount of loans the company has made to its owners.

Solution. Shareholders' Equity comes from exactly two sources: what the owners **paid in** (Common Stock) and what they earned and chose to leave in rather than take out (Retained Earnings). The parenthetical 'including reinvestment' in option (A) is doing precisely that second job. Whatever the owners did not claim, it is *as if* they contributed it as additional financing.

Option (B) is the error I care most about your not making, because it is the error that senior executives and legislators make in public. **Equity is not a resource; it is a claim.** There is no vault of cash sitting somewhere with the shareholders' names on it. The company's cash is on the *left* side of the equation, among the

assets; equity is on the *right*, describing who has a claim on those assets once the creditors are satisfied. A company can have $600k of Shareholders' Equity and $0 of cash. Option (C) is the definition of net income—a flow over a period—not equity, which is a level at a point in time. So the answer is **(A)**.

14. What is the name of the account that indicates the total amount that owners have paid into the company to finance it?

    (A) Cash

    (B) None of the other answers

    (C) Common Stock

    (D) Dividends

    (E) Revenue

    Solution. *Common Stock* is the 'paid-in capital' account: cash that investors directly handed to the company in exchange for shares of ownership. Its partner in Shareholders' Equity is *Retained Earnings*, the 'earned capital'—profits the owners could have taken out and didn't. *Dividends* is the flow in the opposite direction, out to the owners. And *Cash* is an asset—the money itself, not the record of who provided it. So the answer is **(C)**.

15. Which of the following organizations would be required by the SEC, as a U.S. public-company issuer, to report under U.S. GAAP?

    (A) Your neighbor's dog-grooming business, which your neighbor owns as sole proprietor.

    (B) The Coca-Cola Company

    (C) The Gates Foundation

    (D) Multiple other answers

    (E) None of the other answers

    Solution. Trace the chain of authority. Congress created the SEC in 1934 and empowered it to regulate the disclosures of **publicly-traded corporations**; the SEC delegated the setting of the standards to the FASB, which issues U.S. GAAP. So the requirement attaches to public companies, and Coca-Cola is about as public as a company gets.

    Your neighbor keeps books—she had better, if only for the IRS—but nobody is buying her shares on the NYSE, so the SEC public-company requirement does not apply merely because she runs that business. Other legal, contractual, or reporting requirements are separate questions. The Gates Foundation is a private

foundation, not a publicly-traded corporation; it files and discloses plenty, but not because the SEC compels GAAP financial statements from it. Since exactly one option qualifies, (D) is out. So the answer is **(B)**.

16. The organization that has been delegated the role of setting accounting standards for publicly-listed companies in the U.S. is...:

    (A) GAAP

    (B) The FASB

    (C) The IASB

    (D) IFRS

    (E) None of the other answers.

    Solution. Two of these are *bodies* and two are *sets of standards*, and the question asks for an organization. The FASB (in Norwalk, Connecticut) issues U.S. GAAP; the IASB issues IFRS. Since we want the U.S. standard setter, it is the FASB—which holds that role through recognition by the SEC, which in turn holds it by statute from Congress. GAAP and IFRS are the outputs, not the organizations. So the answer is **(B)**.

17. From our company's perspective, the Dividends we pay are a(n):

    (A) Expense

    (B) Liability

    (C) Asset

    (D) Revenue

    (E) None of the other answers

    Solution. Run down the list and rule each one out on definitions. Not an **asset**: paying a dividend does not obtain a future economic benefit; it hands cash away. Not a **revenue**: nothing was earned from a customer. Not a **liability**: dividends are discretionary—the company is under no contractual obligation to pay them, which is exactly what distinguishes owners from creditors. And not an **expense**, which is the answer most people reach for first: an expense is a cost incurred in *generating* value for the owners, whereas a dividend is a *distribution* of value already generated. Cash simply crosses the line from the company's bank account into its owners'.

    So dividends are none of the four—they are their own category, a distribution, and they show up as a reduction in Retained Earnings. So the answer is **(E)**.

18. When a company pays dividends, all else equal, that will [increase / decrease / have no effect on] its ("book" / accounting) Shareholders' Equity:

    (A) Increase

    (B) Decrease

    (C) Have no effect

    (D) Not enough information.

    (E) None of the other answers.

    Solution. Two routes to the same place. Mechanically: $\Delta Retained\ Earnings = Net\ Income - Dividends$, so dividends pull Retained Earnings down, and Retained Earnings is a component of Shareholders' Equity. Or, from the Accounting Equation: cash goes out, so A falls; no liability changed; so E must fall to keep $A = L + E$.

    Here is the thing that bothers people, and it is worth saying out loud, since the previous question just established that dividends are *good* for owners. How can the thing owners want most reduce their equity? Because Shareholders' Equity measures the owners' claim on **the company**, and the dividend moves value *out of the company* and into the owners' own pockets. It is the bathtub being scooped, not sprung a leak: the level in the tub goes down, and the owner is holding the bucket. So the answer is **(B)**.

A.1.2 Practice Problems: Open Prompts

1. We go to a local shop to buy *Supplies* that we plan to use over several reporting periods. The "list price" of the *Supplies* we buy is $12,000. However, the company is having a special sale, selling the *Supplies* for a discount of $10,000. Indeed, the owner tells us that this deal is even better than it may seem, because the *true "market value"* of the *Supplies* is *really* $14,000. After the make this purchase, what value would we record the *Supplies* at on our books?

    Solution. $10,000—the price we actually paid.

    Three numbers are dangled in front of you, and only one of them is a *transaction price*. The $12,000 list price is a number on a sign; nobody paid it. The $14,000 'true market value' is the seller's opinion, delivered by a person with an obvious interest in our feeling good about the purchase. The $10,000 is what left our bank account, and it is the one number an auditor could verify from a receipt.

    In accounting, assets are recorded at **historical cost**—the verifiable purchase price—rather than at someone's appraisal. In accounting, you get what you pay

for. And note that if we *were* allowed to book the supplies at \$14,000, the accounting equation would demand a \$4,000 increase somewhere on the right-hand side; we would be recognizing \$4,000 of value out of thin air, on a stranger's say-so. This is exactly the door that conservatism is meant to keep shut. (Note also that the 'discount' framing is a red herring twice over: we are not recording a \$2,000 bargain, and the fact that the supplies will be used over several periods only tells us they are an *asset* rather than an immediate expense—it says nothing about the *amount*.)

2. MattCorp does annual, calendar-year reporting. At the end of FY2020, MattCorp reported a balance of *Common Stock* of \$1m and *Retained Earnings* of \$200k. Over FY2021, MattCorp earned total *Net Income* of \$150k. Additionally, MattCorp issued 1,000 shares of *Common Stock* for \$50 each on February 17, 2021; and paid total *Dividends* of \$100k on November 18, 2021. **Fill out MattCorp's *Statement of Shareholders' Equity* for FY2021.**

   Solution. The Statement of Shareholders' Equity is laid out differently from the other statements: instead of listing accounts down the side, it lists *events* down the side and the equity accounts across the top, and shows how each event moved each account. So the recipe is: start with the opening balances, apply the year's events one at a time, and total.

   The only arithmetic that isn't handed to us is the stock issuance: $1{,}000\ \textit{shares} \times \$50 = \$50{,}000$ of new Common Stock. Everything else we can place directly. Net Income and Dividends both land in the Retained Earnings column—income adding, dividends subtracting—per $\Delta \textit{Retained Earnings} = \textit{Net Income} - \textit{Dividends}$.

MattCorp: Statement of Shareholders' Equity, FY 2021

	Common Stock	Retained Earnings	Total Shareholders' Equity
Balance, Jan. 1, 2021	\$1,000,000	\$200,000	\$1,200,000
Issue Common Stock (Feb. 17)	50,000		50,000
Total Dividends Paid (Nov. 18)		(100,000)	(100,000)
Net Income		150,000	150,000
Balance, Dec. 31, 2021	**\$1,050,000**	**\$250,000**	**\$1,300,000**

   Checking the two columns independently: Common Stock, $\$1{,}000{,}000 + \$50{,}000 = \$1{,}050{,}000$; Retained Earnings, $\$200{,}000 + \$150{,}000 - \$100{,}000 =$

$250,000. Their sum, $1,300,000, is the Total Shareholders' Equity that will appear on the December 31, 2021 Balance Sheet.

Two things to notice. First, the dates—February 17, November 18—do no work at all. They are there to see whether they spook you. Because we report annually, everything that happened between January 1 and December 31 lands in this one statement, in whatever order we like. Second, look at what this statement *is*: it is the bridge between the Income Statement and the Balance Sheet. The $150,000 of net income arrives from the Income Statement, walks through Retained Earnings, and leaves as part of the $1,300,000 equity balance on the Balance Sheet.

3. What are the equations for the **core identities** (accounting equation, net income definition, retained earnings identity)?

   Solution. Know these three cold—every chapter of this book leans on them.

   $$Assets = Liabilities + Shareholders' \ Equity$$

   $$Revenues - Expenses = Net \ Income$$

   $$Retained \ Earnings = Total \ N.I. \ Earned - Total \ Dividends \ Paid$$

   And it is worth being able to say what each one is *for*, not just what it says. The first governs the **Balance Sheet**: stock measures, financial *position*, at a point in time. The second governs the **Income Statement**: flow measures, financial *performance*, over a period. The third is the link between them, and it is often more useful in its change form:

   $$\Delta Retained \ Earnings = Net \ Income - Dividends$$

   That is the sentence 'this period's profit, minus what we paid out, is what got added to the pile'—the bathtub's level rising by the inflow less what was scooped out. It is why performance 'lives on' in position.

4. List the names of the **four financial statements** from memory.

   Solution.

   1. The **Balance Sheet** (B.S.)
   2. The **Income Statement** (I.S.)
   3. The **Statement of Shareholders' Equity** (S.S.E.)
   4. The **Statement of Cash Flows** (S.C.F.)

   If you can also attach one phrase to each, you are in much better shape than someone who has only memorized the list. Position at a point in time; performance over a period, on an accrual basis; the owners' accounts and how they changed; and performance over a period, on a cash basis, split into operating,

investing, and financing. The first two are where we will spend nearly all of our time, in this book and in the world.

5. Track down the **latest 10-K report filed by Apple**, by searching "EDGAR," the SEC site on which U.S. companies' mandatory filings are distributed and archived, here: https://www.sec.gov/edgar/searchedgar/companysearch.html. **What was Apple's *Net Income* for its most recent fiscal year?**

   Solution. The number changes every autumn, so let me give you both the number and—more usefully—the route, because the route is the transferable skill.

   On EDGAR, search for 'Apple Inc.' (ticker AAPL) and filter the filing type to `10-K`. The 10-K is the annual report; the `10-Q`s are the quarterly ones. Open the most recent 10-K and go to **Item 8, Financial Statements and Supplementary Data**. The first statement there is the **Consolidated Statements of Operations**—which is what Apple, like most large firms, calls its Income Statement. Read the bottom line: *Net income*. (Amounts are 'in millions,' as the column heading will tell you, so do not misplace three zeros. Three years are presented side by side; you want the leftmost, most recent column.)

   To make that concrete, here is what I found when I pulled the filing while writing this: the most recent 10-K then available was for fiscal 2024, the year ended September 28, 2024, and the line read **$93,736 million—about $93.7 billion**, on $391.0 billion of net sales. Your number will be different, and it should be. Apple files a fresh 10-K every autumn, so by the time you are reading this there is at least one more on the site, and the figure above has quietly become a historical fact rather than a current one. That is exactly why the route matters more than the number.

   Two things worth noticing while you are in there. First, Apple's fiscal year ends in late September, not December—a reminder that 'fiscal year' and 'calendar year' are not synonyms. Second, this is a real filing that anyone on earth can read for free, and it contains the actual numbers behind the largest company most of you own a piece of. Get comfortable in EDGAR now. Equity analysts live there.

A.1.3 Essay and Discussion Questions

1. What is the purpose of financial accounting in general? What is this all about? In other words, why are we doing this?

   Solution. Here is how I would think it through, in three moves.

   Start with the problem, not the solution. People have to make consequential decisions about businesses. An investor decides whether to buy the shares.

A bank decides whether the firm will still exist in five years. A supplier decides whether to ship on credit. An employee decides whether the stock options in the offer letter are worth the pay cut. A regulator decides whether to intervene. Every one of those is, at bottom, a **measurement** question about a business—and none of those people can simply walk through the firm and *see* the answer. The full reality of a company is effectively infinite in complexity; no human mind holds it. That is the core problem: **too much information**.

So the purpose of financial accounting is to **measure, compress, summarize, and communicate** that reality in a form a decision-maker can actually use. Accounting is a **map**. And the map is not the territory—real information is lost, deliberately, in the compression. But if you cannot take in the whole territory at once, you need a map, and a map that omits things on purpose is far better than no map at all. Much of what feels arbitrary in accounting is really an argument about what belongs on the map.

Then add the second word: accountability. Information is only half of it. Companies are run by people, and people are not indifferent to how they are measured. Owners hold the ultimate lever of control over a firm, and they use accounting numbers to hold managers **accountable**—to evaluate them, pay them, and, if need be, replace them. This is why 'profit' is not just a statistic but a governance instrument. It is also why the accounting system is so preoccupied with **verifiability**: if managers could simply *declare* what their assets were worth, the measure would bend to the interests of the people being measured, and it would stop being worth anything at all. Notice how much of what looks like accounting's excessive caution—historical cost, conservatism, the whole audit profession—follows from that single worry.

Then close the loop back to Econ 101. You spent a semester assuming firms maximize profits, without ever being told what profit *is* or why anyone would maximize it. Both gaps are filled by the same answer. Profit is the accounting system's attempt to measure the value realized for the **owners**; and owners are assumed to get their way because they hold the residual control rights. That is not a moral claim about what firms *ought* to do—it is a claim about power and who pulls the strings.

If I had to compress the whole thing into one sentence: financial accounting exists because the people who need to judge a business cannot see it, and the people who *can* see it have every incentive to describe it flatteringly.

2. We've talked about how it's common to divide the sources of Shareholders' Equity into two categories: 'Paid-in capital' vs. 'Earned capital.' What's the big idea

here? What are the accounts associated with these two categories? Why are they together in Shareholders' Equity, and how are they different?

Solution. The big idea is a question about **origin**: given some quantity of owner-financing sitting in the business, *how did it get there?* There are exactly two ways.

Paid-in capital—the account is **Common Stock**—is money the owners took out of their own pockets and handed to the company in exchange for shares. **Earned capital**—the account is **Retained Earnings**—is profit the business generated by selling to customers, which the owners *could have* taken out as dividends and chose to leave in. Together:

$$Shareholders' \ Equity = Common \ Stock + Retained \ Earnings$$

Why are they together? Because from the standpoint of the accounting equation, they are the same *kind* of thing: both are owner-provided financing, and both are residual claims. This is the interpretation of Retained Earnings I find most clarifying. The owners had a claim on those profits. They didn't take them. So it is *as if* they turned around and re-contributed that money to the firm—a reinvestment, made by not withdrawing. Economically, a dollar of retained profit and a dollar of new stock issuance finance the company identically: they buy the same equipment, and they sit behind the same creditors in line. Neither carries a repayment obligation, and neither is owed on any schedule. That is what makes both of them equity rather than debt.

How are they different? Two ways, and both matter. First, in what they tell you about the business. A company whose equity is overwhelmingly paid-in capital is one that investors have funded but that has not yet made much money—a startup, or a firm with a long history of losses. A company whose equity is overwhelmingly earned capital has funded itself out of its own operations. That is a genuinely different animal, and you can read it right off the balance sheet.

Second, in their character as measures. Common Stock is essentially a historical record: cash came in on certain dates, at certain prices, and the number sits there. Retained Earnings is a *running* balance, updated every period by the identity $\Delta RE = Net \ Income - Dividends$. It is the account through which the entire Income Statement flows into the Balance Sheet—which is precisely why the Statement of Shareholders' Equity exists.

Let me close with the application I find most illuminating. Critics of Wall Street often point out that large mature firms raise almost no new money by issuing stock, and in aggregate pay out far more to shareholders than they take in—so what, exactly, is the stock market *for*? Set the politics aside and notice that the pattern is nearly forced by this very distinction. How does a company get

into the S&P 500? By being extraordinarily successful—which means generating enormous *earned* capital. A firm like that has no reason to keep going back to investors for paid-in capital and diluting its existing owners; its own operations throw off everything it needs. Paid-in capital gets firms started; earned capital carries the successful ones from there.

3. We've talked about the "two ways of looking at the accounting equation." What are these? What does the accounting equation tell us, under each view?

   Solution. One equation, $A = L + E$; two ways of reading it, depending on which side you rearrange it toward.

   View 1: 'Sources of Financing,' $L + E = A$**.** Read right-to-left. The company has acquired a pile of resources (A), and every one of them had to be paid for somehow. Every dollar of financing came from either creditors or owners—there is no third source. So the right-hand side is the *itemized answer* to the question 'where did the money come from?' Under this view, the equation tells you **how much the company has, and who supplied it.**

   View 2: 'What's Left Over,' $E = A - L$**.** Read as a subtraction. The company controls benefits worth A and owes obligations worth L. The obligations are contractual and come first; whatever survives them belongs to the owners, who are the **residual claimants**. Under this view, the equation tells you **what the owners would actually walk away with** if the business were wound up today. This is the mortgage intuition: your home equity is the value of the house minus what you still owe the bank, and if the house appreciates, the bank gets nothing extra.

   A thought experiment makes the second view concrete. Suppose a company has $20k of cash, and nothing else, financed by a $5k bank loan and $15k of stock. I decide entrepreneurship isn't for me and shut it down. What happens? I *must* pay the bank its $5k first—that is what a liability *is*. The $15k that remains is mine. That $15k is Shareholders' Equity.

   Now, the thing I most want you to take away: **these are not two equations, or two competing theories. They are one identity, seen from two angles**, and they are useful for different purposes. View 1 is the natural way to think when you are asking how a firm funded itself, or whether it is leaning heavily on debt. View 2 is the natural way to think when you are asking what the owners are actually entitled to—and it is the view that explains why equity is called 'residual' at all.

   Let me push on why the second view is worth the effort, since it is the less intuitive one. There is a deep argument, running back to Oliver Hart's Nobel-winning work, that residual claims are not a quirk of our corporate law but a **logical necessity**. Wherever there are contracts and uncertainty, the contracts cannot pos-

sibly specify everything—so *some* party must have the final say, and the leftover claim, on everything unspecified. As a matter of arithmetic, the remainder has to go somewhere and be controlled by someone. That party is the owner, by definition. This is true of a corporation, and it is true of you: your hours are your assets, your obligations are your liabilities, and the residual is yours to allocate.

One warning that follows from holding both views at once. Under View 1 it is tempting to picture L and E as two piles of money the company is sitting on. They are not. **The left side lists what the company has; the right side describes claims on it.** A company can raise $1m, spend all of it on a store, and have $600k of equity and $0 of cash. Equity is not a resource; it is a claim. Get that straight and you will already be thinking more clearly than a number of bank CEOs did during the last financial crisis.

A.2 Chapter 2

A.2.1 Practice Problems: MCQ's

1. Suppose our company borrows $10,000 from a bank. How do we account for this?

	Assets	Liabilities	Sharehldr Equity
(A)	+$10k Cash		
(B)		+$10k Note payable	
(C)	+$10k Cash		+$10k Note payable
(D) None of the above			

Solution. Start with cash. Cash comes in, so that asset account rises by $10k. Now, what keeps the equation in balance? We have to pay the bank back, so we have taken on an obligation—a liability, which we call *Notes Payable*. The correct accounting is therefore **+$10k Cash in the Assets column, and +$10k Notes Payable in the Liabilities column**:

Transaction	Assets	Liabilities	Sh.'s Equity
Borrow $10k from the bank	+$10k (Cash)	+$10k (Notes Payable)	

No option offers that. Option (A) records only half the transaction, leaving the equation out of balance by $10k—and the accounting equation is an *identity*; it is never allowed to be out of balance, not even briefly. Option (B) has the same disease from the other side. Option (C) is the interesting one, and it is the one most people pick: it has both of the right *amounts* and both of the right *account names*, but it puts *Notes Payable* in the **Shareholders' Equity** column. That is not a clerical slip; it is the whole distinction the chapter is built on. A bank is a creditor, not an owner. It has a contractual claim that must be satisfied *before* the owners get anything—which is precisely what it means for something to be a liability rather than equity. So the answer is **(D)**.

2. Our company buys equipment for $24k in cash. How do we account for this?

	Assets	Liabilities	Sharehldr Equity
(A)	-$24k Cash		+$24k Equipment
(B)	-$24k Cash, -$24k Equipment		
(C)	-$24k Cash, +$24k Equipment		
(D) None of the above			

Solution. Start with cash: it goes out, so that asset falls by $24k. What did we get for it? Equipment—a resource the company now controls and will use for years. It does not get consumed on the spot like a bag of candy; it endures and keeps serving the firm. So it satisfies the FASB definition of an asset, and the

whole transaction happens *inside* the asset column ($A\uparrow, A\downarrow$). We have simply converted one asset into another.

Look at the distractors, because each is a distinct error. Option (A) puts *Equipment* in the Shareholders' Equity column—but equipment is a thing the company *has*, and things the company has live on the left side of the equation. Option (B) decreases both Cash *and* Equipment, which would knock the equation \$48k out of balance and also imply we somehow gave away equipment we never had. Option (D) is not needed here. So the answer is **(C)**.

One further point, since it is the theme of the chapter: notice that spending \$24k did **not** produce a \$24k expense. Nothing has been used up yet. The equipment will be expensed later, bit by bit, as it wears out—that is Chapter 4's business. Assets become expenses *later*.

3. Our company has done the following transactions, and only these, to date.

Transaction	Assets	Liabilities	Sharehldr Equity
#1: Sell Shares for \$25k	+\$25k Cash		+\$25k Common Stock
#2: Get loan of \$10k	+\$10k Cash	+\$10k Note payable	
#3: Buy equipment for \$24k	-\$24k Cash, +\$24k Equipment		
#4: Pre-pay 1 yr rent, \$6k	-\$6k Cash, +\$6k Prepaid rent		

What is our company's Total Assets (balance) at this point, after Transaction #4?

(A) \$6k

(B) \$30k

(C) \$35k

(D) \$65k

(E) None of the other answers.

Solution. You can grind through the asset column: $+\$25k+\$10k-\$24k+\$24k-\$6k+\$6k=\$35k$. That works, and it is six chances to make a sign error.

Here is the faster route, and the one I want you to acquire the habit of. Have the courage of your convictions that $A=L+E$ *always* holds. So instead of adding up the left side, add up the right side, which is much cleaner: \$10k of Notes Payable, plus \$25k of Common Stock, and nothing else. $L+E=\$35k$. Therefore $A=\$35k$. Two numbers instead of six.

The distractors are the partial sums you get by losing your place: \$6k is just the prepaid rent, and \$65k is what you get by adding up only the *positive* entries in the asset column, forgetting that Transactions #3 and #4 each *spent* cash to acquire something. So the answer is **(C)**.

4. And, what is our company's *Cash* balance at this point, after Transaction #4?

(A) \$5k

(B) \$10k

(C) \$25k

(D) \$35k

(E) None of the other answers.

Solution. Now you *do* have to walk the column—but only the Cash entries:

$\$25k + \$10k - \$24k - \$6k = \$5k$

And notice what just happened between these two questions, because this trips up more people than it should. Total Assets are \$35k; Cash is \$5k. Those are different questions with different answers. There is an axiom that lets you shortcut total assets ($A = L + E$); there is **no** axiom that says 'Cash equals something simple.' *Cash* is one account sitting inside Assets, alongside *Equipment* (\$24k) and *Prepaid Rent* (\$6k)—and \$5k + \$24k + \$6k = \$35k, as it must. Option (D), \$35k, is total assets; option (C), \$25k, is the common stock. Slow down and match your arithmetic to the quantity actually being asked for. So the answer is **(A)**.

5. Suppose our company buys a factory from another company for \$15m in Cash. (We would put this factory in our *Plant* account.) However, our management is *extremely confident* that, with our superior management practices, and world-class employees, we can actually *double* the productivity and output of that factory. What would be the initial value/amount of *Plant* that we would record in our company's accounting (that is, "on our books") after this purchase?

   (A) \$0

   (B) \$15m

   (C) \$30m

   (D) Not enough information.

   Solution. \$15m. Full stop. And what do we do with management's confidence? For measurement purposes, we ignore it entirely.

   Assets are initially recorded at their **purchase price**—the historical cost—because that is the number an auditor can verify. There is a bank statement, an invoice, a receipt. There is no receipt for a belief. Management is allowed to hold that belief, and they may even turn out to be right in an economic sense; but if firms could write their assets up to whatever they expect to accomplish with them, the balance sheet would become a collection of press releases. Recall the rule of thumb: **in accounting, you get what you pay for.**

Option (C), $30m, is what you get by taking management at their word. Option (D) is worth a moment's sympathy—there is a real sense in which we don't know what the factory is 'truly worth'—but that is exactly why we fall back on the verifiable transaction price rather than throwing up our hands. So the answer is **(B)**.

6. Which of the following represents, conceptually, *credit financing provided by our suppliers*?

   (A) *Notes Payable (L)*

   (B) *Accounts Payable (L)*

   (C) *Supplies (A)*

   (D) Multiple other answers

   (E) None of the other answers

   Solution. When we buy on account, the supplier hands over goods today and lets us pay later. That is a liability, and because the counterparty is a supplier rather than a bank, we call it *Accounts Payable*.

   And it really is credit financing, not merely something that looks like it. Recall the gestalt shift: imagine the supplier first lends us $1k in cash, and we then immediately hand that same $1k back to buy the goods. The cash makes a round trip in one day, so we net it out—and we land exactly on 'Supplies up, Payable up.' Buying on account *is* getting a loan; the only difference is that the lender happens to deliver the loan in the form of merchandise.

   Notes Payable is also credit financing, but from a bank, not a supplier—which is why (D) is wrong. *Supplies* is the asset we received, not the financing that paid for it: it belongs on the left side of the equation, the side that says what we *have*, not the side that says who provided it. So the answer is **(B)**.

7. Suppose that this customer pays cash to us in November of 2021, for a soccer camp that will take place in February 2022. Our firm does **annual reporting** aligned with the **calendar year**. Assuming things go as expected, in what Fiscal Year would we report *Revenue* from this sale?

   (A) 2021

   (B) 2022

   (C) *Listen to your heart.*

   (D) None of the above.

Solution. Under accrual accounting we recognize revenue when it is **earned**—when we actually provide the good or service—not when the check arrives. We run the camp in February 2022. So the revenue belongs in FY 2022, even though the cash was in our bank account three months earlier—and, decisively, in the *prior fiscal year*.

This is the third of the 'three timings of cash,' and it is the trickiest, because it is the one where cash *precedes* the earning. It is also the one where the temptation to record revenue early is strongest—it feels like we've made a sale. We haven't yet; we've made a promise. (If you want the reason this rule is enforced rather than merely suggested, consider the Fyre Festival: everyone paid up front, and no festival ever happened.) So the answer is **(B)**.

8. But, continuing with the example from the question above, should we *account for something* about this transaction (that is, record *something* on our books) on the date on which the customer pays us cash?

   (A) Yes—because we have earned profit.

   (B) No—because we haven't earned profit yet.

   (C) Yes—because we got cash.

   (D) *Listen to your heart.*

   Solution. Yes, absolutely—and the reasoning in (C) is exactly the right reasoning. Start with cash: the customer's money has landed in our bank account, and *Cash* is an asset, so the left side of the equation has moved. It must be balanced by something.

   What balances it is not revenue, but a **liability**: having taken the customer's money, we now have a present obligation to deliver the camp. We call it *Deferred Revenue* (or, equivalently, *Unearned Revenue*).

   Option (B) is the seductive one, and I want to be blunt about it: "do nothing, because we haven't earned it yet" is a *non sequitur*. It correctly notes that we have no revenue, and then illegitimately concludes that we have nothing to record. But the previous question asked about *Revenue*; this one asks about *anything*. Cash moved; the equation must be kept in balance; something gets recorded. File away the general rule for this book: "record nothing" is essentially never the right answer when something real has changed hands. Option (A) contradicts the previous question. So the answer is **(C)**.

9. Which of the following purchases / expenditures would be "fully and immediately expensed" under GAAP accounting?

(A) Paying for a 2-year lease on a building

(B) Purchasing Inventory

(C) Paying the electricity bill for the company's headquarters

(D) Purchasing supplies on account

(E) None of the other answers.

Solution. Ask of each expenditure: did we acquire a resource that will deliver benefits in future periods, under our control? If yes, it is capitalized as an asset now and expensed later. If no, it is expensed immediately.

(A) The two-year lease buys us the right to use a building over multiple periods. That is an asset—*Prepaid Rent*—which becomes *Rent Expense* gradually as we use it up. (B) Inventory sits on the shelf until it is sold, at which point its cost is matched against the sale as Cost of Goods Sold. Asset now, expense later. (D) Supplies are a resource we will draw down over time; the fact that we bought them *on account* rather than for cash affects only *what balances* it—*Accounts Payable* instead of *Cash*—not whether the thing is an asset. Do not let 'on account' distract you—how we paid has nothing to do with what we got.

(B) is different, and the word **headquarters** is a deliberate clue. Electricity is not storable: the current we bought has already been consumed keeping the lights on. There is no stock of it we control going forward, so it fails the asset definition and is expensed on the spot. (Had the bill been for a *factory's* power, we could trace those kilowatt-hours to specific units of product, and matching would send that cost into *Inventory* instead—but nobody can trace the lighting in the CEO's office to any particular future sale.) So the answer is **(C)**.

10. When a customer pays up front for a good/service that will be provided at a later date, this is a(n):

(A) Accounts receivable transaction

(B) Deferred revenue transaction

(C) Prepaid expense

(D) Asset capitalization

(E) Accrual

Solution. Cash in, service still owed: that is *Deferred Revenue*, a liability. Both halves of the name are doing work—we *defer* recognizing the revenue, because we have not yet *earned* it.

Option (A) is the exact mirror image of what happened, and getting the direction straight is the point of the question: an *Accounts Receivable* arises when we provide the service first and get paid later. Option (C) is the same reversal from the other side of the market—a *prepaid expense* is when *we* pay someone up front (our Prepaid Rent), not when a customer pays *us*. Option (D) misfires because nothing was capitalized: yes, our Cash went up, but the balancing entry was a liability, not the acquisition of a long-lived asset. So the answer is **(B)**.

11. Which of the following can be interpreted to represent, conceptually, credit financing provided to our company by our customers?

    (A) Deferred Revenue

    (B) Accounts Receivable

    (C) Accounts Payable

    (D) Notes Payable

    (E) None of the other answers.

    Solution. When customers pay us in advance, they have handed us cash today in exchange for a promise of future performance. That is *exactly* the shape of a loan—cash now, obligation later—except that we will repay it in soccer camps rather than in dollars. So *Deferred Revenue* is, conceptually, **credit financing provided by our customers**.

    This is not an academic curiosity. It is the entire economic logic of Kickstarter, of concert pre-sales, of season tickets: future customers finance the development of a product and are 'repaid' with the product itself. For a young company with no collateral and no track record, this can be the only financing available—a bank cannot evaluate a niche product, but the niche can.

    Run the distractors quickly: *Accounts Receivable* is credit we extend *to* customers, and it is an asset, not financing we received. *Accounts Payable* is credit from **suppliers**. *Notes Payable* is credit from **banks**. Right idea, wrong counterparty. So the answer is **(A)**.

12. If our company purchases *Equipment* for *Cash*, our total asset balance (increases, decreases, is unchanged):

    (A) increases

    (B) decreases

    (C) is unchanged

    (D) none of the other answers.

Solution. One asset down, another asset up, by the same amount: the transaction happens entirely inside the asset column, and total assets are unchanged. If you bought $24k of equipment for $24k of cash, then $-24k + $24k = $0.

Here is a cross-check that costs nothing. The right-hand side of the equation never moved—no new liability, no new equity—so the left-hand side *cannot* have moved either. That is not a coincidence; it is the identity doing your work for you. Option (B) is the instinct that spending money must make you poorer, which is precisely the instinct this chapter is trying to retrain: we did not lose $24k, we *converted* $24k. So the answer is **(C)**.

13. If we provide a service to a customer *on account*, our company's total asset balance (increases, decreases, is unchanged):

    (A) increases

    (B) decreases

    (C) is unchanged

    (D) none of the other answers.

    Solution. We provided the service, so we recognize Revenue, which is a positive entry in the Shareholders' Equity column. What balances it? The customer's promise to pay—*Accounts Receivable*, an asset. So total assets **increase**.

    The worry lurking behind option (C) is: 'but no cash came in, so surely nothing changed.' Cash and assets are not the same thing. A creditworthy customer's enforceable promise to pay us is a genuine economic resource obtained in a past transaction—an asset by the FASB definition, and, if you push on it, not so different in kind from the number your bank shows you, which is also just somebody's enforceable promise to pay. Wealth mostly consists of claims, not of green paper. So the answer is **(A)**.

14. If one of our customers pays down their past account (that is, pays cash for services we had provided to them on credit at an earlier date), our company's total asset balance (increases, decreases, is unchanged):

    (A) increases

    (B) decreases

    (C) is unchanged

    (D) none of the other answers.

Solution. Cash comes in, so that asset rises. But the customer's obligation to us is now extinguished, so *Accounts Receivable* falls by the same amount. One asset converted into another; total assets unchanged.

Option (A) is the trap, and it encodes the most common error at this point in the course: treating the collection as a *second* revenue event. It is not. We earned—and recorded—that revenue back when we provided the service. Recording it again now would be double-counting the same sale. Nothing new was earned today; the check simply arrived in the mail. Structurally, this transaction is the twin of buying equipment for cash: $A\uparrow, A\downarrow$. So the answer is **(C)**.

15. During its first month of operations, SoccerCorp: -borrowed $100,000 from a bank, -bought a piece of equipment for $30,000 in cash, -bought supplies that it planned to use for multiple quarters for $50,000 on account, -provided services to customers for $20,000 on account, -paid out $30,000 cash to pay down one of its accounts payable, -and paid $15,000 cash for utilities expense. What is the company's total asset balance at the end of the month?

    (A) $125k

    (B) $105k

    (C) $150k

    (D) $165k

    (E) None of the other answers

    Solution. Six transactions, so let us just lay them all out rather than trying to hold them in our heads.

Transaction	Assets	Liabilities	Sh.'s Equity
Borrow $100k from bank	+$100k (Cash)	+$100k (Notes Payable)	
Buy equipment, $30k cash	-$30k (Cash), +$30k (Equipment)		
Buy supplies on account, $50k	+$50k (Supplies)	+$50k (Accounts Payable)	
Services on account, $20k	+$20k (Accounts Receivable)		+$20k (Revenue)
Pay down accounts payable, $30k	-$30k (Cash)	-$30k (Accounts Payable)	
Pay utilities, $15k	-$15k (Cash)		-$15k (Utilities Expense)

    Now total the asset accounts:

    Cash: $\$100k - \$30k - \$30k - \$15k = \$25k$

    Equipment: $30k. Supplies: $50k. Accounts Receivable: $20k.

    Total Assets $= \$25k + \$30k + \$50k + \$20k = \$125k$.

    And verify from the right-hand side, as always: Notes Payable $100k, plus Accounts Payable $\$50k - \$30k = \$20k$, plus equity of $\$20k - \$15k = \$5k$. That is

$125k. The two sides agree, which is a good sign we did not fumble a sign along the way.

A note on the traps in the setup. The supplies were bought *on account*, which does not stop them from being an asset—it only changes what balances them. The services were provided *on account*, which does not stop them from being revenue—accrual accounting recognizes revenue when the work is done. The $30k payment on the account payable reduces cash *and* the liability, so it does not touch equity. Only the utilities are an immediate expense. So the answer is **(A)**.

16. If our company sells to a customer on account, this will affect the measures reported on the:

    (A) Income Statement

    (B) Balance Sheet

    (C) Statement of Shareholders' Equity

    (D) Multiple other answers.

    (E) None of the other answers

    Solution. Follow the single transaction outward and watch it touch three statements. We recognize **Revenue**—that is a line on the *Income Statement*. Revenue feeds Net Income, which feeds Retained Earnings, which is reported and reconciled on the *Statement of Shareholders' Equity*. And the balancing entry is an increase in **Accounts Receivable**, an asset on the *Balance Sheet*—where the ending Retained Earnings balance also lands.

    That is three of the four statements from one sale. (Only the Statement of Cash Flows is untouched, since no cash moved—which is precisely the situation accrual accounting was built to handle.) This is worth pausing on: the statements are not four independent reports that happen to be stapled together. They are four views of the same underlying set of transactions, and the core identities are what stitch them into one system. So the answer is **(D)**.

17. If our company collects on an account from a customer from a credit sale from a previous period, this will affect measures reported on our:

    (A) Income Statement

    (B) Balance Sheet

    (C) Statement of Shareholders' Equity

    (D) Multiple other answers

(E) None of the other answers

Solution. Compare this carefully with the question just above—they are deliberately paired.

Collecting on an account is *Cash* up, *Accounts Receivable* down. Both are asset accounts, so the only statement that moves is the **Balance Sheet** (and the composition of its asset side at that; the total doesn't even change). No revenue is recognized, because the revenue was earned and recorded *last period*, when we provided the service. So the Income Statement is untouched, and with it the Statement of Shareholders' Equity, since neither net income nor dividends changed.

The contrast between #16 and #17 is the whole idea of accrual accounting in miniature: the *earning* event moves the Income Statement, and the *cash* event does not. So the answer is **(B)**.

18. When a company purchases something in cash and 'capitalizes' that purchase as an asset:

(A) total assets will remain the same

(B) stockholders equity will increase

(C) total assets will increase

(D) total assets will decrease

(E) none of the other answers.

Solution. 'Capitalizing' is the jargon for recording an expenditure as an **asset** rather than running it through the Income Statement as an expense. If we capitalize a cash purchase, then Cash goes down and the new asset goes up by the same amount: total assets are unchanged, and the right-hand side of the equation never moves at all.

It is worth seeing why the alternative—*expensing*—would look different. If we had expensed the $X instead, Cash would still fall by $X, but nothing on the asset side would replace it; the balance would come from a negative $X in the Shareholders' Equity column. So total assets would fall, and so would equity. That contrast is the whole practical significance of the capitalize-versus-expense decision, and it is why managers are not indifferent to which one applies. Option (C) is the reflex of seeing '+Asset' and forgetting the cash that paid for it. So the answer is **(A)**.

19. Which of the following can partly determine whether an expenditure on something will be initially recognized as an asset or as an expense in the company's accounting:

(A) Whether the managers expect to use the thing over multiple reporting periods

(B) Whether the company received a discount on the thing

(C) Whether the company pays in cash or buys the thing on account

(D) Whether the thing can reliably be matched to revenues in future periods

(E) Two of the other options.

Solution. The asset-versus-expense decision turns on two criteria, and both are in this list.

(A) is the FASB definition applied directly: an asset is a probable *future* economic benefit, controlled by the firm. If the thing will be used up over multiple reporting periods, it is a candidate for capitalization; if it is consumed on the spot, like headquarters electricity, it is expensed now.

(D) is the matching principle. If we can reliably trace the cost to particular future revenues, we want to hold it in an asset account until those revenues arrive, so the two hit the Income Statement together. This is why a factory's electricity gets folded into *Inventory* while the head office's does not: the tracing is feasible in one case and hopeless in the other.

The other two are irrelevant, and each is irrelevant for a reason worth naming. **(B)**, the discount, affects the *amount* we record—we record what we paid—but has nothing to do with *which category* it goes in. **(C)** confuses the two halves of a transaction: how we *paid* determines whether Cash or Accounts Payable does the balancing, while what we *received* determines whether we have an asset or an expense. Those are independent questions. Since exactly two options qualify, the answer is **(E)**.

A.2.2 Practice Problems: Open Prompts

1. Account for the following transactions with the Accounting Equation format

   - 2/14/2022: Issue 1000 shares of *Common Stock* for $50 each.
   - 3/01/2022: Get a $200k loan from a bank.
   - 3/16/2022: Provide swimming lessons to customers for $10k in cash.
   - 3/17/2022: Pay quarterly electricity bill, $5k.
   - 3/22/2022: Receive payment of $8k from customers in advance, for lessons to be provided in April.
   - 04/01/2022: Pay biweekly salaries to employees, a total of $20k.

- 04/10/2022: Provide the swimming lessons as contractually agreed and paid for in the March 22 transaction above.
- 04/12/2022: Purchase *Supplies* on account, with a promise to pay the $2k invoice later.
- 04/15/2022: Pay biweekly salaries to employees, a total of $20k.
- 04/20/2022: Pay off the account from the April 12th purchase of *Supplies*.

Solution. Take them one at a time, and where there is cash in the transaction, start with cash.

Date	Assets	Liabilities	Shareholders' Equity
2/14/2022	+$50k (Cash)		+$50k (Common Stock)
3/01/2022	+$200k (Cash)	+$200k (Notes Payable)	
3/16/2022	+$10k (Cash)		+$10k (Revenue)
3/17/2022	-$5k (Cash)		-$5k (Utilities Expense)
3/22/2022	+$8k (Cash)	+$8k (Deferred Revenue)	
04/01/2022	-$20k (Cash)		-$20k (Salaries Expense)
04/10/2022		-$8k (Deferred Revenue)	+$8k (Revenue)
04/12/2022	+$2k (Supplies)	+$2k (Accounts Payable)	
04/15/2022	-$20k (Cash)		-$20k (Salaries Expense)
04/20/2022	-$2k (Cash)	-$2k (Accounts Payable)	

Now the reasoning, transaction by transaction—and note that six of these ten are variations on things we did in the chapter.

2/14. $1{,}000 \times \$50 = \$50k$ of cash comes in from investors, who receive shares of ownership. Owner-provided financing, so *Common Stock*, in the equity column. This is Transaction #1.

3/01. Cash in from a bank, which must be repaid: *Notes Payable*, a liability. Transaction #2.

3/16. We performed the service and were paid the same day—timing case one. Revenue is earned, and *Revenue* is a positive entry in the equity column.

3/17. Electricity that has already been consumed. Nothing is controlled going forward, so no asset; it is immediately expensed. ('Quarterly' is describing the billing cycle, not a future benefit we are buying.)

3/22. Cash in, service not yet provided—timing case three. No revenue yet. What we have taken on is an obligation to give lessons in April: *Deferred Revenue*, a liability.

04/01. Salaries are the classic immediate expense. Why not an asset? Because employees can walk away; they are not resources 'controlled by' the firm, and last month's paycheck buys us no enforceable claim on next month's work.

04/10. This is the one worth slowing down for. We now provide the lessons, so we finally earn the $8k of revenue. But what balances it? No cash arrives—we were already paid in March. No receivable arises—the customer owes us nothing. What has changed is that our *obligation* is discharged: *Deferred Revenue* goes to zero. So a liability falls, equity rises, and **assets do not move at all**. We earned revenue by decreasing a liability. (Which points at a rather elegant general definition: revenue is an increase in *net* assets, $A - L$, arising from operations.)

04/12. Supplies will be drawn down over time, so they are an asset. No cash moves, but an economic event has plainly occurred—we have more resources than we did this morning—and we have promised to pay: *Accounts Payable*.

04/20. Settling that promise. Cash out, liability out. Note what this is *not*: it is not an expense. The expense will come later, in Chapter 4, as we actually use the supplies up. Paying a bill and incurring a cost are two different events, and accrual accounting insists on keeping them apart.

Finally, a check on the whole set. Cash: $\$50 + \$200 + \$10 - \$5 + \$8 - \$20 - \$20 - \$2 = \$221k$; plus $2k of Supplies gives Total Assets of **$223k**. Liabilities: $200k of Notes Payable (the Deferred Revenue and the Accounts Payable both went back to zero). Equity: $\$50 + \$10 - \$5 - \$20 + \$8 - \$20 = \$23k$. And $\$200k + \$23k = \$223k$. It balances, as it must.

2. Our company has done the following transactions, and only these, to date.

Transaction	Assets	Liabilities	Sharehldr Equity
#1: Sell Shares for $25k	+$25k Cash		+$25k Common Stock
#2: Get loan of $10k	+$10k Cash	+$10k Note payable	
#3: Buy equipment for $24k	-$24k Cash, +$24k Equipment		
#4: Pre-pay 1 yr rent, $6k	-$6k Cash, +$6k Prepaid rent		
#5: Buy supplies on account, $2.3k	+$2.3k Supplies	+$2.3k Accounts Payable	

Suppose that we ended our reporting period here, with no more transactions of any kind. What would our *Balance Sheet* look like at this point? (That is, fill out a Balance Sheet, based on what the account balances would be, as of this point in time, after Transaction #5.)

Solution. Recall the reductive definition: a Balance Sheet is just a **list** of the company's Asset, Liability, and Shareholders' Equity accounts and their balances, as of a point in time. So the job is two steps. First, convert the *incremental effects* in the table above into *balances*, account by account. Second, write the list down in the standard two-column format.

Take the accounts one at a time, running down the columns:

Cash: $\$25k + \$10k - \$24k - \$6k = \$5k$. Equipment: \$24k. Prepaid Rent: \$6k. Supplies: \$2.3k. Notes Payable: \$10k. Accounts Payable: \$2.3k. Common Stock: \$25k.

And Retained Earnings? **Zero.** The company has financed itself and acquired resources, but it has not yet *earned* anything—there is not a single Revenue or Expense entry in that table. Do not skip the line; report it as \$0.

Balance Sheet, after Transactions #1–#5

Assets	**Liabilities**
Cash: \$5k	Accounts Payable: \$2.3k
Equipment: \$24k	Notes Payable: \$10k
Prepaid Rent: \$6k	**Total Liabilities: \$12.3k**
Supplies: \$2.3k	
	Shareholders' Equity:
	Common Stock: \$25k
	Retained Earnings: \$0
	Total Shareholders' Equity: \$25k
Total Assets: \$37.3k	**Total Liabilities and S.E.: \$37.3k**

$\$5k + \$24k + \$6k + \$2.3k = \$37.3k$ on the left; $\$12.3k + \$25k = \$37.3k$ on the right. It balances—and it *had* to, because we kept $A = L + E$ in balance on every single transaction, and the balances are nothing but the cumulation of those transactions. The Balance Sheet cannot fail to balance if the bookkeeping was done right; when it doesn't, that is a message about your arithmetic, not about the company.

3. Account for the following transactions with the Accounting Equation format

   - 05/15/2022: Customer pays us up front, \$5k, for lessons to be provided at a later date.
   - 06/15/2022: We provide the lessons to the customer above, from the May 15 transaction.
   - 07/15/2022: We provide lessons to another customer, on account—the customer accepts our invoice for \$6k
   - 08/15/2022: The customer from the July 15 service pays off their account in full.

Solution. This little sequence is beautifully constructed, so let me point out its architecture before doing the work: it walks you through **two complete sales**, run in opposite orders. In the first, cash comes *before* the service; in the second, cash comes *after*. In both, the revenue is recognized on exactly one date—the date we teach the lessons—and the cash date is bookkeeping around it.

Date	Assets	Liabilities	Shareholders' Equity
05/15/2022	+$5k (Cash)	+$5k (Deferred Revenue)	
06/15/2022		-$5k (Deferred Revenue)	+$5k (Revenue)
07/15/2022	+$6k (Accounts Receivable)		+$6k (Revenue)
08/15/2022	+$6k (Cash), -$6k (Accounts Receivable)		

05/15. Cash in, nothing earned. We owe lessons, so we record the liability *Deferred Revenue*. Resist the pull to call this revenue; the name of the account is there to remind you.

06/15. We teach the lessons, so now the $5k is earned: *Revenue*, positive in the equity column. The balancing entry is the *disappearance of the obligation*—Deferred Revenue drops to zero. Assets are completely untouched on this date. Nothing came in and nothing went out; we simply stopped owing something. That is revenue arising from a decrease in liabilities.

07/15. We teach first this time, so revenue is recognized immediately, balanced by the customer's promise to pay: *Accounts Receivable*, an asset.

08/15. The check arrives. Cash up, receivable down—one asset converted into another, entirely inside the asset column. **No revenue.** We already recorded it on July 15; recording it again would count the same $6k sale twice.

Step back and notice the symmetry. Each sale generated exactly $5k or $6k of revenue, on exactly one date, and in each case the revenue landed on the date the *lessons happened*—not on the date the money moved. Two sales, four transactions, two revenue events. And observe the shape of the 'other' two dates: in one, a liability went down; in the other, an asset was swapped. Either way, the accounting equation held on every line.

4. Suppose our company has the following transactions with our customers, and only these, in Fiscal Year 2021:

   - We provide tutoring to Dane for $400 in cash.
   - Alison pays us $100 in cash up front, for tutoring that we promise to give her next year, in Fiscal Year 2022.

- We provide tutoring to Avery, who promises/agrees to pay us $300 next year, in Fiscal Year 2022.
- We receive $200 in cash from Declan, for tutoring that we had provided to him in Fiscal Year 2020.

What are our (accrual) revenues for FY 2021?

Solution. $700.

One question, asked four times: *did we provide the tutoring in FY 2021?*

Dane: yes. Count the $400. The cash arriving the same day is incidental.

Alison: no. We have her $100, but we owe her a year of tutoring. Count $0. That $100 sits on our Balance Sheet as *Deferred Revenue*, a liability, and becomes revenue in FY 2022 when we actually teach her.

Avery: yes. The work is done; the revenue is earned. Count the $300. The unpaid invoice sits on our books as *Accounts Receivable*, an asset, and will simply be converted into cash next year.

Declan: no. We taught Declan in **FY 2020**, and FY 2020 is where that revenue was recognized. This year's $200 merely collects a receivable: Cash up, Accounts Receivable down, no revenue. Counting it now would be double-counting a sale from a prior year.

$\$400 + \$300 = \$700$.

Now compare that with the cash we collected this year: $\$400 + \$100 + \$200 = \$700$. The same number—by coincidence, and the coincidence is the point of the problem. Two very different sets of items happen to sum to the same total, and if you had taken the lazy route and added up the cash, you would have gotten full marks for entirely the wrong reason. Revenue and cash receipts are different quantities that occasionally collide.

A.2.3 Essay and Discussion Questions

1. It's common to hear CEOs say things like 'The reason we weren't profitable this quarter is that we're investing for the future.' For example, after missing Wall Street expected earnings, CEO "CEO Michelle Gass said in a statement that the company is 'investing in the short-term' in order to 'drive profitable growth over the long-term' " (L. Thomas, CNBC, May 2021). Based on what we've learned so far, what do you make of this? (Hint: This question is prompting you to first reference the specific technical thing we've learned that suggests why this excuse is not obviously legitimate.) Once you've covered that technical bit, give your own opinion according to your own reasoning and judgment.

Solution. Here is how I would think this through: first the technical point, which is sharper than most people realize; then the honest qualifications; then my own view.

The technical point. The excuse presupposes a theory of accounting that accrual accounting explicitly rejects—namely, that spending money reduces profit. It doesn't, not necessarily. This whole chapter has been an extended demonstration of the opposite. When a company spends $24k on equipment it will use for years, the effect on Net Income is **zero**: Cash falls, Equipment rises, and the Income Statement never hears about it. Same for prepaid rent, same for supplies, same for inventory. The accounting system already contains, at its very core, a distinction between spending-as-consumption and spending-as-investment, and it *already* keeps genuine investment out of current earnings. That is precisely what capitalization is for.

So when a CEO says 'we would have been profitable, but we're investing for the future,' the technically informed response is: if you were really investing, the accountants would have **capitalized** it, and your earnings would not have suffered. The fact that these expenditures *did* hit the Income Statement is itself evidence about their character. Something failed the asset test—either the benefits do not extend past this period under the firm's control, or they cannot be reliably matched to future revenues. Whatever the CEO calls it, the accounting system looked at it and concluded: consumed, not invested.

The honest qualifications—because if I only gave you the paragraph above, I would be selling you a slogan. Three of them, and they are real.

First, some genuinely investment-like spending is *not permitted* to be capitalized under U.S. GAAP. The prime example is **research and development**, which is expensed as incurred even though its entire purpose is to generate future benefits. Advertising and brand-building are largely the same. Employee training too. Conservatism keeps these off the balance sheet because their payoffs are too uncertain to verify—which is a defensible rule, but it does mean a company that is really pouring money into R&D will show depressed current earnings for a genuinely forward-looking reason. If that is what the CEO means, the excuse has teeth.

Second, capitalizing does not make the cost vanish; it *defers* it. A new store or distribution center starts depreciating right away, and a big investment program does eventually weigh on reported earnings.

Third, some real investments are not expenditures at all. Cutting prices to gain market share, or absorbing costs to keep customers happy through a bad quarter,

shows up as lower revenue or higher expense with no asset anywhere—and can be a perfectly rational long-horizon play.

My own view. Notice, first, the position the statement is being made from. It is offered by the person whose performance the number measures, in the moment the number came in below expectations, to an audience that can fire her. That does not make it false. It does mean the claim carries approximately zero information *on its own*, because it is exactly what a CEO would say whether or not it were true—and, in the Kohl's example, note that the company had in fact just *beaten* expectations, which should make you read that quote as forward-looking positioning rather than as an excuse for a miss.

So my test is: does the claim make contact with anything checkable? A CEO who says 'we opened forty stores and R&D spending rose 60%, here are the line items' is saying something with content, and I can verify it in the statements and judge the projects on their merits. A CEO who says 'we are investing in the long term' with no line items attached is saying something unfalsifiable, and I discount it entirely. The most useful reflex this chapter gives you is the follow-up question: **'invested in *what*, exactly—and why isn't it on your balance sheet?'** The answer to that second half is usually the interesting part.

And this connects to why Chapter 1 insisted that accounting is about *accountability*, not just information. If every shortfall could be relabeled 'investment,' performance measurement would collapse, because no manager would ever have a bad quarter—they would only ever have quarters in which they were building the future. The rules about what may be capitalized exist precisely so that the classification is not the manager's to choose.

2. What are the two main principles that define accrual-basis accounting? And what is the idea behind them? What is the logic behind this complexity? In other words, can you give some examples or explanation of why these two principles seem to make economic sense?

   Solution. The two principles are **revenue recognition** and **expense matching**. Both are departures from the simpler alternative—cash accounting—and both departures are in the service of one goal.

   Revenue recognition: we record revenue when it is **earned**, that is, when we deliver the good or provide the service—not when the cash arrives. **Expense matching:** we record a cost as an expense in the period in which it helps generate the corresponding revenue—not when the cash is paid out.

   The single idea behind both. Accrual accounting is trying to measure the profitability of the company's real **activities** over a period. Cash accounting mea-

sures something else: the movement of money in and out of a bank account. Those are different questions, and the timing of the money is often a poor proxy for the timing of the activity. Every complication that follows is downstream of that one commitment.

Why revenue recognition makes economic sense. My favorite argument here is a thought experiment. Suppose we ship a huge order to WalMart on December 18. Two universes diverge. In one, WalMart's check arrives December 30. In the other, a snowstorm delays the mail and it arrives January 2. Under cash accounting, our reported profit for the year differs between those universes—materially—because of the *weather*. Does that seem like a measure of how well the business performed? Obviously not. We did exactly the same thing in both worlds.

There is a second argument, about timeliness, and the delays are longer than people guess. Retail is seasonal: a retailer stocks up in November, sells in December, and pays its suppliers in January or February. So if our young company lands its first transformative contract and ships in November 2022 with payment due February 2023, cash accounting would put that sale in the FY 2023 statements—which, after preparation and audit, reach investors in early 2024. Our investors would learn about the deal that defined our company roughly *fifteen months* after we did it. The whole story of our year would be missing from our own annual report.

And if the suspicion nags—'but we don't have the *money* yet'—it is worth worrying about what having money even means. The number your bank shows you is not a drawer of bills with your name on it; it is the bank's legally enforceable obligation to pay you on demand. A creditworthy customer's promise to pay is the same species of thing. Wealth mostly consists of enforceable claims, and accrual accounting simply takes that seriously.

Why matching makes economic sense. Same style of argument. Suppose I pay $100k in January 2019 for T-shirts, sell half in 2019 for $75k cash and half in 2020 for $75k cash. On a cash basis, 2019 shows a $25k loss and 2020 a $75k gain. But those were two identical years of the identical business, selling identical shirts to identical customers. Two identical years should produce two identical measures of performance. Matching does exactly that: apportion $50k of cost to each year, and report $25k of profit in each.

The everyday intuition is that there are two very different kinds of spending in your own life. A bag of Sour Patch Kids at the 7-Eleven: money spent, benefit consumed, episode over. Tuition, or renovating a house: money spent now to create benefits over many years. It would be strange to describe the second as

making you poorer in the year you paid. Matching is the bookkeeping recognition of that distinction—which is also why, mechanically, expenditures on multi-period resources are first recorded as **assets** and only later expensed as they are used up. Assets are expenses in waiting.

Is the complexity worth it? In my view, plainly yes—but the honest way to hold this is to notice that we bought something and paid for it. What we bought is a measure that tracks activity rather than the mail. What we paid is **discretion**: accrual accounting requires judgments about when something is earned and how costs should be apportioned, and judgment can be abused. A firm can ship product aggressively to customers unlikely to pay, and book the revenue. That is not a hypothetical failure mode; it is a recurring one.

Which is exactly why the Cash Flow Statement exists alongside the Income Statement—not because cash accounting was right after all, but as a check on the estimates that make accrual accounting worth having. Net income rising while cash from operations falls is a red flag. And note the shape of that arrangement: we did not choose one system over the other. We built the better measure, and then kept the cruder one around to keep it honest.

A.3 Chapter 3

A.3.1 Practice Problems: MCQ's

1. Which of the following "tools" reports the final balances of all of our company's accounts, in one list/table?

   (A) A trial balance

   (B) A T-account

   (C) A journal entry

   (D) An income statement

   (E) None of the other answers

   Solution. Read the stem for its two load-bearing words: *all* of the accounts, and *final balances*. That is the definition of the **Trial Balance**—literally just a list of where every account stands at a point in time. A T-account gives you the full history of *one* account; a journal entry gives you the incremental effect of *one* transaction. The income statement is the tempting one, but it reports only the revenue and expense accounts, and it is a financial statement, not one of our three organizing tools—indeed, the Trial Balance is the thing we will *use* to fill out the income statement. So the answer is **(A)**.

2. Which tool reports the incremental effect of a particular transaction–i.e., the *changes* in the accounts affected?

   (A) A trial balance

   (B) A T-account

   (C) A journal entry

   (D) An income statement

   (E) None of the other answers

   Solution. "Incremental effect of a particular transaction" is the job description of the **journal entry**. Recall Transaction #2 from the chapter: when we borrowed \$5k, the entry read *Cash (+A)* \$5k, *Notes Payable (+L)* \$5k—and it said nothing at all about the \$15k of cash already sitting in the account from Transaction #1. A journal entry is a report of a *change*, not of a *balance*. So the answer is **(C)**.

3. Which tool reports the full sequence of updates to, and final balance of, a *particular* account?

   (A) A trial balance

(B) A T-account

(C) A journal entry

(D) An income statement

(E) None of the other answers

Solution. One account, its whole history, netted to a balance—that is the **T-account**, and it is the thing your banking app has been showing you your whole life. The Trial Balance is the wrong answer here for a precise reason: it reports final balances for *all* accounts but throws away the sequence that produced them. So the answer is **(B)**.

4. What does debit *mean*?

(A) *Good.*

(B) "On the right."

(C) The company made a bad choice.

(D) Shareholders' Equity has increased.

(E) None of the other answers

Solution. Debit means **"on the left."** That is the whole content of the word. Option (B) has the right *kind* of answer—purely geographic—but points the wrong way; that's credit. Options (A) and (C) are the moralized folk theory the chapter warns against: debits and credits carry no moral content whatsoever. And (D) is exactly backwards—Shareholders' Equity *increases* with a credit, which is precisely the observation that collapses the "assets are good, so debits are good" story. Since none of the four is right, the answer is **(E)**.

5. If a liability account increases, that's a (credit / debit)

(A) Credit

(B) Debit

(C) Not enough information

(D) None of the other answers.

Solution. Reason it out from the one anchor: assets sit on the *left* of the Accounting Equation, and asset increases are recorded on the *left* of their accounts (debits). Liabilities sit on the *other* side of the equation, so their increases go on the *other* side of their accounts. That's a credit. Note that no further information is needed—this is a rule about the *type* of account, not about the size of the balance or what else happened. So the answer is **(A)**.

6. If the *Common Stock* account increases, that's a (credit / debit)

   (A) Credit

   (B) Debit

   (C) Not enough information

   (D) None of the other answers.

   Solution. *Common Stock* is a Shareholders' Equity account, and L and E behave identically under the rules: increases are credits. Transaction #1 in the chapter is exactly this—debit *Cash*, credit *Common Stock*, $15k. And this is the case that kills the moralized reading of the words: more equity in the corporation sounds like unambiguously good news, and it is nonetheless a credit. So the answer is **(A)**.

7. Which of the below is a **debit account**?

   (A) *Common Stock*

   (B) *Equipment*

   (C) *Notes Payable*

   (D) Multiple other answers

   (E) None of the other answers.

   Solution. "Debit account" is the natural-balance jargon: an account is a debit account if *increases* in it are recorded as debits, so that its ordinary, healthy balance sits on the debit side. *Equipment* is an asset, so increases are debits—it is a debit account. *Common Stock* is equity and *Notes Payable* is a liability, and both increase with credits. Exactly one of the three qualifies, so (D) is out. So the answer is **(B)**.

8. Which of the below is a **credit account**?

   (A) *Common Stock*

   (B) *Equipment*

   (C) *Revenue*

   (D) Multiple other answers

   (E) None of the other answers.

   Solution. Work through them: *Common Stock* is equity, increases with a credit—credit account. *Equipment* is an asset, increases with a debit—debit account. *Revenue* increases Shareholders' Equity, so revenues are credits, and since there

is no such thing as an anti-revenue, its natural balance is always a credit—credit account. That's two of the three, which is what (D) says. So the answer is **(D)**.

9. *Salaries Expense* is a (debit account / credit account):

   (A) Debit account

   (B) Credit account

   (C) Not enough information

   (D) None of the other answers.

   Solution. Expenses are anti-revenues. Revenues raise Net Income, which raises Retained Earnings, which raises Shareholders' Equity—so revenues are credits; expenses run the chain in reverse and are therefore debits. And expense increases are debits, so its ordinary balance before closing is a debit. Closing and reversal entries can move it in the opposite direction. That's what Transaction #9 in the chapter looks like: debit *Salaries Expense*, credit *Cash*. So the answer is **(A)**.

10. Indicate the correct journal entry for this transaction: **Get a $10k loan from a bank**.

    (A) Debit *Cash* $10k; Credit *Notes Payable* $10k

    (B) Debit *Notes Payable* $10k; Credit *Cash* $10k

    (C) Debit *Notes Payable* $10k; Credit *Liability* $10k

    (D) Debit *Cash* $10k; Credit *Accounts Payable* $10k

    (E) Not enough information—depends on prior cash and liability balances.

    Solution. Start with cash: we are receiving $10k, an asset going up, so debit *Cash*. What balances it? We now owe the bank, so a liability goes up, and that is a credit.

	Debit	**Credit**
Cash (+A)	$10k	
Notes Payable (+L)		$10k

Option (B) is the same entry run backwards—that would be *paying off* the loan. Option (C) credits an account called "Liability," which is a category, not an account; and it leaves the cash unaccounted for entirely. Option (D) gets the direction right but the wrong creditor: *Accounts Payable* is financing from our suppliers, *Notes Payable* is financing from a bank. And (E) is the standing temptation

of this whole problem set—a journal entry records the *change*, so prior balances never matter. So the answer is **(A)**.

11. Indicate the correct journal entry for this transaction: **Customer pays $10k for services that we deliver immediately**.

    (A) Debit *Cash* $10k; Credit *Deferred Revenue* $10k

    (B) Debit *Revenue* $10k; Credit *Cash* $10k

    (C) Debit *Cash* $10k; Credit *Revenue* $10k

    (D) Debit *Accounts Receivable* $10k; Credit *Revenue* $10k

    (E) Not enough information—depends on prior account balances.

    Solution. We provided the service, and "provide the service" is the cue to record revenue—and revenue is always a credit. The balancing debit is the cash coming in the door.

	Debit	**Credit**
Cash (+A)	$10k	
Revenue (+S.E.)		$10k

    Option (A) would be right if we had taken the money and *not yet* delivered—that's Transaction #8, where the liability *Deferred Revenue* records our obligation to perform later. Here we performed immediately, so no obligation ever exists. Option (D) records a receivable, but the customer paid; nobody owes us anything. Option (B) has both accounts on the wrong sides. So the answer is **(C)**.

12. Indicate the correct journal entry for this transaction: **We provide services on account, to a customer who accepts an invoice for $10k**.

    (A) Debit *Cash* $10k; Credit *Deferred Revenue* $10k

    (B) Debit *Revenue* $10k; Credit *Cash* $10k

    (C) Debit *Cash* $10k; Credit *Revenue* $10k

    (D) Debit *Accounts Receivable* $10k; Credit *Revenue* $10k

    (E) Not enough information—depends on prior account balances.

    Solution. This is Transaction #7. The service is provided, so the revenue is earned and recorded now, credit side. But no cash arrived—what we got instead was the customer's promise to pay, an asset, and assets going up are debits.

	Debit	Credit
Accounts Receivable (+A)	$10k	
Revenue (+S.E.)		$10k

Notice the contrast with the previous problem: the credit side is identical in both, because the revenue trigger—delivery—is identical. All that changed is *what we received in exchange*, and that only touches the debit. So the answer is **(D)**.

13. Indicate the correct journal entry for this transaction: **The customer from the transaction above, now weeks later, pays down their outstanding account of $10k, in full.**

    (A) Debit *Cash* $10k; Credit *Revenue* $10k

    (B) Debit *Revenue* $10k; Credit *Cash* $10k

    (C) Debit *Cash* $10k; Credit *Accounts Receivable* $10k

    (D) Debit *Accounts Receivable* $10k; Credit *Revenue* $10k

    (E) None of the above.

    Solution. This is Transaction #11, the deceptively tricky one. Cash comes in, so debit *Cash* $10k. The balancing credit is *not* revenue: we recorded that revenue weeks ago, when we provided the service, and recording it again now would count one sale twice. What is actually happening is that the customer's promise—the receivable—is being extinguished.

	Debit	Credit
Cash (+A)	$10k	
Accounts Receivable (-A)		$10k

If the "good news, so why does my asset go *down*?" feeling is nagging at you, work backward from the T-account. *Accounts Receivable* means "what our customers still owe us." After today they owe us nothing, so the balance must be zero; and what takes a $10k debit balance to zero is a $10k credit.

Accounts Receivable

Debits	Credits
$10k	$10k
$0	

Option (A) is the classic error: answering with the *combined* effect of both dates, as if the sale and the collection were one event. So the answer is **(C)**.

14. Indicate the correct journal entry for this transaction: **Purchase Supplies that we plan to use for several periods, for $10k in cash**.

    (A) Debit *Cash* $10k; Credit *Supplies* $10k

    (B) Debit *Supplies Expense* $10k; Credit *Accounts Payable* $10k

    (C) Debit *Supplies* $10k; Credit *Cash* $10k

    (D) Not enough information—depends on prior account balances.

    Solution. Start with cash: $10k goes out the door, an asset down, so credit *Cash*. What did we get for it? A resource we expect to use over several periods—which is what an asset *is*—so debit *Supplies*.

	Debit	**Credit**
Supplies (+A)	$10k	
Cash (-A)		$10k

    Option (B) encodes the genuinely interesting confusion: shouldn't buying supplies be an *expense*? Not yet. The phrase "plan to use for several periods" is doing deliberate work in the stem. We have not used the supplies up; we have merely converted one asset (cash) into another (supplies). The expense comes later, as we consume them—which is exactly the machinery of Chapter 4. (Option (B) also credits *Accounts Payable*, when we paid cash.) So the answer is **(C)**.

15. Indicate the correct journal entry for this transaction: **Pay $10k salaries to our employees, on an ordinary bi-weekly payday in July** (that is, in the middle of our annual calendar-year reporting period).

    (A) Debit *Cash* $10k; Credit *Salaries Expense* $10k

    (B) Debit *Employee Assets* $10k; Credit *Cash* $10k

    (C) Debit *Salaries Expense* $10k; Credit *Cash* $10k

    (D) Debit *Salaries Expense* $10k; Credit *Salaries Payable* $10k

    Solution. This is Transaction #9. Cash goes out, so credit *Cash*; and the money bought labor that has already been consumed, which is an expense, and expenses are debits.

	Debit	**Credit**
Salaries Expense (-S.E.)	$10k	
Cash (-A)		$10k

Option (B) is the trap Chapter 2 set up and dismissed: we do not own our employees, so there is no "Employee Assets" account. Option (D) is worth dwelling on, because it will be right *in a different problem*—it's the entry for salaries the employees have **earned but not been paid**. The stem rules that out twice over: it is an *ordinary* payday, so we are settling up in cash, and it is the middle of the period, so we are not doing end-of-period adjusting entries. So the answer is **(C)**.

16. What will our *Cash* **T-account** look after our company begins its life and does the following 3 transactions: #1: Sell Shares of Common Stock for $25k, #2: Get a bank loan for $10k, #2: Buy Equipment for $24k.

(A)

Cash (A)

Debits	Credits
	$24k

(B)

Cash (A)

Debits	Credits
$25k	$24k
$10k	

(C)

Cash (A)

Debits	Credits
$24k	$25k
	$10k

Solution. Walk the three transactions and ask, each time, only what happened *to Cash*. Sell stock: cash in $25k, an asset up, so a debit. Bank loan: cash in $10k, debit again. Buy equipment: cash out $24k, an asset down, so a credit. Two debits and one credit, netting to a $11k debit balance:

Cash

Debits	Credits
$25k	$24k
$10k	
$11k	

Option (A) is the "only the most recent change" error—it treats the T-account as if it were a journal entry, showing just the latest transaction. Option (C) has the whole history but the sides flipped, which would mean the company got poorer by selling stock and richer by buying equipment. So the answer is **(B)**.

17. Previously, on January 3, 2022, our customer bought from us on account, promising to pay $20k later. Now, today, on February 23rd, they pay off half of their account, paying $10k in cash to us. What is the journal entry for this February 23rd transaction?

    (A) Debit *Cash* $10k; Credit *Accounts Receivable* $10k

    (B) Debit *Cash* $10k; Credit *Revenue* $10k

    (C) Debit *Cash* $10k, Debit *Accounts Receivable* $10k; Credit *Revenue* $20k

    (D) Debit *Cash* $10k; Credit *Accounts Receivable* $10k, Credit *Revenue* $20k

    (E) No journal entry—not an external transaction.

    Solution. A journal entry is always tied to a specific event on a specific date, and today's event is a collection, not a sale. The $20k of revenue belongs to January 3. Today, $10k of cash arrives and $10k of the customer's promise is extinguished:

	Debit	**Credit**
Cash (+A)	$10k	
Accounts Receivable (-A)		$10k

    The receivable is only *half* settled, so its balance goes from a $20k debit to a $10k debit—it does not go to zero, and that is fine; the customer still owes us $10k. Option (C) is the January 3 entry, offered to see whether you'll answer with the wrong date's transaction. Option (D) isn't even a legal entry: $10k of debits against $30k of credits. And (E) is wrong on its own terms—cash changed hands with an outside party, which is as external as a transaction gets. So the answer is **(A)**.

18. A new customer comes in and purchases services of $20k, but can only pay $10k in cash today. We accept the $10k in cash from the customer, and the customer accepts a $10k invoice to pay the remainder later. Indicate the correct journal entry for this transaction:

    (A) Debit *Cash* $20k; Credit *Revenue* $20k

    (B) Debit *Cash* $10k; Credit *Revenue* $10k

    (C) Debit *Cash* $10k, Debit *Accounts Receivable* $10k; Credit *Revenue* $20k

(D) Debit *Cash* \$10k; Credit *Deferred Revenue* \$10k

(E) No journal entry—transaction not completed yet.

Solution. Your first compound entry—three lines instead of two. Start where you are most confident: the services were *provided*, all \$20k of them, so we credit *Revenue* \$20k. Now the algorithm demands \$20k of debits, and the stem tells you where they came from: \$10k arrived as cash, and \$10k arrived as the customer's promise to pay.

	Debit	**Credit**
Cash (+A)	\$10k	
Accounts Receivable (+A)	\$10k	
Revenue (+S.E.)		\$20k

Debits of \$10k + \$10k = \$20k equal credits of \$20k, so the Accounting Equation is undisturbed. Option (B) is the pull of the cash-basis intuition—recognizing only the part that was paid—and it is precisely what accrual accounting forbids: we earned \$20k, so we report \$20k. Option (D) has the deferral running the wrong way; *Deferred Revenue* is for cash received *before* delivery, and here the delivery is complete and it is the *cash* that is running late. So the answer is **(C)**.

19. Frankie owns a landscaping and lawn maintenance company, while Riley owns a machine repair shop. For the month of July, the following transactions occurred.

    - July 3: Frankie provides lawn services to Riley's repair shop on account, \$370.
    - July 6: One of Frankie's mowers malfunctions. Riley provides repair services to Frankie on account, \$320.
    - July 9: Riley pays \$370 to Frankie for lawn services provided on July 3.

20. What is the correct journal entry for the July 3rd transaction, from Frankie's perspective?

    (A) Debit Operating Expenses \$370, Credit Cash \$370

    (B) Debit Accounts Receivable \$370, Credit Cash \$370

    (C) Debit Cash \$370, Credit Accounts Payable \$370

    (D) Debit Accounts Payable \$370, Credit Cash \$370

    (E) None of the other answers

Solution. The phrase "from Frankie's perspective" is the whole problem. Frankie is the *seller* here: she mowed the lawn, so she has provided the service and earned $370 of revenue, and she was paid in a promise rather than in cash. Her entry is:

	Debit	**Credit**
Accounts Receivable (+A)	$370	
Revenue (+S.E.)		$370

Every offered option is wrong in the same structural way—each one credits or debits *Cash*, and no cash moved on July 3. That is the point of "on account." So the answer is **(E)**.

21. Continuing with the setup of the problem above, what is the correct journal entry for the July 3rd transaction, from Riley's perspective?

    (A) Debit Operating Expenses $370, Credit Cash $370

    (B) Debit Accounts Receivable $370, Credit Cash $370

    (C) Debit Cash $370, Credit Accounts Payable $370

    (D) Debit Accounts Payable $370, Credit Cash $370

    (E) None of the other answers

    Solution. Same event, opposite side of the table. Riley is the *customer*: he received lawn maintenance, consumed it, and owes $370 for it. So he records an expense—debit—and the obligation to pay, a liability going up, which is a credit:

	Debit	**Credit**
Operating Expenses (-S.E.)	$370	
Accounts Payable (+L)		$370

    Option (A) is the near-miss designed to catch you: it has the correct debit, and it fails only on the credit, because Riley hasn't paid yet. Notice how Frankie's and Riley's entries mirror each other—her *Accounts Receivable* is his *Accounts Payable*, the same promise viewed from the two ends. But (A) is still wrong, so the answer is **(E)**.

22. Continuing with the setup of the problem above, what is the correct journal entry for the July 9th transaction, from Frankie's perspective?

    (A) Debit Operating Expenses $370, Credit Cash $370

(B) Debit Accounts Receivable $370, Credit Cash $370

(C) Debit Cash $370, Credit Accounts Payable $370

(D) Debit Accounts Payable $370, Credit Cash $370

(E) None of the other answers

Solution. Frankie collects. Cash in, so debit *Cash* $370; and Riley's promise is now extinguished, so the receivable she set up on July 3 comes off the books:

	Debit	**Credit**
Cash (+A)	$370	
Accounts Receivable (-A)		$370

No revenue today—that was July 3's business. Option (D) is Riley's July 9 entry, not Frankie's; whenever a problem hands you two parties, check which set of books you are writing in before you check anything else. So the answer is **(E)**.

23. CoyoteCorp is a small company that provides nature tours, hiring nature guides as temps during the summer. At the beginning of FY2017, CoyoteCorp has a balance of $12k in the Retained Earnings account. Over the FY2017 reporting period, CoyoteCorp had the following transactions and only these:

- Issued an additional $10k in Common Stock
- Provided services to customers on account, $20k
- Purchased Equipment for Cash, $8k
- Paid employee salaries in full for the period, $10k
- Pay dividends to stockholders, $6k

24. What would CoyoteCorp's reported Retained Earnings be as of the end of FY2017?

(A) $4k

(B) $10k

(C) $16k

(D) $26k

(E) None of the other answers.

Solution. Use the identities from Chapter 1, and sort the five transactions by whether they touch Retained Earnings at all.

Two of them don't. Issuing $10k of *Common Stock* raises Shareholders' Equity, but through the *Common Stock* account, not Retained Earnings—those are two separate components of equity. And purchasing $8k of *Equipment* for cash is one asset swapped for another, with no effect on equity of any kind. (Yes, the equipment will eventually generate depreciation expense—but that is Chapter 4's business, and this problem gives us no useful-life assumption.)

The other three do. Revenue of $20k less *Salaries Expense* of $10k gives Net Income of $20k - $10k = $10k. Dividends of $6k are then paid out of that, so Retained Earnings grows by $10k - $6k = $4k over the year. And the company started the year with $12k already retained, so it ends with $12k + $4k = $16k.

Note that the $20k of revenue counts in full even though it was earned *on account* and no cash has arrived—we provided the service, so we earned it. The distractors map onto the three ways to go wrong: (A) $4k is the *change* in Retained Earnings, forgetting the $12k the company brought in with it; (B) $10k stops at Net Income, forgetting the dividends; and (D) $26k is what you get by dumping the $10k stock issuance into Retained Earnings on top of everything else. So the answer is **(C)**.

A.3.2 Practice Problems: Open Prompts

1. Explain why expenses are debit accounts (2-3 sentences).

   Solution. Follow the chain of identities: expenses reduce Net Income, Net Income flows into Retained Earnings, and Retained Earnings is a component of Shareholders' Equity—so an expense is ultimately a *decrease* in equity, and decreases in equity are debits. Equivalently, and faster: revenues increase equity and are therefore credits, and an expense is just an anti-revenue, so it must sit on the opposite side. Either route lands you in the same place, which is a good sign that the rules are a system rather than a list.

2. Explain to me, as if I am a newbie, what the difference between a journal entry and a T-account is. Why is it that, for example, when we *journalized* Transaction #2, we just accounted for the $5k in *Cash* from the bank loan, and ignored the *Cash* from the previous *Common Stock* issuance (2-4 sentences).

   Solution. A journal entry answers the question "what did *this one transaction* do to the accounts?"—it reports a **change**, and it is organized around a single event on a single date. A T-account answers a different question: "what has happened to *this one account*, over its whole life, and where does it stand now?"—it reports a **history and a balance**, and it is organized around a single account across all events. So when we journalized the bank loan, the entry recorded only the $5k

that the loan itself moved; the earlier \$15k from the stock issuance wasn't part of that event, and a journal entry that reported it would be answering a question nobody asked. The prior balance is not lost, of course—the whole sequence of transactions "lives on" in the *Cash* T-account, which is exactly the tool built to remember it.

3. "Journalize" (that is, write out the journal entries) the following transactions:

   - January 2: Corporation begins its life and sells 5000 shares of Common Stock for \$1,000 a share.
   - January 3: Purchase a large factory (*Plant*) for \$2m.
   - January 5: Purchase *Supplies* for \$1m, on account.
   - January 12: Pay off the account from the January 5th transaction, in full.

   Solution. Take them one at a time, and in each case start with the account you are surest about.

   January 2. First, the arithmetic the stem is hiding: $5{,}000 \text{ shares} \times \$1{,}000/\text{share} = \$5{,}000{,}000$. Cash comes in, so debit *Cash*; *Common Stock* is an equity account going up, so credit it.

	Debit	**Credit**
Cash (+A)	\$5m	
Common Stock (+S.E.)		\$5m

   January 3. Start with cash: \$2m goes out, an asset down, a credit. What did we get for it? A factory—a resource we will use for many periods, hence an asset.

	Debit	**Credit**
Plant (+A)	\$2m	
Cash (-A)		\$2m

   January 5. "On account" means no cash moves. We get an asset, *Supplies*, and we take on an obligation to our supplier—financing from our suppliers, which is *Accounts Payable*.

	Debit	**Credit**
Supplies (+A)	\$1m	
Accounts Payable (+L)		\$1m

January 12. Now the cash finally moves, and *this* is the event that discharges the obligation. Use the work-backward-from-the-balance trick: *Accounts Payable* carried a $1m credit balance, and after paying in full we owe nothing, so the balance must return to zero. What takes a $1m credit balance to zero is a $1m debit.

	Debit	**Credit**
Accounts Payable (-L)	$1m	
Cash (-A)		$1m

Note that no *Supplies Expense* appears anywhere in these four entries, and no expense of any kind. Nothing here has been used up yet—we have only bought things and paid for them. That, too, is Chapter 4's business.

4. Write out all the **T-accounts** (including netting out final balances) for the company above, after the January 12th transaction.

   Solution. Five accounts got touched. Go back through the four journal entries and sort every line by *account* rather than by *transaction*—that is the whole operation, and it is worth doing slowly once so that you trust it later.

Cash

Debits	Credits
$5m	$2m
	$1m
$2m	

Plant

Debits	Credits
$2m	
$2m	

Supplies

Debits	Credits
$1m	
$1m	

Accounts Payable

Debits	Credits
\$1m	\$1m
	\$0

Common Stock

Debits	Credits
	\$5m
	\$5m

Only two accounts here actually required netting—*Cash*, hit three times, and *Accounts Payable*, hit twice. The rest were touched exactly once, so their "final balance" is just the single amount from their one journal entry. And *Accounts Payable* nets to zero, which is precisely what we want after paying an account in full: we owe our supplier nothing, so the account says nothing.

The *Cash* account is worth a second look, because it is the one account here that narrates something rather than just reporting a number. Read down the debit column, then the credit column: \$5m came in from the owners on January 2, then \$2m went out for the factory and \$1m to the supplier, leaving \$2m still on hand. That is the first two weeks of this company's life in one account—and notice the ordering, which is not an accident: the spending was only possible because the financing came first. That is Chapter 1's *sources of financing* reading of the Accounting Equation, showing up in miniature. The \$5m of *Common Stock*, over on the $L + E$ side, is where the money came *from*; the *Plant*, the *Supplies*, and the \$2m of *Cash* still sitting there, on the A side, are what we did *with* it.

5. Write out the **Trial Balance** at that point in time (that is, immediately after the January 12th transaction)

   Solution. A Trial Balance is nothing more than a list of the balances we just netted out, sorted into a debit column and a credit column. Each account's balance goes in the column its balance sits on—the three assets on the debit side, *Common Stock* on the credit side.

Account:	**Debit**	**Credit**
Cash:	\$2m	
Plant:	\$2m	
Supplies:	\$1m	
Accounts Payable:		\$0
Common Stock:		\$5m
Total:	\$5m	\$5m

Trial Balance immediately after the January 12th transaction

Total debits of \$5m equal total credits of \$5m, which is the check we run every single time. Two details worth noting. First, there is no *Retained Earnings* row with a balance—the corporation began its life on January 2, so it has no earnings retained from any previous period, and in any case none of these four transactions touched revenue, expense, or dividends. Second, *Accounts Payable* is carried at zero. You could just as well drop a zero-balance account from the list; I've left it in to make visible that it existed and was settled.

6. "Journalize" the following transactions:

   - February 1: Provide services to customers for \$10k in cash.
   - February 5: Accept payment up front of \$5k, from customers who plan to return for services later.
   - February 12: The customers from the February 5th transaction come back, and we provide the agreed services to them.
   - February 15: Pay \$10k salaries to our employees, an ordinary biweekly payday.
   - February 20: We provide lessons to customers who accept an invoice for \$5k and promise to pay us later.
   - February 27: the customers from the February 20 transaction come back and pay off their account in full.

   Solution. This sequence is built as three matched pairs, and it is easier if you see the structure: February 5 and 12 are cash-before-service, February 20 and 27 are service-before-cash, and February 1 is the simple case where the two coincide.

   February 1. Service provided, cash received, both on the same day.

	Debit	Credit
Cash (+A)	$10k	
Revenue (+S.E.)		$10k

February 5. Cash arrives, but we have not provided anything—so no revenue. What we have instead is an *obligation* to perform later, a liability. This is customer financing: they have effectively lent us $5k, to be repaid in services.

	Debit	Credit
Cash (+A)	$5k	
Deferred Revenue (+L)		$5k

February 12. Now we deliver. No cash moves today—it moved a week ago—which is what makes this entry feel abstract. Anchor on the thing you know: we provided the service, so we credit *Revenue* $5k. The balancing debit has to be the obligation we just discharged, and a liability going down is a debit.

	Debit	Credit
Deferred Revenue (-L)	$5k	
Revenue (+S.E.)		$5k

February 15. An ordinary payday, mid-period: the labor is consumed, so the expense is a debit, and the cash goes out.

	Debit	Credit
Salaries Expense (-S.E.)	$10k	
Cash (-A)		$10k

February 20. The mirror image of February 5: service now, cash later. Revenue is earned and credited today; what we receive in exchange is a promise, an asset.

	Debit	Credit
Accounts Receivable (+A)	$5k	
Revenue (+S.E.)		$5k

February 27. Collection. Cash in, receivable extinguished, and emphatically *no* second helping of revenue.

	Debit	Credit
Cash (+A)	\$5k	
Accounts Receivable (-A)		\$5k

One check worth running at the end: total *Revenue* for February is $\$10k + \$5k + \$5k = \$20k$, recognized on the three dates we actually *delivered* something—the 1st, the 12th, and the 20th. Total cash collected from customers is \$20k as well, but on entirely different dates—the 1st, the 5th, and the 27th. Same total, different calendar. That gap between the two is accrual accounting in one line.

A.3.3 Essay and Discussion Questions

1. Explain to me, as if I am a newbie, what debits-and-credits are all about. That is, explain how this is an algorithm that keeps The Accounting Equation in balance.

   Solution. Here is how I would think it through, in four moves.

   Start by keeping the division of labor straight, because almost everyone who is confused about debits and credits is confused about this first. **The Accounting Equation is the axiom**: Assets = Liabilities + Shareholders' Equity, always, without exception. **Debits and credits are the procedure**—a little bookkeeping algorithm with the property that, if you follow its rules, the axiom stays satisfied automatically. They are not a second theory of accounting competing with the first. They are a technique for implementing it, and the reason they exist is ergonomic: writing out the full three-column A/L/E table for every transaction is unmanageable once you have thousands of transactions and dozens of accounts.

   Second, strip the words of their connotations. **Debit means "on the left." Credit means "on the right."** That is the entire semantic content. The words feel like they should mean *good* and *bad*, or *in* and *out*, and this is the single most common way people get lost. My own folk theory, when I first learned this, was that assets are good and liabilities are bad, so debits must be the good ones—and then I noticed that Shareholders' Equity, which sounds like a fine thing to have more of, *increases with a credit.* The moralized picture collapses on contact with that one fact. What survives is purely geographic.

   Third, here is the actual algorithm, and notice that it has only two rules. Rule one: an increase in an **asset** is recorded on the **left** (a debit), and, because Liabilities and Shareholders' Equity live on the *opposite side of the equation,* an increase in either of them is recorded on the **right** (a credit). Decreases are the reverse in

each case. Rule two: for every transaction, **total debits must equal total credits**, in dollars. Full stop, no exceptions, ever.

And now the payoff—why those two rules are sufficient. Ask what it actually *takes* to keep $A = L + E$ balanced. Suppose an asset goes up. Exactly three things could restore the equation: another asset goes down, or a liability goes up, or equity goes up. Now translate each into the new vocabulary. The asset going up was a *debit*. And all three of the permissible balancing moves—asset down, liability up, equity up—are *credits*. That is not a coincidence; it is the design. Run it in the other direction and the same thing happens: a credit (asset down, or liability or equity up) can only be balanced by a debit (asset up, or liability or equity down). So the rule "debits must equal credits" is not an additional constraint layered on top of the Accounting Equation. It *is* the Accounting Equation, restated in a shorthand that happens to be much easier to write down:

$$Debits\ (\$) = Credits\ (\$) \Longleftrightarrow A = L + E$$

A concrete demonstration, using the first three transactions of our little company, each of which involved *Cash*. In #1, we sold stock: Cash up (debit), Common Stock up (credit, because equity). In #2, we borrowed from the bank: Cash up (debit), Notes Payable up (credit, because liability). In #3, we bought equipment: Equipment up (debit), Cash down (credit, because an asset went down). Three different transactions, three economically different balancing moves—and every one of them landed on the credit side, without our having to think about the equation at all. That is what it means to say the procedure *keeps* the axiom satisfied: the balancing happens whether or not you are paying attention to it.

One last piece, and it is the piece that trips people up, because the two rules above are stated in terms of A, L, and E—and Revenues, Expenses, and Dividends are none of those. They are handled by inheritance. Revenue increases Net Income, which increases Retained Earnings, which is a component of Shareholders' Equity—so revenue increases equity, and increases in equity are credits. **Revenues are credits.** Expenses are anti-revenues, so they run the chain backwards and are **debits**; dividends reduce Retained Earnings, so they are **debits** too. You do not need a separate rule for these accounts; you need only remember that they are equity in disguise.

And if all of that feels like a lot to hold in your head, here is the good news: I don't hold it in my head. I memorize exactly one thing—**an asset going up is a debit, and debit means left**—and regenerate everything else in about thirty seconds. Assets are on the left of the equation and their increases go on the left of the account; L and E are on the other side of the equation, so their increases

go on the other side of the account; decreases are the opposite of increases in every case; and revenues are equity, so they are credits. One anchor, and the rest is derivation.

A.4 Chapter 4

A.4.1 Practice Problems: MCQ's

1. Hugo's, a specialty clothing store, does annual, calendar-year reporting. On **July 1**, 2021, Hugo's pays $48k for a 2-year lease of a space in a local mall. What will be the related adjusting entry Hugo's will make before closing the books on December 31, 2021?

   (A) Debit Rent Expense $24k; Credit Prepaid Rent $24k

   (B) Debit Rent Expense $12k; Credit Prepaid Rent $12k

   (C) Debit Rent Expense $8k; Credit Prepaid Rent $8k

   (D) Debit Rent Expense $6k; Credit Prepaid Rent $6k

   (E) None of the other answers

   Solution. Always start these by asking what we *already recorded*. On July 1 there was a real external transaction—cash out, a multi-period resource in—and we would have written:

Prepaid Rent (+A)	$48k	
Cash (-A)		$48k

External transaction: signing the lease, 7/1/2021

Nothing has happened since. Hugo's simply occupied the space. But December 31 arrives, and the calendar forces us to admit that some of that resource is gone: of the 24 months of occupancy we bought, **6 have been used up**. The lease is the easy case of the three in this chapter, because it comes with a built-in clock—we don't have to count anything or assume anything, we just read the term off the contract.

Six months out of twenty-four is one quarter of the contract, so one quarter of the cost has become expense: $\$48k \times \frac{6}{24} = \$12k$.

Rent Expense (-S.E.)	$12k	
Prepaid Rent (-A)		$12k

End-of-period adjusting entry for Rent Expense, 12/31/2021

Which leaves \$36k of *Prepaid Rent* on the Balance Sheet—18 months of occupancy still owed to us. The distractor to take seriously is (A), \$24k: that is a full year of rent, and it is what you get by noticing "annual reporting period" and reflexively expensing a year's worth. But the lease only started halfway through the year; Hugo's cannot have consumed twelve months of a space it has occupied for six. So the answer is **(B)**.

2. Hugo's, a specialty clothing store, does annual, calendar-year reporting. On **September 1**, 2021, Hugo's pays \$48k for a 2-year lease of a space in a local mall. What will be the related adjusting entry Hugo's will make before closing the books on December 31, 2021?

   (A) Debit Rent Expense \$24k; Credit Prepaid Rent \$24k

   (B) Debit Rent Expense \$12k; Credit Prepaid Rent \$12k

   (C) Debit Rent Expense \$8k; Credit Prepaid Rent \$8k

   (D) Debit Rent Expense \$6k; Credit Prepaid Rent \$6k

   (E) None of the other answers

   Solution. Identical to the previous problem except for the one thing that matters: the clock started September 1, so only **4 months** of the 24-month term have elapsed by December 31.

   So $\$48k \times \frac{4}{24} = \$8k$.

Rent Expense (-S.E.)	\$8k	
Prepaid Rent (-A)		\$8k

End-of-period adjusting entry for Rent Expense, 12/31/2021

   Here is the useful way to hold this: the "straight line" is a constant rate *per unit of time*—\$2k per month, in this lease—and the amount recognized in any one adjusting entry is that rate multiplied by however much time has elapsed since we last updated the books. The date of the *external* transaction never appears in the entry itself; it only tells us where to start the stopwatch. So the answer is **(C)**.

3. ExpoCorp reported Prepaid Rent of \$500k on its FY2021 Balance Sheet, and Prepaid Rent of \$400k on its FY2022 Balance Sheet. During FY2022, ExpoCorp signed one additional long-term lease, paying \$200k up front. What was ExpoCorp's Rent Expense in FY2022?

   (A) \$200k,

(B) $400k

(C) $300k

(D) $500k

(E) None of the other answers.

Solution. This is the "we are outside the company" exercise: we never see ExpoCorp's journal entries, only the aggregated balances it reports each period. But we know the general form of an operating asset—it goes *up* with purchases and *down* with its related expense, and nothing else touches it. So the three quantities are jointly determined, and knowing two gives us the third.

Prepaid Rent	
FY2021 end bal.: $500k	FY2022 expense: $X
FY2022 purchase: $200k	
FY2022 end bal.: $400k	

Solving for Rent Expense

Setting up the arithmetic, $\$500k + \$200k - \textit{Rent Expense} = \$400k$, so *Rent Expense* must have been $300k.

Or, purely verbally: ExpoCorp started the year with $500k of prepaid rent and added $200k more, so it had $700k of occupancy "stored up" to consume. It ended with $400k still stored up. The $300k difference is what got used, and what gets used is the expense.

The distractor worth naming is the one that isn't even on the list: "the balance fell by $100k, so Rent Expense was $100k." That reasoning ignores the $200k purchase, which was pushing the balance *up* the whole time. Of the options actually offered, (A) is the purchase, (B) is the ending balance, and (D) is the beginning balance—each a number handed to you in the stem, none of them the answer. So the answer is **(C)**.

4. On July 1, 2021, we bought Equipment, paying $100k. We assume that we will use the Equipment for 10 years, and it will have 0 salvage value at the end of that time, and use the straight-line depreciation method. What would be the *Depreciation Expense* we would record for **FY 2021**?

(A) $5k

(B) $10k

(C) $25k

(D) $75k

(E) None of the other answers.

Solution. Two steps, and it is worth keeping them separate in your head.

First, the annual rate. Straight-line depreciation apportions the *depreciable* amount—cost less salvage value—evenly over the useful life: $\frac{\$100k - \$0}{10\ years} = \$10k$ per year.

Second, how much of a year we actually get. We bought the Equipment on July 1, so as of December 31 we have owned and used it for **six months**, not twelve. Half a year at $10k per year is $5k.

Depreciation Expense (-S.E.)	$5k	
Equipment (-A)		$5k

End-of-period adjusting entry for Equipment, 12/31/2021

Option (B), $10k, is the annual rate—correct arithmetic, wrong question. It is the number you write down if you compute the rate and then forget to ask how much of the year has elapsed, which is exactly the mistake the July 1 date is fishing for. So the answer is **(A)**.

5. On July 1, 2021, we bought Equipment, paying $100k. We assume that we will use the Equipment for 10 years, and it will have 0 salvage value at the end of that time, and use the straight-line depreciation method. What would be the *Depreciation Expense* we would record for **FY 2022**?

   (A) $5k

   (B) $10k

   (C) $25k

   (D) $75k

   (E) None of the other answers.

   Solution. Same asset, same $10k-per-year rate—but now a *full* reporting period has elapsed. FY2022 runs January 1 to December 31 and we owned the Equipment for every day of it, so we recognize the full annual amount, $10k.

Depreciation Expense (-S.E.)	$10k	
Equipment (-A)		$10k

End-of-period adjusting entry for Equipment, 12/31/2022

The pair of problems is making one point: only the *first* and *last* years of an asset's life get partial amounts. The stub half-year in 2021 doesn't change the rate; it just means the \$100k gets spread across eleven reporting periods (\$5k, then nine full years of \$10k, then a final \$5k) rather than ten. Add those up and you get \$100k, as you must—the whole expenditure, and no more, ends up on the income statement eventually. So the answer is **(B)**.

6. On July 1, 2021, we bought Equipment, paying \$100k. We assume that we will use the Equipment for 10 years, and it will have \$20k salvage value at the end of that time, and use the straight-line depreciation method. What would be the *Depreciation Expense* we would record for **FY 2022**?

   (A) \$4k

   (B) \$5k

   (C) \$8k

   (D) \$10k

   (E) None of the other answers.

   Solution. The one change from the previous problem is the **salvage value**: we now expect to be able to sell the Equipment for \$20k at the end of its useful life. So we are not going to consume the whole \$100k over ten years—we are going to consume \$80k of it, and hand the remaining \$20k back to the market.

   So the *depreciable* amount is \$80k, not \$100k: $\frac{\$100k - \$20k}{10\ years} = \$8k$ per year.

   FY2022 is a full period, so the entry is for the full \$8k:

Depreciation Expense (-S.E.)	\$8k	
Equipment (-A)		\$8k

End-of-period adjusting entry for Equipment, 12/31/2022

   Option (D), \$10k, is what you get by dividing the full cost by the life and forgetting the salvage value—which would drive the book value all the way to \$0 by 2031, when we have said all along that we expect the thing to still be worth \$20k. Option (A), \$4k, is the correct \$8k rate applied to the *wrong year*—it's the half-year stub for FY2021. So the answer is **(C)**.

A.4.2 Practice Problems: Open Prompts

1. ExpoCorp reports Supplies of \$100k on its FY2021 Balance Sheet, and Supplies of \$120k on its FY2022 Balance Sheet. On its FY2022 Income Statement, ExpoCorp

reports a Supplies Expense of $40k. How much did ExpoCorp purchase in Supplies in FY2022? **(Answer: $60k)**

Solution. Same machinery as the Prepaid Rent problem above, run in the other direction: here we are given the expense and asked for the purchases.

Supplies	
FY2021 end bal.: $100k	FY2022 expense: $40k
FY2022 purchases: $X	
FY2022 end bal.: $120k	

Solving for purchases

Setting up the arithmetic, $\$100k + Purchases - \$40k = \$120k$, so *Purchases* must have been $60k.

And verbally, which is how I actually think about it: ExpoCorp started the year with $100k in the supply closet. If it had bought nothing at all, then after using up $40k it would have finished the year with $60k. It finished with $120k instead. The only thing that pushes a *Supplies* balance up is buying supplies, so it must have bought $60k worth.

Notice what we just did. ExpoCorp never told anyone what it spent on supplies—that figure appears nowhere in its financial statements, and as outside readers we never see the underlying journal entries. We *inferred* it from two balance-sheet numbers and one income-statement number, using nothing but the general form of an operating asset. **$60k.**

2. ExpoCorp reported Prepaid Rent of $700k on its FY2021 Balance Sheet, and Prepaid Rent of $500k on its FY2022 Balance Sheet. During FY2022, ExpoCorp signed one additional long-term lease, paying $200k up front. What was ExpoCorp's Rent Expense in FY2022? **(Answer: $400k)**

   Solution. The "purchase" of a lease is the cash paid up front for it, so the $200k is a debit to *Prepaid Rent*, and the only thing that can be pushing the account back down is *Rent Expense*.

Prepaid Rent	
FY2021 end bal.: $700k	FY2022 expense: $X
FY2022 purchase: $200k	
FY2022 end bal.: $500k	

Solving for Rent Expense

Setting up the arithmetic, $\$700k + \$200k - \textit{Rent Expense} = \$500k$, so *Rent Expense* must have been \$400k.

ExpoCorp had \$700k of occupancy stored up, added \$200k more, and finished the year with \$500k still stored up—so it consumed \$400k. Note again how badly the lazy answer fails: the balance only fell by \$200k, but the expense was double that, because \$200k of new leasing was pushing the balance up at the same time. **\$400k.**

A.4.3 Essay and Discussion Questions

1. In this chapter, we made the end-of-period adjusting entry for the *Supplies* we had used over the period, like so:

	Debit	**Credit**
Supplies Expense (-S.E.)	\$.4k	
Supplies (-A)		\$.4k

... So that the *Supplies* T-account at the end of the period looked like this:

Supplies	
\$1,000	\$400
\$600	

Explain this adjusting entry from the "income statement perspective" vs. balance sheet perspective." That is, explain what we mean by the two perspectives, and how to think about / articulate the adjusting entry from each perspective.

Solution. Here is how I would answer this. Start with what the two perspectives *are*, then narrate the same entry twice.

The two perspectives are two views about **what accounting is fundamentally for**, and therefore about which half of a journal entry is the "real" half and which is the "plug." Under the **balance-sheet perspective**, the point of accounting is to measure and report the firm's assets and liabilities correctly at the end of the period; revenues and expenses are then just the bookkeeping residue that explains how net assets got from where they were to where they are. Under the **income-statement perspective**, the point is to measure the *flow* of profitability from op-

erations in a way that makes economic sense; the balance-sheet accounts are then storage—places to park a cost until the periods it belongs in.

The essential thing to see is that this is a **duckrabbit**. There is exactly one journal entry here, and it is not in dispute. Both perspectives debit *Supplies Expense* $400 and credit *Supplies* $400. Nothing on the page changes. What changes is which line you experience as the *motive* and which as the *consequence*.

The balance-sheet narration. "It is December 31 and I have to report this company's assets. So I walk down the hall to the supply closet and I look. I count what is physically there, and I find that 60% of what we bought is still on the shelf. That means the *Supplies* asset, which the books currently say is $1,000, is really only worth $600—we don't have $1,000 of paper anymore. So I credit *Supplies* $400 to bring the reported balance to the right number, $600, and that is the number that goes on the Balance Sheet. Now, debits have to equal credits, so something has to absorb the other $400. We call it *Supplies Expense*, and it shows up on the Income Statement, where it does the useful job of *explaining* why our net assets went down." Notice the direction of reasoning: the asset measurement is the objective, the expense is the plug.

The income-statement narration. "Back in July we spent $1,000 on printer paper, and we knew at the time that we would be using it across several reporting periods. It would have been silly to charge the whole $1,000 against July's profitability, because most of that paper had nothing to do with earning this period's revenue—so we *stored* the cost in an asset account, to be apportioned out as we consumed it. Over this period we consumed 40% of it. To get a sensible, matched measure of what it actually cost us to operate this year, I recognize $400 of *Supplies Expense*. And, of course, debits must equal credits, so I credit *Supplies* for the $400 I have just apportioned out—leaving $600 still stored up, waiting for the periods it belongs to." Same entry; now the expense is the objective and the asset balance is the leftover.

Two closing observations, because I think this is where the question gets genuinely interesting rather than merely definitional.

First, the two narrations even suggest different *methods*. The balance-sheet narration reaches for the physical count—go look, measure what is there, derive the expense. The income-statement narration reaches for the apportionment—decide how much of the resource this period consumed, and let the remaining balance fall out. With supplies these converge, because the count is easy and reliable. With *Equipment* they do not converge at all: nobody can walk down the hall and observe how much of a machine is left, so we are forced into pure apportionment, and the resulting "book value" is an accounting residue rather than a

measurement of anything.

Second, and this is why the jargon is worth learning now: for the rest of this book, we will repeatedly be offered *two allowable methods* for accounting for the same thing, and the fault line will almost always run along this seam. So it pays to be able to name which side of it you are standing on. My own bias is the income-statement side—I care most about producing a measure of profitability that is persistent and predictive, because that is what anyone valuing the company will use—but I would not want you to take that as the right answer. The right answer is that you can see both animals in the drawing.

2. What do you think about the "weirdness" of accounting depreciation? Is it problematic that *Depreciation Expense*, which is ultimately determined by unverifiable managerial expectations and assumptions, affects the "bottom line" profitability? Is it a problem that the accounting depreciation of an asset doesn't necessarily track the real decline in its market/economic value?

   The answer to this question surely can't be simple. If you think these things *are* a problem, then why do the smart people at the *FASB* have such standards? So a judicious and solid answer also has to think through the positives here, the reason why things are done the way they are...

   Solution. Here is how I would think it through. There are two separate complaints packed into the question, and they deserve separate treatment, because I think one of them is a real concern and the other is mostly a misunderstanding.

   Complaint two first, because it is the easier one: "accounting depreciation doesn't track market value." My answer is that this is not a bug, and it is barely even a criticism—it is a category error, generated by the fact that we all learned the word "depreciation" in ordinary English before we learned it in accounting. In accounting, depreciation is *defined* as the apportionment of the cost of a tangible long-term asset over the periods in which it is used. It is not, and never claimed to be, a running estimate of resale value. The baseball dispute makes this vivid: the players' union heard "amortization" and understood the owners to be calling them washed-up, when the asset being amortized was never the player—it was the *contract*, a finite bundle of rights with less time left on it every day. The player can improve and the contract can still be running out; both things are true, and only one of them is what the accounting is about.

   And even if you did want market-value tracking, ask the practical question: how? A share of Apple has an observable price every second. A specialized crane bolted to your factory floor does not. Historical cost has the enormous virtue of being *verifiable*—the number came from a transaction that actually happened, with a

counterparty, and there is a receipt. Trading that away for a parade of unobservable appraisals would be a poor bargain. (And note that the books are not indifferent to reality forever: the moment we actually *sell* the equipment, a real external transaction occurs, and any difference between proceeds and book value is recognized as a Gain or Loss. Hypothetical value changes don't touch the books; realized ones do.)

Complaint one is the serious one: the bottom line depends on unverifiable forecasts. Here I want to grant the point fully before defending the practice, because the point is genuinely unsettling and I don't think it should be smoothed over. Take our $100k machine on a 10-year life: $10k of expense per year. Now suppose management simply announces that it now expects 40 years of use. Depreciation Expense drops to $2.5k, net income rises by $7.5k, and *not one thing about the company's real operations has changed.* Nobody sold more, nobody spent less, no machine ran differently. That is a strange property for a number that ordinary speech treats as the very emblem of objectivity—"the bottom line." It should bother you a little. It bothers me.

But now ask the question the prompt correctly insists on: **compared to what?** The alternatives are worse, and there are only really three of them.

Alternative one: expense the whole $100k in the year of purchase. This is objective and verifiable—and it produces exactly the distortions we motivated accrual accounting with in Chapter 1. A company that buys a factory in one year would look catastrophically unprofitable that year and implausibly profitable for the next nine, with the reported numbers telling you almost nothing about the underlying business in any of the ten. You would have traded a small amount of subjectivity for a large amount of noise.

Alternative two: don't expense it at all until the asset is sold or scrapped. Then the income statement systematically overstates profitability for the entire life of every asset, because we would be recognizing the revenues the machine helps generate while pretending the machine costs nothing. That is a clean violation of matching, and it is much more misleading than a slightly-off useful-life estimate.

Alternative three: mark assets to market each period. Unobservable for most operating assets, as above—and if you think management's useful-life estimate is unverifiable, wait until you see management's estimate of what its half-obsolete equipment would fetch.

So the FASB's position is not naïveté about the subjectivity; it is a judgment that apportioning cost over the periods of benefit, even with estimated inputs, produces a more decision-useful measure of profitability than any of the objective-

but-useless alternatives. And the discretion is not unbounded. Auditors are industry specialists who know what normal useful lives look like for that kind of asset in that kind of business, and they have to put their own names on the statements; a company whose assumptions are wildly out of line with its peers' will get pushback. Beyond that, the useful-life assumption is disclosed, so analysts can and do adjust for it. The discipline comes from norms, auditors, and market scrutiny rather than from a rulebook—which is looser than one might like, but is not nothing.

Where I land. I think the correct posture is neither "this is fine" nor "this is a scandal," but something more like informed wariness. Depreciation is an *allocation*, not a *measurement*, and the honest thing to say is that a company's reported net income is a considered estimate produced under a set of disclosed assumptions—not a fact of nature. That is the price of admission for accrual accounting, and the alternative (cash-basis reporting, which *is* objective) would be far less useful for the thing we actually want financial statements for: forecasting future profitability. What follows is not that we should abandon the method, but that we should read income statements the way a professional does—checking the assumptions, comparing them to peers, and asking who benefits if they are wrong. Which, as it happens, is a decent one-sentence summary of what this whole book is training you to do.

A.5 Chapter 5

A.5.1 Practice Problems: MCQ's

1. On July 1, 2021, we bought Equipment, paying $100k. We assume that we will use the Equipment for 10 years, and it will have 0 salvage value at the end of that time, and use the straight-line depreciation method. What would be the balance of the *Accumulated Depreciation* (*contra-A*) account, after making the adjusting entries and closing the books, as of December 31, 2023?

   (A) $5k

   (B) $10k

   (C) $25k

   (D) $75k

   (E) None of the other answers.

   Solution. Start with the annual rate. Straight-line depreciation apportions the cost evenly over the useful life: $\frac{\$100k - \$0}{10\ years} = \$10k$ per year. So the adjusting entry, in a *full* year of use, is:

Depreciation Expense (-S.E.)	$10k	
Accumulated Depreciation (+contra-A)		$10k

End-of-period adj. entry for Depreciation Expense, full year

   Now the only thing left is to count the years — and here is where the problem has its teeth. We bought the equipment on **July 1**, halfway through 2021, so fiscal 2021 gets only *half* a year of use, and therefore only $5k of depreciation. Fiscal 2022 and fiscal 2023 are each full years, at $10k apiece. *Accumulated Depreciation* is exactly what its name says — the running total of those credits, never reset, because it is a permanent contra-Asset account riding alongside a permanent asset:

   $\$5k + \$10k + \$10k = \$25k$

   The tempting wrong answer is $10k, which is one year's depreciation rather than the accumulation, or $30k (not offered), which is what you get if you forget the half-year at the front. And $75k is the *book value*, not the accumulated depreciation — that is the next problem. So the answer is **(C)**.

2. On July 1, 2021, we bought Equipment, paying $100k. We assume that we will use the Equipment for 10 years, and it will have 0 salvage value at the end of that time, and use the straight-line depreciation method. What would be the **book value** of the *Equipment*, after making the adjusting entries and closing the books, as of December 31, 2023?

   (A) $100k

   (B) $10k

   (C) $25k

   (D) $75k

   (E) None of the other answers.

   Solution. Book value is the balance of the account *net* of the reductions tracked in its contra-account. The *Equipment* T-account itself is untouched by depreciation — it still sits at the $100k purchase price, because we credited *Accumulated Depreciation* rather than *Equipment* itself. And from the previous problem, *Accumulated Depreciation* stands at $25k as of December 31, 2023. So:

   $Book\ Value = \$100k - \$25k = \$75k$

   Note that this is the number that ends up in Total Assets either way the company chooses to present it — whether it reports the single line "Equipment (at book value): $75k," or reports "Equipment: $100k" with "*less* Accumulated Depreciation: ($25k)" underneath. The reporting format is discretionary; the $75k is not. And notice that (A), $100k, is the trap for anyone who takes "the Equipment account is unaffected by depreciation" too literally — that is the *gross* balance, not the book value. So the answer is **(D)**.

3. What is the general term that is used to indicate the time horizon of assets and liabilities—i.e., the speed with which they could be used or converted into cash or settled?

   (A) Verifiability

   (B) Depreciation

   (C) Capitalization

   (D) Liquidity

   (E) Conservatism

   Solution. This is **liquidity** — how long until the 'end of the story' for that asset or liability. *Cash* is the most liquid asset by definition (its time-to-cash is zero); *Accounts Receivable* is highly liquid; *Equipment* is not. On the liability side, *Salaries*

Payable will be settled within days of year-end, while a bank note might run a decade. Two of the distractors are real accounting terms doing different jobs: *depreciation* is the apportionment of a long-term asset's cost over its useful life, and *capitalization* is the decision to record an expenditure as an asset rather than expensing it immediately. Neither is about time horizon *per se*. So the answer is **(D)**.

4. On the Balance Sheet, Assets and Liabilities are roughly listed in order of...:

   (A) Verifiability

   (B) Size

   (C) Liquidity

   (D) Solvency

   (E) Equity

   Solution. Within the asset section and within the liability section, accounts are listed in order of **liquidity**: Cash first, then receivables, with long-term items like Property, Plant, and Equipment down at the bottom; and among liabilities, the short-term payables high and long-term debt low. Note the word "roughly" in the stem — it is doing honest work. Whether *Supplies* goes above or below *Prepaid Rent* is a coin flip, and no reasonable reader will fault you either way. What is not a coin flip is the gradient. Option (B), size, is the plausible-sounding wrong answer, and it is worth killing explicitly: a company with $500 of Cash and $40m of Equipment still lists Cash first. So the answer is **(C)**.

5. *Accumulated Depreciation* is a...:

   (A) debit account

   (B) credit account

   (C) none of the other answers.

   Solution. *Accumulated Depreciation* is a **contra-Asset** account. It runs *contra* to the Asset it references, tracking that asset's reductions separately — and since Assets are debit accounts, a contra-Asset must be a **credit** account, so that its balance nets *against* the asset it shadows. You can also just read it off the adjusting entry: we debit *Depreciation Expense* and credit *Accumulated Depreciation*, period after period, so credits accumulate in it. So the answer is **(B)**.

6. *Accounts Payable* is a:

   (A) Current asset

(B) Non-current asset

(C) Current liability

(D) Non-current liability

(E) None of the other answers.

Solution. Two questions, answered in sequence. First, asset or liability? *Accounts Payable* is what we owe our *suppliers* for goods and services we bought on account — an obligation, so a **liability**. (The mirror image, *Accounts Receivable*, is what our customers owe *us*, and is an asset. Keeping those two straight is a matter of asking who owes whom.) Second, current or non-current? Supplier bills get settled in weeks, not years — comfortably under the one-year cutoff — so it is **current**. In fact, among current liabilities, *Accounts Payable* typically sits at or near the top of the list, as one of the most liquid. So the answer is **(C)**.

A.5.2 Practice Problems: Open Prompts

1. RustiCorp owns one major tangible asset: a large plant. In its most recent financial statement, it reported depreciation expenses of $1.3m. It further disclosed that the*Accumulated Depreciation (contra-A)* on the plant was $31.85m, and it was using straight-line depreciation. How long ago did Rusticorp buy this asset? **(Answer: 31.85/1.3 = 24.5 years ago.)**

   Solution. The whole problem turns on one observation: under **straight-line** depreciation, the annual *Depreciation Expense* is the *same every year* — and *Accumulated Depreciation* is nothing but the running total of those identical annual amounts. So the accumulated balance divided by the annual amount gives you the number of years the company has been depreciating the asset:

   $$\frac{Accumulated\ Depreciation}{Annual\ Depreciation\ Expense} = \frac{\$31.85m}{\$1.3m} = 24.5\ years$$

   So RustiCorp bought the plant **24.5 years ago**. And the half-year is not a rounding artifact — it is the same wrinkle we met in the chapter's worked example: the company evidently bought the plant in the *middle* of a fiscal year, took a half-year of depreciation in that first period, and full years ever since.

   Two things worth noticing about what this little calculation buys you. First, it is exactly the kind of inference the two-account presentation makes possible. If RustiCorp reported *Plant* at book value on a single line, you could not do this at all. Second, it tells you something an investor cares about: RustiCorp's one major productive asset is roughly 24.5 years into its life. If the plant's assumed useful life were, say, 25 years, then this company is about to face a very large replacement expenditure — and its reported book value is telling you almost nothing

about what it will cost to keep operating. That is precisely the diagnostic the *Accumulated Depreciation*-to-gross-asset ratio is used for in practice.

2. **Core Identities / Classification of Balance Sheet**: ExpoCorp reports Total Liabilities of $60m, Common Stock of $30m, and Retained Earnings of $10m on its FY2021 Balance Sheet. Its long-term assets total $60m. What are ExpoCorp's total current assets ($)?

   Solution. No new formula is needed here — just three identities you already own, applied in order.

   First, Shareholders' Equity has exactly two components, paid-in capital and earned capital:

   $E = Common\ Stock + Retained\ Earnings = \$30m + \$10m = \$40m$

   Second, The Accounting Equation gives us total assets:

   $A = L + E = \$60m + \$40m = \$100m$

   Third, every asset is either current or long-term — the two categories are disjoint and exhaustive — so they must sum to total assets:

   $Current\ Assets = \$100m - \$60m = \$40m$

   So ExpoCorp's total **current assets are $40m**. The thing to internalize is the *shape* of the reasoning: you were handed pieces from both sides of the Balance Sheet, and the way through was to complete one side entirely, cross over via $A = L + E$, and then partition. Any problem of this family yields to the same three moves.

3. In its Quarter 1 financial statements, TestCorp reported a cash balance of $50m, other assets of $240m, Shareholders' Equity of $80m, and current liabilities of $40m. Its Statement of Shareholders' Equity indicated that it had paid out dividends of $10m, and ended the quarter with a retained earnings balance of $30m. What were its long-term/non-current liabilities at the end of Quarter 1? **(Answer: 290m - 80m - 40m = 170m. In words: Total assets is $290m (cash of 50m and 'other assets' of $240m. Total S.E. is $80m. Therefore, total liabilities must be $210m. So, long-term /non-current liabilities must be $170m ($210m - $40m). Everything else is a distraction.)**

   Solution. Same three moves as the previous problem, run in a slightly different order — and this time with some deliberate chaff thrown in.

   Total assets first: $\$50m + \$240m = \$290m$. Then The Accounting Equation, rearranged to solve for the liabilities:

   $L = A - E = \$290m - \$80m = \$210m$

And then the partition, this time on the liability side: every liability is either current or long-term, so

$$\textit{Long-term Liabilities} = \$210m - \$40m = \$170m$$

Now, the chaff. You were also told about $10m of dividends and a $30m *Retained Earnings* balance, and neither one enters the calculation. Why not? Because *Retained Earnings* is already *inside* the $80m of Shareholders' Equity you were given — using it again would be double-counting — and the dividends have already done their work in producing that $30m balance via the Retained Earnings identity. The quantities are real; they are just not on the path from what you have to what you were asked for. Part of the skill this problem is testing is the willingness to look at a given number and decide, deliberately, not to use it.

4. Find the **latest 10-K report filed by Apple**, by searching "EDGAR," the SEC site on which U.S. companies' mandatory filings are distributed and archived, here: https://www.sec.gov/edgar/searchedgar/companysearch.html. What are Apple's total **current assets** (vs. long-term / or total assets)? How does this jibe with what you know about the company?

   Solution. Your figures will differ from mine depending on which year's 10-K you pull — that is the nature of an open-ended "go look it up" prompt, and it is the point. I will work Apple's fiscal 2024 10-K (fiscal year ended September 28, 2024) so that you have a concrete set of numbers to compare your own reasoning against.

Apple Inc., Consolidated Balance Sheet (asset side), September 28, 2024
$ millions, condensed

Current assets	
Cash and cash equivalents	$29,943
Marketable securities	$35,228
Accounts receivable, net	$33,410
Vendor non-trade receivables	$32,833
Inventories	$7,286
Other current assets	$14,287
Total current assets	**$152,987**
Total non-current assets	**$211,993**
Total assets	**$364,980**

So roughly $153bn of Apple's $365bn in total assets — about **42%** — is current. Three things about that number are worth your attention.

First, look at what is *in* the current bucket. The two largest single items are not the things you would guess for a company that sells phones. *Inventories* — the actual

iPhones and Macs sitting in warehouses and on shelves — are only $7.3bn, under 2% of total assets. Meanwhile *Cash and cash equivalents* plus current *Marketable securities* come to $65bn. Apple, the hardware manufacturer, holds nine times as much in short-term financial instruments as it does in product.

Second, look at the non-current side, which is where the real surprise is. Apple's net Property, Plant, and Equipment is about $45.7bn — modest for a company that ships hundreds of millions of physical devices a year. Why so little? Because Apple largely does not *own* the factories. It designs the products and contracts the manufacturing out to firms like Foxconn, whose balance sheets carry the assembly plants. Apple's most valuable resources — the brand, the operating system, the design and engineering talent, the App Store ecosystem — are internally generated intangibles, and by accounting **conservatism** (Chapter 1), those mostly do not appear on the Balance Sheet as assets at all. The most economically important things Apple owns are, in a literal accounting sense, invisible here. Another roughly $91bn of non-current assets is, again, marketable securities: longer-dated investments in the same enormous financial portfolio.

Third, the honest lesson. A first reaction to "42% current" might be to reach for a story about liquidity management. But the more interesting reading is what this balance sheet *fails* to show. Apple's assets look like the assets of a large investment fund with a small consumer-electronics business attached — because the consumer-electronics business itself, the part that generates nearly $400bn of annual revenue, is barely capitalized. Keep that in mind: the Balance Sheet is a rigorous, audited, extremely useful document, and it is also systematically silent about some of the most valuable things a modern company has.

A.5.3 Essay and Discussion Questions

(No essay topics from Chapter 5, since it is so "mechanical.")

A.6 Chapter 6

A.6.1 Practice Problems: MCQ's

1. Which of the following is a measure of the riskiness of a company's financing?

   (A) Leverage

   (B) Margin

   (C) Asset Turnover

   (D) Short-term focus

   (E) Liquidity

   Solution. Read the question carefully: it asks about the riskiness of the company's **financing** — the right-hand side of The Accounting Equation, the mix of creditor money and owner money. That is **leverage**, A/E, which we can also write as $\frac{L}{E}+1$. The higher it goes, the more of the company is funded by obligations that *must* be paid on schedule rather than by claims that can simply go unrewarded in a bad year, and so the higher the risk of default and bankruptcy.

   The distractors are all real, useful measures — they are just measuring something else. *Profit margin* (NI/S) and *asset turnover* (S/A) are the other two legs of the DuPont decomposition, and both are about **operations**: how well the company manages the Income Statement, and how well it employs its assets. Neither says anything about who financed those assets. *Liquidity* is the tempting one, because it is genuinely about risk — but liquidity is about whether the company can make specific cash payments coming due in the short term, which is a question about the *assets* side and about timing, not about capital structure. So the answer is **(A)**.

2. Which of the following ratios represents the calculation of leverage:

   (A) (NI/S)

   (B) (Current Assets / Current Liabilities)

   (C) (NI/A)

   (D) (S/A)

   (E) (A/E)

   Solution. Leverage is $\frac{A}{E}$ — total assets over shareholders' equity. The cleanest way to see why that is the right *form* is the DuPont algebra: $ROE = \frac{NI}{E} = \frac{NI}{A} \times \frac{A}{E} = ROA \times Leverage$. Leverage is literally the scalar sitting between the company's

operational return and the owners' return. And using $A = L+E$, $\frac{A}{E} = \frac{L+E}{E} = \frac{L}{E}+1$, which makes plain that it is just a measure of credit financing relative to equity financing, shifted by one.

Worth naming the others, since they are all ratios you should recognize on sight: (A) is **profit margin**, (C) is **ROA**, (D) is **asset turnover**, and (B) is the **current ratio**, our one liquidity measure. So the answer is **(E)**.

3. Which of the below represents Profit Margin:

   (A) NI/S

   (B) NI/E

   (C) $NI/((A_t + A_{t-1})/2)$

   (D) S/A

   (E) None of the other answers.

   Solution. Profit margin is $\frac{NI}{S}$ — bottom-line profit relative to top-line sales, or how much of each dollar of revenue survives all the way down the Income Statement. Equivalently, $\frac{NI}{S} = \frac{S-Expenses}{S} = 1 - \frac{Expenses}{S}$, so it is one minus the company's 'expense ratio.'

   Two of the distractors deserve a word, because they are testing whether you were paying attention to the *averaging* rule as well as the formulas. (B) is ROE. (C) is not some exotic new ratio — it is just **ROA**, written out properly with average total assets in the denominator, exactly as you are supposed to compute it in practice. Do not let the subscripts spook you into thinking it is something unfamiliar. (D) is asset turnover. So the answer is **(A)**.

A.6.2 Practice Problems: Open Prompts

1. Calculate **ROE, ROA, Profit Margin, Leverage, and Asset Turnover** for MattCorp for FY2021, based on the following financial statements:

Revenue:	**$200k**
Salaries expenses:	**$60k**
Depreciation expenses:	**$40k**
Interest expense:	**$20k**
Income tax expense:	**$20k**
Net Income:	**$60k**

Table A.1: MattCorp Income Statement, For the Year Ended December 31, 2021:

	2021	2020
Assets:		
Cash	**$200k**	**$80k**
Supplies	**$100k**	**$20k**
Land	**$200k**	**$200k**
Total assets:	**$500k**	**$300k**
Liabilities:		
Accounts payable	**$50k**	**$50k**
Salaries payable	**$30k**	**$50k**
Notes payable	**$300k**	**$100k**
Shareholders' Equity:		
Common Stock	**$100k**	**$100k**
Retained Earnings	**$20k**	**$0k**
Total Liabilities and S.E.:	**$500k**	**$300k**

Table A.2: MattCorp Balance Sheets, Years Ended December 31:

Solution. Before touching a single ratio, gather the inputs. Two of them come straight off the Income Statement, and two of them require the averaging rule.

From the Income Statement: $NI = \$60k$ and $S = \$200k$. Those are flows, measured over FY2021, and we take them as they are.

From the Balance Sheets, we need the two stock quantities, and here we must use the **average** of this year's and last year's figures — because we are scaling a year-long flow by a stock, and it would make no sense to scale the whole year's profit by where the balance happened to land on one particular December 31. Total assets are given directly:

$$\bar{A} = \frac{\$500k + \$300k}{2} = \$400k$$

Shareholders' Equity is not given as a subtotal, so build it from its two components, paid-in capital and earned capital. For 2021: $\$100k + \$20k = \$120k$. For 2020: $\$100k + \$0k = \$100k$. So:

$$\bar{E} = \frac{\$120k + \$100k}{2} = \$110k$$

(A quick sanity check that the statements foot: 2021 liabilities are $\$50k + \$30k + \$300k = \$380k$, and $\$380k + \$120k = \$500k$. Good.)

Now the five ratios are just division.

MattCorp, FY2021 ratio analysis

Ratio	Computation	Value
Profit Margin	\$60k / \$200k	30.0%
Asset Turnover	\$200k / \$400k	0.50
Leverage	\$400k / \$110k	3.64
ROA	\$60k / \$400k	15.0%
ROE	\$60k / \$110k	54.5%

And now the part that matters, because a ratio you have computed but not interpreted is not yet worth anything.

Profit margin of 30% means that thirty cents of every sales dollar survives the trip from the top line to the bottom line; the other seventy cents is consumed by salaries, depreciation, interest, and taxes. That is a *high* margin by the standards of most businesses — grocers live at 2%. It tells you MattCorp is not in a commodity business.

Asset turnover of 0.50 means each dollar of assets generated only fifty cents of sales over the year. That is *low*. And notice that the high margin and the low turnover are not two independent facts — that is the equilibrium pattern from this chapter, the luxury-jeweler profile rather than the grocer profile. Multiply them and you get $30\% \times 0.50 = 15\%$, which is exactly ROA, as the decomposition promises.

ROA of 15% is the summary of those two: the return the company earns on its total capital base, agnostic about who supplied that capital. This is the number to use if you want to compare MattCorp's *operating* performance against a differently-financed competitor.

Leverage of 3.64 says total assets are 3.64 times shareholders' equity — equivalently, a debt-to-equity ratio of 2.64. That is meaningfully levered, and the Balance Sheets tell you the story of how it got that way: *Notes Payable* tripled, from \$100k to \$300k, while Common Stock did not move at all. MattCorp funded essentially all of this year's growth with borrowing. Its owners avoided dilution, and they picked up fixed obligations in exchange.

ROE of 54.5% is the payoff to the owners, and the decomposition tells us precisely where it came from: $30\% \times 0.50 \times 3.64 = 54.5\%$. Read that left to right as a diagnosis. Roughly speaking, MattCorp turns a strong margin into a mediocre 15% return on its asset base, and then multiplies that 15% by 3.64 to deliver a spectacular-looking 54.5% to shareholders. **More than two-thirds of the headline ROE is manufactured by the capital structure, not by the operations.**

That is the single most useful sentence you can say about this company, and it is a sentence you can only produce because you decomposed.

So: is 54.5% good? An owner would be delighted this year. But you now know exactly what they are exposed to. Leverage multiplies the bad as well as the good — it is a scalar, and it does not care about the sign of what it multiplies. Run MattCorp through a bad year in which ROA turns slightly negative, and that same 3.64 turns a small operating loss into a savage loss for the owners. And with interest expense already at \$20k against EBIT of \$100k, and \$300k of notes outstanding, a downturn is not a hypothetical concern.

2. Find the **latest 10-K report filed by Apple**, by searching "EDGAR," the SEC site on which U.S. companies' mandatory filings are distributed and archived, here: https://www.sec.gov/edgar/searchedgar/companysearch.html. Then, **calculate Apple's ROA, ROE, and Profit Margin** for the most recent fiscal year, and the previous, using the numbers from the official, audited financial statements within.

Solution. Your numbers will differ from mine depending on which year's 10-K you pull. I will work Apple's fiscal 2024 10-K, which reports fiscal 2024 and fiscal 2023 on the face of the statements — and, conveniently, gives the fiscal 2022 Balance Sheet figures we need in order to compute proper *averages* for the fiscal 2023 ratios. (That is a practical point worth noting: to compute a flow-over-stock ratio for two years, you need *three* years of Balance Sheets. Sometimes that means pulling the prior 10-K as well.)

The raw inputs, in \$ millions:

Apple Inc., selected figures from the FY2024 Form 10-K (\$ millions)

	FY2024	FY2023	FY2022
Net sales	\$391,035	\$383,285	
Net income	\$93,736	\$96,995	
Total assets	\$364,980	\$352,583	\$352,755
Total shareholders' equity	\$56,950	\$62,146	\$50,672

Averaging the stocks: $\bar{A}_{2024} = \frac{364{,}980+352{,}583}{2} = \$358{,}782$ and $\bar{E}_{2024} = \frac{56{,}950+62{,}146}{2} = \$59{,}548$. For fiscal 2023: $\bar{A}_{2023} = \frac{352{,}583+352{,}755}{2} = \$352{,}669$ and $\bar{E}_{2023} = \frac{62{,}146+50{,}672}{2} = \$56{,}409$.

Apple Inc., ratio analysis

Ratio	FY2024	FY2023
Profit Margin (NI/S)	24.0%	25.3%
ROA (NI / avg. A)	26.1%	27.5%
ROE (NI / avg. E)	157.4%	172.0%

Now, what do these mean? Take them in order of how surprising they are.

A **profit margin of 24%** is extraordinary for a company that manufactures physical objects. Apple keeps roughly a quarter of every dollar of sales after *all* expenses — cost of goods, R&D, selling and administrative costs, interest, and taxes. This is the quantitative signature of a company that can command premium prices, and of a business mix increasingly weighted toward high-margin Services rather than hardware.

An **ROA of 26%** is likewise excellent. And notice that it comes with an asset turnover of only about $391{,}035/358{,}782 \approx 1.09$ — roughly a dollar of sales per dollar of assets. Apple is not turning over its asset base especially fast; it is a *margin* story, not a *volume* story. Compare that against the equilibrium logic from this chapter: an ROA in the mid-twenties, sustained for years, is precisely what equilibrium says should *not* persist. Capital ought to flow in and compete it away. That it hasn't is the quantitative shadow of Apple's competitive moat.

And then there is the **ROE of 157%**, which, on first encounter, looks like an arithmetic error. It is not. Run the decomposition: $ROE = ROA \times Leverage$, and Apple's leverage is $358{,}782/59{,}548 \approx 6.0$. So $26.1\% \times 6.0 \approx 157\%$. Apple's equity base is tiny — \$57bn of book equity supporting \$365bn of assets. Why? Because for over a decade Apple has been returning enormous sums to shareholders through dividends and, above all, share repurchases, both of which *reduce* book Shareholders' Equity. The company has, quite deliberately, shrunk the denominator.

This is exactly the situation the chapter warns about, and it is worth being blunt: **the 157% is not telling you that Apple earns 157 cents per dollar the owners put in.** Book equity has drifted far away from any economic measure of what owners have committed; it is a residual accounting number that buybacks have ground down. The honest reading is that ROE here is being driven overwhelmingly by capital structure, and ROA and profit margin are the more informative measures of what the business actually does. Which is the chapter's whole thesis, arriving on schedule: no one number tells you whether a company is performing well, and the moment a ratio produces an absurd-looking value, the right

response is to decompose it and find out which component is responsible.

A.6.3 Essay and Discussion Questions

1. Everyone in the business world agrees and says that **"leverage means risk."** But explain why, very deeply, as if I am a complete newbie. What exactly is leverage? And why would changing ***who we get our financing from*** (creditors vs. owners) change the riskiness of our company?

   Solution. Here is how I would build this up for someone starting from zero, in three moves: what leverage *is*, why the two kinds of financing are fundamentally different in *kind*, and then the arithmetic that shows the risk.

   First, what leverage is. Go back to The Accounting Equation, $A = L + E$. The left side tells you what the company *has* — its resources. The right side tells you *where the money came from* to get them: creditors (L) or owners (E). Leverage is just $\frac{A}{E}$: the size of the asset pile relative to the owners' slice of the funding. Using the identity, $\frac{A}{E} = \frac{L+E}{E} = \frac{L}{E} + 1$. So leverage is nothing more exotic than a measure of how much of the company's financing came from lenders rather than from owners. A company with $100 of assets funded by $50 of debt and $50 of equity has leverage of 2. Fund the same $100 of assets with $90 of debt and $10 of equity, and leverage is 10.

   Notice something important before we go further: **in that comparison, the company's actual business did not change at all.** Same $100 of assets, same factories, same products, same customers. We changed only who wrote the checks. That is what makes the question a good one. Why on earth should the riskiness of a business depend on the composition of the people who funded it?

   Second, the asymmetry between the two kinds of financing. The answer is that debt and equity are not two flavors of the same thing. They differ in one respect that governs everything: **a debt payment is a legal obligation, and a dividend is a discretionary choice.**

   If you lend a company $1m at 8%, you have a contract. On the payment date, the company owes you $80,000, and later it owes you the principal. It does not matter whether the company had a good year. It does not matter whether the CEO thinks now is an inconvenient time. If the company fails to pay, it is in **default** — and default is not a slap on the wrist. Creditors can seize collateral, accelerate the loan, force the company into bankruptcy, and, in bankruptcy, take control of it away from the owners entirely. And note what you *don't* get for lending: no matter how spectacularly the company does, you still get 8%. Your upside is capped at the contract.

If instead you buy $1m of the company's stock, you have no contract at all. You get dividends when the board declares them. In a bad year, the board can simply cut the dividend to zero — as boards routinely do — and nothing legally happens. No default, no seizure, no bankruptcy. You are unhappy, and you may eventually vote to replace the board, but you cannot force cash out of the company. In exchange for bearing that, you own the entire upside: if the company doubles its profits, all of that accrues to you.

So the two are genuinely different instruments: debt is a **fixed claim** with real teeth and no upside; equity is a **residual claim** with no teeth and all the upside. And that is what makes the mix a risk decision. Financing yourself with debt means committing to pay out a *fixed* amount out of an *uncertain* stream of earnings. Financing yourself with equity means the payouts flex with the earnings. The more of your financing is fixed, the more of your operating uncertainty gets concentrated onto the residual claim.

The limiting case makes it vivid: **a company financed 100% with equity has no debt to default on.** There is no contractual payment schedule it can miss. It can perform terribly, disappoint everyone, and lose money for years — but it cannot default, because there is no obligation to default on. Every bit of bankruptcy risk a company carries is risk it *chose* by borrowing.

Third, the arithmetic. Two symmetric ways to see it.

Start with the algebra, which is compact and complete: $ROE = ROA \times \frac{A}{E}$. Leverage is literally the multiplier sitting between the company's operating return and the owners' return. Take a company with leverage of 6. If ROA is $+5\%$, the owners get 30%. If ROA is -5%, the owners get -30%. The multiplier is blind to the sign of what it multiplies. It scales up the good outcomes and the bad ones by exactly the same factor. Which is where the word comes from: a lever multiplies whatever force you apply, including a force in the wrong direction.

Then run a concrete scenario, because the algebra understates how violent the effect is once interest and bankruptcy enter. Take two versions of the same company, both with $100 of assets, borrowing at 20%, and both taxed at 21%. Version 1 is $50 debt / $50 equity. Version 2 is $90 debt / $10 equity. Operating costs are $75 in both. In a good year, revenue is $100; in a bad year, $90 — a decline of only 10%.

In the good year, revenue of $100 less operating costs of $75 gives EBIT of $25 for both, since operations are identical. Now walk each one down the rest of the income statement. **Version 1** owes 20% on $50 of debt, so $10 of interest, leaving $15 of pre-tax income; tax at 21% is $3.15; net income is $11.85; and spread over

\$50 of equity that is an ROE of $\$11.85/\$50 = 23.7\%$. **Version 2** owes 20% on \$90 of debt, so \$18 of interest, leaving \$7 of pre-tax income; tax is \$1.47; net income is \$5.53; and spread over \$10 of equity that is an ROE of $\$5.53/\$10 =$ **55.3%** — more than double Version 1's, on a *smaller* dollar profit, purely because the profit is spread over a fifth as much equity. Shareholders love this, and this is the honest reason companies borrow at all.

Now the bad year. Revenue falls by 10%, to \$90, so EBIT falls to \$15. **Version 1** still owes only \$10 of interest, leaving \$5 of pre-tax income; tax of \$1.05 leaves \$3.95 of net income; and $\$3.95/\$50$ is a respectable ROE of 7.9%. It got poorer, but it is fine. **Version 2** owes \$18 of interest against \$15 of EBIT. **It does not earn enough to cover the interest on its debt.** Pre-tax income is $-\$3$, there is no tax on a loss, and $-\$3/\$10$ is an ROE of -30% — and the company is heading toward default. A 10% revenue decline — a mild recession, not a catastrophe — was enough to move Version 2 from "doubling the shareholders' return" to "can't pay the bank."

That contrast *is* the answer to the question. The two versions of the company are identical in every operational respect. Their fortunes diverge entirely because of who financed them.

A last point, and the one that generalizes furthest. It would be reasonable to conclude from all this that leverage is a private matter — my company, my capital structure, my problem. It isn't, and this is where the topic stops being an accounting exercise. Debt obligations are *interlinked*: one firm's payable is another firm's receivable. When a highly levered firm defaults, it doesn't merely fail; it takes down the value of the assets on somebody else's balance sheet, and if that somebody is also levered, the failure propagates. Add that the bad states of the world tend to arrive together — recessions hit revenues, employment, and asset prices simultaneously — and you have the mechanism behind most financial crises, including 2008. Households levered 5-to-1 on houses, banks levered far more than that, and a modest crack in one place ran through the whole structure.

So when someone says "leverage means risk," they are compressing all of the above into three words. Unpacked: leverage is the ratio of the assets to the owners' funding; it rises when you substitute fixed, enforceable claims for flexible, residual ones; and because those fixed claims must be honored precisely when it is hardest to honor them, leverage multiplies bad outcomes exactly as faithfully as it multiplies good ones.

A.7 Chapter 7

A.7.1 Practice Problems: MCQ's

1. If our company earns *Revenue* on a particular transaction / date, which of the following might that be linked to (i.e., the other account affected as a result / the other line of the journal entry)?

   (A) An asset decreasing

   (B) A liability decreasing

   (C) A liability increasing

   (D) Multiple of the other answers

   (E) None of the other answers

   Solution. Use the fully general definition of a Revenue from this chapter: "an increase in *net assets* — an increase in assets and/or a decrease in liabilities — earned from our operations." *Revenue* is a credit, so the balancing side must be a debit, and the only debits available on the Balance Sheet are an asset going **up** or a liability going **down**.

   Walk the options against that. (A) An asset *decreasing* is a credit, so it cannot balance a credit to *Revenue* — and economically it is backwards: earning revenue does not consume your resources, it adds to them. (B) A liability *decreasing* is exactly the *Deferred Revenue* case: the customer paid in advance, we now provide the service, and we debit the liability away as we credit *Revenue*. That works. (C) A liability *increasing* is a credit, so it cannot pair with *Revenue* either; that is the shape of an **expense** transaction, not a revenue one.

   So exactly one of (A)–(C) is right, which rules out (D). What may make this feel wrong is the nagging sense that a cash sale or a credit sale ought to be on the list — and of course they are the *most common* revenue transactions. But those are assets *increasing*, and no option offered that. So the answer is **(B)**.

2. If our company incurs an *Expense*, which of the following might that be linked to (i.e., the other account affected as a result / the other line of the journal entry)?

   (A) An asset decreasing

   (B) A liability decreasing

   (C) A liability increasing

   (D) Multiple of the other answers

   (E) None of the other answers

Solution. Expenses are "anti-revenues," and everything is beautifully symmetric. An *Expense* is a decrease in net assets incurred from our operations — a decrease in assets *and/or* an increase in liabilities. An expense is a debit, so the balancing side is a credit: an asset going **down** or a liability going **up**. Both of those are on the list.

Option (A) is the asset case: paying the utilities bill in cash (credit *Cash*), or 'using up' *Prepaid Rent* or *Supplies* in an adjusting entry (credit the asset). (C) is the liability case: accruing *Salaries Expense* against *Salaries Payable* at period-end, accruing *Interest Expense* against *Interest Payable*, or buying something on account and crediting *Accounts Payable*. Both are genuine, common expense transactions, so more than one option is right. (B), a liability decreasing, is a debit and belongs to the revenue side of the symmetry.

Note the contrast with the previous problem, which is the whole point of asking them back-to-back: the revenue question had one right answer among the options and this one has two, purely because of which side of the symmetry the options happened to sample. So the answer is **(D)**.

3. FinCorp does annual calendar-year reporting. At the end of FY2021, FinCorp reported \$60m of cash, \$30m of Accounts Receivable, and \$40m of Revenue. At the end of FY2022, FinCorp reported \$80m of cash, \$60m of Accounts Receivable, and \$60m of Revenue. FinCorp never had any *Deferred Revenue* at any point. How much cash did FinCorp actually collect in cash from customers in 2022?

   (A) \$0m

   (B) \$20m

   (C) \$30m

   (D) \$60m

   (E) None of the other answers

   Solution. This is the reconciliation formula, with a pile of decoys around it. The identity, given no *Deferred Revenue*:

   $$Cash\ Collected\ from\ Customers = Revenue - \Delta Accounts\ Receivable$$

   FY2022 *Revenue* is \$60m. *Accounts Receivable* went from \$30m to \$60m, so $\Delta A/R = +\$30m$. Therefore:

   $$Cash\ Collected = \$60m - \$30m = \$30m$$

   In words: A/R rose by \$30m over the year, which means that, on net, \$30m of this year's revenue was earned without being collected. Only the other \$30m came in as cash.

Now the decoys, because there are three of them and each encodes a real confusion. The $40m of FY2021 *Revenue* is irrelevant — last year's flow has nothing to do with this year's collections. The **cash balances** are the nastier trap: cash went from $60m to $80m, and it is tempting to say "so they collected $20m," which is option (B). But the change in *Cash* reflects *everything* the company did — paying salaries, buying equipment, borrowing, paying dividends — not just customer collections. A company can collect $30m from customers and still see its cash balance move by any amount whatsoever. And (D), $60m, is the answer you get by ignoring the receivable entirely and assuming every sale was a cash sale, which is the very assumption accrual accounting exists to relax. So the answer is **(C)**.

4. What is an example of a liability that will not be settled in cash?

   (A) Accounts payable

   (B) Salaries payable

   (C) Interest receivable

   (D) Depreciation expense

   (E) Deferred revenue

   Solution. *Deferred Revenue* is the obligation you discharge by **doing something**, not by paying. The customer has already given us the cash; what we owe them is the good or service. So when the obligation is settled, the journal entry has no *Cash* line in it at all:

Deferred revenue (-L)	$0.8k	
Revenue (+S.E.)		$0.8k

   We provide the agreed service to those customers

   That is what makes it the odd one out. *Accounts Payable* (A) and *Salaries Payable* (B) are both liabilities, and both are settled by writing a check — to a supplier and to an employee respectively. The other two options are not liabilities at all, and it is worth naming why: *Interest Receivable* (C) is an **asset** — it is money owed *to* us, which is the flip side; and *Depreciation Expense* (D) is an **expense**, an Income Statement account, not a Balance Sheet account of any kind. (D) in particular is a nice trap for anyone pattern-matching on "non-cash" without checking what kind of account they are looking at. So the answer is **(E)**.

5. **Setup**: ExpoCorp does annual, calendar year reporting, and pays its employees monthly, on the first Friday of each month. Its employees earn $700 per week.

On Friday, December 3rd, 2021, it pays its employees their accrued salaries in full. On December 31, 2021, ExpoCorp makes its adjusting entries for the fiscal year. On Friday, January 7th, 2022, it is another payday, and ExpoCorp pays its employees in full. **Question**: What would be the balance of ExpoCorp's *Salaries Payable* account, after paying salaries on January 7th, 2022?

(A) $0

(B) $2,800

(C) $3,500

(D) $700

(E) None of the other answers

Solution. You can grind out the full sequence of entries here, and it is good practice — but you don't have to, and recognizing *that* is the point of the question. Read the stem: ExpoCorp "pays its employees **in full**" on January 7. Paying in full means the lagging obligation has been completely discharged. And a liability that has been discharged in full has a balance of $0. Full stop.

For completeness, here is the sequence. On December 31, four weeks have elapsed since the December 3 payday, so the adjusting entry accrues $4 \times \$700 = \$2{,}800$:

Salaries expense (-S.E.)	$2,800	
Salaries Payable (+L)		$2,800

Adjusting entry for accrued Salaries, 12/31/2021

Then on the January 7 payday, five weeks after the last one, we pay $5 \times \$700 = \$3{,}500$ in cash, recognize the one additional week of expense earned since the adjusting entry, and clear the payable:

Salaries expense (-S.E.)	$700	
Salaries payable (-L)	$2,800	
Cash (-A)		$3,500

Next payday, 1/7/2022

Salaries Payable was credited $2,800 and then debited $2,800: balance zero. The distractors are the two intermediate quantities — $2,800 is the balance *before* the

payment, and $3,500 is the cash paid, which the payable never equalled because we never recorded the fifth week's accrual as a liability. (And on that last point, recall the 'moral accounting' discussion: had we scrupulously accrued that fifth week, the payable would have hit $3,500 and then been cleared in full — and every account would have ended in exactly the same place. Either way, zero.) So the answer is **(A)**.

6. Which of the below would be an example of an **accrued expense** for our company?

   (A) Interest on loans

   (B) Customer paying us on account

   (C) Customer paying us up front, before we provide a service

   (D) Prepaid rent

   (E) None of the other answers.

   Solution. Put each option in the matrix. An **accrued expense** is an expense recognized *before* the cash is paid — the lower-left cell.

   Option (A), **interest on loans**, is the textbook case, and it is the one place in ordinary life where you already use the word correctly: your student loans *accrue* interest. The obligation builds up continuously with the passage of time, and the cash payment comes later, on the scheduled date. We debit *Interest Expense* and credit *Interest Payable* at period-end, before a dollar changes hands.

   Now the distractors, which are well-chosen because each lands in a *different* cell of the matrix. (B), a customer paying us on account, is **accrued revenue** — correct column of the matrix, wrong row: it is a revenue, not an expense. (C), a customer paying up front, is **deferred revenue** — wrong row *and* wrong column. And (D), *Prepaid Rent*, is a **deferred expense**: we paid cash first and will recognize *Rent Expense* later as we use up the lease. It is the exact mirror image of an accrual, which is what makes it the most tempting wrong answer for anyone who registered "expense" and stopped reading. So the answer is **(A)**.

7. **Setup**: ExpoCorp reported *Salaries Payable* of $120k on its FY2021 Balance Sheet, and $140k for FY2022. On its Income Statement, it reported *Salaries Expense* of $800k in FY2022. How much did ExpoCorp actually pay in salaries in FY2022?

   (A) $920k

   (B) $660k

   (C) None of the other answers.

(D) \$780k

Solution. ExpoCorp *incurred* \$800k of *Salaries Expense* in FY2022 — that is what its employees earned. But *Salaries Payable*, which is by definition salaries earned by employees but not yet paid to them, went **up** by \$20k over the year. An increase in the payable is precisely the portion of the expense that, on net, went unpaid. So:

$Cash\ Paid\ for\ Salaries = Salaries\ Expense - \Delta Salaries\ Payable = \$800k - \$20k = \$780k$

Two ways to go wrong, and both are represented. \$920k is what you get from adding the change instead of subtracting it — worth a moment of sign-checking intuition: if you owe your employees *more* at the end of the year than at the start, you must have paid out *less* than you racked up, not more. And \$660k is the result of subtracting the ending *balance* rather than the *change* ($\$800k - \$140k$), which is the single most common error on this family of problems. The formula runs on Δ, always. So the answer is **(D)**.

8. Setup: ExpoCorp does annual, calendar year reporting, and pays its employees monthly, on the first Friday of each month. Its employees earn \$700 per week. On Friday, December 3rd, 2021, it pays its employees their accrued salaries in full. On December 31, 2021, ExpoCorp makes its adjusting entries for the fiscal year. On Friday, January 7th, 2022, it is another payday, and ExpoCorp pays its employees in full, for the last 5 weeks of work. What would be the journal entry for the January 7th payment to employees?

   (A) Debit Salaries Expense \$3,500; Credit Cash \$3,500

   (B) Debit Salaries Payable \$3,500; Credit Cash \$3,500

   (C) Debit Salaries Payable \$2,800; Credit Cash \$2,800

   (D) Debit Salaries Expense \$700; Credit Cash \$700

   (E) None of the other answers

   Solution. Build the entry line by line, the way the chapter does — and note in advance that every option offered has only *two* lines, while the correct entry has three. That is the trap.

   Start with cash. Cash is cash, and it is written into the employment contract: five weeks have elapsed since the December 3 payday, so we pay $5 \times \$700 = \$3{,}500$. Credit *Cash* \$3,500.

Then the expense. *Salaries Expense* captures the activity — what the employees have earned *since the last time we recorded salaries*, which was the December 31 adjusting entry. That is one week. Debit *Salaries Expense* $700.

Then the liability. On December 31 we accrued four weeks, $4 \times \$700 = \$2{,}800$, into *Salaries Payable*. Paying in full discharges that obligation entirely, so it must come back to zero. Debit *Salaries Payable* $2,800.

Check: $\$700 + \$2{,}800 = \$3{,}500$. Debits equal credits.

Salaries expense (-S.E.)	$700	
Salaries payable (-L)	$2,800	
Cash (-A)		$3,500

Next payday, 1/7/2022

Now see what each distractor gets wrong, because each one is a coherent mistake. (A) charges the entire $3,500 to *Salaries Expense* — that is **double-counting**: $2,800 of it was already recognized as an expense in *last year's* Income Statement, via the adjusting entry. Booking it again in January would overstate 2022's expenses and understate 2021's. (B) has the cash right and clears $3,500 of *Salaries Payable*, but the payable only ever held $2,800; this entry would drive the account to a $700 debit balance, which is nonsense for a liability. (C) gets the payable right and the cash wrong — it pays only $2,800 when the contract requires $3,500, and it never recognizes the fifth week of expense. (D) recognizes the right expense and pays the wrong amount. So the answer is **(E)**.

A.7.2 Practice Problems: Open Prompts

1. ExpoCorp reported *Salaries Payable* of $120k on its FY2021 Balance Sheet, and $140k for FY2022. On its Income Statement, it reported *Salaries Expense* of $800k in FY2022. How much did ExpoCorp actually pay in salaries?

   Solution. $780k. Reasoning it out rather than reaching for the formula: over FY2022 ExpoCorp's employees earned $800k, and that is what got recorded as *Salaries Expense*. But the company did not settle all of it in cash. We know that because the amount it owed its employees at year-end was $20k *higher* than the amount it owed them at the start ($140k versus $120k). *Salaries Payable* is exactly "earned by employees, not yet paid to them," so a $20k increase in that account means $20k of the year's earnings, on net, stayed unpaid. Therefore $780k went out the door in cash.

As a formula:

$Cash\ Paid\ for\ Salaries = Salaries\ Expense - \Delta Salaries\ Payable = \$800k - (\$140k - \$120k) = \$780k$

And it is worth noticing what this little calculation represents in general. The salaries payable account is a **working capital** account, and its job here is to "plug" the difference between the accrual expense and the cash flow across time. Precisely because it does that job, we can run the logic *backwards*: we are external users of these statements, we never saw ExpoCorp's payroll register, and yet we just recovered its actual cash payroll from two Balance Sheets and one Income Statement line. That is the whole trick of this chapter, and it generalizes — $Cash\ Paid\ for\ X = X\ Expense - \Delta X\ Payable$, whether X is salaries, interest, utilities, or income taxes.

2. FinCorp does annual calendar-year reporting. At the end of FY2021, FinCorp reported \$80m of Accounts Receivable. At the end of FY2022, FinCorp reported \$100m of Accounts Receivable, and \$60m of Revenue. FinCorp never had any Deferred Revenue at any point. How much cash did FinCorp actually collect in cash from customers in 2022?

   Solution. \$40m. The identity, on the revenue side of the symmetry:

   $Revenue = Cash\ Collected\ from\ Customers + \Delta Accounts\ Receivable$

   $\$60m = Cash\ Collected\ from\ Customers + (\$100m - \$80m)$

   $Cash\ Collected\ from\ Customers = \$60m - \$20m = \$40m$

   In words: FinCorp's receivables rose by \$20m over 2022, and the only thing that pushes A/R up is **credit sales** — goods or services delivered but not yet paid for. So, on net, \$20m of the year's \$60m of *Revenue* was not collected. The remaining \$40m was.

   The "on net" is doing real work in that sentence, and it is worth being precise about it. In reality FinCorp was doing both things all year long: making new credit sales (which push A/R up) *and* collecting on old accounts (which push A/R down, and which absolutely count as cash collected from customers, even though they are not this year's revenue). We have no idea how much of each happened. We don't need to. The formula runs on the **net change**, which handles both directions at once — and the chapter's algebra shows the gross credit-sales term cancels out exactly.

   One further comment, since this problem and the last one look like different problems and are really the same one. In the salaries problem, an increase in a

working-capital **liability** meant cash out was *less* than the expense. Here, an increase in a working-capital **asset** means cash in was *less* than the revenue. Same structure, opposite side of the Balance Sheet, and in both cases the working-capital account is storing the timing difference. When you get to the Statement of Cash Flows in Chapter 16 and see a column of $\pm\Delta$ working-capital adjustments underneath Net Income, that statement will be doing nothing but these two problems, in aggregate.

A.7.3 Essay and Discussion Questions

1. You're doing your first finance course, and the professor says that "working capital accounts represent customer and supplier financing." Your friend, who never came to office hours for BUAD 280, does not understand this statement. "Wait, what?," they ask. "Financing comes from banks... customer financing? Supplier financing?" Explain to your friend the meaning of the professor's statement. What is working capital, what are the specific accounts in play here, and why does it make sense to see them as forms of customer/supplier financing?

   Solution. Here is how I would talk your friend through it.

   Start with the vocabulary, so the sentence at least parses. "Working capital accounts" just means a company's **current assets and current liabilities** — the short-horizon stuff, the things with a time-to-cash of under a year. *Accounts Receivable*, *Prepaid Rent*, *Inventory* on the asset side; *Accounts Payable*, *Salaries Payable*, *Deferred Revenue*, *Interest Payable* on the liability side. And "net working capital" is just current assets minus current liabilities.

   Then fix the underlying misconception, which is about what 'financing' means. Your friend's objection is "financing comes from banks," and that instinct is understandable — it is the picture everyone starts with. But go back to what financing actually *is*, stripped of the institutional trappings. Financing is somebody providing you resources now, in exchange for something later. That is it. There is no requirement that the somebody have a marble lobby, or that a formal loan document exist, or that interest be charged. A bank loan is the most *visible* form of financing, not the only one.

   So ask the diagnostic question of each working-capital account: **who is out of pocket, and who is holding the money, between now and settlement?** Whoever is out of pocket is financing whoever is holding the money. Once you ask it that way, the whole class of accounts lights up.

   Now walk the specific accounts.

Accounts Payable. Our supplier ships us goods in March and lets us pay in May. For those two months, we have the goods and they don't have the money. They have handed us resources now in exchange for cash later — which is *exactly* the structure of a bank loan, minus the paperwork and the interest. **Our suppliers are lending to us.** This is what "supplier financing" means. It is sometimes called 'trade credit,' and in aggregate it is enormous; for many companies, *Accounts Payable* is a larger source of funding than any bank facility.

Deferred Revenue. A customer pays us $10,000 in January for a service we will deliver over the coming year. We have their cash *now*; they have nothing yet but a promise. They are out of pocket, and we are holding the money. **Our customers are lending to us.** This is "customer financing," and it is the reason subscription and prepayment business models are so attractive: your customers fund your operations before you incur the costs of serving them. (Think of a Kickstarter campaign, which is customer financing in its purest, most naked form.)

Salaries Payable. Your employees work all of April and get paid at the beginning of May. For that month they have provided real labor and received nothing. **Your employees are, in effect, extending you a short-term, interest-free loan** — one they almost certainly do not think of that way, but that is the economic structure. And it is not small: for a services firm, a month of payroll is a lot of money.

Accounts Receivable. Here the arrow reverses. We deliver to a customer in March and let them pay in May. Now *we* are the ones out of pocket. **We are financing our customers.** Every dollar sitting in A/R is a dollar we have advanced to somebody else's business.

Then step back and give your friend the payoff, because the reframe is not a curiosity — it is why finance people obsess over this.

First, it tells you *who is financing whom* in the company's day-to-day operations. Line up the accounts and you get a map of the informal credit web the firm sits inside: customers funding it here, suppliers funding it there, and it funding its own customers over on the asset side. Net working capital is, in a rough sense, the score — how much of that web the company is funding on net, versus how much of it is funding the company.

Second, and this is the part that makes CFOs care: **this financing is generally free.** No interest, no covenants, no underwriting. So a company that manages it aggressively — collecting from customers fast, paying suppliers slowly, taking money up front where it can — gets a meaningful chunk of its operations bankrolled at zero cost. The classic case is a big-box retailer that sells inventory to shoppers for cash in about three weeks and pays its suppliers in sixty days. No-

tice what that means: the supplier's money funded the inventory, the shoppers' money arrived first, and the retailer had *negative* net working capital and a pile of float, without ever borrowing a dime. Some famously successful businesses are, structurally, working-capital machines with a retail operation attached.

Third — and this is where your friend's finance course is actually heading — this is the same set of accounts that explains the difference between accrual net income and operating cash flow, which is why the Statement of Cash Flows walks from one to the other via changes in exactly these accounts. Your professor's sentence and that statement are the same idea in two costumes.

The one-sentence version, if your friend's eyes have glazed: financing is just "somebody gives you resources now and gets paid later," and if you go through the working-capital accounts asking who is out of pocket in the meantime, you will find that customers, suppliers, and employees are all quietly doing that for the company, every single day, for free.

2. One of the most confusing things about the course so far was understanding the relationship between income statement and balance sheet accounts. They measure different things (performance vs. position), using different accounts and line items, and yet somehow are kept in sync—the income statement effects shareholders' equity, and shareholders' equity is reported on the balance sheet. Can you help me understand this? Illustrate the idea in words and but feel free to reference journal entry examples.

   Solution. This is worth answering carefully, because the confusion is a *good* one — it comes from noticing something real. Let me build the answer in layers.

   Layer one: what the two statements are actually measuring.

   The Balance Sheet is a **stock** measure: a photograph of the company's position at a single instant. What do we have, what do we owe, and what is left over for the owners? The Income Statement is a **flow** measure: a film of what happened *over* an interval. How much did we earn, what did it cost, and what was the net?

   That distinction is not a formatting convention; it goes all the way down. Ask: what is our company's *Cash* balance one minute into the new year? The same as it was a minute earlier — Cash is a stock, and stocks persist across the boundary. Now ask: what is our *Revenue* one minute into the new year? Zero, necessarily. Revenue is a flow, and the new period has only just begun. **Flows reset; stocks accumulate.** That single line explains why we have two statements at all, and why one set of accounts is 'temporary' and the other 'permanent.'

   Layer two: why they must stay in sync — the mechanical answer.

Here is the short answer: they stay in sync because **every journal entry balances, and there is no such thing as an entry that touches only the Income Statement.**

That second clause is the load-bearing one, and it is easy to miss. Look at any revenue or expense entry you have ever made. There is always a Balance Sheet account on the other line. Always. Consider a cash sale:

Cash (+A)	$3.9k	
Revenue (+S.E.)		$3.9k

Cash Sale

Or a credit sale, or an accrued expense:

Salaries expense (-S.E.)	$2.8k	
Salaries Payable (+L)		$2.8k

Adjusting entry for accrued Salaries, 12/31/2021

Revenue up, an asset up. Expense up, a liability up. There is no way to record performance without simultaneously recording its effect on position, because there is no way to write a one-line journal entry. Performance and position are not two independent ledgers that we labor to reconcile — they are two *readings* of the same set of entries.

So the Income Statement is not really a separate book. It is a **report on the subset of accounts that happen to be temporary**. The Balance Sheet is a report on the subset that are permanent. Every entry that hit a temporary account also hit a permanent one.

Layer three: the seam, and how *Retained Earnings* stitches it.

Now, the two statements are still reported separately, and the totals have to tie. Where does the flow figure get absorbed into the stock world? *Retained Earnings*.

The reason this works is that revenues and expenses are not really their own species of account at all. **They are subsidiary accounts of Shareholders' Equity** — sub-ledgers we maintain during the year so that we can produce a detailed Income Statement, rather than dumping everything straight into one enormous *Retained Earnings* account and losing all the detail. That is why the annotations read *(+S.E.)* and *(-S.E.)*: a revenue *is* an increase in equity, and an expense *is* a

decrease in it, from the first moment they are recorded. They were never outside the Balance Sheet; they were being tracked in labelled bins alongside it.

At period-end, we empty the bins:

$\Delta Retained\ Earnings = Net\ Income - Dividends = Revenues - Expenses - Dividends$

Or, if you like seeing it as an actual entry — debit every revenue to zero, credit every expense and dividend to zero, and the plug lands in *Retained Earnings*:

	Debit	**Credit**
Revenues:	\$7k	
Salaries Expense:		\$2.7k
Supplies Expense:		\$.4k
Depreciation Expense:		\$1k
Rent Expense:		\$.6k
Dividends:		\$.1k
Retained Earnings:		\$2.2k

'Closing process', update to Retained Earnings

That entry is worth staring at, because you can *see* the two worlds touching in a single journal entry: every debit and credit above the last line is an Income Statement account, and the last line is a Balance Sheet account. Merle Hopkins used to tell students that revenues, expenses, and dividends "swim like a fish" into Retained Earnings. That is exactly the picture.

Layer four: the proof that it worked.

Here is the demonstration I find most convincing, and it is the one from Chapter 5. Take our toy company at the end of its first year. Total assets came to \$24.9k. Total liabilities plus Common Stock came to \$22.7k. If we had lazily copied the *Retained Earnings* figure straight off the adjusted trial balance — where it still sits at \$0, because the period's profits are all still parked in the temporary accounts — the Balance Sheet **would not balance**. It is off by exactly \$2.2k. And \$2.2k is exactly this year's net income less dividends.

Sit with that for a second, because it is not a coincidence and it is not a lucky cancellation. The \$2.2k of value the company created is *physically there* in the net assets — in the cash collected, the receivables outstanding, the equipment on hand, net of what is owed. The left side of the Balance Sheet already knows about it. The only open question was where it gets *claimed* on the right side.

Retained Earnings is the answer to that question. The Income Statement is the detailed explanation of how the number got that big.

Layer five: the deeper way to hold it, which is where this chapter is going.

Once you have the above, there is a further move available. Rather than thinking "the Income Statement is primary and the Balance Sheet absorbs its residue," you can flip it. This chapter gives the fully general definitions:

> **Revenue:** an increase in net assets (assets up and/or liabilities down) earned from operations.
>
> **Expense:** a decrease in net assets (assets down and/or liabilities up) incurred from operations.

Read those closely. Revenues and expenses are *defined in Balance Sheet terms*. Net income is, on this reading, just the change in net assets arising from operations — which, since $E = A - L$, is a change in Shareholders' Equity by construction. On this view the two statements cannot possibly get out of sync, because the Income Statement is not measuring an independent thing; it is a classified account of *why* net assets moved.

So: the same transactions push on both sides at once, the definitions of revenue and expense are already stated in terms of assets and liabilities, and *Retained Earnings* is the single seam where the flows are deposited into the stock. Nothing is being reconciled after the fact. It stays in sync because it was never two systems. It really is a beautiful system.

3. What is the difference between an internal transaction and an external transaction? When you 'earn down' deferred revenue liability, is that internal or external? (Hint: If the answer is that 'it depends,' provide some examples of when it would be either, and why.)

   Solution. The definitions first. An **external transaction** is a "real" exchange of goods, services, or cash between our company and some outside party. An **internal transaction**, or **adjusting entry**, is everything *else* we have to do to get our accounts right before closing the books at period-end. Note that the second definition is deliberately residual — it is a catch-all, and I want you to hold it that way rather than looking for some crisper principle underneath it.

   Two structural facts before we get to the question. First, **all cash transactions are external**, necessarily: cash changing hands *is* a real exchange with an outside party. Second, running that backwards, **all internal transactions are non-cash**, and therefore all of them are creatures of accrual accounting. If we only recorded cash receipts and payments, there would be nothing to adjust. So every adjusting

entry is, in some sense, accrual accounting cleaning up after itself. But — and students routinely get this backwards — the relationship does not run both ways. Plenty of accruals involve no adjusting entry at all. A credit sale accrues revenue before the cash arrives, and both the sale and the later collection are external transactions with real exchanges in them. It is a rectangle-square situation.

Now the question, and yes: it depends. And I want to be clear that this is not a dodge — the answer genuinely turns on the facts, and the fact that it does is the actual lesson. *Deferred Revenue* is an account, and **accounts do not carry a classification**. The classification attaches to the *transaction*, and you have to look at what happened in the world.

Case one: it is external. Recall Transaction #8 and its sequel. Customers paid $0.8k up front in October for a service we deliver on January 5.

Deferred revenue (-L)	$0.8k	
Revenue (+S.E.)		$0.8k

We provide the agreed service to those customers, 1/5/2022

On January 5 we actually showed up and did the thing. Somebody delivered a service; somebody received it. That is a real exchange with an outside party, on an identifiable date — the very paradigm of an **external transaction**. No cash moved, but cash is not the criterion; a real exchange is. And notice the practical tell: the delivery *itself* triggered the entry. We had an event to book.

Case two: it is internal. Now take the retainer contract. You are a lawyer, and on July 1 a client pays $10k to have you on retainer for a year — meaning they have bought the right to call on you and the assurance that you will not represent their adversaries.

Cash (+A)	$10k	
Deferred revenue (+L)		$10k

Customer Pays for 1-year retainer contract, 7/1/2022

That first entry is obviously external — $10k in cash changed hands. But now: when do you earn it? Your performance obligation is *to be available*. There is no delivery date. There is no moment at which anything is handed over. In fact, if the client never calls you at all, you have still performed perfectly — availability was the product. So there is no event in the world to trigger an entry.

But December 31 arrives, you must close your books, and half the contract term has elapsed, so half the obligation has been satisfied. You have to recognize $5k of *Revenue*, and the only occasion for doing so is the period-end adjustment:

Deferred revenue (-L)	$5k	
Revenue (+S.E.)		$5k

Adjusting entry for half year of Retainer Contract, 12/31/2022

Same account, same direction, same kind of 'earning down' — and this one is unmistakably an **internal transaction / adjusting entry**.

What actually distinguishes the two cases. Compare them and the pattern falls out: it depends on whether the performance obligation is discharged at an **identifiable moment** or **ratably over time**. Discrete delivery — the soccer camp is held, the software is shipped, the concert happens — gives you a date, an event, and an external transaction. Continuous availability — a retainer, an insurance policy, a year of gym membership, a Netflix subscription — gives you nothing to point at, so the recognition has to be swept up at period-end as an adjustment. It is the same distinction, incidentally, that separates *Accounts Payable* (which only moves when you incur or pay a bill, both external events) from *Interest Payable* (which accrues silently with the passage of time and must be adjusted).

And now the part I most want you to take away, which is that in the end it does not much matter. Some students get genuinely hung up on the philosophy here, and there are borderline cases where reasonable people would classify differently. Fine. Here is why you should not lose sleep: the company does not really *have* financial statements until the end of the period. Nobody outside the firm ever sees your classification, or your trial balance, or which bucket you filed an entry under. What the world sees is the reported Income Statement and Balance Sheet as of the reporting date. So the only thing that has to be true is this: **if you did not account for something as an external transaction, then you must account for it as an adjusting entry before you close the books.** The taxonomy is a checklist for making sure nothing falls through the crack between those two, not a metaphysical claim about the nature of events.

Which suggests the right practical habit. Do not ask "is this internal or external?" as your first question. Ask "as of the reporting date, is this account correctly stated?" If it is, and every real exchange got recorded when it happened, you are done — whatever you want to call the entries. And if you find yourself at year-end staring at an *Accounts Payable* balance wondering whether you are supposed to

do something to it: no. Some accounts get adjusted because they are used up or accrue over time; others don't. In those cases, don't just do something — stand there.

4. You see the following journal entry:

	Debit	**Credit**
Cash (+A)	\$3.5k	
Accounts Receivable (+A)	\$1.5k	
Revenue (+S.E.)		\$5k

Explain what has happened in this transaction, and explain the quantities here from the "Income Statement perspective" (focusing first on the measurement of activities that earn revenues and incur expenses, and thinking of the B/S as a plug) AND the "Balance Sheet perspective" (which focuses first on the measurement of assets and liabilities, and thinks of the I/S as a plug).

Solution. First, the narration. We provided a good or service worth \$5k — the soccer camp from the opening of this chapter — and the customer settled part of the bill on the spot and promised the rest. \$3.5k came in as cash; the remaining \$1.5k is the customer's promise to pay, which we record as *Accounts Receivable*. Since we have fully delivered, we have fully **earned** the revenue, so all \$5k is recognized now, regardless of how much of it was collected.

A few things to notice about the entry before we interpret it. It has three lines, which is legal and unremarkable: the rule is never "one debit and one credit," it is that total debit dollars equal total credit dollars. And this is a mixed transaction — part cash sale, part credit sale — sitting in two cells of our timing matrix at once. Do not let that trouble you; the matrix classifies each *dollar*, not each entry.

Now the two perspectives — which is the real point of the question.

The Income Statement perspective. Fix the day and ask first: *what did we earn?* We ran a soccer camp; we delivered \$5k of service; therefore we earned \$5k of *Revenue*, full stop. That is the primary economic fact, and it is settled by the activity itself, without any reference to money. Only then do we ask the secondary question: how did the \$5k arrive? \$3.5k of it showed up as cash. The other \$1.5k did not. So something on the Balance Sheet has to **plug** the gap and carry it forward until the customer pays:

$$Revenue\ Earned - Cash\ Collected = \Delta Accounts\ Receivable$$

$$\$5k - \$3.5k = \$1.5k$$

On this reading, *Accounts Receivable* is not really a *thing* — it is a **timing device**. It exists to store the difference between what we earned and what we collected, and it will run back to zero when the customer pays. It flips on when the revenue is earned and off when the cash arrives, and while it is on, its balance tells you precisely how much revenue has been earned but not yet collected.

(This is, I confess, how I naturally think — I am an income-statement guy. Start with what happened in the business; treat the Balance Sheet as the bookkeeping that carries the consequences across time.)

The Balance Sheet perspective. Now run it in the other direction. Ask first: *what do we own that we did not own this morning?* Two things. \$3.5k of *Cash*, which is plainly an asset. And a customer's enforceable promise to pay \$1.5k — which is *also* plainly an asset, a resource with future economic benefit, no less real for being a promise rather than a banknote. So our net assets went up by \$5k. And *that* is the revenue:

$$Revenue\ Earned = Cash\ Collected + \Delta Accounts\ Receivable$$

$$\$5k = \$3.5k + \$1.5k$$

On this reading, the primary job of accounting is measuring the assets and liabilities — reporting the company's position correctly. *Revenue* is then the **plug**: the Income Statement line that explains *why* net assets rose by \$5k this period, and labels the cause as 'operations' rather than, say, 'the owners put in more money.' This is the older, more classical view, and it is the one that makes the chapter's fully general definition of revenue sing: *an increase in net assets earned from our operations.*

Same entry, same numbers, two readings. And notice that the two 'formulas' above are literally the same equation rearranged — which is itself the tidiest possible demonstration of the point. In fact both are just the journal entry read sideways: the credit must equal the sum of the debits, so $Revenue = Cash + \Delta A/R$ is not analysis bolted onto the accounting, it *is* the accounting, transposed. Neither perspective is correct and neither is incorrect. It is a duckrabbit: the useful skill is being able to flip between them at will.

Why bother learning both? Because it will not stay decorative. Later in this book we will hit gnarly issues where companies have a genuine **choice of method**, and those choices are almost always, at bottom, a disagreement about which side is primary. Should we measure the asset at what it is currently worth and let the income statement report whatever falls out? Or should we measure this period's performance carefully and let the balance sheet carry the residue? Standard-setters have argued about that for decades, and modern GAAP has drifted notice-

ably toward the Balance Sheet view. When we get there, you will find the debates unintelligible without this framework — and, with it, mostly obvious.

A.8 Chapter 8

Model convention: Percentage-of-credit-sales questions use the deliberately simplified, unrevised-expense model defined in Chapter 8. That model is used to study an information tradeoff; it is not a complete current U.S. GAAP reporting method. In the allowance-method problems, the stated percentages are assumed estimates, and a bankruptcy indicates a fully uncollectible account only where the problem stipulates that outcome.

A.8.1 Practice Problems: MCQ's

1. Which approach is consistent with the current U.S. GAAP requirement to maintain an allowance reflecting updated expected credit losses?

   (A) Wait until an account is known to be uncollectible, without estimating material expected losses beforehand.

   (B) Record a fixed percentage of current credit sales and never revise the allowance for later information.

   (C) Estimate expected credit losses and update the allowance as relevant information changes.

   (D) All of the above.

   Solution. The answer is **(C)**. The deliberately simplified model in (B) makes the matching-versus-remeasurement tradeoff visible, but it does not permit a company to leave materially misstated receivables in published financial statements. A sales-based estimate can be an input to the loss estimate; the allowance must still reflect the applicable reporting requirements.

2. What is the journal entry we would make on the date on which we write off an account of $X under the Direct Writeoff Method?

   (A) Debit Bad Debt Expense $X, Credit Allowance for Doubtful Accounts $X

   (B) Debit Bad Debt Expense $X, Credit Accounts Receivable $X

   (C) Debit Allowance for Doubtful Accounts $X, Credit Accounts Receivable $X

   (D) It depends on our previous assumptions about the amount uncollectible.

   (E) None of the above

   Solution. Under the *Direct Writeoff Method* there is no *Allowance*—that account does not exist in this method's world. We made no provision in advance, so when we finally learn that a particular account is dead, the entire loss shows up at that

moment. The receivable has to come off the books (credit *A/R*), and the balancing debit is the new expense account, *Bad Debt Expense*:

Bad Debt Expense (-S.E.):	$X	
Accounts Receivable (-A):		$X

Writeoff of $X under the Direct Writeoff Method

Option (C) is the *Allowance* method's writeoff entry—that one draws down a provision we had already made. Here there is no provision to draw down. Option (D) is the trap for readers who remember that *some* method depends on a prior percentage assumption; this method depends on no assumption whatsoever, which is exactly its defining feature and exactly why GAAP rejects it. So the answer is **(B)**.

3. What is the journal entry we would make on the date on which we write off an account of $X under the Allowance Method?

   (A) Debit Bad Debt Expense $X, Credit Allowance for Doubtful Accounts $X

   (B) Debit Bad Debt Expense $X, Credit Accounts Receivable $X

   (C) Debit Allowance for Doubtful Accounts $X, Credit Accounts Receivable $X

   (D) It depends on our previous assumptions about the amount uncollectible.

   (E) None of the above

   Solution. This is Rule #2 of the *Allowance* method. On the writeoff date we do two things and only two things: extinguish the account (credit *A/R*), and "use up" the *Allowance* we had already set aside (debit—remember the *Allowance* is a contra-asset, so it lives on the credit side, and reducing it is a debit).

Allowance for Doubtful Accounts (-contra-A):	$X	
Accounts Receivable (-A):		$X

Writeoff of $X under the Allowance Method

Notice what is *not* in that entry: *Bad Debt Expense*. Nothing touches the income statement on the writeoff date under this method. That is the single most-missed fact in the chapter, and it is what makes the last essay question in this set worth writing out. The expense arrives later, indirectly, at the 12/31 adjusting entry, because the writeoff has drawn down the *Allowance* and the adjusting entry has

to refill it. Option (A) is the *adjusting* entry, not the writeoff entry—right entry, wrong date. Option (B) is the *Direct Writeoff* method's entry. So the answer is **(C)**.

4. Under the *percentage-of-credit-sales* method, all else equal, the greater our *writeoffs during the period*, the [higher / lower / no change] will be our *Bad Debt Expense*?

   (A) Higher

   (B) Lower

   (C) Unchanged

   (D) Not enough information.

   (E) None of the other answers.

   Solution. Under this method, computing the *Bad Debt Expense* requires exactly two pieces of information: the **credit sales made this period**, and the **assumed percentage**. Writeoffs are not one of them. Per Rule #3, when a customer declares bankruptcy we do *nothing*—we already allowed for expected losses, we made our best guess, and we keep moving. So the answer is **(C)**.

5. Under the *percentage-of-credit-sales* method, all else equal, the greater our *collections on past periods' accounts during this period*, the [higher / lower / no change] will be our *Bad Debt Expense*?

   (A) Higher

   (B) Lower

   (C) Unchanged

   (D) Not enough information.

   (E) None of the other answers.

   Solution. Same answer, same reason, different distraction. Collections on *past* periods' accounts are not credit sales made *this* period, so they enter the calculation nowhere. Your economic instinct is not wrong—collecting more of last year's accounts genuinely is good news, and it genuinely ought to count for something. It is just that this method, by design, refuses to let it count. That refusal is precisely what the *Allowance* method fixes, and precisely why the *percentage-of-credit-sales* method's *Net Accounts Receivable* drifts out of true over time. So the answer is **(C)**.

6. Under the *Allowance* method, all else equal, the greater our *writeoffs during the period*, the [higher / lower / no change] will be our *Net Income*?

(A) Higher

(B) Lower

(C) Unchanged

(D) Not enough information.

(E) None of the other answers.

Solution. Note first that this question asks about *Net Income*, not *Bad Debt Expense*—so whatever happens to the expense, flip the sign. Now trace the mechanics. A writeoff debits the *Allowance*, drawing it down. But the ending balance of the *Allowance* is pinned by the ending *A/R*—and the writeoff reduced *A/R* by the full amount while reducing the *required Allowance* by only p times that amount. So the adjusting entry has to credit the *Allowance* back up by more than it otherwise would, and the matching debit is *Bad Debt Expense*. The general formula makes it explicit:

$$Bad\ Debt\ Expense_t = p * (Credit\ Sales_t - Collections_t) + (1-p)Writeoffs_t$$

The coefficient on writeoffs is $(1-p)$, which is positive for any sane p. Higher writeoffs → higher *Bad Debt Expense* → lower *Net Income*. So the answer is **(B)**.

7. Under the *Allowance* method, all else equal, the greater our *collections on past periods' accounts during this period*, the [higher / lower / no change] will be our *Net Income*?

(A) Higher

(B) Lower

(C) Unchanged

(D) Not enough information.

(E) None of the other answers.

Solution. Collections carry a coefficient of $-p$ in formula (8.7), so more collections means a *lower Bad Debt Expense*, and therefore a *higher Net Income*. The intuition: collecting an old account converts a doubtful receivable into cash, which shrinks the pool of outstanding accounts we still have to be doubtful about, which shrinks the required ending *Allowance*, which shrinks the adjusting entry, which shrinks the expense. This is Goal #2 in action—"surprise" good news on past accounts actually reaches the income statement under this

method. Push it far enough and the *Bad Debt Expense* goes **negative**, as in the chapter's variation where StudentCo pays after all. So the answer is **(A)**.

8. Under the *Allowance* method, holding all else equal (i.e., total cash sales and unrelated expenses), the greater our *credit sales during this period*, the [higher / lower / no change] will be our *Net Income*?

   (A) Higher

   (B) Lower

   (C) Unchanged

   (D) Not enough information.

   (E) None of the other answers.

   Solution. Here you have to hold both ends of the transaction in view at once, because a credit sale hits the income statement twice. Sell an extra \$1 on account, and *Revenue* rises by \$1. That same \$1 sits in ending *A/R*, so the required ending *Allowance* rises by p, so the *Bad Debt Expense* rises by p—which is exactly the coefficient on $Credit\ Sales_t$ in formula (8.7). Net effect on the bottom line: $\$1 - p$, which is positive so long as we do not assume that 100% of our customers are deadbeats.

   So credit sales are still profitable—just less profitable than cash sales, by exactly the expected credit loss. That is the whole economic content of the *Bad Debt Expense*: it is the price of doing business on credit. So the answer is **(A)**.

9. Our company accounts for bad debts / receivables using the ***percentage-of-credit-sales method***, assuming 10% of credit sales will be uncollectible. We began FY2020 with a balance of \$100k in *Accounts Receivable* and \$20k in our *Allowance for Doubtful Accounts*. During FY2020, we made an additional \$200k in credit sales, and collected \$40k on accounts from past credit sales. Customers with accounts totaling \$10k declared bankruptcy. What would be our FY2020 *Bad Debt Expense*?

   (A) \$20k

   (B) \$10k

   (C) \$30k

   (D) \$5k

   (E) None of the other answers.

Solution. Read the method first, then read the numbers—never the other way around. This is the *percentage-of-credit-sales* method, so the *Bad Debt Expense* is $10\% * \$200k = \$20k$. Full stop.

The beginning *A/R* of \$100k, the beginning *Allowance* of \$20k, the \$40k of collections, and the \$10k of bankruptcies are all deliberate red herrings—every one of them is a quantity that the *Allowance* method would need and this method ignores. The problem is engineered so that your confidence gets shaken by the sheer volume of information you are declining to use. Decline to use it anyway. So the answer is **(A)**.

10. Consider the following scenario: MarshallCo uses the income-statement / percentage-of-credit sales method for accounting for bad debts, and assumes 10% of credit sales each period will be uncollectible. In 2019, MarshallCo starts up its life and sells \$1m on account. In 2020, MarshallCo (i.) collects all \$1m from 2019, and (ii.) makes an additional \$1m in new credit sales **Question:** What would MarshallCo's *Net Accounts Receivable* for FY 2020 be?

    (A) \$900k

    (B) \$1m

    (C) \$800k

    (D) \$100k

    (E) None of the other answers.

    Solution. Track both accounts as *stocks*—they accumulate everything ever posted to them.

    Accounts Receivable: \$1m of 2019 sales, less the \$1m collected in 2020, plus \$1m of 2020 sales, or $\$1m - \$1m + \$1m = \$1m$.

    Allowance: under this method it only ever gets credited, by 10% of each period's credit sales, and never gets drawn down. So $\$100k + \$100k = \$200k$.

    Therefore $Net\ Accounts\ Receivable = \$1m - \$200k = \$800k$.

    Now, option (A) is the interesting distractor, because \$900k is the answer that *makes economic sense*: as of the end of 2020 we have \$1m of outstanding accounts and expect 10% of them to go bad. The gap between the \$800k we report and the \$900k that means what the line is supposed to mean is exactly the misstatement this method produces—an *Allowance* carried over from a prior year's estimate that reality has already contradicted. That misstatement is the standing complaint against the income-statement method, and I argue it washes out over time (by the end of 2021, in the chapter's version of this scenario, *NAR* is back to \$900k). But the question asks what we *report*. So the answer is **(C)**.

11. Our company accounts for bad debts / receivables using the **allowance method**, assuming 10% of outstanding accounts will be uncollectible. We began FY2020 with a balance of \$150k in Accounts Receivable and \$15k in our Allowance for Doubtful Accounts. During FY2020, we made an additional \$30k in credit sales, collected \$20k on accounts from past credit sales, and wrote off \$10k in accounts. What would be our FY2020 *Bad Debt Expense*?

    (A) \$2k

    (B) \$3k

    (C) \$4k

    (D) Not enough information.

    (E) None of the other answers.

    Solution. Do the mechanics, then check with the formula. Ending *Accounts Receivable* is $\$150k + \$30k - \$20k - \$10k = \$150k$. So the *Allowance* must **end** at 10% of \$150k, or \$15k. Where does it stand *before* adjustment? It began at \$15k (credit), and the \$10k writeoff debited it down to \$5k. So the adjusting entry is a \$10k credit to the *Allowance*, and the matching debit—the *Bad Debt Expense*, by definition—is **\$10k**:

Bad Debt Expense (-S.E.):	\$10k	
Allowance for Doubtful Accounts (+contra-A):		\$10k

12/31/2020 Adjusting Entry to update the Allowance

    Checking against formula (8.7): $.1 * (\$30k - \$20k) + .9 * (\$10k) = \$1k + \$9k = \$10k$. The two paths agree, as they must.

    And it is worth seeing it a third way, straight from the T-accounts, because that is the representation that makes the plug visible:

Accounts Receivable (A)

Debits	Credits
Beg. Bal: \$150k	Collections: \$20k
Credit sales: \$30k	Writeoffs: \$10k
Bal: \$150k	

Allowance (contra-A)

Debits	Credits
Writeoffs: \$10k	Beg. Bal: \$15k
	12/31 adj.: \$10k
	End. Bal: \$15k

Ending *A/R* of \$150k requires an ending *Allowance* of \$15k; the writeoff had knocked the pre-adjustment balance down to \$5k; and the plug that closes that gap is a \$10k credit, whose matching debit is the *Bad Debt Expense*. \$10k is not among the numerical options, and we plainly had enough information. So the answer is **(E)**.

12. Our company accounts for bad debts / receivables using the **allowance method**, assuming 10% of outstanding accounts will be uncollectible. We began FY2020 with a balance of \$200k in Accounts Receivable. During FY2020, we wrote off \$15k in accounts, and ended the period with total *Accounts Receivable* of \$220k. What would be our FY2020 *Bad Debt Expense*?

    (A) \$2k

    (B) \$3k

    (C) \$4k

    (D) Not enough information.

    (E) None of the other answers.

    Solution. Notice what you are *not* given here: credit sales and collections. That is fine—you do not need them, because you are handed the ending *A/R* directly, which is all those two quantities were ever going to be used for. Use the intermediate form of the formula:

$$Bad\ Debt\ Expense_t = p * (\Delta AR) + Writeoffs_t = .1 * (\$220k - \$200k) + \$15k = \$17k$$

    Or, in mechanics: the beginning *Allowance* is 10% of the beginning \$200k of receivables, or \$20k—that is what the standing assumption means, since last period's adjusting entry left it there. The \$15k writeoff draws it down to \$5k. The required ending balance is 10% of \$220k, or \$22k. So the adjusting entry is a \$17k credit, and the *Bad Debt Expense* is \$17k.

Option (D) is the tempting one, because the problem withholds two inputs you are used to having. But withholding an input you do not need is not the same as withholding information. Compare item 13, where something you genuinely *do* need is missing. So the answer is **(E)**.

13. Our company accounts for bad debts / receivables using the **allowance method**, assuming 10% of outstanding accounts will be uncollectible. We began FY2020 with a balance of \$300k in Accounts Receivable, and \$30k in our Allowance for Doubtful Accounts. We ended the FY2020 period with total *Accounts Receivable* of \$220k. What would be our FY2020 *Bad Debt Expense*?

    (A) \$2k

    (B) \$3k

    (C) \$4k

    (D) Not enough information.

    (E) None of the other answers.

    Solution. This one *is* underdetermined, and here is exactly why. Write the formula:

$$Bad\ Debt\ Expense_t = p * (\Delta AR) + Writeoffs_t = .1 * (\$220k - \$300k) + Writeoffs_t = -\$8k + Writeoffs_t$$

    We are told the *ending* balance of *A/R*, so the ΔAR term is pinned down. But the writeoffs term stands alone, and we are never told what the writeoffs were. And ΔAR cannot back them out for us: *A/R* fell by \$80k, and that \$80k could be \$80k of collections and no writeoffs (giving a *Bad Debt Expense* of $-\$8k$), or \$80k of writeoffs and no collections (giving \$72k), or any mix in between. The *Allowance* method distinguishes those two cases sharply—that is the entire virtue claimed for it—and the problem refuses to tell us which one we are in.

    Contrast item 12, which handed us the writeoffs and withheld the sales and collections. Harmless, because those two only ever mattered *through* ΔAR, which we were given. Here, the missing quantity enters the answer on its own. So the answer is **(D)**.

14. Our company uses the Allowance method, and estimates the Allowance using the *aging-of-receivables* method. We began FY2020 with a balance in the *Allowance* of \$20k. We end FY2020 with \$50k in Accounts outstanding less than 30 days, and \$40k in Accounts outstanding 30 days or more. We assume 10% of the former

category and 15% of the latter category will be uncollectible. What will be our *Net Accounts Receivable* for FY2020?

(A) $2k

(B) $3k

(C) $4k

(D) Not enough information.

(E) None of the other answers.

Solution. Read the question: it asks for *Net Accounts Receivable*, which is the **easy** half of an aging problem. The *Bad Debt Expense* is the hard half, and it is not being asked—which is why the $20k beginning *Allowance* is sitting there doing nothing.

Total *Accounts Receivable* is $\$50k + \$40k = \$90k$. The estimated uncollectible amount is the "dot product" of the buckets and their percentages: $(\$50k * .10) + (\$40k * .15) = \$5k + \$6k = \$11k$. That $11k *is* the required ending balance of the *Allowance*. So $Net\ Accounts\ Receivable = \$90k - \$11k = \$79k$.

Not among the numerical options, and we had everything we needed. So the answer is **(E)**.

15. Our company accounts for bad debts / receivables using the **percentage-of credit-sales method**, assuming 15% of credit sales will be uncollectible. We began FY2020 with a balance of $150k in Accounts Receivable and $30k in our Allowance for Doubtful Accounts. During FY2020, we made an additional $100k in credit sales, and collected $60k on accounts from past credit sales. Customers with accounts totaling $20k declared bankruptcy. What would be our FY2020 Bad Debt Expense?

(A) $11k

(B) $15k

(C) $28.5k

(D) Not enough information.

(E) None of the other answers.

Solution. Percentage-of-credit-sales, so: 15% of this period's credit sales, and nothing else. $15\% * \$100k = \$15k$.

Option (C) is the instructive distractor, because you can see exactly how someone lands on it. Ending *A/R*, if you go computing it, is $\$150k + \$100k - \$60k = \$190k$

before the bankruptcies, and 15% of \$190k is \$28.5k—the right percentage applied to the wrong base, which is to say, the *Allowance* method's base smuggled into the income-statement method's formula. That confusion is the single most common error on this topic, and every extraneous figure in the stem is there to invite it. So the answer is **(B)**.

16. Our company accounts for bad debts / receivables using the **allowance method**, assuming 10% of outstanding accounts will be uncollectible. We began FY2020 with a balance of \$50k in Accounts Receivable and \$5k in our Allowance for Doubtful Accounts. During FY2020, we made an additional \$30k in credit sales, collected \$20k on accounts from past credit sales, and wrote off \$3k in accounts. What would be our FY2020 *Bad Debt Expense*?

    (A) \$2k

    (B) \$3k

    (C) \$4k

    (D) Not enough information.

    (E) None of the other answers.

    Solution. Ending *Accounts Receivable* is $\$50k + \$30k - \$20k - \$3k = \$57k$. So the *Allowance* must end at 10% of \$57k, or \$5.7k. It began at \$5k, and the \$3k writeoff drew it down to \$2k. The adjusting entry must therefore credit $\$5.7k - \$2k = \$3.7k$, and the *Bad Debt Expense* is **\$3.7k**:

Bad Debt Expense (-S.E.):	\$3.7k	
Allowance for Doubtful Accounts (+contra-A):		\$3.7k

12/31/2020 Adjusting Entry to update the Allowance

    Formula check: $.1 * (\$30k - \$20k) + .9 * (\$3k) = \$1k + \$2.7k = \$3.7k$. Option (B) is the trap for readers who see "wrote off \$3k" and reach for it—but under this method the writeoff is never the expense; it only shapes the expense indirectly, through the *Allowance*. So the answer is **(E)**.

17. Under the Allowance method, all else equal, the greater our collections from past period's credit sales during this period, the [higher / lower / no change] will be our Bad Debt Expense in this period?

    (A) Higher

    (B) Lower

(C) Unchanged

(D) Not enough information.

(E) None of the other answers.

Solution. Now the method has changed, and so does the answer. Collections reduce ending *A/R*, which reduces the required ending *Allowance*, which shrinks the credit needed at 12/31, which shrinks the matching debit to *Bad Debt Expense*. Formally, the coefficient on $Collections_t$ in formula (8.7) is $-p$. This is the *Allowance* method delivering on Goal #2: good news about past accounts actually reaches the income statement. So the answer is **(B)**.

18. Under the Allowance method, all else equal, the lower our collections from past period's credit sales during this period, the [higher / lower / no change] will be our Bad Debt Expense in this period?

(A) Higher

(B) Lower

(C) Unchanged

(D) Not enough information.

(E) None of the other answers.

Solution. Item 19 with the sign flipped on the input, so flip the sign on the output. Fewer collections means more accounts still outstanding at year end, a larger required *Allowance*, a larger adjusting credit, and a larger *Bad Debt Expense*. So the answer is **(A)**.

19. Under the Allowance method, all else equal, the greater our writeoffs of past period's credit sales during this period, the [higher / lower / no change] will be our Bad Debt Expense in this period?

(A) Higher

(B) Lower

(C) Unchanged

(D) Not enough information.

(E) None of the other answers.

Solution. Higher—and this is the one worth slowing down on, because the mechanism is genuinely indirect. The writeoff itself never touches *Bad Debt Expense*; it debits the *Allowance*. But that debit "overdraws" the *Allowance* relative to where it needs to end up, and the 12/31 adjusting entry has to credit

it back, with *Bad Debt Expense* as the matching debit. So the writeoff's effect on the expense arrives one step later, through the back door.

Formula (8.7) prices it exactly: the coefficient on writeoffs is $(1-p)$. Why $(1-p)$ rather than the full dollar? Because a writeoff of \$1 pulls \$1 out of the provision, which has to be replenished—but it also removes \$1 from *A/R*, which lowers the required ending *Allowance* by p. Net replenishment required: $1-p$. So the answer is **(A)**.

20. Under the Allowance method, all else equal, the lower our writeoffs of past period's credit sales during this period, the [higher / lower / no change] will be our Bad Debt Expense in this period?

    (A) Higher

    (B) Lower

    (C) Unchanged

    (D) Not enough information.

    (E) None of the other answers.

    Solution. Item 21 run backwards: fewer writeoffs means less of the *Allowance* gets used up during the period, so less has to be replenished at 12/31, so a smaller *Bad Debt Expense*. So the answer is **(B)**.

21. Trojan Company uses the Balance Sheet / Allowance Method for accounting for bad debts, assuming that 14% of outstanding receivables will not be collectible. The balance of Allowance for Uncollectible Accounts is \$380 (credit) before adjustment on December 31, 2019. The balance of Accounts Receivable is \$34,000. What is the adjusting entry that Trojan Company should record for Allowance for Uncollectible Accounts?

    (A) Debit Allowance for Uncollectible Accounts 4,380; Credit Bad Debt Expense 4,380

    (B) Debit Bad Debt Expense 5,140; Credit Allowance for Uncollectible Accounts 5,140

    (C) Debit Allowance for Uncollectible Accounts 5,140; Credit Bad Debt Expense 5,140

    (D) Debit Bad Debt Expense 4,380; Credit Allowance for Uncollectible Accounts 4,380

    (E) None of the answers are correct

Solution. This is a problem of the type the chapter warned about at the end: you are handed the *pre-adjustment* balance of the *Allowance* and the ending *A/R*, and no sales, collections, or writeoffs at all. The formula is useless here. Go straight to the definition.

Required ending balance: $.14 * \$34{,}000 = \$4{,}760$ (credit). Pre-adjustment balance: \$380 (credit). The adjustment is whatever gets us from one to the other, $\$4{,}760 - \$380 = \$4{,}380$, as a credit—and the matching debit is the *Bad Debt Expense*, by definition.

Bad Debt Expense (-S.E.):	\$4,380	
Allowance for Uncollectible Accounts (+contra-A):		\$4,380

12/31/2019 Adjusting Entry, Trojan Company

Now the distractors. The \$5,140 in options (B) and (C) is $\$4{,}760 + \$380$—what you get if you read the existing \$380 as a *debit* balance and therefore have to climb from $-\$380$ all the way up to \$4,760. The stem says "(credit)" for exactly this reason: it is telling you which side of the ledger you are starting from, and that is the whole question. Options (A) and (C) reverse the debit and credit, which would *reduce* the *Allowance* and record a negative *Bad Debt Expense*. That is a real phenomenon under this method—but not this year, since our required balance sits well above where we currently are. So the answer is **(D)**.

A.8.2 Practice Problems: Open Prompts

1. Journalize the following transactions—both the external transactions and the implied related adjusting entries at the end of the reporting period—using the **percentage-of-credit sales** method, assuming 15% of credit sales will be written off. (Assume, as usual and by default, annual calendar-year reporting)

    - February 15, 2022: Sell to Customer A for \$100k in cash.
    - March 28, 2022: Sell to Customer B for \$10k on account.
    - September 25, 2022: Sell to Customer C for \$120k on account.
    - November 11, 2022: Customer B pays off their account in full.
    - December 31, 2022: End of the year, close the books!
    - February 15, 2023: Customer C pays off their account in full.
    - May 11, 2023: Sell to Customer D for \$5k on account.

- September 11, 2023: Sell to Customer E for $80k on account.
- November 11, 2023: Customer D declares bankruptcy—we realize we'll never collect on their account.
- December 31, 2023: End of the year, close the books!

Solution. Under this method the external transactions are all reassuringly familiar—Rule #1 says account for credit sales and collections exactly as we always have—and the only real work is the December 31 adjusting entry, which is 15% of *the credit sales made during that period.* Two things to keep your eye on: the February 15, 2022 sale to Customer A is a **cash** sale, so it never enters the base; and Customer D's bankruptcy in 2023 gets **no entry at all**, per Rule #3.

FY 2022.

Cash (+A):	$100k	
Revenue (+S.E.):		$100k

2/15/2022 Cash Sale to Customer A

Accounts Receivable (+A):	$10k	
Revenue (+S.E.):		$10k

3/28/2022 Credit Sale to Customer B

Accounts Receivable (+A):	$120k	
Revenue (+S.E.):		$120k

9/25/2022 Credit Sale to Customer C

Cash (+A):	$10k	
Accounts Receivable (-A):		$10k

11/11/2022 Customer B Pays Off Their Account

Credit sales during 2022 were $\$10k + \$120k = \$130k$. The $100k cash sale is excluded, because this method keys off *credit* sales—and a customer who has already handed us the cash cannot fail to pay us. So the adjusting entry is $15\% * \$130k = \$19.5k$:

Bad Debt Expense (-S.E.):	$19.5k	
Allowance for Doubtful Accounts (+contra-A):		$19.5k

12/31/2022 Adjusting Entry for Expected Bad Debts

At the end of FY2022, *A/R* stands at $120k and the *Allowance* at $19.5k, so *Net Accounts Receivable* is $100.5k.

FY 2023.

Cash (+A):	$120k	
Accounts Receivable (-A):		$120k

2/15/2023 Customer C Pays Off Their Account

Accounts Receivable (+A):	$5k	
Revenue (+S.E.):		$5k

5/11/2023 Credit Sale to Customer D

Accounts Receivable (+A):	$80k	
Revenue (+S.E.):		$80k

9/11/2023 Credit Sale to Customer E

November 11, 2023: Customer D declares bankruptcy. **No journal entry.** Under the percentage-of-credit-sales method there is effectively no such thing as a writeoff—we already allowed for expected losses, and we do not revisit that guess when the news arrives. It feels wrong to leave a dead $5k receivable sitting on the books, and it *is* the thing that is wrong with this method. But it is the method.

Credit sales during 2023 were $\$5k + \$80k = \$85k$, so the adjusting entry is $15\% * \$85k = \$12.75k$:

Bad Debt Expense (-S.E.):	$12.75k	
Allowance for Doubtful Accounts (+contra-A):		$12.75k

12/31/2023 Adjusting Entry for Expected Bad Debts

Where does that leave us? *A/R* has accumulated to $\$10k + \$120k + \$5k + \$80k - \$10k - \$120k = \$85k$, including Customer D's worthless \$5k. The *Allowance* has accumulated to $\$19.5k + \$12.75k = \$32.25k$, and never gets drawn down. So *Net Accounts Receivable* is \$52.75k. Hold onto that number—the next prompt runs the same facts through the *Allowance* method, and comparing the two is the whole point of the pairing.

2. Now, journalize the same transactions—both the external transactions and the implied related adjusting entries at the end of the reporting period—using the **Allowance Method**, assuming 10% of outstanding accounts will not be collected.

   Solution. The external transactions are identical—Rule #1 is the same rule under both methods. Two things change: on 11/11/2023 we now *do* something (Rule #2, the writeoff), and each December 31 we adjust the *Allowance* to a target set by the **outstanding balance** of *A/R* rather than by the period's credit sales (Rule #3).

   FY 2022.

Cash (+A):	\$100k	
Revenue (+S.E.):		\$100k

2/15/2022 Cash Sale to Customer A

Accounts Receivable (+A):	\$10k	
Revenue (+S.E.):		\$10k

3/28/2022 Credit Sale to Customer B

Accounts Receivable (+A):	\$120k	
Revenue (+S.E.):		\$120k

9/25/2022 Credit Sale to Customer C

Cash (+A):	\$10k	
Accounts Receivable (-A):		\$10k

11/11/2022 Customer B Pays Off Their Account

Now the 12/31/2022 adjustment. Outstanding *A/R* is $\$10k + \$120k - \$10k = \$120k$—only Customer C's account is still open. So the *Allowance* must **end** at 10% of \$120k, or \$12k. Nothing has touched the *Allowance* yet, so it sits at \$0, and the plug is a \$12k credit:

Bad Debt Expense (-S.E.):	\$12k	
Allowance for Doubtful Accounts (+contra-A):		\$12k

12/31/2022 Adjusting Entry to update the Allowance

Net Accounts Receivable at the end of FY2022 is $\$120k - \$12k = \$108k$.

FY 2023.

Cash (+A):	\$120k	
Accounts Receivable (-A):		\$120k

2/15/2023 Customer C Pays Off Their Account

Accounts Receivable (+A):	\$5k	
Revenue (+S.E.):		\$5k

5/11/2023 Credit Sale to Customer D

Accounts Receivable (+A):	\$80k	
Revenue (+S.E.):		\$80k

9/11/2023 Credit Sale to Customer E

Now Customer D goes bankrupt. Under *this* method we write the account off—extinguish the receivable, and draw down the provision we had already set aside for exactly this sort of event. No expense today:

Allowance for Doubtful Accounts (-contra-A):	\$5k	
Accounts Receivable (-A):		\$5k

11/11/2023 Write Off Customer D

Finally, 12/31/2023. It is worth drawing the T-accounts *before* the adjusting entry, so you can see where both accounts actually stand:

Accounts Receivable (A)

Debits	Credits
3/28/22: $10k	11/11/22: $10k
9/25/22: $120k	2/15/23: $120k
5/11/23: $5k	11/11/23: $5k
9/11/23: $80k	
Bal: $80k	

Allowance (contra-A)

Debits	Credits
11/11/23: $5k	12/31/22: $12k
	Bal: $7k

Outstanding *A/R* is $80k—Customer E's account, and nothing else—so the *Allowance* must end at $8k. It currently sits at $7k: the $12k from last year's adjustment, less the $5k the writeoff consumed. So the adjusting entry is a $1k credit:

Bad Debt Expense (-S.E.):	$1k	
Allowance for Doubtful Accounts (+contra-A):		$1k

12/31/2023 Adjusting Entry to update the Allowance

Checking against formula (8.7), with 2023's credit sales of $85k, collections of $120k, and writeoffs of $5k: $.1 * (\$85k - \$120k) + .9 * (\$5k) = -\$3.5k + \$4.5k = \$1k$. Agreed.

Now compare the two methods on the same facts. *Net Accounts Receivable* at the end of FY2023 is $\$80k - \$8k = \$72k$ here, versus $52.75k under the income-statement method—and the $72k is plainly the honest number, since the only account still open is Customer E's $80k and we expect 10% of it to go bad. That is the *Allowance* method's Goal #2 virtue, visible in one line. But look at what it cost on the income statement. FY2023's *Bad Debt Expense* is a mere $1k despite $85k of new credit sales, because a large positive surprise (Customer C paid in full, having been provisioned against) nearly offset a negative one (Customer D did not),

and both got dumped into a line that is supposed to be about *this* year's selling. Under the income-statement method the FY2023 charge was a clean \$12.75k—15% of \$85k, meaning one thing and one thing only. Neither column is free. That tradeoff is the chapter.[1]

3. Our company accounts for bad debts / receivables using the *percentage-of-credit-sales method*, assuming 10% of credit sales will be uncollectible. We began FY2020 with a balance of \$100k in *Accounts Receivable* and \$20k in our *Allowance for Doubtful Accounts*. During FY2020, we made an additional \$50k in credit sales, and collected \$40k on accounts from past credit sales. Customers with accounts totaling \$10k declared bankruptcy. What would be our FY2020 *Bad Debt Expense*?

   Solution. $10\% * \$50k = \$5k$, full stop. Everything else is a distraction.

   I mean that literally, and it is worth being blunt about why this problem exists. Under the percentage-of-credit-sales method, the *Bad Debt Expense* requires exactly two inputs: this period's credit sales, and the assumed percentage. You have both in the first sentence. The beginning *A/R*, the beginning *Allowance*, the collections, and the bankruptcies are four separate invitations to lose your nerve and start "adjusting" for things this method tells you to ignore.

Bad Debt Expense (-S.E.):	\$5k	
Allowance for Doubtful Accounts (+contra-A):		\$5k

12/31/2020 Adjusting Entry for Expected Bad Debts

4. Our company accounts for bad debts / receivables using the *allowance method*, assuming 10% of outstanding accounts will be uncollectible. We began FY2020 with a balance of \$100k in *Accounts Receivable* and \$10k in our *Allowance for Doubtful Accounts*. During FY2020, we made an additional \$50k in credit sales, and collected \$40k on accounts from past credit sales, and wrote off \$5k in accounts. What would be our FY2020 *Bad Debt Expense*?

   Solution. Nearly the same numbers as the previous prompt, and a completely different answer, because the method changed.

   We started with \$100k in *A/R*, then made an additional \$50k in credit sales, collected \$40k, and wrote off \$5k. So our ending balance of *A/R* is $\$100k + \$50k -$

[1] The two methods use different assumed percentages here—15% of credit *sales* versus 10% of outstanding *accounts*—which is not sloppiness on the problem's part. As the chapter notes, those are substantively different assumptions about different quantities, one a flow and one a stock, and there is no reason the two numbers should coincide.

$\$40k - \$5k = \$105k$. So our ending balance of the *Allowance* must be $10\% * \$105k = \$10.5k$. Our beginning balance of the *Allowance* at the start of the period *was* \$10k, but we "used up" \$5k of that in the writeoff. So, prior to adjustment, the balance of the *Allowance* is \$5k. So the adjustment to the *Allowance* must be \$5.5k ($\$10.5k - \$5k$), which is our *Bad Debt Expense* by definition.

Bad Debt Expense (-S.E.):	\$5.5k	
Allowance for Doubtful Accounts (+contra-A):		\$5.5k

12/31/2020 Adjusting Entry to update the Allowance

And the formula agrees: $.1 * (\$50k - \$40k) + .9 * (\$5k) = \$1k + \$4.5k = \$5.5k$. Again, I solved this in words, but I recommend you track your work with T-accounts.

5. Under the Allowance method, when we write off an account worth \$10k, what is the correct journal entry?

   Solution.

Allowance for Doubtful Accounts (-contra-A):	\$10k	
Accounts Receivable (-A):		\$10k

Writeoff of a \$10k Account under the Allowance Method

The customer is never going to pay, so the receivable comes off the books. That is the credit to *A/R*, and it is the obvious half. The debit is the half worth explaining. The *Allowance* is a **provision**: last period we set aside a bucket of expected losses, saying "about this much of our receivables will eventually go bad." Now a specific loss has materialized, of exactly the kind we were provisioning for—so we draw down the bucket. And since the *Allowance* is a contra-asset, a credit-balance account, drawing it down is a debit.

Note what is absent: *Bad Debt Expense*. Nothing here touches the income statement. That expense was recognized *in advance*, when we built the provision, and it will be topped back up at the next adjusting entry. Writing off an account under this method is not a moment of loss recognition; it is a moment of bookkeeping cleanup.

6. Our company uses the Allowance method, and estimates the Allowance using the *aging-of-receivables* method. We began FY2020 with a balance in the *Allowance*

of $20k. We end FY2020 with $50k in Accounts outstanding less than 30 days, and $40k in Accounts outstanding 30 days or more. We assume 10% of the former category and 15% of the latter category will be uncollectible. What will be our *Net Accounts Receivable* for FY2020?

Solution. *Total* Accounts Receivable is $90k. Of the $50k outstanding less than 30 days, we assume 10%, or $5k total, is uncollectible; of the $40k outstanding more than 30 days, we assume 15%, or $6k total, is uncollectible. So $11k is uncollectible. ***Net*** *Accounts Receivable* is $\$90k - \$11k = \$79k$.

The aging method changes nothing about the *Allowance* method itself. It is just a better-informed way of arriving at the required balance of the *Allowance*—a "dot product" of buckets and percentages instead of one flat rate—and everything downstream of that number is unchanged. Note also that the $20k beginning balance plays no role in *this* question. It would matter if you were asked for the *Bad Debt Expense*, which is always the harder half: the *Allowance* must end at $11k, and to know the size of the adjustment you would also need the writeoffs during the period, to see where the balance stood before adjustment.

A.8.3 Essay and Discussion Questions

1. Compare and contrast the Percentage-of-Credit-Sales Method and the Allowance Method for accounting for receivables. What are the strengths and limitations of each approach? Why is one of them called the "Income Statement method" and the other called the "Balance Sheet method"? Which do you think is the better approach overall?

   Solution. Here is how I would think it through.

   Start with what the two methods have in common, because they share a commitment that the *Direct Writeoff Method* abandons. Both accept that when we sell on account, we know *today* that some of those sales will never turn into cash, and that this expected loss belongs in the period of the sale. That is Goal #1, and both methods achieve it. Where they part company is Goal #2: what to do when reality subsequently disagrees with our estimate.

   The **percentage-of-credit-sales** method computes the *Bad Debt Expense* directly, as a fixed percentage of the period's credit sales, and credits that amount to the *Allowance*. The *Allowance* balance is then whatever it happens to be—a byproduct. Writeoffs are ignored entirely. It is the "make our best guess and keep moving" method.

   The **Allowance** method inverts the direction of the calculation. It fixes the required *ending balance of the Allowance* first, from the outstanding receivables,

and then the *Bad Debt Expense* is whatever adjustment gets us there. Now the *expense* is the byproduct. Writeoffs matter, because they draw down the *Allowance* during the period and thereby change the size of the adjustment needed at year end.

That inversion is the entire explanation for the nicknames. Under the first method, the number we compute deliberately and directly is an income-statement number—a flow, matched to the same period's flow of sales—so it is the **Income Statement method**. Under the second, the number we compute deliberately and directly is a balance-sheet number, the *Allowance* balance, and hence *Net Accounts Receivable*—so it is the **Balance Sheet method**. In each case the *other* statement gets whatever falls out, through the linkages we know well by now.

Now the tradeoff, and I do not think there is a way to escape it. The *Allowance* method keeps *Net Accounts Receivable* honest: the line means what it is supposed to mean, the amount we currently expect to collect. The cost is that the *Bad Debt Expense* becomes a mongrel. In any given year it mixes together expected losses on *this* year's sales and revisions of estimates about *prior* years' receivables. A perfectly healthy company—revenues stable, margins stable, operations humming—can report a bad year because one large old customer went under. It can even report a *negative* expense, boosting net income, when old accounts come in better than expected. That is not hypothetical; it happens.

The percentage-of-credit-sales method gives the mirror image. Its *Bad Debt Expense* means exactly one thing, is proportional to this period's selling, and is therefore interpretable and forecastable. The cost is that *Net Accounts Receivable* drifts, because the *Allowance* accumulates estimates that reality has already contradicted, with no mechanism for correction.

Which is better? I lean toward the income-statement method, and the reasoning comes out of valuation. When you buy a share of stock you are buying a claim on future profits, so you are implicitly forecasting them—and you forecast them by anchoring on current operating performance, which is empirically persistent. That makes a clean, single-meaning measure of *this* period's profitability the most valuable thing an income statement can give you. Meanwhile the balance-sheet cost strikes me as mild: if our percentage is well calibrated, the errors run in both directions and wash out over time, and total assets are never wrong anyway—only the composition between *Cash* and *Net A/R*.

I hold that view with real humility, for two reasons. First, most major U.S. corporations use the *Allowance* method, and the standard setters have marched steadily in the balance-sheet direction for decades—in receivables, in pensions,

and elsewhere. A great many serious people disagree with me. Second, I have changed my own mind on this every time I have taught it. The honest summary is that this is a genuine tradeoff rather than a puzzle with a hidden right answer—and a good essay here is one that names the tradeoff precisely and *then* commits to a side, with reasons.

2. Under the *Allowance Method*, when we "write off" an account during the reporting period, we don't debit *Bad Debt Expense* right away. And yet, all else equal, the higher our writeoffs during the period, the higher our ultimate *Bad Debt Expense* will be. Explain how this can be, and precisely how this works.

   Solution. The apparent paradox dissolves once you see that under this method the *Bad Debt Expense* is never recorded *directly*. It is always a residual—the plug that forces the *Allowance* to the balance it needs to have. So the right question is not "does the writeoff create an expense?" (it does not) but "does the writeoff change the size of the plug?" (it does).

   Walk it through in three moments, and keep them rigidly separate, because conflating them is the whole difficulty here.

   The writeoff date. We credit *A/R* and debit the *Allowance*. Two balance-sheet accounts; nothing crosses over to the income statement. Conceptually, we are drawing down a provision that was already set aside for exactly this kind of event. The loss was recognized in advance, back when we built the provision; recognizing it again now would be double-counting it.

   The state of the *Allowance* afterward. The debit leaves the *Allowance* lower than it was—"overdrawn," relative to where it will need to sit at year end.

   The 12/31 adjusting entry. Now we ask what the *Allowance* must end at: p times ending *A/R*. We compare that target to the depleted pre-adjustment balance, and the gap is the credit we must post, with *Bad Debt Expense* as the matching debit, by definition. Because the writeoff dug the hole deeper, the credit needed to climb out of it is larger. That is the mechanism, whole: the writeoff's effect on the expense arrives one step later, through the back door of the adjusting entry.

   Now the precise part, because "higher writeoffs, higher expense" understates what we can actually say. Read it straight off the general-form T-account:

Allowance for Doubtful Accounts (contra-A)

$Writeoffs_t$	Beg. Bal: $(p * AR_{t-1})$
	12/31: $Bad\ Debt\ Expense_t$
	End. Bal.: $(p * (AR_t))$

Beginning balance, less writeoffs, plus the adjustment, equals the ending balance—which rearranges to $Bad\ Debt\ Expense_t = p * (\Delta AR) + Writeoffs_t$ and, decomposing ΔAR into its underlying events, to formula (8.7):

$$Bad\ Debt\ Expense_t = p * (Credit\ Sales_t - Collections_t) + (1-p)Writeoffs_t$$

So the sensitivity of the expense to a dollar of writeoffs is exactly $(1-p)$. Not a full dollar, and not zero. Why? Because the writeoff does two things at once. It pulls $1 out of the provision, which has to be replenished. But it also removes $1 from *A/R*, which lowers the required ending *Allowance* by p. Net replenishment required: $1-p$. Notice the limiting case: if we had assumed $p = 100\%$—that every receivable was doomed—writeoffs would cost us nothing at year end, because they would merely be confirming what we had already fully provided for.

Step back, and this is the *Allowance* method's deepest feature rather than a quirk of its bookkeeping. Writeoffs *are* the bad news: reality informing us that our prior estimate was too rosy. This method lets that news reach the income statement, in the period in which the news arrives. That is Goal #2, achieved. Whether it is a *good* thing for news about last year's sales to land in this year's profit is the debate in the previous essay. But the machinery by which it happens is precisely this.

A.9 Chapter 9

A.9.1 Practice Problems: MCQ's

1. From our company's perspective, the interest we owe, prior to the date on which we actually pay it, is a(n):

   (A) accrued revenue

   (B) deferred revenue

   (C) accrued expense

   (D) deferred expense

   (E) None of the other answers

   Solution. Sort this with the two questions we always ask: is it ours to *pay* or ours to *receive*, and has the cash moved yet? Interest we *owe* is a cost to us, so it is on the expense side—which knocks out (A) and (B) immediately, since those are revenue-side items and would be the *lender's* view of the same contract, not ours.

   Then, timing. **Accrued** means the economic event has happened but the cash has not yet moved; **deferred** means the cash moved first and the economic event comes later. Interest accretes continuously with the passage of time—we have had the use of the borrowed money every day since the loan was made—but we typically pay it only on the stated payment dates. So on any date before payment, we have incurred the cost without paying it. That is an **accrued expense**, and it is why the December 31 adjusting entry for a loan is a debit to *Interest Expense* and a credit to *Interest Payable*: the expense is real, and the matching credit is a liability, not a cash outflow.

   Option (D), a deferred expense, would be the reverse pattern—cash out today for a benefit consumed later, like prepaid rent. So the answer is **(C)**.

2. Which of the below is the best characterization of what corporate bankruptcy is?

   (A) the death of the corporation

   (B) a legal process for publicly shaming and condemning bad companies, so that they cannot continue to do their bad stuff

   (C) the process by which the government steps in and pays the debts of corporations

   (D) the legal process for deciding what to do (and especially, reorganizing) when a corporation simply cannot meet its obligations

Solution. Bankruptcy is, in the United States, mostly a technical legal proceeding for determining **which creditors get paid how much** when a company cannot meet *all* of its legal obligations—and, under Chapter 11, for negotiating "The Plan" by which the company reorganizes itself so as to pay back as much as is feasible. So the answer is **(D)**.

Option (A) is worth dwelling on, because it is not merely wrong—it is the specific popular misconception that this chapter exists to dislodge. "Corporate death" is an inapt and misleading metaphor. Companies keep operating in bankruptcy: they keep running their factories, keep selling, keep paying employees. American Airlines spent 2011 to 2014 in bankruptcy court, and if you flew American in those years you almost certainly never noticed. Indeed, the *point* of filing for bankruptcy **protection** is to stop creditors from seizing collateral, precisely so that the company *can* continue operating—because a company whose assets have been repossessed generates no cash flow, which makes every creditor worse off.

Getting this right has real consequences. The 2020 argument that the airlines had to be "bailed out" so that they could continue providing essential transportation was a **non sequitur**, and you can only see that it is a non sequitur if you know that (A) is false. Option (C) is a description of a bailout, which is a *substitute* for bankruptcy, not a description of it. And option (B) is not how any of this works: bankruptcy court is not a morality play, and being unable to pay is not the same as being wicked. Who actually loses in a classic bankruptcy? The **equity investors**—the residual claimants—who get wiped out. Which is a useful thing to keep in mind when you are listening to executives with large stock holdings explain why bankruptcy must be avoided at all costs.

A.9.2 Practice Problems: Open Prompts

1. Suppose that on September 1, 2022, our company receives a $10,000 loan, with an 8% interest rate, paid annually, and a 3-year maturity, from a bank. Journalize each external transaction and adjusting entry, over the entire life of this loan (that is, until the maturity date on which we pay down the principal and face value).

   Solution. First, get the shape of the thing before touching a single journal entry. *We* are the borrower here, so this is a **Note Payable** for us—the mirror image of the chapter's *Notes Receivable* example. The face value is $10,000, the annual interest payment is $8\% * \$10{,}000 = \$800$, and a three-year maturity from September 1, 2022 means the principal comes due on **September 1, 2025**.

   Now, the calendar. Interest is *paid* every September 1—in 2023, 2024, and 2025—

but our reporting year ends every December 31. Those two calendars do not line up, and that misalignment is the whole substance of the problem. Each September-to-December stub is **four months**, and each January-to-September stub is **eight months**:

- Accrued at each 12/31: $\$10{,}000 * 8\% * \frac{4}{12} = \$266.67$
- Accrued between 1/1 and each 9/1: $\$10{,}000 * 8\% * \frac{8}{12} = \$533.33$

Those two sum to the $800 actually paid each year, as they must. Here is the full life of the loan.

Cash (+A):	$10,000	
Note Payable (+L):		$10,000

Sep. 1, 2022: We receive the $10,000 loan

Interest Expense (-S.E.):	$266.67	
Interest Payable (+L):		$266.67

Dec. 31, 2022: adjusting entry, **four months of *accrued* interest**

The September 1, 2023 payment is the interesting one, and it is the same three-part structure you will see twice more—the second time with the principal bolted onto it. **Start with the Cash**: $800 goes out the door, so credit *Cash* $800. What is that $800 settling? Part of it—$266.67—is a liability we already put on the books at the last adjusting entry, so we **extinguish** the *Interest Payable*. The rest—$533.33—is interest that accrued over the eight months since that entry and has never been recorded at all, so it becomes *Interest Expense* now.

Interest Payable (-L):	$266.67	
Interest Expense (-S.E.):	$533.33	
Cash (-A):		$800

Sep. 1, 2023: First annual interest payment

Interest Expense (-S.E.):	$266.67	
Interest Payable (+L):		$266.67

Dec. 31, 2023: adjusting entry, four months of accrued interest

Interest Payable (-L):	$266.67	
Interest Expense (-S.E.):	$533.33	
Cash (-A):		$800

Sep. 1, 2024: Second annual interest payment

Interest Expense (-S.E.):	$266.67	
Interest Payable (+L):		$266.67

Dec. 31, 2024: adjusting entry, four months of accrued interest

Finally, maturity. This is the four-part entry, and it is just the September 1 payment entry with the principal bolted on. **Start with the Cash**: we pay the last $800 of interest *and* the $10,000 face value, so $10,800 leaves. Then account for what that settles: the $266.67 *Interest Payable* accrued at the last 12/31 is extinguished; $533.33 of new interest expense is recognized for the final eight months; and the $10,000 *Note Payable* is extinguished, because the borrower—us—has now fully discharged the obligation.

Note Payable (-L):	$10,000	
Interest Payable (-L):	$266.67	
Interest Expense (-S.E.):	$533.33	
Cash (-A):		$10,800

Sep. 1, 2025: Final interest payment and repayment of principal

Two things to notice, now that it is all laid out. First, the *Interest Payable* account returns to zero at every payment date and finishes at zero—as it must, since we have paid everything we owed. Second, total interest expense over the life of the loan is $\$266.67 + \$533.33$, three times over, or $2,400—exactly the three $800

payments. Accruals and cash flows reconcile in the end. They always do; that is the point of accrual accounting, not an accident of it.[2]

A.9.3 Essay and Discussion Questions

1. In your opinion, does it ever make sense for the government to "bail out" companies that are at risk of bankruptcy? If not, why? If so, when and why?

   Solution. Here is how I would think it through.

   Begin by disposing of the bad argument, because it dominates the public conversation and it is simply invalid. The claim goes: this company provides an essential service and employs thousands of people, therefore it must be rescued from bankruptcy. That is a **non sequitur**. Companies continue operating in bankruptcy—that is the entire purpose of bankruptcy *protection*, which stops creditors from seizing the collateral the company needs in order to keep running. American Airlines flew planes throughout its 2011–2014 bankruptcy. So "the planes must keep flying" is not, by itself, an argument for a bailout; it is an argument for the bankruptcy system we already have.

   Then ask the question the chapter trains you to ask about any business situation: **who has what power, and what do those people want?** In a classic bankruptcy, the equity holders—the residual claimants—get wiped out. Creditors take a haircut but generally recover something. Employees mostly keep working. So the party with the most to lose, and the loudest incentive to lobby, is precisely the party whose losses have the weakest public-interest claim: shareholders, prominently including executives holding large stock-based compensation. When you hear a bailout argued for on the grounds of jobs and essential services, it is worth checking whose balance sheet the money would actually land on.

   So: is a bailout *ever* justified? I think yes, but on narrower grounds than are usually offered, and the grounds have to be about something bankruptcy court genuinely cannot handle. Two cases seem real to me.

   The first is **systemic** failure. Bankruptcy is a court proceeding designed for one firm at a time. It is slow, it is negotiated case by case, and it presumes the surrounding economy is functioning well enough for the reorganized firm to be worth reorganizing. When a large fraction of the economy is failing *simultaneously*—2008, the early months of 2020—the machinery does not scale, and the failures propagate: my default becomes your uncollectible receivable becomes

[2] The $266.67 and $533.33 are rounded thirds of $800. If a problem's figures are ugly this way, do not let it rattle you—just keep the two stubs summing to the full annual payment, and the accounts will close out cleanly.

your default. That is a genuine externality, and it is not one that any individual bankruptcy court is positioned to price. Note that this is an argument about *correlation*, not about size or sympathy.

The second is a **pure liquidity shock with no solvency problem underneath**: a firm whose business is sound but whose revenue has been extinguished by an exogenous event—a government-ordered shutdown, say—that everyone expects to reverse. Here a *loan* on hard terms can be defensible, since the government is doing what a credit market would do if credit markets were functioning. But the word doing the work is *loan*. Terms matter enormously, and if the taxpayer is bearing the downside risk, the taxpayer should get the upside too: senior claims, warrants or equity stakes, restrictions on dividends and buybacks and executive compensation while the money is outstanding. A bailout structured as a gift to shareholders is a transfer; a bailout structured as an expensive, senior, dilutive loan is closer to insurance.

And the cost side of the ledger is not small. Bailouts create **moral hazard**: if the downside is socialized, the rational move is to run thin balance sheets and heavy debt loads, because leverage is free money when someone else absorbs the tail. That is not hypothetical—several of the airlines seeking rescue in 2020 had spent the preceding years converting cash into share buybacks rather than into a buffer. A bailout regime that is expected will *change behavior in advance*, and generally not for the better. Bankruptcy, by contrast, is the mechanism that makes credit risk real, and credit risk being real is what makes lenders do the work of scrutinizing borrowers.

So my position, stated plainly: bailouts are justified rarely, on systemic or genuinely exogenous-liquidity grounds rather than on "essential services" grounds, and when they happen they should be structured as expensive senior credit that leaves the existing equity holders bearing their losses. The default answer should be: let it go to bankruptcy court. That is what the court is for. And if you find that answer harsh, notice that it is harsh mainly to the shareholders—which is exactly the group that signed up, back in Chapter 1, to be the residual claimant in exchange for the residual upside.

A.10 Chapter 10

Operating-income convention for the exercises below: Where a question separates a disposal Gain or Loss from operating costs, assume that the disposed asset was held as a nonoperating investment. This assumption preserves the stated comparison with EBIT; a disposal gain or loss on an operating asset need not be excluded from GAAP Operating Income.

A.10.1 Practice Problems: MCQ's

1. Consider the following scenario: In FY2020, MarshallCo starts its life, and has the following events related to the purchase and sale of *Inventory*. January 1: Purchases 100 units of Inventory for $10 each. January 5: Sells 50 units of Inventory for $20 each. January 10th: Purchases 100 units of Inventory for $15 each. January 12: Sells 120 units of Inventory for $20 each. **Calculate the Gross Profit for the January 12 sale, assuming FIFO.**

   (A) $0k

   (B) $100k

   (C) $200k

   (D) $900k

   (E) None of the other answers.

   Solution. The whole trick here is that there are *two* sales, and the first one has already eaten part of the first batch. You cannot expense out units that have already been expensed out. So let's keep the layers explicit, unit by unit.

   After the two purchases, the layers are: **Batch 1**, 100 units at $10 (Jan. 1), and **Batch 2**, 100 units at $15 (Jan. 10).

   January 5 sale, 50 units. As of that date, only Batch 1 exists. Under FIFO we take the oldest units first, so all 50 come from Batch 1: COGS of $50 \times \$10 = \$500$. That leaves **50 units of Batch 1** on the shelf. Then Batch 2 arrives on January 10.

   January 12 sale, 120 units. Our pac-man resumes where he left off—50 units still remaining in Batch 1, then on to Batch 2:

Layer	**Units taken**	**Cost/unit**	**Cost**
Batch 1 (Jan. 1) remainder	50	$10	$500
Batch 2 (Jan. 10)	70	$15	$1,050
Total COGS	**120**		**$1,550**

January 12 sale, FIFO

Revenue on the sale is $120 \times \$20 = \$2,400$, so:

$$Gross\ Profit = \$2,400 - \$1,550 = \$850$$

None of the listed figures is \$850 (and none is \$850k, if you prefer to read the whole problem in thousands—the units scale, the answer doesn't change). So the answer is **(E)**.

2. Consider the following scenario: In FY2020, MarshallCo starts its life, and has the following events related to the purchase and sale of *Inventory*. January 1: Purchases 100 units of Inventory for \$10 each. January 5: Sells 50 units of Inventory for \$20 each. January 10th: Purchases 100 units of Inventory for \$15 each. January 12: Sells 120 units of Inventory for \$20 each. **Calculate the COGS Expense for the January 12 sale, assuming LIFO.**

   (A) \$0k

   (B) \$100k

   (C) \$200k

   (D) \$900k

   (E) None of the other answers.

   Solution. Same facts, opposite pac-man. And note the wrinkle the January 5 sale creates: "last in" means the last batch *in our Inventory as of the date of that sale.*

   January 5 sale, 50 units. As of January 5, Batch 2 has not been purchased yet—it arrives on the 10th. So the "last in" batch available is Batch 1, and all 50 units come from it, at \$10 each. That leaves **50 units of Batch 1** at \$10. Notice that FIFO and LIFO agree on this first sale, because there is only one layer to take from.

   January 12 sale, 120 units. Now Batch 2 (100 units at \$15) is on the books, and it is the most recent. Working backwards:

Layer	**Units taken**	**Cost/unit**	**Cost**
Batch 2 (Jan. 10)	100	\$15	\$1,500
Batch 1 (Jan. 1) remainder	20	\$10	\$200
Total COGS	**120**		**\$1,700**

January 12 sale, LIFO

So COGS on the January 12 sale is $1,700—which, sure enough, is higher than the $1,550 FIFO gave us in the previous problem, exactly as you should expect when costs are rising. $1,700 is not among the listed figures. So the answer is **(E)**.

3. If our inventory costs are rising monotonically (that is, increasing and only increasing) over time, which inventory costing method will cause us to report the highest *Net Income*?

   (A) FIFO

   (B) LIFO

   (C) Weighted Average Cost

   (D) Specific Identification

   (E) Not enough information / ambiguous

   Solution. Rising costs means the oldest units in inventory are the cheapest. FIFO expenses out the *oldest*—that is, the cheapest—units first. Lowest COGS means highest Gross Profit, and so highest Net Income.

   LIFO is the mirror image: it picks up the most recent, most expensive units, giving the highest COGS and the lowest Net Income. Weighted-Average Cost blends old and new, so it lands somewhere in between—which, as the chapter notes, is precisely why it never attracted LIFO's constituency: when costs are rising, it captures only part of the tax benefit LIFO delivers in full. And Specific Identification depends on which particular units the customer happened to grab, so it cannot be ranked without more information. So the answer is **(A)**.

4. Which method is the most commonly used by U.S. public companies for financial reporting?

   (A) FIFO

   (B) LIFO

   (C) Weighted Average Cost

   (D) Specific Identification

   (E) None of the above

   Solution. FIFO. LIFO was once the dominant method—nearly 70% of large U.S. firms used it during the inflationary 1970s and early 1980s—but its use has collapsed to something like 7% of U.S. public companies with inventories today, surviving mostly in oil and gas, metals, auto dealers, and supermarkets. Decades of low inflation, the 2017 cut in the corporate rate from 35% to 21%, the IFRS prohibition on LIFO, and the sheer administrative cost of maintaining LIFO layers

all shrank its appeal. Weighted-Average Cost has a real following but a smaller one; Specific Identification is essentially never used for interchangeable goods. So the answer is **(A)**.

5. At the end of FY2020, our company reported $100k in *Inventory*. Over FY2021, we spent another $80k purchasing *Inventory*. At the end of FY2021, we reported $70k in *Inventory* on our Balance Sheet. What was our company's *Cost of Goods Sold* for FY2021?

   (A) $70k

   (B) $90k

   (C) $110k

   (D) Not enough information

   (E) None of the other answers

   Solution. No costing method required, and none given—because none is needed. This is just the general form of an asset, wearing a new label:

   $$\textit{Asset Ending Balance} = \textit{Asset Beginning Balance} + \textit{Purchases} - \textit{Related Expense}$$

   Purchases are debited to *Inventory* at the purchase price, and the only thing that credits it is the related expense, *COGS*. We began with $100k and added $80k of purchases, so $180k passed through the account. We ended with $70k. Whatever took the balance from $180k down to $70k must have been the 'using up' of the asset:

   $$COGS = \$100k + \$80k - \$70k = \$110k$$

   Option (D) is the tempting one, because we were never told whether the company uses FIFO, LIFO, or anything else. But the costing method determines *how the pie is sliced*, not how big it is: whichever method is in use, the T-account still has to balance. So the answer is **(C)**.

6. A company reports this *Income Statement:*

Total Revenues	\$120k
Cost of Goods Sold	(\$40k)
Salaries Expense	(\$20k)
Rent Expense	(\$10k)
Gain upon Sale of Equipment	\$5k
Interest Expense	(\$10k)
Tax Expense	(\$9k)
Net Income	**\$36k**

What is its *Operating Income*?

(A) 36k

(B) 80k

(C) 60k

(D) 55k

(E) 50k

(F) None of the other answers.

Solution. *Operating Income* is revenue less all the *operating* expenses. The reliable way to get it is the one from the chapter: start at the bottom and work up, adding back everything that is not operational. Here that list has three items—*Interest Expense* (financing), *Tax Expense*, and the *Gain upon Sale of Equipment*. The question's stated classification treats this Gain as nonoperating, so it comes out. Its cash-flow classification as investing does not, by itself, determine that income-statement treatment. Note that the Gain is *added* to Net Income on the statement, so to remove it we *subtract* it:

$$Operating\ Income = \$36k + \$10k + \$9k - \$5k = \$50k$$

Or, equivalently, working top-down: $\$120k - \$40k - \$20k - \$10k = \$50k$.

The distractors are worth naming, because each is a real quantity. \$80k is the **Gross Profit** ($\$120k - \$40k$). \$55k is **EBIT** under its strict definition—Net Income plus Interest plus Tax—which adds back financing and taxes but makes no judgment about operating versus non-operating items, and so leaves the Gain in. This is exactly the case where *Operating Income* and *EBIT* come apart, as promised in the chapter: \$50k versus \$55k. And \$36k is just the bottom line itself. \$60k is the one with no name—revenue less COGS less salaries, an arbitrary stopping point partway down the statement, and it is there precisely to catch someone who quits subtracting one line early. So the answer is **(E)**.

7. **Setup:** Suppose our company starts its life in FY2022, and, on January 10, purchases **100 units** from a wholesaler at a price of **$10,000** and separately pays **$1,000** to ship the goods to our store. Then, on January 25, we sell 25 units. What would be our *Inventory* **balance** after the January 25 sale?

   (A) $7,500

   (B) $7,750

   (C) $7,250

   (D) Not enough information.

   (E) None of the other answers.

   Solution. Freight-in is capitalized into the batch. So the January 10 entry is a single debit for the full $11,000:

Inventory (+A):	$11,000	
Cash (-A):		$11,000

Purchase of 100 units ($10,000) plus $1,000 of shipping, 1/10

   That gives us 100 units at a cost of $\$11{,}000/100 = \$110$ per unit. Selling 25 of them expenses out $25 \times \$110 = \$2{,}750$:

Cost of Goods Sold (-S.E.):	$2,750	
Inventory (-A):		$2,750

COGS on the sale of 25 units, 1/25

   So the *Inventory* balance is $\$11{,}000 - \$2{,}750 = \$8{,}250$—or, equivalently, the 75 units still on the shelf at $110 each.

   Option (A), $7,500, is the answer you get if you expense the shipping immediately and carry the goods at their $100 sticker price. That is the error the capitalization rule exists to prevent: the shipping was a cost of getting these goods into a condition to be sold, so it belongs in *Inventory*, to be matched against revenue as the units go out the door. None of the listed figures is $8,250, so the answer is **(E)**.

8. Which inventory costing method is characterized as the "income statement method"?

   (A) FIFO

(B) LIFO

(C) Weighted Average Cost

(D) Specific Identification

(E) Not enough information / ambiguous

Solution. LIFO. The label comes from what each method does *well*. LIFO charges the most recent acquisition costs against today's revenues, so the income statement answers the question an analyst actually cares about—*can this firm still earn a margin at today's input prices?*—right away, rather than after the cheap old batches run out. FIFO is the "balance-sheet method," for the mirror-image reason: it leaves the most recent purchases sitting in the asset, so the *Inventory* balance more closely approximates current value.

One thing the labels do **not** mean: that a company gets to use one method for one statement and another for the other. You pick one method, and it flows through both. So the answer is **(B)**.

9. Which of the following is a reason why many companies elect to use the LIFO method for Inventory costing?

(A) To lower their tax burden by showing lower expenses.

(B) To lower their tax burden by showing lower tax profit.

(C) To show higher GAAP profit, as inventory costs rise over time due to inflation.

(D) Multiple of the other answers.

(E) None of the other answers.

Solution. The tax story is the real one, but you have to state it in the right direction. When costs are rising, LIFO charges the newest and most expensive units to COGS, producing **higher** expenses, **lower** profit, and therefore a lower tax bill. That is (B).

Option (A) has the mechanism backwards: LIFO shows *higher* expenses, not lower. It is a nice trap, because the conclusion is right and only the reasoning is wrong—and in accounting, getting the direction right is most of the work. Option (C) is backwards too: in an inflationary environment LIFO produces *lower* GAAP profit, not higher. Indeed, that is the whole tradeoff—firms accept lower reported *Net Income* in exchange for real cash tax savings, and, because of the **LIFO conformity rule**, they have no choice about it. LIFO is the one exception in this book to the general rule that tax reporting and financial reporting are separate systems. So the answer is **(B)**.

10. MattCorp is a retailer. At the end of FY2015, MattCorp reported Inventory of \$100k. At the end of FY2016, MattCorp reported Inventory of \$80k. Additionally, we are able to determine that MattCorp earned Revenue of \$90k and Gross Profit of \$60k over FY2016. How much did MattCorp purchase in Inventory over 2016?

    (A) \$10k

    (B) None of the other answers.

    (C) \$30k

    (D) \$60k

    (E) Not enough information. Depends on Inventory Costing Method (FIFO vs. LIFO, etc.)

    Solution. Two steps, and the first one is the income statement.

    Step 1: back out COGS from the gross profit. Gross Profit is Revenue minus COGS, so $COGS = \$90k - \$60k = \$30k$.

    Step 2: run the asset identity. Rearranged for purchases:

    $$Purchases = \Delta Inventory\ Balance + COGS = (\$80k - \$100k) + \$30k = \$10k$$

    MattCorp sold \$30k of goods at cost, but replenished only \$10k—which is why its inventory shrank by \$20k over the year. And, once again, we never needed to know the costing method: the identity holds whichever one is in use, so (E) is wrong. \$30k is COGS, not purchases. So the answer is **(A)**.

11. A company reports Revenue of \$100K, COGS of \$60k, Other Operating Expenses of \$20k, Interest Expense of \$10k, and Tax Expense of \$4k, for a bottom-line income of \$6k. What is its Gross Profit Margin?

    (A) 60%

    (B) 40%

    (C) 20%

    (D) 6%

    (E) None of the other answers are correct

    Solution. Gross Profit is the very first 'level'—revenue less COGS only, before any overhead:

$$Gross\ Profit\ Margin = \frac{Revenue - COGS}{Revenue} = \frac{\$100k - \$60k}{\$100k} = 40\%$$

The distractors are the other numbers you can build out of this statement, and it is worth knowing which is which. 60% is the *COGS* as a percentage of revenue—the complement, since $Gross\ Profit\ Margin = 1 - COGS/Revenue$. 20% is the *operating* margin ($\$100k - \$60k - \$20k = \$20k$, over \$100k). And 6% is the plain *profit margin*, the bottom line over revenue. Lined up, those four numbers *are* the common-size income statement. So the answer is **(B)**.

A.10.2 Practice Problems: Open Prompts

1. MattCorp starts its life in FY2021, and makes the following Inventory purchases and Sales, and only these.

   Purchases: January 20, pays \$10k for 100 units, and an additional \$500 for shipping. March 15, pays \$8k for 60 units, and an additional \$300 for shipping.

   Sales: February 15, sells 50 units.

   Question: What would be the *COGS* MattCorp would record upon the February 15th sale, assuming **FIFO**.

   Solution. There are two things to get right here, and the second one is the trap.

   First, capitalize the freight. The January 20 batch cost \$10,000 for the goods plus \$500 to ship them, so \$10,500 goes into *Inventory* for 100 units—\$105 per unit. (The March batch, similarly, is \$8,300 for 60 units, or $\$138.\overline{3}$ per unit, but see below.)

   Second, mind the calendar. The sale is on **February 15**. The March 15 batch does not exist yet. FIFO or LIFO, it makes no difference: you cannot expense out units you have not bought. So all 50 units come from the January 20 batch:

$$COGS,\ Feb.\ 15,\ FIFO = 50 \times \$105 = \$5{,}250$$

Cost of Goods Sold (-S.E.):	\$5,250	
Inventory (-A):		\$5,250

COGS on the February 15 sale of 50 units

Everything about the March purchase is a distraction, deliberately placed. And notice the size of the shipping effect: had you ignored the \$500, you would have

recorded $5,000 of COGS and overstated your gross profit on this sale by $250.

2. MattCorp starts its life in FY2021, and makes the following Inventory purchases and Sales, and only these. **Purchases**: January 15, 2000 units at $30 each; March 15, 2000 units at $40 each; June 15, 1000 units at $45 each. **Sales**: March 20th, 3000 units, sold for $80 each; July 4th, 1500 units sold for $80 each. Calculate the *Gross Profit* for the July 4th sale, using the **LIFO** method.

   Solution. This is the hard kind of LIFO problem—a sale sandwiched between purchases—so the discipline is to track, layer by layer, exactly what remains on the shelf after each event. Diligence, not genius.

   Purchases through March 20. Batch 1: 2,000 units at $30 (Jan. 15). Batch 2: 2,000 units at $40 (Mar. 15). Batch 3 has not arrived yet.

   March 20 sale, 3,000 units, LIFO. The most recent batch in inventory is Batch 2, so we take all 2,000 of its units, then reach back for 1,000 more from Batch 1:

$$COGS,\ Mar.\ 20,\ LIFO = (2{,}000 \times \$40) + (1{,}000 \times \$30) = \$110{,}000$$

   What survives: **1,000 units of Batch 1, at $30.** Batch 2 is gone entirely.

   June 15 purchase. Batch 3 arrives: 1,000 units at $45. The shelf now holds 1,000 units at $30 and 1,000 units at $45—2,000 units in all, at a total *Inventory* cost of $75,000.

   July 4 sale, 1,500 units, LIFO. Pac-man starts again from the right—Batch 3—and then must skip straight over the fully-consumed Batch 2 back to what is left of Batch 1:

Layer	**Units taken**	**Cost/unit**	**Cost**
Batch 3 (Jun. 15)	1,000	$45	$45,000
Batch 2 (Mar. 15)	—	—	*already fully expensed*
Batch 1 (Jan. 15) remainder	500	$30	$15,000
Total COGS	**1,500**		**$60,000**

July 4 sale, LIFO

   Revenue on the sale is $1{,}500 \times \$80 = \$120{,}000$. So:

$$Gross\ Profit,\ July\ 4 = \$120{,}000 - \$60{,}000 = \$60{,}000$$

   (The two $60,000s are a coincidence of the numbers, not a rule—don't read anything into it.) And a check on the remaining balance: 500 units of Batch 1 are left,

at $30 each, so *Inventory* stands at $15,000. The entry for the sale, in full:

Cash (+A):	$120,000	
Revenue (+S.E.)		$120,000
Cost of Goods Sold (-S.E.):	$60,000	
Inventory (-A):		$60,000

July 4 sale of 1,500 units at $80

3. MattCorp starts its life in FY2021, and makes the following Inventory purchases and Sales, and only these. **Purchases**: January 15, 2000 units at $30 each; March 15, 2000 units at $40 each; June 15, 1000 units at $45 each. **Sales**: March 20th, 3000 units, sold for $80 each; July 4th, 1500 units sold for $80 each. Calculate *COGS* for the July 4th sale, using the **LIFO** method.

   Solution. This is the same scenario as the previous problem, asking for the other half of it. From the layer analysis there: after the March 20 sale consumed all of Batch 2 and 1,000 units of Batch 1, the July 4 sale takes all 1,000 units of Batch 3 at $45, and then the 500 remaining units of Batch 1 at $30:

$$COGS,\ July\ 4,\ LIFO = (1{,}000 \times \$45) + (500 \times \$30) = \$45{,}000 + \$15{,}000 = \$60{,}000$$

   The tempting wrong answer is $1{,}000 \times \$45 + 500 \times \$40 = \$65{,}000$—reaching for Batch 2 because it is 'next' in line, and forgetting that our pac-man already ate every unit of it back in March.

4. MattCorp is a retailer. At the end of Fiscal Year 2015, MattCorp reported Inventory of $100k. At the end of Fiscal Year 2016, MattCorp reported Revenue of $100k and Inventory of $80k. We are additionally able to determine that MattCorp spent $40k on purchasing inventory during FY 2016. What was MattCorp's Gross Profit for FY 2016?

   Solution. Run the asset identity to get COGS, then take it to the income statement.

   Step 1: COGS. $100k of beginning inventory plus $40k of purchases is $140k that passed through the account; $80k remains at year end; so the credits—the COGS—were:

$$COGS = \$100k + \$40k - \$80k = \$60k$$

Step 2: Gross Profit.

$$Gross\ Profit = Revenue - COGS = \$100k - \$60k = \$40k$$

So MattCorp's Gross Profit Margin for FY2016 was 40%. Note again what we never needed: the costing method, the number of units, or the unit prices. The relationship between an asset and its related expense does all the work.

5. A company reports this *Income Statement:*

Total Revenues	\$120k
Cost of Goods Sold	(\$40k)
Salaries Expense	(\$20k)
Rent Expense	(\$10k)
Loss (upon Sale of Equipment)	(\$5k)
Interest Expense	(\$10k)
Tax Expense	(\$9k)
Net Income	**\$26k**

What is its *Operating Income*? **Answer:\$50k.**

Solution. Start at the bottom and add back everything that is not operational: *Interest Expense*, *Tax Expense*, and the *Loss* on the sale of equipment. Because a Loss *reduces* Net Income, removing it means adding it back:

$$Operating\ Income = \$26k + \$10k + \$9k + \$5k = \$50k$$

Or top-down: $\$120k - \$40k - \$20k - \$10k = \$50k$. Notice that this is the identical income statement to MCQ #6 above, with the \$5k Gain flipped to a \$5k Loss—and *Operating Income* is \$50k in both cases. That is the point of the line: the operating business is doing exactly the same thing in both scenarios, and the disposal is assumed to be nonoperating in these particular exercises. What *does* change is EBIT, strictly defined: \$55k in the Gain case, $\$26k + \$10k + \$9k = \$45k$ here.

A.10.3 Essay and Discussion Questions

1. What are the factors companies consider when they choose between LIFO and FIFO? E.g., what are the normative arguments about which method makes the most sense, economically? And which method would financially self-interested companies choose if their costs are rising over time, and why?

Solution. I'd break this into two genuinely separate questions, because they pull in different directions and conflating them is the usual source of confusion. The first is *normative*: which method better represents the economics of the firm? The second is *positive and self-interested*: which method makes the company better off in cash? Let me take them in order.

The normative argument. The honest answer is that there is a real tradeoff, and it is the same tradeoff we met with the allowance for uncollectibles: an **income-statement perspective** versus a **balance-sheet perspective**.

LIFO charges the *most recent* acquisition costs against today's revenues. If what you want from an income statement is an answer to the question *is this firm selling its product for more than it currently costs to acquire?*, then LIFO gives it to you promptly. That is why LIFO is called the **income-statement method**. The vividness of this argument depends on the environment: in the 2021–2022 inflation, a FIFO firm could match today's selling prices against last year's pre-surge costs and look wonderfully profitable while its true current-cost margin had evaporated—and the FIFO income statement would not tell you so until the cheap old layers ran out. The flip side is LIFO's cost: the oldest layers never get expensed, so the *Inventory* balance on the Balance Sheet goes stale, potentially by decades.

FIFO is exactly the mirror image, and is called the **balance-sheet method** for exactly that reason: the units left in the asset are the most recently purchased ones, so the reported *Inventory* balance is a decent approximation of current value—at the cost of an income statement matched against stale costs.

Two further arguments get made for FIFO, and I'd grade them differently. The first—that FIFO is mechanically tidier, since pac-man only ever moves left to right—is minor but true. The second—that FIFO "tracks the real physical flow of goods," since grocers really do sell the old milk first—I don't find persuasive, because **it proves too much**. If tracking physical flow is the goal, nothing beats specific identification, which tracks it *literally*. And we have already rejected specific identification: the shovel thought experiment shows that letting a customer's random reach for one of two identical shovels determine our reported gross profit is measuring the wobble of a shopping cart, not the economics of the business.

For what it's worth, my own sympathies are with the income-statement view, and so with LIFO—I come from valuation, and what I want is a measure of ongoing profitability I can forecast from. But this is a judgment call about which lens is more useful, not a question with a right answer like $2+2$.

The self-interested argument. Here the answer is sharp: if costs are rising, a financially self-interested company chooses **LIFO**, and the reason is **taxes**.

Rising costs means LIFO produces higher COGS, hence lower profit. Lower *tax* profit means a lower tax bill—real cash that stays in the firm. Ordinarily this would be no argument at all for the *financial* reporting choice, because tax reporting and financial reporting are separate systems: that is the general rule, and you should be suspicious of anyone who invokes "tax benefits" to explain a GAAP choice. But inventory is the one exception in this book. Under the **LIFO conformity rule**, a firm using LIFO for tax *must* use LIFO for its financial statements. So the tax tail wags the financial-reporting dog, and the firm must accept lower reported *Net Income*—and, through it, lower reported ROA, ROE, and profit margin—as the price of the cash savings.

Historically, firms took that deal in droves: nearly 70% of large U.S. companies used LIFO during the high-inflation era. And they were right to, for a reason worth internalizing: **investors and analysts are not stupid.** The method is disclosed in the footnotes, and a competent analyst can unwind its effects. A dollar not paid to the IRS is a dollar available for reinvestment, dividends, or buybacks—so a good analyst might well assign a *higher* valuation to the firm reporting *lower* earnings. This is, incidentally, one of my favorite counterexamples to the claim that firms are myopically fixated on reported earnings. If that story were literally true, no firm would ever have adopted LIFO.

A final wrinkle worth mentioning. The exodus from LIFO since the 1980s is the best evidence for everything above. Inflation fell; the corporate rate dropped from 35% to 21%, shrinking the value of any deduction; IFRS banned LIFO outright, making life awkward for multinationals; repeal bills kept surfacing in Congress; and LIFO's layer-by-layer recordkeeping is genuinely expensive. Firms abandoned LIFO precisely as its tax benefit shrank—which tells you the tax motive was doing the work all along. And note that all of this comparative statics reverses in an industry where costs *fall* over time: there, FIFO is the method with the tax advantage, while the normative case for LIFO is entirely unchanged, because the most recent costs are the most relevant either way.

A.11 Chapter 11

A.11.1 Practice Problems: MCQ's

1. MattCorp purchases a factory from a seller for a price of $10m. MattCorp pays fees of $500k to a law firm to secure title; pays another $400k commission to the realtor who brokered the deal; pays its first annual state property tax of $100k; pays an annual $50k insurance premium to insure the Equipment; and salvages copper from the equipment that it then sells for $250k in Cash. What would be the initial carrying value of the Equipment on MattCorp's books (i.e., after all these events but before any subsequent depreciation expense)?

   (A) $10m

   (B) $10.6m

   (C) $10.7m

   (D) $10.9m

   (E) None of the other answers.

   Solution. Walk down the list and ask 'in or out?' for each item, using the two tests from the chapter: is the cost **non-recurring**, and is it **a condition of initiating the use of the asset**? If both, capitalize.

Item	**Treatment**	**Effect**
$10m purchase price	In	$10.00m
$500k legal fees, for title	In: one-time, condition of legal use	$0.50m
$400k realtor's commission	In: one-time cost of acquiring	$0.40m
$100k first annual property tax	**Out**: recurring periodic cost	—
$50k annual insurance premium	**Out**: recurring periodic cost	—
$250k salvaged copper, sold	In, but *negatively*	($0.25m)
Initial carrying value		**$10.65m**

In or out?

   The two recurring items are the ones the problem is really testing. It is tempting to capitalize that first property-tax payment, since you do have to pay it—but the word **annual** gives it away. A recurring periodic cost is *already* being matched to the periods of its benefit simply by being paid and expensed as it recurs; there is nothing to be gained by routing it into the asset and then dripping it back out again.

The copper is the other test. Getting $250k of cash for salvaged material does not create a Gain—the salvageable copper was reasonably anticipated and therefore already priced into what MattCorp agreed to pay. Selling it is a re-allocation, one asset up and another down, per the pot-of-gold thought experiment:

Cash (+A):	$250,000	
Equipment (-A):		$250,000

Sale of salvaged copper

So the carrying value is $\$10m + \$0.5m + \$0.4m - \$0.25m = \$10.65m$. That figure is not among the options. Look carefully at what *is* offered: $10.6m and $10.7m sit on either side of $10.65m, exactly $50k away in each direction—the size of the insurance premium. They are there to make you doubt yourself and round to the nearest listed figure. Do not. $10.65m is the number, and the fact that the list brackets it without containing it is the whole point of the question. ($10.9m, meanwhile, is what you get by forgetting the salvaged copper.) So the answer is **(E)**.

2. **Setup:** On July 1, 2020, our company bought **Equipment** for $20m, and assumed a useful life of 10 years, zero salvage value, and straight-line depreciation. On December 31, 2024, we sold the Equipment, and recorded a $2m *Loss* upon the sale. What was the sale price?

   (A) $2m

   (B) $10m

   (C) $12m

   (D) $14m

   (E) None of the other answers.

   Solution. This one runs the machinery backwards, so let's make the chain visible link by link. We want the sale price; the formula that gets us there is

   $$Sale\ Price = Book\ Value + Gain \quad (or\ minus\ the\ Loss)$$

   so the real work is reconstructing the book value on the sale date.

   Link 1: the annual depreciation. $20m of depreciable value (zero salvage) over 10 years is $2m per year.

Link 2: how much time actually elapsed. Here is the trap, and it is the whole point of the problem. From July 1, 2020 to December 31, 2024 is **four and a half years**, not five—the asset was in service for only half of FY2020.

Period	**Depreciation**
FY2020 (Jul. 1 – Dec. 31, half year)	$1m
FY2021	$2m
FY2022	$2m
FY2023	$2m
FY2024	$2m
Accumulated Depreciation	**$9m**

Accumulated Depreciation at 12/31/2024

Link 3: the book value. $\$20m - \$9m = \$11m$.

Link 4: undo the Loss. A Loss means we sold *below* book value, so we subtract: $\$11m - \$2m = \$9m$.

And, as a check, the full entry has to balance:

Cash (+A):	$9m	
Loss (-S.E.):	$2m	
Accumulated Depreciation (-contra-A):	$9m	
Equipment (-A):		$20m

Sale of the Equipment, 12/31/2024

$20m of debits, $20m of credits. $9m is not among the options. If you had assumed five full years, you would have gotten a $10m book value and an $8m sale price—also not listed—and if you had added the Loss instead of subtracting it, you would have landed on $12m, option (C). So the answer is **(E)**.

3. What does a *Loss* on the sale of a long-term fixed asset establish?

   (A) The original depreciation estimate was necessarily unreasonable.

   (B) The sale proceeds were below the asset's carrying amount at the sale date.

   (C) Management necessarily made a poor decision to sell.

   (D) All of the above.

Solution. The answer is **(B)**. A loss is the shortfall of sale proceeds below book value. It can reflect depreciation assumptions, changes in market prices, or unexpected events. The loss alone does not establish an accounting error or a poor selling decision.

4. Which of the following are true of *Gains* in accounting?

   (A) We would have a *Gain* if we determined that the fair value of our *Equipment* is greater than book value

   (B) They are reported on the Income Statement

   (C) They are *Revenues*

   (D) Multiple other answers

   (E) None of the other answers.

   Solution. Take them one at a time.

   Option (A) is false, and this is the most instructive of the three. Merely *determining*—believing, appraising, knowing in your bones—that an asset is worth more than its book value is not a transaction. Assets are carried at verifiable historical cost, and belief is not verifiable. A Gain is recognized when there is a real exchange with an outside party that reveals a price. No sale, no Gain.

   Option (B) is true. A Gain increments Net Income, and it *has* to: our net assets rose, no liability was incurred, and the increase was not a contribution from our own shareholders—so by $A = L + E$ it must be earned capital, and the only route into *Retained Earnings* is through *Net Income*.

   Option (C) is false. Mechanically a Gain behaves just like a Revenue, and it is even presented as a positive number alongside revenues. But we decline to *call* it one, precisely because the label carries information: revenues are recurring and operational, and a one-off sale of a long-term asset is neither. A gain's inclusion in Operating Income depends on its nature; a gain on an asset used in operations may be included.

   Exactly one statement is true, so (D) is out. The answer is **(B)**.

5. Which of the following is a reason why a company might elect to use an Accelerated Depreciation method for depreciating its asset for its GAAP financial reporting?

   (A) To show higher profitability sooner, rather than later.

   (B) To show higher expenses, for the tax advantages.

(C) To "store up" the ability to yield a 'Gain' in the future if/when desired, for "earnings management" purposes.

(D) Multiple other answers.

(E) None of the other answers.

Solution. (C), and it is the cynical story from the chapter. Charging heavier depreciation early drives the asset's book value down faster than economic reality warrants. Book value is the benchmark a future sale price gets compared against—so a lower book value means a larger Gain whenever the manager chooses to sell. That Gain sits on the shelf like a stored battery, available to buffer reported earnings in some future year when operations disappoint.

Option (A) is backwards: accelerated depreciation means *higher* expenses early, so *lower* profitability sooner. (The name confuses people; 'accelerated' means we get to the depreciation earlier, the way an accelerated math program gets you to the hard material earlier.)

Option (B) is the tempting non-reason, and it is worth being emphatic about. Reasoning by analogy to LIFO, you might expect a tax motive here too. But the LIFO conformity rule is the one exception in this book to the general rule that **tax reporting and financial reporting are separate systems**. There is no depreciation conformity rule: a company can use accelerated methods for tax and straight-line for GAAP, and many do. The tax game is played on a different board. So the answer is **(C)**.

6. Our company purchases a piece of Equipment for $4m on July 1, 2020. In FY2020, we record a *Depreciation Expense* of $.5m on the Equipment. If our company is using an accelerated depreciation method, which of the below would be a possible *Depreciation Expense* for FY2021?

   (A) $.4m

   (B) $.8m

   (C) $1.2m

   (D) Multiple other answers

   (E) None of the other answers.

   Solution. Everything turns on the July 1 purchase date, so let's be careful about what FY2020's $0.5m actually represents.

   The asset was in service for only *half* of FY2020. So $0.5m is a half-year charge, and the depreciation for the asset's **first full year of life** is $1m. Call the succes-

sive full-year charges $d_1, d_2, d_3, \ldots$; the definition of accelerated depreciation is that they shrink: $d_1 > d_2 > d_3 > \cdots$. We have just established that $d_1 = \$1m$.

Now, what does FY2021 contain? It is a full calendar year, but it straddles the asset's life: January through June is the *second half* of the asset's first year, and July through December is the *first half* of its second year. So:

$$FY2021\ Depreciation = \tfrac{1}{2}d_1 + \tfrac{1}{2}d_2 = \$0.5m + \tfrac{1}{2}d_2$$

And since $0 < d_2 < d_1 = \$1m$, the FY2021 expense must fall strictly between **$0.5m and $1.0m**. Only $0.8m qualifies.

Option (A), $0.4m, is the answer you get from the natural-but-wrong shortcut: "accelerated means the expense shrinks, so FY2021 must be below FY2020's $0.5m." That reasoning compares a half-year to a full year. The comparison the definition actually licenses is between comparable full years of the asset's life. Option (C), $1.2m, would require the second year to be heavier than the first—decelerated, not accelerated. So the answer is **(B)**.

7. Do we amortize an acquired finite-lived *Patent* and an acquired *Trademark* assessed as having an indefinite useful life, respectively?

   (A) Yes and yes

   (B) Yes and no

   (C) No and yes

   (D) No and no

   (E) None of the other answers

   Solution. The finite-lived Patent is amortized over its expected useful life, which cannot exceed the remaining legal protection. A U.S. utility patent generally expires 20 years from the relevant filing date, subject to adjustments; the company's expected useful life may be shorter.

   The Trademark in this question is explicitly **indefinite-lived**, so it is not amortized. A trademark with a finite expected useful life would be amortized. Legal renewability alone does not settle that assessment. Both assets remain subject to the appropriate impairment rules. The answer is **(B)**.

8. Company A developed a *Patent* internally, through hiring scientists 10 years ago and paying them to do research—the *Patent* was granted by the PTO 5 years ago ("build" strategy), at which point Company A divested its research division and began selling the product. Company B competes in the same product market,

and is very similar, but instead purchased its *Patent* 5 years ago from a startup (pursuing a "buy" strategy), and began selling the product. All else equal, based on this information alone, would we expect Company A to have [higher / lower / the same] *ROE* compared to Company B, at this point? Assume comparable acquisition costs and financing, positive equity and current net income for both firms, and no capitalized patent-registration costs for Company A.

(A) Higher

(B) Lower

(C) The same

(D) Not enough information

(E) None of the other answers

Solution. This is the build-vs.-buy asymmetry, and $ROE = NI/E$ is distorted in the *same direction* by both its numerator and its denominator. Take them separately.

The denominator. Company A never recorded a *Patents* asset. Its scientists' salaries were expensed as *R&D Expense* year by year, and the PTO's grant of the patent is not a transaction—no cash changes hands, no verifiable price exists—so nothing was ever capitalized. Those expenses ran through Net Income and reduced *Retained Earnings*. Company B, by contrast, has the patent sitting on its balance sheet at its purchase price. So A's book equity is **lower**.

The numerator. Today, both companies earn the same revenue from the same patent. But Company B records *Amortization Expense* against it each year, while Company A records nothing—there is no asset on A's books to amortize; the cost was fully expensed years ago. So A's Net Income is **higher**.

Higher numerator over lower denominator: A's ROE is unambiguously higher. And note the punchline—the two firms are, from this point forward, doing *exactly the same thing*. This illustrates one influence on measured ROE. It does not isolate the causes of Apple's historical ROE; share repurchases and other factors also affect its book-equity denominator. So the answer is **(A)**.

9. MattCorp purchases Equipment from a seller for a price of $10m. MattCorp pays legal fees of $500k to secure title, as a part of the transaction; pays a nonrefundable purchase tax of $250k on the equipment; pays a recurring annual $1m insurance premium to insure the Equipment; and salvages scrap materials from the equipment that it then sells for $1m in Cash. What would be the initial carrying value of the Equipment on MattCorp's books (i.e., after all these events but before any subsequent depreciation expense).

(A) \$10.25m

(B) \$9.75m

(C) \$10m

(D) \$10.85m

(E) None of the other answers.

Solution. Same drill as problem #1: for each item, is it directly attributable to acquiring and preparing the asset for use?

- The \$10m purchase price: in, obviously.
- The \$500k of legal fees to secure title: in. One-time, and a condition of getting the asset into a legally usable state.
- The \$250k nonrefundable purchase tax: in, as part of the acquisition cost.
- The recurring annual \$1m insurance premium: out. Periodic, expensed year by year as it recurs.
- The \$1m of salvaged scrap sold: in, but negatively—it reduces the net cost of the Equipment.

$$\textit{Carrying Value} = \$10m + \$0.5m + \$0.25m - \$1m = \$9.75m$$

Every distractor here is a version of dropping or mishandling one line of that walk-down. Option (C), \$10m, is the 'just use the sticker price' answer, which ignores the capitalization rule and the salvage rule alike. The other two figures are near-misses assembled from partial walk-downs—the defense is not to recognize them but to do the walk-down completely, line by line, and trust it. Note also that the salvage adjustment is the only one that pushes *downward*—which is the counterintuitive one, and therefore the one to double-check. So the answer is **(B)**.

10. On January 1, 2017, Joe's Inc. bought a machine for \$100,000. The company assumed a useful life of 20 years, a salvage/residual value of \$20,000, and straight-line depreciation. On January 1, 2022, Joe's Inc. sold the machine and recorded a \$10,000 Gain upon sale. What was the sale price of the machine?

(A) \$10,000

(B) \$70,000

(C) \$90,000

(D) \$110,000

(E) None of the answers are correct

Solution. Backwards again, and the chain is short. Note the salvage value: we do not depreciate the whole $100,000.

Link 1: the depreciable value. $\$100{,}000 - \$20{,}000 = \$80{,}000$.

Link 2: annual depreciation. $\$80{,}000/20\ years = \$4{,}000$ per year.

Link 3: elapsed time and accumulated depreciation. January 1, 2017 to January 1, 2022 is exactly five years, so Accumulated Depreciation is $5 \times \$4{,}000 = \$20{,}000$.

Link 4: book value at the sale date. $\$100{,}000 - \$20{,}000 = \$80{,}000$.

Link 5: undo the Gain. A Gain means we sold *above* book value, so we add: $Sale\ Price = \$80{,}000 + \$10{,}000 = \$90{,}000$.

Cash (+A):	$90,000	
Accumulated Depreciation (-contra-A):	$20,000	
Machine (-A):		$100,000
Gain (+S.E.):		$10,000

Sale of the machine, 1/1/2022

Option (B), $70,000, is subtracting the Gain instead of adding it. Note also the arithmetic coincidence built into this problem: the accumulated depreciation and the salvage value are both $20,000, which is exactly the sort of thing that lets a careless reader use one where the other belongs. So the answer is **(C)**.

11. On July 1, 2012, Topanga Corp bought a machine for $200k. On January 1, 2017, it sold it for $165k, and recorded a $10k Gain upon the sale. Topanga Corp always used straight-line depreciation for its long-term tangible assets, and never recorded an impairment on this asset. What was Topanga Corp's FY2016 depreciation expense for this machine?

(A) $5k

(B) $10k

(C) $55k

(D) Not enough information

(E) None of the answers are correct

Solution. This one runs the chain backwards *twice*: from the Gain to the book value, and from the book value to the annual depreciation rate. We are never told the useful life or the salvage value—and we do not need them.

Link 1: book value at the sale date. $Book\ Value = Sale\ Price - Gain = \$165k - \$10k = \$155k$.

Link 2: total accumulated depreciation. The machine cost \$200k, so the depreciation charged over its life on our books was $\$200k - \$155k = \$45k$.

Link 3: how much time that \$45k covers. July 1, 2012 to January 1, 2017 is **four and a half years**. (This is the same half-year trap as problem #2, and here it bites: assume five full years and you get \$45k/5 = \$9k, which lands you on (E).)

Link 4: the annual rate. Straight-line means a constant charge per full year, x, with a half-year charge in FY2012:

$$4.5x = \$45k \quad \Rightarrow \quad x = \$10k \text{ per year}$$

FY2016 is a full year in which the company still held the machine—it was sold on January 1, 2017, the very first day of FY2017—so FY2016 got a full year's charge of **\$10k**.

Option (D) is the one to resist. It *feels* like we are missing information, since nobody told us the useful life. But the sale price and the Gain jointly reveal the book value, and the book value reveals what the depreciation must have been. This is exactly the detective work the chapter describes: from the outside, you never observe an asset's sale price or depreciation schedule directly—you reconstruct them. So the answer is **(B)**.

12. Our company has only one item in its Equipment account, and we use an accelerated depreciation methodology. In FY2017, the book value of the Equipment was \$900k. In FY2018, its book value was \$600k. (There was no impairment loss.) Based on this information alone, which of the below is a possible Depreciation Expense for the asset in FY2019?

    (A) None of the other answers.

    (B) \$200k

    (C) \$400k

    (D) \$500k

    (E) Multiple of the other answers.

    Solution. Reason it out by remixing two things you already know.

    First: what makes the book value of Equipment fall? Only its related expense. *Equipment* itself sits untouched at historical cost while *Accumulated Depreciation* takes the credits, and the problem has ruled out an impairment. So the FY2018 Depreciation Expense must have been $\$900k - \$600k = \$300k$.

Second: the definition of accelerated depreciation is that the annual charge shrinks each year. So FY2019's expense must be strictly *less than* \$300k. Of the figures offered, only \$200k qualifies—\$400k and \$500k both exceed \$300k, which would be decelerated depreciation.

(A test of whether you have really got this: if the question had instead asked for a possible *book value* at the end of FY2019, the answer would be any figure strictly between \$300k and \$600k—that is, \$600k less some expense between zero and \$300k.) So the answer is **(B)**.

13. Our company owns several different patents. At the end of FY2019, our company reported Patents (at book value) of \$900k. Over FY2020, we recorded total Amortization Expense of \$200k, recorded a \$300k impairment Loss on one of our patents, and purchased an additional Patent for \$100k. What would be the book value of our Patents on our FY2020 Balance Sheet?

    (A) None: Intangible asset not recorded.

    (B) \$800k

    (C) \$500k

    (D) \$700k

    (E) None of the other answers.

    Solution. Nothing new here—it is the general form of an asset, with *Patents* in the asset slot and *Amortization* in the expense slot. Purchases debit the account; the related expense credits it; and an impairment credits it too.

Patents	
Beg.: \$900k	Amortization: \$200k
Purchase: \$100k	Impairment: \$300k
\$500k	

$$\$900k + \$100k - \$200k - \$300k = \$500k$$

Option (A) is the trap for anyone who half-remembers the slogan that "accounting doesn't recognize intangibles." The operative distinction is not tangible versus intangible—it is **purchased versus internally developed**. These patents were purchased, so they are on the books like any other asset. Option (B), \$800k, ignores the impairment; (D), \$700k, ignores the new purchase. So the answer is **(C)**.

14. Topanga Corporation purchased a new automotive manufacturing plant on 50 acres of land. The building is being depreciated using straight line depreciation. What method should the *Land* be depreciated with?

    (A) Straight Line

    (B) Double Declining

    (C) Units of production

    (D) Sum of years digits

    (E) None of the other answers

    Solution. You do not depreciate Land. The question is built so that every listed option is a perfectly respectable depreciation method—and every one of them is wrong, because the premise of the question is false.

    Why is ordinary Land exempt? Its service potential is generally not consumed over a finite, estimable useful life, so there is no finite period over which to allocate its cost as depreciation. This does not require us to assume that its future selling price will equal its acquisition cost. Land improvements and extractable natural resources are separate accounting questions.

    Note that the *building* is a different matter entirely, and it is being depreciated straight-line, as the problem says. Plant wears out; the dirt underneath it does not. So the answer is **(E)**.

15. Tommy Corporation sold its two-year-old machine to Trojan Inc. for $580,000. The machine originally cost $778,000, had an estimated useful life of 10 years, and a salvage value of $48,000. Assuming the machine was depreciated using straight-line depreciation, what was the gain/loss on sale of the machine at the end of the second year?

    (A) Gain of $198,000

    (B) Loss of $198,000

    (C) Loss of $52,000

    (D) Gain of $52,000

    (E) None of the answers are correct

    Solution. This one runs forwards, which makes it a good check on whether you have the pieces in the right order.

    Depreciable value: $\$778{,}000 - \$48{,}000 = \$730{,}000$. Do not skip the salvage value—it is the single most common slip in these problems.

Annual depreciation: $\$730{,}000/10 = \$73{,}000$ per year.

Accumulated depreciation after two years: $2 \times \$73{,}000 = \$146{,}000$.

Book value at the sale date: $\$778{,}000 - \$146{,}000 = \$632{,}000$.

Gain or Loss:

$$Gain\ (Loss) = Sale\ Price - Book\ Value = \$580{,}000 - \$632{,}000 = -\$52{,}000$$

Negative, so it is a **Loss of $52,000**.

Cash (+A):	$580,000	
Loss (-S.E.):	$52,000	
Accumulated Depreciation (-contra-A):	$146,000	
Machine (-A):		$778,000

Sale of the machine to Trojan Inc.

Option (B), a Loss of $198,000, is what you get by comparing the sale price to the *original cost* and forgetting that two years of depreciation have already come off: $778,000 - $580,000. Option (A) is that same figure with the sign flipped—and the sign is the giveaway: $580,000 received against a $632,000 book value cannot possibly be a Gain. What the Loss *means*, per problem #3, is that $73,000 a year turned out to understate how fast this machine actually lost value. So the answer is **(C)**.

16. Glacier Company purchased equipment for $30,000. The equipment is estimated to last five years with the residual value at the end of the service life expected to be $3,100. The equipment was in operation for 3,100 hours in the first year and the company expects the equipment to operate a total of 20,000 hours. What is the *Depreciation Expense* for the first year of the equipment's use, assuming the use of activity-based depreciation?

    (A) $6,750

    (B) $10,000

    (C) $4,185

    (D) Not enough information.

    (E) None of the answers are correct

Solution. Activity-based depreciation allocates the depreciable value evenly *per unit of activity*—here, per hour of operation—rather than evenly per year. So the five-year estimate is a red herring; it tells us nothing we need.

$$\textit{Depreciation per hour} = \frac{\$30{,}000 - \$3{,}100}{20{,}000\ \textit{hours}} = \frac{\$26{,}900}{20{,}000} = \$1.345\ \textit{per hour}$$

$$\textit{Depreciation Expense, Year } 1 = 3{,}100 \times \$1.345 = \$4{,}169.50$$

Now look at option (C), \$4,185. That is $3{,}100 \times \$1.35$—the answer you get by rounding the per-hour rate up from \$1.345 to \$1.35 before multiplying. It is close, and it is wrong, and it is *deliberately* close: with 3,100 hours of operation, a half-cent of rounding error compounds into \$15.50. The rule is the boring one: carry the rate exactly, and round only at the end, if at all.

Option (D) is worth dismissing explicitly too. Everything we need is given—purchase price, residual value, total expected hours, and hours used. The five-year service life is extra information, not missing information. Since \$4,169.50 is not listed, the answer is **(E)**.

A.11.2 Practice Problems: Open Prompts

1. Our company purchases a piece of *Equipment* for \$25m in FY2020. We expect the equipment to produce 500k units over its useful life, and have zero salvage/residual value at the end. We use **activity-based depreciation** for this asset. During FY2020, its produces 150k units; during FY2021, it produces 200k units; and during FY2022 it produces 50k units, before we sell it for \$4m. What is the *Gain / Loss* upon the sale?

   Solution. Two stages: build up the accumulated depreciation year by year, then compare the resulting book value to the \$4m sale price.

   Stage 1: the per-unit rate. Zero salvage, so the depreciable value is the full \$25m, spread over the 500,000 units we expect the machine to produce:

   $$\textit{Depreciation per unit} = \frac{\$25m}{500{,}000\ \textit{units}} = \$50\ \textit{per unit}$$

   Stage 2: the annual charges. The rate is constant per unit; the annual expense is lumpy, because production is lumpy.

Year	Units produced	Depreciation Expense
FY2020	150,000	\$7.5m
FY2021	200,000	\$10.0m
FY2022	50,000	\$2.5m
Total	**400,000**	**\$20.0m**

Accumulated Depreciation, at \$50 per unit

Stage 3: book value at the sale date. We expected 500,000 units and got 400,000 out of it before selling, so \$20m of the \$25m has been expensed:

$$Book\ Value = \$25m - \$20m = \$5m$$

Stage 4: the plug.

$$Gain\ (Loss) = Sale\ Price - Book\ Value = \$4m - \$5m = -\$1m$$

So we record a **Loss of \$1m**. In full, remembering that *Equipment* and *Accumulated Depreciation* must be extinguished separately:

Cash (+A):	\$4m	
Loss (-S.E.):	\$1m	
Accumulated Depreciation (-contra-A):	\$20m	
Equipment (-A):		\$25m

Sale of the Equipment for \$4m, FY2022

And the interpretation: our activity-based schedule assumed each unit consumed \$50 of the machine, which left \$5m of book value standing after 400,000 units. The market disagreed and paid \$4m. The Loss is the one-time catch-up correction—we had under-depreciated by \$1m, and the sale is what revealed it.

2. Our company purchases a plot of *Land* for \$10m. We expect to use it for the next 20 years. In general, our company uses *straight-line depreciation* for all of our depreciable assets. What would be the annual *Depreciation Expense* recorded on the *Land* each year?

   Solution. Zero. You do not depreciate Land.

   The problem is constructed to make the wrong answer easy to compute—\$10m over 20 years is \$500,000, and the words “straight-line” and “20 years” are sitting

right there. But the phrase to notice is *all of our depreciable assets*, and Land is not one of them.

The reason is the indefinite useful life of ordinary Land, not a forecast that its resale price will equal its purchase price. A twenty-year intended holding period does not make the underlying land a wasting asset.

Two caveats keep this from meaning Land is 'free.' First, the parcel must still be tested for **impairment** (next chapter), so the balance can be written down if its fair value falls below cost. Second, that $10m had to be financed—with debt, which brings interest expense, or with equity, which dilutes existing shareholders who expect a return. The absence of a depreciation line is not the absence of a cost.

A.11.3 Essay and Discussion Questions

1. What is a "gain" in accounting? Is it good or bad news? Does it indicate that the company did a good or bad decision or job selling an asset? Why or why not? If it's not good news, why is it added to Net Income / Shareholders' Equity? What is the economic interpretation of a Gain upon sale of an asset?

   Solution. Here is how I'd work through this, in four steps: what a Gain *is*, why it *has* to increment Net Income, what it actually *means*, and only then whether it is good news.

   What it is. A Gain is a **plug**. We sell an asset with a book value of $5m and receive $6m in cash. Two real things happened—cash came in, an asset went out—and they were not the same size. Something must absorb the $1m difference. That something is the Gain. Strip away the bookkeeping and a Gain is nothing more or less than $Sale\ Price - Book\ Value$.

   Why it must hit Net Income. This is derivable, not memorizable, and I think working the derivation is the best way to make it stick. Our net assets rose by $1m. No new liability was incurred. So by $A = L + E$, Shareholders' Equity must rise by $1m. Within Shareholders' Equity there are only two sources: paid-in capital and earned capital. This was not a transaction with our own shareholders, so it is not new *Common Stock*. Therefore it is earned capital—*Retained Earnings*—and the only route into Retained Earnings is through *Net Income*. So the plug has to flow through the income statement. All that remains is what to call it, and we decline to call it *Revenue*, because sales of long-term assets are neither operational nor repeatable. Hence a new category, with a new label whose entire job is to signal *non-recurring*. (Whether a gain is included in Operating Income depends on its nature.)

What it means. A Gain compares realized sale proceeds with the asset's remaining book value. Depreciation assumptions affect that book value, but market changes and unexpected events can also affect the sale price. The Gain does not by itself establish that the prior depreciation estimate was unreasonable, or that management sold especially well.

So: good news? No—or at least, not in the way the word 'gain' invites you to think. Three reasons.

First, it says nothing about whether we sold well. A competent manager sells for the highest price available; the book value played no part in the negotiation, and the buyer never saw our depreciation schedule. Recognizing a Gain does not mean we drove a hard bargain, and recognizing a Loss does not mean we were fleeced.

Second—and this is the argument I find most decisive—you can flip a Gain into a Loss without changing a single real event. Hold the purchase, the use, and the sale price all fixed, and merely assume a 20-year useful life instead of a 10-year one. Now the book value at sale is $7.5m instead of $5m, and our $1m Gain becomes a $1.5m Loss. Same cash, same asset, same buyer, same price—opposite sign on the income statement. Any quantity that can be reversed by a decade-old estimate is not measuring the quality of today's decision.

Third, a Gain does not prove that past depreciation was too high or that prior statements were wrong. Depreciation allocated historical cost using estimates; the sale price can also reflect later market changes. The comparison is informative, but it is not a retrospective verdict on the reasonableness of every past estimate.

The practical upshot. For an analyst, this is a forecasting problem. If net income jumps and the jump came from a one-time Gain on the sale of a building, you must not extrapolate it—'sells another factory' is not a line item for every future year. And you should be alert to the game the chapter describes: managers who over-depreciate in good years are deliberately storing up Gains, to be harvested in a bad year to buffer reported earnings. Savvy analysts adjust; the multi-step income statement helps them, by identifying the item and examining its nature and persistence.

2. It's very common for people to claim that "accounting does not recognize the value of intangible assets." What do you make of this statement? Is it true, false, true-ish, and why, and what are the implications?

   Solution. My answer is: **true-ish, and importantly mis-stated.** The claim gets the *symptom* roughly right and the *mechanism* wrong—and if you have the

mechanism wrong, you cannot make the right adjustments.

Why it is mis-stated. Accounting recognizes intangibles perfectly happily when they meet the FASB's asset criteria. Buy a patent for $1m and you debit *Patents* for $1m, exactly as you would debit *Equipment*; you then amortize it over its expected useful life, which may be shorter than the remaining legal term. Buy a trademark and it goes on the books too; whether it is amortized depends on whether its expected useful life is finite or indefinite. The operative distinction is not **tangible vs. intangible** at all. It is **purchased vs. internally developed.**

Set the two journal entries side by side and the whole issue is visible in four lines. 'Build':

Research and Development Expense (-S.E.):	$1m	
Cash (-A):		$1m

'Build': pay scientists to develop technology internally

...versus 'buy':

Patents (+A):	$1m	
Cash (-A):		$1m

'Buy': purchase a patent on the open market

The same cash leaves the firm. The statements could hardly look more different.

Why the claim nonetheless has bite. Because of a historical accident that is not really an accident: many of the most economically valuable intangibles in the modern economy—brands, software, ecosystems, organizational know-how, drug pipelines—happen to be the internally-developed kind. Apple's design and brand were built, not bought. So the *category* that accounting declines to capitalize overlaps heavily with the *category* people mean when they say 'intangibles.' That overlap is what gives the sloppy version of the claim its surface plausibility.

The FASB's side of the argument, which is not weak. Recognizing an asset requires a verifiable measurement. When we buy a patent, an arm's-length transaction hands us a price. When we hire scientists for ten years and the PTO eventually grants us a patent, there was never a transaction with a price—not even at the grant, since no consideration changes hands—and, more to the point, we cannot verifiably match R&D outlays to future sales. Most R&D fails. If capitalization were permitted, 'we expect this to pay off' would become the most abused

sentence in corporate finance, and the balance sheet would fill with managerial optimism. Recall the 'capitalize it all!' temptation from the getting-ready-for-use discussion: in a well-run firm, *everything* can be argued to serve future sales.

The critics' side, which is also not weak. The matching principle plainly says these costs belong in the periods the resulting products generate revenue. And there is a policy worry: if R&D depresses reported earnings, and managers are judged on reported earnings, we may be discouraging exactly the investment with the largest positive externalities.

The implications, which are the real payoff. Three of them.

First, **understated book equity.** A firm that builds carries none of its most valuable assets on the balance sheet, so the denominator of ROE is hollowed out.

Second, **overstated current earnings.** Once the internally-developed intangible starts producing revenue, that revenue arrives with *no* associated amortization—the investment was expensed years ago. CEOs are eager to point out the first effect and quiet about the second.

Third, and the reason this matters for anyone reading statements: **comparability breaks down.** The build-vs.-buy pair makes this vivid. From year 10 on, Company A and Company B own the same patent and earn the same $2m a year from it. A reports $2m of income from it; B reports $1m, after amortization. A looked terrible for a decade and fabulous ever after; B never looked bad and now looks merely fine. An analyst pulling 'comps' at year 15 sees wildly different margins, ROA, and ROE from the same later revenues but different acquisition-cost histories. Apple's historical example also shows why book-equity returns need interpretation; intangibles and repurchases can both matter, and the ratio alone does not isolate their effects.

Where I'd push further than the standard account. I would add that accounting recognizes intangible value more than the critics allow—just not where they are looking. It shows up on the **income statement, in the margins.** The reason Apple can charge what it charges is its brand, its design, and its ecosystem. Those intangibles are not on the balance sheet as a stock, but their value is being reported every quarter as a flow, in a gross margin that a commodity manufacturer could never sustain. So the fairest version of the claim is narrow: *accounting does not recognize internally-developed intangibles as assets on the Balance Sheet.* Stated that way, it is true, and it is something you can adjust for.

3. We recently learned that when companies incur costs to get their assets ready for use, they capitalize those costs (add them to the asset balance). This would seem odd to accounting novices. Can you explain the logic of this—in particular,

can you relate it to the core accounting principle of matching? (Remember the discussion of Shipping and *Inventory* and subsequent *COGS*.)

Solution. The novice's objection is worth stating plainly, because it is a good objection: we *paid out cash*, which is a bad thing, a cost—and yet the accounting *increases* an asset. How can spending money make you richer on paper?

Here is the short answer: capitalizing a cost is not a claim that it wasn't a cost. It is a decision about **when** to recognize it as an expense. Recall the framing from earlier in the book—**assets are deferred expenses**. Putting the shipping charge into *Inventory* is not denying that it is an expense; it is scheduling it, to be released later, as part of *COGS*, on the day the goods are sold.

Now the matching argument, in its cleanest form. Take the retailer who pays $10,000 for 100 units and $500 to ship them. Ask the question matching always asks: *in which period will these expenditures help generate revenue?* Both of them—the goods and the freight—are helping generate revenue on the day the units are sold, and not a moment before. Neither expenditure has done anything for us yet; the goods are sitting in the stockroom. So both should be expensed then, together, against that revenue. Capitalizing the $500 into the batch is simply what makes that possible: it raises the unit cost from $100 to $105, and when ten units go out the door we recognize $1,050 of COGS.

Consider the alternative, and notice that it is mis-matched at *both* ends. If we expensed the freight immediately, we would recognize $500 of cost in a period with no corresponding revenue—too early, and understating that period's profit—and then, when the sale finally arrived, we would charge only $1,000 of COGS against it—too little, and overstating *that* period's profit. One economic act, split across two periods, mis-stating both.

The same logic, one asset over. The nonrefundable purchase tax on MattCorp's equipment belongs to the acquisition cost. It is capitalized with the equipment and allocated through depreciation over the periods of use. Paying an unrelated tax liability from the buyer's past operations would be a different transaction and would not become an equipment cost merely because payment was demanded before the purchase.

Why the nature of the cost matters. The purchase tax is part of acquiring the asset. Insurance for the following year provides a separate period of coverage, so it is expensed over that coverage period, with any unexpired prepaid coverage recorded separately. Being nonrecurring is not sufficient for capitalization; the cost must be attributable to the asset and meet the applicable recognition rules.

And a reason to be glad of the rule beyond the theory. Imagine there were no

capitalization rule for freight. I am a newly hired CEO who would love to blame my predecessor for everything and then engineer a dramatic turnaround. So I call my supplier: “Rather than $10,000 for the goods and $500 for shipping, could you invoice me $2,000 for the goods and $8,500 for shipping?” I pay $10,500 either way, for identical goods delivered identically—nothing economic has changed. But without capitalization I could expense $8,500 now, sink my predecessor’s final year, carry the inventory at $2,000, and post spectacular margins next year when I sell it. The capitalization rule forces economically identical arrangements to be accounted for identically, and takes the game off the table. That, in the end, is the deepest defense of the rule: matching is the principle, and resistance to presentational gamesmanship is the payoff.

A.12 Chapter 12

A.12.1 Practice Problems: MCQ's

1. Which statement best describes the fair value of an asset under the market-participant measurement framework?

   (A) The price that maximizes overall social well-being.

   (B) The estimated price received to sell the asset in an orderly transaction between market participants at the measurement date.

   (C) Historical cost less accumulated depreciation.

   (D) Whatever value the current owner assigns to the asset.

   Solution. The answer is **(B)**. A discounted-cash-flow model can estimate that price if it uses market-participant assumptions. A company-specific valuation is not automatically fair value, and fair value is not necessarily book value.

2. **Scenario:** Suppose our company buys **equipment and the property** it is on, for $1 million. Our appraisal/valuation experts and advisers estimate the "fair value" of the **Equipment** and **Land** to be **$660k** and **$440k**, **respectively**. What would be the journal entry for this transaction?

   (A) Debit Equipment $440k, Debit Land $660k; Credit Cash $1.1m

   (B) Debit Equipment $440k, Debit Land $660k; Credit Cash $1m, Credit Gain $100k

   (C) Debit Equipment $400k, Debit Land $600k; Credit Cash $1m

   (D) Debit Equipment $500k, Debit Land $500k; Credit Cash $1m

   (E) None of the other answers

   Solution. This is a **basket purchase**: two assets that must live in two different accounts (because we depreciate *Equipment* and do not depreciate *Land*), bought for one lump-sum price. The rule is that the appraisals supply the *proportions* and the purchase price supplies the *total*. The appraisals sum to $660k + $440k = $1.1m, so *Equipment* is 60% and *Land* is 40% of the appraised total. Apply those percentages to the $1m we actually paid:

Equipment (+A):	$600k	
Land (+A):	$400k	
Cash (-A):		$1m

Basket purchase of equipment and land:

Now check the options against that. (A) and (B) commit the classic error of recording the assets at the *appraised* \$1.1m rather than at the \$1m we paid—and (B) then manufactures a \$100k *Gain* to plug the hole. There is no gain here. We did not become \$100k richer by buying something; we stayed **anchored to the purchase price**, and used the appraisals only to split it. (D) splits the price 50/50, ignoring the appraisals entirely. And (C) has the right two dollar amounts—\$600k and \$400k—but attached to the wrong two accounts: it puts \$400k on *Equipment*, the asset our appraisers valued more highly, and \$600k on *Land*. Read the "respectively" carefully; the 60% belongs to *Equipment*. So the answer is **(E)**.

3. Our company owns several different patents. At the end of FY2019, our company reported Patents (at book value) of \$900k. During FY2020, we recorded a \$300k impairment Loss on one of our patents, and recorded total *Amortization* on our patents of \$200k. The total book value of Patents on our FY2020 Balance Sheet was \$700k. How much did we purchase in *Patents* over the period?

   (A) \$200k

   (B) \$300k

   (C) \$400k

   (D) \$700k

   (E) None of the other answers.

   Solution. This is the familiar beginning-balance-and-flows problem, with the one new twist from this chapter: an **impairment loss** reduces the asset without being an expense. So we use the general form of the formula:

$$Asset\ End.\ Bal. = Asset\ Beg.\ Bal. + Purchases - Related\ Expense - Related\ Impairment\ Loss$$

   Everything here is given except purchases, so solve for it:

$\$700k = \$900k + Purchases - \$200k - \$300k$

$Purchases = \$700k - \$900k + \$200k + \$300k = \$300k$

   The trap is (A): if you forget the impairment and use the old, simpler formula ($\$700k = \$900k + P - \$200k$), you get \$0k—or, if you drop the amortization instead and keep the impairment, \$100k. Neither is offered, which is the exam-writer's mercy. The point stands, though: the impairment is a *third* thing draining the account, and you have to add it back when you work the formula in reverse. So the answer is **(B)**.

4. Early on in the pandemic, our company purchased the patent for Remdesivir, an anti-viral drug believed to have some efficacy against SARS-CoV-2. Its current book value is $10m. Due to the development and rollout of Paxlovid, another drug with apparently higher efficacy, we now worry that our patent may be impaired. Our financial experts do an analysis, and determine that the expected net cash flow from the asset is $11m and the discounted expected future cash flows are $10.2 million. What journal entry would we make as a result?

   (A) Debit Loss $.2m; Credit Patents $.2m

   (B) Debit Patent $.2m; Credit Gain $.2m

   (C) Debit Cash $10.2m; Credit Gain $.2m; Credit Patents $.2m

   (D) No journal entry as a result.

   (E) None of the other answers.

   Solution. Line the three numbers up. Book value: $10m. Undiscounted expected net cash flows: $11m. Fair value—the discounted expected future cash flows: $10.2m. Both of the estimates are *above* the $10m book value, so the asset fails the impairment test in the good direction: it is not impaired, and there is nothing to write down.

   Now, the tempting move is (B): fair value is $10.2m, book value is $10m, so why not write the patent *up* by $0.2m and book a gain? Because of **conservatism**, which is asymmetric by design. We are required to recognize revisions in value *downward*, and forbidden from recognizing them *upward*. That asymmetry is not an accident or an oversight—it is the whole posture of the accounting system, which we first met back in Chapter 1. (A) writes the asset down, which is exactly backwards. (C) debits *Cash*, but no cash changed hands; we tested an asset, we did not sell one. So the answer is **(D)**.

5. Which of the following would be implicitly "baked in" to Goodwill after an acquisition?

   (A) The value of the target's outstanding *Accounts Receivable*

   (B) The value of the target's *Patents*

   (C) The value of the target's *Trademarks*

   (D) Multiple other answers.

   (E) None of the other answers.

   Solution. Goodwill is a *residual*—the plug for whatever is left of the purchase price after the target's **identifiable** assets and liabilities have been written up to

fair value and counted. So the question is really: which of these is *not* identifiable? And the rule of thumb serves us well here: for an acquired intangible, look for contractual or legal rights, or separability. *Accounts Receivable*, *Patents*, *Trademarks*—you have seen all three, in this very book. Every one of them goes into **Net Identifiable Assets**, and, by definition, anything in Net Identifiable Assets is *not* in Goodwill.

The trap is (B) and (C), and it is worth naming. The tempting reasoning is: "internally developed patents and trademarks aren't on the target's books at all, so their value must be hiding in Goodwill." That overlooks the first step of acquisition accounting. In an acquisition—the one place in this book where assets get written *up*—we must recognize the target's internally developed intangibles at their estimated fair value. Once written up, they are identifiable, and they are out of Goodwill. So none of (A), (B), (C) qualifies, and (D) is therefore out too. So the answer is **(E)**.

6. ExpoCorp makes an acquisition of SimpleCorp, paying $13m. Immediately prior to the acquisition, SimpleCorp had the following assets and liabilities, at the following book values. Prepaid Rent: $2m, Trademarks: $1, Patents: $0m, Deferred Revenue: $1m, Salaries Payable: $1m, Notes Payable: $3m. Our internal valuation experts estimate that the fair value of SimpleCorps's Trademarks and Patents, respectively, are $4m and $3m. (And the fair value of all other identifiable assets and liabilities is the same as their book value.) What is the *Goodwill* we would record upon the acquisition?

   (A) $5m

   (B) $3m

   (C) $8m

   (D) $2m

   (E) None of the other answers.

   Solution. Two steps, always in this order. **First**, compute Net Identifiable Assets at fair value—every identifiable asset, minus every liability, with the *Trademarks* and *Patents* written up to the experts' numbers and everything else left at book:

$$NIA = \underbrace{\$2m}_{Prepd.\ Rent} + \underbrace{\$4m}_{Trdmks.} + \underbrace{\$3m}_{Patents} - \underbrace{\$1m}_{Def.\ Rev.} - \underbrace{\$1m}_{Sal.\ Pay.} - \underbrace{\$3m}_{Notes\ Pay.} = \$4m$$

   Second, plug the difference:

$$Goodwill = \$PP - NIA = \$13m - \$4m = \$9m$$

   $9m is not on the menu, so the answer is **(E)**.

Two things to watch here. The first is *Deferred Revenue*. It is easy to skim past it, or to treat it as some odd non-item, but it is a **liability**—an obligation to deliver goods or services we have already been paid for—and it comes over to the acquirer along with everything else. Leave it out and you get $5m of NIA and $8m of Goodwill, which is option (C), sitting right there waiting for you. The second is the $1 book value of the *Trademarks*: it is a red herring. Once our valuation experts hand us a fair value of $4m, the prior book value plays no role whatsoever. That is the whole point of the write-up step.

7. Our company purchases *Land* from a seller for non-cash consideration, exchanging 1 million shares of our *Common Stock*. Our company is publicly traded and, on the date of the deal, our stock price is $50 per share. What would be the journal entry for this transaction?

   (A) Debit Cash $50m, Credit Common Stock $50m

   (B) Debit *Land* $50m, Credit *Common Stock* $50m

   (C) Debit *Land* $50m, Credit *Cash* $50m

   (D) Debit *Land* $1m, Credit *Common Stock* $1m

   (E) None—no cash purchase price.

   Solution. No cash changes hands, so there is no purchase price to anchor on—which is our cue that **fair value** must come into play. We need an estimate of what one side or the other of this swap is worth, and for a publicly traded company the easiest, most reliable estimate is right there on the exchange: 1 million shares at $50 per share is $50m of **non-cash consideration**. That gives us both the amount and the entry:

Land (+A):	$50m	
Common Stock (+S.E.):		$50m

   Land acquired for 1 million shares, trading at $50 per share:

   Option (E) encodes the confusion this problem is built to catch: "there was no cash, so there is no number, so there is no entry." But we really did acquire a plot of land, and it really is an accounting asset; refusing to record it would leave the balance sheet simply missing something we own. The absence of a cash price is a *measurement* problem, not a reason to skip the transaction—and fair value is how we solve it.

The other three options are each wrong in exactly one place, which is a useful way to see what the entry is actually recording. Option (A) has the right $50m but debits *Cash*, which is backwards: we gave up stock and received land; no cash came in. Option (C) correctly debits *Land* for $50m but credits *Cash*, which says we paid for the land in cash—and we did not; our cash balance is untouched by this deal. Option (D) debits and credits the right two accounts, but for $1m, which is the *number of shares* rather than their *value*. Shares are not dollars; we have to price them, and $50 each is the price the market is quoting us. So the answer is **(B)**.

8. Our company owns a plot of *Land* in Malibu that is currently on our books for $3m. However, due to a historic drought and unprecedented fire risk, we worry the area has become less desirable, and the *Land* may be impaired. Our valuation experts test for impairment, and determine that the expected net cash flows from the plot of *Land* are $2.6m, and the new fair value of the property is $2.2m. What is the journal entry we would make after determining this?

   (A) Debit Depreciation Expense $.8m; Credit Land $.8m

   (B) Debit Loss $.8m; Credit Land $.8m

   (C) Debit Loss $.4m; Credit Land $.4m

   (D) None, because you do not depreciate *Land*

   (E) None of the above are correct.

   Solution. Both estimates are below the $3m book value—undiscounted net cash flows of $2.6m, fair value of $2.2m—so the *Land* is unambiguously impaired, and the mechanics are the simple ones: write the asset down to its **fair value**, and let a *Loss* absorb the difference. The new book value must end up at $2.2m, so the credit to *Land* is $\$3m - \$2.2m = \$0.8m$:

Loss (-S.E.):	$0.8m	
Land (-A):		$0.8m

Impairment of the Malibu land:

Three distractors, three real confusions. (A) has the right magnitude but calls it *Depreciation Expense*. It is not depreciation—we are not allocating cost over a useful life—and it is not an expense at all; it is a **Loss**, which behaves like an expense on the income statement but is classified separately precisely because it is one-time and not part of core operations. (C) writes the asset down to $2.6m,

the *undiscounted* net cash flow figure. But that number does only one job: it triggers the test. Once the asset is impaired, the writedown is always measured to fair value. And (D) is the sharpest one, because the premise is *true*: you really do not depreciate *Land*. But impairment is not depreciation. Land is a long-term asset like any other, and it must be tested for impairment like any other—a fact that becomes obvious the moment you picture a Malibu hillside that nobody wants to build on anymore. So the answer is **(B)**.

9. Which of the following would be implicitly "baked in" to Goodwill after an acquisition?

   (A) The value of the target's Patents.

   (B) The value of intangibles such as Prepaid Rent.

   (C) The value of the target's Trademarks.

   (D) The value of positive social-media buzz about the target's products.

   (E) Multiple other answers.

   (F) None of the other answers.

   Solution. Apply the recognition criteria: the patents and trademarks arise from legal rights, and Prepaid Rent is an identifiable contractual resource. *Patents*: yes. *Prepaid Rent*: yes, and it is about as identifiable as an asset gets. *Trademarks*: yes. All three are identifiable, all three get written up to fair value in Step 1, and all three therefore land in **Net Identifiable Assets** rather than in Goodwill.

   For this question, "positive social-media buzz" means a general reputational benefit that does not create separate contractual rights and is not separable. It is not separately recognized on those assumptions, so any value paid for it remains within Goodwill. Other customer-related or marketing-related intangibles can be separately identifiable; do not generalize from the label alone. The answer is **(D)**.

10. ExpoCorp makes an acquisition of SimpleCorp, paying $13m. Immediately prior to the acquisition, SimpleCorp had the following assets and liabilities, at the following book values. Prepaid Rent: $2m, Trademarks: $1, Patents: $0m, Salaries Payable: $1m, Notes Payable: $3m. Our internal valuation experts estimate that the fair value of SimpleCorps's Trademarks and Patents, respectively, are $4m and $3m. (And the fair value of all other identifiable assets and liabilities is the same as their book value.) What is SimpleCorp's Net Identifiable Assets?

    (A) $15m

    (B) $5m

(C) $10m

(D) $8m

(E) None of the other answers.

Solution. Net Identifiable Assets is the *net* fair value of the target's identifiable assets and liabilities—so add up the assets at fair value, and subtract the liabilities:

$$NIA = \underbrace{\$2m}_{Prepd.\ Rent} + \underbrace{\$4m}_{Trdmks.} + \underbrace{\$3m}_{Patents} - \underbrace{\$1m}_{Sal.\ Pay.} - \underbrace{\$3m}_{Notes\ Pay.} = \$9m - \$4m = \$5m$$

Notice that the $13m purchase price plays no part in this calculation. NIA is a statement about the target's own assets and liabilities; the purchase price only enters at the *next* step, when we plug for Goodwill. (D), $8m, drops the $3m of *Notes Payable*—a mistake worth guarding against, since if liabilities vanished on acquisition, every indebted company in America would be shopping itself to a buyer. (A), $15m, is not a near-miss of this calculation at all; if you landed there, you have let the $13m purchase price contaminate a figure that should not contain it. So the answer is **(B)**.

11. ExpoCorp makes an acquisition of SimpleCorp, paying $13m. Immediately prior to the acquisition, SimpleCorp had the following assets and liabilities, at the following book values. Prepaid Rent: $2m, Trademarks: $1, Patents: $0m, Salaries Payable: $1m, Notes Payable: $3m. Our internal valuation experts estimate that the fair value of SimpleCorps's Trademarks and Patents, respectively, are $4m and $3m. (And the fair value of all other identifiable assets and liabilities is the same as their book value.) What is the *Goodwill* we would record upon the acquisition?

    (A) $5m

    (B) $3m

    (C) $8m

    (D) $2m

    (E) None of the other answers.

    Solution. This is the previous problem, carried one step further. We already have $NIA = \$5m$. Goodwill is the plug between what we paid and what we can identify:

    $$Goodwill = \$PP - NIA = \$13m - \$5m = \$8m$$

    Option (A) is a trap for the reader who solves for NIA and stops there without doing the subtraction. So the answer is **(C)**.

12. Which of the following is a scenario in which "fair value" could be used in accounting?

(A) Basket purchase

(B) Impairment

(C) An acquisition

(D) Multiple other answers

(E) None of the other answers

Solution. All three, and each for its own distinct reason—which is really the organizing question of this chapter. In a **basket purchase**, the total purchase price is perfectly reliable but does not tell us how to split it between two accounts, so relative fair values supply the proportions. In an **impairment**, historical cost is available but conservatism says we may no longer stand behind it, so we write the asset down to fair value. In an **acquisition**, we write the target's identifiable assets and liabilities *up* to fair value, because the target's own book values were never meant to represent what the acquirer just paid for.

Notice that in none of these three cases did we abandon historical cost because we suddenly decided that chasing true value was a good idea. In each, our hand was forced—the price was unavailable, or unreliable, or the wrong price for the question being asked. So the answer is **(D)**.

A.12.2 Practice Problems: Open Prompts

- ExpoCorp makes an acquisition of SimpleCorp, paying $15m. Immediately prior to the acquisition, SimpleCorp had the following assets and liabilities, at the following book values. *Prepaid Rent: $2m, Trademarks: $0, Patents: $0m, Salaries Payable: $1m, Notes Payable: $8m.* Our valuation experts estimate that the fair value of SimpleCorps's *Trademarks* and *Patents*, respectively, are $3m and $2m. (And the fair value of all other identifiable assets and liabilities is the same as their book value.) What is the journal entry for the acquisition?

  Solution. Work it in the standard two steps, then write the entry. **Step 1**, Net Identifiable Assets, with the intangibles at fair value and everything else at book:

  $$NIA = \underbrace{\$2m}_{Prepd.\ Rent} + \underbrace{\$3m}_{Trdmks.} + \underbrace{\$2m}_{Patents} - \underbrace{\$1m}_{Sal.\ Pay.} - \underbrace{\$8m}_{Notes\ Pay.} = \$7m - \$9m = -\$2m$$

  Step 2, plug for Goodwill:

  $$Goodwill = \$PP - NIA = \$15m - (-\$2m) = \$17m$$

  Now, that minus sign may have given you pause, so let me address it head-on: SimpleCorp's identifiable *liabilities* exceed its identifiable *assets*, so its net identifiable assets are **negative $2m**. Is that a problem? Not at all—and note carefully

that this is not the exotic case of *negative Goodwill*. Negative Goodwill would mean paying *less* than net identifiable assets, which essentially never happens in ordinary M&A. Here it is *NIA itself* that is negative, and Goodwill is correspondingly enormous. Economically, ExpoCorp is paying $15m in cash *and* taking on a company whose identifiable balance sheet is $2m in the hole; so ExpoCorp must believe there is $17m in the residual after separately identifiable items have been measured, reflecting such things as expected synergies and other benefits, or possibly overpayment. Just be careful with the arithmetic: subtracting a negative adds.

Everything that comes over gets recorded, assets debited and liabilities credited, with *Cash* credited for the purchase price:

Prepaid Rent (+A):	$2m	
Trademarks (+A):	$3m	
Patents (+A):	$2m	
Goodwill (+A):	$17m	
Salaries Payable (+L):		$1m
Notes Payable (+L):		$8m
Cash (-A):		$15m

ExpoCorp acquires SimpleCorp

Debits total $24m; credits total $24m. And notice the *Trademarks* and *Patents*, which came onto our books at $3m and $2m despite having book values of zero at SimpleCorp. That is not a mistake—it is Step 1 doing its job. SimpleCorp presumably developed them internally and expensed the costs as incurred, so they were never on its books at all; the acquisition is the one moment when accounting lets them in at fair value.

- Early on in the pandemic, our company purchased the patent for *Remdesivir*, an anti-viral drug believed to have some efficacy against SARS-CoV-2. Its current book value is $10m. Due to the development and rollout of unexpectedly-effective vaccines, we now believe that our patent may be *impaired*. Our financial experts do an analysis, and determine that the expected net cash flow from the asset is $9.5m and the discounted expected future cash flows are $8 million. **What journal entry would we make** as a result?

  Solution. Three numbers, and the order of operations matters. Book value is $10m. The undiscounted expected net cash flows are $9.5m, which is *below* book

value—so the asset fails Step 1 and is deemed **impaired**. Having concluded that, Step 2 measures the writedown to the supplied market-participant **fair value** of \$8m. So the \$9.5m has served its purpose and drops out of the calculation entirely; it never touches the entry.

We need the *Patents* balance to end up at \$8m, so we credit it for the difference, $\$10m - \$8m = \$2m$, and the balancing debit is a *Loss*:

Loss (-S.E.):	\$2m	
Patents (-A):		\$2m

Impairment of the Remdesivir patent:

In T-account form, you can see that the ending balance *is* the fair value, by construction:

Patents

Debits	Credits
\$10m	
	\$2m
\$8m	

Two follow-ups worth having straight. First, the writedown does not take the patent to zero—\$8m of book value survives, and we go on amortizing *that* over the remaining useful life, at a correspondingly lower annual amortization expense. Second, the debit is a *Loss*, not an expense. It reduces Net Income just as an expense would, but it is classified separately because an impairment is a one-time, non-repeatable event, not part of the core operations of the firm.

A.12.3 Essay and Discussion Questions

1. What is Goodwill in accounting? What is the technical definition? And what does it represent, conceptually? What are some examples of "economic assets" that would be implicitly captured by *Goodwill*?

   Solution. Here is how I would work through this, in three passes: the definition, the concept, and the examples.

   Start with the **technical definition**, because it is the part that must be exactly right. *Goodwill* is an intangible asset account that arises only in an acquisition, and it is defined as a residual: the purchase price paid for the target, minus

the net fair value of the target's **identifiable** assets and liabilities. In symbols, $Goodwill = \$PP - NIA$. Notice what that definition does *not* say. It does not say that Goodwill measures customer loyalty, or brand strength, or synergies, or the premium over the target's pre-deal stock price. It is a *plug*—the number that makes the acquisition entry balance once everything identifiable has been written up to fair value and recorded.

Now the **concept**, which is where the interesting part lives. Why does a plug turn out to be necessary at all? Because an acquisition is the one moment in accounting where two incompatible number systems collide head-on. On one side is the target's **book value**—assets at historical cost, conservatively stated, with internally developed intangibles not on the books at all. On the other is the **purchase price**: a negotiated market number, reflecting what the acquirer thinks the whole enterprise is economically worth. Throughout this book we have said, comfortably, that book value is not true value and that this is fine, because the two never have to meet. In an acquisition they meet. The purchase price must be recorded—the cash really left—and the target's assets must be recorded, and the two do not reconcile. Goodwill is the bridge across that gap. So, conceptually, Goodwill represents *everything real that the acquirer paid for and the accounting system cannot separately name.*

Which brings us to the **examples**. Patents and trademarks arise from legal rights; some customer relationships and contracts can also be recognized separately. An assembled workforce and expected synergies generally remain within Goodwill. For the questions here, general "buzz" and undifferentiated reputation are assumed not to meet the separate-identification criteria. The relevant distinction is contractual rights or separability, not whether the item has appeared under a familiar label in a Balance Sheet.

Goodwill can be a substantial part of an acquisitive company's assets, while remaining a residual that cannot be broken into a tidy list of separately recognized items. That is what makes it important to investigate the acquisitions and expectations behind the balance. Its size alone does not tell us whether the deals were wise.

2. What do you make of Impairments in accounting? What are the accounting principles justifying them (i.e., why does the FASB have these rules in place)? Why do Wall Street analysts often "back out" impairment losses, in their "street" earnings?

   Solution. Let me take the FASB's justification first, then the analysts' behavior, and then tell you where I actually come down—because I do not think this rule

is obviously correct, and you should not either.

The principle behind the rule is **conservatism**, applied to the balance sheet. When credible evidence arrives that a long-term asset is worth far less than its carrying amount—a superior rival drug, a technology gone obsolete, a hillside nobody wants to build on—the FASB's position is that the balance sheet should say so, now. Recognize the bad news when you learn it; do not carry an asset at $10m that everyone in the building knows is worth $2m. And there is a real absurdity in the alternative: without the rule, we would solemnly record $1m of amortization expense every year for a decade, on a patent generating exactly nothing, and call the resulting income statement a faithful picture of anything.

Why analysts back it out follows from the *shape* of an impairment. Think of what the rule actually does to reported earnings: it takes bad news about the *future* and pulls all of it into the *current* period, at once. It is **anti-smoothing**, the exact opposite of the matching-principle logic behind depreciation. A $10m impairment is a one-time, non-recurring, non-persistent event. If you are an equity analyst, your job is to forecast next year's earnings and to value the company off its sustainable earning power—often as some multiple of earnings. Including a giant one-time charge in this year's number makes that number *less* informative about next year. So analysts strip it out, and companies help them by reporting "adjusted" or "non-GAAP" earnings that exclude it. The cynical read—that companies exclude losses simply because it makes them look better—is not wrong about the *effect*, but it misses that sophisticated investors, who have no incentive to be fooled, make the very same adjustment in their own models.

But I want to be careful here, because there is a sharp objection and it deserves a straight answer. If analysts always exclude impairments and never do anything else about them, they are systematically ignoring real economic losses. In a world with no impairment rule, the analyst would have been forced to confront that $10m one $1m slice at a time, for ten years running; in the actual world, she excludes the one-time charge and never confronts it at all. That is a genuine failure mode, and it is what separates a good analyst from a lazy one. The right treatment is to view the impairment as a one-time *accounting recognition* of a long-run *economic* loss—strip it out of the earnings figure you are extrapolating from, and then put the bad news back into your forecast and your valuation model, spread across the periods it actually belongs to.

So what do I make of impairments? Candidly: I am not convinced. Go back to the ten-year patent. What exactly was so terrible about letting the loss show up gradually, period by period? The income statement would be smoother, more interpretable, more forecastable—and we would not need an entire parallel uni-

verse of "adjusted" earnings measures whose main function is to undo the rule. The revealed preference of companies *and* investors, who both systematically remove impairments from earnings, strikes me as a quiet vote of no confidence.

That said, I hold this view knowing exactly why a reasonable person disagrees, and the disagreement is instructive: it is really a fight about **which financial statement gets priority**. My instinct is income-statement-oriented—I want a smooth, interpretable measure of ongoing profitability. The FASB's rule is balance-sheet-oriented—if the asset is impaired, say so on the balance sheet immediately. And which statement you care about depends on who you are. An equity investor doing valuation wants smoothing, for the forecasting reasons above. A **creditor** wants the opposite, and from Chapter 1 first principles you can see why: creditors do not share in the upside, only the downside, so they want bad news recognized fast and hard, not smoothed over a decade. Managers, meanwhile, have their own angle—consider the **"big bath,"** where a newly arrived CEO impairs everything in sight in year one, blames the prior regime, and then presents years of glorious "growth" off the depressed base. For that CEO, this rule is not a burden; it is a gift.

Which is the real lesson to take away. The rule is mandatory and the auditor will enforce it in undeniable cases. But most cases are not undeniable—sales are soft, a competitor has appeared, the future is murky—and in that gray zone, whether an impairment "indicator" exists at all is a judgment call made by management. The rule is written in the standards; its application runs on judgment; and judgment runs on incentives.

3. What is "fair value"? And when does it come into play in accounting, based on what we've seen so far in this class—and why?

   Solution. Let me answer the "what" quickly, spend most of the time on the "when," and then say what I think the whole pattern *means*—because the pattern is the real answer here.

   What it is. For an asset, fair value estimates a market-participant exit price in an orderly transaction at the measurement date. Quoted prices, comparable transactions, and valuation models can supply evidence. A discounted-cash-flow estimate must use market-participant assumptions to serve that objective. It is not a uniquely knowable "true value," nor whatever value the present owner expects to obtain.

   When it comes into play. Five situations, and I want to state each one along with the specific reason our hand was forced:

   In a **basket purchase**, we buy a factory and the land under it for one lump sum.

Historical cost is perfectly available in total—the $1m credit to *Cash* is as verifiable as anything gets—but it does not tell us how to split that total between an account we will depreciate and one we will not. So we use *relative* fair values to supply the proportions. Note how minimal the intrusion is: fair value determines the 60/40, and the purchase price still determines the total.

In a transaction with **non-cash consideration**—paying for equipment, or land, or a lease with shares of our own stock—there is simply *no purchase price to be had*. Something real entered our balance sheet, the monetary unit assumption demands a dollar figure, and there is no verifiable transaction price to supply one. Fair value is the only available number.

In a **related-party transaction**, the stated price is not automatically evidence of an arm's-length value. That requires scrutiny and disclosure, but not every such transaction is remeasured at fair value. The applicable accounting depends on the transaction. In a consolidated parent-subsidiary group, internal sales and gains are eliminated. The WeWork example illustrates the governance concern; it does not create a universal fair-value measurement rule.

In an **impairment**, historical cost is both available and reliable—and we override it anyway, because of **conservatism**. And in an **acquisition**, we go further still, writing the target's identifiable assets *up* to fair value: the single place in this entire book where an asset is written up.

And now the "why," which is the part I actually care about. It would be easy to look at that list and conclude that accounting has been gradually losing its nerve—that historical cost was the old religion, fair value is the modern one, and the chapter is documenting a slow surrender. I want to argue against that reading.

Look at the reasons again. In three of the five cases, we did not turn to fair value because we decided true value was a worthier goal. We turned to it because historical cost had *failed*—it was unavailable as a cash amount (noncash consideration) or incomplete as an allocation (basket purchase). Related-party prices raise a separate reliability issue whose accounting depends on the transaction. In each of these, fair value is not the ideal; it is the **lesser evil**, adopted narrowly, for a specific job, and abandoned again the moment historical cost can do the work. That is why the basket-purchase rule is so carefully constructed: it uses the appraisals only for the *split*, and stays anchored to the actual $1m for the *total*. It gives up exactly as much as it has to and not one dollar more.

Impairment is the genuine exception, and it is worth being honest about that—as, in its way, is acquisition, where we write *up*; but there the target's book values

were never evidence about the acquirer's price in the first place, so there is no historical cost being overruled, only one that was never on point. There, historical cost is available and reliable and we overrule it anyway. But notice the *shape* of the override: it is **asymmetric**. We write down, never up. That asymmetry is the tell. A system that had converted to fair value as a matter of principle would mark assets in both directions, quarterly, as values moved. Ours marks only in one direction, only when triggered—which is not a fair-value regime at all. It is a historical-cost regime with a conservatism override.

So here is my position. Fair value in GAAP is not a rival philosophy creeping in at the edges of historical cost; it is a set of **patches**, each one installed at a specific point where the historical-cost machine breaks down, and each one carefully limited in scope. That is why I keep pressing the question, at every one of these situations: *why, exactly, can't we rely on historical cost here?* If you can answer that question in each case, you understand fair value. If you cannot—if your answer is just "because fair value is more accurate"—then you have missed what the accounting system is actually doing, which is guarding its verifiability jealously and giving it up only under duress.

I will add the caveat that this is a live and fast-moving area, and the standards will very likely shift during your career. And there are hard cases at the frontier where I hold my own view loosely: should a manufacturer mark its portfolio of stocks to market every quarter, running the gyrations through earnings? (For a bank, whose business *is* holding financial assets, I think plainly yes. For a widget-maker, I am much less sure the noise is worth the accuracy.) Should we write up the Culver City office we have owned for twenty years? I would say no—and notice that my reason is a historical-cost reason: appraisals of unique commercial real estate are soft, unverifiable, and management-influenced, and letting a company book profit off them invites exactly the abuse conservatism exists to prevent.

A.13 Chapter 13

A.13.1 Practice Problems: MCQ's

1. We are considering purchasing an asset that promises to pay us $600 one year from today, and $10,600 two years from today. Based on our assessment of the riskiness, we determine that the appropriate interest rate is 6%. What is the present value of this asset? (Note: if your calculation is within a couple percent of one of the offered multiple choice answers, you should reasonably assume that's a rounding-error issue, and choose that answer.)

   (A) $10,566.03

   (B) $10,000.00

   (C) $8,833.96

   (D) None of the other answers.

   (E) Not enough information.

   Solution. Two cash flows, so two terms. Discount each by $(1+i)$ raised to the number of periods you have to wait for it: the $600 arrives in one year, so it gets discounted once; the $10,600 arrives in two years, so it gets discounted twice. With $i = .06$:

   $PV = \frac{\$600}{1.06} + \frac{\$10{,}600}{1.06^2} = \$566.04 + \$9{,}433.96 = \$10{,}000.00$

   Option (A) is what you get if you fail to discount the second payment for the *second* period—that is, if you divide the $10,600 by 1.06 rather than by 1.06^2. That is the single most common error in this whole chapter, so it is worth naming: the exponent is the number of periods you wait, not a decoration. (C) is what you get if you *subtract* the first payment instead of adding it—$9,433.96 - $600—which is a slip of a sign rather than of a concept, but it costs the same. (E) is a temptation only if you have not yet noticed that the identity needs exactly three things—the cash flows, the timing, and the rate—and we have all three.

   One more thing is worth noticing here, because it is not a coincidence and it will matter enormously next chapter. This asset is a two-year bond with a $10,000 face value and a 6% coupon: $600 of interest in year one, then $600 of interest plus the $10,000 principal in year two. We discounted it at 6%—exactly its stated coupon rate—and its present value came out to exactly its face value, $10,000. When the market interest rate equals the stated rate, the bond is worth par. So the answer is **(B)**.

2. We are considering purchasing an asset called the "Wild Money Mystery Machine," which we expect will pay out $500k in cash one year from now, $420k

ten years from now, and \$1m twenty years from now. However, based on our uncertainty (i.e., the risk) we determine the appropriate interest rate on this investment is 12%. What is the present value of the Wild Money Machine, under this assumption?

(A) \$103k

(B) \$685k

(C) An amount greater than \$1m

(D) Not enough information

(E) None of the other answers

Solution. The pattern of cash flows is deliberately ridiculous, and it does not matter in the slightest. The identity does not care whether the cash arrives on a tidy annual schedule or in three lumps scattered across two decades. Ask only: what are the cash flows, when does each one arrive, and what discount factor applies to each? With $i = .12$:

$PV = \frac{\$500k}{1.12} + \frac{\$420k}{1.12^{10}} + \frac{\$1m}{1.12^{20}}$

Taking those one at a time: $\$500k/1.12 = \$446.4k$. Then $1.12^{10} \approx 3.106$, so $\$420k/3.106 \approx \$135.2k$. Then $1.12^{20} \approx 9.646$, so $\$1m/9.646 \approx \$103.7k$. Adding them up: $\$446.4k + \$135.2k + \$103.7k \approx \$685k$.

The distractors are the two ways of not really believing in discounting. (C) is the undiscounted instinct: the promised payments total \$1.92m, so surely the thing is worth more than \$1m? No—not remotely. At 12%, twenty years of compounded discounting shrinks that \$1m payment to about \$104k, roughly a *tenth* of its face amount. This is the single most important thing to feel in your bones about present value: over long horizons, discounting is brutal. And (A) is the present value of the twenty-year payment alone. So the answer is **(B)**.

3. All else equal, the riskier an asset is, the [...] is its expected return.

(A) Higher

(B) Lower

(C) Unchanged

(D) Not enough information

(E) None of the other answers.

Solution. This is the "high risk, high return" slogan—but the reason is mechanical, not moralistic. Nothing in the universe rewards boldness as such. What

happens is this: buyers dislike uncertainty, so when an asset's future payoffs are more uncertain, they will only take it at a *lower price today*. And paying a low price today for the same hoped-for future payoff simply *is* a high expected return, since return is, by definition, future value divided by what you paid. Put in the bluntest terms: risky assets are cheap, and cheap is another word for high expected return. So the answer is **(A)**.

4. Holding the expected/promised future payments of an asset fixed, the *riskier* the asset, the [...] is its **present value**.

   (A) Higher

   (B) Lower

   (C) Unchanged

   (D) Not enough information

   (E) None of the other answers.

   Solution. Note the clause doing the work: *holding the expected future payments fixed*. So the numerator of our identity is nailed down, and only the denominator can move. More risk means we discount the promise harder—a higher i—and a bigger denominator over a fixed numerator gives a smaller number:

   $$Present\ Value = \frac{Future\ Value\ (cash)}{(1+i)}$$

   Recall Taylor and Matt. Both promised $1,000 in one year. Taylor's promise was worth $800 to you; Matt's, being the shiftier proposition, only $600. Same promised cash, more risk, lower present value.

   And notice that this is the *same fact* as the previous problem, viewed from the other end. Lower present value for fixed future payments means a higher discount rate—and when discounting expected payoffs with a required expected-return rate, the lower price corresponds to a higher required expected return. A yield calculated from promised debt payments is different: default risk can make the expected return lower than the promised yield. So the answer is **(B)**.

5. All else equal, the riskier some form of credit is (e.g. a loan or a bond), the [...] the market interest rate should be...

   (A) Higher

   (B) Lower

   (C) Unchanged

   (D) Not enough information

(E) None of the other answers.

Solution. Same fact once more, now dressed in credit-market clothing. When we are talking specifically about forms of credit, we call the discount rate the **interest rate**—it is a rectangle-and-square situation, in which a market yield used to discount a bond's promised payments must be distinguished from its contractual coupon rate. So a riskier borrower gets discounted harder, which is to say charged a higher interest rate. The Taylor and Matt example again: your i for Taylor was 25%, and for the shifty Matt it was 67%.

And this is exactly what you observe in the world. A start-up borrows at a higher rate than a blue-chip firm; a firm on the edge of bankruptcy borrows at a higher rate than a healthy one; and the U.S. Treasury, the safest borrower around, pays the least of all. So the answer is **(A)**.

6. Based on the logic of NPV analysis, if we calculate that the NPV of some investment opportunity is greater than zero, we...

   (A) Should make the investment, because the present value of the benefits is greater than that of the costs...

   (B) Should not make the investment, because the present value of the benefits is less than that of the costs...

   (C) It's ambiguous / not enough information

   (D) Not enough information

   (E) None of the other answers.

   Solution. Net Present Value is the present value of the benefits, net of the present value of the costs. So a positive NPV says, in so many words: *in today's dollars, what we get is worth more than what we give up.* If we grant that and then decline the deal, we are contradicting ourselves—we have just said the thing is worth more to us than its price, and then refused to buy it at that price. That is why NPV is *the* metric for rational investment decisions: calculating it is really just a matter of expressing our own beliefs and preferences formally, and then obeying them.

   Option (B) simply has the inequality backwards. And I would flag the temptation lurking behind (C) and (D), which is the feeling that a decision this consequential cannot really turn on one number. But the number already contains everything: the forecast cash flows and the risk-appropriate discount rate. If you distrust the conclusion, what you actually distrust is one of those inputs—and that is a quarrel with your own forecast, not with the logic of NPV. So the answer is **(A)**.

A.13.2 Practice Problems: Open Prompts

- We are considering purchasing an asset that promises to pay us $600 one year from today, and $10,600 two years from today. Based on our assessment of the riskiness, we determine that the appropriate discount / interest rate is 5%. What is the **present value** of this asset?

  Solution. Same two cash flows as the first problem, same two-term calculation, with $i = .05$. The $600 waits one period, so it is discounted once; the $10,600 waits two, so it is discounted twice:

  $PV = \frac{\$600}{1.05} + \frac{\$10{,}600}{1.05^2} = \frac{\$600}{1.05} + \frac{\$10{,}600}{1.1025}$

  $PV \approx \$571.43 + \$9,614.51 \approx \$10,185.94$

  Now, the useful thing here is not the number but the comparison. This is the same asset we valued at 6% and found to be worth exactly $10,000. Lower the rate to 5%, and its value rises to about $10,186. That direction is not an accident, and you should be able to see it straight from the identity: the rate sits in the *denominator*, so a smaller i means a smaller divisor and a larger present value. A less-worried market pays more for the same promise.

  Here is what this means in bond language, since that is where we are headed next chapter. This asset is a $10,000-face, 6%-coupon, two-year bond. Its stated coupon rate—6%—is just text in a contract, and the market is under no obligation to accept it as the true rate. Here the market has decided this is a *5% kind of borrower*, safer than the contract assumed. The promised cash flows cannot be renegotiated, so what adjusts is the price: the bond sells at about $10,186, above its $10,000 face value.

- We are considering purchasing an asset that promises to pay us $600 one year from today, and $10,600 two years from today. Based on our assessment of the riskiness, we determine that the appropriate discount / interest rate is 7%. What is the **present value** of this asset?

  Solution. Identical structure, now with $i = .07$:

  $PV = \frac{\$600}{1.07} + \frac{\$10{,}600}{1.07^2} = \frac{\$600}{1.07} + \frac{\$10{,}600}{1.1449}$

  $PV \approx \$560.75 + \$9,258.45 \approx \$9,819.20$

  Put the three versions of this asset side by side and the whole logic of bond pricing falls out of them. At 5%, it was worth $10,186; at 6%, exactly $10,000; at 7%, about $9,819. The promised cash flows never changed—$600 and then $10,600, fixed by contract. All that changed was how hard the market chose to discount them.

  So: when the market rate is *below* the 6% stated coupon rate, the bond is worth *more* than its $10,000 face value; when the market rate is *above* the coupon rate,

it is worth *less*; and when the two rates coincide, price equals face value exactly. That is not a coincidence to memorize but a consequence of the identity, and next chapter it becomes the foundation of how we account for bonds issued at a discount or a premium.

- We are considering purchasing an asset that promises to pay us \$600 one year from today, and \$10,600 two years from today. Based on our assessment of the riskiness, we determine that the appropriate discount / interest rate is 5%. To purchase this asset today, we would need to pay \$10,000. What is the **net present value** of this investment opportunity?

  Solution. We already have the hard part: at 5%, the present value of the two promised inflows is about \$10,185.94. What remains is to net the cost against it.

  How should the \$10,000 purchase price be discounted? Not at all. It is a sure thing, paid today—no delay, no risk—so there is nothing to discount. In finance terms it is simply a negative cash flow at time zero, and it enters the sum at full face:

  $$NPV = -\$10,000 + \frac{\$600}{1.05} + \frac{\$10,600}{1.05^2} \approx -\$10,000 + \$10,185.94 \approx \$185.94$$

  The NPV is positive, so this is a good deal and we should buy the asset. The reason is exactly the one from the multiple-choice question above: we have just calculated that the entitlement to those future payments is worth about \$10,186 to us, in today's dollars, and we can have it for \$10,000. Declining would be refusing to buy something for less than we ourselves say it is worth.

  And notice how thin the margin is—about \$186 on a \$10,000 outlay, well under 2%. Had we assessed this borrower as a 7% risk rather than a 5% risk, the same asset at the same price would have carried an NPV of roughly $\$9,819.20 - \$10,000 = -\$180.80$, and we should have walked away. The cash flows and the price were identical in both cases. The entire decision turned on the discount rate—which is why estimating risk properly, rather than mechanically turning the crank on the arithmetic, is what the finance profession actually spends its days doing.

A.13.3 Essay and Discussion Questions

1. Explain precisely *why* it is that high risk means high expected return.

   Solution. Here is how I would build the answer: first clear away the wrong explanation, then derive the right one from the identity, and then guard against the two objections that always follow.

The wrong explanation first, because most people arrive holding it. "High risk, high return" sounds like a moral exhortation—as though the market pays out for boldness, and the way to earn great returns is to make big, brave, decisive moves. When I first heard the phrase in college, I pictured Wolf-of-Wall-Street types generating returns through sheer nerve. That is exactly backwards, and noticing why is the whole insight.

The right explanation is mechanical, and it comes straight out of the one identity of finance. Start with what we know: people do not value promised future cash 1-for-1, because they dislike delay and risk. So they discount it. The more uncertain the promised payoff, the harder they discount—which is to say, the *less they are willing to pay today* for the same promised future amount. Now write down what a return *is*: the future value you get from an asset, divided by what you paid for it. Set the two statements next to each other:

$$Present\ Value = \frac{Expected\ Future\ Value}{1+i}$$

$$\frac{Expected\ Future\ Value}{Present\ Value} = 1 + Expected\ Return = 1 + i$$

The expected return and the discount rate i are the *same number*, read from opposite ends of the same equation. So the chain is: more risk $\rightarrow$ buyers are more reluctant $\rightarrow$ they pay a lower price today $\rightarrow$ a lower price today for the same expected future payoff *is*, arithmetically, a higher expected return. That is the entire argument, and note that no step in it involves anyone being rewarded for anything. High expected returns are not a prize for bravery; they are the discount that reluctant buyers extract as the price of holding something they would rather not hold. Or, in the bluntest possible phrasing: **risky assets are cheap**, and cheap is what a high expected return looks like from the buyer's side of the table.

Taylor and Matt make it concrete. Both promise $1,000 in a year. You would pay $800 for Taylor's promise and only $600 for Matt's, because you trust Matt less. Matt's promise therefore carries a 67% expected return, against Taylor's 25%. Did Matt do anything admirable to earn that 67%? No—quite the reverse. The high return exists *because* he is a bad bet; it is the compensation you demanded for putting up with him. Risk is bad, not bold, and the return is what the market pays you to bear something bad.

Now the two objections, since a careful reader will raise them.

The first: "If risky assets are cheap, then I put in *less money*—so haven't I taken on *less* risk, which unravels the whole story?" This confuses two different things. The **risk of an asset** means the uncertainty in that asset's own future payoffs, the spread of possible outcomes; it is a property of the asset, and it determines

the price *per unit*. A wildly volatile stock is a high-risk asset even if you own $10 of it. Your **portfolio risk** is a property of *you*: it depends on which assets you hold and how much you have put into each. When we say higher risk means a higher discount rate, we are talking about the first. How much of the thing you then buy merely scales your own exposure; it does not change the asset's payoff-uncertainty or its price.

The second: "If risky assets reliably deliver higher returns, isn't that free money—just buy the risky ones?" No, and this is where **market efficiency** does its work. *Expected* return is not *realized* return. You are being paid a higher expected return precisely because a real chance exists that you will not be paid at all. If risky assets delivered high returns reliably, they would not be risky, and the price would rise until the extra return vanished. The higher return is not a bonus sitting on the table; it is the exact compensation the market has priced for bearing the possibility of loss—no more, and, in an efficient market, no less.

A closing note in the same spirit: this is also why you should not describe yourself as "high risk, high return" on a dating app. You imagine you are advertising boldness. Given where high expected returns actually come from, what you are announcing is that you are cheap.

A.14 Chapter 14

A.14.1 Practice Problems: MCQ's

1. Our company sells a bond with $100k face value, a three-year maturity, and a 7% coupon, paid annually. Upon issuance to the market, the market-implied interest rate on the bond is 5%. What was the present value / issuance price of the bond upon the sale?

   (A) $81.62k

   (B) $86.38k

   (C) $92.43k

   (D) $105.45k

   (E) None of the above.

   Solution. Two steps, in the order we always do them. **First**, translate the verbal description into a schedule of cash flows. A 7% coupon on a $100k face value is $7k per year, for three years, and the $100k face value comes back at the end. So the bond promises $7k, $7k, and $107k. **Second**, discount those cash flows at the *market* rate — the one formula:

   $$P.V. = \frac{\$7k}{1.05} + \frac{\$7k}{1.05^2} + \frac{\$107k}{1.05^3} = \$105,446.50$$

   Before you even reach for a calculator, though, you should know the *sign* of the answer. The market rate (5%) is *less* than the coupon rate (7%), so the bond must issue at a **premium** — above face value. That single observation kills (A), (B), and (C) instantly, and leaves only (D). So the answer is **(D)**.

2. Our company sells a bond with $100k face value, a three-year maturity, and a 7% coupon, paid annually. Upon issuance to the market, the market-implied interest rate on the bond is 8%. What was the present value / issuance price of the bond upon the sale?

   (A) $84.94k

   (B) $97.42k

   (C) $100k

   (D) $105.45k

   (E) None of the above.

Solution. Same bond, same promised cash flows — \$7k, \$7k, \$107k — but now the market has decided we are an 8%-interest-rate kind of company. Bigger denominators, smaller present value:

$$P.V. = \frac{\$7k}{1.08} + \frac{\$7k}{1.08^2} + \frac{\$107k}{1.08^3} = \$97,422.90$$

Again, the direction was knowable in advance: market rate (8%) $>$ coupon rate (7%), so the bond issues at a **discount**, below the \$100k face value. That leaves only (A) and (B), and (A) is far too deep a discount for a one-percentage-point gap over three years. So the answer is **(B)**.

3. Suppose a bond has a face value of \$10k, and a coupon rate of 6%. But upon being issued to the market, the issuance price is only \$9,200. Then, is the implied market interest rate equal to, greater than, or less than 6%?

   (A) Equal to 6%.

   (B) Greater than 6%.

   (C) Less than 6%.

   (D) It could be any of the above.

   (E) Not enough information.

   Solution. This is the directional relationship, run in reverse. The one fact worth memorizing is: coupon rate = market rate $\leftrightarrow$ price = face value. Here the price (\$9,200) came in *below* the face value (\$10,000) — the bond was issued at a **discount** — so the market must have been discounting those promised cash flows harder than 6%. Note (E) is tempting if you think you need the maturity date. You don't: the direction holds for any maturity. So the answer is **(B)**.

4. Suppose a bond has a face value of \$10k, and a coupon rate of 6%. But upon being issued to the market, the issuance price is \$10,400. Then, is the implied market interest rate equal to, greater than, or less than 6%?

   (A) Equal to 6%.

   (B) Greater than 6%.

   (C) Less than 6%.

   (D) It could be any of the above.

   (E) Not enough information.

   Solution. The mirror image of the previous question. The price (\$10,400) exceeds the face value (\$10,000), so this bond was issued at a **premium**. Investors

were willing to pay more than face value because the 6% coupon looked generous relative to what they required — which is another way of saying that the market rate is below 6%. So the answer is **(C)**.

5. The *Discount on Bonds Payable* is a(n)...

   (A) asset account

   (B) contra-asset account

   (C) liability account

   (D) contra-liability account

   (E) Not enough information

   Solution. *Discount on Bonds Payable* is the first **contra-liability** account in this book. It sits against *Bonds Payable*, carries a **debit** balance, and *reduces* the reported liability: $\textit{Carrying Value} = \textit{Bonds Payable} - \textit{Discount on BP}$. Do not let the debit balance fool you into calling it an asset — *Accumulated Depreciation* carries a credit balance and is not a liability either. What makes an account 'contra' is which account it nets against, not which side its balance sits on. So the answer is **(D)**.

6. The *Premium on Bonds Payable* is a(n)...

   (A) asset account

   (B) contra-asset account

   (C) liability account

   (D) contra-liability account

   (E) Not enough information

   Solution. Here the symmetry breaks, and this is precisely why the two questions are asked back to back. *Premium on Bonds Payable* is an ordinary **liability** account — a credit account — that gets *added* to *Bonds Payable*: $\textit{Carrying Value} = \textit{Bonds Payable} + \textit{Premium on BP}$. It is not 'contra' to anything, because it moves the carrying value in the *same* direction as the account it accompanies. Only the *Discount* is a contra-account. So the answer is **(C)**.

7. Is the following statement true or false? The carrying value of a bond is always equal to what its initial issuance price was.

   (A) true

   (B) false

Solution. False. The carrying value *starts* at the issuance price — that much is true, and is worth holding onto. But the whole point of the amortization algorithm is that the carrying value then *moves*: every interest payment plugs the gap between the *Interest Expense* and the cash coupon into the *Discount* or *Premium*, and the carrying value shifts accordingly. For a bond issued at a discount it climbs toward the face value; for a bond issued at a premium it falls toward the face value. By the maturity date, it equals the face value in both cases. So the answer is **(B)**.

8. Which of the below is the correct formula for determining the annual *Interest Expense* under the bond amortization method we learned in class?

   (A) $\textit{Interest Expense} = \textit{Carrying Value} \times \textit{Original Market Interest Rate}$

   (B) $\textit{Interest Expense} = \textit{Face Value} \times \textit{Original Market Interest Rate}$

   (C) $\textit{Interest Expense} = \textit{Carrying Value} \times \textit{Coupon Rate}$

   (D) $\textit{Interest Expense} = \textit{Face Value} \times \textit{Coupon Rate}$

   (E) None of the above.

   Solution. This is Rule #3, and it is the load-bearing rule of the whole chapter. Both halves matter, and each distractor breaks exactly one of them. We use the **market** rate rather than the coupon rate because the market rate is the real economic cost of the financing — that is the whole reason we can't just let companies report the coupon as their interest cost. And we multiply it by the **carrying value** rather than the face value because the carrying value is what the market actually 'loaned' us, updated for the amortization to date. Option (D) — face value times coupon rate — is the formula for the *cash coupon payment*, not the expense; confusing those two is the single most common error in this material. So the answer is **(A)**.

9. Which of the below is the correct formula for determining the annual *coupon payments* for a typical bond?

   (A) $\textit{Annual Coupon Payment} = \textit{Carrying Value} \times \textit{Original Market Interest Rate}$

   (B) $\textit{Annual Coupon Payment} = \textit{Face Value} \times \textit{Original Market Interest Rate}$

   (C) $\textit{Annual Coupon Payment} = \textit{Carrying Value} \times \textit{Coupon Rate}$

   (D) $\textit{Annual Coupon Payment} = \textit{Face Value} \times \textit{Coupon Rate}$

   (E) None of the above.

   Solution. And here is the other half of the pair. The coupon payment is not an estimate or an allocation — it is **text in a contract**. The indenture says '6% of

$10,000, annually,' and so $600 leaves the company's bank account every year, no matter what the market thinks, and no matter what the carrying value has drifted to. Face value times coupon rate. Notice that (A) is the *Interest Expense* formula from the previous question: these two questions are deliberately the same four options, and the answer moves from one corner of the grid to the opposite one. So the answer is **(D)**.

10. If the market-implied interest rate on a bond upon issuance is *greater* than the coupon rate, the bond will...

    (A) ... issue at a "premium" to face value

    (B) ... issue at a "discount" to face value

    (C) ... not be issued at all by the corporation

    (D) ... find no buyers in the financial market

    (E) ... None of the other answers

    Solution. A higher market rate means bigger denominators in the one formula, means heavier discounting, means a lower present value — below the face value. Below face value is what we *call* a **discount**. Options (C) and (D) encode a real misunderstanding worth naming: the bond does not fail to sell. It sells perfectly well; it just sells for less. The market is not rejecting the bond, it is *pricing* it. So the answer is **(B)**.

11. If the market-implied interest rate on a bond upon issuance is *less* than the coupon rate, the bond will...

    (A) ... issue at a "premium" to face value

    (B) ... issue at a "discount" to face value

    (C) ... not be issued at all by the corporation

    (D) ... find no buyers in the financial market

    (E) ... None of the other answers

    Solution. The flip side. A lower market rate means lighter discounting, means a present value *above* the face value — a **premium**. Concretely: we contractually promised to pay more per year than investors now require, so they bid the price up above $100 per $100 of face. So the answer is **(A)**.

12. For a bond issued at a premium, which of the following can we know to be true?

    (A) The cash interest payment will go down over time.

(B) The cash interest payment will go up over time.

(C) The interest expense recognized will go down over time.

(D) The interest expense recognized will go up over time.

(E) None of the other answers

Solution. Deal with the cash first, because it is the easy half: the cash interest payment is fixed by the contract at (face value $\times$ coupon rate), and it does not budge over the life of the bond. That eliminates (A) and (B) at once. Now the expense. For a premium bond, we *debit* the *Premium on BP* each period, using it up, so the carrying value **falls** toward the face value. And since $Interest\ Expense = Carrying\ Value \times Market\ Rate$, a falling carrying value against a fixed rate means a **falling** expense. The intuition, from the chapter: we are paying out more cash each year than our true interest cost, so — like an installment payment or a mortgage — each payment is partly interest and partly a paydown of principal, and the interest on a shrinking principal shrinks. So the answer is **(C)**.

13. For a bond issued at a discount, which of the following can we know to be true?

(A) The cash interest payment will go down over time.

(B) The cash interest payment will go up over time.

(C) The interest expense recognized will go down over time.

(D) The interest expense recognized will go up over time.

(E) None of the other answers

Solution. Same reasoning, opposite sign. The cash payment is still fixed by contract, so (A) and (B) are out. For a discount bond, we *credit* the *Discount on BP* each period, using it up, so the carrying value **rises** toward the face value — and a rising carrying value times a fixed market rate gives a **rising** *Interest Expense*. The credit-card intuition: the market demands 8% but we are only handing over 6% in cash, so we are falling behind. The shortfall gets capitalized into the balance, and next year's interest is charged on the bigger balance. So the answer is **(D)**.

14. Suppose our company issues a bond with a \$10k face value, and a 6% coupon rate, paid annually. Upon sale to the market, it issues for a premium, with an implied market interest rate of 5%. What would be the annual cash interest payment on this bond?

(A) \$500

(B) \$600

(C) Not enough information

(D) None of the other answers.

Solution. The 5% is a distractor, and a well-aimed one. Cash coupon payments come from the contract and only from the contract: 6% $\times$ \$10,000 = **\$600** per year. The 5% market rate would matter if we were asked for the *issuance price* or the *Interest Expense* — but not for the cash. Option (A), \$500, is what you get by multiplying the face value by the market rate, which is a formula that appears nowhere in this chapter. So the answer is **(B)**.

15. Suppose our company issues a bond with a \$10k face value, and a 6% coupon rate, paid annually. Upon sale to the market, **it issues for a premium**. Based on this information alone, would the annual *Interest Expense* be less than, equal to, or greater than \$600?

(A) Less than

(B) Equal to

(C) Greater than

(D) Not enough information

Solution. \$600 is the cash coupon payment, so the question is really: for a premium bond, is the expense above or below the cash payment? Below. Here is a way to see it that does not require knowing the actual market rate. Each period we *debit* the *Premium on BP* to plug the entry — and a debit to a credit-balance account only works if the debit side is short, which means the *Interest Expense* debit must be smaller than the *Cash* credit. If the expense were ever larger than \$600, the plug would run the *Premium* the wrong way, and it would never amortize down to zero by maturity. Note that (D) is the trap here: it feels as though we need the market rate. We don't — 'issued at a premium' already tells us the market rate is below 6%, and that is all the direction requires. So the answer is **(A)**.

16. Based on the bond amortization method we learned in this class, what determines the *change* in the *Carrying Value* of a bond between periods?

(A) The change in the *market value* of the bond, trading in the bond market everyday

(B) The change in the interest rate assumptions of our financial experts

(C) The difference (plug) between the *Interest Expense* and the actual cash coupon payment

(D) The coupon payment by itself (i.e., $\Delta Carrying\ Value = Coupon\ Payment$)

(E) None of the above.

Solution. This is Rule #4. The carrying value moves by exactly the amount of the plug — the gap between the *Interest Expense* debit and the *Cash* credit — because that plug *is* the amortization of the *Discount* or *Premium*. Option (A) is the most instructive wrong answer: bonds payable are carried at **amortized cost**, not fair value. We do not re-mark the liability as the bond's yield wanders around in the secondary market, any more than we revalue a factory when the market for used factories wiggles. (And recall the deeper point from the chapter: so long as the market rate *does* hold constant, the amortized carrying value tracks the fair value anyway — but that is a happy consequence of the algorithm, not its input.) Option (B) is worse still: the rate we use is frozen at the *original* market rate on the issuance date, precisely so that management cannot manage its interest expense by revising 'assumptions.' So the answer is **(C)**.

17. Suppose our company issues a bond with a $10k face value, and a 6% coupon rate, paid annually. Upon sale to the market, **it issues for a discount**. Based on this information alone, would the annual *Interest Expense* be less than, equal to, or greater than $600?

    (A) Less than

    (B) Equal to

    (C) Greater than

    (D) Not enough information

    Solution. The mirror of question 15. Issued at a discount means the market rate exceeds the 6% coupon rate, so the true cost of the financing exceeds the $600 of cash we hand over. And the entry confirms it: we plug with a **credit** to the *Discount on BP*, which requires the debit side — the *Interest Expense* — to exceed the $600 *Cash* credit. That excess is the initial discount being apportioned out over the life of the bond, on top of the coupons. So the answer is **(C)**.

18. Suppose our company issues a bond with a 2-year maturity, a face value of $100k, and a coupon rate of 4%. The market-implied interest rate upon issuance is 5%. What was the bond's issuance price?

    (A) $98.14k

    (B) $107.44k

    (C) $102.86k

(D) \$94.33k

Solution. Cash flows first: 4% of \$100k is \$4k, so the bond promises \$4k at the end of year one and \$104k at the end of year two. Then discount at the market rate of 5%:

$$P.V. = \frac{\$4k}{1.05} + \frac{\$104k}{1.05^2} = \$3,809.52 + \$94,331.07 = \$98,140.59$$

The market rate exceeds the coupon rate, so we expected a discount — below \$100k — which already rules out (B) and (C). Option (D), \$94.33k, is the *second term alone*: the present value of the final \$104k payment, with the year-one coupon forgotten. So the answer is **(A)**.

19. Suppose our company currently has a callable bond on our books with a face value of \$10,000 and a call price of \$10,000. The balance of the Premium on Bonds Payable account for that bond is \$200. If we were to redeem the bond at the call price, we would have...:

(A) A gain of \$200

(B) A loss of \$200

(C) A gain of \$300

(D) A loss of \$300

(E) None of the other answers

Solution. Get the carrying value first, then compare it to what we pay. A *Premium* is **added**: $CV = \$10,000 + \$200 = \$10,200$. We pay the \$10,000 call price to make a \$10,200 liability disappear — we settled a debt for less than its book value — so that is a **gain** of \$200. Or let the journal entry produce it as a plug:

Bonds Payable (-L):	\$10,000	
Premium on Bonds Payable (-L):	\$200	
Cash (-A):		\$10,000
Gain (+S.E.):		\$200

Redeeming the bond at the \$10,000 call price

Gains are credits, losses are debits, and the plug lands on the correct side automatically. So the answer is **(A)**.

20. Suppose our company currently has a non-callable bond on our books with a face value of $10,000 and a carrying value of $10,200. We go to the market and repurchase it for $10,500. Then, we would have a...:

    (A) A gain of $300

    (B) A loss of $300

    (C) A gain of $500

    (D) A loss of $500

    (E) None of the other answers

    Solution. Here the carrying value is handed to us: $10,200. We pay $10,500 to extinguish it, so we paid $300 *more* than book value to get rid of the liability — a **loss** of $300. If you would rather derive the sign than remember it: liabilities fall by $10,200 (good), assets fall by $10,500 (bad), so net assets — and therefore equity — fall by $300. Option (D), $500, comes from comparing the repurchase price to the *face value* instead of the carrying value; the face value is not what the bond is on our books for. And note that 'non-callable' changes nothing about the accounting — the same four-row entry applies whether we redeem at a call price or repurchase in the open market. So the answer is **(B)**.

21. We are given the following T-accounts related to a company's bond:

Bonds Payable (L)

Debits	Credits
	$10,000

Discount on B/P (-L)

Debits	Credits
$400	$100

What is the **Carrying value** of this bond?

(A) $9,500

(B) $9,600

(C) $9,700

(D) $10,000

(E) None of the other answers

Solution. Two steps. **First**, net out the contra-account. *Discount on B/P* is a debit-balance account, so its balance is $400 − $100 = $300. (The $400 was the discount recorded at issuance; the $100 is one period's worth of amortization, credited away.) **Second**, apply the definition — the *Discount* is **subtracted**:

$$\textit{Carrying Value} = \$10,000 - \$300 = \$9,700$$

Option (B), $9,600, is what you get by subtracting the $400 original discount and ignoring the $100 of amortization already recorded. So the answer is **(C)**.

22. We are given the following T-accounts related to a company's bond:

Bonds Payable (L)

Debits	Credits
	$10,000

Premium on BP (L)

Debits	Credits
$100	$400

What is the **Carrying value** of this bond?

(A) $9,500

(B) $9,600

(C) $9,700

(D) $10,000

(E) None of the other answers

Solution. Same arithmetic, opposite direction — and the trap is that the four numeric options are identical to the previous question's. *Premium on BP* is an ordinary credit-balance liability account, so its balance is \$400 − \$100 = \$300 on the *credit* side, and the premium is **added**:

$$\textit{Carrying Value} = \$10,000 + \$300 = \$10,300$$

\$10,300 is not among the four numeric options — as it shouldn't be, since a bond issued at a premium must carry *above* its face value, and every numeric option here is at or below \$10,000. So the answer is **(E)**.

23. Our company does annual, calendar-year reporting, and issues a bond with an annual coupon on January 1st 2021. The market issuance price is \$95k, implying a market interest rate of 5%. What would be our Interest Expense from this bond for FY 2021 (i.e., the first year).

    (A) \$4750

    (B) \$5000

    (C) \$2500

    (D) Not enough information, without the coupon rate

    (E) None of the other answers

    Solution. The temptation is to reach for (D), because we have not been told the coupon rate. But look at what Rule #3 actually needs: the **carrying value** and the **market rate**. And on the issuance date, the carrying value *is* the issuance price — that is the amount the market really loaned us. Both inputs are in hand:

$$\textit{Interest Expense}_{2021} = \$95,000 \times .05 = \$4,750$$

    Option (B), \$5,000, is the face-value trap: it takes \$100k as the base, but we do not know that the face value is \$100k, and even if we did, the face value is not what we multiply. So the answer is **(A)**.

24. Our company does annual, calendar-year reporting, and issues a bond with an annual coupon on January 1st 2021. The market issuance price is \$95k, implying a market interest rate of 5%. What would be our Interest Expense from this bond for FY 2022 (i.e., the second year).

    (A) \$4750

    (B) \$5000

(C) $2500

(D) Not enough information, without the coupon rate

(E) None of the other answers

Solution. Now the missing coupon rate bites — and this question exists to show you exactly *where* it starts to bite. The second year's expense is the *updated* carrying value times 5%, and the update is Rule #5:

$$CV_{1/1/2022} = \$95,000 + (\$4,750 - \textit{Cash Coupon Payment})$$

The cash coupon payment is (face value $\times$ coupon rate), and we know neither. So the carrying value at the start of 2022 could be almost anything, and so could the expense. Notice the structure: the first year's expense needs only the issuance price, because the amortization hasn't started yet; every year *after* the first depends on the whole schedule of coupon payments. So the answer is **(D)**.

25. All else equal, if our company were to issue a *callable* bond instead of (that is, as compared to the alternative of) a non-callable bond, the initial market interest rate on the bond would be...

(A) Higher, because the call feature means more risk for the bond *buyer*

(B) Lower, because callable bonds are better in every way

(C) Unchanged

(D) Not enough information / ambiguous

Solution. The fruit principle: the deal has to be worth it to both sides. A call provision is an **option** for the *issuer* — we get to buy the bond back at a pre-set price, and we will exercise that right precisely when it hurts the holder, namely when rates have fallen and the bond has become valuable. So our optionality is the buyer's risk: the risk of having a nice fat coupon yanked away exactly when nice fat coupons are hard to find. And by our core finance theory, more risk means a higher required return, which means a higher market interest rate and a lower issuance price. Option (B) gets the direction of the benefit right but forgets that somebody has to be paid for bearing the other side of it — there is no free lunch in a competitive market. (Empirically, this is also just what we observe: callable bonds price at measurably higher yields than comparable non-callable bonds from the same issuers.) So the answer is **(A)**.

26. Under current U.S. GAAP, how does a lessee generally recognize a lease longer than 12 months, assuming no applicable exception?

(A) Recognize an asset and liability only if the contract transfers ownership.

(B) Recognize an asset and liability only if the lease is classified as a finance lease.

(C) Recognize a Right-of-Use Asset and a Lease Liability for both operating and finance leases.

(D) Keep every operating lease off the Balance Sheet.

Solution. The answer is **(C)**. ASC Topic 842 generally brings both operating and finance leases onto the lessee's Balance Sheet. Classification still affects subsequent expense and cash-flow presentation. The older 75% and 90% tests are not conditions that an operating lease must meet to be recognized.

A.14.2 Practice Problems: Open Prompts

1. Suppose our company does annual, calendar-year reporting, Jan 1 to Jan 1, and issues the following bond on January 1st of 2021: Face value of $100k, 3-year maturity, coupon rate of 5%, paid annually. Upon issuance to the market, the market interest rate is 6%. Journalize the transactions for January 1 of 2021 (bond issuance), January 1 of 2022 (first annual interest payment / expense recognition), and January 1 of 2023 (second annual interest payment / expense recognition).

   Solution. Work in the same order every time: cash flows, then price, then the schedule, then the entries.

   Step 1 — the promised cash flows. A 5% coupon on $100k face value is $5,000 per year, for three years, plus the $100,000 face value at maturity. So: $5,000, $5,000, $105,000.

   Step 2 — the issuance price. The market rate (6%) exceeds the coupon rate (5%), so we already know this bond issues at a **discount**. Discounting at 6%:

$$P.V. = \frac{\$5,000}{1.06} + \frac{\$5,000}{1.06^2} + \frac{\$105,000}{1.06^3} = \$97,326.99$$

   So the initial *Discount on Bonds Payable* is $100,000 − $97,326.99 = **$2,673.01**.

   Step 3 — the amortization schedule. Each row applies the same four rules: the *Interest Expense* is the carrying value times 6%; the cash payment is fixed at $5,000; the difference amortizes the *Discount* (a credit, reducing it); and the carrying value rises by that same difference. I carry full precision through the calculation and round only for display — with the odd penny absorbed into a single row, as every real amortization schedule does — so that the columns foot exactly and the last row lands on the $100,000 face value:

Date	Carrying Value	Cash Paid	Interest Exp.	Ending C.V.
1/1/2022	$97,326.99	$5,000.00	$5,839.62	$98,166.61
1/1/2023	$98,166.61	$5,000.00	$5,889.99	$99,056.60
1/1/2024	$99,056.60	$5,000.00	$5,943.40	$100,000.00

Amortization schedule: $100k face, 5% coupon, 6% market rate, 3 years

(The 'amortization' in each row is the fourth column minus the third: $839.62, then $889.99, then $943.40. Those three sum to $2,673.01 — the entire original discount, apportioned out over the life of the bond. That is the whole game.)

Step 4 — the entries. On the issuance date, credit *Bonds Payable* for the full face value, debit *Cash* for what the market actually paid, and plug the difference into the *Discount*:

Cash (+A):	$97,326.99	
Discount on Bonds Payable (+contra-L):	$2,673.01	
Bonds Payable (+L):		$100,000.00

Bond issued at a discount, 1/1/2021

Then the first interest date. The expense is $97,326.99 $\times$ .06 = $5,839.62; the cash is $5,000; the $839.62 gap is credited against the *Discount*:

Interest Expense (-S.E.):	$5,839.62	
Discount on Bonds Payable (-contra-L):		$839.62
Cash (-A):		$5,000.00

First annual interest payment/expense, 1/1/2022

The *Discount* balance is now $2,673.01 − $839.62 = $1,833.39, so the carrying value has climbed to $100,000 − $1,833.39 = $98,166.61. Multiply by 6% again for the second year:

Interest Expense (-S.E.):	$5,889.99	
Discount on Bonds Payable (-contra-L):		$889.99
Cash (-A):		$5,000.00

Second annual interest payment/expense, 1/1/2023

Notice that the expense *grew* between the two years, even though the cash payment did not. We are contractually paying 5% in cash on a debt the market prices at 6%, so we fall a little further behind each year, the shortfall is capitalized into the liability, and the interest compounds on the larger balance.

2. Suppose our company does annual, calendar-year reporting, Jan 1 to Jan 1, and issues the following bond on January 1st of 2021: Face value of $100k, 3-year maturity, coupon rate of 5%, paid annually. Upon issuance to the market, the market interest rate is 4%. Journalize the transactions for January 1 of 2021 (bond issuance), January 1 of 2022 (first annual interest payment / expense recognition), and January 1 of 2023 (second annual interest payment / expense recognition).

   Solution. Identical bond contract to the previous problem — same $5,000, $5,000, $105,000 — but the market now assigns it a 4% rate. Since $4\% < 5\%$, this one issues at a **premium**, and every sign flips.

   Step 1 — the issuance price.

$$P.V. = \frac{\$5,000}{1.04} + \frac{\$5,000}{1.04^2} + \frac{\$105,000}{1.04^3} = \$102,775.09$$

   So the initial *Premium on Bonds Payable* is $102,775.09 − $100,000 = **$2,775.09**, recorded as a credit.

   Step 2 — the amortization schedule. Same four rules, but now the expense is *below* the cash payment, so the plug is a **debit** to the *Premium*, and the carrying value falls toward the face value. Again, full precision carried, rounded only for display, with the odd penny absorbed into a single row so that the columns foot:

Date	Carrying Value	Cash Paid	Interest Exp.	Ending C.V.
1/1/2022	$102,775.09	$5,000.00	$4,111.00	$101,886.09
1/1/2023	$101,886.09	$5,000.00	$4,075.45	$100,961.54
1/1/2024	$100,961.54	$5,000.00	$4,038.46	$100,000.00

Amortization schedule: $100k face, 5% coupon, 4% market rate, 3 years

   (The amortization amounts are $889.00, $924.55, and $961.54, which sum to $2,775.09 — the whole premium, used up exactly by the maturity date.)

   Step 3 — the entries. At issuance, *Bonds Payable* still gets the full face value; the extra cash the market handed us is credited to the *Premium*:

Cash (+A):	$102,775.09	
Premium on Bonds Payable (+L):		$2,775.09
Bonds Payable (+L):		$100,000.00

Bond issued at a premium, 1/1/2021

First interest date: expense of $102,775.09 × .04 = $4,111.00, cash of $5,000, and the $889.00 difference debited against the *Premium*:

Interest Expense (-S.E.):	$4,111.00	
Premium on Bonds Payable (-L):	$889.00	
Cash (-A):		$5,000.00

First annual interest payment/expense, 1/1/2022

The *Premium* balance is now $2,775.09 − $889.00 = $1,886.09, so the carrying value has fallen to $100,000 + $1,886.09 = $101,886.09. Times 4%:

Interest Expense (-S.E.):	$4,075.45	
Premium on Bonds Payable (-L):	$924.55	
Cash (-A):		$5,000.00

Second annual interest payment/expense, 1/1/2023

Here the expense *shrinks* year over year. We are handing over $5,000 in cash against a true annual cost of about $4,100, so each payment is part interest and part paydown of the effective principal — exactly the mortgage logic — and interest on a shrinking principal shrinks with it.

3. Suppose our company currently has a callable bond on our books with a face value of $10,000 and a call price of $10,000. The balance of the Discount on Bonds Payable account for that bond is $300. Suppose we redeem the bond at the **call price**. **Journalize this transaction**.

   Solution. Three rules, in order. **Credit Cash** for the $10,000 call price. **Extinguish** every account attached to this bond: *Bonds Payable* sits at $10,000 on the credit side, so we debit it away; *Discount on Bonds Payable* sits at $300 on the debit side, so we credit it away. **Plug** what's left. Debits so far are $10,000, credits are $10,300, so the plug is a $300 **debit** — and debits are losses:

Bonds Payable (-L):	$10,000.00	
Loss (-S.E.):	$300.00	
Discount on Bonds Payable (-contra-L):		$300.00
Cash (-A):		$10,000.00

Redeeming a callable bond at its $10,000 call price

And the sign should make sense before the plug ever tells you. The bond's carrying value was $10,000 − $300 = **$9,700**, and we paid $10,000 to make it go away. We overpaid relative to book value, by $300. Symmetrically to selling an asset for less than its book value, that is a **loss**. (Which is worth pausing on: calling a bond is not automatically a good deal just because the contract lets you. Here the call price sits above the carrying value, and the income statement duly reports the difference.)

4. Suppose our company currently has a bond on our books with a face value of $10,000. The balance of the Discount on Bonds Payable account for that bond is $100. Suppose that we repurchase the bond in the bond market for the market price of $10,200. **Journalize this transaction.**

   Solution. This bond is not callable, so we are buying it back at whatever the market will take — but as the chapter stresses, the *accounting* is indifferent to that distinction. Same three rules. Credit *Cash* $10,200. Debit *Bonds Payable* $10,000 and credit *Discount on Bonds Payable* $100 to clear the bond off the books. Debits of $10,000 against credits of $10,300 leaves a $300 debit plug:

Bonds Payable (-L):	$10,000.00	
Loss (-S.E.):	$300.00	
Discount on Bonds Payable (-contra-L):		$100.00
Cash (-A):		$10,200.00

Repurchasing our bond in the market for $10,200

Checking it the intuitive way: carrying value was $10,000 − $100 = $9,900; we paid $10,200; we paid $300 more than book value to extinguish the liability, so we booked a $300 **loss**. Or through the accounting equation, if you prefer to derive rather than memorize: liabilities fell by $9,900, assets fell by $10,200, so net assets fell by $300.

A.14.3 Essay and Discussion Questions

1. Explain what the whole 'bond amortization' process is all about. What is the basic problem that requires this elaborate accounting process, and what is the bond amortization algorithm trying to achieve/trying to account for? Why is it that, for a bond issued at a premium, the interest expense will be lower than the cash interest payments?

   Solution. Here is how I'd think it through, in three moves: the problem, the fix, and then the premium case specifically.

   The problem is a timing problem baked into how bonds get sold. A bond is a contract, and the contract has to be *written before it is sold.* The company drafts an indenture with its lawyers, fixes a face value and a coupon rate, runs the regulatory gauntlet, and only then takes the thing to market — where the buyers, in a free market, make up their own minds about what those promises are worth. That price *is* the market interest rate, by the core identity of finance. So the coupon rate is a number our company wrote down in advance, and the market rate is a number the market handed back to us afterward, and there is no mechanism by which they must agree. Two consequences follow, and both are accounting problems. First, the *Cash* we receive on the issuance date does not equal the *face value* we will one day have to repay. Second, the cash coupon payments we make each year do not measure the true economic cost of the financing.

   The fix is to notice that a non-par bond has two costs, not one, and then apportion the second one. Forget discounting for a moment and just count raw dollars, using the chapter's running example—a $10,000 face value bond, 6% annual coupon, five years—issued at a discount for $9,201.46 when the market rate was 8%. We take in $9,201.46, and we pay out $3,000 of coupons plus $10,000 of face value. Our total cost of that financing was $3,798.54 — the $3,000 of coupons *plus* the $798.54 gap between what we received and what we must repay. That gap is a genuine cost of borrowing. It is just a cost that shows up all at once, in a lump, at the two ends of the bond's life. Recognizing all of it at issuance would be wrong (we haven't done the borrowing yet), and recognizing all of it at maturity would be wrong for the same reason in reverse. So we do what accrual accounting always does with a lump that belongs to many periods: we **apportion** it. That is all 'amortization' means here, and it is the same root idea as depreciating a building — the word is doing the same work, on a liability instead of an asset.

   The machinery that performs the apportionment is the **effective-interest** rule: each period, $Interest\ Expense = Carrying\ Value \times Market\ Rate\ upon\ issuance$, with

the difference between that expense and the fixed cash coupon plugged into the *Discount* or *Premium* account. And it has a lovely property, which is my favorite thing about this chapter: it isn't an arbitrary allocation scheme like straight-line depreciation. So long as the market rate holds constant, the carrying value at every date equals the present value of the *remaining* promised cash flows. The rote algorithm is quietly performing a principled valuation of the liability, period after period, without ever being asked to.

Now the premium case. For a premium bond, the coupon rate we wrote into the contract is *higher* than the rate the market ended up requiring. So the cash we hand over each year overshoots our true interest cost — and the overshoot is not a gift to the bondholder, it is us paying down the effective principal. This is exactly a mortgage: a fixed monthly payment, part interest and part principal reduction. The counting-dollars version says the same thing: we collected $10,432.95 at issuance but only owe $10,000 at the end, and that $432.95 we get to keep **offsets** the coupons, so our total cost is *less* than the sum of the coupons. Amortization spreads that offset across the periods, which is why the expense in each year sits below the cash payment. And the bookkeeping enforces it: the plug is a **debit** to the credit-balance *Premium* account, which only balances if the *Interest Expense* debit is smaller than the *Cash* credit. Carrying value falls each period, the expense falls with it, and both land exactly on the face value at maturity.

2. Why don't companies *only* issue callable bonds? That is, if a 'callable' bond is just like a non-callable bond, *plus* some extra 'optionality' for the firm, why would it ever choose to issue non-callable bonds?

   Solution. The premise of the question is right, and that is what makes it a good question. A call provision genuinely is a free-standing benefit to the issuer: the right, with no obligation, to buy the bond back at a pre-specified price. If rates fall from 7% to 3%, the bond becomes expensive for us and valuable to its holder — its market price might climb to $11,500 on a $10,000 face value — and the call lets us take it back for the contractual $10,500 anyway. The holder must accept. We then refinance at 3% and cut our interest payments. That is real, and it is why callability has become so popular: something like 89% of new U.S. corporate issues by 2020, up from about 35% in 2000.

   So why not always? Because **an option is only free to the party who doesn't have to pay for it, and in a market you always have to pay for it.** This is the fruit principle from the top of the chapter. The plant cannot demand that animals disperse its seeds; it has to wrap the seeds in sugar, or the animal walks away. We cannot bolt an option onto our side of a bond contract and expect the buyer not to notice. And what the buyer notices is that our upside is precisely their

downside: we will exercise the call exactly when it hurts them most — when rates have fallen and a 7% coupon has become a wonderful thing to own — so they lose the very cash flows they were counting on, at the worst possible moment, and are handed cash to reinvest in a low-rate world. That is **reinvestment risk**, and it is a real risk, so by our core finance theory they demand a higher return for bearing it.

Which means the call feature is not free: it is paid for, up front and continuously, in the form of a **higher market interest rate** and therefore a lower issuance price. And the empirical evidence lines up neatly — callable bonds are issued at measurably higher yields than comparable non-callable bonds from the same companies.

So the actual decision is a **tradeoff**, not a free lunch, and the answer to 'why would a firm ever choose non-callable?' is: because sometimes the option isn't worth its price. If our management thinks rates are unlikely to fall much, or we expect to hold this debt to maturity regardless, or our capital structure is settled and we simply want the cheapest possible long-term money, then paying a permanently higher coupon for an option we will probably never exercise is a bad deal. Different firms, facing different rate outlooks and different refinancing needs, land differently — which is exactly what a competitive market for contract terms should produce. The general habit worth taking from this: whenever you spot an 'extra feature' in a financial contract that looks like a pure benefit to one side, immediately ask *who is bearing the corresponding risk, and how are they being compensated for it?* Someone always is, and they always are.

3. Why would a bond investor be willing to buy a bond issued at a premium—that is, pay more today than the face value of the bond?

   Solution. The puzzle only feels like a puzzle if you are looking at one of the bond's cash flows and ignoring the rest. So let's put them all on the table.

   Take our running example: a $10,000 face value bond, 6% annual coupon, five years, issuing at a premium of $10,432.95 when the market rate is 5%. It is perfectly true that the investor pays $10,432.95 today and gets only $10,000 back at maturity — a guaranteed $432.95 shortfall on that leg of the deal. But the face value is not the whole contract. The investor also collects $600 a year for five years — $3,000 in coupons — and *that* is where the compensation lives. The reason the coupons are generous is precisely the reason the bond trades above par: the contract promises 6% of face value annually in a world where investors in this credit only require 5%. The buyer is knowingly overpaying at the back end in exchange for being overpaid, relative to the market rate, at every point in between.

And the two effects don't just roughly cancel — they cancel *exactly*, which is the real content of the answer. The price of $10,432.95 is not a number someone negotiated; it is the present value of the whole package at 5%. Paying that price is, by construction, the price at which the investor earns exactly the 5% they required. Pay a dollar less and they'd earn more than 5%; pay a dollar more and they'd earn less. The premium is not a concession the investor made, it is the market **finding the price at which the deal is worth doing** — the same mechanism, running in the other direction, that produces a discount when the coupon is stingy.

Two related points are worth holding onto. First, notice that the investor's 'loss' on the face value is our **premium** — the money we collected and never have to repay — and that it offsets our coupon payments in exactly the way the amortization algorithm records. The buyer's puzzle and the issuer's accounting are two views of one transaction. Second, and more generally: nothing in finance is priced one cash flow at a time. Whenever some feature of a security looks like an obvious bad deal for the person taking it, the answer is nearly always that you are looking at one term of the sum. The market prices the whole contract, and it prices it so that the deal is, at the margin, worth it to both sides.

A.15 Chapter 15

A.15.1 Practice Problems: MCQ's

1. Our company issues 1000 shares Common Stock with a par value of $1 per share, for an issuance price of $20 per share. What would be our journal entry for this transaction?

   (A) Debit Cash $20k; Credit Common Stock $20k

   (B) Debit Cash $20k; Credit APIC $1k, Credit Common Stock $19k

   (C) Debit Cash $20k; Credit Common Stock $1k, Credit APIC $19k

   (D) Debit Common Stock $19k, Credit APIC $1k; Debit Cash $20k

   (E) None of the other answers

   Solution. Start with *Cash*, as always — it is the leg you can never get wrong. We sold 1,000 shares at $20, so we received $20,000: a debit. Now the credits. *Common Stock* gets the **total par value** of the shares issued, and only that: 1,000 shares $\times$ $1 = $1,000. Everything else is **plugged** to *Additional Paid-in Capital*: $20,000 $-$ $1,000 = $19,000.

Cash (+A):	$20,000	
Common Stock (+S.E.):		$1,000
Additional Paid-In Capital (+S.E.):		$19,000

Issuing 1,000 shares of $1 par stock for $20 per share

   Option (A) is how we did it back in Chapter 3, when we first journalized Transaction #1, and it is right only for a company issuing **no-par** stock. Option (B) has the two credits swapped — it treats par value as the 'real' value of the share and the market price as an afterthought, which inverts the whole story of par value. So the answer is **(C)**.

2. A company had previously repurchased 100 shares of its own Common Stock for $25 each. Now, the company re-sells those same shares for $30 each. Which of the below will result from this resale?

   (A) Trading Revenue of $3,000

   (B) A Gain of $500

   (C) A Loss of $500

(D) An increase in APIC of $500

(E) None of the other answers

Solution. The re-sale rule, in three legs. Debit *Cash* for the proceeds: 100 × $30 = $3,000. Credit *Treasury Stock* at the shares' **prior repurchase price**, 100 × $25 = $2,500 — that leg is fixed by what we paid, and does not care what we re-sell for. Plug the $500 difference to *APIC*:

Cash (+A):	$3,000	
Treasury Stock (-contra-S.E.):		$2,500
Additional Paid-in Capital (+S.E.):		$500

Re-selling 100 treasury shares (repurchased at $25) for $30

Options (A) and (B) are the whole point of the question. If this were a piece of *Equipment* bought for $2,500 and sold for $3,000, a $500 Gain would be exactly right. But **companies cannot book gains by trading in their own stock**. Management holds enormous private information and controls the timing of its release, so a firm that could book net income by buying low from its own shareholders and selling high to them would have a standing invitation to manipulate. The cash is real and must land in equity somewhere — so it goes to *APIC*, bypassing net income entirely. So the answer is **(D)**.

3. A company had previously repurchased 100 shares of its own Common Stock for $25 each. Now, it re-sells those shares for $20 each. Assume at least $500 of APIC from prior treasury-stock transactions is available. Which of the below will be in the journal entry for this resale?

   (A) A Gain of $500

   (B) A Loss of $500

   (C) A credit to APIC of $500

   (D) A debit to APIC $500

   (E) None of the other answers

   Solution. Same rule, run in the loss-shaped direction. *Cash* debit of 100 × $20 = $2,000. *Treasury Stock* credit of 100 × $25 = $2,500 — unchanged, because that leg is set by the repurchase price. Now the credits exceed the debits by $500, so the entry can only balance with a $500 **debit** to *APIC*:

Cash (+A):	\$2,000	
Additional Paid-in Capital (-S.E.):	\$500	
Treasury Stock (-contra-S.E.):		\$2,500

Re-selling 100 treasury shares (repurchased at \$25) for \$20

APIC is a normal credit-balance equity account, so a debit shrinks it: we are reducing recorded paid-in capital. And the self-check from the chapter holds — re-sell *above* the repurchase price and the plug is a credit; re-sell *below* it and the plug is a debit. Just as no 'Gain' was allowed in the previous question, no 'Loss' is allowed here, for the same reason: no income-statement effects from trading in your own shares, in either direction. So the answer is **(D)**.

4. ShreyCorp issues common stock with a *par value* of \$.01 per share, and reports \$1.5k in its *Common Stock* account. Based on this information alone, we can infer that ShreyCorp has...

   (A) 1.5k shares issued

   (B) 150k shares outstanding

   (C) 150k shares issued

   (D) Multiple other answers

   (E) None of the other answers

   Solution. The *Common Stock* balance is the total **par value** of every share ever issued, so dividing it by the par value per share recovers the share count:

$$\frac{\$1,500}{\$.01 \textit{ per share}} = 150,000 \textit{ shares issued}$$

   Now, why **issued** and not **outstanding**? Because a repurchase debits *Treasury Stock* and never touches *Common Stock* at all. So the *Common Stock* balance keeps counting shares that have long since been bought back and are sitting in the treasury — 'issued' means the share left the shelf at some point, not that anyone outside the company holds it today. Without a treasury share count, we cannot get to outstanding. That kills (B), and therefore (D) as well. Option (A) is the arithmetic slip of forgetting to divide by par. So the answer is **(C)**.

5. Suppose that we note that a company's Treasury Stock account has increased in magnitude from FY2019 to FY2020. This indicates a...

(A) Financing cash inflow

(B) Financing cash outflow

(C) Investing cash inflow

(D) Investing cash outflow

(E) None of the other answers

Solution. A growing *Treasury Stock* balance means the company repurchased shares — it paid cash to its own shareholders, so cash went **out**. And the category is **financing**, because financing transactions are, by the Chapter 1 definition, transactions between the company and its own investors: issuing stock, repurchasing stock, paying dividends, borrowing and repaying. *Investing* covers buying and selling long-lived assets. The reason (C) and (D) tempt anyone is the word 'stock,' which makes it feel like the company is trading securities as an investment. It isn't — it is buying back its *own* equity, which is financing by definition. So the answer is **(B)**.

6. What determines whether a preferred share is classified as equity or a liability?

(A) Its name: all preferred stock is equity under U.S. GAAP.

(B) Its name: all preferred stock is a liability under IFRS.

(C) Its contractual terms and the applicable classification requirements, including redemption and payment obligations.

(D) Whether its market price has risen.

Solution. The answer is **(C)**. Both systems examine the instrument's rights and obligations. Nonredeemable preferred shares with discretionary distributions can be equity; mandatory cash redemption can produce liability classification. U.S. SEC presentation also distinguishes certain redeemable instruments from permanent equity. There is no blanket GAAP-equity/IFRS-liability rule.

7. When a company engages in a **stock split**, this...

(A) increases its total ("book") Shareholders' Equity

(B) decreases its total ("book") Shareholders' Equity

(C) has no effect on its total ("book") Shareholders' Equity

(D) it depends

Solution. Ask the question that settles every one of these: **what did the company receive, and what did it give up?** In a split, nothing. No cash came in, no asset was acquired, no liability was incurred — there was no transaction with

an outside party at all. The company merely re-denominated its existing equity into more, smaller slices, and every shareholder's proportional claim is exactly what it was. Now, the *market* value of the equity might tick up a percent or two through the liquidity and psychology effects discussed in the chapter — but we do not write book equity up to market value, and never have in this text. So the total is untouched. The practical corollary is worth keeping: if you are handed a list of transactions and asked for ending total Shareholders' Equity, and a stock split appears in the middle of the list, skip straight over it. So the answer is **(C)**.

8. StoneCorp issues Common Stock with a stated par value of $1 per share. Below are some selected amounts reported from StoneCorp's balance sheet for 2020 and 2021.

	2021	2020
Common Stock	$15k	$15k
Additional Paid-in Capital	$350k	$400k
(Treasury Stock)	$100k	$200k

Based on this information alone, which of the below must have happened?

(A) StoneCorp issued more *Common Stock* over FY2020

(B) StoneCorp repurchased shares over FY2020

(C) StoneCorp re-sold shares from its Treasury, for greater than their original repurchase price

(D) StoneCorp re-sold shares from its Treasury, for less than their original repurchase price

(E) None of the other answers

Solution. Read the changes off the two columns, one account at a time. *Common Stock* is flat at $15k, so no new shares were issued. *Treasury Stock* **fell** by $100k, from $200k to $100k — and the only thing that reduces *Treasury Stock* is re-selling treasury shares. So a re-sale happened; the question is at what price. *APIC* also **fell**, by $50k. And we know exactly what a *debit* to *APIC* on a treasury re-sale means: the shares went out for **less** than they came in for.

We can even reconstruct the entry. The $100k credit to *Treasury Stock* is the shares at their repurchase cost; the $50k debit to *APIC* is the plug; so *Cash* received must have been $50k:

Cash (+A):	$50,000	
Additional Paid-in Capital (-S.E.):	$50,000	
Treasury Stock (-contra-S.E.):		$100,000

Reconstructed: StoneCorp's treasury re-sale during FY2021

Shares that had cost $100k were re-sold for $50k — half their repurchase price. Options (A) and (B) are wrong twice over: nothing here indicates an issuance or a repurchase, *and* the changes we can observe span FY2021, not FY2020 (we have no 2019 balance sheet, so we cannot see any FY2020 activity at all). Option (C) has the direction backwards — a re-sale above cost would have *credited* APIC and raised it. So the answer is **(D)**.

9. MarshallCo issues Common Stock with a stated par value of $1 per share. Below are some selected amounts reported from MarshallCo's balance sheet for 2018 and 2019.

	2019	2018
Common Stock	$10k	$10k
Additional Paid-in Capital	$900k	$900k
(Treasury Stock)	$200k	$0

Based on this information alone, which of the below can we determine:

(A) The number of shares outstanding as of the end of FY 2018

(B) The number of shares outstanding as of the end of FY2019

(C) The number of issued shares as of the end of FY 2019

(D) Two of the other answers

(E) Three of the other answers

Solution. Take the three in turn.

(A) — yes. *Common Stock* of $10k at $1 par means 10,000 shares **issued**. And at the end of 2018, *Treasury Stock* is $0 — nothing has been repurchased — so every issued share is still outstanding. **10,000 shares outstanding.**

(B) — no. During 2019, *Treasury Stock* went from $0 to $200k, so shares were certainly repurchased. But *Treasury Stock* records **dollars spent, not a count of shares**. Were those repurchases at $20 per share (10,000 shares — every share the company had!) or at $400 per share (a mere 500)? Nothing on the balance sheet tells us, so we cannot convert dollars into shares, and we cannot get outstanding shares.

(C) — yes. A repurchase debits *Treasury Stock* and leaves *Common Stock* alone, so the *Common Stock* balance still tracks the par value of every share ever issued. It is unchanged at \$10k, and *APIC* is unchanged too, so no new shares went out during 2019 either. **10,000 shares issued**, same as 2018.

Two of the three are determinable. So the answer is **(D)**.

A.15.2 Practice Problems: Open Prompts

1. **Journalize the following sequence of transactions:**

   - January 15, 2022: Issue 10,000 shares of *Common Stock*, with \$1 par value, for \$100 each.
   - February 15, 2022: Repurchase 1,000 of those shares, for \$80 each.
   - March 15, 2022: Re-sell 500 of *those* shares, for \$120 each.

   Solution. The three transactions are the issue–repurchase–re-issue cycle, and each gets its own rule.

   January 15 — issuance. Cash of 10,000 × \$100 = \$1,000,000. *Common Stock* gets the par value only: 10,000 × \$1 = \$10,000. *APIC* takes the \$990,000 remainder.

Cash (+A):	\$1,000,000	
Common Stock (+S.E.):		\$10,000
Additional Paid-in Capital (+S.E.):		\$990,000

Issue 10,000 shares of \$1 par stock at \$100, 1/15/2022

February 15 — repurchase. Debit *Treasury Stock* and credit *Cash* for the **full repurchase cost**, 1,000 × \$80 = \$80,000. Note what we do *not* do: we do not reverse any part of the January entry. The \$100 we originally sold these shares for is irrelevant here, and the \$1 of par is not clawed back out of *Common Stock*.

Treasury Stock (+contra-S.E.):	\$80,000	
Cash (-A):		\$80,000

Repurchase 1,000 shares at \$80, 2/15/2022

March 15 — re-sale. Cash of 500 × \$120 = \$60,000. *Treasury Stock* is credited at those shares' **repurchase** price, 500 × \$80 = \$40,000. The \$20,000 difference is a credit plug to *APIC* — not a Gain.

Cash (+A):	$60,000	
Treasury Stock (-contra-S.E.):		$40,000
Additional Paid-in Capital (+S.E.):		$20,000

Re-sell 500 treasury shares (repurchased at $80) for $120, 3/15/2022

Where does that leave us? *Common Stock* $10,000, *APIC* $1,010,000, and *Treasury Stock* with a $40,000 debit balance — the 500 shares still sitting in the treasury, at their $80 cost. Total Shareholders' Equity from these three transactions is $980,000, which is exactly the net cash that crossed the line: $1,000,000 in, $80,000 out, $60,000 in.

2. **Journalize the following sequence of transactions:**

   - January 15, 2022: Issue 20,000 shares of *Common Stock*, with $.01 par value, for $150 each.
   - February 15, 2022: Repurchase 2,000 of those shares, for $200 each.
   - March 15, 2022: Re-sell 1,000 of *those* shares, for $160 each.

   Solution. Same three rules, but with two twists worth watching for: a nearly-invisible par value, and a re-sale *below* the repurchase price.

   January 15 — issuance. Cash of 20,000 × $150 = $3,000,000. *Common Stock* gets 20,000 × $.01 = **$200**. This is the realistic case — the 'plug' dwarfing the account it plugs, exactly as on any real balance sheet.

Cash (+A):	$3,000,000	
Common Stock (+S.E.):		$200
Additional Paid-in Capital (+S.E.):		$2,999,800

Issue 20,000 shares of $.01 par stock at $150, 1/15/2022

February 15 — repurchase. 2,000 × $200 = $400,000, straight to *Treasury Stock*. Notice that the company is buying back at $200 shares it sold at $150 — which changes nothing about the entry. The repurchase is recorded at what we paid, full stop.

Treasury Stock (+contra-S.E.):	$400,000	
Cash (-A):		$400,000

Repurchase 2,000 shares at $200, 2/15/2022

March 15 — re-sale. Assume no APIC balance from previous treasury-stock transactions. Cash of 1,000 × $160 = $160,000. *Treasury Stock* is credited at the repurchase price, 1,000 × $200 = $200,000. Credits exceed debits by $40,000. With no treasury-stock APIC available, debit *Retained Earnings* for that shortfall:

Cash (+A):	$160,000	
Retained Earnings (-S.E.):	$40,000	
Treasury Stock (-contra-S.E.):		$200,000

Re-sell 1,000 treasury shares (repurchased at $200) for $160, 3/15/2022

The company received $40,000 less than the cost of the shares reissued, but records no Loss in Net Income. With no treasury-stock APIC available, the shortfall reduces Retained Earnings. From these transactions, Common Stock is $200, APIC is $2,999,800, and Treasury Stock has a $200,000 debit balance. Retained Earnings is $40,000 lower than it otherwise would have been.

3. JackCorp is a small company that provides tutoring to the most desperate and hopeless students. JackCorp's corporate charter states that its shares have a *par value* of $1 per share. At the beginning of FY2017, JackCorp has a balance of *Common Stock* of $10k, *Additional Paid-in Capital* of $990k, and *Retained Earnings* of $250k–and no other Shareholders' Equity accounts. Over the FY2017 reporting period, JackCorp had the following transactions and only these:

   - Issued an additional 1000 shares, for $100 each.
   - Provided tutoring to students for cash, $200k.
   - Provided tutoring to students on account, $100k.
   - Repurchase 500 shares, for $120 each.
   - Received pre-payment from customers of $50k, for services to be rendered in FY2018.
   - Purchased Supplies for Cash, $30k.
   - Paid a utilities bill in cash, $20k.

- Re-sell 200 of those shares, from the previous repurchase, for $150 each.
- Paid dividends to stockholders, $40k.

Solution. Before computing anything, journalize the nine transactions — and, as you go, sort them into the ones that touch Shareholders' Equity and the ones that don't. Three of these nine are set up to fool you on exactly that question, in two different directions, and getting them into the right bucket is the whole problem.

Cash (+A):	$100,000	
Common Stock (+S.E.):		$1,000
Additional Paid-in Capital (+S.E.):		$99,000

1. Issue 1,000 shares of $1 par stock at $100

Cash (+A):	$200,000	
Service Revenue (+S.E.):		$200,000

2. Tutoring services provided for cash

Accounts Receivable (+A):	$100,000	
Service Revenue (+S.E.):		$100,000

3. Tutoring services provided on account

Treasury Stock (+contra-S.E.):	$60,000	
Cash (-A):		$60,000

4. Repurchase 500 shares at $120

Cash (+A):	$50,000	
Deferred Revenue (+L):		$50,000

5. Customer pre-payment for FY2018 services

Supplies (+A):	$30,000	
Cash (-A):		$30,000

6. Purchase supplies for cash

Utilities Expense (-S.E.):	$20,000	
Cash (-A):		$20,000

7. Pay utilities bill in cash

Cash (+A):	$30,000	
Treasury Stock (-contra-S.E.):		$24,000
Additional Paid-in Capital (+S.E.):		$6,000

8. Re-sell 200 treasury shares (repurchased at $120) for $150

Dividends (-S.E.):	$40,000	
Cash (-A):		$40,000

9. Pay dividends to stockholders

Three of these are traps, and each is a different flavor of trap. **Transaction #5** is cash received, but for services not yet performed — a **liability** (*Deferred Revenue*), not revenue, and so not equity. **Transaction #6** swaps one asset for another; supplies become an expense only as they are *used*, and no adjusting entry has been given to us here. **Transaction #3** is the opposite trap: no cash at all, but the service *was* performed, so under accrual accounting the revenue is earned and equity rises by the full $100k.

Two more things worth flagging in the entries above. In **#8**, the credit to *Treasury Stock* is 200 × **$120** — the repurchase price, not the $150 re-sale price — and the $6,000 remainder goes to *APIC*, not to a Gain. And notice that nowhere in **#4** or **#8** does *Common Stock* move: repurchasing and re-selling shares never touches it.

4. **What would JackCorp's total Shareholders' Equity be as of the end of FY2017?**

Solution. First, roll up net income from the entries above. Revenues are $200,000 + $100,000 = $300,000; the only expense is the $20,000 utilities bill. So **Net Income = $280,000**. Dividends of $40,000 are not an expense — they are a distribution — so they hit *Retained Earnings* directly, alongside net income:

$$\Delta Retained\ Earnings = \$280,000 - \$40,000 = \$240,000$$

Now the full Statement of Shareholders' Equity, which is really just the four equity accounts marched down the page transaction by transaction:

	Common Stock	Additional Paid-in Capital	Retained Earnings	Treasury Stock	Total Shareholders' Equity
Balance, Jan 1, 2017	$10,000	$990,000	$250,000	$0	$1,250,000
Issue 1,000 shares @ $100	1,000	99,000			100,000
Repurchase 500 shares @ $120				(60,000)	(60,000)
Re-sell 200 treasury shares @ $150		6,000		24,000	30,000
Net Income			280,000		280,000
Dividends			(40,000)		(40,000)
Balance, Dec 31, 2017	**$11,000**	**$1,095,000**	**$490,000**	**($36,000)**	**$1,560,000**

JackCorp, Statement of Shareholders' Equity, FY2017

So **total Shareholders' Equity at the end of FY2017 is $1,560,000**. Reading the ending column down: $11,000 + $1,095,000 + $490,000 − $36,000 = $1,560,000. Note that *Treasury Stock* enters with a **minus sign** — it is a contra-equity account, and the 500 shares repurchased at $120 less the 200 re-sold (relieved at that same $120 cost) leave $36,000 of anti-equity sitting there.

Here is a second route to the same number, worth doing as a check because it is faster and it makes the logic vivid. Total equity changes only when equity-affecting transactions occur, so just net them:

$$\Delta S.E. = \$100,000 - \$60,000 + \$30,000 + \$280,000 - \$40,000 = \$310,000$$

And $1,250,000 + $310,000 = **$1,560,000**. (Beginning equity is $10k + $990k + $250k = $1,250k; there were no other equity accounts.) Notice what this second route makes obvious: the deferred revenue, the supplies purchase, and the accounts receivable balance simply never appear. The first two never touched equity at all, and the third already came in through revenue.

A.15.3 Essay and Discussion Questions

1. Do you think it makes more sense to think of Treasury Stock as a (negative) component of the company's paid-in capital or earned capital? Why? (Note, a good answer would first show the technical knowledge of what Treasury Stock is, how it arises on company's Balance Sheets, and what paid-in-capital and earned capital are, conceptually, before arguing your case...)

   Solution. Let me lay out the machinery first, then argue, and then say what I actually think — which, fair warning, is that the question does not have a clean answer, and that the *reason* it doesn't is the most valuable thing in it.

   The technical picture. Shareholders' Equity decomposes into **paid-in capital** — what shareholders literally handed the corporation in exchange for shares, tracked in *Common Stock* (at par) and *Additional Paid-in Capital* (the rest) — and **earned capital**, the profits the business generated and did not distribute, tracked in *Retained Earnings*. The dividing line is conceptual: money that came *in* from owners versus money the enterprise *made*. **Treasury Stock** arises when the company repurchases its own previously-issued shares from investors. We debit *Treasury Stock* for the full cash cost of the repurchase and credit *Cash* — deliberately *not* reversing the original *Common Stock* and *APIC* credits, because the repurchase price will almost never equal the issuance price, so a straight reversal wouldn't balance. *Treasury Stock* is therefore a **contra-Shareholders'-Equity** account with a debit balance, reported in parentheses, reducing total equity. And, unusually among contra-accounts, it isn't paired against any one account the way *Accumulated Depreciation* is paired against *Equipment*. It just stands there, a lump of anti-equity.

   The case for paid-in capital. A repurchase is, in the most literal sense available, the *reversal of a stock issuance*. The company issued a share and took in cash; now it hands the cash back and takes the share. *Common Stock* and *APIC* are positive components of paid-in capital, so their reversal ought to be a negative component of the same thing. The share-count bookkeeping supports this too: the whole story is about shares moving between the corporation and its owners, which is the paid-in-capital story from beginning to end. And there is a symmetry argument: when the company later re-sells those treasury shares, the plug goes to **APIC** — a paid-in-capital account. If the re-sale lands in paid-in capital, it is strange for the repurchase to have lived somewhere else.

   The case for earned capital. Repurchases and dividends are the same economic act: cash crossing the line from the corporate bank account into shareholders' bank accounts. That is not my editorializing — it is a celebrated theorem of fi-

nance that $X returned via repurchase and $X returned via dividend leave total shareholder wealth identical. And dividends unambiguously reduce **Retained Earnings**. So if two transactions are economically the same thing, why should one reduce earned capital and the other paid-in capital? Substance over form says treat them alike. There is a practical edge to this, too: analysts who care about total shareholder payouts have to add dividends and repurchases together anyway, and the accounts as presented make you go to two different places to do it.

Where I come out. Genuinely ambiguous — and I don't think GAAP draws a bright line either. What is telling is that real companies seem to feel the same ambivalence: look at Pfizer's balance sheet and you will find *Treasury Stock*, all $115 billion of it, sandwiched *between* the paid-in capital lines and *Retained Earnings*, as if the presentation itself is declining to commit. Other firms order it differently.

If pressed, I would lean paid-in, for one reason: what makes earned capital *earned* is that the enterprise generated it through operations, and a repurchase generates nothing — it is a transfer of pre-existing value between the company and its owners, with no productive activity anywhere in it. The dividend analogy is powerful, but notice that dividends reduce retained earnings not because they are cash-out-the-door but because *Retained Earnings* is definitionally the cumulative earnings **not yet distributed**. That is a definition about earnings, and a repurchase is not a distribution of earnings; it is a purchase of shares.

But I hold that loosely, and I would rather you take the meta-lesson. Our clean conceptual categories — paid-in versus earned, debt versus equity, asset versus expense — are enormously useful, and they blur at the edges as the world gets more complicated. That is not a scandal. It is what concepts do.

2. Do you think it makes more sense to think of *Preferred Stock* as a liability, or a part of shareholders' equity? Begin by distinguishing the particular contractual terms; neither IFRS nor U.S. GAAP classifies every preferred share the same way. An excellent answer to this essay prompt would cover *why* reasonable/informed people (such as those two bodies) might disagree (i.e., why *a* case could be made for *either*) before expressing your position and argument.

   Solution. Start with the two definitions we have been using since Chapter 1, because the whole disagreement lives in the gap between them. **Credit financing**: the investor hands over cash today and the company is *contractually obligated* to pay it back, with no upside beyond the contracted payments. **Equity financing**: the investor hands over cash today, gets no contractual entitlement to any cash flow, and instead becomes an *owner* — a residual claimant with voting control,

entitled to whatever is left after the creditors are satisfied.

Now hold preferred stock up against that. Its contract **pre-specifies** the distributions — the 'preferred dividends' — like a bond's coupon. The company is **not unconditionally obligated** to pay them, like equity. But *conditional on* wanting to pay anything to the common stockholders, it must pay the preferred holders first, and often the accumulated arrears from missed years as well. And preferred holders typically get **no votes** — so they lack the element of control we have associated with ownership since Chapter 1.

The analytical case for treating the stated instrument as debt-like. Look at substance over legal form. The distributions are fixed in amount and specified in advance — the defining feature of *fixed income*. The holders have no upside: if the company triples its profits, the preferred dividend does not budge, exactly like a bondholder. They have no votes, so they exercise none of the control that makes ownership ownership. The economic pressure to pay can still be strong, even when there is no unconditional legal obligation. The common stockholders are the owners; sooner or later they will demand a return; and nothing reaches them until the preferred holders are paid in full, arrears and all. So in practice the company will treat these very much like obligations. A claim with fixed payments, no upside, no control, and practical priority over the owners is, in substance, debt.

The case for treating the stated instrument as equity. Look at where the hard legal line falls. What makes a liability a liability is that failing to pay it triggers **default** — creditors can sue, force restructuring, push the firm into bankruptcy. Skipping a preferred dividend does none of that. The company can pass on it indefinitely and no court is involved, no covenant is breached, no acceleration clause fires. That is an enormous difference in a crisis, which is exactly when the debt-versus-equity classification matters most: preferred stock *absorbs* losses instead of *causing* failure, which is the functional signature of equity capital. And in liquidation, preferred holders stand behind every genuine creditor. There is also a rule-design argument: the boundary needs to be **verifiable and hard to game**, and 'is there an enforceable obligation?' is a bright line, whereas 'does it feel practically obligatory?' invites exactly the structuring games we saw with leases.

My view. I think it sits right in between, and I would not pretend otherwise — but if forced to file it in one drawer, I would file it with **liabilities** for reporting purposes, on the substance-over-form principle that runs through this whole book. The classification that matters most to actual users of financial statements is the leverage assessment, and a claim that has fixed payments, no upside, no votes,

and de facto priority over the owners behaves like leverage. Note that this is what sophisticated users already do: analysts and bankers computing leverage ratios from GAAP statements routinely reclassify preferred stock as debt in their own models — 'GAAP calls this equity, but it behaves like a liability, so I'll model it as one.' When the professionals systematically undo a classification before using the numbers, that is decent evidence the classification isn't serving them.

The honest caveat is that GAAP's bright-line rule has the virtue of being **hard to manipulate**, and that is not a small virtue. My preferred solution is not really to relabel it but to *disclose* it prominently enough that the reader can put it wherever their analysis requires — which, in practice, is roughly what happens.

The broader lesson, and the one to carry out of this chapter: classification in accounting is sometimes a judgment call, not a law of nature. A classification debate must start from the actual contractual terms, while distinguishing the reporting requirement from an analyst's preferred treatment. And wherever accounting draws a line between two categories, someone will engineer a contract that sits precisely on top of it — preferred stock is only the tamest example, with things like banks' AT1 bonds straddling the same line far more aggressively.

A.16 Chapter 16

Assumptions for reconstruction problems: Use U.S. GAAP cash-flow classifications. Unless the problem supplies additional transactions, assume the stated operating balances change only through the related revenues, expenses, cash receipts, and cash payments. For simplified investing and financing reconstructions, assume no unstated purchases and disposals that offset each other, noncash acquisitions, debt repayments, share retirements, share-based compensation, or other changes. Treat dividends as cash dividends declared and paid in the same period, with no other direct adjustments to Retained Earnings. These are problem assumptions, not conclusions that can generally be drawn from two Balance Sheets.

A.16.1 Practice Problems: MCQ's

1. Under the categorization scheme used for the **cash flow statement**, cash paid for salaries would be...

   (A) Operating

   (B) Investing

   (C) Financing

   (D) None of the other answers

   Solution. Paying employees is about as close to the beating heart of "normal short-term business operations" as you can get—the workers show up, do the work of selling goods and services to customers, and get paid. There is no long-term asset being bought or sold here (so not investing), and no transaction in our own stock or debt (so not financing). So the answer is **(A)**.

2. Under the categorization scheme used for the **cash flow statement**, cash paid for *Inventory* would be...

   (A) Operating

   (B) Investing

   (C) Financing

   (D) None of the other answers

   Solution. This is the one designed to catch you. *Inventory* is an asset—we *capitalize* the expenditure rather than expensing it immediately—so the word "investing" practically leaps to mind. But the operating/investing line is not drawn between capitalizing and expensing; it follows the nature of the activity. Some

current investments are investing items, while Inventory purchases are operating. *Inventory* is a current asset, so buying it is an **operating** outflow. (Remember: there is *not* a 1-to-1 mapping from capitalizing vs. expensing to investing vs. operating.) So the answer is **(A)**.

3. Under the categorization scheme used for the **cash flow statement**, cash paid (by our company) for *interest* on loans would be...

   (A) Operating

   (B) Investing

   (C) Financing

   (D) None of the other answers

   Solution. Here is the counterintuitive exception we flagged in the chapter. Interest payments are obviously *about* our debt financing, so "financing" is the intuitive guess—and it is wrong. Cash interest paid is classified as **operating**, and the reason is a deliberate design choice: *Interest Expense* sits on the Income Statement, so putting cash interest in the operating section keeps that section comparable to the Income Statement, which is much of what makes it useful as a check on it. (Symmetrically, cash interest *received* is operating too.) So the answer is **(A)**.

4. Under the categorization scheme used for the **cash flow statement**, cash paid for *Equipment* would be...

   (A) Operating

   (B) Investing

   (C) Financing

   (D) None of the other answers

   Solution. *Equipment* is a long-term asset, and investing activities are, by definition, the purchase and sale of long-term assets. So the answer is **(B)**.

5. Under the categorization scheme used for the **cash flow statement**, cash paid for *Trademarks* would be...

   (A) Operating

   (B) Investing

   (C) Financing

   (D) None of the other answers

Solution. The category does not care whether the long-term asset is tangible or intangible. A *Trademark* is a long-term asset, so buying one is an **investing** outflow, exactly like buying *Equipment* or *Land*. So the answer is **(B)**.

6. Under the **direct method** for the operating section of the cash flow statement, in adjusting Salaries Expenses to Cash Paid for Salaries, what would we do with the Salaries Payable reported on the Balance Sheet?

   (A) Add the total Salaries Payable for that year

   (B) Subtract total Salaries Payable for that year

   (C) Add the change in Salaries Payable between that year and the previous

   (D) Subtract the change in Salaries Payable between that year and the previous

   (E) None of the other answers

   Solution. Two decisions here, and the question is testing both. *Which direction?* This is an expense, and our rule for expenses is to **subtract** the change in the related liability. *Which quantity—the level or the change?* Always the **change**. The balance in *Salaries Payable* at year-end is the accumulated stock of unpaid wages, some of it inherited from prior periods; what we want is what happened *this* period, and that is the flow, not the stock. Formally, $Cash\ Paid\ for\ Salaries = Salaries\ Expense - \Delta(Salaries\ Payable)$. So the answer is **(D)**.

7. Under the **direct method** for the operating section of the cash flow statement, in adjusting Interest Expense to Cash Paid for Interest, what would we do with the Interest Payable reported on the Balance Sheet?

   (A) Add the total Interest Payable for that year

   (B) Subtract total Interest Payable for that year

   (C) Add the change in Interest Payable between that year and the previous

   (D) Subtract the change in Interest Payable between that year and the previous

   (E) None of the other answers

   Solution. Identical logic to the previous question, and that is the point—once you have the rule, the particular expense does not matter. *Interest Expense* is an expense with no related asset account, so the whole adjustment runs through the payable, and for an expense we subtract the change in the related liability: $Cash\ Paid\ for\ Interest = Interest\ Expense - \Delta(Interest\ Payable)$. If *Interest Payable* rose, we accrued interest we did not pay, so cash paid was *less* than the expense. So the answer is **(D)**.

8. Under the **indirect method** for the operating section of the cash flow statement, what is the adjustment we make for *Accounts Receivable*?

   (A) Add the total Accounts Receivable for that year

   (B) Subtract total Accounts Receivable for that year

   (C) Add the change in Accounts Receivable between that year and the previous

   (D) Subtract the change in Accounts Receivable between that year and the previous

   (E) None of the other answers

   Solution. Under the indirect method the working-capital rule is: **subtract** changes in non-cash current assets, **add** changes in current liabilities. *A/R* is a non-cash current asset, so we subtract its change. The intuition, if you prefer it in words: a rising *A/R* means we booked revenue in *Net Income* that we have not yet collected in cash, so on the way from NI down to CFO we have to take it back out. And note again that it is the *change*, not the balance—the balance includes receivables generated in earlier periods, which have nothing to do with this year's cash. So the answer is **(D)**.

9. Suppose that, unlike in the example used in the main text above, our company had a $10k *Gain* upon the sale of the land. Then, under the **indirect method** for the operating section of the cash flow statement, we would [add / subtract / no change] that $10k *Gain*?

   (A) Add the $10k Gain

   (B) Subtract the $10k Gain

   (C) No adjustment for Gains under the indirect method

   (D) None of the other answers

   Solution. The indirect method starts at *Net Income*, and the $10k *Gain* is already sitting inside it, having pushed it *up* by $10k. But the cash from selling that land belongs in the **investing** section, not operating—so it must not be left in CFO. To get it out, we subtract it. That is Mental Rotation #1: what we "skip over" under the direct method, we "back out" under the indirect method, and backing out a *Gain* means subtracting. (Option (C) is the trap for anyone who half-remembers "gains and losses aren't real cash flows so we ignore them." They are ignored in the *destination*; they are precisely *not* ignored in the *reconciliation*.) So the answer is **(B)**.

10. Under the **indirect method** for the operating section of the cash flow statement, what is the adjustment we make for *Salaries Payable*?

(A) Add the total *Salaries Payable* for that year

(B) Subtract total *Salaries Payable* for that year

(C) Add the change in *Salaries Payable* between that year and the previous

(D) Subtract the change in *Salaries Payable* between that year and the previous

(E) None of the other answers

Solution. *Salaries Payable* is a current liability, and under the indirect method we **add** the change in current liabilities. In words: an increase in *Salaries Payable* means we recognized *Salaries Expense*—already deducted in arriving at *Net Income*—that we never actually paid in cash, so we add it back on the way to CFO. Compare this to question 6, and notice the sign has flipped: under the *direct* method we subtracted this same change (to compute *Cash Paid for Salaries*, which was then itself subtracted); under the *indirect* method we enter the adjustment directly, so it appears with the opposite sign. That is Mental Rotation #2. So the answer is **(C)**.

11. Suppose that our company had Prepaid Rent reported as a current asset on our Balance Sheet. What adjustment would we make for it under the direct method, in adjusting *Rent Expense* to *Cash Paid for Rent*?

(A) Add the change in Prepaid Rent to *Rent Expense*

(B) Subtract the change in Prepaid Rent from *Rent Expense*

(C) Do nothing with Prepaid Rent.

(D) Debit Prepaid Rent

(E) None of the other answers

Solution. *Prepaid Rent* is to *Rent Expense* exactly what *Inventory* is to *COGS*: the related **asset** account. Our direct-method rule for expenses says to *add* the change in the related asset, so $\textit{Cash Paid for Rent} = \textit{Rent Expense} + \Delta\,\textit{Prepaid Rent}$ (plus a $-\Delta\,\textit{Rent Payable}$ term, if the company had one). Sanity-check it: if *Prepaid Rent* went up \$5k, we handed the landlord \$5k of cash that has not yet been used up and therefore has not yet hit the Income Statement—so cash paid exceeds the expense by \$5k. Option (D) is a category error: a journal-entry instruction has no business in a cash flow statement. So the answer is **(A)**.

12. Suppose that our company had Prepaid Rent reported as a current asset on our Balance Sheet. What adjustment would we make for it under the indirect method for the operating section?

(A) Add the change in Prepaid Rent

(B) Subtract the change in Prepaid Rent.

(C) Do nothing with Prepaid Rent.

(D) Debit Prepaid Rent

(E) None of the other answers

Solution. Under the indirect method you do not even have to know what *Prepaid Rent* is "for." It is a non-cash current asset, and the one working-capital rule says: subtract the change in non-cash current assets. Done. (And note the sign has flipped relative to question 11—the same mental rotation as in question 10.) So the answer is **(B)**.

13. Which of the following could explain why a company might have a much higher *Net Income*, relative to its *Cash Flows from Operations* (i.e., high 'net accruals')?

(A) A large decrease in *Accounts Receivable*–i.e., many accounts from past period's sales paid off

(B) A large buildup in *Inventory*

(C) Making major new investments in long-term assets like *Land* and *Trademarks*

(D) Multiple other answers

(E) None of the other answers

Solution. We want NI to sit *above* CFO, which means we want adjustments that are **negative** in the reconciliation. Take them in turn. (A) A *decrease* in *A/R* is a negative change in a current asset, and we subtract that change—subtracting a negative is adding. That pushes CFO *up*, the wrong way. (B) A buildup in *Inventory* is a positive change in a current asset, which we subtract; CFO falls below NI. That is the warehouse-full-of-unwanted-goods story from the chapter—the company is bleeding cash into inventory while the Income Statement, which only recognizes *COGS* when goods are *sold*, sails along unbothered. That works. (C) Purchases of *Land* and *Trademarks* are **investing** outflows; they never touch the operating section at all, so they cannot open a gap between NI and CFO. (D) requires two or more of the above, and only one qualifies. So the answer is **(B)**.

14. Suppose that every one of a company's Long-Term Liabilities and Shareholders' Equity accounts reported on its Balance Sheet is completely unchanged from last period to this period. The company earned Net Income of $20 over the period. What is its Net Cash Flows from Financing?

(A) $0

(B) –$20

(C) +$20

(D) Not enough information

(E) None of the other answers.

Solution. The tempting answer is $0—nothing moved, so nothing happened. But this question is testing the one financing cash flow that is not *reported* anywhere: *Dividends*. Recall the identity Δ *Retained Earnings* = *Net Income* − *Dividends*. We are told every S.E. account is unchanged, which includes *Retained Earnings*, so $\Delta RE = \$0$, and therefore $Dividends = \$20 - \$0 = \$20$. The company earned $20 for its shareholders and *Retained Earnings* did not budge—so the whole $20 walked out the door to the shareholders. With no bond or stock activity to offset it, net financing cash flow is an outflow of $20. So the answer is **(B)**.[3]

15. In order to calculate the *Dividends* paid (a key financing cash outflow), we would...

(A) Look up the *Dividends* reported on the Income Statement

(B) Look up the *Dividends* reported on the Balance Sheet

(C) *Infer* the *Dividends* using the *Retained Earnings* identity

(D) Calculate the *Dividends* from the change in *Treasury Stock*

(E) None of the other answers

Solution. *Dividends* are not an expense, so they are not on the Income Statement (that has been true since Chapter 1—dividends are a distribution *of* profit, not a cost of earning it). And *Dividends* is a temporary account, closed out at year-end, so it is not on the Balance Sheet either. They are genuinely invisible. But they leave a footprint: *Dividends* = *Net Income* − Δ *Retained Earnings*. Option (D) confuses two different ways of returning cash to shareholders—*Treasury Stock* tracks share **repurchases**, which are a separate financing outflow. So the answer is **(C)**.

16. Suppose that we note that a company's Treasury Stock account has increased in magnitude from FY2019 to FY2020. This indicates a...

(A) Financing cash inflow

[3] Assume, for the sake of this problem—and consistent with the current stage of this undergraduate textbook—that we are preserving the long-term vs. current mapping onto financing vs. operating.

(B) Financing cash outflow

(C) Investing cash inflow

(D) Investing cash outflow

(E) None of the other answers

Solution. *Treasury Stock* records repurchases of the company's **own** shares—a transaction in how it finances itself, so financing, not investing. And buying back your own stock means writing a check, so it is an **outflow**. Conveniently, *Treasury Stock* is the simplest account in this whole exercise: par value plays no role in repurchases, so we debit *Treasury Stock* and credit *Cash* for the full amount, and you can read the cash outflow straight off the change in the account. So the answer is **(B)**.

17. Suppose that we note that a company's *Bonds Payable* account (tracking the total face value of outstanding bonds) has increased in magnitude from FY2019 to FY2020. This indicates a...

(A) Financing cash inflow

(B) Financing cash outflow

(C) Investing cash inflow

(D) Investing cash outflow

(E) None of the other answers

Solution. More bonds outstanding means the company issued bonds, which means investors handed it cash. Transactions in the company's *own* debt are financing, and this one brings money in. (Note the "investing" options are there to catch the confusion between buying *another* company's bonds—investing—and issuing your *own*—financing.) So the answer is **(A)**.

18. Suppose that we note that a company's *Bonds Payable* account (tracking the total face value of outstanding bonds) has decreased in magnitude from FY2019 to FY2020. This indicates a...

(A) Financing cash inflow

(B) Financing cash outflow

(C) Investing cash inflow

(D) Investing cash outflow

(E) None of the other answers

Solution. The mirror image of the previous question. Face value outstanding went down, which means bonds were repaid at maturity, called, or repurchased—in every one of those cases, cash left the company to go to its creditors. Financing, outflow. So the answer is **(B)**.

19. Suppose that our company determines that our *Land* is impaired—triggering us to "write down" the *Land* and record a corresponding *Loss*. Then, that amount would be...

    (A) ... an operating cash inflow

    (B) ... an operating cash outflow

    (C) ... an investing cash inflow

    (D) ... an investing cash outflow

    (E) None of the above.

    Solution. Ask the only question that matters for this statement: **did cash move?** It did not. An impairment is a bookkeeping write-down—debit *Loss*, credit *Land*—and not one dollar left the building. So it appears nowhere on the Statement of Cash Flows as a cash flow of any category. It *does* show up in the operating section under the indirect method, but only as an add-back in the reconciliation—an adjustment removing a non-cash charge from *Net Income*, which is exactly the opposite of being a cash flow. This is the case from Chapter 12 that so annoys analysts: an enormous reported *Loss* in a year when nothing happened to the cash. So the answer is **(E)**.

20. Suppose that our company's Balance Sheet reports a balance of *Cash* of \$700k for FY 2020, and \$950k for FY 2021. We have filled out the first two sections of the cash flow statement (correctly), and calculated that we had net operating cash flows of \$200k and net investing cash flows of -\$300k, for the period. Based on this information, what must our net financing cash flows be?

    (A) \$250k

    (B) \$300k

    (C) \$350k

    (D) Not enough information

    (E) None of the other answers

    Solution. This question runs the statement's foundational identity in reverse. The three sections must decompose the *total* change in cash, so $X + Y + Z =$

$\Delta\,Cash$. Here $\Delta Cash = \$950k - \$700k = \$250k$, and we are given $X = \$200k$ and $Y = -\$300k$. So:

$$Z = \$250k - \$200k - (-\$300k) = \$350k$$

Option (A) is there for anyone who grabs the change in cash itself; option (B) for anyone who hands back the investing figure's magnitude without ever running the identity. And this is the same arithmetic as the check you should run at the bottom of every cash flow statement you build—here you are just solving for a different unknown. So the answer is **(C)**.

21. Is the following statement true or false? A company can increase its Cash Flows From Operations by using a more aggressive depreciation method (i.e., increasing its *Depreciation Expense* for the year).

    (A) True

    (B) False

    Solution. This is the interview question from the chapter, and the answer is **false**—for two reasons that are really one reason. Conceptually: cash is cash. Changing a depreciation *assumption* is a change in an accounting allocation; it cannot conjure or destroy a dollar that customers did or did not hand over. Mechanically: suppose we crank *Depreciation Expense* from \$30k up to \$40k. *Net Income*, the starting line of the reconciliation, falls by \$10k. The add-back on the very next line rises by \$10k. The two moves cancel exactly, and CFO is unchanged—which is, of course, precisely what makes CFO attractive to people who distrust accrual estimates. So the answer is **(B)**.

22. Is the following statement true or false? A company's *net cash from operations* may turn out to be different, depending on whether it uses the direct method or indirect method for the operating section.

    (A) True

    (B) False

    Solution. False. The direct and indirect methods are two *presentations* of the same reconciliation, using the same concepts and—as the side-by-side comparison in the chapter shows—literally the same adjustments, just factored differently. The direct method converts each income-statement line to its cash analogue and nets them; the indirect method starts at *Net Income* and applies those same adjustments at a summary level, with the expense adjustments'

signs flipped. They must arrive at the same number, and if yours don't, one of them is wrong. (In the chapter's example, both give $56k.) So the answer is **(B)**.

A.16.2 Practice Problems: Open Prompts

1. Cash Flow Statement #1

   Fill out a cash flow statement, in its entirety, using the *indirect method* for the operating section, based on the income statement and balance sheets below.

Revenue:	**$200k**
COGS:	**$80k**
Salaries expenses:	**$40k**
Depreciation expenses:	**$30k**
Loss on sale of land:	**$5k**
Interest expense:	**$10k**
Income tax expense:	**$10k**
Net Income:	**$25k**

Table A.3: MattCorp Income Statement, For the Year Ended December 31, 2021:

	2021	**2020**
Assets:		
Cash	**$230k**	**$220k**
Accounts Receivable	**$90k**	**$70k**
Inventory	**$120k**	**$140k**
Investments	**$130k**	**$0**
Land	**$220k**	**$280k**
Equipment	**$250k**	**$200k**
Accumulated Depreciation	**$(80k)**	**$(50k)**
Total assets:	**$960k**	**$860k**
Liabilities:		
Accounts payable	**$80k**	**$90k**
Salaries payable	**$30k**	**$20k**
Interest payable	**$7k**	**$12k**
Income Tax payable	**$16k**	**$15k**
Notes payable	**$260k**	**$220k**
Shareholders' Equity:		
Common Stock	**$432k**	**$383k**
Retained Earnings	**$135k**	**$120k**
Total Liabilities and S.E.:	**$960k**	**$860k**

Table A.4: MattCorp Balance Sheets, Year Ended December 31:

Now, re-do the operating section for the income statement and balance sheet above, using the *direct method*.

Solution. Before touching anything, write down the number we are aiming at: *Cash* went from \$220k to \$230k, so the finished statement must foot to a **net change in cash of \$10k**. That is our check at the end, and you should always take it before you start, so you cannot fool yourself afterwards.

Now work the three sections in order.

Operating (indirect). Start at *Net Income* of \$25k and apply the two rules. Rule #1, back out the non-cash and non-operating items: add *Depreciation Expense* of \$30k (its related cash flows are investing, so it must not survive into CFO), and add the \$5k *Loss on Sale of Land* (same reason—the cash from that sale belongs in the investing section, where it will show up in full). Rule #2, everything else is working capital: subtract changes in non-cash current assets, add changes in current liabilities.

- *Accounts Receivable* rose \$20k (\$70k → \$90k). We booked revenue we did not collect. **Subtract \$20k.**
- *Inventory* fell \$20k (\$140k → \$120k). Goods were expensed through *COGS* that we had paid for in an earlier period. Subtracting a decrease means **adding \$20k.**
- *Accounts Payable* fell \$10k (\$90k → \$80k). We paid down suppliers—cash out, but not this period's expense. Adding a decrease means **subtracting \$10k.**
- *Salaries Payable* rose \$10k. Wages accrued but unpaid. **Add \$10k.**
- *Interest Payable* fell \$5k. We paid off previously accrued interest. **Subtract \$5k.**
- *Income Tax Payable* rose \$1k. **Add \$1k.**

That gives $\$25k + \$30k + \$5k - \$20k + \$20k - \$10k + \$10k - \$5k + \$1k = \$56k$ of CFO.

Investing. Look at the long-term assets, one at a time. *Investments* went from \$0 to \$130k: an increase in a long-term asset is a purchase, and absent any statement to the contrary we assume it was in cash—a **\$130k outflow**. *Land* fell \$60k (\$280k → \$220k), and *Land* is neither depreciated nor (here) impaired, so by elimination the company **sold land with a book value of \$60k**. That is not the cash figure, though; the *Loss* triangulates it for us: $Sale\ Price = Book\ Value + Gain\ (Loss) = \$60k - \$5k = \$55k$, a **\$55k inflow**. *Equipment* rose \$50k; since *Accumulated Depreciation* is tracked separately, depreciation does not touch the

Equipment account, and there is no gain or loss on equipment, so this is cleanly a **$50k purchase**. Net investing: $(125k).

Financing. *Notes Payable* rose $40k—the company borrowed, a **$40k inflow**. *Common Stock* rose $49k, and with no *APIC* account this is no-par stock, so the entire proceeds hit *Common Stock*: a **$49k inflow**. And then the item that is reported nowhere: *Retained Earnings* rose only $15k ($120k → $135k) while the company earned $25k. The missing $10k went out the door as **dividends**. Net financing: $79k.

Operating cash flows:	
Net Income:	**$25k**
... adjustments to reconcile NI to CFO	
+ Depreciation Expense	$30k
+ Loss on Sale of Land	$5k
- Δ Accounts Receivable	($20k)
- Δ Inventory	$20k
+ Δ Accounts Payable	($10k)
+ Δ Salaries Payable	$10k
+ Δ Interest Payable	($5k)
+ Δ Income Tax Payable	$1k
Net change in cash from operations:	**$56k**
Investing Cash Flows:	
Purchase of Investments	($130k)
Cash from Sale of Land	$55k
Purchase of Equipment	($50k)
Net Cash Flows from Investing	**($125k)**
Financing Cash Flows:	
Issuance of Notes Payable	$40k
Issuance of Common Stock	$49k
Dividends Paid Out	($10k)
Net Cash Flows from Financing	**$79k**
Net change in cash:	**$10k**

MattCorp Cash Flow Statement, For the Year Ended December 31, 2021 (indirect method)

The check. $\$56k - \$125k + \$79k = \$10k$, and the Balance Sheets show *Cash* rising

from $220k to $230k. It foots. Sigh of relief.

Now the operating section again, by the direct method. Same adjustments, different packaging: instead of starting from *Net Income*, we walk down the Income Statement converting each line to its cash analogue. Revenues get the change in the related asset *subtracted*; expenses get the change in the related asset *added* and the change in the related payable *subtracted*; *Depreciation* and the *Loss* are simply skipped, because they have no operating cash analogue to be converted into.

$$Cash\ Collected\ from\ Customers = \$200k - (\$90k - \$70k) = \$180k$$

$$Cash\ Paid\ for\ Inventory = \$80k + (\$120k - \$140k) - (\$80k - \$90k) = \$80k - \$20k + \$10k = \$70k$$

$$Cash\ Paid\ for\ Salaries = \$40k - (\$30k - \$20k) = \$30k$$

$$Cash\ Paid\ for\ Interest = \$10k - (\$7k - \$12k) = \$15k$$

$$Cash\ Paid\ for\ Income\ Tax = \$10k - (\$16k - \$15k) = \$9k$$

Cash Collected from Customers:	**$180k**
Cash paid to suppliers for Inv.:	($70k)
Cash paid for salaries:	($30k)
Cash paid for interest:	($15k)
Cash paid for income taxes:	($9k)
Net Cash Flows from Operations:	**$56k**

MattCorp Operating Section (Direct Method), FY ending Dec. 31, 2021

$56k, the same number, as it must be. The investing and financing sections are unchanged—the direct/indirect distinction never touches them.

2. Cash Flow Statement #2

   Fill out a cash flow statement, in its entirety, using the *indirect method* for the operating section, based on the income statement and balance sheets below.

Revenue:	**$100k**
COGS:	**$40k**
Salaries expenses:	**$20k**
Depreciation expenses:	**$15k**
Loss on sale of land:	**$5k**
Interest expense:	**$5k**
Income tax expense:	**$5k**
Net Income:	**$10k**

Table A.5: MattCorp Income Statement, For the Year Ended December 31, 2021:

	2021	2020
Assets:		
Cash	**$105k**	**$120k**
Accounts Receivable	**$45k**	**$35k**
Inventory	**$70k**	**$60k**
Investments	**$65k**	**$0**
Land	**$100k**	**$130k**
Equipment	**$125k**	**$100k**
Accumulated Depreciation	**$(40k)**	**$(25k)**
Total assets:	**$470k**	**$420k**
Liabilities:		
Accounts payable	**$40k**	**$45k**
Salaries payable	**$15k**	**$10k**
Interest payable	**$4k**	**$6k**
Income Tax payable	**$8k**	**$9k**
Notes payable	**$130k**	**$110k**
Shareholders' Equity:		
Common Stock	**$11k**	**$10k**
Additional Paid-in Capital	**$184k**	**$160k**
Retained Earnings	**$78k**	**$70**
Total Liabilities and S.E.:	**$470k**	**$420k**

Table A.6: MattCorp Balance Sheets, Year Ended December 31:

Now, re-do the operating section for the income statement and balance sheet above, using the *direct method*.

Solution. Take the target first: *Cash* fell from $120k to $105k, so this statement

must foot to **negative $15k**. Notice that this company's cash went *down* in a profitable year—a nice illustration of the whole point of the chapter, and something we should be able to explain by the time we are done.

Operating (indirect). Start at *Net Income* of $10k. Rule #1: add back $15k of *Depreciation Expense* and the $5k *Loss on Sale of Land*, both of which reduced NI but whose related cash flows are investing. Rule #2, the working-capital adjustments:

- *Accounts Receivable* rose $10k ($35k → $45k): revenue recognized, cash not collected. **Subtract $10k.**
- *Inventory* rose $10k ($60k → $70k): cash (or credit) went into goods that have not been expensed through *COGS* yet. **Subtract $10k.**
- *Accounts Payable* fell $5k ($45k → $40k): we paid suppliers down. **Subtract $5k.**
- *Salaries Payable* rose $5k: expense recognized, wages unpaid. **Add $5k.**
- *Interest Payable* fell $2k ($6k → $4k): we paid off accrued interest. **Subtract $2k.**
- *Income Tax Payable* fell $1k ($9k → $8k). **Subtract $1k.**

CFO $= \$10k + \$15k + \$5k - \$10k - \$10k - \$5k + \$5k - \$2k - \$1k = \$7k$. And there is our first clue about the cash decline: almost every working-capital line here runs *against* the company. It is financing its customers (*A/R* up), stuffing cash into *Inventory*, and paying down its own payables—so $10k of *Net Income* converts into only $7k of cash.

Investing. *Investments* went $0 → $65k: a **$65k purchase**, assumed in cash. *Land* fell $30k ($130k → $100k); land is not depreciated and there is no impairment here, so the company sold land with a book value of $30k, and the $5k *Loss* tells us the price: $\$30k - \$5k = \$25k$, a **$25k inflow**. *Equipment* rose $25k ($100k → $125k), with *Accumulated Depreciation* tracked separately and no gain or loss on equipment, so that is a clean **$25k purchase**. Net investing: $-\$65k + \$25k - \$25k = \$(65k)$.

Financing. *Notes Payable* rose $20k: a **$20k borrowing inflow**. Now the stock, which is where this problem differs from the last one: this company reports *both Common Stock* and *Additional Paid-in Capital*, so its shares carry a par value and the issuance proceeds are split across the two accounts. From Chapter 15, the entry to issue par stock debits *Cash* for the full proceeds and credits *Common Stock* for par and *APIC* for the rest—so the two accounts *together* track the cash.

Common Stock rose \$1k and *APIC* rose \$24k, so the company collected **\$25k** issuing stock. (If you took only the \$1k change in *Common Stock*, you would have missed 96% of the cash. This is the trap.) Finally, dividends: NI of \$10k against a *Retained Earnings* increase of \$8k (\$70k → \$78k) means **\$2k paid out**. Net financing: $\$20k + \$25k - \$2k = \$43k$.

Operating cash flows:	
Net Income:	**\$10k**
... adjustments to reconcile NI to CFO	
+ Depreciation Expense	\$15k
+ Loss on Sale of Land	\$5k
- Δ Accounts Receivable	(\$10k)
- Δ Inventory	(\$10k)
+ Δ Accounts Payable	(\$5k)
+ Δ Salaries Payable	\$5k
+ Δ Interest Payable	(\$2k)
+ Δ Income Tax Payable	(\$1k)
Net change in cash from operations:	**\$7k**
Investing Cash Flows:	
Purchase of Investments	(\$65k)
Cash from Sale of Land	\$25k
Purchase of Equipment	(\$25k)
Net Cash Flows from Investing	**(\$65k)**
Financing Cash Flows:	
Issuance of Notes Payable	\$20k
Issuance of Common Stock (incl. APIC)	\$25k
Dividends Paid Out	(\$2k)
Net Cash Flows from Financing	**\$43k**
Net change in cash:	**(\$15k)**

MattCorp Cash Flow Statement, For the Year Ended December 31, 2021 (indirect method)

The check. $\$7k - \$65k + \$43k = -\$15k$, and *Cash* did indeed fall from \$120k to \$105k. It foots. And now we can answer the question we posed at the start: the company burned cash not because operations lost money, but because it spent \$90k acquiring investments and equipment while its operations threw off only

\$7k, and it raised \$43k to cover part of the gap. $X > 0, Y \ll 0, Z > 0$—a company in an expansion phase, funding growth partly out of operations and mostly out of the capital markets and its own cash balance.

Now the operating section again, by the direct method.

$$Cash\ Collected\ from\ Customers = \$100k - (\$45k - \$35k) = \$90k$$

$$Cash\ Paid\ for\ Inventory = \$40k + (\$70k - \$60k) - (\$40k - \$45k) = \$40k + \$10k + \$5k = \$55k$$

$$Cash\ Paid\ for\ Salaries = \$20k - (\$15k - \$10k) = \$15k$$

$$Cash\ Paid\ for\ Interest = \$5k - (\$4k - \$6k) = \$7k$$

$$Cash\ Paid\ for\ Income\ Tax = \$5k - (\$8k - \$9k) = \$6k$$

Cash Collected from Customers:	**\$90k**
Cash paid to suppliers for Inv.:	(\$55k)
Cash paid for salaries:	(\$15k)
Cash paid for interest:	(\$7k)
Cash paid for income taxes:	(\$6k)
Net Cash Flows from Operations:	**\$7k**

MattCorp Operating Section (Direct Method), FY ending Dec. 31, 2021

$\$90k - \$55k - \$15k - \$7k - \$6k = \$7k$. Same destination.

3. Cash Flow Statement #3:

   Fill out a cash flow statement, in its entirety, using the *indirect method* for the operating section, based on the income statement and balance sheets below. Note that, in the below, the years for the Balance Sheet are flipped (2020 in the left column, 2021 in the right column).

Revenue:	**$200k**
COGS:	**($80k)**
Salaries expenses:	**($40k)**
Depreciation expenses:	**($30k)**
Gain on sale of land:	**$10k**
Interest expense:	**($10k)**
Income tax expense:	**($10k)**
Net Income:	**$40k**

Table A.7: MattCorp Income Statement, For the Year Ended December 31, 2021:

	2020	**2021**
Assets:		
Cash	**$230k**	**$220k**
Accounts Receivable	**$90k**	**$70k**
Inventory	**$120k**	**$140k**
Land	**$260k**	**$200k**
Equipment	**$250k**	**$250k**
Accumulated Depreciation	**$(50k)**	**$(80k)**
Total assets:	**$X**	**$Y**
Liabilities:		
Accounts payable	**$80k**	**$90k**
Salaries payable	**$30k**	**$20k**
Interest payable	**$7k**	**$12k**
Income Tax payable	**$16k**	**$15k**
Notes payable	**$380k**	**$220k**
Shareholders' Equity:		
Common Stock	**$380k**	**$420k**
Retained Earnings	**$W**	**$Z**
Total Liabilities and S.E.:	**$X**	**$Y**

Table A.8: MattCorp Balance Sheets, Year Ended December 31:

Now, re-do the operating section using the *direct method*.

Solution. This one throws two wrenches at you, so slow down. **Wrench one:** the columns are reversed. Every "change" in this problem runs *left to right*, not right to left as in every other balance sheet you have seen. If you go on autopilot, every single adjustment will carry the wrong sign, and—this is the diabolical part—the

statement will still foot, to exactly the wrong number. So the very first thing to do is write, at the top of your page: **2020 is the left column, 2021 is the right column, $\Delta = right - left$. Wrench two:** *Retained Earnings* is not given, so we cannot read dividends off the Balance Sheet directly. We will have to be detectives.

Start with the target. *Cash* went from \$230k (2020) to \$220k (2021): a **\$10k decrease**. Note in passing that a careless reader who took the left column as the current year would set out to hit \$+10k and would arrive, triumphantly, at a completely wrong statement.

Digging out W, X, Y, and Z. The Balance Sheet balances—that is the whole point of it—so we can use that fact to recover the missing numbers. Total 2020 assets:

$$X = \$230k + \$90k + \$120k + \$260k + \$250k - \$50k = \$900k$$

$$Y = \$220k + \$70k + \$140k + \$200k + \$250k - \$80k = \$800k$$

Now set liabilities plus equity equal to each. For 2020, the named accounts sum to $\$80k + \$30k + \$7k + \$16k + \$380k + \$380k = \$893k$, so $W = \$900k - \$893k = \$7k$. For 2021, they sum to $\$90k + \$20k + \$12k + \$15k + \$220k + \$420k = \$777k$, so $Z = \$800k - \$777k = \$23k$. *Retained Earnings* therefore rose by \$16k while the company earned \$40k—so **dividends were \$24k**.

Operating (indirect). Start at *Net Income* of \$40k. Rule #1: add back \$30k of *Depreciation*, and—note the sign, this is the difference from the previous two problems—**subtract** the \$10k *Gain on Sale of Land*. The *Gain* was added in arriving at *Net Income*, but the cash it represents belongs in investing, so it has to come back out. Rule #2, working capital, remembering the direction of time:

- *Accounts Receivable* **fell** \$20k (\$90k → \$70k): we collected on old accounts, cash in without revenue this period. Subtracting a decrease means **adding \$20k.**
- *Inventory* **rose** \$20k (\$120k → \$140k): cash tied up in goods not yet sold. **Subtract \$20k.**
- *Accounts Payable* **rose** \$10k: expenses incurred, suppliers not paid. **Add \$10k.**
- *Salaries Payable* **fell** \$10k: we paid down what we owed employees. **Subtract \$10k.**
- *Interest Payable* **rose** \$5k: accrued but unpaid interest. **Add \$5k.**
- *Income Tax Payable* **fell** \$1k. **Subtract \$1k.**

$\text{CFO} = \$40k + \$30k - \$10k + \$20k - \$20k + \$10k - \$10k + \$5k - \$1k = \$64k$.

Investing. This company has only two long-term assets. *Land* fell \$60k (\$260k → \$200k)—not depreciated, not impaired, so sold, at a book value of \$60k. With a \$10k *Gain*: $Sale\ Price = \$60k + \$10k = \$70k$, an inflow. *Equipment* is **unchanged** at \$250k in both years, so there were no equipment purchases or sales at all; the movement in *Accumulated Depreciation* from \$50k to \$80k is fully explained by the \$30k *Depreciation Expense*, which is not a cash flow. Net investing: **\$70k**, positive—a company selling off part of its asset base.

Financing. *Notes Payable* **fell** \$160k (\$380k → \$220k): the company paid off debt, a \$160k **outflow**. *Common Stock* rose \$40k, with no *APIC* account, so \$40k of no-par stock was issued—an inflow. And dividends of \$24k, as we inferred above, out the door. Net financing: $-\$160k + \$40k - \$24k = \$(144k)$.

Operating cash flows:	
Net Income:	**\$40k**
... adjustments to reconcile NI to CFO	
+ Depreciation Expense	\$30k
- Gain on Sale of Land	(\$10k)
- Δ Accounts Receivable	\$20k
- Δ Inventory	(\$20k)
+ Δ Accounts Payable	\$10k
+ Δ Salaries Payable	(\$10k)
+ Δ Interest Payable	\$5k
+ Δ Income Tax Payable	(\$1k)
Net change in cash from operations:	**\$64k**
Investing Cash Flows:	
Cash from Sale of Land	\$70k
Net Cash Flows from Investing	**\$70k**
Financing Cash Flows:	
Repayment of Notes Payable	(\$160k)
Issuance of Common Stock	\$40k
Dividends Paid Out	(\$24k)
Net Cash Flows from Financing	**(\$144k)**
Net change in cash:	**(\$10k)**

MattCorp Cash Flow Statement, For the Year Ended December 31, 2021 (indirect method)

The check. $\$64k + \$70k - \$144k = -\$10k$, and *Cash* fell from \$230k to \$220k. It foots—which also retroactively validates the \$7k and \$23k we dug out for *Retained Earnings*, since a wrong dividend figure would have shown up here immediately.

Worth a pause on the *shape* of this statement: $X > 0, Y > 0, Z \ll 0$. Operations are healthy, but the company is selling off land and using the proceeds, along with its operating cash, to retire a large slug of debt. Positive investing cash flows are the unusual signature here—as the chapter noted, a company with persistently positive investing cash flows is a company shrinking its asset base. Once might be deliberate deleveraging; year after year, it is a distress signal.

Now the operating section again, by the direct method. Same discipline about the column order.

$$Cash\ Collected\ from\ Customers = \$200k - (\$70k - \$90k) = \$200k + \$20k = \$220k$$

$$Cash\ Paid\ for\ Inventory = \$80k + (\$140k - \$120k) - (\$90k - \$80k) = \$80k + \$20k - \$10k = \$90k$$

$$Cash\ Paid\ for\ Salaries = \$40k - (\$20k - \$30k) = \$50k$$

$$Cash\ Paid\ for\ Interest = \$10k - (\$12k - \$7k) = \$5k$$

$$Cash\ Paid\ for\ Income\ Tax = \$10k - (\$15k - \$16k) = \$11k$$

Cash Collected from Customers:	**\$220k**
Cash paid to suppliers for Inv.:	(\$90k)
Cash paid for salaries:	(\$50k)
Cash paid for interest:	(\$5k)
Cash paid for income taxes:	(\$11k)
Net Cash Flows from Operations:	**\$64k**

MattCorp Operating Section (Direct Method), FY ending Dec. 31, 2021

$\$220k - \$90k - \$50k - \$5k - \$11k = \$64k$. The *Gain* and the *Depreciation* were

simply skipped, since neither has an operating cash analogue. Same \$64k as the indirect method.

4. Open quantitative problem:

   (Note, you may use whatever method you prefer to get to the number being elicited in this question). FertittaCorp reports net income of \$40 million. We are additionally given the following information: depreciation expense was \$5 million and the company recorded a gain on the sale of land of \$10 million (there were no other gains/losses or amortization). Finally, we are able to determine that accounts receivable decreased by \$3m, accounts payable decreased by \$4m, and inventory increased by \$2m. Those were the only working capital accounts. What were FertittaCorp's **net cash flows from operations**?

   Solution. We are handed *Net Income* and a list of adjustments, which is to say we are handed the indirect method whether we like it or not. Run the two rules.

   Rule #1, back out the non-cash and non-operating items: **add** the \$5m of *Depreciation Expense*, and **subtract** the \$10m *Gain* (the cash from that land sale is investing-section business; it must not be double-counted here).

   Rule #2, working capital—subtract changes in current assets, add changes in current liabilities:

   - *Accounts Receivable* **decreased** \$3m. The change is $-\$3m$; subtracting it **adds \$3m**. (Collections on old accounts: cash in, no revenue this period.)
   - *Inventory* **increased** \$2m. **Subtract \$2m.**
   - *Accounts Payable* **decreased** \$4m. It is a liability, so we add the change—and the change is $-\$4m$, so we **subtract \$4m**. (We paid suppliers down.)

   Putting it together:

   $$CFO = \$40m + \$5m - \$10m + \$3m - \$2m - \$4m = \$32m$$

   So **net cash flows from operations were \$32 million**. The one place to be careful is the two *decreases*: a decrease in an asset helps CFO and a decrease in a liability hurts it, which is exactly the reverse of the increases you see most often. Keep the rule stated in terms of the signed *change* and the signs take care of themselves—do not try to memorize four separate cases.

5. Quantitative problem: Our company issues *Common Stock* with *par value* of \$1 per share. We report the following amounts for 2020 and 2021:

	2020	2021
Common Stock	\$1m	\$1.5m
APIC	\$10m	\$40m
Treasury Stock	\$0	\$0

What was our financing cash inflow from issuing *Common Stock* in FY2021?

Solution. Assume that the only changes in Common Stock and APIC were cash issuances, with no repurchases, retirements, share-based compensation, or other noncash changes. The zero Treasury Stock balances alone do not establish those facts.

Now the key point, which is the whole reason par value was mentioned in the stem. Recall the issuance entry from Chapter 15: debit *Cash* for the **total proceeds**, credit *Common Stock* for the **par value** of the shares issued, and credit *APIC* for the remainder. So neither account alone tells you the cash—but the two of them **together** do. Over FY2021—the period between the two reported Balance Sheet dates—*Common Stock* rose by \$0.5m and *APIC* rose by \$30m, so the company collected:

$$Cash\ from\ Issuing\ Stock = \Delta\, Common\ Stock + \Delta\, APIC = \$0.5m + \$30m = \$30.5m$$

So the financing cash inflow was **\$30.5m**. Anyone who reported \$0.5m—reading the change in *Common Stock* and stopping there—understated the company's stock issuance by about 98%.

And a free bonus inference, since we were told the par value. At \$1 par, a \$0.5m increase in *Common Stock* means **500,000 new shares** were issued. Divide the proceeds by the shares and you have the average issue price: $\$30.5m/500{,}000 =$ \$61 per share. That is the kind of thing you can pull out of a Balance Sheet with nothing but the journal entry in your head.

A.16.3 Essay / Discussion Questions

1. What explains the need and demand for the Statement of Cash Flows—that is, for companies to report their performance on a cash basis? That is, what are the defects or limitations in the Income Statement that make it an inadequate or imperfect way of measuring a company's performance? And how might the cash flow statement therefore help us? In short, what do we get from the Statement of Cash Flows?

Solution. Here is how I would build the answer: start by granting that accrual accounting is *good*, because otherwise the rest of the argument has no shape. Recognizing revenue when earned and matching expenses against it is a genuine intellectual achievement—it tries to measure whether the company is selling above cost and creating value on an ongoing basis, rather than reporting the accident of which checks happened to clear before December 31. Nobody wants to go back to a shoebox of receipts.

But that achievement is purchased with **estimates and discretion**, and that is where the trouble starts. Depreciation lives on assumed useful lives and salvage values. The *Allowance for Uncollectible Accounts*—and therefore *Bad Debt Expense*, and therefore the bottom line—rests on a manager's judgment about who will pay. *Deferred Revenue* rests on judgments about when a service is delivered; recall *Zynga*, where bad news about customer churn was transmuted, via a change in a deferred-revenue estimate, into *higher* reported revenue right before an IPO. Impairments let a company report an enormous *Loss* in a year when not one additional dollar left the building. So there are two distinct problems, and it is worth keeping them apart: managers can deliberately **"massage"** earnings, *and*—even with entirely honest managers—the sheer baroque complexity of accrual accounting can make the Income Statement an **opaque** window onto the business.

Then there is the second, quite separate argument, which is not about manipulation at all: **cash is king**. You cannot pay the bank in receivables. A company can be handsomely profitable on an accrual basis and still default, if its *Accounts Receivable* will not convert to cash in time or if its purchases of *Inventory* have quietly consumed everything. The inventory case is the insidious one, because accrual accounting is *working as designed* while it happens: goods pile up in a warehouse that nobody wants, *COGS* is not recognized because nothing has been sold, and *Net Income* sails along unbothered while cash drains away.

So what do we *get*? Three things, I would say. First, a performance measure—*Cash Flows from Operations*—that is immune to the specific gimmicks accruals are vulnerable to, because cash is cash. Second, and less obviously, an **explicit reconciliation**: the indirect-method operating section is a line-by-line decomposition of the gap between the two bottom lines, so when NI and CFO diverge, the statement itself tells you which accounts are responsible. Third, the decomposition into **operating, investing, and financing**, which answers the question the Balance Sheet's cash line cannot: not *how much* did cash change, but *why*. A company whose cash rose because it issued bonds and one whose cash rose because customers paid it are in wholly different situations; lumping them together

throws away the information we came for.

2. In your opinion, overall, does the Income Statement or the Cash Flow Statement provide the more informative perspective and measure of a company's performance over a period? Why?

   Solution. When I ask this in class right after teaching this chapter, the vote is nearly unanimous for the Cash Flow Statement—and I understand why, since a chapter's worth of net income's defects is the freshest thing in everyone's mind. But I will make the unfashionable case, and I want you to be able to argue both sides regardless of where you land.

   Two arguments against putting all your weight on cash flows. **First, cash measures can be gamed too.** Run the thought experiment: we adopt a policy of systematic evasiveness about paying our employees and suppliers—another month, another quarter, we stall. Our Cash Flow Statement improves *mechanically*, because we are simply not writing the checks. Our Income Statement does not improve at all: the employees still worked, the suppliers still delivered, the expenses are still incurred, and the unpaid amounts just pile up in *Payables*. Accrual accounting sees straight through the maneuver, because seeing through exactly that sort of thing is what it was built for. So the manipulability argument runs in both directions.

   Second, and deeper: CFO makes no charge whatsoever for long-term assets. Look at the structure of the reconciliation. The working-capital adjustments are timing differences—the check hasn't come, then it comes—and over time they reverse and wash out. But the *Depreciation* add-back never reverses. It recurs every period, always with the same sign, for as long as the company owns long-term assets, which for a normal operating company is forever. That is why CFO systematically exceeds NI. And what it means is that CFO grades a company without ever charging it for the factories and machines it consumes—real cash, spent through the investing section, that must be spent again to replace them before a dollar can reach shareholders. The Income Statement charges those costs, smoothly, as *Depreciation* and *Amortization*. (The identical critique applies to **EBITDA**, which is why Buffett has spent decades ridiculing it.)

   So my verdict, if genuinely forced to pick one: old-school accrual *Net Income*. It runs on estimates that can be massaged, but it charges the company for all the resources it consumes, and my reading of the empirical research is that accrual earnings tend to carry more information about future performance and value than raw cash flows do. But notice how artificial the forced choice is. These statements were designed as a **system**, each viewing the same reality from a different

angle and each checking the others. The honest answer is that the *gap* between them is often more informative than either one—which is precisely what the accruals anomaly suggests.

3. Suppose that a company has very solid Net Income but poor Cash Flows from Operations. What are some of the "usual suspects" that could account for this difference—that is, what kinds of accounts/activities, etc. could drive a large wedge between NI and CFO, and why?

   Solution. The elegant thing about this question is that you do not have to guess. Whatever is driving the wedge between NI and CFO *is*, literally, the list of adjustments in the indirect-method operating section. That is what the reconciliation is. So the way to answer is to walk the reconciliation and ask which lines could be large and negative.

   Ballooning *Accounts Receivable.* The company is recognizing revenue on credit sales and not collecting the cash—it is, in effect, financing its customers. Benign interpretation: it is growing fast and receivables scale with sales. Worrying interpretation: it is booking revenue from customers who cannot or will not pay, or channel-stuffing to hit a number. Either way, a rising *A/R* subtracts directly from CFO.

   Ballooning *Inventory.* The chapter's warehouse story. The company is spending cash on goods, capitalizing them rather than expensing them, and *COGS* only hits the Income Statement when something is *sold*. So *Net Income* is untouched by a pile of unsold product, while the cash is very much gone. This is the one I would look at hardest, because it can be invisible in earnings for quite a while.

   Shrinking *Accounts Payable* (and other payables). The company is paying down what it owes suppliers, employees, and the tax authorities. Those payments are cash out, but they relate to expenses recognized in *prior* periods, so they are nowhere in this period's NI. Note that this one also runs in reverse, and that the reverse is a *warning* rather than a comfort: a company whose CFO looks strong because *Accounts Payable* is ballooning is being propped up by stretching its suppliers, which is not a repeatable strategy.

   Large *Depreciation* charges. These push NI down while leaving cash untouched, and so they widen the gap in the *other* direction—NI below CFO. So if you observe the opposite pattern, solid NI with weak CFO, you should be struck by it: the structural tilt of the reconciliation runs the other way, and something in working capital must be overwhelming it.

   Now the necessary caveat, and it is not a hedge. None of these proves anything sinister. Growing companies legitimately build up receivables and inventory;

that is what growth *looks like* on a Balance Sheet. What the pattern establishes is that a wedge exists and that you now know exactly which account is responsible. The reconciliation is a **diagnostic instrument**, not a verdict—it tells the analyst where to start digging.

A.16.4 Technical written-response questions

1. Why is it that when calculating Cash paid to Suppliers (for Inventory) using the Direct Method (in the example in the homework and covered in class), we calculated it as... $Cash\ paid\ to\ suppliers\ (for\ Inventory) = COGS + \Delta Inventory - \Delta Accounts\ Payable$. Why do we make *two* adjustments in this case? What does each one do?

   Solution. Because there are **two gaps** between *COGS* and cash, not one, and each adjustment closes exactly one of them. The cleanest way to see it is to notice that we are not making one leap—we are making two, in sequence, and it is worth doing them slowly and separately before you let the formula collapse them.

   Step one: from *COGS* to *Inventory Purchases*. *COGS* measures inventory **used up** this period. It says nothing about how much we **bought**. Those differ whenever the warehouse level moves, and the gap is stored in the *Inventory* account, per the identity we have leaned on all book long: $Ending\ Balance = Beginning\ Balance + Purchases - Related\ Expense$. Rearranged for inventory:

   $$Inventory\ Purchases = COGS + \Delta\ Inventory$$

   In words: if *Inventory* grew, we must have bought more than we sold, by exactly that amount. That is the "+ Δ Inventory" term.

   Step two: from *Inventory Purchases* to *Cash Paid*. We now know what we bought—but not what we *paid for*. Purchases on account are purchases without cash, and that gap is stored in *Accounts Payable* (which, by the simplifying assumption we adopt throughout this book, relates entirely to inventory purchases):

   $$Cash\ Paid\ for\ Inventory = Inventory\ Purchases - \Delta\ Accounts\ Payable$$

   In words: if *A/P* grew, some of those purchases were financed by our suppliers rather than paid for, so cash out was less than purchases by that amount. That is the "−Δ Accounts Payable" term. Substituting step one into step two gives the formula in the question.

So: **the *Inventory* adjustment converts an expense into a purchase; the *Accounts Payable* adjustment converts a purchase into a payment.** And that also answers the implicit second half of the question—why *COGS* takes two adjustments when *Salaries Expense* and *Interest Expense* take only one. Those expenses have no related asset account, because we do not capitalize them; the moment they are incurred they are on the Income Statement, so step one is trivial and only the payable adjustment remains. *COGS* is different precisely because inventory expenditures are **capitalized first and expensed later**—which inserts a whole extra stage between the cash and the expense, and therefore an extra adjustment on the way back.

2. Why is it that when calculating cash flow from operations (CFO) using the indirect method we add back *Depreciation Expense* but subtract *Gains* at the start? Do *Depreciation Expenses* and *Gains* increase cash flows from operations? If not, why do we make these adjustments at the start of the operating section under the indirect method? Help me make sense of this.

   Solution. Let me answer the middle question first, because it is the one causing the trouble: **no.** Neither *Depreciation* nor a *Gain* increases cash flows from operations by one cent. Nothing you do to a depreciation assumption changes the cash your operations generated, and the cash from selling a piece of land is investing cash, not operating cash. If the adjustments *were* claiming otherwise, the objection in the question would be devastating.

   So what are they doing there? Here is the short answer: **they are in the reconciliation precisely *because* they do not belong in the destination.** That sounds like a koan, so let me unwind it.

   The indirect method does not compute CFO from scratch. It builds a **bridge** from *Net Income* to CFO. And a bridge is defined entirely by the difference between its two ends. *Depreciation Expense* was subtracted on the way down to *Net Income*; it must *not* be reflected in *Cash Flows from Operations*; therefore the bridge between them has to put it back. The add-back is not an assertion that depreciation generated cash—it is the removal of an item that has no business being in the destination. (I find it helps to picture the "ghost of the Income Statement" hovering above the statement, with *Net Income* at its bottom, physically joined to the top line of the operating section. Everything in the reconciliation is undoing something that ghost did.)

   Note the sign, which follows mechanically from that framing rather than from any intuition about cash. *Depreciation* and *Losses* were **subtracted** in getting to NI, so we **add** them back. A *Gain* was **added** in getting to NI, so we **subtract** it.

There is nothing to memorize beyond "undo what the Income Statement did." And notice the exact correspondence with the direct method: there, we simply **skipped over** these lines, because there was no operating cash flow for them to be converted into. Skipping in the direct method and backing out in the indirect method are the same act performed from two different starting points—that is Mental Rotation #1.

Two final points, because they are where the real understanding lives. **First, be suspicious of the popular one-liner** that "we add back depreciation because it's a non-cash expense." It explains less than it appears to. *Everything* in this reconciliation is non-cash—*Salaries Expense* is not cash, *Revenue* is not cash; the entire project is converting non-cash accruals into cash. What actually distinguishes *Depreciation* is that it is non-cash **and non-operating**: its related cash flows, the purchase and eventual sale of the asset, are classified as **investing**. Same for the *Gain*.

Second, and this matters more than it looks: a *Gain* is not a fake number. You will hear it said that gains and losses are mere accounting plugs, not real cash flows. That is wrong, and getting it wrong will cost you later in the problem. Since $Sale\ Price = Book\ Value + Gain\ (Loss)$, the gain is a genuine component of a genuine cash inflow. We back it out of the operating section not because it is imaginary, but because it is **misfiled**—and a couple of sections later, in the investing section, it reappears, doing essential work: it is exactly what lets us triangulate the cash sale price from the asset's prior book value. So a *Gain* removed from operating does not vanish. It gets escorted to the section where it belongs.

B

Misc. Additional Proofs

B.1 Proof that Present Value Equals Face Value When Coupon Rate Equals Market Rate

Note: Credit to Jack Zhou, my Fall 2019 student and research assistant on the original edition, for this.

The following proof is for those of you who are not satisfied by simply pasting the calculations into Google.

We would like to prove the following identity when coupon rate equals market interest rate.

$$Present\ Value = Face\ Value$$

Suppose a 5-year bond with coupon rate at 6% and face value at $10,000, and the market rate is also 6%, the following equation can be set up with the left side representing the equation for present value and the right side representing the face value:

$$\frac{600}{1.06} + \frac{600}{(1.06)^2} + \frac{600}{(1.06)^3} + \frac{600}{(1.06)^4} + \frac{10600}{(1.06)^5} = 10000$$

To generalize this, if we label face value as $\boldsymbol{x}$ and interest rate as $\boldsymbol{i}$, then the equation can be transformed as follows:

$$\frac{xi}{(1+i)}+\frac{xi}{(1+i)^2}+\frac{xi}{(1+i)^3}+\frac{xi}{(1+i)^4}+\frac{x(1+i)}{(1+i)^5}=x$$

which can be generalized to the following for a bond with $\boldsymbol{n}$ years to maturity:

$$\frac{xi}{(1+i)}+\frac{xi}{(1+i)^2}+\ldots+\frac{xi}{(1+i)^{n-2}}+\frac{xi}{(1+i)^{n-1}}+\frac{x(1+i)}{(1+i)^n}=x$$

This can then be rewritten as:

$$xi(1+i)^{n-1}+xi(1+i)^{n-2}+\ldots+xi(1+i)^2+xi(1+i)+x(1+i)=x(1+i)^n$$

The $\boldsymbol{x}$ on the left hand side can be factored out:

$$x(i(1+i)^{n-1}+i(1+i)^{n-2}+\ldots+i(1+i)^2+i(1+i)+(1+i))=x(1+i)^n$$

Which can be further simplified by combining the two $(1+i)$:

$$\begin{aligned}x(i(1+i)^{n-1}+i(1+i)^{n-2}+\ldots+i(1+i)^2+(i+1)(1+i))&=x(1+i)^n\\ x(i(1+i)^{n-1}+i(1+i)^{n-2}+\ldots+i(1+i)^2+(1+i)^2)&=x(1+i)^n\end{aligned}$$

Continuing this operation...

$$\begin{aligned}x(i(1+i)^{n-1}+i(1+i)^{n-2}+\ldots+(i+1)(1+i)^2)&=x(1+i)^n\\ x(i(1+i)^{n-1}+i(1+i)^{n-2}+\ldots+i(1+i)^3+(1+i)^3)&=x(1+i)^n\\ &\ldots\\ x((i+1)(1+i)^{n-1})&=x(1+i)^n\\ x(1+i)^n&=x(1+i)^n\end{aligned}$$

Simplifying, we get the identity that we started with:

$$x=x$$

$$\textit{Present Value}=\textit{Face Value}$$

Therefore, algebraically, when the market's interest rate equals the coupon rate, the present value of the bond equals its face value.

Index

accelerated depreciation, 243
account, 34
accrual accounting, 361
acquisition, 273
activity-based depreciation, 241
additional paid-in capital, 352
adjusted earnings, 270
adjusting entries, 79
aging of receivables, 187
amortization, 247
amortized cost, 322
anti-smoothing, 269
APIC, 352
authorized stock, 341

Balance Sheet, 17
balance sheet, 44
bank run, 322
bonds, 7, 303
book value, 111

cash basis accounting, 361
Cash Flow Statement, 27
cash flows from operations, 361
cash is king, 361
cogs, 197
collection on account, 56
common stock, 35
common-size income statement, 221
conservatism, 19
consideration, 260
contra-account, 111
contra-equity accounts, 65
convergence, 31
cost of equity, 288
coupon payment, 304
coupon rate, 282, 304
covenants, 14
credit, 7, 62
credit sale, 46
credit sales, 161
creditors, 7
current assets, 108
current ratio, 127

days in inventory, 221
debit, 62
debtholders, 7
deferred revenue, 48
depletion, 246
depreciation, 240
dilution, 125
direct writeoff method, 165
discount rate, 285
discounted cash flow (DCF), 291
dividends, 13, 53
DuPont decomposition, 121

earned capital, 338
earnings management, 142
earnings per share, 354
earnings vs. cash flows, 395
EBIT, 104, 239, 376
EBITDA, 397
equipment, 38
equity, 6

face value, 282, 304
fair value, 255
FASB, 31
FIFO, 201
financing, 6
financing section, 387
fixed income, 281, 303
free cash flow, 395
freight, 224
fungibility, 284
future value, 289

GAAP, 31
gain, 234

general ledger, 66
Generally Accepted Accounting Principles, 31
Gestalt shift, 86
goodwill, 273
gross profit, 218
gross profit margin, 221

historical cost, 255
home equity, 331
hostile takeover, 343

IASB, 31
IFRS, 31
Income Statement, 20
indirect method, 376
interest expense, 312
interest rate, 285
interest-rate risk, 322
internal transactions, 79
International Accounting Standards Board, 31
International Financial Reporting Standards, 31
inventory, 197
inventory costing methods, 201
inventory turnover, 221
investing section, 383
issued stock, 341

lease assets, 332
leases, 332
levels of profitability, 217
leverage, 128
leveraged buyout, 220
life cycle, 393
LIFO, 201
liquidity, 107, 108
loans, 7
loss, 234

M&A, 273
manufacturing companies, 198
matching, 21, 134
maturity date, 282
MD&A, 30
merchandising companies, 198
merger, 273
mergers and acquisitions, 220
Modigliani-Miller, 347
mortgage, 130
mortgages, 331

Net Accounts Receivable, 168
net assets, 51, 97
net present value (NPV), 297
Net Sales, 194
non-cash consideration, 260
non-cash liability, 49
non-GAAP earnings, 270
notes payable, 37
notes receivable, 190

off balance sheet liabilities, 332
operating income, 239
other comprehensive income, 353
outstanding stock, 341

paid-in capital, 338
par value, 344
patents, 248
percent-of-credit sales, 168
poison pill, 343
preferred stock, 355
prepaid rent, 40
prepayment, 48
present value, 289

receivables, 161
refinancing, 325
related-party transaction, 264
rent, 40
residual claimant, 9
retailers, 198
Retained Earnings, 25
return, 289
return on assets, 120
return on equity, 117
revenue recognition, 21, 48, 134
risk, 128

salaries, 51
salaries expense, 51
salaries payable, 53
sale on account, 46
sales, 161
sales discount, 193
salvage value, 89
service revenue, 44

SG&A, 218
share repurchases, 347
shareholders' equity, 337
shipping, 224
specific identification, 201
standard setters, 30
Statement of Cash Flows, 27
statement of cash flows, 153, 361
Statement of Shareholders' Equity, 25
statement of shareholders' equity, 337
stock buybacks, 347
stock splits, 357
supplies, 42

T-account, 67
tangible asset, 90
tax accounting, 213
taxes, 213
the direct method, 367
The Financial Accounting Standards Board, 31
trademarks, 248
transfer pricing, 264
Treasury Stock, 349
treasury stock, 341
trial balance, 68

utilities expense, 56

weighted-average cost, 201
wholesalers, 198
working capital, 152, 381

Zynga, 142